Appendix B: Detailed computational results

In this section we present detailed computational results. In Tables 16-135 we report, for every problem instance considered in our campaign, the dual bound (under column DB), primal bound (under column PB), relative gap (under column GAP), and CPU time in seconds (under column CPU). For the solvers CN24 and NAIVE in addition we report the total number of linear pieces (under column NP). For our solver, in addition we report the number of iterations of the main loop of our method (under column NIT). We also report the total number of problems solved and the shifted geometric means of the computing times, and the gaps, in each table.

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
10-1	483.55	994.66	51.39	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	384.64	1180.73	67.42	3600.10	$-\infty$	$+\infty$	100.00	3601.19	73240	980.15	980.15	**0.00**	**494.93**	555	30	⋯
10-10	364.58	744.17	51.01	3600.01	369.70	744.17	50.32	3603.29	273.06	972.63	71.93	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	729.86	729.86	**0.00**	**451.08**	555	24	⋯
10-2	414.07	911.41	54.57	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	306.33	1283.86	76.14	3600.10	$-\infty$	$+\infty$	100.00	3601.16	80145	905.37	905.38	**0.00**	**436.86**	632	15	⋯
10-3	427.48	810.39	47.25	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	322.75	1030.31	68.67	3600.10	$-\infty$	$+\infty$	100.00	3601.17	74058	806.75	806.75	**0.00**	**346.03**	529	25	⋯
10-4	377.74	850.89	55.61	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	292.09	1197.68	75.61	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	846.58	846.59	**0.00**	**649.61**	520	19	⋯
10-5	502.61	820.99	38.78	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	357.34	987.34	63.81	3600.10	820.99	820.99	**0.00**	3205.48	72284	820.99	820.99	**0.00**	**162.35**	466	14	⋯
10-6	356.01	689.89	48.40	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	272.80	854.67	68.08	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	688.54	688.54	**0.00**	**313.81**	546	21	⋯
10-7	350.33	745.18	52.99	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	258.31	908.98	71.58	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	734.51	734.51	**0.00**	**930.46**	557	20	⋯
10-8	346.56	756.03	54.16	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	253.06	947.96	73.30	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	737.09	737.09	**0.00**	**416.52**	569	18	⋯
10-9	362.89	935.98	61.23	3600.01	387.77	998.34	61.16	3605.55	291.70	1310.94	77.75	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	918.78	921.41	**0.29**	3601.02	446	14	⋯
15-1	461.62	1124.56	58.95	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1077.66	1123.89	**4.11**	3601.02	831	8	⋯
15-10	422.89	1013.59	58.28	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	985.34	1016.15	**3.03**	3601.02	830	9	⋯
15-2	459.88	1040.74	55.81	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1035.62	1037.40	**0.17**	3601.01	871	19	⋯
15-3	426.93	1008.56	57.67	3600.01	477.21	1009.95	52.75	3602.94	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	968.65	1004.32	**3.55**	3601.02	805	7	⋯
15-4	384.31	960.70	60.00	3600.01	398.95	954.71	58.21	3605.47	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	901.14	949.48	**5.09**	**3599.24**	850	12	⋯
15-5	602.40	1357.00	55.61	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1356.85	1357.00	**0.01**	3601.02	886	13	⋯
15-6	369.98	839.01	55.90	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	832.78	835.64	**0.34**	3601.02	832	13	⋯
15-7	425.85	1021.00	58.29	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	995.88	1020.36	**2.40**	3601.02	817	11	⋯
15-8	469.59	1068.77	56.06	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1053.83	1060.43	**0.62**	3601.02	860	14	⋯
15-9	369.82	791.31	53.26	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	790.71	791.32	**0.08**	3601.02	929	19	⋯
20-1	611.42	1253.17	51.21	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	452.80	3005.20	84.93	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1042.03	1227.22	**15.09**	3601.02	1325	2	⋯
20-10	632.49	1489.65	57.54	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	484.30	3393.08	85.73	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1400.16	1467.06	**4.56**	3601.12	1288	4	⋯
20-2	548.83	1255.24	56.28	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	403.32	2414.98	83.30	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1184.38	1270.72	**6.79**	3601.03	1354	4	⋯
20-3	574.30	1373.94	58.20	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	437.42	3003.36	85.44	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1159.19	1389.76	**16.59**	3601.02	1340	2	⋯
20-4	509.93	1065.57	52.14	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	351.28	2539.49	86.17	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1018.52	1080.36	**5.72**	3601.02	1378	4	⋯
20-5	585.77	1314.76	55.45	3600.01	641.48	1390.64	53.87	3603.04	452.68	2566.83	82.36	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1125.40	1320.37	**14.77**	3601.03	1314	2	⋯
20-6	526.43	1337.63	60.64	3600.01	574.39	1385.94	58.56	3602.95	408.70	2803.08	85.42	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1182.15	1302.16	**9.22**	3601.02	1331	4	⋯
20-7	474.69	919.51	48.38	3600.01	506.74	938.22	45.99	3603.06	345.71	2049.29	83.13	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	905.15	923.64	**2.00**	3601.01	1403	8	⋯
20-8	591.34	1450.79	59.24	3600.01	639.04	1471.73	56.58	3608.78	456.65	2644.20	82.73	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1281.52	1434.40	**10.66**	3601.02	1307	3	⋯
20-9	574.12	1426.00	59.74	3600.01	621.77	1474.18	57.82	3603.12	432.41	3164.83	86.34	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1201.35	1420.55	**15.43**	3601.02	1347	2	⋯
5-1	417.40	417.40	**0.00**	141.85	417.40	417.40	**0.00**	59.21	417.40	417.40	**0.00**	991.70	417.40	417.40	**0.00**	289.96	19657	417.40	417.40	**0.00**	**42.91**	177	20	⋯
5-10	577.69	577.69	**0.00**	66.62	577.69	577.69	**0.00**	278.64	505.63	605.19	16.45	3600.10	577.69	577.69	**0.00**	179.14	18304	577.69	577.69	**0.00**	**38.49**	171	13	⋯
5-2	279.62	279.62	**0.00**	**13.47**	279.63	279.63	**0.00**	23.30	279.63	279.63	**0.00**	99.32	279.63	279.63	**0.00**	252.24	21167	279.63	279.63	**0.00**	50.93	203	11	⋯
5-3	369.94	506.87	27.01	3600.01	494.66	494.66	**0.00**	3508.55	318.82	535.14	40.42	3600.10	494.66	494.66	**0.00**	182.18	20911	494.66	494.66	**0.00**	**64.49**	180	11	⋯
5-4	432.34	501.91	13.86	3600.01	501.91	501.91	**0.00**	418.97	398.73	531.98	25.05	3600.10	501.91	501.91	**0.00**	170.91	21104	501.91	501.91	**0.00**	**68.76**	180	14	⋯
5-5	477.46	477.46	**0.00**	**19.40**	477.47	477.47	**0.00**	43.16	477.47	477.47	**0.00**	284.87	477.47	477.47	**0.00**	167.06	18717	477.47	477.47	**0.00**	46.72	155	11	⋯
5-6	311.37	422.95	26.38	3600.01	422.95	422.95	**0.00**	745.12	271.48	486.90	44.24	3600.10	422.95	422.95	**0.00**	203.36	21534	422.95	422.95	**0.00**	**63.84**	180	12	⋯
5-7	460.14	537.30	14.36	3600.01	537.30	537.30	**0.00**	536.18	381.91	592.26	35.52	3600.10	537.30	537.30	**0.00**	143.89	21140	537.30	537.30	**0.00**	**50.38**	302	13	⋯
5-8	371.86	479.43	22.44	3600.01	479.43	479.43	**0.00**	679.47	348.77	483.72	27.90	3600.10	479.43	479.43	**0.00**	301.08	21576	479.43	479.43	**0.00**	**85.73**	220	19	⋯
5-9	308.21	427.71	27.94	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	261.33	475.90	45.09	3600.10	427.71	427.71	**0.00**	238.94	21591	427.71	427.71	**0.00**	**72.76**	175	12	⋯
#S			4				9				3				11					**19**				⋯
SGM			31.40	8220.25			30.29	5980.94			50.80	10816.56			27.39	4852.85	46304054408			**1.20**	**1643.19**	587	11	⋯

Table 16 Detailed results for problem NLTP, cost functions f_1

	GUROBI				SCIP				COUENNE				NAIVE					CN24						...
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	...
10-1	365.56	365.56	**0.00**	**28.73**	$-\infty$	$+\infty$	100.0	$+\infty$	-1176.43	656.70	155.82	3600.10	$-\infty$	$+\infty$	100.00	3601.22	157912	365.56	365.56	**0.00**	2512.22	736	29	...
10-10	288.17	288.17	**0.00**	**142.05**	-664.94	376.78	156.66	3603.28	-1017.68	550.42	154.09	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	287.27	289.39	0.73	3601.02	445	11	...
10-2	327.60	327.60	**0.00**	**44.05**	$-\infty$	$+\infty$	100.0	$+\infty$	-1188.32	701.89	159.07	3600.10	$-\infty$	$+\infty$	100.00	3601.18	98100	327.60	327.60	**0.00**	2156.90	722	25	...
10-3	296.89	296.89	**0.00**	**25.34**	$-\infty$	$+\infty$	100.0	$+\infty$	-1151.22	621.97	154.03	3600.10	$-\infty$	$+\infty$	100.00	3601.21	129580	296.89	296.89	**0.00**	2275.68	698	26	...
10-4	316.96	316.96	**0.00**	**54.52**	$-\infty$	$+\infty$	100.0	$+\infty$	-1233.65	688.59	155.82	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	316.95	316.96	0.00	3601.02	643	18	...
10-5	328.82	328.82	**0.00**	**29.35**	$-\infty$	$+\infty$	100.0	$+\infty$	-1382.75	790.84	157.19	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	328.82	328.82	**0.00**	2234.79	730	27	...
10-6	270.75	270.76	**0.00**	**46.89**	$-\infty$	$+\infty$	100.0	$+\infty$	-1103.76	587.65	153.24	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	270.76	270.76	**0.00**	2819.50	781	26	...
10-7	261.96	261.96	**0.00**	**3.04**	$-\infty$	$+\infty$	100.0	$+\infty$	-1211.91	665.66	154.93	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	261.96	261.96	**0.00**	676.44	838	18	...
10-8	268.18	268.18	**0.00**	**19.47**	$-\infty$	$+\infty$	100.0	$+\infty$	-1167.40	661.92	156.70	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	268.18	268.18	**0.00**	2081.85	845	27	...
10-9	335.59	335.59	**0.00**	**1162.64**	-634.80	488.33	176.93	3603.06	-994.50	519.95	152.28	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	328.13	341.02	3.78	3601.01	382	4	...
15-1	394.24	394.24	**0.00**	**1699.13**	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	383.83	400.28	4.11	3601.02	785	8	...
15-10	352.24	353.20	**0.27**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	342.56	356.05	3.79	3601.02	776	6	...
15-2	383.05	383.05	**0.00**	**1794.64**	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	374.21	387.11	3.33	3601.02	793	7	...
15-3	354.97	362.40	**2.05**	3600.01	-1114.15	991.75	189.01	3603.12	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	347.43	382.88	9.26	3601.02	751	3	...
15-4	335.26	335.26	**0.00**	**1371.27**	-1141.97	639.19	155.97	3607.02	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	324.14	345.39	6.15	3601.01	733	4	...
15-5	497.03	497.03	**0.00**	**3539.07**	-1380.87	$+\infty$	100.00	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	483.78	512.15	5.54	3601.02	779	5	...
15-6	312.89	312.89	**0.00**	**2277.88**	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	303.40	317.43	4.42	3601.02	777	6	...
15-7	360.70	365.19	**1.23**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	352.91	373.29	5.46	3601.02	780	5	...
15-8	378.29	378.29	**0.00**	**1079.07**	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	365.67	387.18	5.56	3601.02	768	4	...
15-9	291.32	291.32	**0.00**	**2266.43**	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	285.60	293.24	2.61	3601.01	792	6	...
20-1	435.30	451.63	**3.62**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	428.49	464.29	7.71	3601.02	1262	3	...
20-10	525.13	541.96	**3.11**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	524.51	552.76	5.11	3601.02	1309	4	...
20-2	443.75	443.75	**0.00**	**2450.08**	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	432.84	449.05	3.61	3601.02	1303	5	...
20-3	461.43	487.93	**5.43**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	455.00	509.66	10.73	3601.02	1270	3	...
20-4	373.88	378.09	**1.11**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	370.71	381.97	2.95	3601.02	1353	4	...
20-5	462.42	478.39	**3.34**	3600.01	-1607.35	1343.44	183.58	3602.88	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	447.90	503.78	11.09	3601.02	1255	3	...
20-6	438.03	457.05	**4.16**	3600.01	-1628.94	1442.72	188.57	3603.10	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	434.05	477.89	9.17	3601.03	1278	4	...
20-7	345.72	351.20	**1.56**	3600.01	-1561.31	1059.59	167.87	3603.07	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	342.43	357.67	4.26	3601.02	1311	7	...
20-8	494.80	504.75	**1.97**	3600.01	-1843.71	1192.70	164.69	3603.04	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	474.27	529.40	10.41	3601.02	1274	3	...
20-9	470.08	490.32	**4.13**	3600.01	-1622.13	1292.86	179.70	3603.06	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	460.44	523.59	12.06	3601.03	1246	2	...
5-1	183.17	183.17	**0.00**	**0.25**	$-\infty$	$+\infty$	100.0	$+\infty$	-320.12	310.24	196.92	3600.10	183.17	183.17	**0.00**	483.20	24520	183.17	183.17	**0.00**	162.10	230	19	...
5-10	235.88	235.88	**0.00**	**0.27**	$-\infty$	$+\infty$	100.0	$+\infty$	-174.61	295.07	159.17	3600.10	235.88	235.88	**0.00**	493.14	24525	235.88	235.88	**0.00**	205.35	265	23	...
5-2	129.82	129.82	**0.00**	**0.20**	$-\infty$	$+\infty$	100.0	$+\infty$	-285.17	236.83	183.05	3600.10	129.82	129.82	**0.00**	490.85	24525	129.82	129.82	**0.00**	154.59	312	20	...
5-3	182.06	182.06	**0.00**	**0.29**	$-\infty$	$+\infty$	100.0	$+\infty$	-248.68	240.20	196.59	3600.10	182.06	182.06	**0.00**	504.15	24525	182.06	182.06	**0.00**	146.02	292	23	...
5-4	205.21	205.21	**0.00**	**0.20**	$-\infty$	$+\infty$	100.0	$+\infty$	-292.53	264.55	190.44	3600.10	205.21	205.21	**0.00**	553.40	24525	205.21	205.21	**0.00**	135.52	191	20	...
5-5	192.92	192.92	**0.00**	**0.28**	$-\infty$	$+\infty$	100.0	$+\infty$	-171.84	281.10	161.13	3600.10	192.92	192.92	**0.00**	735.75	40265	192.92	192.92	**0.00**	91.65	164	11	...
5-6	181.58	181.58	**0.00**	**0.27**	-227.47	$+\infty$	100.00	$+\infty$	-262.35	221.95	184.60	3600.10	181.58	181.58	**0.00**	545.38	24525	181.58	181.58	**0.00**	184.12	315	22	...
5-7	207.70	207.70	**0.00**	**0.14**	$-\infty$	$+\infty$	100.0	$+\infty$	-302.22	311.09	197.15	3600.10	207.70	207.70	**0.00**	444.50	24525	207.69	207.70	**0.00**	116.73	263	17	...
5-8	186.62	186.62	**0.00**	**0.28**	-242.84	$+\infty$	100.00	$+\infty$	-315.88	261.56	182.80	3600.10	186.62	186.62	**0.00**	527.39	24525	186.62	186.62	**0.00**	172.02	400	29	...
5-9	167.58	167.58	**0.00**	**0.40**	$-\infty$	$+\infty$	100.0	$+\infty$	-227.57	210.92	192.69	3600.10	167.58	167.58	**0.00**	511.12	24525	167.58	167.58	**0.00**	188.37	291	25	...
#S			**28**				0				0				10					17				...
SGM			**0.44**	**392.79**			100.00	14400.00			100.00	14400.00			30.86	6313.46	86177467015			1.75	3301.89	648	11	...

Table 17 Detailed results for problem NLTP, cost functions f_2

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-1	241.31	294.82	18.15	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	294.82	294.82	**0.00**	**533.26**	2464	47	···
10-10	149.27	198.42	24.77	3600.01	155.59	198.42	21.59	3620.76	122.00	199.34	38.80	3600.10	−∞	+∞	100.00	3600.01	+∞	198.42	198.42	**0.00**	**336.87**	2573	50	···
10-2	224.09	266.97	16.06	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	266.96	266.97	**0.00**	3601.02	2112	36	···
10-3	156.03	197.54	21.01	3600.01	−∞	+∞	100.0	+∞	126.07	197.54	36.18	3600.10	−∞	+∞	100.00	3600.01	+∞	197.54	197.54	**0.00**	**512.30**	2483	44	···
10-4	167.50	221.15	24.26	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	221.15	221.15	**0.00**	2262.86	170110	221.15	221.15	**0.00**	**423.96**	2598	52	···
10-5	191.74	241.30	20.54	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	241.29	241.30	**0.01**	3601.02	1938	34	···
10-6	152.29	193.44	21.27	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	193.44	193.44	**0.00**	2590.56	170104	193.44	193.44	**0.00**	**2252.64**	2322	42	···
10-7	168.73	202.75	16.78	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	202.75	202.75	**0.00**	3601.02	2208	47	···
10-8	155.01	200.39	22.65	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	200.39	200.39	**0.00**	2923.16	170124	200.39	200.39	**0.00**	**1480.58**	2628	51	···
10-9	144.93	194.46	25.47	3600.01	154.33	194.46	20.64	3603.08	129.20	197.22	34.49	3600.10	194.46	194.46	**0.00**	1355.61	170117	194.46	194.46	**0.00**	**178.42**	2533	45	···
15-1	179.89	248.60	27.64	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	248.60	248.60	**0.00**	3601.02	5372	59	···
15-10	157.66	228.32	30.95	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	228.32	228.33	**0.00**	3601.02	4887	50	···
15-2	175.60	245.30	28.41	3600.01	175.71	245.30	28.37	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	245.30	245.30	**0.00**	3601.02	4622	41	···
15-3	123.71	194.78	36.49	3600.01	128.26	194.78	34.15	3603.15	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	194.77	194.78	**0.00**	3601.07	5613	62	···
15-4	112.65	178.34	36.84	3600.01	120.79	178.34	32.27	3606.82	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	178.34	178.35	**0.00**	3601.02	5190	56	···
15-5	285.36	377.50	24.41	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	377.49	377.52	**0.01**	3601.03	4103	44	···
15-6	137.41	194.95	29.52	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	194.95	194.96	**0.00**	3601.03	4851	51	···
15-7	164.36	238.05	30.96	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	238.05	238.05	**0.00**	3601.03	5025	56	···
15-8	143.55	224.53	36.06	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	224.53	224.53	**0.00**	3601.03	5644	57	···
15-9	134.60	195.88	31.29	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	195.88	195.89	**0.00**	3601.02	4569	48	···
20-1	140.62	239.96	41.40	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	239.94	239.97	**0.01**	3601.02	7521	46	···
20-10	257.39	356.85	27.87	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	356.82	356.87	**0.01**	3601.03	7004	46	···
20-2	166.08	262.47	36.73	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	262.44	262.49	**0.02**	3601.03	6817	43	···
20-3	118.96	214.64	44.58	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	214.60	214.65	**0.02**	3601.03	7160	48	···
20-4	114.79	201.71	43.09	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	201.69	201.72	**0.02**	3601.03	7538	52	···
20-5	189.45	282.77	33.00	3600.01	194.07	282.77	31.37	3602.88	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	282.54	282.90	**0.13**	3601.03	5718	30	···
20-6	134.06	225.75	40.62	3600.01	138.70	225.75	38.56	3616.39	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	225.73	225.76	**0.01**	3601.04	7305	42	···
20-7	113.27	189.94	40.36	3600.01	118.36	189.94	37.69	3602.96	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	189.90	189.97	**0.03**	3601.03	6763	45	···
20-8	190.26	298.36	36.23	3600.01	190.16	298.36	36.26	3603.12	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	298.13	298.62	**0.17**	3601.03	5566	32	···
20-9	120.86	215.97	44.04	3600.01	124.86	215.97	42.19	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	215.96	215.97	**0.01**	3601.04	8094	54	···
5-1	170.48	170.48	**0.00**	**472.27**	−∞	+∞	100.0	+∞	170.36	170.48	0.07	3612.36	−∞	+∞	100.00	3601.08	42520	170.48	170.48	**0.00**	819.45	597	35	···
5-10	251.56	251.90	0.14	3600.01	−∞	+∞	100.0	+∞	248.98	251.92	1.16	3600.10	−∞	+∞	100.00	3601.29	175150	251.90	251.90	**0.00**	**1994.98**	672	36	···
5-2	102.02	102.02	**0.00**	**16.86**	102.02	102.02	**0.00**	1335.19	−∞	+∞	100.0	3600.10	102.02	102.02	**0.00**	389.29	42534	102.02	102.02	**0.00**	82.61	515	37	···
5-3	184.98	185.85	0.47	3600.01	−∞	+∞	100.0	+∞	180.40	185.85	2.93	3600.10	185.85	185.85	**0.00**	733.40	42534	185.85	185.85	**0.00**	**213.65**	629	34	···
5-4	195.67	195.78	0.06	3600.01	194.92	+∞	100.00	+∞	194.56	195.78	0.62	3613.88	195.78	195.78	**0.00**	839.10	42525	195.78	195.78	**0.00**	**386.37**	636	33	···
5-5	232.14	232.14	**0.00**	**110.53**	232.14	232.14	**0.00**	3603.01	232.09	232.15	0.02	3622.13	−∞	+∞	100.00	3600.01	+∞	232.14	232.14	**0.00**	804.48	532	34	···
5-6	175.55	175.99	0.25	3600.01	−∞	+∞	100.0	+∞	170.34	176.10	3.27	3600.10	175.99	175.99	**0.00**	645.28	42525	175.99	175.99	**0.00**	**215.35**	656	36	···
5-7	227.32	227.32	**0.00**	2812.32	227.15	227.32	0.08	3602.93	226.17	227.33	0.51	3611.53	227.32	227.32	**0.00**	493.76	42530	227.32	227.32	**0.00**	**371.98**	578	33	···
5-8	172.38	172.39	**0.00**	1238.20	172.26	172.39	0.07	3603.09	172.01	172.39	0.22	3612.23	172.39	172.39	**0.00**	664.42	42525	172.39	172.39	**0.00**	**212.85**	564	31	···
5-9	173.65	174.35	0.40	3600.01	−∞	+∞	100.0	+∞	168.49	174.36	3.36	3600.10	174.35	174.35	**0.00**	612.74	42525	174.35	174.35	**0.00**	**252.60**	660	32	···
#S			5				1				0				11					**17**				···
SGM			12.26	9054.43			47.14	13570.54			37.81	14400.00			27.39	6876.10	107131269046			**0.01**	**3307.27**	2652	43	···

Table 18 Detailed results for problem NLTP, cost functions f_3

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-1	267.66	362.99	26.26	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	362.60	362.60	**0.00**	3601.02	2643	67	···
10-10	178.21	265.88	32.98	3600.01	183.72	253.90	27.64	3603.09	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	253.90	253.90	**0.00**	3601.01	2476	39	···
10-2	227.16	321.87	29.42	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	300.57	300.57	**0.00**	2658.86	220484	300.57	300.57	**0.00**	**1602.29**	2685	56	···
10-3	184.16	289.01	36.28	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	134.12	287.30	53.32	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	268.44	268.44	**0.00**	**2163.84**	2414	41	···
10-4	194.40	309.98	37.29	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	130.04	415.59	68.71	3600.10	294.92	294.92	**0.00**	3473.03	220513	294.92	294.92	**0.00**	**3450.25**	2685	50	···
10-5	218.09	302.73	27.96	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	287.45	287.45	**0.00**	**835.83**	2487	50	···
10-6	168.07	230.75	27.16	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	227.18	227.18	**0.00**	**1177.87**	2460	52	···
10-7	204.05	270.10	24.45	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	132.80	352.46	62.32	3600.10	257.23	257.23	**0.00**	**2582.05**	220488	257.23	257.23	**0.00**	3601.02	2264	46	···
10-8	177.76	240.60	26.12	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	127.18	303.32	58.07	3600.10	240.55	240.55	**0.00**	**2155.72**	220498	240.55	240.55	**0.00**	2778.40	2536	52	···
10-9	167.58	336.50	50.20	3600.01	187.32	316.66	40.84	3608.81	129.01	397.48	67.54	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	309.09	309.61	**0.17**	3601.02	1654	34	···
15-1	208.35	407.44	48.86	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	360.97	368.65	**2.08**	3601.02	2413	13	···
15-10	172.80	338.57	48.96	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	322.01	327.69	**1.73**	3601.02	2416	15	···
15-2	194.33	377.09	48.47	3600.01	200.02	350.81	42.98	3607.60	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	341.33	341.77	**0.13**	3601.02	3270	29	···
15-3	135.57	340.53	60.19	3600.01	142.44	333.39	57.28	3614.32	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	310.30	315.12	**1.53**	**3598.72**	2406	14	···
15-4	131.90	279.79	52.86	3600.01	134.21	277.09	51.57	3606.61	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	268.38	268.69	**0.11**	3601.02	3410	30	···
15-5	277.61	478.11	41.94	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	452.13	453.79	**0.37**	3601.02	2920	25	···
15-6	157.35	288.16	45.39	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	269.19	269.47	**0.10**	3601.02	3239	29	···
15-7	173.03	366.19	52.75	3600.01	183.17	345.75	47.02	3603.10	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	328.00	330.49	**0.75**	3601.02	2542	17	···
15-8	156.89	342.98	54.26	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	334.66	339.13	**1.32**	3601.02	2588	15	···
15-9	142.28	263.79	46.06	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	250.43	250.57	**0.06**	3601.02	3449	26	···
20-1	148.06	408.01	63.71	3600.01	147.35	404.95	63.61	3603.00	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	366.74	404.16	**9.26**	3601.02	3667	3	···
20-10	252.06	484.15	47.94	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	456.31	478.10	**4.56**	3601.02	3924	7	···
20-2	183.66	443.34	58.57	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	380.04	386.58	**1.69**	3601.02	4109	16	···
20-3	132.51	420.34	68.47	3600.01	135.95	419.19	67.57	3602.89	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	376.04	439.56	**14.45**	3601.02	3650	3	···
20-4	120.37	358.77	66.45	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	316.23	350.35	**9.74**	3601.02	3655	3	···
20-5	182.49	460.89	60.41	3600.01	194.84	429.14	54.60	3603.10	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	390.38	418.53	**6.73**	3601.02	3723	4	···
20-6	151.36	411.59	63.23	3600.02	156.32	403.86	61.29	3602.89	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	365.81	430.30	**14.99**	3601.03	3621	2	···
20-7	126.82	339.97	62.70	3600.01	125.89	308.27	59.16	3602.94	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	289.74	311.10	**6.87**	3601.02	3780	7	···
20-8	204.18	492.36	58.53	3600.01	200.17	442.12	54.73	3607.47	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	430.97	436.69	**1.31**	3601.02	4196	10	···
20-9	131.53	411.50	68.04	3600.01	137.92	411.50	66.48	3603.10	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	369.93	413.06	**10.44**	3601.02	3697	3	···
5-1	152.40	152.40	**0.00**	190.44	152.40	152.40	**0.00**	707.91	152.31	152.40	0.06	3626.00	152.40	152.40	**0.00**	1436.29	239429	152.40	152.40	**0.00**	**111.21**	622	29	···
5-10	223.23	223.23	**0.00**	201.17	223.23	223.23	**0.00**	858.09	222.99	223.24	0.11	3623.26	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	223.23	223.23	**0.00**	**95.30**	623	32	···
5-2	100.36	100.36	**0.00**	**19.27**	100.36	100.36	**0.00**	259.16	100.35	100.36	0.01	3629.61	100.36	100.36	**0.00**	336.74	55130	100.36	100.36	**0.00**	92.64	585	29	···
5-3	198.05	198.05	**0.00**	378.88	198.05	198.05	**0.00**	2631.90	197.75	198.05	0.15	3622.60	198.05	198.05	**0.00**	523.24	55135	198.05	198.05	**0.00**	**165.98**	656	39	···
5-4	184.06	184.06	**0.00**	267.81	184.06	184.06	**0.00**	398.59	183.97	184.06	0.05	3617.30	184.06	184.06	**0.00**	359.77	55133	184.06	184.06	**0.00**	**130.14**	629	28	···
5-5	203.48	203.48	**0.00**	**66.08**	203.48	203.48	**0.00**	190.82	203.39	203.48	0.05	3635.68	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	203.48	203.48	**0.00**	121.86	582	30	···
5-6	175.47	175.47	**0.00**	821.81	175.47	175.47	**0.00**	3602.95	175.14	176.45	0.74	3617.20	175.47	175.47	**0.00**	386.55	55128	175.47	175.47	**0.00**	**161.57**	631	37	···
5-7	227.47	227.47	**0.00**	538.93	227.47	227.47	**0.00**	2218.06	227.34	227.47	0.06	3622.61	227.47	227.47	**0.00**	383.41	55127	227.47	227.47	**0.00**	**132.34**	624	30	···
5-8	185.74	185.74	**0.00**	**113.21**	185.74	185.74	**0.00**	3602.72	185.66	185.74	0.04	3620.15	185.74	185.74	**0.00**	345.19	55131	185.74	185.74	**0.00**	145.11	583	30	···
5-9	171.25	171.25	**0.00**	306.43	171.25	171.25	0.00	3603.07	171.06	171.25	0.11	3617.67	171.25	171.25	**0.00**	356.59	55134	171.25	171.25	**0.00**	**162.99**	602	35	···
#S			10				7				0				12					**16**				···
SGM			16.86	5007.14			24.86	8474.23			29.90	14400.00			24.30	6107.52	198874967870			**0.90**	**3185.89**	2001	23	···

Table 19 Detailed results for problem NLTP, cost functions f_4

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-1	59.83	86.27	30.65	3600.01	−∞	+∞	100.0	+∞	43.82	100.72	56.50	3600.10	−∞	+∞	100.00	3600.01	+∞	75.03	85.60	**12.35**	3601.02	1082	4	···
10-10	40.50	64.20	36.92	3600.01	−∞	+∞	100.0	+∞	29.07	85.28	65.91	3600.10	−∞	+∞	100.00	3600.01	+∞	57.31	63.31	**9.47**	3601.01	1124	6	···
10-2	68.80	104.37	34.08	3600.01	−∞	+∞	100.0	+∞	54.06	122.19	55.76	3600.10	−∞	+∞	100.00	3601.78	437400	78.76	96.76	**18.60**	3601.02	1026	2	···
10-3	45.51	69.30	34.33	3600.01	−∞	+∞	100.0	+∞	31.71	97.61	67.51	3600.10	−∞	+∞	100.00	3600.01	+∞	61.31	70.46	**12.99**	3601.02	1080	4	···
10-4	46.51	77.52	40.00	3600.01	−∞	+∞	100.0	+∞	32.30	99.52	67.54	3600.10	−∞	+∞	100.00	3600.01	+∞	60.72	73.59	**17.49**	3601.02	1026	2	···
10-5	62.00	88.54	29.98	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	78.50	89.18	**11.98**	3601.02	1075	4	···
10-6	46.27	69.44	33.37	3600.01	−∞	+∞	100.0	+∞	35.14	89.25	60.62	3600.10	−∞	+∞	100.00	3600.01	+∞	61.32	67.74	**9.48**	3601.02	1111	6	···
10-7	45.41	64.85	29.98	3600.01	−∞	+∞	100.0	+∞	30.58	93.06	67.14	3600.10	−∞	+∞	100.00	3600.01	+∞	59.50	63.25	**5.93**	3601.02	1145	8	···
10-8	43.31	68.62	36.88	3600.01	−∞	+∞	100.0	+∞	30.23	88.17	65.71	3600.10	−∞	+∞	100.00	3600.01	+∞	57.13	65.47	**12.74**	3601.02	1078	4	···
10-9	39.02	77.74	**49.81**	3600.01	30.65	75.29	59.29	3608.71	28.36	87.08	67.44	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	1000	1	···
15-1	49.76	107.61	**53.76**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	2250	1	···
15-10	43.90	104.47	**57.98**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	2250	1	···
15-2	47.30	114.01	**58.52**	3600.01	38.26	109.86	65.18	3603.25	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	2250	1	···
15-3	34.47	100.29	**65.63**	3600.01	27.90	87.48	68.10	3603.11	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	2250	1	···
15-4	35.15	106.35	**66.95**	3600.01	27.30	92.79	70.58	3603.28	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	2250	1	···
15-5	72.97	153.35	**52.42**	3600.01	64.02	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	2250	1	···
15-6	45.68	84.47	**45.92**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	2250	1	···
15-7	44.43	104.66	**57.55**	3600.01	35.58	108.25	67.13	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.04	2250	1	···
15-8	41.40	98.21	**57.84**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	2250	1	···
15-9	36.46	83.61	**56.39**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	2250	1	···
20-1	39.12	161.20	**75.73**	3600.01	31.67	137.93	77.04	3603.18	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	4000	1	···
20-10	63.22	163.90	**61.43**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	4000	1	···
20-2	46.84	163.51	**71.35**	3600.01	−∞	+∞	100.0	+∞	35.03	264.45	86.75	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	4000	1	···
20-3	36.07	111.51	**67.65**	3600.01	28.97	136.46	78.77	3602.97	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	4000	1	···
20-4	33.19	121.68	**72.72**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	4000	1	···
20-5	62.75	171.82	**63.48**	3600.01	54.30	164.68	67.03	3603.10	50.60	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	4000	1	···
20-6	40.02	144.94	**72.39**	3600.01	31.22	136.61	77.15	3603.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	4000	1	···
20-7	33.36	120.91	**72.41**	3600.01	27.59	124.24	77.80	3603.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	4000	1	···
20-8	51.24	145.10	**64.69**	3600.01	41.62	134.13	68.97	3607.24	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	4000	1	···
20-9	35.97	145.62	**75.30**	3600.01	30.13	155.65	80.64	3700.43	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.04	4000	1	···
5-1	53.84	53.84	**0.00**	539.82	−∞	+∞	100.0	+∞	53.79	53.84	0.10	3609.11	−∞	+∞	100.00	3600.01	+∞	53.84	53.84	**0.00**	**322.37**	769	45	···
5-10	102.49	102.49	**0.00**	147.35	−∞	+∞	100.0	+∞	102.47	102.49	0.02	3643.99	−∞	+∞	100.00	3600.01	+∞	102.49	102.49	**0.00**	**126.89**	655	36	···
5-2	38.83	38.83	**0.00**	558.00	−∞	+∞	100.0	+∞	38.83	38.83	0.02	3621.31	38.83	38.83	**0.00**	1521.22	91618	38.83	38.83	**0.00**	**188.62**	682	35	···
5-3	55.23	55.23	**0.00**	839.86	−∞	+∞	100.0	+∞	52.63	55.23	4.72	3600.10	55.23	55.23	**0.00**	2842.54	91602	55.23	55.23	**0.00**	**316.26**	720	39	···
5-4	56.59	56.59	**0.00**	513.70	56.59	56.59	0.01	3603.04	56.55	56.59	0.07	3618.60	56.59	56.59	**0.00**	2656.12	91603	56.59	56.59	**0.00**	**416.49**	720	41	···
5-5	106.84	106.84	**0.00**	**17.58**	106.83	106.84	0.00	3603.07	106.83	106.84	0.00	3607.24	−∞	+∞	100.00	3600.01	+∞	106.84	106.84	**0.00**	191.65	579	34	···
5-6	43.04	43.04	**0.00**	**244.98**	−∞	+∞	100.0	+∞	37.19	43.05	13.63	3600.10	43.04	43.04	**0.00**	3135.00	91620	43.04	43.04	**0.00**	789.83	778	47	···
5-7	60.30	60.30	**0.00**	1089.95	60.01	60.30	0.48	3602.92	60.19	60.31	0.19	3610.06	60.30	60.30	**0.00**	2846.21	91596	60.30	60.30	**0.00**	**237.19**	727	38	···
5-8	47.05	47.05	**0.00**	**77.48**	46.61	47.05	0.92	3604.04	47.04	47.05	0.01	3627.37	47.05	47.05	**0.00**	2492.21	91617	47.05	47.05	**0.00**	221.24	706	36	···
5-9	38.62	38.73	0.27	3600.01	38.73	38.73	0.01	3602.99	38.71	38.73	0.06	3624.65	38.73	38.73	**0.00**	1730.59	91614	38.73	38.73	**0.00**	**265.02**	715	35	···
#S			9				0				0				7					**10**				···
SGM			**18.74**	5977.16			51.67	14400.00			31.44	14400.00			44.04	10518.40	1619461089071			19.02	5379.96	1617	7	···

Table 20 Detailed results for problem NLTP, cost functions f_5

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
10-1	35.99	467.47	92.30	3600.01	-59.25	477.41	112.41	3603.19	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	444.08	444.46	**0.09**	3601.02	836	21	⋯
10-10	-21.10	345.16	106.11	3600.01	-94.41	359.18	126.29	3602.99	-61.39	411.41	114.92	3600.10	−∞	+∞	100.00	3600.01	+∞	328.97	329.41	**0.13**	3601.03	807	17	⋯
10-2	-6.57	418.30	101.57	3600.01	-98.69	423.09	123.33	3603.17	-28.05	580.14	104.83	3600.10	−∞	+∞	100.00	3600.01	+∞	383.51	383.51	**0.00**	3601.03	1006	35	⋯
10-3	-25.83	375.00	106.89	3600.01	-121.82	388.35	131.37	3602.90	-183.53	527.75	134.78	3600.10	−∞	+∞	100.00	3600.01	+∞	358.95	364.35	**1.48**	3601.02	765	11	⋯
10-4	-44.72	401.57	111.14	3600.01	-134.07	425.38	131.52	3603.04	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	383.80	393.98	**2.58**	3601.02	686	7	⋯
10-5	-28.19	382.49	107.37	3600.01	-121.68	415.01	129.32	3603.03	-197.81	616.48	132.09	3600.10	−∞	+∞	100.00	3600.01	+∞	381.17	381.43	**0.07**	3601.02	914	20	⋯
10-6	-28.01	311.11	109.00	3600.01	-113.98	333.33	134.20	3603.04	-164.85	529.88	131.11	3600.10	−∞	+∞	100.00	3600.01	+∞	300.82	300.82	**0.00**	**1586.43**	910	29	⋯
10-7	-21.36	314.82	106.78	3600.01	-124.35	334.15	137.21	3603.02	-183.25	549.00	133.38	3600.10	−∞	+∞	100.00	3600.01	+∞	313.19	313.19	**0.00**	**1536.50**	1082	37	⋯
10-8	-40.07	328.78	112.19	3600.01	-123.18	353.42	134.85	3603.05	-171.51	506.48	133.86	3600.10	−∞	+∞	100.00	3600.01	+∞	316.18	316.23	**0.01**	3601.02	922	24	⋯
10-9	-56.69	435.60	113.01	3600.01	-115.83	440.19	126.31	3603.05	-79.53	548.42	114.50	3600.10	−∞	+∞	100.00	3600.01	+∞	393.20	423.32	**7.12**	3601.02	634	3	⋯
15-1	-167.40	524.16	131.94	3600.01	-308.38	602.19	151.21	3602.95	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	467.44	487.74	**4.16**	3601.02	1423	6	⋯
15-10	-208.77	463.89	145.00	3600.01	-295.61	515.70	157.32	3603.01	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	412.73	426.23	**3.17**	3601.02	1437	6	⋯
15-2	-213.47	487.92	143.75	3600.02	-311.41	587.49	153.01	3603.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	442.04	454.37	**2.71**	3601.02	1442	6	⋯
15-3	-230.81	446.81	151.66	3600.01	-298.29	507.16	158.82	3603.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	410.56	442.16	**7.15**	3601.03	1391	3	⋯
15-4	-244.71	441.38	155.44	3600.01	-305.37	555.29	154.99	3603.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	381.89	393.70	**3.00**	3601.02	1408	6	⋯
15-5	-148.84	652.01	122.83	3600.01	-256.60	769.08	133.36	3603.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	582.49	596.61	**2.37**	3601.02	1446	6	⋯
15-6	-198.85	388.95	151.12	3600.01	-292.47	492.81	159.35	3603.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	354.81	364.97	**2.78**	3601.03	1469	8	⋯
15-7	-182.02	461.16	139.47	3600.01	-268.51	569.75	147.13	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	427.49	439.33	**2.70**	3601.02	1451	7	⋯
15-8	-247.71	484.99	151.08	3600.01	-331.29	550.87	160.14	3602.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	442.77	461.73	**4.11**	3601.02	1421	5	⋯
15-9	-227.85	360.57	163.19	3600.01	-292.81	471.60	162.09	3602.97	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	335.74	338.64	**0.85**	3601.02	1495	7	⋯
20-1	-486.20	633.20	176.79	3600.01	-538.52	753.09	171.51	3603.26	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	506.08	536.21	**5.62**	3601.02	2464	3	⋯
20-10	-343.90	730.18	147.10	3600.01	-438.86	830.84	152.82	3603.18	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	616.55	642.35	**4.02**	3601.03	2450	4	⋯
20-2	-431.89	630.57	168.49	3600.01	-531.32	713.92	174.42	3603.13	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	518.14	530.04	**2.25**	3601.02	2534	7	⋯
20-3	-461.72	653.44	170.66	3600.02	-521.07	779.79	166.82	3603.15	-756.64	1301.51	158.14	3600.10	−∞	+∞	100.00	3600.01	+∞	534.51	582.95	**8.31**	3601.02	2428	2	⋯
20-4	-448.36	521.99	185.89	3600.01	-512.38	672.08	176.24	3603.15	-742.06	1110.89	166.80	3600.10	−∞	+∞	100.00	3600.01	+∞	439.84	454.70	**3.27**	3601.03	2495	4	⋯
20-5	-378.87	693.49	154.63	3600.01	-428.60	702.67	161.00	3603.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	526.56	573.60	**8.20**	3601.03	2427	2	⋯
20-6	-413.31	712.10	158.04	3600.02	-485.72	689.26	170.47	3603.14	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	2400	1	⋯
20-7	-385.82	481.98	180.05	3600.01	-462.30	902.81	151.21	3603.17	-663.26	1118.69	159.29	3600.10	−∞	+∞	100.00	3600.01	+∞	396.24	411.68	**3.75**	3601.02	2463	5	⋯
20-8	-412.08	724.07	156.91	3600.01	-511.49	1162.79	143.99	3603.14	-745.89	1320.50	156.49	3600.10	−∞	+∞	100.00	3600.01	+∞	567.85	603.58	**5.92**	3601.02	2440	3	⋯
20-9	-431.28	735.42	158.64	3600.01	-498.00	773.54	164.38	3602.72	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	2400	1	⋯
5-1	187.46	187.46	**0.00**	265.09	−∞	+∞	100.0	+∞	49.54	192.19	74.22	3600.10	−∞	+∞	100.00	3600.01	+∞	187.46	187.46	**0.00**	**162.88**	243	17	⋯
5-10	244.08	266.13	8.28	3600.01	−∞	+∞	100.0	+∞	122.35	270.14	54.71	3600.10	−∞	+∞	100.00	3600.01	+∞	266.13	266.13	**0.00**	**251.67**	242	15	⋯
5-2	134.08	134.08	**0.00**	232.13	−∞	+∞	100.0	+∞	12.93	137.69	90.61	3600.10	−∞	+∞	100.00	3600.01	+∞	134.08	134.08	**0.00**	**191.94**	244	16	⋯
5-3	174.12	224.36	22.39	3600.01	−∞	+∞	100.0	+∞	57.65	232.68	75.22	3600.10	−∞	+∞	100.00	3600.01	+∞	224.16	224.16	**0.00**	**330.47**	444	34	⋯
5-4	181.06	234.24	22.70	3600.01	−∞	+∞	100.0	+∞	53.00	237.06	77.64	3600.10	−∞	+∞	100.00	3600.01	+∞	233.47	233.47	**0.00**	**318.70**	350	36	⋯
5-5	220.29	220.29	**0.00**	406.65	220.29	220.29	**0.00**	3227.15	126.24	223.99	43.64	3600.10	−∞	+∞	100.00	3600.01	+∞	220.29	220.29	**0.00**	**155.86**	263	17	⋯
5-6	186.19	214.86	13.35	3600.01	−∞	+∞	100.0	+∞	43.61	219.96	80.17	3600.10	−∞	+∞	100.00	3600.01	+∞	212.53	212.53	**0.00**	**214.17**	249	14	⋯
5-7	231.69	262.77	11.83	3600.01	−∞	+∞	100.0	+∞	76.21	267.10	71.47	3600.10	−∞	+∞	100.00	3600.01	+∞	262.07	262.07	**0.00**	**315.17**	363	35	⋯
5-8	217.52	217.52	**0.00**	3003.34	−∞	+∞	100.0	+∞	39.87	233.49	82.93	3600.10	−∞	+∞	100.00	3600.01	+∞	217.52	217.52	**0.00**	**186.52**	260	21	⋯
5-9	163.28	201.77	19.07	3600.01	−∞	+∞	100.0	+∞	52.70	203.59	74.11	3600.10	−∞	+∞	100.00	3600.01	+∞	199.40	199.40	**0.00**	**238.36**	267	18	⋯
#S			4				1				0				0					**12**				⋯
SGM			47.35	10361.85			88.99	13871.98			91.85	14400.00			100.00	14400.00	+∞			**1.85**	4617.97	965	11	⋯

Table 21 Detailed results for problem NLTP, cost functions f_6

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
10-1	536.22	536.22	**0.00**	**1.40**	536.22	536.22	**0.00**	10.92	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	536.22	536.22	**0.00**	362.30	1001	27	⋯
10-10	387.22	387.22	**0.00**	**3.65**	387.22	387.22	**0.00**	21.32	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	387.22	387.22	**0.00**	727.75	924	24	⋯
10-2	465.46	465.46	**0.00**	**1.32**	465.46	465.46	**0.00**	10.93	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	465.46	465.46	**0.00**	2311.09	906	29	⋯
10-3	419.65	419.65	**0.00**	**1.03**	419.65	419.65	**0.00**	8.80	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	419.65	419.65	**0.00**	632.94	992	21	⋯
10-4	447.67	447.67	**0.00**	**7.10**	447.67	447.67	**0.00**	28.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	447.67	447.67	**0.00**	2956.73	1087	25	⋯
10-5	433.10	433.10	**0.00**	**0.68**	433.10	433.10	**0.00**	4.04	433.10	433.10	**0.00**	1617.63	−∞	+∞	100.00	3600.01	+∞	433.10	433.10	**0.00**	92.27	819	16	⋯
10-6	367.53	367.53	**0.00**	**2.08**	367.53	367.53	**0.00**	18.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	367.53	367.53	**0.00**	1059.64	893	19	⋯
10-7	398.88	398.88	**0.00**	**3.27**	398.88	398.88	**0.00**	17.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	398.88	398.88	**0.00**	761.71	1048	25	⋯
10-8	389.41	389.41	**0.00**	**2.98**	389.41	389.41	**0.00**	20.61	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	389.41	389.41	**0.00**	1706.50	962	20	⋯
10-9	505.34	505.34	**0.00**	**54.44**	505.34	505.34	**0.00**	395.80	497.09	505.95	1.75	3600.10	−∞	+∞	100.00	3600.01	+∞	477.37	505.64	5.59	3601.02	424	7	⋯
15-1	605.07	605.07	**0.00**	**415.45**	605.07	605.07	**0.00**	2652.06	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	576.83	605.59	4.75	3601.02	921	7	⋯
15-10	540.87	540.87	**0.00**	**315.82**	540.87	540.87	**0.00**	2510.72	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	506.12	541.26	6.49	3601.02	922	5	⋯
15-2	563.67	563.67	**0.00**	**78.06**	563.67	563.67	**0.00**	722.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	544.88	564.04	3.40	3601.02	942	6	⋯
15-3	509.47	509.47	**0.00**	**134.36**	509.47	509.47	**0.00**	630.23	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	491.07	509.86	3.69	3601.02	926	7	⋯
15-4	479.24	479.24	**0.00**	**190.21**	479.24	479.24	**0.00**	1595.75	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	446.11	479.66	6.99	3601.02	912	7	⋯
15-5	703.43	703.43	**0.00**	**176.25**	703.43	703.43	**0.00**	1016.54	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	684.36	707.44	3.26	3601.02	941	7	⋯
15-6	429.74	429.74	**0.00**	**33.71**	429.74	429.74	**0.00**	318.97	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	427.83	429.99	0.50	3601.01	1013	21	⋯
15-7	516.90	516.90	**0.00**	**91.74**	516.90	516.90	**0.00**	487.29	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	497.42	519.44	4.24	3601.02	925	7	⋯
15-8	567.65	567.65	**0.00**	**902.06**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	518.46	568.09	8.74	3601.02	911	5	⋯
15-9	414.29	414.29	**0.00**	**54.05**	414.29	414.29	**0.00**	203.43	400.73	418.35	4.21	3600.10	−∞	+∞	100.00	3600.01	+∞	409.77	414.74	1.20	3601.01	982	12	⋯
20-1	634.80	646.03	**1.74**	3600.01	624.69	646.96	3.44	3603.05	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	528.87	646.85	18.24	3601.03	1605	2	⋯
20-10	759.61	759.61	**0.00**	**2042.82**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	695.79	760.37	8.49	3601.02	1627	3	⋯
20-2	619.52	640.98	**3.35**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	1600	1	⋯
20-3	673.42	683.42	**1.46**	3600.01	668.68	684.18	2.27	3602.99	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	551.71	684.81	19.44	3601.03	1603	2	⋯
20-4	545.63	560.29	**2.62**	3600.01	539.47	565.44	4.59	3602.95	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	456.16	561.07	18.70	3601.02	1609	2	⋯
20-5	650.42	677.93	**4.06**	3600.01	650.77	679.10	4.17	3603.00	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	554.26	682.78	18.82	3601.03	1606	2	⋯
20-6	651.25	687.98	**5.34**	3600.01	650.90	700.62	7.10	3603.12	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	1600	1	⋯
20-7	487.33	487.33	**0.00**	**212.78**	487.33	487.33	**0.00**	1916.69	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	447.39	491.27	8.93	3601.02	1619	4	⋯
20-8	726.03	744.48	**2.48**	3600.01	720.16	750.88	4.09	3603.01	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	607.19	752.25	19.28	3601.02	1602	2	⋯
20-9	681.70	729.12	**6.50**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	1600	1	⋯
5-1	237.14	237.14	**0.00**	**0.04**	237.14	237.14	**0.00**	3.11	237.14	237.14	**0.00**	4.89	−∞	+∞	100.00	3600.01	+∞	237.14	237.14	**0.00**	9.14	246	16	⋯
5-10	293.34	293.34	**0.00**	**0.03**	293.34	293.34	**0.00**	3.24	293.34	293.34	**0.00**	0.95	−∞	+∞	100.00	3600.01	+∞	293.34	293.34	**0.00**	8.25	248	16	⋯
5-2	177.70	177.70	**0.00**	**0.03**	177.70	177.70	**0.00**	3.26	177.70	177.70	**0.00**	2.82	−∞	+∞	100.00	3600.01	+∞	177.70	177.70	**0.00**	10.32	258	16	⋯
5-3	273.38	273.38	**0.00**	**0.04**	273.38	273.38	**0.00**	3.05	273.38	273.38	**0.00**	2.41	−∞	+∞	100.00	3600.01	+∞	273.38	273.38	**0.00**	10.96	295	17	⋯
5-4	269.68	269.68	**0.00**	**0.05**	269.68	269.68	**0.00**	3.45	269.68	269.68	**0.00**	2.62	−∞	+∞	100.00	3600.01	+∞	269.68	269.68	**0.00**	10.65	218	18	⋯
5-5	258.54	258.54	**0.00**	**0.03**	258.54	258.54	**0.00**	3.31	258.54	258.54	**0.00**	2.81	−∞	+∞	100.00	3600.01	+∞	258.54	258.54	**0.00**	7.73	237	16	⋯
5-6	242.05	242.05	**0.00**	**0.05**	242.05	242.05	**0.00**	3.30	242.05	242.05	**0.00**	4.81	−∞	+∞	100.00	3600.01	+∞	242.05	242.05	**0.00**	9.63	283	16	⋯
5-7	333.74	333.74	**0.00**	**0.07**	333.74	333.74	**0.00**	3.52	333.74	333.74	**0.00**	28.61	−∞	+∞	100.00	3600.01	+∞	333.74	333.74	**0.00**	14.93	284	18	⋯
5-8	286.79	286.79	**0.00**	**0.06**	286.79	286.79	**0.00**	3.14	286.79	286.79	**0.00**	3.00	−∞	+∞	100.00	3600.01	+∞	286.79	286.79	**0.00**	13.65	318	23	⋯
5-9	242.02	242.02	**0.00**	**0.05**	242.02	242.02	**0.00**	3.22	242.02	242.02	**0.00**	84.57	−∞	+∞	100.00	3600.01	+∞	242.02	242.02	**0.00**	10.36	216	16	⋯
#S			**32**				30				11				0					19				⋯
SGM			**0.33**	**107.89**			1.02	263.42			23.09	2542.99			100.00	14400.00	+∞			2.55	1459.45	768	11	⋯

Table 22 **Detailed results for problem NLTP, cost functions f_7**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
10-1	130.46	130.46	0.00	0.02	130.46	130.46	0.00	3.31	130.46	130.46	0.00	7.84	130.46	130.46	0.00	146.83	149643	130.46	130.46	0.00	87.38	2512	50	⋯
10-10	85.18	85.18	0.00	0.01	85.18	85.18	0.00	8.15	85.18	85.18	0.00	22.92	85.18	85.18	0.00	148.39	149680	85.18	85.18	0.00	176.51	2596	51	⋯
10-2	134.71	134.71	0.00	0.01	134.71	134.71	0.00	3.03	134.71	134.71	0.00	61.28	134.71	134.71	0.00	149.06	149700	134.71	134.71	0.00	113.35	2532	46	⋯
10-3	89.52	89.52	0.00	0.01	89.52	89.52	0.00	3.16	89.52	89.52	0.00	19.41	89.52	89.52	0.00	155.16	149644	89.52	89.52	0.00	67.49	2558	53	⋯
10-4	95.42	95.42	0.00	0.02	95.42	95.42	0.00	2.94	95.42	95.42	0.00	3613.15	95.42	95.42	0.00	154.75	149690	95.42	95.42	0.00	80.72	2600	53	⋯
10-5	112.91	112.91	0.00	0.01	112.91	112.91	0.00	2.97	112.91	112.91	0.00	12.29	112.91	112.91	0.00	150.86	149653	112.91	112.91	0.00	63.23	2619	49	⋯
10-6	89.80	89.80	0.00	0.01	89.80	89.80	0.00	3.23	89.80	89.80	0.00	18.20	89.80	89.80	0.00	145.89	149660	89.80	89.80	0.00	88.34	2556	50	⋯
10-7	87.86	87.86	0.00	0.01	87.86	87.86	0.00	3.29	87.86	87.86	0.00	88.98	87.86	87.86	0.00	152.70	149681	87.86	87.86	0.00	158.65	2505	49	⋯
10-8	88.52	88.52	0.00	0.01	88.52	88.52	0.00	3.09	88.52	88.52	0.00	14.18	88.52	88.52	0.00	155.60	149690	88.52	88.52	0.00	70.61	2538	46	⋯
10-9	80.99	80.99	0.00	0.01	80.99	80.99	0.00	38.50	80.99	80.99	0.00	508.13	80.99	80.99	0.00	142.92	149700	80.99	80.99	0.00	59.54	2645	52	⋯
15-1	104.73	104.73	0.00	0.02	104.73	104.73	0.00	3.60	104.67	104.73	0.07	3615.45	104.73	104.73	0.00	343.54	336738	104.73	104.73	0.00	3555.10	5794	65	⋯
15-10	99.16	99.16	0.00	0.02	99.16	99.16	0.00	3.47	99.04	99.17	0.14	3616.48	99.16	99.16	0.00	346.15	336767	99.16	99.16	0.00	1174.64	5775	66	⋯
15-2	107.59	107.59	0.00	0.02	107.59	107.59	0.00	3.71	107.62	107.62	0.00	134.61	107.59	107.59	0.00	340.33	336780	107.59	107.59	0.00	1155.86	5718	69	⋯
15-3	81.77	81.77	0.00	0.02	81.77	81.77	0.00	3.79	81.60	81.79	0.23	3623.17	81.77	81.77	0.00	316.07	336753	81.77	81.77	0.00	945.40	5787	67	⋯
15-4	77.09	77.09	0.00	0.02	77.09	77.09	0.00	3.56	76.96	77.16	0.26	3616.00	77.09	77.09	0.00	340.03	336767	77.09	77.09	0.00	1389.32	5853	71	⋯
15-5	178.69	178.69	0.00	0.02	178.69	178.69	0.00	5.14	178.64	178.70	0.04	3615.31	178.69	178.69	0.00	335.34	336708	178.69	178.69	0.00	3601.03	5663	68	⋯
15-6	88.91	88.91	0.00	0.02	88.91	88.91	0.00	12.85	88.89	88.92	0.03	3614.55	88.91	88.91	0.00	343.61	336780	88.91	88.91	0.00	830.78	5704	63	⋯
15-7	104.06	104.06	0.00	0.02	104.05	104.05	0.00	7.86	104.01	104.06	0.05	3616.93	104.05	104.06	0.00	346.72	336739	104.05	104.06	0.00	3601.02	5418	64	⋯
15-8	95.00	95.00	0.00	0.02	95.00	95.00	0.00	3.54	94.92	95.01	0.09	3618.46	95.00	95.00	0.00	334.24	336726	95.00	95.00	0.00	775.26	5692	64	⋯
15-9	86.14	86.14	0.00	0.02	86.14	86.14	0.00	3.59	86.06	86.15	0.11	3614.55	86.14	86.14	0.00	338.03	336752	86.14	86.14	0.00	1910.19	5822	67	⋯
20-1	100.57	100.57	0.00	0.02	100.57	100.57	0.00	4.42	100.21	100.64	0.43	3616.38	100.57	100.57	0.00	634.00	598547	100.57	100.57	0.00	3601.03	8597	68	⋯
20-10	165.30	165.30	0.00	0.02	165.30	165.30	0.00	7.66	164.57	165.43	0.52	3618.62	165.30	165.30	0.00	643.30	598607	165.30	165.30	0.00	3601.03	9671	82	⋯
20-2	111.26	111.26	0.00	0.02	111.26	111.26	0.00	4.54	110.97	111.32	0.32	3615.11	111.26	111.26	0.00	611.46	598518	111.26	111.26	0.00	3601.03	8464	63	⋯
20-3	88.78	88.78	0.00	0.02	88.78	88.78	0.00	3.98	88.07	88.93	0.97	3632.51	88.78	88.78	0.00	625.53	598683	88.78	88.78	0.00	3601.04	8730	66	⋯
20-4	83.96	83.96	0.00	0.02	83.96	83.96	0.00	4.14	83.29	84.06	0.91	3624.55	83.96	83.96	0.00	664.87	598702	83.95	83.96	0.00	3601.03	8184	59	⋯
20-5	137.64	137.64	0.00	0.02	137.64	137.64	0.00	4.20	137.20	137.73	0.38	3619.51	137.64	137.64	0.00	652.04	598490	137.64	137.64	0.00	3601.03	8468	67	⋯
20-6	94.44	94.44	0.00	0.02	94.44	94.44	0.00	4.39	93.91	94.55	0.67	3621.37	94.44	94.44	0.00	643.52	598646	94.44	94.44	0.00	3601.02	8857	70	⋯
20-7	79.79	79.79	0.00	0.02	79.79	79.79	0.00	4.51	79.46	79.89	0.53	3616.39	79.79	79.79	0.00	603.21	598575	79.79	79.79	0.00	3601.02	9212	69	⋯
20-8	126.37	126.37	0.00	0.02	126.37	126.37	0.00	4.35	$-\infty$	$+\infty$	100.0	3600.10	126.37	126.37	0.00	662.60	598541	126.37	126.37	0.00	3601.03	8522	67	⋯
20-9	89.45	89.45	0.00	0.02	89.45	89.45	0.00	13.10	88.67	89.64	1.07	3600.10	89.45	89.45	0.00	676.07	598664	89.45	89.45	0.00	3601.03	10311	77	⋯
5-1	94.04	94.04	0.00	0.01	94.04	94.04	0.00	2.98	94.04	94.04	0.00	0.92	94.04	94.04	0.00	40.96	37415	94.04	94.04	0.00	9.02	631	34	⋯
5-10	154.37	154.37	0.00	0.01	154.37	154.37	0.00	2.91	154.37	154.37	0.00	0.13	154.37	154.37	0.00	41.49	37410	154.37	154.37	0.00	7.68	628	36	⋯
5-2	60.23	60.23	0.00	0.01	60.23	60.23	0.00	3.00	60.23	60.23	0.00	0.81	60.23	60.23	0.00	39.78	37425	60.23	60.23	0.00	13.80	677	34	⋯
5-3	90.27	90.27	0.00	0.01	90.27	90.27	0.00	2.87	90.27	90.27	0.00	0.14	90.27	90.27	0.00	44.57	37425	90.27	90.27	0.00	12.99	642	32	⋯
5-4	97.04	97.04	0.00	0.01	97.04	97.04	0.00	3.08	97.04	97.04	0.00	0.49	97.04	97.04	0.00	40.68	37425	97.04	97.04	0.00	13.05	647	36	⋯
5-5	153.04	153.04	0.00	0.02	153.04	153.04	0.00	3.05	153.04	153.04	0.00	0.16	153.04	153.04	0.00	37.64	37410	153.04	153.04	0.00	11.00	588	31	⋯
5-6	81.57	81.57	0.00	0.01	81.57	81.57	0.00	3.18	81.57	81.57	0.00	1.44	81.57	81.57	0.00	39.53	37425	81.57	81.57	0.00	10.48	646	34	⋯
5-7	117.37	117.37	0.00	0.01	117.37	117.37	0.00	3.03	117.37	117.37	0.00	0.59	117.37	117.37	0.00	40.84	37425	117.37	117.37	0.00	12.16	635	35	⋯
5-8	82.62	82.62	0.00	0.01	82.62	82.62	0.00	2.81	82.62	82.62	0.00	0.60	82.62	82.62	0.00	40.34	37425	82.62	82.62	0.00	11.45	609	30	⋯
5-9	81.75	81.75	0.00	0.01	81.75	81.75	0.00	3.11	81.75	81.75	0.00	1.03	81.75	81.75	0.00	37.72	37425	81.75	81.75	0.00	16.97	656	32	⋯
#S			39				39				19				39					27				⋯
SGM			0.00	0.02			0.00	4.66			0.29	592.48			0.00	194.00	180477			0.00	487.01	2975	52	⋯

Table 23 Detailed results for problem NLTP, cost functions f_8

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-1	524.89	524.89	**0.00**	**1.61**	524.89	524.89	**0.00**	296.47	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	524.89	524.89	**0.00**	476.98	1335	26	···
10-10	385.17	385.17	**0.00**	**15.14**	385.17	385.17	**0.00**	85.81	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	385.17	385.17	**0.00**	2808.07	1258	30	···
10-2	454.47	454.47	**0.00**	**6.74**	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	454.47	454.47	**0.00**	2936.02	1226	20	···
10-3	415.22	415.22	**0.00**	**6.13**	415.22	415.22	**0.00**	24.30	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	415.22	415.22	**0.00**	1617.32	1331	28	···
10-4	438.14	438.14	**0.00**	**8.33**	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	438.14	438.15	0.00	3601.02	1325	23	···
10-5	423.61	423.61	**0.00**	**0.33**	423.61	423.61	**0.00**	31.81	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	423.61	423.61	**0.00**	146.92	1165	19	···
10-6	354.01	354.01	**0.00**	**2.32**	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	354.01	354.01	**0.00**	1506.63	1137	19	···
10-7	388.31	388.31	**0.00**	**7.99**	388.31	388.31	**0.00**	93.62	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	388.31	388.31	**0.00**	1726.73	1298	30	···
10-8	382.82	382.82	**0.00**	**9.19**	382.82	382.82	**0.00**	331.13	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	382.80	382.82	0.00	3601.02	1244	22	···
10-9	495.30	495.30	**0.00**	**81.07**	$-\infty$	$+\infty$	100.0	$+\infty$	478.67	504.49	5.12	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	467.87	507.26	7.76	3601.02	621	6	···
15-1	586.64	586.64	**0.00**	**551.33**	$-\infty$	$+\infty$	100.0	$+\infty$	531.18	605.92	12.34	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	570.11	608.06	6.24	3601.02	1361	7	···
15-10	522.51	522.51	**0.00**	**505.72**	$-\infty$	$+\infty$	100.0	$+\infty$	481.79	535.87	10.09	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	503.15	548.74	8.31	3601.02	1368	5	···
15-2	550.69	550.69	**0.00**	**263.90**	520.03	556.82	6.61	3603.02	515.66	564.74	8.69	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	542.17	571.33	5.10	3601.02	1394	9	···
15-3	502.75	502.75	**0.00**	**389.08**	478.70	520.33	8.00	3603.01	480.30	509.68	5.76	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	491.93	518.03	5.04	3601.08	1365	6	···
15-4	469.61	469.61	**0.00**	**540.55**	443.81	471.88	5.95	3608.67	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	438.51	488.31	10.20	3601.02	1365	5	···
15-5	688.17	688.17	**0.00**	**156.76**	$-\infty$	$+\infty$	100.0	$+\infty$	659.10	704.62	6.46	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	649.52	711.66	8.73	3601.02	1359	4	···
15-6	422.59	422.59	**0.00**	**82.42**	414.52	$+\infty$	100.00	$+\infty$	399.64	433.55	7.82	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	419.09	434.16	3.47	3601.02	1408	13	···
15-7	505.52	505.52	**0.00**	**53.67**	$-\infty$	$+\infty$	100.0	$+\infty$	490.04	515.43	4.93	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	495.32	519.67	4.69	3601.02	1362	5	···
15-8	551.18	551.18	**0.00**	**1039.71**	$-\infty$	$+\infty$	100.0	$+\infty$	514.16	571.59	10.05	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	523.72	567.90	7.78	3601.08	1372	5	···
15-9	400.40	400.40	**0.00**	**49.93**	$-\infty$	$+\infty$	100.0	$+\infty$	393.32	401.83	2.12	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	397.03	419.09	5.26	3601.02	1639	8	···
20-1	617.76	627.45	**1.54**	3600.01	599.24	643.66	6.90	3603.13	580.87	646.42	10.14	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	2400	1	···
20-10	732.81	741.53	**1.18**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	689.07	760.45	9.39	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	621.90	773.29	19.58	3601.04	2409	2	···
20-2	601.74	621.80	**3.23**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	569.30	637.34	10.68	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	517.88	656.04	21.06	3601.02	2405	2	···
20-3	659.67	674.38	**2.18**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	552.16	700.01	21.12	3601.03	2403	2	···
20-4	525.98	541.51	**2.87**	3600.01	500.53	542.25	7.69	3602.90	498.44	542.25	8.08	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	456.71	572.61	20.24	3601.02	2414	2	···
20-5	$-\infty$	$+\infty$	100.0	$+\infty$	623.85	674.84	**7.56**	3603.01	605.64	680.12	10.95	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	2400	1	···
20-6	640.26	669.82	**4.41**	3600.01	607.32	684.22	11.24	3602.97	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	2400	1	···
20-7	477.83	477.83	**0.00**	**1184.35**	468.77	482.66	2.88	3603.13	443.67	486.71	8.84	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	434.30	503.92	13.82	3601.02	2411	3	···
20-8	708.66	721.51	**1.78**	3600.01	668.08	732.69	8.82	3603.08	647.81	761.49	14.93	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	2400	1	···
20-9	671.97	717.50	**6.35**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	641.68	743.50	13.69	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	2400	1	···
5-1	227.66	227.66	**0.00**	**0.05**	227.66	227.66	**0.00**	3.51	227.66	227.66	**0.00**	11.83	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	227.66	227.66	**0.00**	6.20	310	18	···
5-10	290.96	290.96	**0.00**	**0.04**	290.96	290.96	**0.00**	3.50	290.96	290.96	**0.00**	9.11	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	290.96	290.96	**0.00**	6.70	309	16	···
5-2	165.29	165.29	**0.00**	**0.04**	165.29	165.29	**0.00**	3.75	165.29	165.29	**0.00**	9.39	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	165.29	165.29	**0.00**	9.55	313	20	···
5-3	268.34	268.34	**0.00**	**0.04**	268.34	268.34	**0.00**	3.99	268.34	268.34	**0.00**	17.53	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	268.34	268.34	**0.00**	15.29	372	18	···
5-4	264.56	264.56	**0.00**	**0.06**	264.56	264.56	**0.00**	5.88	264.56	264.56	**0.00**	18.63	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	264.56	264.56	**0.00**	15.03	317	28	···
5-5	258.89	258.89	**0.00**	**0.04**	258.89	258.89	**0.00**	3.84	258.89	258.89	**0.00**	6.51	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	258.89	258.89	**0.00**	4.49	282	17	···
5-6	235.51	235.51	**0.00**	**0.05**	235.51	235.51	**0.00**	4.41	235.51	235.51	**0.00**	28.20	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	235.51	235.51	**0.00**	14.00	383	18	···
5-7	319.77	319.77	**0.00**	**0.07**	319.77	319.77	**0.00**	3.90	319.77	319.77	**0.00**	12.89	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	319.77	319.77	**0.00**	17.32	338	19	···
5-8	273.43	273.43	**0.00**	**0.05**	273.43	273.43	**0.00**	3.45	273.43	273.43	**0.00**	21.18	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	273.43	273.43	**0.00**	12.77	354	21	···
5-9	235.51	235.51	**0.00**	**0.05**	235.51	235.51	**0.00**	3.51	235.51	235.51	**0.00**	21.49	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	235.51	235.51	**0.00**	10.55	291	17	···
#S			**31**				16				10				0					17				···
SGM			**0.45**	**142.40**			8.01	1208.21			9.89	2926.96			100.00	14400.00	$+\infty$			3.48	1794.72	1067	10	···

Table 24 Detailed results for problem NLTP, cost functions f_9

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-1	5.69	113.73	95.00	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.43	332.27	99.57	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	113.72	113.72	**0.00**	**317.69**	2511	48	···
10-10	3.17	45.84	93.09	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.08	335.82	99.68	3600.10	45.83	45.83	**0.00**	1355.36	213481	45.83	45.83	**0.00**	**346.50**	2432	48	···
10-2	4.62	122.27	96.22	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.41	434.87	99.68	3600.10	$-\infty$	$+\infty$	100.00	3601.26	213490	122.22	122.22	**0.00**	**268.88**	2049	50	···
10-3	3.56	54.83	93.51	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.08	447.08	99.76	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	54.81	54.81	**0.00**	**263.50**	2572	54	···
10-4	2.84	56.37	94.96	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.20	331.41	99.64	3600.10	56.33	56.33	**0.00**	1394.15	213480	56.33	56.33	**0.00**	**257.80**	2527	48	···
10-5	5.82	75.11	92.25	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.33	398.16	99.67	3600.10	75.07	75.07	**0.00**	1778.93	213463	75.07	75.07	**0.00**	**336.17**	2537	50	···
10-6	3.28	56.14	94.15	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.08	301.08	99.64	3600.10	56.14	56.14	**0.00**	1626.88	213470	56.14	56.14	**0.00**	**334.37**	2389	49	···
10-7	2.71	48.49	94.42	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.04	267.19	99.61	3600.10	48.47	48.47	**0.00**	1006.17	213481	48.47	48.47	**0.00**	**324.21**	2464	56	···
10-8	3.08	56.97	94.60	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.05	303.78	99.65	3600.10	56.93	56.93	**0.00**	1336.08	213473	56.93	56.93	**0.00**	**255.84**	2555	48	···
10-9	2.15	34.46	93.77	3600.01	1.44	63.42	97.73	3609.32	1.17	195.75	99.40	3600.10	34.46	34.46	**0.00**	625.21	213500	34.46	34.46	**0.00**	**162.25**	2591	48	···
15-1	1.53	52.27	97.07	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	51.91	51.92	**0.00**	3601.02	4290	64	···
15-10	1.22	67.66	98.19	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	64.14	64.14	**0.00**	**2196.28**	480320	64.14	64.14	0.00	3601.03	4363	63	···
15-2	1.39	64.81	97.86	3600.01	1.59	122.06	98.70	3602.88	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	64.45	64.45	**0.00**	3601.03	3924	56	···
15-3	1.26	42.36	97.03	3600.01	1.52	112.34	98.65	3603.03	$-\infty$	$+\infty$	100.0	3600.10	41.87	41.87	**0.00**	**1785.51**	480306	41.87	41.87	**0.00**	3601.02	4631	56	···
15-4	1.42	46.62	96.96	3600.01	1.27	95.11	98.67	3612.99	$-\infty$	$+\infty$	100.0	3600.10	46.10	46.10	**0.00**	**2695.79**	480332	46.10	46.10	0.01	3601.02	3997	51	···
15-5	2.45	183.94	98.67	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	156.93	161.21	**2.65**	3601.02	2081	10	···
15-6	1.50	60.22	97.51	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	54.34	54.44	**0.17**	3601.02	3055	33	···
15-7	1.69	73.30	97.69	3600.01	1.49	136.83	98.91	3605.45	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	71.64	72.67	**1.42**	3601.02	2394	21	···
15-8	1.42	51.44	97.24	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	50.81	50.81	**0.01**	3601.02	4119	57	···
15-9	0.96	56.57	98.31	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	54.19	56.95	**4.85**	3601.02	2173	10	···
20-1	1.41	58.20	97.57	3600.01	1.83	132.02	98.61	3603.21	1.48	254.61	99.42	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	53.99	57.93	**6.79**	3601.03	3618	8	···
20-10	1.73	157.61	98.90	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	1.92	1065.74	99.82	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	125.82	130.70	**3.74**	3601.02	3388	13	···
20-2	1.36	63.51	97.86	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.53	664.83	99.77	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	58.83	63.85	**7.87**	3601.02	3361	7	···
20-3	1.45	48.61	97.02	3600.01	1.77	128.38	98.62	3602.92	1.53	130.69	98.83	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	40.08	48.82	**17.91**	3601.02	3204	3	···
20-4	1.21	46.92	97.43	3600.01	1.61	110.78	98.54	3602.78	1.31	141.07	99.07	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	40.59	46.72	**13.12**	3601.02	3311	4	···
20-5	1.46	114.52	98.73	3600.01	1.92	203.68	99.06	3603.09	1.65	566.87	99.71	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	104.57	107.97	**3.15**	3601.02	3614	10	···
20-6	1.41	52.41	97.30	3600.01	1.70	126.51	98.66	3603.10	1.45	209.29	99.30	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	47.46	52.18	**9.05**	3601.02	3462	7	···
20-7	1.20	44.67	97.32	3600.01	1.36	116.25	98.83	3602.99	1.20	181.34	99.34	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	39.35	44.84	**12.24**	3601.02	3196	4	···
20-8	1.55	73.52	97.89	3600.01	1.87	141.56	98.68	3603.04	1.64	212.24	99.23	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	69.64	73.30	**5.00**	3601.02	3536	11	···
20-9	1.40	49.86	97.19	3600.01	1.93	136.05	98.58	3607.17	1.54	131.27	98.83	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	40.66	49.84	**18.42**	3601.03	3183	3	···
5-1	34.15	112.64	69.68	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.27	130.30	99.02	3600.10	112.64	112.64	**0.00**	799.56	53370	112.64	112.64	**0.00**	**70.31**	552	32	···
5-10	113.63	202.69	43.94	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	65.75	228.83	71.27	3600.10	202.69	202.69	**0.00**	1139.27	53365	202.69	202.69	**0.00**	**217.71**	650	45	···
5-2	16.51	68.29	75.82	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.28	84.44	98.49	3600.10	68.29	68.29	**0.00**	492.19	53375	68.29	68.29	**0.00**	**56.43**	553	34	···
5-3	27.21	100.63	72.96	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.67	129.41	98.71	3600.10	100.63	100.63	**0.00**	675.00	53375	100.63	100.63	**0.00**	**57.96**	578	29	···
5-4	25.55	99.83	74.41	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.37	157.41	99.13	3600.10	99.83	99.83	**0.00**	424.66	53375	99.83	99.83	**0.00**	**58.33**	573	31	···
5-5	108.63	196.07	44.60	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	61.87	212.89	70.94	3600.10	$-\infty$	$+\infty$	100.00	3601.93	922394	196.07	196.07	**0.00**	**54.03**	555	31	···
5-6	13.85	73.75	81.22	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.04	153.38	99.32	3600.10	73.75	73.75	**0.00**	344.74	53375	73.75	73.75	**0.00**	**62.97**	620	32	···
5-7	31.85	151.38	78.96	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.41	190.82	99.26	3600.10	151.38	151.38	**0.00**	629.06	53375	151.38	151.38	**0.00**	**60.54**	557	29	···
5-8	22.09	78.24	71.76	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1.30	108.42	98.80	3600.10	78.24	78.24	**0.00**	301.81	53375	78.24	78.24	**0.00**	**54.50**	571	31	···
5-9	16.91	105.61	83.99	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	0.99	157.05	99.37	3600.10	105.61	105.61	**0.00**	646.16	53375	105.61	105.61	**0.00**	**117.22**	621	35	···
#S			0				0				0				19					**20**				···
SGM			88.50	14400.00			99.55	14400.00			97.86	14400.00			10.28	3939.13	2263498161			**0.94**	**1485.86**	2012	28	···

Table 25 **Detailed results for problem NLTP, cost functions f_{10}**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						…
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	…
10-20-1	593.46	593.46	**0.00**	**0.11**	593.46	593.46	**0.00**	2.56	593.46	593.46	**0.00**	230.55	593.45	593.46	**0.00**	204.98	79600	593.45	593.46	**0.00**	59.61	680	11	…
10-20-10	604.19	604.19	**0.00**	**0.25**	604.19	604.19	**0.00**	2.67	604.19	604.19	**0.00**	1814.62	604.18	604.19	**0.00**	195.07	79600	604.18	604.19	**0.00**	71.74	683	14	…
10-20-2	479.56	479.56	**0.00**	**0.16**	479.56	479.56	**0.00**	3.09	479.56	479.56	**0.00**	379.39	479.56	479.56	**0.00**	209.10	79600	479.56	479.56	**0.00**	60.19	680	12	…
10-20-3	552.59	552.59	**0.00**	**0.22**	552.59	552.59	**0.00**	2.85	552.59	552.59	**0.00**	654.30	552.59	552.59	**0.00**	162.48	79600	552.59	552.59	**0.00**	62.50	686	13	…
10-20-4	539.23	539.23	**0.00**	**0.14**	539.23	539.23	**0.00**	2.16	539.23	539.23	**0.00**	305.69	539.23	539.23	**0.00**	159.02	79600	539.23	539.23	**0.00**	74.36	678	11	…
10-20-5	628.69	628.69	**0.00**	**0.11**	628.69	628.69	**0.00**	2.53	628.69	628.69	**0.00**	273.48	628.69	628.69	**0.00**	211.30	79600	628.69	628.69	**0.00**	57.48	682	14	…
10-20-6	600.06	600.06	**0.00**	**0.12**	600.06	600.06	**0.00**	2.19	600.06	600.06	**0.00**	228.72	600.06	600.06	**0.00**	197.24	79600	600.06	600.06	**0.00**	69.72	680	12	…
10-20-7	471.42	471.42	**0.00**	**0.11**	471.42	471.42	**0.00**	2.49	471.42	471.42	**0.00**	155.22	471.42	471.42	**0.00**	191.44	79600	471.42	471.42	**0.00**	58.92	680	10	…
10-20-8	594.92	594.92	**0.00**	**0.14**	594.92	594.92	**0.00**	2.47	594.92	594.92	**0.00**	397.40	594.92	594.92	**0.00**	165.99	79600	594.92	594.92	**0.00**	59.21	680	13	…
10-20-9	597.30	597.30	**0.00**	**0.24**	597.30	597.30	**0.00**	3.23	597.30	597.30	**0.00**	362.03	597.30	597.30	**0.00**	173.04	79600	597.30	597.30	**0.00**	60.73	681	14	…
20-40-1	859.01	859.01	**0.00**	**0.63**	859.01	859.01	**0.00**	12.31	801.53	936.90	14.45	3600.10	859.01	859.01	**0.00**	926.33	318400	859.01	859.01	**0.00**	267.30	2561	14	…
20-40-10	876.42	876.42	**0.00**	**0.74**	876.42	876.42	**0.00**	8.00	811.30	992.45	18.25	3600.10	876.42	876.42	**0.00**	903.80	318400	876.42	876.42	**0.00**	259.00	2564	13	…
20-40-2	760.27	760.27	**0.00**	**1.11**	760.27	760.27	**0.00**	11.15	697.12	1221.47	42.93	3600.10	760.27	760.27	**0.00**	944.32	318400	760.27	760.27	**0.00**	233.99	2567	14	…
20-40-3	815.18	815.18	**0.00**	**0.68**	815.18	815.18	**0.00**	11.00	749.19	854.41	12.31	3600.10	815.18	815.18	**0.00**	943.54	318400	815.18	815.18	**0.00**	262.07	2563	16	…
20-40-4	895.08	895.08	**0.00**	**1.19**	895.08	895.08	**0.00**	12.77	797.27	939.71	15.16	3600.10	895.08	895.08	**0.00**	958.06	318400	895.08	895.08	**0.00**	269.38	2569	18	…
20-40-5	779.62	779.62	**0.00**	**0.72**	779.62	779.62	**0.00**	10.83	717.89	945.34	24.06	3600.10	779.62	779.62	**0.00**	859.34	318400	779.62	779.62	**0.00**	254.08	2565	15	…
20-40-6	866.44	866.44	**0.00**	**0.50**	866.44	866.44	**0.00**	11.60	809.03	976.45	17.15	3600.10	866.44	866.44	**0.00**	902.19	318400	866.44	866.44	**0.00**	270.07	2561	12	…
20-40-7	884.33	884.33	**0.00**	**0.70**	884.33	884.33	**0.00**	12.31	813.94	994.54	18.16	3600.10	884.33	884.33	**0.00**	901.46	318400	884.33	884.33	**0.00**	290.58	2560	12	…
20-40-8	848.12	848.12	**0.00**	**0.39**	848.12	848.12	**0.00**	11.17	791.39	947.15	16.45	3600.10	848.12	848.12	**0.00**	866.66	318400	848.12	848.12	**0.00**	332.50	2559	11	…
20-40-9	806.01	806.01	**0.00**	**0.44**	806.01	806.01	**0.00**	13.27	750.88	910.13	17.50	3600.10	806.01	806.01	**0.00**	782.45	318400	806.01	806.01	**0.00**	286.75	2563	13	…
30-60-1	1052.14	1052.14	**0.00**	**1.23**	1052.14	1052.14	**0.00**	126.80	963.43	1162.13	17.10	3600.10	1052.14	1052.14	**0.00**	2201.56	716400	1052.14	1052.14	**0.00**	590.52	5650	13	…
30-60-10	796.30	796.30	**0.00**	**0.89**	796.30	796.30	**0.00**	86.84	729.82	904.46	19.31	3600.10	796.30	796.30	**0.00**	1997.44	716400	796.30	796.30	**0.00**	527.75	5641	13	…
30-60-2	928.11	928.11	**0.00**	**1.30**	928.11	928.11	**0.00**	151.04	852.88	1035.28	17.62	3600.10	928.11	928.11	**0.00**	2191.76	716400	928.11	928.11	**0.00**	765.46	5642	14	…
30-60-3	983.59	983.59	**0.00**	**1.18**	983.59	983.59	**0.00**	39.58	902.63	1192.74	24.32	3600.10	983.59	983.59	**0.00**	2132.73	716400	983.59	983.59	**0.00**	742.94	5640	12	…
30-60-4	1036.67	1036.67	**0.00**	**1.73**	1036.67	1036.67	**0.00**	127.86	942.53	1190.32	20.82	3600.10	1036.67	1036.67	**0.00**	2160.40	716400	1036.67	1036.67	**0.00**	544.85	5640	13	…
30-60-5	975.03	975.03	**0.00**	**1.01**	975.03	975.03	**0.00**	123.27	$-\infty$	$+\infty$	100.0	3600.10	975.03	975.03	**0.00**	2177.53	716400	975.03	975.03	**0.00**	624.05	5640	13	…
30-60-6	1036.47	1036.47	**0.00**	**1.13**	1036.47	1036.47	**0.00**	141.51	947.69	1244.34	23.84	3600.10	1036.47	1036.47	**0.00**	2173.35	716400	1036.47	1036.47	**0.00**	555.83	5644	14	…
30-60-7	1163.30	1163.30	**0.00**	**2.21**	1163.30	1163.30	**0.00**	143.53	1038.19	1363.68	23.87	3600.10	1163.30	1163.30	**0.00**	2149.18	716400	1163.30	1163.30	**0.00**	615.54	5642	14	…
30-60-8	1075.28	1075.28	**0.00**	**1.73**	1075.28	1075.28	**0.00**	103.10	966.67	1265.01	23.58	3600.10	1075.28	1075.28	**0.00**	2247.03	716400	1075.28	1075.28	**0.00**	521.08	5657	17	…
30-60-9	1005.58	1005.58	**0.00**	**1.89**	1005.58	1005.58	**0.00**	138.00	905.58	1176.10	23.00	3600.10	1005.58	1005.58	**0.00**	2199.36	716400	1005.58	1005.58	**0.00**	611.29	5640	14	…
40-80-1	1261.54	1261.54	**0.00**	**3.87**	1261.54	1261.54	**0.00**	574.27	1119.17	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1261.54	1261.54	**0.00**	1243.28	9925	14	…
40-80-10	1244.18	1244.18	**0.00**	**3.75**	1244.18	1244.18	**0.00**	120.85	1113.63	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1244.18	1244.18	**0.00**	1116.47	9938	17	…
40-80-2	1271.85	1271.85	**0.00**	**3.23**	1271.85	1271.85	**0.00**	117.22	1123.53	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1271.85	1271.85	**0.00**	1134.63	9958	16	…
40-80-3	1133.70	1133.70	**0.00**	**3.51**	1133.70	1133.70	**0.00**	283.30	1024.08	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1133.70	1133.70	**0.00**	1081.23	9933	17	…
40-80-4	1197.66	1197.66	**0.00**	**3.39**	1197.66	1197.66	**0.00**	159.83	1072.95	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1197.66	1197.66	**0.00**	1079.33	9933	16	…
40-80-5	1224.79	1224.79	**0.00**	**2.70**	1224.79	1224.79	**0.00**	102.44	1111.90	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1224.79	1224.79	**0.00**	1004.04	9921	13	…
40-80-6	1141.46	1141.46	**0.00**	**3.96**	1141.46	1141.46	**0.00**	258.71	1040.94	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1141.46	1141.46	**0.00**	1211.24	9926	13	…
40-80-7	1310.33	1310.33	**0.00**	**4.23**	1310.33	1310.33	**0.00**	124.16	1176.35	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1310.33	1310.33	**0.00**	1200.45	9954	17	…
40-80-8	1169.33	1169.33	**0.00**	**2.74**	1169.33	1169.33	**0.00**	285.38	1055.24	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1169.33	1169.33	**0.00**	1318.12	9931	15	…
40-80-9	1177.80	1177.80	**0.00**	**3.78**	1177.80	1177.80	**0.00**	471.64	1076.08	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1177.80	1177.80	**0.00**	1204.51	9921	14	…
50-100-1	1383.93	1383.93	**0.00**	**3.72**	1383.93	1383.93	**0.00**	317.98	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1383.92	1383.93	**0.00**	2028.49	15402	13	…
50-100-10	1443.91	1443.91	**0.00**	**5.92**	1443.91	1443.91	**0.00**	804.16	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1443.91	1443.91	**0.00**	2106.71	15439	21	…
50-100-2	1440.02	1440.02	**0.00**	**9.79**	1440.02	1440.02	**0.00**	690.92	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1440.02	1440.02	**0.00**	1716.87	15406	13	…
50-100-3	1379.38	1379.39	**0.00**	**9.48**	1379.39	1379.39	**0.00**	725.86	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1379.38	1379.39	**0.00**	1927.39	15441	17	…
50-100-4	1285.89	1285.89	**0.00**	**5.74**	1285.89	1285.89	**0.00**	613.73	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1285.89	1285.89	**0.00**	2058.81	15404	16	…
50-100-5	1264.63	1264.63	**0.00**	**6.42**	1264.63	1264.63	**0.00**	272.05	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1264.62	1264.63	**0.00**	1867.61	15409	16	…
50-100-6	1321.17	1321.17	**0.00**	**8.36**	1321.17	1321.17	**0.00**	333.31	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1321.17	1321.17	**0.00**	1776.97	15419	17	…
50-100-7	1525.00	1525.00	**0.00**	**10.32**	1525.00	1525.00	**0.00**	1284.64	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1525.00	1525.00	**0.00**	1998.02	15400	14	…
50-100-8	1327.70	1327.70	**0.00**	**8.18**	1327.70	1327.70	**0.00**	613.27	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1327.70	1327.70	**0.00**	1724.34	15415	16	…
50-100-9	1445.28	1445.28	**0.00**	**6.65**	1445.28	1445.28	**0.00**	556.64	1286.56	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1445.28	1445.28	**0.00**	2011.65	15405	17	…

Table 26 Detailed results for problem NLUFLP-A-gunluk, cost functions f_1

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	313.93	313.93	**0.00**	**46.79**	-399.50	341.42	185.46	3602.02	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	313.93	313.93	**0.00**	3601.01	1278	22	···
10-20-10	277.15	277.15	**0.00**	**711.08**	-372.94	396.65	194.02	3601.97	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	277.15	277.15	**0.00**	3117.87	1478	23	···
10-20-2	251.56	251.56	**0.00**	**3.05**	-357.28	277.07	177.55	3602.01	-750.85	295.65	139.38	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	251.56	251.56	**0.00**	368.14	1190	22	···
10-20-3	276.69	276.69	**0.00**	**6.06**	-453.83	404.58	189.15	3602.37	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	276.69	276.69	**0.00**	718.17	1355	22	···
10-20-4	309.24	309.24	**0.00**	**29.02**	-410.28	369.45	190.05	3601.92	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	309.24	309.24	**0.00**	1226.19	1272	21	···
10-20-5	331.03	331.03	**0.00**	**248.11**	-517.71	452.75	187.45	3602.27	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	331.03	331.03	**0.00**	1967.52	1385	23	···
10-20-6	364.69	364.69	**0.00**	**10.72**	-385.06	383.29	199.54	3601.90	-843.10	444.10	152.67	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	364.69	364.69	**0.00**	602.86	1165	21	···
10-20-7	256.73	256.73	**0.00**	**2.36**	-392.94	289.66	173.72	3602.00	-806.82	320.47	139.72	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	256.73	256.73	**0.00**	359.38	1123	20	···
10-20-8	356.91	356.91	**0.00**	**28.45**	-350.66	421.44	183.20	3602.06	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	356.91	356.91	**0.00**	1317.96	1222	20	···
10-20-9	314.47	314.47	**0.00**	**4.61**	-462.74	377.49	181.58	3602.08	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	314.47	314.48	**0.00**	354.13	1208	20	···
20-40-1	439.54	442.36	**0.64**	3600.01	-1060.41	869.00	181.95	3602.12	-2141.38	781.39	136.49	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	4000	1	···
20-40-10	454.17	454.23	**0.01**	3600.01	-1072.13	1117.74	195.92	3601.94	-2156.23	629.97	129.22	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	450.82	477.21	5.53	3601.03	4038	2	···
20-40-2	403.77	403.80	**0.01**	3600.01	-1014.85	690.62	168.05	3601.99	-2019.75	667.79	133.06	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	402.79	404.50	0.42	3601.13	4098	4	···
20-40-3	432.01	436.05	**0.93**	3600.01	-1040.24	772.78	174.29	3601.95	-2128.30	686.57	132.26	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	4000	1	···
20-40-4	432.80	435.84	**0.70**	3600.01	-1086.64	627.18	157.72	3602.10	-2153.53	725.68	133.70	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	4000	1	···
20-40-5	399.89	399.92	**0.01**	3600.01	-1192.96	813.88	168.22	3602.16	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	398.13	402.29	1.04	3601.02	4075	3	···
20-40-6	466.81	468.23	**0.30**	3600.02	-1054.17	782.63	174.24	3602.11	-2169.48	608.41	128.04	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	465.43	486.60	4.35	3601.02	4038	2	···
20-40-7	463.82	463.92	**0.02**	3600.02	-1066.12	686.10	164.35	3602.03	-2217.95	728.18	132.83	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	4000	1	···
20-40-8	463.48	463.74	**0.05**	3600.01	-1026.92	802.15	178.11	3602.19	-2161.94	741.64	134.30	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	460.65	485.62	5.14	3601.03	4038	2	···
20-40-9	441.78	441.95	**0.04**	3600.01	-1143.73	1011.31	188.42	3602.14	-2307.20	810.08	135.11	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	441.07	442.69	0.37	3601.03	4110	4	···
30-60-1	583.42	589.09	**0.96**	3600.03	-1619.71	1215.45	175.04	3602.26	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	9000	1	···
30-60-10	420.93	421.86	**0.22**	3600.03	-1722.89	1543.61	189.59	3602.16	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	420.48	423.67	0.75	3601.03	9091	3	···
30-60-2	519.38	524.23	**0.93**	3600.02	-1741.22	1253.34	171.98	3602.37	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.19	9000	1	···
30-60-3	522.27	525.87	**0.68**	3600.02	-1687.70	1378.61	181.69	3602.24	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.15	9000	1	···
30-60-4	560.21	563.97	**0.67**	3600.02	-1743.97	1633.53	193.67	3602.30	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.19	9000	1	···
30-60-5	495.01	499.34	**0.87**	3600.03	-1812.95	1436.17	179.22	3602.44	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.26	9000	1	···
30-60-6	538.25	542.00	**0.69**	3600.01	-1802.33	1324.16	173.47	3602.67	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	9000	1	···
30-60-7	626.81	639.19	**1.94**	3600.04	-1787.36	1295.18	172.46	3602.31	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.19	9000	1	···
30-60-8	572.20	587.71	**2.64**	3600.03	-1834.81	1438.75	178.41	3602.41	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.25	9000	1	···
30-60-9	525.31	532.87	**1.42**	3600.02	-1821.29	1260.10	169.19	3602.33	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.18	9000	1	···
40-80-1	664.03	685.48	**3.13**	3600.16	-2369.65	1658.95	170.01	3608.24	-4762.40	1658.95	134.83	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.15	16000	1	···
40-80-10	650.89	667.36	**2.47**	3600.04	-2459.55	1706.61	169.39	3607.74	-4774.57	1869.56	139.16	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	16000	1	···
40-80-2	661.49	681.16	**2.89**	3600.02	-2446.67	2082.60	185.12	3609.62	-4783.96	2082.60	143.53	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	16000	1	···
40-80-3	600.81	611.33	**1.72**	3600.08	-2411.36	1999.13	182.90	3605.90	-4738.07	1999.13	142.19	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	16000	1	···
40-80-4	614.09	635.44	**3.36**	3600.03	-2393.05	2307.23	196.41	3602.73	-4722.70	2311.36	148.94	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.23	16000	1	···
40-80-5	664.45	673.79	**1.39**	3600.05	-2618.81	1883.72	171.93	3604.70	-5010.88	1883.72	137.59	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.19	16000	1	···
40-80-6	585.38	595.59	**1.71**	3600.02	-2440.97	2377.77	197.41	3604.04	-4819.46	2377.77	149.34	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.21	16000	1	···
40-80-7	709.44	729.68	**2.77**	3600.02	-2399.85	2104.65	187.70	3620.71	-4891.65	2104.65	143.03	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	16000	1	···
40-80-8	614.53	627.17	**2.01**	3600.03	-2631.72	2466.78	193.73	3602.74	-5049.43	2663.36	152.75	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	16000	1	···
40-80-9	634.38	648.97	**2.25**	3600.07	-2578.32	2272.29	188.13	3605.54	$-\infty$	2545.52	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	16000	1	···
50-100-1	734.20	769.51	**4.59**	3600.18	-3179.99	2800.73	188.07	3612.47	$-\infty$	2800.73	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.22	25000	1	···
50-100-10	724.53	755.87	**4.15**	3600.03	-3145.38	2759.69	187.74	3620.51	$-\infty$	2759.68	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	25000	1	···
50-100-2	776.42	829.17	**6.36**	3600.07	-3109.46	3009.70	196.79	3631.68	$-\infty$	3138.05	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.20	25000	1	···
50-100-3	701.40	734.15	**4.46**	3600.04	-3072.90	3211.28	195.69	3646.22	$-\infty$	3363.66	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.10	25000	1	···
50-100-4	665.83	682.59	**2.45**	3600.03	-3260.16	2291.56	170.29	3620.60	$-\infty$	2291.56	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.19	25000	1	···
50-100-5	666.35	686.76	**2.97**	3600.08	-3207.48	1995.46	162.21	3645.72	$-\infty$	2067.62	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	25000	1	···
50-100-6	668.42	691.14	**3.29**	3600.04	-3349.14	2226.02	166.47	3620.12	$-\infty$	2226.02	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	25000	1	···
50-100-7	767.94	795.43	**3.46**	3600.47	-3230.50	2224.01	168.84	3630.67	$-\infty$	2224.01	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	25000	1	···
50-100-8	718.58	751.80	**4.42**	3600.03	-3408.11	2092.62	161.40	3611.51	$-\infty$	2092.62	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.09	25000	1	···
50-100-9	755.82	800.31	**5.56**	3600.03	-3235.11	2570.35	179.45	3622.48	$-\infty$	2570.35	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.09	25000	1	···

Table 27 Detailed results for problem NLUFLP-A-gunluk, cost functions f_2

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	185.32	260.37	28.82	3600.01	188.62	260.37	27.56	3602.31	111.78	297.23	62.39	3600.10	−∞	+∞	100.00	3600.01	+∞	260.37	260.37	**0.00**	3601.02	3601	42	···
10-20-10	191.39	264.27	27.58	3600.01	193.55	264.27	26.76	3602.07	97.78	300.45	67.45	3600.10	−∞	+∞	100.00	3600.01	+∞	263.61	264.63	**0.39**	3601.02	2382	11	···
10-20-2	211.31	259.43	18.55	3600.01	207.72	257.58	19.36	3602.01	105.07	264.88	60.33	3600.10	−∞	+∞	100.00	3600.01	+∞	254.37	258.00	**1.40**	3601.02	2146	5	···
10-20-3	199.15	269.80	26.19	3600.01	196.30	269.80	27.24	3602.05	105.97	335.86	68.45	3600.10	−∞	+∞	100.00	3600.01	+∞	266.71	270.69	**1.47**	3601.01	2227	6	···
10-20-4	232.68	301.36	22.79	3600.01	232.78	301.36	22.76	3601.92	115.76	328.36	64.74	3600.10	−∞	+∞	100.00	3600.01	+∞	299.16	302.49	**1.10**	3601.01	2318	9	···
10-20-5	236.19	292.06	19.13	3600.01	232.54	292.06	20.38	3601.88	109.46	443.17	75.30	3600.10	−∞	+∞	100.00	3600.01	+∞	292.04	292.07	**0.01**	3601.01	2818	24	···
10-20-6	335.92	373.02	9.95	3600.01	−∞	+∞	100.0	+∞	165.75	443.19	62.60	3600.10	−∞	+∞	100.00	3600.01	+∞	370.89	374.00	**0.83**	3601.02	2129	7	···
10-20-7	206.82	255.30	18.99	3600.01	202.67	255.30	20.62	3602.23	96.91	325.58	70.24	3600.10	−∞	+∞	100.00	3600.01	+∞	249.89	256.94	**2.74**	3601.02	2097	4	···
10-20-8	285.99	347.87	17.79	3600.01	−∞	+∞	100.0	+∞	147.56	394.74	62.62	3600.10	−∞	+∞	100.00	3600.01	+∞	339.03	351.31	**3.50**	3601.02	2082	3	···
10-20-9	265.52	318.99	16.76	3600.01	258.61	318.99	18.93	3602.37	125.59	442.67	71.63	3600.10	−∞	+∞	100.00	3600.01	+∞	316.22	320.03	**1.19**	3601.02	2145	7	···
20-40-1	226.13	369.70	38.83	3600.01	220.52	368.49	40.16	3602.35	100.29	734.31	86.34	3600.10	−∞	+∞	100.00	3600.01	+∞	349.95	374.73	**6.61**	3601.02	8136	2	···
20-40-10	202.75	342.67	40.83	3600.01	205.06	342.47	40.12	3602.15	97.96	661.22	85.18	3600.10	−∞	+∞	100.00	3600.01	+∞	341.04	343.56	**0.73**	3601.03	9127	10	···
20-40-2	224.80	357.27	37.08	3600.01	221.52	356.54	37.87	3601.99	86.94	667.75	86.98	3600.10	−∞	+∞	100.00	3600.01	+∞	349.97	359.92	**2.76**	3601.02	8390	5	···
20-40-3	237.51	373.60	36.43	3600.01	236.56	372.60	36.51	3602.34	101.32	702.01	85.57	3600.10	−∞	+∞	100.00	3600.01	+∞	357.46	380.03	**5.94**	3601.08	8138	2	···
20-40-4	235.18	372.57	36.87	3600.01	233.38	370.11	36.94	3601.93	97.61	652.38	85.04	3600.10	−∞	+∞	100.00	3600.01	+∞	354.83	375.48	**5.50**	3601.03	8260	3	···
20-40-5	190.87	325.15	41.30	3600.01	188.46	324.02	41.84	3602.28	87.76	614.43	85.72	3600.10	−∞	+∞	100.00	3600.01	+∞	307.40	329.89	**6.82**	3601.03	8152	2	···
20-40-6	244.57	401.08	39.02	3600.01	238.32	400.12	40.44	3602.01	98.14	649.71	84.89	3600.10	−∞	+∞	100.00	3600.01	+∞	382.38	405.92	**5.80**	3601.02	8235	3	···
20-40-7	254.47	394.78	35.54	3600.01	254.57	393.46	35.30	3602.03	106.97	720.33	85.15	3600.10	−∞	+∞	100.00	3600.01	+∞	378.24	399.02	**5.21**	3601.02	8355	3	···
20-40-8	265.62	402.65	34.03	3600.01	257.50	401.70	35.90	3602.12	103.74	778.60	86.68	3600.10	−∞	+∞	100.00	3600.01	+∞	381.22	406.69	**6.26**	3601.03	8130	2	···
20-40-9	259.62	385.02	32.57	3600.01	253.29	383.14	33.89	3602.19	112.11	812.01	86.19	3600.10	−∞	+∞	100.00	3600.01	+∞	368.13	389.32	**5.44**	3601.03	8123	2	···
30-60-1	235.94	468.89	49.68	3600.02	273.66	459.33	40.42	3602.26	−∞	1058.03	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	439.17	467.46	**6.05**	3601.03	18220	2	···
30-60-10	201.48	342.26	41.13	3600.02	199.18	338.69	41.19	3602.15	83.46	963.74	91.34	3600.10	−∞	+∞	100.00	3600.01	+∞	322.88	344.68	**6.32**	3601.05	18216	2	···
30-60-2	198.83	429.87	53.75	3600.02	257.49	417.39	38.31	3602.41	77.61	871.19	91.09	3600.10	−∞	+∞	100.00	3600.01	+∞	399.22	423.68	**5.77**	3601.06	18185	2	···
30-60-3	236.59	412.53	42.65	3600.01	235.09	410.60	42.75	3602.45	−∞	1148.83	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	389.18	418.14	**6.93**	3601.03	18225	2	···
30-60-4	235.26	421.52	44.19	3600.01	247.20	420.60	41.23	3602.40	82.78	1008.84	91.79	3600.10	−∞	+∞	100.00	3600.01	+∞	403.01	429.92	**6.26**	3601.08	18358	2	···
30-60-5	218.53	376.90	42.02	3600.02	213.80	376.05	43.15	3602.38	80.61	886.39	90.91	3600.10	−∞	+∞	100.00	3600.01	+∞	357.03	383.57	**6.92**	3601.10	18218	2	···
30-60-6	236.98	426.48	44.43	3600.02	234.07	415.61	43.68	3602.36	87.18	952.06	90.84	3600.10	−∞	+∞	100.00	3600.01	+∞	396.11	424.33	**6.65**	3601.08	18250	2	···
30-60-7	217.23	521.87	58.37	3600.02	308.31	506.33	39.11	3602.30	−∞	1107.58	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	488.08	516.05	**5.42**	3601.04	18593	3	···
30-60-8	220.69	393.44	43.91	3600.01	227.44	392.62	42.07	3602.42	83.71	1078.60	92.24	3600.10	−∞	+∞	100.00	3600.01	+∞	374.41	399.30	**6.23**	3601.05	18423	2	···
30-60-9	228.92	423.74	45.98	3600.02	245.00	417.05	41.25	3602.70	85.12	1032.82	91.76	3600.10	−∞	+∞	100.00	3600.01	+∞	400.07	424.53	**5.76**	3601.03	18292	2	···
40-80-1	199.58	525.68	62.04	3600.02	273.24	483.17	43.45	3603.10	8.87	1218.41	99.27	3600.10	−∞	+∞	100.00	3600.01	+∞	462.62	492.55	**6.08**	3601.08	32322	2	···
40-80-10	197.60	447.90	55.88	3600.02	254.84	446.67	42.95	3602.63	7.54	1541.65	99.51	3600.10	−∞	+∞	100.00	3600.01	+∞	421.25	449.99	**6.39**	3601.06	32581	2	···
40-80-2	177.00	506.95	65.09	3600.02	287.49	496.66	**42.12**	3603.01	11.44	1217.52	99.06	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	32000	1	···
40-80-3	197.08	455.31	56.71	3600.02	246.88	443.53	**44.34**	3603.03	7.80	1399.65	99.44	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	32000	1	···
40-80-4	181.05	450.02	59.77	3600.02	246.04	436.13	43.59	3602.91	6.73	1070.92	99.37	3600.10	−∞	+∞	100.00	3600.01	+∞	415.42	443.75	**6.38**	3601.04	32407	2	···
40-80-5	174.38	491.73	64.54	3600.02	265.01	461.67	**42.60**	3603.04	8.70	1237.11	99.30	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	32000	1	···
40-80-6	197.22	417.77	52.79	3600.04	229.53	409.23	43.91	3602.56	5.80	1150.70	99.50	3600.10	−∞	+∞	100.00	3600.01	+∞	388.45	414.03	**6.18**	3601.11	32478	2	···
40-80-7	189.80	582.74	67.43	3600.04	327.25	539.58	39.35	3602.66	15.26	1349.81	98.87	3600.10	−∞	+∞	100.00	3600.01	+∞	517.26	548.05	**5.62**	3601.10	32428	2	···
40-80-8	173.80	447.91	61.20	3600.03	249.50	436.97	42.90	3603.14	7.50	1363.29	99.45	3600.10	−∞	+∞	100.00	3600.01	+∞	416.02	445.69	**6.66**	3601.14	32276	2	···
40-80-9	177.32	487.35	63.61	3600.03	262.32	466.23	**43.73**	3602.73	7.62	1246.45	99.39	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.19	32000	1	···
50-100-1	155.34	533.54	70.88	3600.04	287.38	519.15	**44.64**	3603.42	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.27	50000	1	···
50-100-10	150.48	507.33	70.34	3600.02	281.32	498.94	**43.62**	3603.33	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.24	50000	1	···
50-100-2	155.12	611.23	74.62	3600.03	321.09	576.26	**44.28**	3603.40	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.23	50000	1	···
50-100-3	148.77	489.33	69.60	3600.04	258.78	470.43	**44.99**	3603.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.23	50000	1	···
50-100-4	135.50	486.92	72.17	3600.03	263.22	468.59	**43.83**	3603.01	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.22	50000	1	···
50-100-5	140.81	486.93	71.08	3600.03	271.02	467.94	**42.08**	3603.37	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	50000	1	···
50-100-6	143.74	494.42	70.93	3600.04	263.07	472.06	**44.27**	3603.29	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	50000	1	···
50-100-7	157.37	607.87	74.11	3600.04	331.71	568.74	**41.68**	3603.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.20	50000	1	···
50-100-8	148.65	532.45	72.08	3600.03	283.81	502.47	**43.52**	3603.14	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.34	50000	1	···
50-100-9	152.20	565.36	73.08	3600.03	297.15	536.91	**44.66**	3602.80	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	50000	1	···

Table 28 **Detailed results for problem NLUFLP-A-gunluk, cost functions f_3**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	192.93	271.11	28.84	3600.01	175.68	271.11	35.20	3602.01	105.68	297.07	64.43	3600.10	−∞	+∞	100.00	3600.01	+∞	271.11	271.11	**0.00**	**1025.75**	3800	48	···
10-20-10	194.53	253.31	23.21	3600.01	184.11	253.31	27.32	3602.07	93.99	258.61	63.66	3600.10	−∞	+∞	100.00	3600.01	+∞	253.31	253.31	**0.00**	**785.78**	3177	43	···
10-20-2	213.68	224.11	4.66	3600.01	209.83	224.11	6.37	3602.09	94.73	237.88	60.18	3600.10	−∞	+∞	100.00	3600.01	+∞	224.11	224.11	**0.00**	**619.46**	3119	36	···
10-20-3	203.63	246.27	17.31	3600.01	203.55	246.27	17.35	3601.98	102.28	285.90	64.23	3600.10	−∞	+∞	100.00	3600.01	+∞	246.27	246.27	**0.00**	**531.19**	3471	47	···
10-20-4	238.88	278.31	14.17	3600.01	233.11	278.48	16.29	3601.83	112.66	346.57	67.49	3600.10	−∞	+∞	100.00	3600.01	+∞	278.31	278.31	**0.00**	**931.79**	3152	42	···
10-20-5	244.86	293.19	16.48	3600.01	244.85	292.32	16.24	3601.93	111.32	335.59	66.83	3600.10	−∞	+∞	100.00	3600.01	+∞	292.32	292.32	**0.00**	**571.79**	3752	45	···
10-20-6	334.93	339.62	1.38	3600.01	326.45	339.62	3.88	3601.92	139.86	365.67	61.75	3600.10	−∞	+∞	100.00	3600.01	+∞	339.62	339.62	**0.00**	**1845.41**	3109	42	···
10-20-7	210.03	226.43	7.24	3600.01	208.16	226.43	8.07	3602.15	100.45	297.15	66.20	3600.10	−∞	+∞	100.00	3600.01	+∞	226.43	226.43	**0.00**	**566.31**	3152	45	···
10-20-8	286.88	326.25	12.07	3600.01	286.58	326.25	12.16	3601.86	138.58	362.37	61.76	3600.10	−∞	+∞	100.00	3600.01	+∞	326.25	326.25	**0.00**	**1214.95**	3181	42	···
10-20-9	263.29	279.57	5.82	3600.01	266.10	279.57	4.82	3601.99	122.38	387.64	68.43	3600.10	−∞	+∞	100.00	3600.01	+∞	279.57	279.57	**0.00**	**629.38**	3114	37	···
20-40-1	224.93	438.75	48.73	3600.01	221.08	379.02	41.67	3602.06	99.20	706.86	85.97	3600.10	−∞	+∞	100.00	3600.01	+∞	376.67	380.11	**0.91**	3601.03	9300	12	···
20-40-10	203.80	405.64	49.76	3600.02	202.55	384.88	47.37	3601.98	95.27	520.31	81.69	3600.10	−∞	+∞	100.00	3600.01	+∞	381.19	383.55	**0.62**	3601.02	9446	18	···
20-40-2	223.04	372.76	40.17	3600.01	224.09	350.80	36.12	3601.94	85.03	615.64	86.19	3600.10	−∞	+∞	100.00	3600.01	+∞	350.67	350.88	**0.06**	3601.03	9808	24	···
20-40-3	235.75	398.42	40.83	3600.01	227.85	374.98	39.24	3602.16	99.60	587.81	83.06	3600.10	−∞	+∞	100.00	3600.01	+∞	374.45	375.46	**0.27**	3601.02	9587	17	···
20-40-4	233.68	391.55	40.32	3600.02	224.55	390.62	42.52	3602.03	94.38	629.22	85.00	3600.10	−∞	+∞	100.00	3600.01	+∞	387.80	391.82	**1.03**	3601.03	9202	13	···
20-40-5	194.64	420.94	53.76	3600.01	191.47	353.40	45.82	3602.30	86.54	471.03	81.63	3600.10	−∞	+∞	100.00	3600.01	+∞	344.47	345.93	**0.42**	3601.03	9591	15	···
20-40-6	248.91	439.40	43.35	3600.01	235.38	415.53	43.36	3602.03	96.66	640.28	84.90	3600.10	−∞	+∞	100.00	3600.01	+∞	404.77	407.27	**0.62**	3601.03	9347	14	···
20-40-7	250.80	430.94	41.80	3600.01	247.27	397.84	37.85	3602.17	95.61	685.58	86.05	3600.10	−∞	+∞	100.00	3600.01	+∞	394.98	399.05	**1.02**	3601.03	9301	11	···
20-40-8	265.35	451.77	41.26	3600.01	257.61	411.37	37.38	3602.16	101.43	697.19	85.45	3600.10	−∞	+∞	100.00	3600.01	+∞	409.52	412.02	**0.61**	3601.02	9391	18	···
20-40-9	259.52	389.05	33.29	3600.01	248.52	385.11	35.47	3602.43	107.72	676.64	84.08	3600.10	−∞	+∞	100.00	3600.01	+∞	380.92	387.21	**1.63**	3601.03	8974	9	···
30-60-1	198.26	673.11	70.55	3600.03	259.70	516.65	49.73	3602.26	92.40	1075.09	91.41	3600.10	−∞	+∞	100.00	3600.01	+∞	487.88	524.94	**7.06**	3601.03	19806	2	···
30-60-10	190.62	430.35	55.71	3600.02	194.62	352.41	**44.77**	3602.12	82.59	1181.29	93.01	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	19800	1	···
30-60-2	186.88	557.33	66.47	3600.02	244.77	469.50	47.87	3602.42	79.84	914.74	91.27	3600.10	−∞	+∞	100.00	3600.01	+∞	436.89	471.80	**7.40**	3601.03	19804	2	···
30-60-3	216.48	533.35	59.41	3600.02	225.37	453.34	**50.29**	3602.25	91.31	1348.53	93.23	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
30-60-4	199.12	612.45	67.49	3600.01	236.97	506.34	53.20	3602.37	83.82	1048.80	92.01	3600.10	−∞	+∞	100.00	3600.01	+∞	463.24	497.84	**6.95**	3601.08	19808	2	···
30-60-5	199.19	467.12	57.36	3600.01	205.70	423.30	**51.41**	3602.33	82.43	987.94	91.66	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	19800	1	···
30-60-6	227.13	516.39	56.02	3600.02	226.10	445.87	**49.29**	3602.89	−∞	994.25	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.23	19800	1	···
30-60-7	205.32	658.49	68.82	3600.02	297.55	560.01	**46.87**	3602.29	88.83	1111.44	92.01	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.18	19800	1	···
30-60-8	177.03	660.59	73.20	3600.02	218.50	501.59	**56.44**	3602.66	−∞	1142.81	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
30-60-9	197.16	560.05	64.80	3600.03	237.16	452.31	**47.57**	3602.35	87.09	1009.07	91.37	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.22	19800	1	···
40-80-1	146.01	853.28	82.89	3600.02	259.52	616.83	**57.93**	3603.16	13.86	1336.66	98.96	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
40-80-10	135.60	835.22	83.76	3600.03	241.82	609.73	**60.34**	3602.62	10.04	1672.54	99.40	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
40-80-2	106.80	1035.86	89.69	3600.03	271.34	595.06	**54.40**	3602.86	14.93	1326.78	98.87	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
40-80-3	149.51	726.65	79.42	3600.03	234.79	530.02	**55.70**	3603.05	12.22	1490.75	99.18	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
40-80-4	127.79	732.42	82.55	3600.02	232.95	551.58	**57.77**	3602.88	9.05	1207.24	99.25	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
40-80-5	138.06	792.82	82.59	3600.02	253.44	577.90	**56.14**	3602.74	11.87	1426.65	99.17	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
40-80-6	144.33	672.91	78.55	3600.03	219.76	525.48	**58.18**	3602.61	8.20	1307.13	99.37	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
40-80-7	146.11	836.18	82.53	3600.03	310.25	646.80	**52.03**	3603.00	18.64	1444.56	98.71	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
40-80-8	102.44	931.69	89.01	3600.03	239.29	566.98	**57.79**	3603.09	10.46	1498.44	99.30	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
40-80-9	100.09	711.69	85.94	3600.03	250.44	571.00	**56.14**	3602.74	10.61	1359.52	99.22	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
50-100-1	101.09	983.33	89.72	3600.02	275.21	712.69	**61.38**	3603.11	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
50-100-10	94.70	938.40	89.91	3600.03	259.97	718.18	**63.80**	3603.45	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
50-100-2	91.56	1069.06	91.44	3600.09	289.38	847.46	**65.85**	3603.05	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
50-100-3	91.59	898.58	89.81	3600.04	248.29	678.07	**63.38**	3603.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
50-100-4	90.97	812.93	88.81	3600.06	250.23	601.49	**58.40**	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
50-100-5	90.23	869.01	89.62	3600.04	259.00	618.58	**58.13**	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
50-100-6	134.05	812.94	83.51	3600.02	254.03	607.53	**58.19**	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
50-100-7	101.31	937.81	89.20	3600.08	318.53	706.51	**54.91**	3603.29	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
50-100-8	93.82	962.52	90.25	3600.03	264.44	698.54	**62.14**	3603.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
50-100-9	94.99	1014.61	90.64	3600.06	283.62	794.32	**64.29**	3602.80	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···

Table 29 Detailed results for problem NLUFLP-A-gunluk, cost functions f_4

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	77.68	103.57	**25.00**	3600.01	70.98	109.97	35.46	3602.29	51.86	109.54	52.66	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.04	3000	1	···
10-20-10	72.32	81.44	11.19	3600.01	66.04	88.16	25.09	3602.02	48.71	85.08	42.74	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	79.44	81.76	**2.84**	3601.03	3110	4	···
10-20-2	92.09	97.82	5.85	3600.01	85.21	97.82	12.89	3602.04	61.15	97.82	37.49	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	97.82	97.82	**0.00**	**3243.69**	3757	32	···
10-20-3	87.23	97.38	10.42	3600.01	80.93	107.67	24.84	3602.24	60.32	105.02	42.56	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	95.53	97.97	**2.49**	3601.02	3125	5	···
10-20-4	104.07	117.67	11.56	3600.01	97.37	125.46	22.39	3601.92	66.39	118.87	44.14	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	112.11	118.47	**5.37**	3601.02	3048	2	···
10-20-5	95.84	112.23	**14.61**	3600.01	82.18	121.60	32.42	3602.32	61.69	117.37	47.44	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.04	3000	1	···
10-20-6	150.42	150.42	**0.00**	**198.38**	149.22	150.42	0.80	3602.08	96.62	181.74	46.84	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	150.42	150.42	**0.00**	560.80	3655	26	···
10-20-7	93.96	99.77	5.82	3600.01	87.13	99.77	12.66	3602.12	64.01	103.00	37.86	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	99.77	99.77	**0.00**	3601.02	3637	27	···
10-20-8	131.19	131.19	**0.00**	**439.36**	130.52	131.19	0.51	3602.14	96.93	136.81	29.15	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	131.19	131.19	**0.00**	618.47	3669	27	···
10-20-9	107.42	121.36	11.49	3600.02	98.43	121.81	19.20	3602.08	77.43	121.81	36.44	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	120.30	121.97	**1.37**	3601.02	3204	9	···
20-40-1	101.53	139.29	**27.11**	3600.01	96.19	140.12	31.35	3602.12	55.67	193.58	71.24	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	12000	1	···
20-40-10	86.88	128.82	**32.56**	3600.02	81.28	133.37	39.05	3602.02	49.43	179.10	72.40	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	12000	1	···
20-40-2	96.20	129.49	**25.71**	3600.01	90.00	138.90	35.20	3601.97	49.82	175.35	71.59	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	12000	1	···
20-40-3	101.44	135.94	25.38	3600.01	95.45	138.94	31.30	3602.09	58.50	231.40	74.72	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	127.75	137.24	**6.92**	3601.02	12069	2	···
20-40-4	115.86	150.70	**23.12**	3600.01	109.75	153.01	28.27	3602.21	55.50	310.17	82.11	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	12000	1	···
20-40-5	83.83	123.83	**32.30**	3600.02	77.71	134.70	42.31	3602.11	47.09	179.78	73.80	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.18	12000	1	···
20-40-6	100.98	137.68	**26.66**	3600.01	94.47	149.34	36.75	3602.01	57.47	181.83	68.40	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	12000	1	···
20-40-7	121.55	162.72	**25.30**	3600.02	115.59	169.33	31.74	3602.11	58.00	266.41	78.23	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	12000	1	···
20-40-8	125.42	165.07	**24.02**	3600.01	119.43	172.39	30.72	3602.10	61.81	280.34	77.95	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	12000	1	···
20-40-9	122.55	164.18	**25.36**	3600.01	115.15	168.85	31.80	3602.15	66.53	201.50	66.98	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	12000	1	···
30-60-1	124.60	196.28	**36.52**	3600.02	119.09	200.33	40.55	3602.29	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.09	27000	1	···
30-60-10	93.19	161.98	**42.47**	3600.02	88.24	160.41	44.99	3602.36	$-\infty$	4600836.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	27000	1	···
30-60-2	127.49	193.02	**33.95**	3600.02	122.29	200.51	39.01	3602.29	47.55	530.25	91.03	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	27000	1	···
30-60-3	104.64	164.82	**36.51**	3600.02	99.03	163.54	39.45	3602.25	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	27000	1	···
30-60-4	115.36	206.67	44.18	3600.02	113.02	190.73	**40.74**	3602.35	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.09	27000	1	···
30-60-5	100.64	175.72	42.73	3600.02	95.53	165.96	**42.44**	3602.56	46.25	493.98	90.64	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.09	27000	1	···
30-60-6	99.90	165.95	**39.80**	3600.02	94.77	168.51	43.76	3602.89	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.10	27000	1	···
30-60-7	155.08	230.47	**32.71**	3600.03	151.79	246.33	38.38	3602.35	54.58	566.42	90.36	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.12	27000	1	···
30-60-8	109.94	188.02	**41.53**	3600.01	104.79	185.80	43.60	3602.43	45.00	330.89	86.40	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.09	27000	1	···
30-60-9	115.53	178.64	**35.33**	3600.01	108.89	178.69	39.06	3602.35	$-\infty$	770.62	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	27000	1	···
40-80-1	95.43	257.28	62.91	3600.17	113.97	219.06	**47.97**	3602.49	8.73	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.10	48000	1	···
40-80-10	93.11	223.64	58.37	3600.02	116.69	211.90	**44.93**	3602.59	7.44	1315.10	99.43	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	48000	1	···
40-80-2	95.02	251.50	62.22	3600.03	134.17	248.79	**46.07**	3602.56	11.32	975.11	98.84	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.16	48000	1	···
40-80-3	101.40	239.25	57.62	3600.02	109.98	213.05	**48.38**	3603.37	7.71	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.24	48000	1	···
40-80-4	98.03	257.77	61.97	3600.02	115.14	209.68	**45.09**	3603.24	6.68	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.11	48000	1	···
40-80-5	104.55	246.93	57.66	3600.02	120.96	223.11	**45.79**	3602.43	8.58	966.11	99.11	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.11	48000	1	···
40-80-6	78.54	227.48	65.47	3600.03	98.98	196.85	**49.72**	3605.39	5.71	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.13	48000	1	···
40-80-7	112.14	302.96	62.98	3600.04	166.76	283.93	**41.27**	3603.29	15.13	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.16	48000	1	···
40-80-8	84.43	280.59	69.91	3600.03	116.16	210.46	**44.81**	3602.50	7.43	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.23	48000	1	···
40-80-9	90.80	253.02	64.12	3600.03	113.81	222.19	**48.78**	3603.20	7.51	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.11	48000	1	···
50-100-1	78.50	312.34	74.87	3600.02	122.62	247.11	**50.38**	3615.87	6.93	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
50-100-10	64.40	327.01	80.31	3600.04	129.06	244.19	**47.15**	3606.70	6.90	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
50-100-2	67.17	344.87	80.52	3600.05	145.43	329.39	**55.85**	3604.15	8.79	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
50-100-3	64.87	342.89	81.08	3600.07	115.23	273.65	**57.89**	3608.44	5.62	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
50-100-4	64.45	295.69	78.20	3600.10	124.93	237.79	**47.46**	3606.06	6.68	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
50-100-5	63.04	327.22	80.74	3600.07	126.40	279.36	**54.75**	3656.50	6.90	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
50-100-6	64.79	318.35	79.65	3600.07	121.27	248.19	**51.14**	3606.49	5.71	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
50-100-7	77.62	305.88	74.63	3600.07	165.16	301.16	**45.16**	3611.06	12.97	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
50-100-8	70.76	317.97	77.75	3600.02	136.18	249.26	**45.37**	3622.77	7.43	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
50-100-9	72.23	287.88	74.91	3600.05	130.13	247.16	**47.35**	3612.14	7.51	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···

Table 30 Detailed results for problem NLUFLP-A-gunluk, cost functions f_5

	GUROBI				SCIP				COUENNE				NAIVE					CN24						...
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	...
10-20-1	74.73	340.57	78.06	3600.01	47.10	359.05	86.88	3602.26	-19.59	395.80	104.95	3600.10	$-\infty$	$+\infty$	100.00	3601.39	288600	336.10	336.11	**0.00**	**1696.66**	1349	21	...
10-20-10	108.87	310.92	64.98	3600.01	100.74	334.69	69.90	3602.18	3.47	315.18	98.90	3600.10	$-\infty$	$+\infty$	100.00	3601.49	288600	310.92	310.92	**0.00**	**3493.45**	1357	21	...
10-20-2	123.24	262.51	53.05	3600.01	101.97	267.54	61.89	3602.22	11.72	270.34	95.67	3600.10	262.51	262.51	**0.00**	2433.10	288600	262.51	262.51	**0.00**	**569.65**	1307	12	...
10-20-3	101.34	300.82	66.31	3600.01	74.59	309.61	75.91	3602.69	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.37	288600	300.82	300.82	**0.00**	**2710.03**	1368	22	...
10-20-4	129.56	329.26	60.65	3600.01	108.02	329.26	67.19	3602.68	1.15	382.17	99.70	3600.10	$-\infty$	$+\infty$	100.00	3601.36	288600	329.26	329.26	**0.00**	**911.94**	1348	19	...
10-20-5	106.57	358.31	70.26	3600.01	67.88	377.80	82.03	3602.62	-21.46	477.71	104.49	3600.10	353.87	353.87	**0.00**	3562.65	288600	353.87	353.87	**0.00**	**1029.71**	1352	25	...
10-20-6	242.22	383.16	36.78	3600.01	235.64	384.06	38.64	3602.68	25.86	416.14	93.79	3600.10	$-\infty$	$+\infty$	100.00	3601.40	288600	383.16	383.16	**0.00**	**763.49**	1327	18	...
10-20-7	126.85	265.20	52.17	3600.01	112.08	270.77	58.61	3602.65	17.38	278.95	93.77	3600.10	265.20	265.20	**0.00**	2327.06	288600	265.20	265.20	**0.00**	**573.46**	1327	22	...
10-20-8	198.15	370.99	46.59	3600.01	188.40	384.06	50.95	3602.67	38.70	422.63	90.84	3600.10	$-\infty$	$+\infty$	100.00	3601.47	288600	370.99	370.99	**0.00**	**881.13**	1354	16	...
10-20-9	162.97	330.41	50.68	3600.01	144.18	339.11	57.48	3602.66	-119.01	343.57	134.64	3600.10	330.41	330.41	**0.00**	2579.55	288600	330.41	330.41	**0.00**	**649.88**	1319	14	...
20-40-1	-55.28	502.21	111.01	3600.02	-104.08	522.62	119.92	3603.58	$-\infty$	1030.43	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	475.85	481.11	**1.09**	3601.03	4912	7	...
20-40-10	-109.87	538.27	120.41	3600.01	-159.28	546.02	129.17	3603.57	-640.34	1106.18	157.89	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	483.09	496.78	**2.75**	3601.03	4862	4	...
20-40-2	-51.78	473.96	110.92	3600.01	-105.97	487.15	121.75	3603.29	-606.04	1045.69	157.96	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	430.14	436.58	**1.48**	3601.03	4897	6	...
20-40-3	-48.24	473.87	110.18	3600.03	-111.63	532.28	120.97	3603.27	-611.31	1206.04	150.69	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	448.50	478.12	**6.20**	3601.03	4833	2	...
20-40-4	-87.64	502.47	117.44	3600.01	-140.11	687.07	120.39	3603.32	-643.17	1052.60	161.10	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	463.28	496.78	**6.74**	3601.03	4844	2	...
20-40-5	-141.63	449.58	131.50	3600.02	-190.19	554.23	134.32	3603.36	$-\infty$	919.78	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	427.32	428.02	**0.16**	3601.03	4941	13	...
20-40-6	-1.48	519.11	100.28	3600.01	-61.57	563.65	110.92	3603.24	-633.77	1286.19	149.27	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	490.46	500.07	**1.92**	3601.03	4870	4	...
20-40-7	-36.03	503.10	107.16	3600.01	-90.32	539.68	116.74	3603.30	$-\infty$	882.30	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	484.66	492.99	**1.69**	3601.03	4855	4	...
20-40-8	-33.84	524.66	106.45	3600.02	-92.81	561.86	116.52	3603.80	$-\infty$	1100.83	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	496.56	499.72	**0.63**	3601.02	4883	9	...
20-40-9	-52.46	475.59	111.03	3600.01	-110.79	505.17	121.93	3603.28	$-\infty$	1075.08	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	467.26	467.55	**0.06**	3601.03	4934	12	...
30-60-1	-193.69	686.11	128.23	3600.02	-215.37	767.98	128.04	3603.56	-986.84	1486.04	166.41	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	619.20	639.10	**3.11**	**3576.12**	10848	2	...
30-60-10	-347.54	476.20	172.98	3600.02	-359.11	862.18	141.65	3606.91	-1204.21	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	439.37	443.21	**0.87**	3601.03	10906	7	...
30-60-2	-251.77	650.71	138.69	3600.01	-270.04	814.76	133.14	3603.55	-1161.32	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	10800	1	...
30-60-3	-279.49	594.46	147.02	3600.02	-296.91	869.20	134.16	3604.23	-983.55	2131.55	146.14	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	553.37	560.87	**1.34**	3601.05	10897	5	...
30-60-4	-333.07	645.84	151.57	3600.02	-344.18	983.38	135.00	3603.53	-1011.82	2208.53	145.81	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	575.38	607.52	**5.29**	3601.04	10845	2	...
30-60-5	-318.37	542.31	158.71	3600.03	-320.70	826.59	138.80	3603.14	-1095.14	2475.87	144.23	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	515.55	546.03	**5.58**	3601.03	10843	2	...
30-60-6	-308.33	643.19	147.94	3600.02	-330.47	708.04	146.67	3603.93	-1051.73	2453.16	142.87	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	559.93	575.59	**2.72**	3601.04	10872	3	...
30-60-7	-215.48	758.45	128.41	3600.02	-228.95	984.74	123.25	3603.08	-987.65	1473.14	167.04	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	10800	1	...
30-60-8	-348.45	702.03	149.63	3600.02	-365.81	990.27	136.94	3603.95	-1221.62	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	10800	1	...
30-60-9	-321.59	603.85	153.26	3600.02	-337.21	872.33	138.66	3603.97	-1026.94	163799.50	100.63	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	542.31	572.38	**5.25**	3601.04	10848	2	...
40-80-1	-457.01	908.62	150.30	3600.02	-464.74	1353.52	134.34	3606.42	-1746.26	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
40-80-10	-583.01	957.70	160.88	3600.03	-590.82	1103.04	153.56	3604.17	-1742.58	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
40-80-2	-526.42	992.55	153.04	3600.03	-534.53	1544.77	134.60	3603.79	-1723.94	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
40-80-3	-504.79	982.05	151.40	3600.04	-512.54	1447.50	135.41	3605.60	-1737.80	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
40-80-4	-569.01	863.74	165.88	3600.06	-577.20	1467.63	139.33	3603.79	-1734.80	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
40-80-5	-537.64	952.82	156.43	3600.03	-546.68	1604.33	134.08	3603.80	-1819.08	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
40-80-6	-599.05	794.70	175.38	3600.03	-606.77	1532.53	139.59	3603.79	-1747.86	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
40-80-7	-446.53	1077.65	141.44	3600.06	-454.84	1743.88	126.08	3607.93	-1805.67	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
40-80-8	-599.08	829.28	172.24	3600.03	-607.84	1800.12	133.77	3603.80	-1822.61	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
40-80-9	-633.31	885.35	171.53	3600.03	-642.11	1788.75	135.90	3603.70	-1854.85	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
50-100-1	-735.60	1322.07	155.64	3600.07	-743.14	2176.98	134.14	3605.32	-2400.37	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
50-100-10	-833.75	1114.57	174.80	3600.05	-842.01	2561.16	132.88	3605.39	-2350.65	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
50-100-2	-776.67	1216.60	163.84	3600.05	-785.21	2279.46	134.45	3605.31	-2284.79	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
50-100-3	-769.83	945.96	181.38	3600.03	-776.87	2082.58	137.30	3605.31	-2294.88	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
50-100-4	-828.12	1239.44	166.81	3600.03	-830.87	2213.78	137.53	3605.18	-2309.38	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
50-100-5	-840.10	1078.62	177.89	3600.07	-847.16	1919.49	144.13	3605.21	-2402.81	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
50-100-6	-870.35	1196.72	172.73	3600.03	-879.15	2051.46	142.85	3605.44	-2358.10	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
50-100-7	-678.34	1178.32	157.57	3600.05	-689.07	2199.43	131.33	3605.19	-2424.83	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
50-100-8	-815.51	1260.82	164.68	3600.03	-823.71	2092.70	139.36	3605.34	-2456.33	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...
50-100-9	-818.60	1404.43	158.29	3600.04	-826.54	2422.77	134.12	3605.35	-2413.87	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	...

Table 31 Detailed results for problem NLUFLP-A-gunluk, cost functions f_6

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	336.09	336.09	**0.00**	**0.02**	336.09	336.09	**0.00**	2.01	336.09	336.09	**0.00**	0.63	−∞	+∞	100.00	3600.01	+∞	336.09	336.09	**0.00**	10.02	1284	13	···
10-20-10	310.91	310.91	**0.00**	**0.02**	310.91	310.91	**0.00**	2.07	310.91	310.91	**0.00**	0.55	−∞	+∞	100.00	3600.01	+∞	310.91	310.91	**0.00**	8.36	1207	13	···
10-20-2	262.50	262.50	**0.00**	**0.02**	262.50	262.50	**0.00**	2.47	262.50	262.50	**0.00**	0.73	−∞	+∞	100.00	3600.01	+∞	262.50	262.50	**0.00**	8.66	1284	12	···
10-20-3	300.81	300.81	**0.00**	**0.02**	300.81	300.81	**0.00**	1.89	300.81	300.81	**0.00**	0.74	−∞	+∞	100.00	3600.01	+∞	300.81	300.81	**0.00**	12.06	1363	14	···
10-20-4	329.25	329.25	**0.00**	**0.02**	329.25	329.25	**0.00**	2.30	329.25	329.25	**0.00**	0.55	−∞	+∞	100.00	3600.01	+∞	329.25	329.25	**0.00**	8.79	1208	13	···
10-20-5	353.86	353.86	**0.00**	**0.01**	353.86	353.86	**0.00**	2.01	353.86	353.86	**0.00**	0.04	−∞	+∞	100.00	3600.01	+∞	353.86	353.86	**0.00**	9.97	1359	14	···
10-20-6	383.15	383.15	**0.00**	**0.02**	383.15	383.15	**0.00**	1.94	383.15	383.15	**0.00**	0.74	−∞	+∞	100.00	3600.01	+∞	383.15	383.15	**0.00**	9.46	1206	13	···
10-20-7	265.20	265.20	**0.00**	**0.02**	265.20	265.20	**0.00**	1.98	265.20	265.20	**0.00**	0.63	−∞	+∞	100.00	3600.01	+∞	265.20	265.20	**0.00**	8.24	1280	13	···
10-20-8	371.64	371.64	**0.00**	**0.02**	371.64	371.64	**0.00**	2.41	371.64	371.64	**0.00**	0.85	−∞	+∞	100.00	3600.01	+∞	371.64	371.64	**0.00**	10.30	1293	15	···
10-20-9	330.40	330.40	**0.00**	**0.02**	330.40	330.40	**0.00**	2.00	330.40	330.40	**0.00**	0.51	−∞	+∞	100.00	3600.01	+∞	330.40	330.40	**0.00**	11.71	1290	13	···
20-40-1	480.17	480.17	**0.00**	**0.04**	480.17	480.17	**0.00**	2.31	480.17	480.17	**0.00**	6.79	−∞	+∞	100.00	3600.01	+∞	480.17	480.17	**0.00**	97.03	5045	15	···
20-40-10	492.03	492.03	**0.00**	**0.03**	492.03	492.03	**0.00**	2.59	492.03	492.03	**0.00**	7.33	−∞	+∞	100.00	3600.01	+∞	492.03	492.03	**0.00**	65.16	4880	13	···
20-40-2	435.42	435.42	**0.00**	**0.03**	435.42	435.42	**0.00**	2.59	435.42	435.42	**0.00**	5.89	−∞	+∞	100.00	3600.01	+∞	435.42	435.42	**0.00**	85.93	4893	15	···
20-40-3	474.33	474.33	**0.00**	**0.04**	474.33	474.33	**0.00**	2.46	474.33	474.33	**0.00**	10.91	−∞	+∞	100.00	3600.01	+∞	474.33	474.33	**0.00**	73.86	4891	15	···
20-40-4	492.96	492.96	**0.00**	**0.04**	492.96	492.96	**0.00**	2.46	492.96	492.96	**0.00**	8.39	−∞	+∞	100.00	3600.01	+∞	492.96	492.96	**0.00**	111.76	4919	19	···
20-40-5	428.83	428.83	**0.00**	**0.03**	428.83	428.83	**0.00**	2.39	428.83	428.83	**0.00**	7.02	−∞	+∞	100.00	3600.01	+∞	428.83	428.83	**0.00**	65.50	5039	13	···
20-40-6	498.29	498.29	**0.00**	**0.03**	498.29	498.29	**0.00**	2.36	498.29	498.29	**0.00**	8.48	−∞	+∞	100.00	3600.01	+∞	498.29	498.29	**0.00**	79.45	4885	14	···
20-40-7	490.89	490.89	**0.00**	**0.03**	490.89	490.89	**0.00**	2.38	490.89	490.89	**0.00**	8.99	−∞	+∞	100.00	3600.01	+∞	490.89	490.89	**0.00**	101.57	4880	12	···
20-40-8	499.41	499.41	**0.00**	**0.04**	499.41	499.41	**0.00**	2.42	499.41	499.41	**0.00**	5.41	−∞	+∞	100.00	3600.01	+∞	499.41	499.41	**0.00**	102.21	4890	14	···
20-40-9	467.54	467.54	**0.00**	**0.04**	467.54	467.54	**0.00**	2.53	467.54	467.54	**0.00**	8.63	−∞	+∞	100.00	3600.01	+∞	467.54	467.54	**0.00**	98.86	4887	13	···
30-60-1	634.78	634.78	**0.00**	**0.07**	634.78	634.78	**0.00**	3.22	634.78	634.78	**0.00**	35.74	−∞	+∞	100.00	3600.01	+∞	634.78	634.78	**0.00**	219.01	10802	15	···
30-60-10	442.37	442.37	**0.00**	**0.08**	442.37	442.37	**0.00**	2.80	442.37	442.37	**0.00**	34.75	−∞	+∞	100.00	3600.01	+∞	442.37	442.37	**0.00**	219.88	11063	14	···
30-60-2	565.71	565.71	**0.00**	**0.09**	565.71	565.71	**0.00**	3.29	565.71	565.71	**0.00**	31.83	−∞	+∞	100.00	3600.01	+∞	565.71	565.71	**0.00**	350.28	11322	14	···
30-60-3	559.35	559.35	**0.00**	**0.10**	559.35	559.35	**0.00**	3.13	559.35	559.35	**0.00**	35.43	−∞	+∞	100.00	3600.01	+∞	559.35	559.35	**0.00**	177.79	11047	14	···
30-60-4	602.17	602.17	**0.00**	**0.07**	602.17	602.17	**0.00**	3.37	602.17	602.17	**0.00**	41.30	−∞	+∞	100.00	3600.01	+∞	602.17	602.17	**0.00**	212.14	10822	14	···
30-60-5	542.29	542.29	**0.00**	**0.07**	542.29	542.29	**0.00**	3.00	542.29	542.29	**0.00**	39.33	−∞	+∞	100.00	3600.01	+∞	542.29	542.29	**0.00**	166.63	11048	13	···
30-60-6	571.87	571.87	**0.00**	**0.11**	571.87	571.87	**0.00**	3.15	571.87	571.87	**0.00**	39.37	−∞	+∞	100.00	3600.01	+∞	571.87	571.87	**0.00**	170.76	10805	14	···
30-60-7	690.49	690.49	**0.00**	**0.14**	690.49	690.49	**0.00**	4.79	690.49	690.49	**0.00**	148.26	−∞	+∞	100.00	3600.01	+∞	690.49	690.49	**0.00**	470.17	10609	16	···
30-60-8	630.72	630.72	**0.00**	**0.10**	630.72	630.72	**0.00**	3.60	630.72	630.72	**0.00**	36.81	−∞	+∞	100.00	3600.01	+∞	630.72	630.72	**0.00**	330.09	11336	17	···
30-60-9	569.33	569.33	**0.00**	**0.10**	569.33	569.33	**0.00**	3.32	569.33	569.33	**0.00**	46.53	−∞	+∞	100.00	3600.01	+∞	569.33	569.33	**0.00**	270.91	11060	16	···
40-80-1	736.31	736.31	**0.00**	**0.14**	736.31	736.31	**0.00**	4.82	736.31	736.31	**0.00**	155.68	−∞	+∞	100.00	3600.01	+∞	736.31	736.31	**0.00**	794.45	19421	15	···
40-80-10	712.62	712.62	**0.00**	**0.16**	712.62	712.62	**0.00**	4.87	712.62	712.62	**0.00**	117.08	−∞	+∞	100.00	3600.01	+∞	712.62	712.62	**0.00**	534.13	19073	14	···
40-80-2	733.78	733.78	**0.00**	**0.15**	733.78	733.78	**0.00**	6.61	733.78	733.78	**0.00**	127.10	−∞	+∞	100.00	3600.01	+∞	733.78	733.78	**0.00**	1202.01	19194	25	···
40-80-3	653.70	653.70	**0.00**	**0.16**	653.70	653.70	**0.00**	5.05	653.70	653.70	**0.00**	125.95	−∞	+∞	100.00	3600.01	+∞	653.70	653.70	**0.00**	554.23	19384	15	···
40-80-4	678.05	678.05	**0.00**	**0.17**	678.05	678.05	**0.00**	5.76	678.05	678.05	**0.00**	118.41	−∞	+∞	100.00	3600.01	+∞	678.05	678.05	**0.00**	678.78	19085	17	···
40-80-5	722.26	722.26	**0.00**	**0.23**	722.26	722.26	**0.00**	4.36	722.26	722.26	**0.00**	120.13	−∞	+∞	100.00	3600.01	+∞	722.26	722.26	**0.00**	618.77	19059	15	···
40-80-6	634.77	634.77	**0.00**	**0.24**	634.77	634.77	**0.00**	4.74	634.77	634.77	**0.00**	118.17	−∞	+∞	100.00	3600.01	+∞	634.77	634.77	**0.00**	632.90	19693	15	···
40-80-7	768.39	768.39	**0.00**	**0.22**	768.39	768.39	**0.00**	5.16	768.39	768.39	**0.00**	138.73	−∞	+∞	100.00	3600.01	+∞	768.39	768.39	**0.00**	941.99	18436	16	···
40-80-8	676.84	676.84	**0.00**	**0.27**	676.84	676.84	**0.00**	5.64	676.84	676.84	**0.00**	121.79	−∞	+∞	100.00	3600.01	+∞	676.84	676.84	**0.00**	767.04	19380	15	···
40-80-9	706.91	706.91	**0.00**	**0.14**	706.91	706.91	**0.00**	5.76	706.91	706.91	**0.00**	130.40	−∞	+∞	100.00	3600.01	+∞	706.91	706.91	**0.00**	820.78	19388	15	···
50-100-1	819.00	819.00	**0.00**	**0.53**	819.00	819.00	**0.00**	14.49	819.00	819.00	**0.00**	952.75	−∞	+∞	100.00	3600.01	+∞	819.00	819.00	**0.00**	1442.57	30019	14	···
50-100-10	820.61	820.61	**0.00**	**0.59**	820.61	820.61	**0.00**	13.10	820.61	820.61	**0.00**	1250.52	−∞	+∞	100.00	3600.01	+∞	820.61	820.61	**0.00**	1834.26	29637	16	···
50-100-2	885.03	885.03	**0.00**	**0.55**	885.03	885.03	**0.00**	11.30	885.03	885.03	**0.00**	1265.70	−∞	+∞	100.00	3600.01	+∞	885.03	885.03	**0.00**	2008.48	30088	15	···
50-100-3	792.09	792.09	**0.00**	**0.51**	792.09	792.09	**0.00**	12.06	792.09	792.09	**0.00**	1210.87	−∞	+∞	100.00	3600.01	+∞	792.09	792.09	**0.00**	2169.08	29275	17	···
50-100-4	733.58	733.58	**0.00**	**0.56**	733.58	733.58	**0.00**	11.16	733.58	733.58	**0.00**	1270.82	−∞	+∞	100.00	3600.01	+∞	733.58	733.58	**0.00**	1277.39	30013	14	···
50-100-5	733.23	733.23	**0.00**	**0.55**	733.23	733.23	**0.00**	12.02	733.23	733.23	**0.00**	760.61	−∞	+∞	100.00	3600.01	+∞	733.23	733.23	**0.00**	1182.70	30797	14	···
50-100-6	741.44	741.44	**0.00**	**0.49**	741.44	741.44	**0.00**	9.56	741.44	741.44	**0.00**	1236.46	−∞	+∞	100.00	3600.01	+∞	741.44	741.44	**0.00**	1344.06	30022	15	···
50-100-7	857.60	857.60	**0.00**	**0.54**	857.60	857.60	**0.00**	10.41	857.60	857.60	**0.00**	1250.04	−∞	+∞	100.00	3600.01	+∞	857.60	857.60	**0.00**	1552.06	28834	17	···
50-100-8	797.78	797.78	**0.00**	**0.53**	797.78	797.78	**0.00**	11.94	797.78	797.78	**0.00**	1273.00	−∞	+∞	100.00	3600.01	+∞	797.78	797.78	**0.00**	1474.71	30014	14	···
50-100-9	857.67	857.67	**0.00**	**0.66**	857.67	857.67	**0.00**	11.37	857.67	857.67	**0.00**	905.97	−∞	+∞	100.00	3600.01	+∞	857.67	857.67	**0.00**	2245.36	30101	18	···

Table 32 Detailed results for problem NLUFLP-A-gunluk, cost functions f_7

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	140.04	140.04	**0.00**	**0.05**	140.04	140.04	**0.00**	2.39	112.76	159.78	29.43	3600.10	140.04	140.04	**0.00**	1542.17	298800	140.04	140.04	**0.00**	219.01	3770	44	···
10-20-10	141.08	141.08	**0.00**	**0.06**	141.08	141.08	**0.00**	2.76	100.97	159.78	36.81	3600.10	141.08	141.08	**0.00**	1591.23	298800	141.08	141.08	**0.00**	163.13	3410	43	···
10-20-2	158.96	158.96	**0.00**	**0.08**	158.96	158.96	**0.00**	2.88	116.32	159.17	26.93	3600.10	158.96	158.96	**0.00**	2810.73	298800	158.96	158.96	**0.00**	582.87	3490	45	···
10-20-3	152.41	152.41	**0.00**	**0.06**	152.41	152.41	**0.00**	2.36	113.68	175.55	35.24	3600.10	152.41	152.41	**0.00**	1834.79	298800	152.41	152.41	**0.00**	149.97	3340	39	···
10-20-4	185.02	185.02	**0.00**	**0.08**	185.02	185.02	**0.00**	3.09	168.40	185.02	8.98	3600.10	−∞	+∞	100.00	3601.36	298800	185.02	185.02	**0.00**	611.60	3300	42	···
10-20-5	177.41	177.41	**0.00**	**0.07**	177.41	177.41	**0.00**	3.44	111.38	213.51	47.83	3600.10	−∞	+∞	100.00	3601.30	298800	177.41	177.41	**0.00**	318.67	3713	42	···
10-20-6	260.69	260.69	**0.00**	**0.12**	260.69	260.69	**0.00**	10.53	129.89	317.83	59.13	3600.10	−∞	+∞	100.00	3601.42	298800	260.69	260.69	**0.00**	335.94	3100	43	···
10-20-7	159.14	159.14	**0.00**	**0.07**	159.14	159.14	**0.00**	2.80	100.82	230.50	56.26	3600.10	159.14	159.14	**0.00**	2139.69	298800	159.14	159.14	**0.00**	334.56	3348	40	···
10-20-8	227.64	227.64	**0.00**	**0.12**	227.64	227.64	**0.00**	4.89	146.25	284.79	48.65	3600.10	−∞	+∞	100.00	3601.31	298800	227.64	227.64	**0.00**	501.27	3015	33	···
10-20-9	193.46	193.46	**0.00**	**0.06**	193.46	193.46	**0.00**	3.22	128.87	278.05	53.65	3600.10	193.46	193.46	**0.00**	3091.73	298800	193.46	193.46	**0.00**	199.71	3011	37	···
20-40-1	202.58	202.58	**0.00**	**1.50**	202.58	202.58	**0.00**	7.40	106.33	416.80	74.49	3600.10	−∞	+∞	100.00	3600.01	+∞	202.17	202.80	0.31	3601.03	9391	18	···
20-40-10	181.88	181.88	**0.00**	**1.92**	181.88	181.88	**0.00**	5.90	99.80	315.51	68.37	3600.10	−∞	+∞	100.00	3600.01	+∞	181.48	182.22	0.41	3601.03	9366	15	···
20-40-2	197.16	197.16	**0.00**	**1.27**	197.16	197.16	**0.00**	6.63	87.62	380.13	76.95	3600.10	−∞	+∞	100.00	3600.01	+∞	197.16	197.16	**0.00**	2266.88	11450	51	···
20-40-3	203.48	203.48	**0.00**	**0.80**	203.48	203.48	**0.00**	5.80	105.70	422.35	74.97	3600.10	−∞	+∞	100.00	3600.01	+∞	203.48	203.48	**0.00**	1794.08	11265	50	···
20-40-4	210.93	210.93	**0.00**	**0.59**	210.92	210.93	**0.00**	5.52	112.98	295.82	61.81	3600.10	−∞	+∞	100.00	3600.01	+∞	210.92	210.93	0.00	3601.03	11149	43	···
20-40-5	169.89	169.89	**0.00**	**0.73**	169.89	169.89	**0.00**	6.26	88.87	278.45	68.08	3600.10	−∞	+∞	100.00	3600.01	+∞	169.89	169.89	**0.00**	2092.11	12076	56	···
20-40-6	211.28	211.28	**0.00**	**0.69**	211.28	211.28	**0.00**	4.83	99.37	432.08	77.00	3600.10	−∞	+∞	100.00	3600.01	+∞	211.28	211.28	**0.00**	2180.87	11422	56	···
20-40-7	230.48	230.48	**0.00**	**1.02**	230.48	230.48	**0.00**	7.53	99.23	422.89	76.53	3600.10	−∞	+∞	100.00	3600.01	+∞	229.51	231.01	0.65	3601.04	9217	12	···
20-40-8	238.27	238.27	**0.00**	**1.40**	238.27	238.27	**0.00**	9.45	106.57	453.23	76.49	3600.10	−∞	+∞	100.00	3600.01	+∞	237.57	238.75	0.50	3601.04	9091	14	···
20-40-9	231.17	231.17	**0.00**	**2.17**	231.17	231.17	**0.00**	10.90	116.69	410.91	71.60	3600.10	−∞	+∞	100.00	3600.01	+∞	231.10	231.23	0.06	3601.03	9779	28	···
30-60-1	262.83	262.83	**0.00**	**1.21**	262.83	262.83	**0.00**	28.34	99.04	549.54	81.98	3600.10	−∞	+∞	100.00	3600.01	+∞	257.86	265.75	2.97	3601.05	18734	5	···
30-60-10	192.37	192.37	**0.00**	**0.96**	192.37	192.37	**0.00**	21.01	85.90	444.08	80.66	3600.10	−∞	+∞	100.00	3600.01	+∞	189.13	194.10	2.56	3601.05	18907	6	···
30-60-2	246.94	246.94	**0.00**	**2.31**	246.94	246.94	**0.00**	30.09	81.02	533.17	84.80	3600.10	−∞	+∞	100.00	3600.01	+∞	242.22	249.70	3.00	3601.04	18716	5	···
30-60-3	226.85	226.85	**0.00**	**1.50**	226.85	226.85	**0.00**	22.58	94.07	434.71	78.36	3600.10	−∞	+∞	100.00	3600.01	+∞	222.58	229.06	2.83	3601.04	18945	6	···
30-60-4	238.26	238.26	**0.00**	**1.08**	238.26	238.26	**0.00**	27.53	92.51	550.09	83.18	3600.10	−∞	+∞	100.00	3600.01	+∞	232.67	240.60	3.30	3601.04	19035	4	···
30-60-5	207.17	207.17	**0.00**	**0.77**	207.17	207.17	**0.00**	17.44	85.18	560.71	84.81	3600.10	−∞	+∞	100.00	3600.01	+∞	202.21	209.53	3.50	3601.04	18714	4	···
30-60-6	226.62	226.62	**0.00**	**1.78**	226.62	226.62	**0.00**	20.15	87.71	345.89	74.64	3600.10	−∞	+∞	100.00	3600.01	+∞	224.29	227.94	1.60	3601.05	19627	8	···
30-60-7	300.29	300.29	**0.00**	**1.30**	300.29	300.29	**0.00**	31.14	100.19	625.29	83.98	3600.10	−∞	+∞	100.00	3600.01	+∞	293.97	303.17	3.03	3601.03	18791	4	···
30-60-8	220.28	220.28	**0.00**	**1.20**	220.28	220.28	**0.00**	23.97	94.24	515.37	81.71	3600.10	−∞	+∞	100.00	3600.01	+∞	214.06	222.08	3.61	3601.03	18909	4	···
30-60-9	239.07	239.07	**0.00**	**2.31**	239.07	239.07	**0.00**	34.92	88.87	599.79	85.18	3600.10	−∞	+∞	100.00	3600.01	+∞	233.60	241.76	3.38	3601.04	18709	4	···
40-80-1	274.14	274.14	**0.00**	**5.12**	274.14	274.14	**0.00**	97.08	87.35	884.60	90.13	3600.10	−∞	+∞	100.00	3600.01	+∞	262.32	279.99	6.31	3601.04	32439	2	···
40-80-10	253.34	253.34	**0.00**	**3.12**	253.34	253.34	**0.00**	93.15	83.74	778.36	89.24	3600.10	−∞	+∞	100.00	3600.01	+∞	243.19	257.92	5.71	3601.04	32518	2	···
40-80-2	288.49	288.49	**0.00**	**5.00**	288.49	288.49	**0.00**	109.88	86.08	672.73	87.20	3600.10	−∞	+∞	100.00	3600.01	+∞	275.77	294.38	6.32	3601.11	32457	2	···
40-80-3	243.27	243.27	**0.00**	**2.69**	243.26	243.26	**0.00**	60.99	91.85	911.37	89.92	3600.10	−∞	+∞	100.00	3600.01	+∞	236.69	246.23	3.87	3601.06	32965	4	···
40-80-4	241.84	241.83	**0.00**	**1.99**	241.83	241.83	**0.00**	65.49	85.54	727.94	88.25	3600.10	−∞	+∞	100.00	3600.01	+∞	237.13	244.11	2.86	3601.12	33624	6	···
40-80-5	262.98	262.98	**0.00**	**4.61**	262.98	262.98	**0.00**	86.69	89.01	773.32	88.49	3600.10	−∞	+∞	100.00	3600.01	+∞	251.69	267.74	5.99	3601.04	32488	2	···
40-80-6	228.98	228.98	**0.00**	**4.23**	228.98	228.98	**0.00**	75.33	79.73	806.76	90.12	3600.10	−∞	+∞	100.00	3600.01	+∞	219.64	235.00	6.54	3601.07	32455	2	···
40-80-7	323.70	323.70	**0.00**	**3.85**	323.70	323.70	**0.00**	86.26	99.10	961.40	89.69	3600.10	−∞	+∞	100.00	3600.01	+∞	311.21	329.66	5.60	3601.06	32487	2	···
40-80-8	245.55	245.55	**0.00**	**2.88**	245.55	245.55	**0.00**	66.38	80.57	698.85	88.47	3600.10	−∞	+∞	100.00	3600.01	+∞	238.72	248.61	3.98	3601.05	32952	4	···
40-80-9	262.47	262.47	**0.00**	**5.66**	262.47	262.47	**0.00**	170.27	80.20	745.80	89.25	3600.10	−∞	+∞	100.00	3600.01	+∞	251.16	268.57	6.48	3601.05	32866	3	···
50-100-1	289.23	289.23	**0.00**	**9.84**	289.23	289.23	**0.00**	275.50	76.90	1170.48	93.43	3600.10	−∞	+∞	100.00	3600.01	+∞	276.69	296.58	6.71	3601.07	50574	2	···
50-100-10	282.23	282.23	**0.00**	**6.60**	282.23	282.23	**0.00**	237.57	75.88	1244.34	93.90	3600.10	−∞	+∞	100.00	3600.01	+∞	271.80	289.88	6.23	3601.06	50618	2	···
50-100-2	325.47	325.47	**0.00**	**11.65**	325.47	325.47	**0.00**	509.69	67.98	913.05	92.55	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	50000	1	···
50-100-3	258.90	258.90	**0.00**	**7.12**	258.90	258.90	**0.00**	220.24	76.11	973.11	92.18	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.18	50000	1	···
50-100-4	261.35	261.35	**0.00**	**7.10**	261.35	261.35	**0.00**	256.23	73.03	1132.14	93.55	3600.10	−∞	+∞	100.00	3600.01	+∞	250.43	266.37	5.98	3601.06	50950	3	···
50-100-5	271.97	271.97	**0.00**	**14.33**	271.97	271.97	**0.00**	307.29	70.42	966.52	92.71	3600.10	−∞	+∞	100.00	3600.01	+∞	261.46	276.62	5.48	3601.08	50581	2	···
50-100-6	263.85	263.85	**0.00**	**7.09**	263.85	263.85	**0.00**	225.71	73.54	1006.99	92.70	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.20	50000	1	···
50-100-7	332.59	332.59	**0.00**	**5.39**	332.59	332.59	**0.00**	176.07	79.49	1389.32	94.28	3600.10	−∞	+∞	100.00	3600.01	+∞	319.90	339.19	5.69	3601.06	50544	2	···
50-100-8	283.47	283.47	**0.00**	**9.40**	283.47	283.47	**0.00**	199.97	77.88	1204.04	93.53	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	50000	1	···
50-100-9	298.74	298.74	**0.00**	**11.29**	298.74	298.74	**0.00**	269.16	74.49	948.37	92.15	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.16	50000	1	···

Table 33 Detailed results for problem NLUFLP-A-gunluk, cost functions f_8

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	336.09	336.09	**0.00**	**0.02**	336.09	336.09	**0.00**	3.17	336.09	336.09	**0.00**	15.45	−∞	+∞	100.00	3600.01	+∞	336.09	336.09	**0.00**	8.22	1872	12	···
10-20-10	310.91	310.91	**0.00**	**0.02**	310.91	310.91	**0.00**	2.53	310.91	310.91	**0.00**	11.90	−∞	+∞	100.00	3600.01	+∞	310.91	310.91	**0.00**	11.39	1791	12	···
10-20-2	262.50	262.50	**0.00**	**0.02**	262.50	262.50	**0.00**	2.20	262.50	262.50	**0.00**	21.14	−∞	+∞	100.00	3600.01	+∞	262.50	262.50	**0.00**	9.30	1875	12	···
10-20-3	300.81	300.81	**0.00**	**0.02**	300.81	300.81	**0.00**	2.75	300.81	300.81	**0.00**	32.99	−∞	+∞	100.00	3600.01	+∞	300.81	300.81	**0.00**	11.89	1959	12	···
10-20-4	329.25	329.25	**0.00**	**0.03**	329.25	329.25	**0.00**	2.62	329.25	329.25	**0.00**	10.04	−∞	+∞	100.00	3600.01	+∞	329.25	329.25	**0.00**	12.27	1786	11	···
10-20-5	353.86	353.86	**0.00**	**0.02**	353.86	353.86	**0.00**	2.75	353.86	353.86	**0.00**	15.08	−∞	+∞	100.00	3600.01	+∞	353.86	353.86	**0.00**	9.59	1948	12	···
10-20-6	383.15	383.15	**0.00**	**0.02**	383.15	383.15	**0.00**	2.25	383.15	383.15	**0.00**	6.34	−∞	+∞	100.00	3600.01	+∞	383.15	383.15	**0.00**	12.45	1788	12	···
10-20-7	265.20	265.20	**0.00**	**0.02**	265.20	265.20	**0.00**	2.44	265.20	265.20	**0.00**	7.05	−∞	+∞	100.00	3600.01	+∞	265.20	265.20	**0.00**	7.56	1876	12	···
10-20-8	371.64	371.64	**0.00**	**0.02**	371.64	371.64	**0.00**	2.52	371.64	371.64	**0.00**	26.30	−∞	+∞	100.00	3600.01	+∞	371.64	371.64	**0.00**	11.76	1860	13	···
10-20-9	330.40	330.40	**0.00**	**0.02**	330.40	330.40	**0.00**	2.78	330.40	330.40	**0.00**	10.64	−∞	+∞	100.00	3600.01	+∞	330.40	330.40	**0.00**	12.08	1875	12	···
20-40-1	480.17	480.17	**0.00**	**0.12**	480.17	480.17	**0.00**	2.80	480.17	480.17	**0.00**	60.14	−∞	+∞	100.00	3600.01	+∞	480.17	480.17	**0.00**	97.55	7427	13	···
20-40-10	492.03	492.03	**0.00**	**0.07**	492.03	492.03	**0.00**	3.18	492.03	492.03	**0.00**	37.39	−∞	+∞	100.00	3600.01	+∞	492.03	492.03	**0.00**	106.29	7265	12	···
20-40-2	435.42	435.42	**0.00**	**0.12**	435.42	435.42	**0.00**	4.76	435.42	435.42	**0.00**	53.71	−∞	+∞	100.00	3600.01	+∞	435.42	435.42	**0.00**	113.51	7259	12	···
20-40-3	474.33	474.33	**0.00**	**0.12**	474.33	474.33	**0.00**	5.70	474.33	474.33	**0.00**	317.79	−∞	+∞	100.00	3600.01	+∞	474.33	474.33	**0.00**	134.18	7268	13	···
20-40-4	492.96	492.96	**0.00**	**0.12**	492.96	492.96	**0.00**	5.57	492.96	492.96	**0.00**	73.11	−∞	+∞	100.00	3600.01	+∞	492.96	492.96	**0.00**	135.37	7271	14	···
20-40-5	428.83	428.83	**0.00**	**0.08**	428.83	428.83	**0.00**	4.69	428.83	428.83	**0.00**	26.40	−∞	+∞	100.00	3600.01	+∞	428.83	428.83	**0.00**	75.03	7415	12	···
20-40-6	498.29	498.29	**0.00**	**0.09**	498.29	498.29	**0.00**	25.84	498.29	498.29	**0.00**	39.03	−∞	+∞	100.00	3600.01	+∞	498.29	498.29	**0.00**	101.98	7269	12	···
20-40-7	490.89	490.89	**0.00**	**0.08**	490.89	490.89	**0.00**	3.20	490.89	490.89	**0.00**	42.39	−∞	+∞	100.00	3600.01	+∞	490.89	490.89	**0.00**	134.35	7264	12	···
20-40-8	499.41	499.41	**0.00**	**0.13**	499.41	499.41	**0.00**	10.91	499.41	499.41	**0.00**	37.23	−∞	+∞	100.00	3600.01	+∞	499.41	499.41	**0.00**	109.41	7256	13	···
20-40-9	467.54	467.54	**0.00**	**0.09**	467.54	467.54	**0.00**	5.66	467.54	467.54	**0.00**	148.52	−∞	+∞	100.00	3600.01	+∞	467.54	467.54	**0.00**	123.68	7259	13	···
30-60-1	634.78	634.78	**0.00**	**0.30**	634.78	634.78	**0.00**	44.32	634.78	634.78	**0.00**	541.31	−∞	+∞	100.00	3600.01	+∞	634.78	634.78	**0.00**	498.78	16167	12	···
30-60-10	442.37	442.37	**0.00**	**0.31**	442.37	442.37	**0.00**	31.85	442.37	442.37	**0.00**	532.36	−∞	+∞	100.00	3600.01	+∞	442.37	442.37	**0.00**	366.23	16421	14	···
30-60-2	565.71	565.71	**0.00**	**0.38**	565.71	565.71	**0.00**	70.31	565.71	565.71	**0.00**	1590.30	−∞	+∞	100.00	3600.01	+∞	565.71	565.71	**0.00**	402.21	16600	13	···
30-60-3	559.35	559.35	**0.00**	**0.21**	559.35	559.35	**0.00**	17.49	559.35	559.35	**0.00**	2179.41	−∞	+∞	100.00	3600.01	+∞	559.35	559.35	**0.00**	425.18	16387	12	···
30-60-4	602.17	602.17	**0.00**	**0.43**	602.17	602.17	**0.00**	79.14	602.17	602.17	**0.00**	495.23	−∞	+∞	100.00	3600.01	+∞	602.17	602.17	**0.00**	431.53	16202	13	···
30-60-5	542.29	542.29	**0.00**	**0.32**	542.29	542.29	**0.00**	33.96	542.29	542.29	**0.00**	312.67	−∞	+∞	100.00	3600.01	+∞	542.29	542.29	**0.00**	305.20	16404	13	···
30-60-6	571.87	571.87	**0.00**	**0.42**	571.87	571.87	**0.00**	16.25	571.87	571.87	**0.00**	603.69	−∞	+∞	100.00	3600.01	+∞	571.87	571.87	**0.00**	453.10	16175	12	···
30-60-7	690.49	690.49	**0.00**	**0.65**	690.49	690.49	**0.00**	21.84	690.49	690.49	**0.00**	323.74	−∞	+∞	100.00	3600.01	+∞	690.49	690.49	**0.00**	665.65	15942	16	···
30-60-8	630.72	630.72	**0.00**	**0.47**	630.72	630.72	**0.00**	28.35	630.72	630.72	**0.00**	1047.04	−∞	+∞	100.00	3600.01	+∞	630.72	630.72	**0.00**	417.59	16653	14	···
30-60-9	569.33	569.33	**0.00**	**0.27**	569.33	569.33	**0.00**	131.10	569.33	569.33	**0.00**	81.77	−∞	+∞	100.00	3600.01	+∞	569.33	569.33	**0.00**	396.26	16413	13	···
40-80-1	736.31	736.31	**0.00**	**1.12**	736.31	736.31	**0.00**	212.99	736.31	736.31	**0.00**	2317.10	−∞	+∞	100.00	3600.01	+∞	736.31	736.31	**0.00**	1213.44	28907	14	···
40-80-10	712.62	712.62	**0.00**	**1.00**	712.62	712.62	**0.00**	136.48	712.62	712.62	**0.00**	1689.76	−∞	+∞	100.00	3600.01	+∞	712.62	712.62	**0.00**	878.78	28621	13	···
40-80-2	733.78	733.78	**0.00**	**0.92**	733.78	733.78	**0.00**	157.87	733.79	733.79	**0.00**	2865.56	−∞	+∞	100.00	3600.01	+∞	733.78	733.78	**0.00**	1710.96	28567	19	···
40-80-3	653.70	653.70	**0.00**	**1.09**	653.70	653.70	**0.00**	201.55	653.71	653.71	**0.00**	1246.92	−∞	+∞	100.00	3600.01	+∞	653.70	653.70	**0.00**	1054.37	28932	14	···
40-80-4	678.05	678.05	**0.00**	**0.63**	678.05	678.05	**0.00**	197.33	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	678.05	678.05	**0.00**	1473.30	28618	15	···
40-80-5	722.26	722.26	**0.00**	**1.83**	722.26	722.26	**0.00**	1211.35	722.26	722.26	**0.00**	1414.96	−∞	+∞	100.00	3600.01	+∞	722.26	722.26	**0.00**	1330.03	28595	13	···
40-80-6	634.77	634.77	**0.00**	**1.24**	634.77	634.77	**0.00**	57.44	634.78	634.78	**0.00**	3289.72	−∞	+∞	100.00	3600.01	+∞	634.77	634.77	**0.00**	1012.18	29199	12	···
40-80-7	768.39	768.39	**0.00**	**1.44**	768.39	768.39	**0.00**	51.10	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	768.39	768.39	**0.00**	1712.10	27975	16	···
40-80-8	676.84	676.84	**0.00**	**1.00**	676.84	676.84	**0.00**	1625.41	676.85	676.85	**0.00**	2491.21	−∞	+∞	100.00	3600.01	+∞	676.84	676.84	**0.00**	1135.94	28903	13	···
40-80-9	706.91	706.91	**0.00**	**1.43**	706.91	706.91	**0.00**	58.93	706.91	706.91	**0.00**	818.00	−∞	+∞	100.00	3600.01	+∞	706.91	706.91	**0.00**	948.46	28919	13	···
50-100-1	819.00	819.00	**0.00**	**2.02**	819.00	819.00	**0.00**	291.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	819.00	819.00	**0.00**	1607.57	38405	15	···
50-100-10	820.61	820.61	**0.00**	**2.48**	820.61	820.61	**0.00**	123.73	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	820.61	820.61	**0.00**	2011.18	38032	15	···
50-100-2	885.03	885.03	**0.00**	**5.57**	885.03	885.03	**0.00**	134.87	885.03	885.03	**0.00**	1446.86	−∞	+∞	100.00	3600.01	+∞	885.03	885.03	**0.00**	2837.87	44999	14	···
50-100-3	792.09	792.09	**0.00**	**5.49**	792.09	792.09	**0.00**	152.71	792.09	792.09	**0.00**	1506.56	−∞	+∞	100.00	3600.01	+∞	792.09	792.09	**0.00**	3412.09	44244	17	···
50-100-4	733.58	733.58	**0.00**	**4.06**	733.58	733.58	**0.00**	1540.62	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	733.58	733.58	**0.00**	1752.81	38408	14	···
50-100-5	733.23	733.23	**0.00**	**3.53**	733.23	733.23	**0.00**	963.29	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	733.23	733.23	**0.00**	2361.18	45702	13	···
50-100-6	741.44	741.44	**0.00**	**2.55**	741.44	741.44	**0.00**	367.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	741.44	741.44	**0.00**	1957.71	44963	13	···
50-100-7	857.60	857.60	**0.00**	**4.59**	857.60	857.60	**0.00**	495.68	857.60	857.60	**0.00**	1728.80	−∞	+∞	100.00	3600.01	+∞	857.60	857.60	**0.00**	1600.24	43715	14	···
50-100-8	797.78	797.78	**0.00**	**4.15**	797.77	841.88	5.24	3637.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	797.78	797.78	**0.00**	1704.51	38401	13	···
50-100-9	857.67	857.67	**0.00**	**4.09**	857.67	970.72	11.65	3607.13	857.67	857.67	**0.00**	1377.49	−∞	+∞	100.00	3600.01	+∞	857.67	857.67	**0.00**	1952.07	38623	15	···

Table 34 Detailed results for problem NLUFLP-A-gunluk, cost functions f_9

Instance	GUROBI DB	PB	GAP	CPU	SCIP DB	PB	GAP	CPU	COUENNE DB	PB	GAP	CPU	NAIVE DB	PB	GAP	CPU	NP	CN24 DB	PB	GAP	CPU	NP	NIT	⋯
10-20-1	22.71	92.72	75.51	3600.01	18.72	212.38	91.19	3602.29	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	92.72	92.72	**0.00**	**257.98**	3509	41	⋯
10-20-10	17.72	105.90	83.26	3600.01	20.35	173.01	88.24	3602.20	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	105.90	105.90	**0.00**	**165.33**	3180	36	⋯
10-20-2	36.91	135.13	72.68	3600.01	39.82	193.13	79.38	3602.52	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	129.81	129.81	**0.00**	**506.21**	3148	39	⋯
10-20-3	21.35	119.19	82.09	3600.01	27.42	119.19	77.00	3602.05	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	119.19	119.19	**0.00**	**374.66**	3164	44	⋯
10-20-4	51.69	146.64	64.75	3600.01	41.61	157.81	73.64	3602.48	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	146.64	146.64	**0.00**	**426.33**	3204	38	⋯
10-20-5	21.69	138.72	84.37	3600.01	19.89	145.71	86.35	3602.06	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	132.67	132.67	**0.00**	**1938.80**	3603	41	⋯
10-20-6	89.77	264.99	66.12	3600.01	77.72	292.58	73.44	3602.38	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	261.65	261.65	**0.00**	**1060.84**	3049	37	⋯
10-20-7	48.12	133.04	63.83	3600.01	41.09	134.37	69.42	3602.19	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	131.29	131.29	**0.00**	**375.81**	3134	35	⋯
10-20-8	61.62	206.38	70.14	3600.01	71.01	251.12	71.72	3602.40	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	204.00	204.01	**0.00**	**608.74**	3160	37	⋯
10-20-9	42.15	233.37	81.94	3600.01	42.37	230.25	81.60	3602.36	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	179.24	179.24	**0.00**	**548.23**	3144	38	⋯
20-40-1	14.64	175.01	91.64	3600.02	13.48	380.09	96.45	3602.35	11.86	521.58	97.73	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	127.80	127.80	**0.00**	3601.03	10389	47	⋯
20-40-10	7.40	209.33	96.47	3600.02	7.46	373.12	98.00	3602.46	4.79	363.48	98.68	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	106.32	106.34	**0.01**	3601.02	9612	30	⋯
20-40-2	12.05	198.73	93.93	3600.01	10.78	261.08	95.87	3601.86	7.98	464.21	98.28	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	130.61	130.73	**0.09**	3601.02	8760	21	⋯
20-40-3	24.57	140.04	82.46	3600.02	15.13	306.17	95.06	3602.36	11.82	487.31	97.58	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	134.21	134.21	**0.00**	3601.02	10359	52	⋯
20-40-4	21.34	195.20	89.07	3600.01	21.20	194.91	89.12	3602.38	13.98	330.98	95.78	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	141.93	144.08	**1.49**	3601.03	8074	6	⋯
20-40-5	12.51	114.41	89.07	3600.01	11.77	149.93	92.15	3602.46	7.44	442.64	98.32	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	100.80	100.80	**0.00**	3601.02	11256	53	⋯
20-40-6	12.34	137.17	91.00	3600.01	6.79	416.34	98.37	3602.24	4.75	482.28	99.01	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	137.06	137.06	**0.00**	3601.03	10128	44	⋯
20-40-7	21.06	250.43	91.59	3600.01	20.42	281.16	92.74	3602.72	15.96	421.40	96.21	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	156.26	156.46	**0.12**	3601.05	8714	17	⋯
20-40-8	27.00	205.15	86.84	3600.02	20.68	394.83	94.76	3601.97	15.75	447.35	96.48	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	157.48	157.48	**0.00**	3601.03	10284	43	⋯
20-40-9	20.39	265.92	92.33	3600.02	17.13	395.06	95.67	3602.16	11.93	574.41	97.92	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	157.34	160.94	**2.23**	3601.02	7783	5	⋯
30-60-1	13.93	363.07	96.16	3600.01	14.12	491.23	97.13	3602.35	8.16	529.23	98.46	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	154.51	160.71	**3.86**	3601.04	16963	4	⋯
30-60-10	7.87	168.11	95.32	3600.01	7.39	223.73	96.70	3602.73	$-\infty$	406.86	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	116.22	119.81	**3.00**	3601.05	17155	4	⋯
30-60-2	10.06	268.83	96.26	3600.02	12.41	496.81	97.50	3602.35	7.88	542.88	98.55	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	156.49	160.11	**2.27**	3601.03	17205	5	⋯
30-60-3	11.36	235.78	95.18	3600.01	11.84	215.79	94.51	3602.30	6.75	536.05	98.74	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	131.39	131.86	**0.36**	3601.04	18275	17	⋯
30-60-4	9.18	318.49	97.12	3600.02	10.14	243.82	95.84	3602.73	7.05	398.86	98.23	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	134.91	136.59	**1.23**	3601.04	17837	7	⋯
30-60-5	11.47	214.67	94.66	3600.03	11.56	211.56	94.54	3602.37	6.87	529.94	98.70	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	122.55	125.64	**2.46**	3601.04	17572	6	⋯
30-60-6	5.88	226.72	97.41	3600.02	6.46	225.76	97.14	3602.34	4.66	419.93	98.89	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	127.77	131.35	**2.73**	3601.04	17660	6	⋯
30-60-7	20.17	393.51	94.87	3600.02	17.80	475.73	96.26	3602.37	11.33	533.53	97.88	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	190.09	192.91	**1.46**	3601.04	17380	6	⋯
30-60-8	12.22	205.13	94.04	3600.02	13.79	264.66	94.79	3602.41	8.72	413.39	97.89	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	127.72	130.97	**2.49**	3601.04	17463	5	⋯
30-60-9	12.15	254.79	95.23	3600.02	12.77	222.49	94.26	3602.58	7.57	492.81	98.46	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	143.40	147.26	**2.62**	3601.04	17183	5	⋯
40-80-1	7.57	503.58	98.50	3600.02	8.99	289.67	96.90	3602.63	6.65	489.11	98.64	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	143.96	150.03	**4.04**	3601.04	29979	4	⋯
40-80-10	6.48	574.43	98.87	3600.03	8.37	289.97	97.11	3602.73	5.73	447.93	98.72	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	148.65	153.61	**3.23**	3601.05	30444	5	⋯
40-80-2	9.00	756.16	98.81	3600.03	11.67	290.49	95.98	3602.60	7.61	740.42	98.97	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	169.35	175.22	**3.35**	3601.04	29990	4	⋯
40-80-3	10.03	397.18	97.47	3600.03	12.31	264.07	95.34	3603.10	7.41	630.74	98.82	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	138.90	141.98	**2.17**	3601.07	30690	6	⋯
40-80-4	9.55	353.06	97.30	3600.02	13.43	334.42	95.98	3603.09	8.20	407.97	97.99	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	142.36	147.17	**3.27**	3601.05	30266	4	⋯
40-80-5	10.13	607.85	98.33	3600.05	12.47	288.35	95.67	3602.73	8.03	531.86	98.49	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	153.82	156.10	**1.46**	3601.05	30906	7	⋯
40-80-6	6.41	434.14	98.52	3600.03	7.76	282.89	97.26	3602.46	5.30	476.64	98.89	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	130.62	136.19	**4.09**	3601.05	30122	4	⋯
40-80-7	17.47	564.06	96.90	3600.04	17.67	743.17	97.62	3602.71	12.34	632.73	98.05	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	200.07	205.58	**2.68**	3601.06	30129	4	⋯
40-80-8	12.04	650.57	98.15	3600.04	13.26	309.27	95.71	3602.71	9.35	412.72	97.73	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	143.77	152.51	**5.73**	3601.05	29764	3	⋯
40-80-9	10.12	622.54	98.37	3600.03	11.52	316.91	96.37	3602.71	8.72	395.12	97.79	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	147.61	148.75	**0.76**	3601.05	31218	11	⋯
50-100-1	9.03	1027.43	99.12	3600.05	9.85	372.33	97.36	3603.17	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	151.75	162.25	**6.47**	3601.10	45598	2	⋯
50-100-10	7.42	1030.63	99.28	3600.03	10.31	381.40	**97.30**	3603.42	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.10	45000	1	⋯
50-100-2	7.40	1047.64	99.29	3600.03	11.51	374.66	96.93	3603.13	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	179.62	191.18	**6.05**	3601.11	45595	2	⋯
50-100-3	9.76	747.72	98.69	3600.04	12.54	353.05	96.45	3603.11	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	146.69	156.50	**6.27**	3601.06	45635	2	⋯
50-100-4	9.43	772.73	98.78	3600.04	14.11	400.11	96.47	3603.52	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	155.60	165.29	**5.86**	3601.07	45541	2	⋯
50-100-5	11.43	1044.43	98.91	3600.03	13.03	354.14	96.32	3603.54	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	161.65	172.79	**6.44**	3601.07	45591	2	⋯
50-100-6	5.60	882.30	99.37	3600.04	9.13	368.35	97.52	3603.45	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	152.25	162.33	**6.21**	3601.09	45700	2	⋯
50-100-7	19.15	995.49	98.08	3600.04	16.24	385.59	95.79	3603.81	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	201.21	212.15	**5.16**	3601.11	45600	2	⋯
50-100-8	12.88	916.77	98.60	3600.03	13.60	372.46	96.35	3604.03	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	168.30	178.71	**5.82**	3601.06	45699	2	⋯
50-100-9	11.07	1037.11	98.93	3600.05	12.72	384.31	96.69	3603.39	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	161.44	172.34	**6.33**	3601.07	45600	2	⋯

Table 35 Detailed results for problem NLUFLP-A-gunluk, cost functions f_{10}

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
cap101	1471358.65	1471358.65	**0.00**	**2.13**	1471357.28	1471358.65	**0.00**	22.20	1320614.00	1484522.00	11.04	3600.10	1471357.58	1471358.65	**0.00**	1545.60	497103	1471357.62	1471358.65	**0.00**	432.26	3949	19	⋯
cap102	1557008.07	1557008.08	**0.00**	**2.39**	1557007.82	1557008.08	**0.00**	7.87	1378010.00	1595330.00	13.62	3600.10	1557006.98	1557008.08	**0.00**	1598.12	497103	1557006.86	1557008.08	**0.00**	438.70	3950	16	⋯
cap103	1627452.66	1627452.66	**0.00**	**3.14**	1627451.87	1627452.66	**0.00**	28.10	1423081.00	1710330.00	16.79	3600.10	1627451.53	1627452.66	**0.00**	1518.08	497103	1627451.28	1627452.66	**0.00**	517.25	3959	20	⋯
cap104	1709340.86	1709340.86	**0.00**	**3.64**	1709340.86	1709340.86	**0.00**	33.76	1476386.00	1861943.00	20.71	3600.10	1709339.69	1709340.86	**0.00**	1480.38	497103	1709339.65	1709340.86	**0.00**	462.43	3965	21	⋯
cap111	1469371.10	1469371.10	**0.00**	**6.86**	1469370.99	1469371.10	**0.00**	108.43	1275623.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.31	994603	1469370.06	1469371.10	**0.00**	1014.01	7707	22	⋯
cap112	1551183.18	1551183.72	**0.00**	**7.91**	1551182.61	1551183.72	**0.00**	178.79	1337724.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.27	994603	1551182.57	1551183.72	**0.00**	746.84	7702	18	⋯
cap113	1621035.21	1621035.21	**0.00**	**7.66**	1621034.40	1621035.21	**0.00**	265.96	1387950.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.28	994603	1621034.05	1621035.21	**0.00**	1041.76	7719	23	⋯
cap114	1702923.41	1702923.41	**0.00**	**10.88**	1702923.41	1702923.41	**0.00**	520.83	1445680.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.36	994603	1702922.20	1702923.41	**0.00**	963.74	7719	22	⋯
cap121	1469371.10	1469371.10	**0.00**	**6.31**	1469370.99	1469371.10	**0.00**	117.58	1275623.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.39	994603	1469370.06	1469371.10	**0.00**	880.24	7707	22	⋯
cap122	1551183.18	1551183.72	**0.00**	**6.67**	1551182.61	1551183.72	**0.00**	168.97	1337724.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.38	994603	1551182.57	1551183.72	**0.00**	1056.55	7702	18	⋯
cap123	1621035.21	1621035.21	**0.00**	**7.45**	1621034.40	1621035.21	**0.00**	235.94	1387950.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.86	994603	1621034.05	1621035.21	**0.00**	1083.27	7719	23	⋯
cap124	1702923.41	1702923.41	**0.00**	**9.86**	1702923.41	1702923.41	**0.00**	422.08	1445680.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.47	994603	1702922.20	1702923.41	**0.00**	907.98	7719	22	⋯
cap131	1469371.10	1469371.10	**0.00**	**5.93**	1469370.99	1469371.10	**0.00**	120.39	1275623.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.33	994603	1469370.06	1469371.10	**0.00**	1058.90	7707	22	⋯
cap132	1551183.18	1551183.72	**0.00**	**7.00**	1551182.61	1551183.72	**0.00**	190.34	1337724.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.30	994603	1551182.57	1551183.72	**0.00**	872.83	7702	18	⋯
cap133	1621035.21	1621035.21	**0.00**	**6.84**	1621034.40	1621035.21	**0.00**	284.62	1387950.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.29	994603	1621034.05	1621035.21	**0.00**	950.10	7719	23	⋯
cap134	1702923.41	1702923.41	**0.00**	**9.70**	1702923.41	1702923.41	**0.00**	440.41	1445680.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.42	994603	1702922.20	1702923.41	**0.00**	756.60	7719	22	⋯
cap41	1783194.18	1783194.18	**0.00**	**1.59**	1783194.19	1783194.19	**0.00**	17.94	1591708.00	1788363.00	11.00	3600.10	1783192.83	1783194.19	**0.00**	742.67	318003	1783192.88	1783194.19	**0.00**	301.18	2600	20	⋯
cap42	1839559.03	1839559.03	**0.00**	**1.51**	1839559.02	1839559.03	**0.00**	9.01	1622249.00	1863363.00	12.94	3600.10	1839557.66	1839559.03	**0.00**	755.92	318003	1839557.39	1839559.03	**0.00**	247.61	2597	19	⋯
cap43	1890152.11	1890152.11	**0.00**	**1.58**	1890151.28	1890152.13	**0.00**	16.52	1655012.00	1924778.00	14.02	3600.10	1890150.74	1890152.13	**0.00**	756.52	318003	1890150.79	1890152.13	**0.00**	301.25	2608	22	⋯
cap44	1955517.56	1955517.56	**0.00**	**1.41**	1955516.71	1955517.56	**0.00**	26.72	1695702.00	2011016.00	15.68	3600.10	1955516.14	1955517.56	**0.00**	781.10	318003	1955515.98	1955517.56	**0.00**	282.17	2605	19	⋯
cap51	1890152.11	1890152.11	**0.00**	**1.78**	1890151.28	1890152.13	**0.00**	14.63	1655012.00	1924778.00	14.02	3600.10	1890150.74	1890152.13	**0.00**	774.50	318003	1890150.79	1890152.13	**0.00**	279.95	2608	22	⋯
cap61	1783194.18	1783194.18	**0.00**	**1.87**	1783194.19	1783194.19	**0.00**	17.84	1591888.00	1788363.00	10.99	3600.10	1783192.83	1783194.19	**0.00**	745.86	318003	1783192.88	1783194.19	**0.00**	286.98	2600	20	⋯
cap62	1839559.03	1839559.03	**0.00**	**1.24**	1839559.02	1839559.03	**0.00**	10.16	1622249.00	1863363.00	12.94	3600.10	1839557.66	1839559.03	**0.00**	744.66	318003	1839557.39	1839559.03	**0.00**	249.67	2597	19	⋯
cap63	1890152.11	1890152.11	**0.00**	**1.37**	1890151.28	1890152.13	**0.00**	18.31	1655012.00	1924778.00	14.02	3600.10	1890150.74	1890152.13	**0.00**	755.08	318003	1890150.79	1890152.13	**0.00**	311.76	2608	22	⋯
cap64	1955517.56	1955517.56	**0.00**	**1.32**	1955516.71	1955517.56	**0.00**	21.52	1695344.00	2011016.00	15.70	3600.10	1955516.14	1955517.56	**0.00**	768.79	318003	1955515.98	1955517.56	**0.00**	252.81	2605	19	⋯
cap71	1783194.18	1783194.18	**0.00**	**1.54**	1783194.19	1783194.19	**0.00**	19.23	1591888.00	1788363.00	10.99	3600.10	1783192.83	1783194.19	**0.00**	745.93	318003	1783192.88	1783194.19	**0.00**	243.42	2600	20	⋯
cap72	1839559.03	1839559.03	**0.00**	**1.44**	1839559.02	1839559.03	**0.00**	10.70	1622249.00	1863363.00	12.94	3600.10	1839557.66	1839559.03	**0.00**	744.01	318003	1839557.39	1839559.03	**0.00**	305.96	2597	19	⋯
cap73	1890152.11	1890152.11	**0.00**	**1.55**	1890151.28	1890152.13	**0.00**	15.03	1655012.00	1924778.00	14.02	3600.10	1890150.74	1890152.13	**0.00**	755.98	318003	1890150.79	1890152.13	**0.00**	281.73	2608	22	⋯
cap74	1955517.56	1955517.56	**0.00**	**1.30**	1955516.71	1955517.56	**0.00**	21.80	1695344.00	2011016.00	15.70	3600.10	1955516.14	1955517.56	**0.00**	763.56	318003	1955515.98	1955517.56	**0.00**	276.86	2605	19	⋯
cap81	1471358.65	1471358.65	**0.00**	**2.04**	1471357.28	1471358.65	**0.00**	21.66	1320614.00	1484522.00	11.04	3600.10	1471357.58	1471358.65	**0.00**	1509.28	497103	1471357.62	1471358.65	**0.00**	500.43	3949	19	⋯
cap82	1557008.07	1557008.08	**0.00**	**2.24**	1557007.82	1557008.08	**0.00**	8.42	1378010.00	1595330.00	13.62	3600.10	1557006.98	1557008.08	**0.00**	1532.18	497103	1557006.86	1557008.08	**0.00**	447.64	3950	16	⋯
cap83	1627452.66	1627452.66	**0.00**	**2.38**	1627451.87	1627452.66	**0.00**	31.63	1423081.00	1710330.00	16.79	3600.10	1627451.53	1627452.66	**0.00**	1459.40	497103	1627451.28	1627452.66	**0.00**	447.19	3959	20	⋯
cap84	1709340.86	1709340.86	**0.00**	**3.57**	1709340.86	1709340.86	**0.00**	33.14	1476386.00	1861943.00	20.71	3600.10	1709339.69	1709340.86	**0.00**	1571.12	497103	1709339.65	1709340.86	**0.00**	475.47	3965	21	⋯
cap91	1471358.65	1471358.65	**0.00**	**1.90**	1471357.28	1471358.65	**0.00**	18.89	1320614.00	1484522.00	11.04	3600.10	1471357.58	1471358.65	**0.00**	1432.45	497103	1471357.62	1471358.65	**0.00**	421.19	3949	19	⋯
cap92	1557008.07	1557008.08	**0.00**	**2.08**	1557007.82	1557008.08	**0.00**	8.17	1378010.00	1595330.00	13.62	3600.10	1557006.98	1557008.08	**0.00**	1570.62	497103	1557006.86	1557008.08	**0.00**	506.07	3950	16	⋯
cap93	1627452.66	1627452.66	**0.00**	**2.60**	1627451.87	1627452.66	**0.00**	25.49	1423081.00	1710330.00	16.79	3600.10	1627451.53	1627452.66	**0.00**	1600.69	497103	1627451.28	1627452.66	**0.00**	451.24	3959	20	⋯
cap94	1709340.86	1709340.86	**0.00**	**3.54**	1709340.86	1709340.86	**0.00**	32.16	1476386.00	1861943.00	20.71	3600.10	1709339.69	1709340.86	**0.00**	1558.07	497103	1709339.65	1709340.86	**0.00**	458.24	3965	21	⋯
#S			**37**				**37**				0				25					**37**				⋯
SGM			**0.00**	**3.63**			**0.00**	46.50			27.08	14400.00			3.47	2479.14	532068			**0.00**	487.35	4241	20	⋯

Table 36 Detailed results for problem NLUFLP-A-orlib, cost functions f_1

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
cap101	667593.99	710680.52	**6.06**	3600.02	-4125072.00	1151380.23	127.91	3603.43	-7225243.00	1799536.00	124.91	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	6247	1	⋯
cap102	707391.97	754108.48	**6.19**	3600.02	-4096375.16	1140808.50	127.85	3603.41	-7203265.00	1804536.00	125.05	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	6247	1	⋯
cap103	739227.72	790898.58	**6.53**	3600.02	-4064893.46	1161380.23	128.57	3603.10	-7192135.00	1809536.00	125.16	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	6247	1	⋯
cap104	781097.58	830603.69	**5.96**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-7179658.00	1817036.00	125.31	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	6247	1	⋯
cap111	640641.91	700874.23	**8.59**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	2276187.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	12497	1	⋯
cap112	685683.25	765911.04	**10.47**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	-8033113.00	2281187.00	128.40	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	12497	1	⋯
cap113	718823.69	792168.07	**9.26**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	2286187.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	12497	1	⋯
cap114	756971.28	830603.68	**8.86**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	2293687.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	12497	1	⋯
cap121	647085.22	700874.23	**7.67**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	2276187.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	12497	1	⋯
cap122	687442.75	765911.04	**10.25**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	-8033113.00	2281187.00	128.40	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	12497	1	⋯
cap123	719858.48	792168.07	**9.13**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	2286187.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	12497	1	⋯
cap124	757687.78	830603.68	**8.78**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	2293687.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	12497	1	⋯
cap131	646193.39	700874.23	**7.80**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	2276187.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	12497	1	⋯
cap132	687074.72	765911.04	**10.29**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	-8033113.00	2281187.00	128.40	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	12497	1	⋯
cap133	720859.41	792168.07	**9.00**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	2286187.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	12497	1	⋯
cap134	758260.14	830603.68	**8.71**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	2293687.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	12497	1	⋯
cap41	771681.16	812767.32	**5.06**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	3997	1	⋯
cap42	797870.17	853648.13	**6.53**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-5342825.00	2389681.00	144.73	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.04	3997	1	⋯
cap43	824222.34	880227.40	**6.36**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-5329318.00	2408930.00	145.20	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	3997	1	⋯
cap44	851525.62	906446.53	**6.06**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	3997	1	⋯
cap51	824548.59	880227.40	**6.33**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-5324045.00	2408930.00	145.25	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	3997	1	⋯
cap61	771503.07	812767.32	**5.08**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	3997	1	⋯
cap62	797415.19	853648.13	**6.59**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	3997	1	⋯
cap63	823822.82	880227.40	**6.41**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-5324045.00	2408930.00	145.25	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	3997	1	⋯
cap64	850065.27	906446.54	**6.22**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	3997	1	⋯
cap71	771698.69	812767.32	**5.05**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.04	3997	1	⋯
cap72	797690.77	853648.13	**6.56**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-5342825.00	2389681.00	144.73	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	3997	1	⋯
cap73	824460.99	880227.40	**6.34**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	-5324045.00	2408930.00	145.25	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	3997	1	⋯
cap74	850542.97	906446.54	**6.17**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	3997	1	⋯
cap81	667577.92	710680.52	**6.06**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-7225243.00	1799536.00	124.91	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	6247	1	⋯
cap82	707306.46	754108.48	**6.21**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-7203265.00	1804536.00	125.05	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	6247	1	⋯
cap83	739835.80	790898.58	**6.46**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-7192135.00	1809536.00	125.16	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	6247	1	⋯
cap84	780604.23	830603.69	**6.02**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-7179658.00	1817036.00	125.31	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	6247	1	⋯
cap91	667166.90	710680.52	**6.12**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-7225243.00	1799536.00	124.91	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	6247	1	⋯
cap92	706879.29	754108.48	**6.26**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-7203265.00	1804536.00	125.05	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	6247	1	⋯
cap93	739273.65	790898.58	**6.53**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-7192135.00	1809536.00	125.16	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	6247	1	⋯
cap94	780032.98	830603.69	**6.09**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	-7179658.00	1817036.00	125.31	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	6247	1	⋯
#S			0				0				0				0					0				⋯
SGM			**6.95**	14400.00			100.00	14400.00			100.00	14400.00			100.00	14400.00	$+\infty$			100.00	14400.00	6688	1	⋯

Table 37 Detailed results for problem NLUFLP-A-orlib, cost functions f_2

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	136250.21	337542.48	59.63	3600.01	199543.64	331208.22	39.75	3603.17	77711.76	462466.60	83.20	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	323889.72	333166.77	**2.78**	3601.03	13283	9	···
cap102	166461.80	418590.11	60.23	3600.01	251417.65	410153.22	38.70	3603.09	83126.77	577407.40	85.60	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	403726.74	413559.92	**2.38**	3601.03	13035	9	···
cap103	189740.21	484660.82	60.85	3600.01	292783.46	474689.88	38.32	3603.04	89718.83	632224.30	85.81	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	464760.54	479824.72	**3.14**	3601.03	12872	6	···
cap104	225136.06	561099.96	59.88	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	94792.73	798991.20	88.14	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	545043.01	561785.81	**2.98**	3601.03	12855	8	···
cap111	67783.08	356499.15	**80.99**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	6517.96	523392.30	98.75	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.23	24991	1	···
cap112	92291.42	432922.87	**78.68**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	10863.27	633392.30	98.28	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	24991	1	···
cap113	106646.53	501525.86	**78.74**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	24962.60	743392.30	96.64	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.28	24991	1	···
cap114	128869.44	583970.76	**77.93**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	31616.64	908392.30	96.52	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.26	24991	1	···
cap121	69080.39	356499.15	**80.62**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	6517.96	523392.30	98.75	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	24991	1	···
cap122	92326.77	432922.87	**78.67**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	10863.27	633392.30	98.28	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.25	24991	1	···
cap123	107903.97	501525.86	**78.48**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	24962.60	743392.30	96.64	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.27	24991	1	···
cap124	127747.00	583970.76	78.12	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	31616.64	908392.30	96.52	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	530939.81	563903.07	**5.85**	3601.06	25032	2	···
cap131	69307.22	356499.15	**80.56**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	6517.96	523392.30	98.75	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	24991	1	···
cap132	92685.48	432922.87	**78.59**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	10863.27	633392.30	98.28	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	24991	1	···
cap133	108916.33	501525.86	**78.28**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	17534.68	743392.30	97.64	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.26	24991	1	···
cap134	128264.50	583970.76	**78.04**	3600.02	339271.09	$+\infty$	100.00	$+\infty$	31616.64	908392.30	96.52	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.19	24991	1	···
cap41	214992.71	367143.03	41.44	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	123915.70	550504.60	77.49	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	364268.17	368390.43	**1.12**	3601.03	10014	14	···
cap42	251116.78	446707.78	43.79	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	138639.60	610504.60	77.29	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	434782.66	440682.54	**1.34**	3601.04	9306	11	···
cap43	288647.09	504777.47	42.82	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	155880.30	700163.50	77.74	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	497335.96	503294.02	**1.18**	3601.03	9452	17	···
cap44	337020.21	580554.96	41.95	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	180588.60	789692.80	77.13	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	573030.36	582526.38	**1.63**	3601.03	8851	13	···
cap51	288593.13	504778.45	42.83	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	159154.10	700163.50	77.27	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	498492.43	503294.02	**0.95**	3601.02	9458	18	···
cap61	213198.26	367171.07	41.93	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	123915.70	550504.60	77.49	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	364268.17	368390.43	**1.12**	3601.03	10014	14	···
cap62	254813.63	441913.19	42.34	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	138639.60	610504.60	77.29	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	434782.66	440682.54	**1.34**	3601.03	9306	11	···
cap63	285738.83	504778.45	43.39	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	159154.10	700163.50	77.27	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	497335.96	503294.02	**1.18**	3601.04	9452	17	···
cap64	341872.62	580542.35	41.11	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	180588.60	789692.80	77.13	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	574219.42	582526.38	**1.43**	3601.03	8859	14	···
cap71	214408.68	367153.17	41.60	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	123915.70	550504.60	77.49	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	364268.17	368390.43	**1.12**	3601.04	10014	14	···
cap72	253914.11	441913.19	42.54	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	138639.60	610504.60	77.29	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	434782.66	440682.54	**1.34**	3601.03	9306	11	···
cap73	286344.20	504778.45	43.27	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	159154.10	700163.50	77.27	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	497335.96	503294.02	**1.18**	3601.03	9452	17	···
cap74	340147.12	580554.96	41.41	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	180588.60	789692.80	77.13	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	573030.36	583072.77	**1.72**	3601.04	8639	12	···
cap81	136147.29	337542.48	59.67	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	77711.76	462466.60	83.20	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	324958.50	333088.61	**2.44**	3601.03	13409	11	···
cap82	167151.27	418590.11	60.07	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	83126.77	577407.40	85.60	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	403506.33	413559.92	**2.43**	3601.04	13032	8	···
cap83	193003.37	484660.82	60.18	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	89718.83	632224.30	85.81	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	464760.54	479824.72	**3.14**	3601.04	12872	6	···
cap84	223971.88	561099.96	60.08	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	94792.73	798991.20	88.14	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	545043.01	561785.81	**2.98**	3601.05	12855	8	···
cap91	135791.42	337542.48	59.77	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	77711.76	462466.60	83.20	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	323889.72	333166.77	**2.78**	3601.04	13283	9	···
cap92	166111.13	418595.74	60.32	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	83126.77	577407.40	85.60	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	403506.33	413559.92	**2.43**	3601.03	13032	8	···
cap93	192076.85	484660.82	60.37	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	89718.83	632224.30	85.81	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	464760.54	479824.72	**3.14**	3601.03	12872	6	···
cap94	226280.05	560430.30	59.62	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	94792.73	798991.20	88.14	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	545043.01	561785.81	**2.98**	3601.03	12855	8	···
#S			0				0				0				0					0				···
SGM			58.08	14400.00			92.68	14400.00			86.22	14400.00			100.00	14400.00	$+\infty$			**7.54**	14400.00	14339	7	···

Table 38 Detailed results for problem NLUFLP-A-orlib, cost functions f_3

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
cap101	129926.41	595022.69	78.16	3600.02	191393.39	557508.15	**65.67**	3603.00	74649.44	766304.80	90.26	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	13740	1	⋯
cap102	152930.63	712351.69	78.53	3600.01	241296.98	672654.92	**64.13**	3603.03	80750.20	851293.90	90.51	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	13740	1	⋯
cap103	176893.60	747591.33	76.34	3600.01	280627.52	769892.11	**63.55**	3602.96	86280.04	865126.60	90.03	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	13740	1	⋯
cap104	210975.13	778283.05	**72.89**	3600.02	−∞	+∞	100.0	+∞	94789.88	1003642.00	90.56	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	13740	1	⋯
cap111	53742.00	757742.09	**92.91**	3600.02	−∞	+∞	100.0	+∞	6781.40	823465.10	99.18	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
cap112	64008.75	868410.11	**92.63**	3600.02	−∞	+∞	100.0	+∞	11302.32	933465.10	98.79	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
cap113	75295.17	871672.54	**91.36**	3600.02	−∞	+∞	100.0	+∞	15823.25	1043465.00	98.48	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
cap114	90253.96	910858.33	**90.09**	3600.02	−∞	+∞	100.0	+∞	22604.65	1698870.00	98.67	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
cap121	53742.00	643341.47	**91.65**	3600.02	−∞	+∞	100.0	+∞	6781.40	823465.10	99.18	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
cap122	64008.75	806387.48	**92.06**	3600.01	−∞	+∞	100.0	+∞	11302.32	933465.10	98.79	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
cap123	76026.38	871672.54	**91.28**	3600.02	−∞	+∞	100.0	+∞	15823.25	1043465.00	98.48	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
cap124	90253.96	910858.33	**90.09**	3600.01	−∞	+∞	100.0	+∞	22604.65	1698870.00	98.67	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
cap131	53742.00	643341.47	**91.65**	3600.01	−∞	+∞	100.0	+∞	6781.40	823465.10	99.18	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
cap132	64008.75	806387.48	**92.06**	3600.02	−∞	+∞	100.0	+∞	11302.32	933465.10	98.79	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
cap133	77202.82	829360.71	**90.69**	3600.02	−∞	+∞	100.0	+∞	15823.25	1043465.00	98.48	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
cap134	90253.96	910858.33	**90.09**	3600.01	−∞	+∞	100.0	+∞	22604.65	1698870.00	98.67	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
cap41	217014.57	730753.37	**70.30**	3600.01	−∞	+∞	100.0	+∞	115734.40	829201.60	86.04	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	8790	1	⋯
cap42	247943.61	843459.77	**70.60**	3600.01	−∞	+∞	100.0	+∞	126189.80	897667.50	85.94	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	8790	1	⋯
cap43	270651.21	849104.49	68.13	3600.01	−∞	+∞	100.0	+∞	137756.80	930098.50	85.19	3600.10	−∞	+∞	100.00	3600.01	+∞	730348.82	797402.01	**8.41**	3601.03	8797	3	⋯
cap44	344365.28	836049.91	58.81	3600.01	−∞	+∞	100.0	+∞	151382.40	936215.50	83.83	3600.10	−∞	+∞	100.00	3600.01	+∞	767143.38	812508.72	**5.58**	3601.03	8803	4	⋯
cap51	275311.41	849104.49	67.58	3600.01	−∞	+∞	100.0	+∞	137756.80	930098.50	85.19	3600.10	−∞	+∞	100.00	3600.01	+∞	730348.82	797402.01	**8.41**	3601.14	8790	2	⋯
cap61	215600.28	730753.37	**70.50**	3600.01	−∞	+∞	100.0	+∞	115734.40	829201.60	86.04	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	8790	1	⋯
cap62	248017.80	843459.77	**70.60**	3600.01	−∞	+∞	100.0	+∞	126189.80	897667.50	85.94	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	8790	1	⋯
cap63	272403.41	849104.49	67.92	3600.01	−∞	+∞	100.0	+∞	137756.80	930098.50	85.19	3600.10	−∞	+∞	100.00	3600.01	+∞	730348.82	797402.01	**8.41**	3601.19	8797	3	⋯
cap64	343578.10	836049.91	58.90	3600.01	−∞	+∞	100.0	+∞	151382.40	936215.50	83.83	3600.10	−∞	+∞	100.00	3600.01	+∞	782033.73	811298.67	**3.61**	3601.03	8836	6	⋯
cap71	219859.70	729452.97	**69.86**	3600.01	−∞	+∞	100.0	+∞	115734.40	829201.60	86.04	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	8790	1	⋯
cap72	252631.45	843459.77	**70.05**	3600.01	−∞	+∞	100.0	+∞	126189.80	897667.50	85.94	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	8790	1	⋯
cap73	273255.79	849104.49	67.82	3600.01	−∞	+∞	100.0	+∞	137756.80	930098.50	85.19	3600.10	−∞	+∞	100.00	3600.01	+∞	730348.82	797402.01	**8.41**	3601.03	8797	3	⋯
cap74	350986.67	836049.91	58.02	3600.01	−∞	+∞	100.0	+∞	151382.40	936215.50	83.83	3600.10	−∞	+∞	100.00	3600.01	+∞	767143.38	811298.67	**5.44**	3601.02	8832	5	⋯
cap81	132371.15	595022.69	77.75	3600.01	−∞	+∞	100.0	+∞	74649.44	766304.80	90.26	3600.10	−∞	+∞	100.00	3600.01	+∞	528113.26	563566.64	**6.29**	3601.03	13798	2	⋯
cap82	154509.42	712351.69	**78.31**	3600.02	−∞	+∞	100.0	+∞	80750.20	851293.90	90.51	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	13740	1	⋯
cap83	176999.92	747591.33	**76.32**	3600.01	−∞	+∞	100.0	+∞	86280.04	865126.60	90.03	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.24	13740	1	⋯
cap84	211704.19	778283.05	**72.80**	3600.02	−∞	+∞	100.0	+∞	94789.88	1003642.00	90.56	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	13740	1	⋯
cap91	129765.69	595022.69	78.19	3600.02	−∞	+∞	100.0	+∞	74649.44	766304.80	90.26	3600.10	−∞	+∞	100.00	3600.01	+∞	528113.26	563566.64	**6.29**	3601.13	13798	2	⋯
cap92	152555.76	712351.69	**78.58**	3600.01	−∞	+∞	100.0	+∞	80750.20	851293.90	90.51	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	13740	1	⋯
cap93	177234.12	747591.33	**76.29**	3600.01	−∞	+∞	100.0	+∞	86280.04	865126.60	90.03	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	13740	1	⋯
cap94	211626.37	778283.05	**72.81**	3600.02	−∞	+∞	100.0	+∞	94789.88	1003642.00	90.56	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	13740	1	⋯
#S			0				0				0				0					0				⋯
SGM			77.19	14400.00			96.51	14400.00			91.12	14400.00			100.00	14400.00	+∞			**52.78**	14400.00	18571761	1	⋯

Table 39 Detailed results for problem NLUFLP-A-orlib, cost functions f_4

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
cap101	66992.53	198914.51	66.32	3600.02	102285.09	299294.15	**65.82**	3603.03	−∞	1498975.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	18736	1	⋯
cap102	91250.61	217871.74	58.12	3600.02	129273.27	229093.85	**43.57**	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.20	18736	1	⋯
cap103	110111.71	239225.60	53.97	3600.02	150558.76	269611.89	**44.16**	3603.05	15179.18	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	18736	1	⋯
cap104	142045.21	257462.95	**44.83**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	18736	1	⋯
cap111	35425.06	229697.74	**84.58**	3600.02	−∞	+∞	100.0	+∞	6505.36	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.13	37486	1	⋯
cap112	50652.91	217772.11	**76.74**	3600.02	−∞	+∞	100.0	+∞	10842.27	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.20	37486	1	⋯
cap113	67269.97	240104.25	**71.98**	3600.02	−∞	+∞	100.0	+∞	15179.18	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.23	37486	1	⋯
cap114	100238.25	254758.47	**60.65**	3600.03	−∞	+∞	100.0	+∞	21684.54	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	37486	1	⋯
cap121	35425.06	229313.24	**84.55**	3600.02	−∞	+∞	100.0	+∞	6505.36	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	37486	1	⋯
cap122	51123.60	217772.11	**76.52**	3600.01	−∞	+∞	100.0	+∞	10842.27	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	37486	1	⋯
cap123	67865.99	240104.25	**71.73**	3600.01	−∞	+∞	100.0	+∞	15179.18	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	37486	1	⋯
cap124	100418.32	254758.47	**60.58**	3600.02	−∞	+∞	100.0	+∞	21684.54	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	37486	1	⋯
cap131	35425.06	229313.24	**84.55**	3600.02	−∞	+∞	100.0	+∞	6505.36	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	37486	1	⋯
cap132	51387.42	217718.14	**76.40**	3600.01	−∞	+∞	100.0	+∞	10842.27	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	37486	1	⋯
cap133	67998.08	240104.25	**71.68**	3600.02	−∞	+∞	100.0	+∞	15179.18	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	37486	1	⋯
cap134	101452.62	254757.67	**60.18**	3600.02	−∞	+∞	100.0	+∞	21684.54	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	37486	1	⋯
cap41	100424.84	206295.26	**51.32**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	11986	1	⋯
cap42	118101.86	231694.57	**49.03**	3600.02	−∞	+∞	100.0	+∞	−∞	1426395.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.04	11986	1	⋯
cap43	144037.62	243559.66	**40.86**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	11986	1	⋯
cap44	195220.57	259736.46	**24.84**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	11986	1	⋯
cap51	145462.55	243260.55	**40.20**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	11986	1	⋯
cap61	101103.95	206295.26	**50.99**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.18	11986	1	⋯
cap62	117874.25	231694.57	**49.13**	3600.01	−∞	+∞	100.0	+∞	−∞	1426395.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	11986	1	⋯
cap63	142825.53	243559.66	**41.36**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.23	11986	1	⋯
cap64	194370.80	259736.46	**25.17**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	11986	1	⋯
cap71	100872.24	206295.26	**51.10**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	11986	1	⋯
cap72	119534.86	231694.57	**48.41**	3600.01	−∞	+∞	100.0	+∞	−∞	1426395.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	11986	1	⋯
cap73	143196.09	243559.66	**41.21**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	11986	1	⋯
cap74	193828.34	259736.46	**25.37**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	11986	1	⋯
cap81	66946.07	198914.51	**66.34**	3600.01	−∞	+∞	100.0	+∞	−∞	1498975.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	18736	1	⋯
cap82	90037.40	217871.74	**58.67**	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	18736	1	⋯
cap83	109683.25	239225.60	**54.15**	3600.02	−∞	+∞	100.0	+∞	15179.18	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	18736	1	⋯
cap84	141200.02	257462.95	**45.16**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	18736	1	⋯
cap91	67379.84	198914.51	**66.13**	3600.02	−∞	+∞	100.0	+∞	−∞	1498975.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	18736	1	⋯
cap92	91087.31	217871.74	**58.19**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	18736	1	⋯
cap93	112136.63	239225.60	**53.13**	3600.01	−∞	+∞	100.0	+∞	15179.18	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	18736	1	⋯
cap94	141778.32	257462.95	**44.93**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	18736	1	⋯
#S			0				0				0				0					0				⋯
SGM			**54.05**	14400.00			94.59	14400.00			100.00	14400.00			100.00	14400.00	+∞			100.00	14400.00	20055	1	⋯

Table 40 **Detailed results for problem NLUFLP-A-orlib, cost functions f_5**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	-702701.54	816667.62	186.04	3600.02	-1032684.94	1181176.40	187.43	3602.64	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	7496	1	···
cap102	-618222.94	894588.47	169.11	3600.02	-958206.74	1248191.12	176.77	3602.71	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	7496	1	···
cap103	-546501.86	912340.73	159.90	3600.02	-894716.78	1195364.67	174.85	3602.68	-1824647.00	63315840.00	102.88	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	7496	1	···
cap104	-452226.27	1037327.20	143.60	3600.02	-810069.81	1115148.48	172.64	3602.72	$-\infty$	29260510.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	7496	1	···
cap111	-1092400.73	838887.67	176.79	3600.03	-1325141.77	1248191.11	194.19	3603.88	-2574760.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
cap112	-1001217.94	885999.61	188.49	3600.03	-1242758.09	1248191.13	199.56	3603.88	-2565150.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	14996	1	···
cap113	-902690.12	909350.11	199.27	3600.02	-1175068.47	1248191.13	194.14	3603.45	-2555667.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
cap114	-782976.29	970500.21	180.68	3600.02	-1082229.01	1200242.50	190.17	3603.44	-2541788.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	14996	1	···
cap121	-1090178.54	838887.67	176.95	3600.02	-1325141.77	1248191.11	194.19	3604.66	-2574760.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
cap122	-990486.51	885999.61	189.45	3600.02	-1242758.09	1248191.13	199.56	3605.35	-2565150.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
cap123	-895376.15	909350.11	198.46	3600.01	-1175068.47	1248191.13	194.14	3605.77	-2555667.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
cap124	-790559.43	970500.21	181.46	3600.02	-1082228.53	1200242.50	190.17	3604.69	-2541788.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	14996	1	···
cap131	-1090178.54	838887.67	176.95	3600.03	-1324739.57	1248191.11	194.22	3602.81	-2574760.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
cap132	-990486.51	885999.61	189.45	3600.02	-1242756.25	1248191.13	199.56	3602.78	-2565150.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
cap133	-890137.36	909350.11	197.89	3600.03	-1174970.18	1248191.13	194.13	3602.78	-2555667.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
cap134	-778545.38	970500.21	180.22	3600.01	-1082227.53	1200242.50	190.17	3602.77	-2541788.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
cap41	-480494.58	960119.46	150.05	3600.01	-757378.24	1248191.13	160.68	3602.09	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	4796	1	···
cap42	-394058.60	1000135.75	139.40	3600.02	-697207.98	1248191.13	155.86	3602.40	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	4796	1	···
cap43	-329351.07	1016662.99	132.40	3600.02	-641991.66	1248191.11	151.43	3602.08	-856386.80	15577310.00	105.50	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	4796	1	···
cap44	-277846.06	1038214.64	126.76	3600.01	-562679.45	1114694.62	150.48	3602.80	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	4796	1	···
cap51	-328966.14	1016662.99	132.36	3600.02	-641991.66	1248191.11	151.43	3602.79	-856386.80	15577310.00	105.50	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	4796	1	···
cap61	-480894.70	960441.95	150.07	3600.02	-757378.55	1248191.13	160.68	3602.80	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	4796	1	···
cap62	-402130.51	1000135.76	140.21	3600.01	-697208.18	1248191.13	155.86	3602.81	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	4796	1	···
cap63	-329073.92	1016662.99	132.37	3600.01	-641991.66	1248191.11	151.43	3602.76	-856386.80	15577310.00	105.50	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	4796	1	···
cap64	-280889.43	1038214.64	127.06	3600.02	-562679.45	1114694.62	150.48	3602.76	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	4796	1	···
cap71	-476823.82	960107.00	149.66	3600.01	-757378.55	1248191.13	160.68	3602.81	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	4796	1	···
cap72	-395236.87	1000135.75	139.52	3600.01	-697207.98	1248191.13	155.86	3602.41	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	4796	1	···
cap73	-322894.20	1016269.68	131.77	3600.01	-641991.90	1248191.11	151.43	3602.79	-856386.80	15577310.00	105.50	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	4796	1	···
cap74	-283229.07	1038214.64	127.28	3600.03	-562679.45	1114694.62	150.48	3602.81	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	4796	1	···
cap81	-708296.51	816667.62	186.73	3600.03	-1032684.94	1181176.40	187.43	3602.72	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	7496	1	···
cap82	-616714.64	894478.91	168.95	3600.02	-958206.74	1248191.12	176.77	3602.71	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	7496	1	···
cap83	-549693.26	912406.40	160.25	3600.02	-894716.78	1195364.67	174.85	3602.70	-1824647.00	63315840.00	102.88	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	7496	1	···
cap84	-449276.77	1037327.20	143.31	3600.01	-810069.81	1115148.48	172.64	3602.70	$-\infty$	29260510.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	7496	1	···
cap91	-704459.59	816667.62	186.26	3600.02	-1032684.94	1181176.40	187.43	3603.01	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	7496	1	···
cap92	-617444.85	894478.91	169.03	3600.01	-958206.74	1248191.12	176.77	3602.68	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	7496	1	···
cap93	-545974.72	912340.73	159.84	3600.02	-894716.78	1195364.67	174.85	3602.73	-1824647.00	63315840.00	102.88	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.19	7496	1	···
cap94	-448346.63	1037327.20	143.22	3600.01	-810069.81	1115148.48	172.64	3602.71	$-\infty$	29260510.00	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	7496	1	···
#S			0				0				0				0					0				···
SGM			100.00	14400.00			100.00	14400.00			100.00	14400.00			100.00	14400.00	$+\infty$			100.00	14400.00	1970191	1	···

Table 41 **Detailed results for problem NLUFLP-A-orlib, cost functions f_6**

	GUROBI				**SCIP**				**COUENNE**				**NAIVE**					**CN24**						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	796648.44	796648.44	**0.00**	**0.06**	796648.44	796648.44	**0.00**	3.79	796648.40	796648.40	**0.00**	77.85	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	796648.20	796648.46	**0.00**	160.97	9327	20	···
cap102	854704.20	854704.20	**0.00**	**0.06**	854704.20	854704.20	**0.00**	3.83	854704.20	854704.20	**0.00**	55.27	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	854703.88	854704.41	**0.00**	295.38	8567	22	···
cap103	893782.11	893782.11	**0.00**	**0.06**	893782.11	893782.11	**0.00**	3.87	893782.10	893782.10	**0.00**	69.70	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	893781.85	893782.11	**0.00**	250.03	8050	25	···
cap104	928941.75	928941.75	**0.00**	**0.07**	928941.75	928941.75	**0.00**	3.73	928941.80	928941.80	**0.00**	82.76	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	928941.41	928941.77	**0.00**	241.50	7209	22	···
cap111	793439.56	793439.56	**0.00**	**0.18**	793439.56	793439.56	**0.00**	6.46	793439.60	793439.60	**0.00**	365.12	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	793439.25	793439.73	**0.00**	443.04	15583	19	···
cap112	851495.32	851495.33	**0.00**	**0.16**	851495.32	851495.32	**0.00**	6.34	851495.30	851495.30	**0.00**	261.12	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	851495.02	851495.37	**0.00**	913.61	14866	24	···
cap113	893076.71	893076.71	**0.00**	**0.16**	893076.71	893076.71	**0.00**	14.51	893076.70	893076.70	**0.00**	105.38	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	893076.41	893077.11	**0.00**	944.00	14301	25	···
cap114	928941.75	928941.75	**0.00**	**0.19**	928941.75	928941.75	**0.00**	15.24	928941.80	928941.80	**0.00**	1.19	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	928941.20	928942.06	**0.00**	916.68	13488	25	···
cap121	793439.56	793439.56	**0.00**	**0.15**	793439.56	793439.56	**0.00**	6.37	793439.60	793439.60	**0.00**	362.52	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	793439.25	793439.73	**0.00**	476.22	15583	19	···
cap122	851495.32	851495.33	**0.00**	**0.15**	851495.32	851495.32	**0.00**	6.25	851495.30	851495.30	**0.00**	263.54	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	851495.02	851495.37	**0.00**	698.02	14866	24	···
cap123	893076.71	893076.71	**0.00**	**0.16**	893076.71	893076.71	**0.00**	5.54	893076.70	893076.70	**0.00**	104.06	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	893076.41	893077.11	**0.00**	951.89	14301	25	···
cap124	928941.75	928941.75	**0.00**	**0.15**	928941.75	928941.75	**0.00**	6.59	928941.80	928941.80	**0.00**	1.17	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	928941.20	928942.06	**0.00**	1057.93	13488	25	···
cap131	793439.56	793439.56	**0.00**	**0.15**	793439.56	793439.56	**0.00**	14.95	793439.60	793439.60	**0.00**	362.24	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	793439.25	793439.73	**0.00**	460.17	15583	19	···
cap132	851495.32	851495.33	**0.00**	**0.15**	851495.32	851495.32	**0.00**	21.72	851495.30	851495.30	**0.00**	265.36	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	851495.02	851495.37	**0.00**	742.28	14866	24	···
cap133	893076.71	893076.71	**0.00**	**0.16**	893076.71	893076.71	**0.00**	6.00	893076.70	893076.70	**0.00**	105.59	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	893076.41	893077.11	**0.00**	850.51	14301	25	···
cap134	928941.75	928941.75	**0.00**	**0.16**	928941.75	928941.75	**0.00**	6.38	928941.80	928941.80	**0.00**	1.18	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	928941.20	928942.06	**0.00**	736.69	13488	25	···
cap41	932615.75	932615.75	**0.00**	**0.05**	932615.75	932615.75	**0.00**	3.05	932615.70	932615.70	**0.00**	28.81	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	932615.36	932616.15	**0.00**	70.41	6237	17	···
cap42	977799.40	977799.40	**0.00**	**0.04**	977799.40	977799.40	**0.00**	3.28	977799.40	977799.40	**0.00**	31.33	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	977798.96	977799.86	**0.00**	129.14	5870	19	···
cap43	1010641.45	1010641.45	**0.00**	**0.04**	1010641.45	1010641.45	**0.00**	10.89	1010641.00	1010641.00	**0.00**	25.39	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1010641.04	1010641.71	**0.00**	147.72	5133	22	···
cap44	1034976.97	1034976.97	**0.00**	**0.05**	1034976.97	1034976.97	**0.00**	16.10	1034977.00	1034977.00	**0.00**	32.40	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1034976.48	1034977.17	**0.00**	145.27	4920	20	···
cap51	1010641.45	1010641.45	**0.00**	**0.04**	1010641.45	1010641.45	**0.00**	19.52	1010641.00	1010641.00	**0.00**	24.69	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1010641.04	1010641.71	**0.00**	138.73	5133	22	···
cap61	932615.75	932615.75	**0.00**	**0.04**	932615.75	932615.75	**0.00**	3.35	932615.70	932615.70	**0.00**	27.27	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	932615.36	932616.15	**0.00**	58.72	6237	17	···
cap62	977799.40	977799.40	**0.00**	**0.04**	977799.40	977799.40	**0.00**	3.53	977799.40	977799.40	**0.00**	31.42	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	977798.96	977799.86	**0.00**	132.97	5870	19	···
cap63	1010641.45	1010641.45	**0.00**	**0.04**	1010641.45	1010641.45	**0.00**	3.37	1010641.00	1010641.00	**0.00**	26.03	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1010641.04	1010641.71	**0.00**	182.45	5133	22	···
cap64	1034976.97	1034976.97	**0.00**	**0.04**	1034976.97	1034976.97	**0.00**	3.48	1034977.00	1034977.00	**0.00**	33.00	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1034976.48	1034977.17	**0.00**	139.14	4920	20	···
cap71	932615.75	932615.75	**0.00**	**0.04**	932615.75	932615.75	**0.00**	3.49	932615.70	932615.70	**0.00**	28.20	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	932615.36	932616.15	**0.00**	64.04	6237	17	···
cap72	977799.40	977799.40	**0.00**	**0.04**	977799.40	977799.40	**0.00**	3.02	977799.40	977799.40	**0.00**	31.57	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	977798.96	977799.86	**0.00**	115.04	5870	19	···
cap73	1010641.45	1010641.45	**0.00**	**0.04**	1010641.45	1010641.45	**0.00**	3.43	1010641.00	1010641.00	**0.00**	25.55	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1010641.04	1010641.71	**0.00**	161.28	5133	22	···
cap74	1034976.97	1034976.97	**0.00**	**0.04**	1034976.97	1034976.97	**0.00**	21.85	1034977.00	1034977.00	**0.00**	33.18	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1034976.48	1034977.17	**0.00**	135.88	4920	20	···
cap81	796648.44	796648.44	**0.00**	**0.06**	796648.44	796648.44	**0.00**	38.67	796648.40	796648.40	**0.00**	75.76	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	796648.20	796648.46	**0.00**	144.59	9327	20	···
cap82	854704.20	854704.20	**0.00**	**0.06**	854704.20	854704.20	**0.00**	21.96	854704.20	854704.20	**0.00**	55.91	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	854703.88	854704.41	**0.00**	276.98	8567	22	···
cap83	893782.11	893782.11	**0.00**	**0.06**	893782.11	893782.11	**0.00**	4.17	893782.10	893782.10	**0.00**	68.42	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	893781.85	893782.11	**0.00**	282.57	8050	25	···
cap84	928941.75	928941.75	**0.00**	**0.06**	928941.75	928941.75	**0.00**	4.12	928941.80	928941.80	**0.00**	81.08	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	928941.41	928941.77	**0.00**	220.80	7209	22	···
cap91	796648.44	796648.44	**0.00**	**0.06**	796648.44	796648.44	**0.00**	3.85	796648.40	796648.40	**0.00**	76.84	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	796648.20	796648.46	**0.00**	160.55	9327	20	···
cap92	854704.20	854704.20	**0.00**	**0.07**	854704.20	854704.20	**0.00**	3.73	854704.20	854704.20	**0.00**	55.35	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	854703.88	854704.41	**0.00**	264.14	8567	22	···
cap93	893782.11	893782.11	**0.00**	**0.07**	893782.11	893782.11	**0.00**	4.05	893782.10	893782.10	**0.00**	70.74	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	893781.85	893782.11	**0.00**	263.14	8050	25	···
cap94	928941.75	928941.75	**0.00**	**0.06**	928941.75	928941.75	**0.00**	3.84	928941.80	928941.80	**0.00**	84.87	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	928941.41	928941.77	**0.00**	200.56	7209	22	···
#S			**37**				**37**				**37**				0					**37**				···
SGM			**0.00**	**0.09**			**0.00**	7.32			**0.00**	58.47			100.00	14400.00	$+\infty$			**0.00**	266.41	8591	22	···

Table 42 Detailed results for problem NLUFLP-A-orlib, cost functions f_7

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	198198.98	198198.74	**0.00**	**135.81**	198198.89	198198.94	**0.00**	358.32	93524.14	228724.80	59.11	3600.10	−∞	+∞	100.00	3600.01	+∞	195154.99	199552.54	2.20	3601.03	13334	10	···
cap102	249809.19	249809.19	**0.00**	**727.31**	249803.16	249809.19	0.00	3603.31	108281.70	291026.60	62.79	3600.10	−∞	+∞	100.00	3600.01	+∞	246922.40	251421.40	1.79	3601.02	13182	11	···
cap103	290945.90	290945.87	**0.00**	702.61	290945.88	290945.89	**0.00**	**571.71**	123064.20	339818.60	63.79	3600.10	−∞	+∞	100.00	3600.01	+∞	286628.00	293240.71	2.26	3601.03	12897	9	···
cap104	**341671.75**	**341672.49**	**0.00**	**1580.24**	**341163.10**	**341163.12**	**0.00**	**61.87**	**150049.70**	**393689.30**	**61.89**	**3600.10**	**−∞**	**+∞**	**100.00**	**3600.01**	**+∞**	**337543.40**	**343262.87**	**1.67**	**3601.03**	**13007**	**12**	···
cap111	196699.35	196697.95	**0.00**	2133.11	196692.04	196692.07	**0.00**	**1139.25**	39873.82	270927.10	85.28	3600.10	−∞	+∞	100.00	3600.01	+∞	190739.93	198654.16	3.98	3601.04	25156	4	···
cap112	248406.39	250100.57	0.68	3600.02	248407.24	248407.26	**0.00**	**895.06**	44539.27	343857.60	87.05	3600.10	−∞	+∞	100.00	3600.01	+∞	243787.10	250996.00	2.87	3601.04	25289	8	···
cap113	289640.55	293207.02	1.22	3600.03	289641.47	289641.49	**0.00**	**1063.45**	49915.76	447031.20	88.83	3600.10	−∞	+∞	100.00	3600.01	+∞	283835.68	292656.68	3.01	3601.07	25264	7	···
cap114	340489.12	347757.88	2.09	3600.02	340489.87	340489.89	**0.00**	**828.44**	59717.97	493779.90	87.91	3600.10	−∞	+∞	100.00	3600.01	+∞	332873.39	343169.64	3.00	3601.04	25181	7	···
cap121	196699.35	196697.95	**0.00**	1902.46	196692.04	196692.07	**0.00**	**909.62**	42629.48	270927.10	84.27	3600.10	−∞	+∞	100.00	3600.01	+∞	191282.09	198654.16	3.71	3601.06	25222	5	···
cap122	248407.26	248407.22	**0.00**	3458.68	248407.24	248407.26	**0.00**	**931.89**	44539.27	343857.60	87.05	3600.10	−∞	+∞	100.00	3600.01	+∞	243787.10	250996.00	2.87	3601.04	25289	8	···
cap123	289640.55	291923.50	0.78	3600.01	289641.47	289641.49	**0.00**	**1305.49**	49915.76	447031.20	88.83	3600.10	−∞	+∞	100.00	3600.01	+∞	284500.38	292656.68	2.79	3601.03	25281	8	···
cap124	340489.12	347757.88	2.09	3600.02	340489.87	340489.89	**0.00**	**745.83**	59717.97	493779.90	87.91	3600.10	−∞	+∞	100.00	3600.01	+∞	333287.92	343169.64	2.88	3601.03	25195	9	···
cap131	196699.35	196697.95	**0.00**	1746.66	196692.04	196692.07	**0.00**	**1058.24**	42629.48	270927.10	84.27	3600.10	−∞	+∞	100.00	3600.01	+∞	192328.93	198654.16	3.18	3601.03	25290	6	···
cap132	248407.26	248407.22	**0.00**	3066.45	248407.24	248407.26	**0.00**	**1020.24**	44539.27	343857.60	87.05	3600.10	−∞	+∞	100.00	3600.01	+∞	243292.17	251322.64	3.20	3601.05	25219	7	···
cap133	289640.55	291923.50	0.78	3600.02	289641.47	289641.49	**0.00**	**1314.53**	49915.76	447031.20	88.83	3600.10	−∞	+∞	100.00	3600.01	+∞	283835.68	292656.68	3.01	3601.06	25264	7	···
cap134	340489.12	347757.88	2.09	3600.02	340489.87	340489.89	**0.00**	**914.41**	59717.97	493779.90	87.91	3600.10	−∞	+∞	100.00	3600.01	+∞	332873.39	343169.64	3.00	3601.07	25181	7	···
cap41	209069.54	209147.51	0.04	790.88	209147.47	209147.51	**0.00**	**171.27**	136278.90	223621.30	39.06	3600.10	−∞	+∞	100.00	3600.01	+∞	207210.16	209950.99	1.31	3601.03	9347	18	···
cap42	258383.76	258383.82	**0.00**	283.09	258383.98	258384.01	**0.00**	**31.87**	174930.90	268970.50	34.96	3600.10	−∞	+∞	100.00	3600.01	+∞	256838.49	259035.75	0.85	3601.03	9390	20	···
cap43	297468.89	298499.20	0.35	340.07	297469.72	297469.74	**0.00**	**41.71**	189309.00	313279.60	39.57	3600.10	−∞	+∞	100.00	3600.01	+∞	295397.73	298410.26	1.01	3601.03	8950	16	···
cap44	346436.47	346437.30	**0.00**	286.06	346437.28	346437.30	**0.00**	**30.77**	186605.70	379405.80	50.82	3600.10	−∞	+∞	100.00	3600.01	+∞	345778.83	346834.19	0.30	3601.02	9687	21	···
cap51	297468.89	298499.20	0.35	333.04	297469.72	297469.74	**0.00**	**31.41**	189371.30	313279.60	39.55	3600.10	−∞	+∞	100.00	3600.01	+∞	295901.69	298410.26	0.84	3601.02	8969	17	···
cap61	209069.54	209147.51	0.04	804.78	209147.47	209147.51	**0.00**	**108.81**	136363.60	223621.30	39.02	3600.10	−∞	+∞	100.00	3600.01	+∞	207210.16	209950.99	1.31	3601.03	9347	18	···
cap62	258383.76	258383.82	**0.00**	280.00	258383.98	258384.01	**0.00**	**33.80**	174849.50	268970.50	34.99	3600.10	−∞	+∞	100.00	3600.01	+∞	256838.49	259035.75	0.85	3601.03	9390	20	···
cap63	297468.89	298499.20	0.35	329.23	297469.72	297469.74	**0.00**	**29.74**	189278.30	313279.60	39.58	3600.10	−∞	+∞	100.00	3600.01	+∞	295397.73	298410.26	1.01	3601.03	8950	16	···
cap64	346436.47	346437.30	**0.00**	292.19	346437.28	346437.30	**0.00**	**24.14**	186498.20	379405.80	50.84	3600.10	−∞	+∞	100.00	3600.01	+∞	345193.77	347138.08	0.56	3601.03	9143	18	···
cap71	209069.54	209147.51	0.04	802.18	209147.47	209147.51	**0.00**	**136.40**	136249.40	223621.30	39.07	3600.10	−∞	+∞	100.00	3600.01	+∞	207210.16	209950.99	1.31	3601.03	9347	18	···
cap72	258383.76	258383.82	**0.00**	295.97	258383.98	258384.01	**0.00**	**34.57**	174770.40	268970.50	35.02	3600.10	−∞	+∞	100.00	3600.01	+∞	256838.49	259035.75	0.85	3601.04	9390	20	···
cap73	297468.89	298499.20	0.35	329.97	297469.72	297469.74	**0.00**	**30.43**	189119.30	313279.60	39.63	3600.10	−∞	+∞	100.00	3600.01	+∞	295397.73	298410.26	1.01	3601.02	8950	16	···
cap74	346436.47	346437.30	**0.00**	296.96	346437.28	346437.30	**0.00**	**50.11**	186557.10	379405.80	50.83	3600.10	−∞	+∞	100.00	3600.01	+∞	345677.76	346951.14	0.37	3601.03	9440	20	···
cap81	198198.98	198198.74	**0.00**	**133.64**	198198.89	198198.94	**0.00**	384.44	93524.14	228724.80	59.11	3600.10	−∞	+∞	100.00	3600.01	+∞	195154.99	199552.54	2.20	3601.03	13334	10	···
cap82	249809.19	249809.19	**0.00**	**738.58**	−∞	+∞	100.0	+∞	108279.80	291026.60	62.79	3600.10	−∞	+∞	100.00	3600.01	+∞	246922.40	251421.40	1.79	3601.03	13182	11	···
cap83	290945.90	290945.87	**0.00**	697.62	290945.88	290945.89	**0.00**	**558.70**	123100.90	339818.60	63.77	3600.10	−∞	+∞	100.00	3600.01	+∞	286024.72	294290.05	2.81	3601.03	12837	7	···
cap84	**341671.75**	**341672.49**	**0.00**	**1734.91**	**341163.10**	**341163.12**	**0.00**	**55.51**	**150132.50**	**393689.30**	**61.87**	**3600.10**	**−∞**	**+∞**	**100.00**	**3600.01**	**+∞**	**337543.40**	**343262.87**	**1.67**	**3601.05**	**13007**	**12**	···
cap91	198198.98	198198.74	**0.00**	**118.11**	198198.89	198198.94	**0.00**	328.58	93524.14	228724.80	59.11	3600.10	−∞	+∞	100.00	3600.01	+∞	195154.99	199552.54	2.20	3601.05	13334	10	···
cap92	249809.19	249809.19	**0.00**	**723.29**	−∞	+∞	100.0	+∞	108281.70	291026.60	62.79	3600.10	−∞	+∞	100.00	3600.01	+∞	246922.40	251421.40	1.79	3601.04	13182	11	···
cap93	290945.90	290945.87	**0.00**	694.92	290945.88	290945.89	**0.00**	**620.45**	123064.20	339818.60	63.79	3600.10	−∞	+∞	100.00	3600.01	+∞	286276.77	293240.71	2.37	3601.03	12887	8	···
cap94	**341671.75**	**341672.49**	**0.00**	**1466.47**	**341163.10**	**341163.12**	**0.00**	**57.11**	**149984.00**	**393689.30**	**61.90**	**3600.10**	**−∞**	**+∞**	**100.00**	**3600.01**	**+∞**	**335900.34**	**343262.87**	**2.14**	**3601.02**	**12987**	**11**	···
#S			23				**31**				0				0					0				···
SGM			**0.23**	1082.20			0.31	**361.87**			59.49	14400.00			100.00	14400.00	+∞			1.83	14400.00	14464	11	···

Table 43 **Detailed results for problem NLUFLP-A-orlib, cost functions f_8**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
cap101	796648.44	796648.44	**0.00**	**0.20**	796648.44	796648.44	**0.00**	11.26	796649.00	796649.00	**0.00**	247.34	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	796647.65	796648.44	**0.00**	262.31	13002	17	⋯
cap102	854704.20	854704.20	**0.00**	**0.20**	854704.20	854704.20	**0.00**	7.76	854704.80	854704.80	**0.00**	202.37	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	854704.01	854704.20	**0.00**	292.50	12228	19	⋯
cap103	893782.11	893782.11	**0.00**	**0.18**	893782.11	893782.11	**0.00**	9.09	893782.70	893782.70	**0.00**	232.65	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	893781.81	893782.11	**0.00**	617.95	11688	22	⋯
cap104	928941.75	928941.75	**0.00**	**0.21**	928941.75	928941.75	**0.00**	8.47	928942.30	928942.30	**0.00**	283.98	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	928941.52	928941.75	**0.00**	525.74	10881	18	⋯
cap111	793439.56	793439.56	**0.00**	**0.85**	793439.56	793439.56	**0.00**	22.82	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	793439.08	793439.56	**0.00**	1002.72	20290	28	⋯
cap112	851495.32	851495.33	**0.00**	**0.97**	851495.32	851495.32	**0.00**	24.82	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	851494.87	851495.32	**0.00**	1385.62	19077	31	⋯
cap113	893076.71	893076.71	**0.00**	**0.90**	893076.71	893076.71	**0.00**	32.29	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	893076.48	893076.71	**0.00**	2244.13	21711	23	⋯
cap114	928941.75	928941.75	**0.00**	**0.89**	928941.75	928941.75	**0.00**	10.11	928943.00	928943.00	**0.00**	1711.69	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	928941.70	928941.75	**0.00**	2470.09	20955	21	⋯
cap121	793439.56	793439.56	**0.00**	**0.80**	793439.56	793439.56	**0.00**	22.29	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	793439.08	793439.56	**0.00**	1010.62	20290	28	⋯
cap122	851495.32	851495.33	**0.00**	**0.90**	851495.32	851495.32	**0.00**	25.07	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	851494.87	851495.32	**0.00**	1394.09	19077	31	⋯
cap123	893076.71	893076.71	**0.00**	**0.82**	893076.71	893076.71	**0.00**	19.88	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	893076.48	893076.71	**0.00**	2195.08	21711	23	⋯
cap124	928941.75	928941.75	**0.00**	**0.85**	928941.75	928941.75	**0.00**	15.82	928943.00	928943.00	**0.00**	1761.96	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	928941.70	928941.75	**0.00**	2228.60	20955	21	⋯
cap131	793439.56	793439.56	**0.00**	**0.76**	793439.56	793439.56	**0.00**	22.13	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	793439.08	793439.56	**0.00**	1007.85	20290	28	⋯
cap132	851495.32	851495.33	**0.00**	**0.88**	851495.32	851495.32	**0.00**	37.38	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	851494.87	851495.32	**0.00**	1388.13	19077	31	⋯
cap133	893076.71	893076.71	**0.00**	**0.85**	893076.71	893076.71	**0.00**	27.01	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	893076.48	893076.71	**0.00**	1794.73	21711	23	⋯
cap134	928941.75	928941.75	**0.00**	**0.89**	928941.75	928941.75	**0.00**	9.80	928943.00	928943.00	**0.00**	1751.99	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	928941.70	928941.75	**0.00**	1518.38	20955	21	⋯
cap41	932615.75	932615.75	**0.00**	**0.08**	932615.75	932615.75	**0.00**	15.66	932616.10	932616.10	**0.00**	286.26	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	932615.36	932615.75	**0.00**	129.60	8620	18	⋯
cap42	977799.40	977799.40	**0.00**	**0.09**	977799.40	977799.40	**0.00**	6.45	977799.70	977799.70	**0.00**	108.08	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	977799.09	977799.40	**0.00**	201.37	8234	20	⋯
cap43	1010641.45	1010641.45	**0.00**	**0.09**	1010641.45	1010641.45	**0.00**	8.76	1010642.00	1010642.00	**0.00**	253.51	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1010641.15	1010641.45	**0.00**	219.92	7484	23	⋯
cap44	1034976.97	1034976.97	**0.00**	**0.11**	1034976.97	1034977.15	**0.00**	10.78	1034977.00	1034977.00	**0.00**	284.84	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1034976.57	1034976.98	**0.00**	245.85	7266	20	⋯
cap51	1010641.45	1010641.45	**0.00**	**0.09**	1010641.45	1010641.45	**0.00**	12.05	1010642.00	1010642.00	**0.00**	221.05	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1010641.15	1010641.45	**0.00**	221.24	7484	23	⋯
cap61	932615.75	932615.75	**0.00**	**0.07**	932615.75	932615.75	**0.00**	8.62	932616.10	932616.10	**0.00**	275.00	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	932615.36	932615.75	**0.00**	116.80	8620	18	⋯
cap62	977799.40	977799.40	**0.00**	**0.08**	977799.40	977799.40	**0.00**	7.31	977799.70	977799.70	**0.00**	108.32	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	977799.09	977799.40	**0.00**	192.77	8234	20	⋯
cap63	1010641.45	1010641.45	**0.00**	**0.10**	1010641.45	1010641.45	**0.00**	9.07	1010642.00	1010642.00	**0.00**	230.02	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1010641.15	1010641.45	**0.00**	281.38	7484	23	⋯
cap64	1034976.97	1034976.97	**0.00**	**0.09**	1034976.97	1034977.15	**0.00**	7.57	1034977.00	1034977.00	**0.00**	289.51	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1034976.57	1034976.98	**0.00**	290.31	7266	20	⋯
cap71	932615.75	932615.75	**0.00**	**0.09**	932615.75	932615.75	**0.00**	40.52	932616.10	932616.10	**0.00**	285.99	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	932615.36	932615.75	**0.00**	137.94	8620	18	⋯
cap72	977799.40	977799.40	**0.00**	**0.08**	977799.40	977799.40	**0.00**	7.32	977799.70	977799.70	**0.00**	110.52	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	977799.09	977799.40	**0.00**	211.14	8234	20	⋯
cap73	1010641.45	1010641.45	**0.00**	**0.09**	1010641.45	1010641.45	**0.00**	9.44	1010642.00	1010642.00	**0.00**	232.44	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1010641.15	1010641.45	**0.00**	255.77	7484	23	⋯
cap74	1034976.97	1034976.97	**0.00**	**0.10**	1034976.97	1034977.15	**0.00**	14.74	1034977.00	1034977.00	**0.00**	290.28	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	1034976.57	1034976.98	**0.00**	245.04	7266	20	⋯
cap81	796648.44	796648.44	**0.00**	**0.17**	796648.44	796648.44	**0.00**	28.40	796649.00	796649.00	**0.00**	244.52	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	796647.65	796648.44	**0.00**	273.73	13002	17	⋯
cap82	854704.20	854704.20	**0.00**	**0.18**	854704.20	854704.20	**0.00**	10.13	854704.80	854704.80	**0.00**	204.87	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	854704.01	854704.20	**0.00**	444.68	12228	19	⋯
cap83	893782.11	893782.11	**0.00**	**0.18**	893782.11	893782.11	**0.00**	7.23	893782.70	893782.70	**0.00**	235.06	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	893781.81	893782.11	**0.00**	548.46	11688	22	⋯
cap84	928941.75	928941.75	**0.00**	**0.19**	928941.75	928941.75	**0.00**	8.40	928942.30	928942.30	**0.00**	277.08	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	928941.52	928941.75	**0.00**	439.68	10881	18	⋯
cap91	796648.44	796648.44	**0.00**	**0.18**	796648.44	796648.44	**0.00**	10.66	796649.00	796649.00	**0.00**	244.16	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	796647.65	796648.44	**0.00**	345.63	13002	17	⋯
cap92	854704.20	854704.20	**0.00**	**0.20**	854704.20	854704.20	**0.00**	7.21	854704.80	854704.80	**0.00**	202.90	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	854704.01	854704.20	**0.00**	429.35	12228	19	⋯
cap93	893782.11	893782.11	**0.00**	**0.17**	893782.11	893782.11	**0.00**	8.16	893782.70	893782.70	**0.00**	240.39	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	893781.81	893782.11	**0.00**	457.41	11688	22	⋯
cap94	928941.75	928941.75	**0.00**	**0.21**	928941.75	928941.75	**0.00**	7.94	928942.30	928942.30	**0.00**	281.95	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	928941.52	928941.75	**0.00**	499.15	10881	18	⋯
#S			**37**				**37**				28				0					**37**				⋯
SGM			**0.00**	**0.37**			**0.00**	13.56			2.07	747.10			100.00	14400.00	$+\infty$			**0.00**	499.56	12272	21	⋯

Table 44 Detailed results for problem NLUFLP-A-orlib, cost functions f_9

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	13432.27	149615.09	91.02	3600.02	10729.30	865050.41	98.76	3603.59	8420.08	202313.20	95.84	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	117871.56	118653.80	**0.66**	3601.03	12828	23	···
cap102	20542.49	181878.21	88.71	3600.02	15895.63	323963.27	95.09	3603.06	7854.80	286055.90	97.25	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	149126.12	149509.06	**0.26**	3601.02	12834	20	···
cap103	25807.96	254911.08	89.88	3600.02	20950.78	946805.74	97.79	3603.49	9118.42	381734.70	97.61	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	176636.86	178706.04	**1.16**	3601.03	12028	15	···
cap104	39737.43	302719.78	86.87	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	10767.37	450317.70	97.61	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	214464.26	216206.04	**0.81**	3601.02	12080	16	···
cap111	9733.68	282978.43	96.56	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	5870.08	315259.50	98.14	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	116322.89	117927.78	**1.36**	3601.04	23105	12	···
cap112	14194.30	278215.74	94.90	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	7535.35	439924.50	98.29	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	147280.65	149697.19	**1.61**	3601.03	22883	10	···
cap113	19002.17	410117.89	95.37	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	8771.67	555759.10	98.42	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	174267.70	178201.70	**2.21**	3601.05	22810	10	···
cap114	25098.85	447765.34	94.39	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	8778.24	718160.40	98.78	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	211131.01	215907.48	**2.21**	3601.04	22635	6	···
cap121	9733.68	276674.68	96.48	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	5870.08	315259.50	98.14	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	116054.95	117927.78	**1.59**	3601.03	23051	9	···
cap122	14194.30	274381.69	94.83	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	7535.35	439924.50	98.29	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	146157.00	151120.85	**3.28**	3601.03	22633	4	···
cap123	19002.17	404126.18	95.30	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	8771.67	555759.10	98.42	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	173631.01	178407.48	**2.68**	3601.03	22635	6	···
cap124	25029.11	447765.34	94.41	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	8778.24	718160.40	98.78	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	211131.01	215907.48	**2.21**	3601.03	22635	6	···
cap131	9733.68	280763.37	96.53	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	5870.08	315259.50	98.14	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	114807.23	118059.94	**2.76**	3601.04	22737	5	···
cap132	14194.30	274381.69	94.83	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	7535.35	439924.50	98.29	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	146932.70	150518.05	**2.38**	3601.04	22803	7	···
cap133	19002.17	404126.18	95.30	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	8771.67	555759.10	98.42	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	175426.58	178176.98	**1.54**	3601.03	22995	13	···
cap134	25098.85	447765.34	94.39	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	8778.24	718160.40	98.78	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	211131.01	215907.48	**2.21**	3601.05	22635	6	···
cap41	19911.65	156757.49	87.30	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	10020.56	441204.80	97.73	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	120934.83	121751.56	**0.67**	3601.03	8548	17	···
cap42	24674.92	189675.03	86.99	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	7829.01	296604.40	97.36	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	151646.00	152253.87	**0.40**	3601.03	8606	23	···
cap43	37781.86	218845.06	82.74	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	9118.38	368265.90	97.52	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	178284.89	179653.59	**0.76**	3601.03	8046	15	···
cap44	49910.58	305943.78	83.69	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	10767.40	443454.00	97.57	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	215784.89	217153.59	**0.63**	3601.02	8045	15	···
cap51	37863.29	218845.06	82.70	3600.06	$-\infty$	$+\infty$	100.0	$+\infty$	9118.38	368265.90	97.52	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	178284.89	179653.59	**0.76**	3601.03	8045	15	···
cap61	19751.96	156757.49	87.40	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	10020.56	441204.80	97.73	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	120934.83	121751.56	**0.67**	3601.11	8547	17	···
cap62	24400.51	189675.03	87.14	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	7829.01	296604.40	97.36	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	151497.10	152214.73	**0.47**	3601.03	8638	22	···
cap63	37865.45	218845.06	82.70	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	9118.38	368265.90	97.52	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	178284.89	179653.59	**0.76**	3601.03	8049	15	···
cap64	50455.30	305943.78	83.51	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	10767.40	443454.00	97.57	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	215784.89	217153.59	**0.63**	3601.03	8045	15	···
cap71	20383.40	156757.49	87.00	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	10020.56	441204.80	97.73	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	120816.74	121751.56	**0.77**	3601.03	8542	16	···
cap72	25739.36	189675.03	86.43	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	7829.01	296604.40	97.36	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	151500.38	152220.83	**0.47**	3601.03	8641	22	···
cap73	37847.66	218845.06	82.71	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	9118.38	368265.90	97.52	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	178465.09	179626.67	**0.65**	3601.02	8151	16	···
cap74	49897.33	305943.78	83.69	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	10767.40	443454.00	97.57	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	215785.70	217153.68	**0.63**	3601.03	8045	15	···
cap81	12536.50	149615.09	91.62	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	8420.08	202313.20	95.84	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	118130.76	118623.82	**0.42**	3601.03	13061	26	···
cap82	20893.68	181878.21	88.51	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	7854.80	286055.90	97.25	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	149110.44	149603.39	**0.33**	3601.04	12593	19	···
cap83	25991.55	254911.08	89.80	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	9118.42	381734.70	97.61	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	176636.86	178706.04	**1.16**	3601.04	12028	15	···
cap84	39816.46	302610.78	86.84	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	10767.37	450317.70	97.61	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	214136.86	216206.04	**0.96**	3601.04	12028	15	···
cap91	14241.27	149615.09	90.48	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	8420.08	202313.20	95.84	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	117871.56	118653.80	**0.66**	3601.03	12828	23	···
cap92	20657.58	181878.21	88.64	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	7854.80	286055.90	97.25	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	149183.13	149517.03	**0.22**	3601.03	12751	21	···
cap93	26146.60	254911.08	89.74	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	9118.42	381734.70	97.61	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	176636.86	178706.04	**1.16**	3601.02	12028	15	···
cap94	40037.01	301371.27	86.72	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	10767.37	450317.70	97.61	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	213529.50	216409.71	**1.33**	3601.04	11870	13	···
#S			0				0				0				0					0				···
SGM			89.51	14400.00			99.77	14400.00			97.67	14400.00			100.00	14400.00	$+\infty$			**1.05**	14400.00	13122	14	···

Table 45 Detailed results for problem NLUFLP-A-orlib, cost functions f_{10}

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	13742.54	13742.54	**0.00**	**0.41**	13742.54	13742.54	**0.00**	5.83	13348.01	13765.56	3.03	3600.10	13742.53	13742.54	**0.00**	546.04	199000	13742.53	13742.54	**0.00**	157.62	1702	14	···
p10	12451.56	12451.56	**0.00**	**0.25**	12451.56	12451.56	**0.00**	5.08	12235.79	12451.56	1.73	3600.10	12451.55	12451.56	**0.00**	530.30	199000	12451.56	12451.56	**0.00**	173.17	1702	13	···
p11	14395.54	14395.54	**0.00**	**0.31**	14395.54	14395.55	**0.00**	5.01	13973.23	14425.54	3.14	3600.10	14395.53	14395.54	**0.00**	546.30	199000	14395.53	14395.54	**0.00**	175.79	1702	14	···
p12	15859.49	15859.49	**0.00**	**0.25**	15859.49	15859.49	**0.00**	5.48	15241.91	16297.47	6.48	3600.10	15859.48	15859.49	**0.00**	560.42	199000	15859.48	15859.49	**0.00**	173.04	1705	14	···
p13	12619.59	12619.59	**0.00**	**1.10**	12619.59	12619.59	**0.00**	5.01	11680.38	14414.70	18.97	3600.10	12619.59	12619.59	**0.00**	1217.80	398000	12619.59	12619.59	**0.00**	364.54	3205	15	···
p14	11363.61	11363.61	**0.00**	**0.75**	11363.61	11363.61	**0.00**	5.67	10468.16	12359.71	15.30	3600.10	11363.61	11363.61	**0.00**	1452.64	398000	11363.61	11363.61	**0.00**	352.05	3211	16	···
p15	13283.59	13283.59	**0.00**	**0.81**	13283.59	13283.59	**0.00**	14.23	12280.13	15567.70	21.12	3600.10	13283.59	13283.59	**0.00**	1300.59	398000	13283.59	13283.59	**0.00**	332.80	3200	13	···
p16	15069.56	15069.56	**0.00**	**1.31**	15069.56	15069.56	**0.00**	5.83	13934.45	25457.21	45.26	3600.10	15069.55	15069.56	**0.00**	1235.88	398000	15069.55	15069.56	**0.00**	364.22	3220	17	···
p17	12619.59	12619.59	**0.00**	**1.17**	12619.59	12619.59	**0.00**	4.69	11680.38	14414.70	18.97	3600.10	12619.59	12619.59	**0.00**	1208.03	398000	12619.59	12619.59	**0.00**	395.22	3205	15	···
p18	11363.61	11363.61	**0.00**	**0.78**	11363.61	11363.61	**0.00**	5.49	10468.16	12359.71	15.30	3600.10	11363.61	11363.61	**0.00**	1470.80	398000	11363.61	11363.61	**0.00**	393.46	3211	16	···
p19	13283.59	13283.59	**0.00**	**0.91**	13283.59	13283.59	**0.00**	13.35	12280.13	15567.70	21.12	3600.10	13283.59	13283.59	**0.00**	1329.23	398000	13283.59	13283.59	**0.00**	377.08	3200	13	···
p2	12451.56	12451.56	**0.00**	**0.20**	12451.56	12451.56	**0.00**	5.20	12227.16	12451.56	1.80	3600.10	12451.55	12451.56	**0.00**	525.60	199000	12451.56	12451.56	**0.00**	171.66	1702	13	···
p20	15069.56	15069.56	**0.00**	**1.20**	15069.56	15069.56	**0.00**	5.72	13934.45	25457.21	45.26	3600.10	15069.55	15069.56	**0.00**	1229.78	398000	15069.55	15069.56	**0.00**	367.47	3220	17	···
p21	12619.59	12619.59	**0.00**	**1.18**	12619.59	12619.59	**0.00**	4.93	11680.38	14414.70	18.97	3600.10	12619.59	12619.59	**0.00**	1209.36	398000	12619.59	12619.59	**0.00**	372.01	3205	15	···
p22	11363.61	11363.61	**0.00**	**0.84**	11363.61	11363.61	**0.00**	5.20	10468.16	12359.71	15.30	3600.10	11363.61	11363.61	**0.00**	1467.20	398000	11363.61	11363.61	**0.00**	361.35	3211	16	···
p23	13283.59	13283.59	**0.00**	**0.93**	13283.59	13283.59	**0.00**	13.52	12280.13	15567.70	21.12	3600.10	13283.59	13283.59	**0.00**	1341.86	398000	13283.59	13283.59	**0.00**	367.54	3200	13	···
p24	15069.56	15069.56	**0.00**	**1.42**	15069.56	15069.56	**0.00**	5.88	13934.45	25457.21	45.26	3600.10	15069.55	15069.56	**0.00**	1227.84	398000	15069.55	15069.56	**0.00**	372.26	3220	17	···
p25	19760.22	19760.22	**0.00**	**5.43**	19760.20	19760.22	**0.00**	259.50	18181.16	20520.27	11.40	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	19760.20	19760.22	**0.00**	1870.06	14124	24	···
p26	18787.24	18787.24	**0.00**	**4.16**	18787.23	18787.24	**0.00**	350.12	17263.82	19173.27	9.96	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	18787.23	18787.24	**0.00**	1815.51	14134	24	···
p27	20385.22	20385.22	**0.00**	**4.19**	20385.20	20385.22	**0.00**	221.36	18763.00	20967.26	10.51	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	20385.20	20385.22	**0.00**	1878.67	14146	25	···
p28	21785.22	21785.22	**0.00**	**3.70**	21785.21	21785.22	**0.00**	282.69	20028.30	22967.26	12.80	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	21785.20	21785.22	**0.00**	1759.83	14139	22	···
p29	19760.22	19760.22	**0.00**	**3.90**	19760.20	19760.22	**0.00**	252.47	18181.16	20520.27	11.40	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	19760.20	19760.22	**0.00**	1869.84	14124	24	···
p3	14395.54	14395.54	**0.00**	**0.23**	14395.54	14395.55	**0.00**	8.99	13973.23	14425.54	3.14	3600.10	14395.53	14395.54	**0.00**	541.21	199000	14395.53	14395.54	**0.00**	178.35	1702	14	···
p30	18787.24	18787.24	**0.00**	**5.00**	18787.23	18787.24	**0.00**	363.65	17263.82	19173.27	9.96	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	18787.23	18787.24	**0.00**	1898.45	14134	24	···
p31	20385.22	20385.22	**0.00**	**3.95**	20385.20	20385.22	**0.00**	225.32	18763.00	20967.26	10.51	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	20385.20	20385.22	**0.00**	1827.40	14146	25	···
p32	21785.22	21785.22	**0.00**	**4.37**	21785.21	21785.22	**0.00**	294.36	20028.30	22967.26	12.80	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	21785.20	21785.22	**0.00**	2013.06	14139	22	···
p33	19760.22	19760.22	**0.00**	**4.33**	19760.20	19760.22	**0.00**	350.04	18181.16	20520.27	11.40	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	19760.20	19760.22	**0.00**	1901.97	14124	24	···
p34	18787.24	18787.24	**0.00**	**4.80**	18787.23	18787.24	**0.00**	406.11	17263.82	19173.27	9.96	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	18787.23	18787.24	**0.00**	1839.11	14134	24	···
p35	20385.22	20385.22	**0.00**	**5.20**	20385.20	20385.22	**0.00**	292.48	18763.00	20967.26	10.51	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	20385.20	20385.22	**0.00**	1976.65	14146	25	···
p36	21785.22	21785.22	**0.00**	**4.69**	21785.21	21785.22	**0.00**	267.09	20028.30	22967.26	12.80	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	21785.20	21785.22	**0.00**	1858.81	14139	22	···
p37	19760.22	19760.22	**0.00**	**4.66**	19760.20	19760.22	**0.00**	299.07	18181.16	20520.27	11.40	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	19760.20	19760.22	**0.00**	1903.99	14124	24	···
p38	18787.24	18787.24	**0.00**	**4.72**	18787.23	18787.24	**0.00**	402.07	17263.82	19173.27	9.96	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	18787.23	18787.24	**0.00**	1911.22	14134	24	···
p39	20385.22	20385.22	**0.00**	**4.20**	20385.20	20385.22	**0.00**	243.19	18763.00	20967.26	10.51	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	20385.20	20385.22	**0.00**	1901.23	14146	25	···
p4	15859.49	15859.49	**0.00**	**0.29**	15859.49	15859.49	**0.00**	9.66	15226.25	16297.47	6.57	3600.10	15859.48	15859.49	**0.00**	562.05	199000	15859.48	15859.49	**0.00**	183.29	1705	14	···
p40	21785.22	21785.22	**0.00**	**4.66**	21785.21	21785.22	**0.00**	305.32	20028.30	22967.26	12.80	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	21785.20	21785.22	**0.00**	1801.71	14139	22	···
p41	9914.68	9914.68	**0.00**	**1.06**	9914.68	9914.68	**0.00**	12.09	$-\infty$	10020.72	100.00	3600.10	9914.67	9914.68	**0.00**	1256.77	358200	9914.67	9914.68	**0.00**	352.60	3083	14	···
p42	8761.72	8761.72	**0.00**	**2.96**	8761.71	8761.72	**0.00**	32.34	7769.16	10786.80	27.98	3600.10	8761.71	8761.72	**0.00**	3236.50	636800	8761.71	8761.72	**0.00**	680.52	5160	20	···
p43	7996.75	7996.75	**0.00**	**4.88**	7996.75	7996.75	**0.00**	64.94	6880.16	9899.83	30.50	3600.10	$-\infty$	$+\infty$	100.00	3602.25	835800	7996.74	7996.75	**0.00**	857.56	6640	24	···
p44	9910.72	9910.72	**0.00**	**0.46**	9910.72	9910.72	**0.00**	4.50	$-\infty$	9918.72	100.00	3600.10	9910.72	9910.72	**0.00**	969.93	358200	9910.72	9910.72	**0.00**	352.53	3060	13	···
p45	9127.75	9127.75	**0.00**	**1.82**	9127.75	9127.75	**0.00**	8.22	8327.09	11446.81	27.25	3600.10	9127.74	9127.75	**0.00**	2270.84	636800	9127.74	9127.75	**0.00**	635.74	5181	24	···
p46	7923.78	7923.78	**0.00**	**3.30**	7923.78	7923.78	**0.00**	30.82	7134.42	8719.82	18.18	3600.10	7923.77	7923.78	**0.00**	3506.00	835800	7923.77	7923.78	**0.00**	864.87	6583	13	···
p47	7468.83	7468.83	**0.00**	**0.13**	7468.83	7468.83	**0.00**	4.95	7468.83	7468.83	**0.00**	1.92	7468.83	7468.83	**0.00**	918.28	358200	7468.83	7468.83	**0.00**	328.97	3060	10	···
p48	7104.84	7104.84	**0.00**	**0.69**	7104.84	7104.84	**0.00**	7.10	6821.20	7716.86	11.61	3600.10	7104.84	7104.84	**0.00**	1910.40	636800	7104.84	7104.84	**0.00**	606.66	5135	12	···
p49	6657.87	6657.87	**0.00**	**1.17**	6657.86	6657.87	**0.00**	8.63	6356.84	6767.87	6.07	3600.10	6657.86	6657.87	**0.00**	2830.26	835800	6657.86	6657.87	**0.00**	889.25	6580	12	···
p5	13742.54	13742.54	**0.00**	**0.29**	13742.54	13742.54	**0.00**	9.81	13391.61	13765.56	2.72	3600.10	13742.53	13742.54	**0.00**	538.19	199000	13742.53	13742.54	**0.00**	171.19	1702	14	···
p50	12839.50	12839.50	**0.00**	**0.60**	12839.49	12839.50	**0.00**	5.32	$-\infty$	13106.51	100.00	3600.10	12839.49	12839.50	**0.00**	1079.77	398000	12839.49	12839.50	**0.00**	365.84	3399	12	···
p51	11722.56	11722.56	**0.00**	**1.73**	11722.56	11722.56	**0.00**	49.79	10737.31	12956.61	17.13	3600.10	11722.56	11722.56	**0.00**	2812.84	796000	11722.56	11722.56	**0.00**	822.77	6404	17	···
p52	13920.47	13920.47	**0.00**	**0.85**	13920.47	13920.47	**0.00**	25.54	$-\infty$	14065.48	100.00	3600.10	13920.46	13920.47	**0.00**	1087.89	398000	13920.46	13920.47	**0.00**	380.60	3405	18	···
p53	13122.52	13122.52	**0.00**	**1.55**	13122.52	13122.52	**0.00**	65.47	12377.66	13819.54	10.43	3600.10	13122.51	13122.52	**0.00**	2230.03	796000	13122.51	13122.52	**0.00**	784.34	6413	18	···
p54	13524.46	13524.46	**0.00**	**0.71**	13524.45	13524.46	**0.00**	12.17	$-\infty$	14618.50	100.00	3600.10	13524.45	13524.46	**0.00**	1067.51	398000	13524.45	13524.46	**0.00**	381.41	3424	19	···

Table 46 Detailed results for problem NLUFLP-A-holmberg, cost functions f_1

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	8004.44	8014.94	**0.13**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7971.90	8315.85	4.14	3601.03	2544	2	···
p10	7274.31	7274.95	**0.01**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7236.43	7485.11	3.32	3601.03	2538	2	···
p11	8474.81	8493.04	**0.21**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8449.89	8724.02	3.14	3601.03	2534	2	···
p12	9388.32	9493.04	**1.10**	3600.01	−∞	+∞	100.0	+∞	-37373.73	19803.44	152.99	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	2500	1	···
p13	7654.35	7704.28	**0.65**	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	5000	1	···
p14	6734.54	6843.48	**1.59**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	5000	1	···
p15	8170.56	8429.04	**3.07**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	5000	1	···
p16	9300.07	9751.73	**4.63**	3600.01	−∞	+∞	100.0	+∞	-42463.74	16371.98	138.56	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	5000	1	···
p17	7656.39	7704.28	**0.62**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	5000	1	···
p18	6739.47	6843.48	**1.52**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	5000	1	···
p19	8163.38	8429.04	**3.15**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	5000	1	···
p2	7274.40	7274.95	**0.01**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7251.99	7309.19	0.78	3601.03	2589	3	···
p20	9309.09	9751.73	**4.54**	3600.01	−∞	+∞	100.0	+∞	-42455.57	16371.98	138.56	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	5000	1	···
p21	7660.23	7704.28	**0.57**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	5000	1	···
p22	6736.67	6843.48	**1.56**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	5000	1	···
p23	8169.68	8429.04	**3.08**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	5000	1	···
p24	9304.31	9751.73	**4.59**	3600.02	−∞	+∞	100.0	+∞	-42463.74	16371.98	138.56	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	5000	1	···
p25	10475.67	10595.63	**1.13**	3600.05	−∞	+∞	100.0	+∞	−∞	68678.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	22500	1	···
p26	9745.73	9869.15	**1.25**	3600.03	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.23	22500	1	···
p27	11022.63	11146.81	**1.11**	3600.02	−∞	+∞	100.0	+∞	−∞	68775.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	22500	1	···
p28	12201.99	12347.82	**1.18**	3600.03	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	22500	1	···
p29	10476.56	10595.62	**1.12**	3600.03	−∞	+∞	100.0	+∞	−∞	68678.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	22500	1	···
p3	8474.24	8493.04	**0.22**	3600.01	-19299.81	12821.96	166.44	3603.21	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8468.52	8530.46	0.73	**3592.32**	2534	2	···
p30	9745.52	9869.15	**1.25**	3600.02	-151370.28	68575.00	145.30	3603.44	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	22500	1	···
p31	11022.78	11146.81	**1.11**	3600.02	-150610.21	68775.00	145.66	3603.42	−∞	68775.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	22500	1	···
p32	12201.27	12347.82	**1.19**	3600.02	-149624.98	68975.00	146.10	3603.61	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	22500	1	···
p33	10476.43	10595.62	**1.12**	3600.03	-151375.91	68678.00	145.37	3609.17	−∞	68678.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	22500	1	···
p34	9744.83	9869.15	**1.26**	3600.02	-151370.28	68575.00	145.30	3606.80	−∞	68575.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	22500	1	···
p35	11021.45	11146.81	**1.12**	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	22500	1	···
p36	12199.94	12347.82	**1.20**	3600.04	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.22	22500	1	···
p37	10476.00	10595.62	**1.13**	3600.12	−∞	+∞	100.0	+∞	−∞	68678.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	22500	1	···
p38	9743.43	9869.27	**1.28**	3600.04	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	22500	1	···
p39	11021.44	11146.81	**1.12**	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	22500	1	···
p4	9389.58	9493.04	**1.09**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	2500	1	···
p40	12202.31	12347.82	**1.18**	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	22500	1	···
p41	5682.27	5855.44	**2.96**	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	4500	1	···
p42	5065.52	5180.05	**2.21**	3600.05	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	8000	1	···
p43	4469.08	4632.02	**3.52**	3600.02	-10292.00	8887.00	186.35	3603.36	-21549.77	8887.00	141.24	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	10500	1	···
p44	6255.87	6316.13	**0.95**	3600.02	-15841.19	7873.45	149.70	3602.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	4500	1	···
p45	5552.87	5650.97	**1.74**	3600.02	-14591.47	12921.00	188.55	3603.27	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	8000	1	···
p46	4921.86	5021.35	**1.98**	3600.02	-13449.73	10659.55	179.25	3605.67	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	10500	1	···
p47	5655.09	5655.10	**0.00**	**364.42**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5654.53	5655.11	0.01	3601.02	4609	10	···
p48	4951.08	5043.98	**1.84**	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	8000	1	···
p49	4668.92	4870.79	**4.14**	3600.02	-12326.57	12398.86	199.42	3603.33	-25910.33	13113.00	150.61	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	10500	1	···
p5	7993.14	8014.94	**0.27**	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7971.90	8315.85	4.14	3601.02	2544	2	···
p50	7049.06	7053.54	**0.06**	3600.01	-14048.05	10249.00	172.96	3603.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7049.00	7054.99	0.09	3601.03	5227	6	···
p51	6481.03	6502.99	**0.34**	3600.02	-14599.24	15693.00	193.03	3603.29	-31173.73	15693.00	150.34	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	10000	1	···
p52	7839.41	7851.50	0.15	3600.02	-19114.18	17186.02	189.91	3612.10	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7846.04	7853.48	**0.09**	3601.03	5243	6	···
p53	7419.35	7436.36	**0.23**	3600.02	-19626.01	20745.00	194.61	3607.64	-40887.55	20745.00	150.74	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	10000	1	···
p54	7376.85	7376.85	**0.00**	**141.89**	-16888.05	17356.06	197.30	3608.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7376.79	7376.89	0.00	3601.02	5860	17	···

Table 47 Detailed results for problem NLUFLP-A-holmberg, cost functions f_2

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	4890.66	6958.10	29.71	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	6681.00	7069.38	**5.49**	3601.04	5144	2	···
p10	4186.72	6047.30	30.77	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5824.61	6119.61	**4.82**	3601.03	5173	2	···
p11	5440.26	7645.24	28.84	3600.01	−∞	+∞	100.0	+∞	3155.70	9356.93	66.27	3600.10	−∞	+∞	100.00	3600.01	+∞	7373.96	7744.12	**4.78**	3601.03	5121	2	···
p12	6486.08	8963.40	**27.64**	3600.01	−∞	+∞	100.0	+∞	3612.60	10667.83	66.14	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	5000	1	···
p13	3545.52	6682.30	46.94	3600.01	−∞	+∞	100.0	+∞	1804.92	10451.26	82.73	3600.10	−∞	+∞	100.00	3600.01	+∞	6441.23	6748.29	**4.55**	3601.03	10148	2	···
p14	2847.33	5982.74	**52.41**	3600.01	−∞	+∞	100.0	+∞	1501.27	8279.49	81.87	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	10000	1	···
p15	3827.50	7638.98	**49.90**	3600.01	−∞	+∞	100.0	+∞	1915.64	10139.23	81.11	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	10000	1	···
p16	4573.02	9102.50	**49.76**	3600.01	−∞	+∞	100.0	+∞	2256.85	12379.32	81.77	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	10000	1	···
p17	3577.89	6682.30	46.46	3600.01	−∞	+∞	100.0	+∞	1804.92	10451.26	82.73	3600.10	−∞	+∞	100.00	3600.01	+∞	6441.23	6748.29	**4.55**	3601.03	10148	2	···
p18	2885.47	5977.83	**51.73**	3600.01	−∞	+∞	100.0	+∞	1501.27	8279.49	81.87	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	10000	1	···
p19	3763.23	7640.68	**50.75**	3600.01	−∞	+∞	100.0	+∞	1915.64	10139.23	81.11	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	10000	1	···
p2	4175.20	6047.65	30.96	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5824.61	6119.61	**4.82**	3601.03	5173	2	···
p20	4675.03	9087.68	**48.56**	3600.01	−∞	+∞	100.0	+∞	2256.85	12379.32	81.77	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	10000	1	···
p21	3567.92	6682.30	46.61	3600.02	−∞	+∞	100.0	+∞	1839.36	10451.26	82.40	3600.10	−∞	+∞	100.00	3600.01	+∞	6441.23	6748.29	**4.55**	3601.03	10148	2	···
p22	2884.18	5977.83	**51.75**	3600.01	−∞	+∞	100.0	+∞	1501.27	8279.49	81.87	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	10000	1	···
p23	3794.69	7640.68	**50.34**	3600.01	−∞	+∞	100.0	+∞	1915.64	10139.23	81.11	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	10000	1	···
p24	4637.31	9087.68	**48.97**	3600.01	−∞	+∞	100.0	+∞	2256.85	12379.32	81.77	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	10000	1	···
p25	3925.03	10991.89	**64.29**	3600.09	−∞	+∞	100.0	+∞	273.08	29441.76	99.07	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.29	45000	1	···
p26	3508.12	9613.53	**63.51**	3600.03	−∞	+∞	100.0	+∞	300.00	27853.76	98.92	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.30	45000	1	···
p27	4380.55	10875.05	**59.72**	3600.02	−∞	+∞	100.0	+∞	500.00	30253.76	98.35	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.12	45000	1	···
p28	4679.89	13275.26	**64.75**	3600.03	−∞	+∞	100.0	+∞	700.00	32653.76	97.86	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.27	45000	1	···
p29	3870.46	10991.89	**64.79**	3600.02	−∞	+∞	100.0	+∞	273.08	29441.76	99.07	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.24	45000	1	···
p3	5446.10	7643.71	28.75	3600.01	5347.89	7637.28	29.98	3603.40	3155.70	9356.93	66.27	3600.10	−∞	+∞	100.00	3600.01	+∞	7373.96	7744.12	**4.78**	3601.03	5121	2	···
p30	3419.49	9613.53	64.43	3600.03	5328.18	8507.07	**37.37**	3603.23	300.00	27853.76	98.92	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.22	45000	1	···
p31	4378.48	10875.05	59.74	3600.03	6757.97	10382.25	**34.91**	3603.16	500.00	30253.76	98.35	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	45000	1	···
p32	4190.53	16499.88	74.60	3600.04	7975.05	11782.25	**32.31**	3603.19	700.00	32653.76	97.86	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.33	45000	1	···
p33	3841.42	10991.89	65.05	3600.02	6155.65	9625.86	**36.05**	3609.16	273.08	29441.76	99.07	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.12	45000	1	···
p34	3483.34	9613.53	63.77	3600.03	5331.77	8507.07	**37.33**	3606.73	300.00	27853.76	98.92	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.13	45000	1	···
p35	4316.22	10875.05	**60.31**	3600.03	−∞	+∞	100.0	+∞	500.00	30253.76	98.35	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.15	45000	1	···
p36	4583.14	13275.26	**65.48**	3600.04	−∞	+∞	100.0	+∞	700.00	32653.76	97.86	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	45000	1	···
p37	3886.42	10991.89	**64.64**	3600.03	−∞	+∞	100.0	+∞	273.08	29441.76	99.07	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.13	45000	1	···
p38	3484.17	9613.53	**63.76**	3600.03	−∞	+∞	100.0	+∞	300.00	27853.76	98.92	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.14	45000	1	···
p39	4181.63	10875.05	**61.55**	3600.03	−∞	+∞	100.0	+∞	500.00	30253.76	98.35	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.13	45000	1	···
p4	6508.44	8957.35	**27.34**	3600.01	−∞	+∞	100.0	+∞	3610.36	10812.63	66.61	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	5000	1	···
p40	4163.44	18062.04	**76.95**	3600.03	−∞	+∞	100.0	+∞	700.00	32653.76	97.86	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.29	45000	1	···
p41	3587.05	5310.81	**32.46**	3600.01	−∞	+∞	100.0	+∞	1759.88	6738.98	73.89	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	9000	1	···
p42	2345.35	5012.46	**53.21**	3600.02	−∞	+∞	100.0	+∞	902.33	7205.75	87.48	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	16000	1	···
p43	1734.99	4630.91	62.53	3600.02	2843.48	4315.39	**34.11**	3603.09	591.99	7747.26	92.36	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	21000	1	···
p44	4008.00	5788.15	**30.76**	3600.01	3958.59	5745.11	31.10	3603.02	2108.24	8818.38	76.09	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	9000	1	···
p45	2659.32	5295.72	49.78	3600.01	3612.64	5190.58	**30.40**	3603.28	1075.91	9005.00	88.05	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	16000	1	···
p46	1877.05	4960.52	62.16	3600.02	3206.84	4599.33	**30.28**	3605.88	675.27	7955.01	91.51	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	21000	1	···
p47	3801.61	5273.39	**27.91**	3600.01	−∞	+∞	100.0	+∞	2184.20	6905.77	68.37	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	9000	1	···
p48	2367.57	4795.62	**50.63**	3600.02	−∞	+∞	100.0	+∞	1024.33	7029.09	85.43	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.22	16000	1	···
p49	1827.09	4734.33	61.41	3600.02	2951.26	4376.96	**32.57**	3603.29	631.28	7797.91	91.90	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	21000	1	···
p5	4880.93	6959.64	29.87	3600.01	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	6681.00	7069.38	**5.49**	3601.03	5144	2	···
p50	4306.09	6568.10	**34.44**	3600.01	4270.13	6533.56	34.64	3602.89	2145.69	9141.91	76.53	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	10000	1	···
p51	2444.79	6303.00	61.21	3600.02	3970.95	5994.76	**33.76**	3603.36	1160.12	10908.18	89.36	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	20000	1	···
p52	4811.39	7203.97	**33.21**	3600.02	4769.58	7158.10	33.37	3610.79	2528.41	11171.45	77.37	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	10000	1	···
p53	2740.52	7343.88	62.68	3600.02	4446.11	6822.60	**34.83**	3607.48	1398.64	12571.08	88.87	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.29	20000	1	···
p54	4563.54	6953.02	**34.37**	3600.01	4542.49	6922.08	34.38	3605.20	2565.09	8662.57	70.39	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	10000	1	···

Table 48 Detailed results for problem NLUFLP-A-holmberg, cost functions f_3

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
p1	4829.84	7069.97	31.69	3600.01	−∞	+∞	100.0	+∞	2679.88	8537.14	68.61	3600.10	−∞	+∞	100.00	3600.01	+∞	6954.59	6990.14	**0.51**	3601.03	6360	17	⋯
p10	4103.61	6571.17	37.55	3600.01	−∞	+∞	100.0	+∞	2548.08	7760.77	67.17	3600.10	−∞	+∞	100.00	3600.01	+∞	6234.40	6252.21	**0.28**	3601.03	6666	18	⋯
p11	5316.94	7658.47	30.57	3600.01	−∞	+∞	100.0	+∞	3007.67	9061.03	66.81	3600.10	−∞	+∞	100.00	3600.01	+∞	7433.03	7468.66	**0.48**	3601.03	6251	15	⋯
p12	6330.26	8603.91	26.43	3600.01	−∞	+∞	100.0	+∞	3556.95	10171.58	65.03	3600.10	−∞	+∞	100.00	3600.01	+∞	8405.07	8484.81	**0.94**	3601.03	5926	11	⋯
p13	3238.66	7941.29	59.22	3600.01	−∞	+∞	100.0	+∞	1657.14	10006.65	83.44	3600.10	−∞	+∞	100.00	3600.01	+∞	6709.83	6910.10	**2.90**	**3499.27**	11068	4	⋯
p14	2612.91	6798.16	61.56	3600.02	−∞	+∞	100.0	+∞	1453.27	9012.26	83.87	3600.10	−∞	+∞	100.00	3600.01	+∞	5762.58	6102.17	**5.57**	3601.04	11051	3	⋯
p15	3521.11	8333.06	57.75	3600.02	−∞	+∞	100.0	+∞	1858.35	11630.22	84.02	3600.10	−∞	+∞	100.00	3600.01	+∞	7350.17	7565.33	**2.84**	3601.03	11078	4	⋯
p16	4028.48	9610.61	58.08	3600.01	−∞	+∞	100.0	+∞	2252.02	12102.06	81.39	3600.10	−∞	+∞	100.00	3600.01	+∞	8492.49	8978.02	**5.41**	3601.03	11051	4	⋯
p17	3272.02	7624.57	57.09	3600.01	−∞	+∞	100.0	+∞	1657.14	10006.65	83.44	3600.10	−∞	+∞	100.00	3600.01	+∞	6709.83	6910.10	**2.90**	3601.04	11126	5	⋯
p18	2692.02	6798.16	60.40	3600.01	−∞	+∞	100.0	+∞	1453.27	9012.26	83.87	3600.10	−∞	+∞	100.00	3600.01	+∞	5828.39	6053.93	**3.73**	3601.16	11075	4	⋯
p19	3437.66	8556.23	59.82	3600.01	−∞	+∞	100.0	+∞	1858.35	11630.22	84.02	3600.10	−∞	+∞	100.00	3600.01	+∞	7264.54	7666.53	**5.24**	3601.04	11004	2	⋯
p2	4100.74	6571.17	37.59	3600.01	−∞	+∞	100.0	+∞	2527.10	7760.77	67.44	3600.10	−∞	+∞	100.00	3600.01	+∞	6209.12	6268.09	**0.94**	3601.04	6028	12	⋯
p20	4221.83	9482.65	**55.48**	3600.01	−∞	+∞	100.0	+∞	2252.02	12102.06	81.39	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p21	3266.95	7624.57	57.15	3600.02	−∞	+∞	100.0	+∞	1657.14	10006.65	83.44	3600.10	−∞	+∞	100.00	3600.01	+∞	6709.83	6910.10	**2.90**	3601.05	11126	5	⋯
p22	2614.38	6798.16	61.54	3600.02	−∞	+∞	100.0	+∞	1453.27	9012.26	83.87	3600.10	−∞	+∞	100.00	3600.01	+∞	5828.39	6053.93	**3.73**	3601.04	11073	4	⋯
p23	3538.05	8333.06	57.54	3600.01	−∞	+∞	100.0	+∞	1858.35	11630.22	84.02	3600.10	−∞	+∞	100.00	3600.01	+∞	7264.54	7644.97	**4.98**	3601.03	11051	3	⋯
p24	4151.23	9482.65	**56.22**	3600.02	−∞	+∞	100.0	+∞	2252.02	12102.06	81.39	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p25	2712.49	11625.39	**76.67**	3600.03	−∞	+∞	100.0	+∞	302.23	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p26	2312.66	11007.49	**78.99**	3600.04	−∞	+∞	100.0	+∞	300.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p27	2790.71	12578.63	**77.81**	3600.02	−∞	+∞	100.0	+∞	500.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p28	3248.81	16285.80	**80.05**	3600.03	−∞	+∞	100.0	+∞	700.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p29	2649.17	11625.39	**77.21**	3600.02	−∞	+∞	100.0	+∞	302.23	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p3	5328.66	7637.42	30.23	3600.01	5184.60	7497.22	30.85	3603.41	3007.67	9061.03	66.81	3600.10	−∞	+∞	100.00	3600.01	+∞	7433.03	7468.66	**0.48**	3601.03	6251	15	⋯
p30	2312.66	11007.49	78.99	3600.05	5097.98	8736.94	**41.65**	3603.15	300.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p31	2790.71	12578.63	77.81	3600.03	6434.39	10313.39	**37.61**	3603.36	500.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p32	3248.81	16285.80	80.05	3600.04	7576.96	11262.45	**32.72**	3607.71	700.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p33	2649.17	11625.39	77.21	3600.03	5869.39	9451.55	**37.90**	3606.54	302.23	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p34	2312.66	11007.49	78.99	3600.04	5097.98	8736.94	**41.65**	3610.11	300.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p35	2790.71	12578.63	**77.81**	3600.03	−∞	+∞	100.0	+∞	500.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p36	3242.28	16285.80	**80.09**	3600.05	−∞	+∞	100.0	+∞	700.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p37	2649.17	11625.39	**77.21**	3600.03	−∞	+∞	100.0	+∞	302.23	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p38	2312.66	11007.49	**78.99**	3600.05	−∞	+∞	100.0	+∞	300.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p39	2790.71	12578.63	**77.81**	3600.03	−∞	+∞	100.0	+∞	500.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p4	6340.64	8552.17	25.86	3600.01	−∞	+∞	100.0	+∞	3554.26	10171.58	65.06	3600.10	−∞	+∞	100.00	3600.01	+∞	8405.07	8484.81	**0.94**	3601.03	5925	11	⋯
p40	3248.81	16285.80	**80.05**	3600.02	−∞	+∞	100.0	+∞	700.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p41	3486.18	5495.31	36.56	3600.02	−∞	+∞	100.0	+∞	1691.32	6071.40	72.14	3600.10	−∞	+∞	100.00	3600.01	+∞	4994.92	5266.56	**5.16**	3601.05	10079	5	⋯
p42	1986.03	5451.26	**63.57**	3600.01	−∞	+∞	100.0	+∞	890.72	6997.24	87.27	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	17600	1	⋯
p43	1492.69	4490.06	66.76	3600.02	2720.37	4376.23	**37.84**	3603.43	651.81	7692.70	91.53	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p44	3893.89	6239.49	37.59	3600.02	3872.85	5782.40	33.02	3603.25	2005.37	8352.02	75.99	3600.10	−∞	+∞	100.00	3600.01	+∞	5652.90	5728.33	**1.32**	3601.04	10182	8	⋯
p45	2319.65	6434.95	63.95	3600.01	3489.41	5325.29	**34.47**	3605.73	1023.19	8528.35	88.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.24	17600	1	⋯
p46	1550.99	5841.36	73.45	3600.02	3064.42	4979.99	**38.47**	3605.48	728.43	8013.10	90.91	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p47	3659.46	6011.84	39.13	3600.01	−∞	+∞	100.0	+∞	2105.82	6737.85	68.75	3600.10	−∞	+∞	100.00	3600.01	+∞	5158.99	5278.42	**2.26**	**3596.56**	9982	3	⋯
p48	2125.59	5102.88	**58.35**	3600.02	−∞	+∞	100.0	+∞	962.60	7196.27	86.62	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	17600	1	⋯
p49	1533.98	4862.09	68.45	3600.01	2850.36	4469.18	**36.22**	3603.28	691.11	10329.58	93.31	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p5	4817.66	7077.69	31.93	3600.01	−∞	+∞	100.0	+∞	2756.24	8537.14	67.71	3600.10	−∞	+∞	100.00	3600.01	+∞	6954.59	6990.14	**0.51**	3601.03	6360	17	⋯
p50	4166.64	6733.77	38.12	3600.01	4098.59	6178.64	33.67	3603.20	2054.86	9203.65	77.67	3600.10	−∞	+∞	100.00	3600.01	+∞	6007.72	6219.21	**3.40**	3601.03	11210	5	⋯
p51	2187.02	7716.63	71.66	3600.02	3802.51	5899.09	**35.54**	3603.14	1138.27	11125.29	89.77	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p52	4666.36	7754.65	39.83	3600.01	4602.64	6885.51	33.15	3609.71	2418.39	12478.14	80.62	3600.10	−∞	+∞	100.00	3600.01	+∞	6787.80	6926.13	**2.00**	3601.03	11376	8	⋯
p53	2546.08	8917.61	71.45	3600.02	4294.03	6527.97	**34.22**	3611.07	1362.96	12653.19	89.23	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
p54	4431.49	7404.81	**40.15**	3600.01	−∞	+∞	100.0	+∞	2284.37	9893.26	76.91	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	11000	1	⋯

Table 49 Detailed results for problem NLUFLP-A-holmberg, cost functions f_4

Instance	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	2604.02	3348.29	**22.23**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1574.74	4244.20	62.90	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	7500	1	···
p10	2349.72	2993.62	**21.51**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1313.78	3606.72	63.57	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	7500	1	···
p11	3094.59	3658.82	**15.42**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	1925.03	4093.04	52.97	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	7500	1	···
p12	3692.52	4266.52	**13.45**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	2444.18	5018.40	51.30	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	7500	1	···
p13	2371.28	3240.04	**26.81**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1020.76	5008.91	79.62	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	15000	1	···
p14	1768.46	3118.08	**43.28**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	860.69	28348.36	96.96	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	15000	1	···
p15	2686.44	3703.62	**27.46**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1236.69	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	15000	1	···
p16	3500.90	4312.28	**18.82**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	1641.61	6415.73	74.41	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	15000	1	···
p17	2347.38	3240.04	**27.55**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1020.76	5008.91	79.62	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	15000	1	···
p18	1831.24	3118.08	**41.27**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	860.69	28348.36	96.96	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	15000	1	···
p19	2608.36	3726.22	**30.00**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	1236.69	5979.00	79.32	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	15000	1	···
p2	2352.53	2993.62	**21.42**	3600.01	2250.04	3210.58	29.92	3603.31	1256.66	3606.72	65.16	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.24	7500	1	···
p20	3430.27	4312.28	**20.45**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	1641.61	6415.73	74.41	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.23	15000	1	···
p21	2350.51	3240.04	**27.45**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	1020.76	5008.91	79.62	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.18	15000	1	···
p22	1803.91	3118.08	**42.15**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	860.69	28348.36	96.96	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	15000	1	···
p23	2657.80	3703.62	**28.24**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1236.69	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	15000	1	···
p24	3418.23	4312.28	**20.73**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1641.61	6415.73	74.41	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	15000	1	···
p25	1763.36	5769.15	**69.43**	3600.06	$-\infty$	$+\infty$	100.0	$+\infty$	271.76	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p26	1507.26	5266.67	**71.38**	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	300.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p27	2023.87	5914.96	**65.78**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	500.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p28	2642.27	6049.37	**56.32**	3600.06	$-\infty$	$+\infty$	100.0	$+\infty$	700.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p29	1763.36	5769.15	**69.43**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	271.76	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p3	3095.96	3658.82	**15.38**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1925.03	4093.04	52.97	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	7500	1	···
p30	1507.26	5266.67	71.38	3600.05	2899.27	4035.73	**28.16**	3603.25	300.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p31	2023.87	5914.96	65.78	3600.05	3762.81	5340.52	**29.54**	3603.20	500.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p32	2642.27	6049.37	56.32	3600.08	4517.68	6511.06	**30.62**	3608.90	700.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p33	1763.36	5769.15	69.43	3600.03	3416.44	4918.39	**30.54**	3606.66	271.76	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p34	1507.26	5266.67	**71.38**	3600.04	$-\infty$	$+\infty$	100.0	$+\infty$	300.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p35	2023.87	5914.96	**65.78**	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	500.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p36	2642.27	6043.66	**56.28**	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	700.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p37	1763.36	5769.15	**69.43**	3600.06	$-\infty$	$+\infty$	100.0	$+\infty$	271.76	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p38	1507.26	5266.67	**71.38**	3600.04	$-\infty$	$+\infty$	100.0	$+\infty$	300.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p39	2023.87	5914.96	**65.78**	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	500.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p4	3698.27	4265.42	**13.30**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	2114.67	5018.40	57.86	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.21	7500	1	···
p40	2642.27	6049.37	**56.32**	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	700.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	0	···
p41	2015.05	2444.91	**17.58**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1004.80	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	13500	1	···
p42	1793.53	2140.70	**16.22**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	5586.87	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.09	24000	1	···
p43	1451.46	1912.63	24.11	3600.02	1619.81	1912.64	**15.31**	3603.08	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.10	31500	1	···
p44	2265.37	2858.73	**20.76**	3600.02	2207.96	3055.18	27.73	3603.12	1121.35	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.23	13500	1	···
p45	1794.50	2644.57	32.14	3600.02	2062.90	2995.84	**31.14**	3603.18	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.09	24000	1	···
p46	1453.86	2368.62	38.62	3600.08	1868.69	2368.62	**21.11**	3606.08	$-\infty$	104903.50	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	31500	1	···
p47	2196.43	2749.91	**20.13**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	1069.03	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	13500	1	···
p48	1716.74	2334.25	**26.45**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	5532.57	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.09	24000	1	···
p49	1451.89	2135.80	32.02	3600.03	1719.68	2185.21	**21.30**	3603.26	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.15	31500	1	···
p5	2602.72	3348.29	**22.27**	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	1574.95	4244.20	62.89	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.07	7500	1	···
p50	2399.10	3039.49	**21.07**	3600.02	2345.95	3315.45	29.24	3603.18	$-\infty$	6919.31	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	15000	1	···
p51	1789.03	3015.29	40.67	3600.02	2234.71	3271.72	**31.70**	3603.33	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	30000	1	···
p52	2675.34	3499.30	**23.55**	3600.02	2597.78	3654.40	28.91	3607.39	1270.53	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.06	15000	1	···
p53	1993.46	3275.67	39.14	3600.03	2390.69	3409.59	**29.88**	3604.19	$-\infty$	42289.14	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.24	30000	1	···
p54	2525.10	3319.92	**23.94**	3600.02	2453.96	3679.16	33.30	3607.38	1227.96	7445.02	83.51	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.05	15000	1	···

Table 50 Detailed results for problem NLUFLP-A-holmberg, cost functions f_5

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	-1180.86	8673.60	113.61	3600.01	-2003.67	9873.79	120.29	3602.20	-624.73	14931.35	104.18	3600.10	−∞	+∞	100.00	3600.01	+∞	8320.72	8325.48	**0.06**	3601.02	3150	13	···
p10	-2337.68	7729.98	130.24	3600.01	-2986.90	11258.75	126.53	3602.39	-6939.43	16630.86	141.73	3600.10	−∞	+∞	100.00	3600.01	+∞	7577.77	7580.21	**0.03**	3601.02	3149	17	···
p11	-510.75	9435.15	105.41	3600.01	-1215.78	11028.97	111.02	3602.38	-224.68	15440.53	101.46	3600.10	−∞	+∞	100.00	3600.01	+∞	8796.91	8799.32	**0.03**	3601.02	3201	13	···
p12	1176.90	10214.43	88.48	3600.02	176.76	11418.77	98.45	3602.37	-190.22	18659.23	101.02	3600.10	−∞	+∞	100.00	3600.01	+∞	9796.09	9800.20	**0.04**	3601.02	3145	11	···
p13	-2030.92	8334.37	124.37	3600.01	-2495.58	12617.06	119.78	3603.05	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7975.23	8063.39	**1.09**	3601.03	6100	6	···
p14	-3491.79	7456.90	146.83	3600.02	-3809.14	11344.26	133.58	3604.48	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7044.59	7090.44	**0.65**	3601.03	6103	7	···
p15	-1246.68	9729.78	112.81	3600.01	-1724.60	13216.77	113.05	3602.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8585.83	8756.44	**1.95**	3601.03	6062	3	···
p16	490.14	11205.58	**95.63**	3600.01	-490.68	12352.12	103.97	3602.92	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	6000	1	···
p17	-2041.56	8506.94	124.00	3600.01	-2495.97	12617.06	119.78	3602.93	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7962.20	8063.39	**1.25**	3601.03	6094	5	···
p18	-3469.03	7456.90	146.52	3600.02	-3809.16	11963.31	131.84	3602.60	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7044.59	7090.44	**0.65**	3601.02	6103	7	···
p19	-1274.48	9868.75	112.91	3600.02	-1724.60	13216.77	113.05	3602.94	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8585.83	8756.44	**1.95**	**3561.35**	6047	2	···
p2	-2340.41	7729.98	130.28	3600.01	-2987.01	11258.75	126.53	3602.18	-6939.43	16630.86	141.73	3600.10	−∞	+∞	100.00	3600.01	+∞	7579.23	7580.21	**0.01**	3601.03	3247	21	···
p20	496.37	11204.43	**95.57**	3600.01	-490.68	12352.12	103.97	3602.95	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	6000	1	···
p21	-2041.56	8506.94	124.00	3600.02	-2495.97	12617.06	119.78	3602.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7975.23	8063.39	**1.09**	3601.03	6100	6	···
p22	-3475.76	7456.90	146.61	3600.02	-3809.16	11963.31	131.84	3602.93	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7044.59	7090.44	**0.65**	3601.03	6103	7	···
p23	-1266.62	9868.64	112.83	3600.02	-1724.60	13216.77	113.05	3602.96	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8585.83	8756.44	**1.95**	3601.03	6062	3	···
p24	500.63	11204.43	**95.53**	3600.02	-490.68	12352.12	103.97	3602.90	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	6000	1	···
p25	-41680.55	13664.75	132.78	3600.05	-42066.76	66159.53	163.58	3605.14	-79928.81	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p26	-43122.31	11894.22	127.58	3600.03	-43510.71	67670.79	164.30	3605.61	-80095.62	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p27	-40211.15	13219.39	132.87	3600.05	-40608.69	68777.64	159.04	3605.09	-79799.76	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p28	-37627.14	14787.33	139.30	3600.07	-37998.21	68977.64	155.09	3605.14	-79515.27	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p29	-41680.55	13664.75	132.78	3600.05	-42066.76	66159.53	163.58	3605.13	-79928.81	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p3	-513.12	9435.15	105.44	3600.01	-1220.72	11028.97	111.07	3602.28	-223.27	15440.53	101.45	3600.10	−∞	+∞	100.00	3600.01	+∞	8796.58	8799.41	**0.03**	3601.03	3145	12	···
p30	-43122.31	11894.22	127.58	3600.03	-43510.71	67670.79	164.30	3605.22	-80095.62	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p31	-40211.15	13289.40	133.05	3600.07	-40608.69	68777.64	159.04	3605.15	-79799.76	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p32	-37627.14	14791.92	139.31	3600.03	-37998.21	68977.64	155.09	3605.16	-79515.27	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p33	-41680.55	13664.75	132.78	3600.06	-42066.76	66159.53	163.58	3605.25	-79928.81	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p34	-43122.31	11898.99	127.59	3600.08	-43510.71	67670.79	164.30	3605.22	-80095.62	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p35	-40211.15	13289.40	133.05	3600.04	-40608.69	68777.64	159.04	3605.11	-79799.76	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p36	-37627.14	14791.92	139.31	3600.05	-37998.21	68977.64	155.09	3605.18	-79515.27	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p37	-41680.55	13673.75	132.81	3600.07	-42066.76	66159.53	163.58	3605.36	-79928.81	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p38	-43122.31	11901.69	127.60	3600.08	-43510.71	67670.79	164.30	3606.12	-80095.62	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p39	-40211.15	13289.40	133.05	3600.03	-40608.69	68777.64	159.04	3605.28	-79799.76	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p4	1206.46	10214.43	88.19	3600.01	178.22	11418.77	98.44	3602.37	-190.22	18659.23	101.02	3600.10	−∞	+∞	100.00	3600.01	+∞	9793.22	9800.69	**0.08**	3601.02	3135	10	···
p40	-37627.14	14787.33	139.30	3600.04	-37997.59	68977.64	155.09	3605.75	-79515.27	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
p41	116.58	6227.38	**98.13**	3600.03	-334.64	6613.78	105.06	3602.66	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.05	5400	1	···
p42	14.43	5904.34	99.76	3600.02	-321.29	6262.02	105.13	3603.91	-3567.72	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	5165.82	5403.60	**4.40**	3601.03	9670	2	···
p43	266.85	5475.44	**95.13**	3600.02	-259.38	6964.25	103.72	3603.78	-6837.48	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	12600	1	···
p44	-769.05	6686.08	111.50	3600.02	-1112.28	7897.14	114.08	3602.66	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	6406.02	6671.26	**3.98**	3601.03	5474	2	···
p45	-723.72	6478.47	111.17	3600.03	-954.85	7558.94	112.63	3603.42	-8746.40	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.20	9600	1	···
p46	-554.66	7286.81	107.61	3600.02	-820.30	8360.66	109.81	3603.82	-8680.98	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	12600	1	···
p47	-383.28	5673.07	106.76	3600.04	-773.15	7376.03	110.48	3602.66	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5673.07	5673.07	**0.00**	**1913.03**	5850	9	···
p48	-305.82	5771.94	105.30	3600.02	-623.21	6521.43	109.56	3603.48	-8107.24	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	9600	1	···
p49	-229.76	6916.59	103.32	3600.03	-619.02	7426.19	108.34	3603.88	-7664.32	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	12600	1	···
p5	-1200.99	8673.60	113.85	3600.01	-2003.32	9873.79	120.29	3602.40	-624.73	14931.35	104.18	3600.10	−∞	+∞	100.00	3600.01	+∞	8320.08	8326.69	**0.08**	3601.02	3139	11	···
p50	-753.29	7677.22	109.81	3600.01	-1099.66	9791.63	111.23	3603.13	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7278.64	7308.65	**0.41**	3601.02	6199	7	···
p51	-1301.55	7435.02	117.51	3600.03	-1423.90	11067.16	112.87	3605.53	-9647.74	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	6425.32	6728.83	**4.51**	3601.15	12082	2	···
p52	-2150.47	8209.24	126.20	3600.01	-2450.01	10843.80	122.59	3604.31	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8118.01	8169.51	**0.63**	3601.03	6204	8	···
p53	-2960.37	8032.24	136.86	3600.02	-3068.98	17407.67	117.63	3605.40	-12894.30	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	7597.16	7765.28	**2.17**	3601.03	12097	3	···
p54	-1527.64	8066.22	118.94	3600.02	-1791.16	10781.40	116.61	3602.62	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7662.93	7754.24	**1.18**	3601.03	6196	6	···

Table 51 **Detailed results for problem NLUFLP-A-holmberg, cost functions f_6**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	8332.00	8332.00	0.00	0.02	8332.00	8332.00	0.00	3.61	8332.00	8332.00	0.00	2.93	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	8332.00	8332.00	0.00	29.95	3603	14	···
p10	7580.00	7580.00	0.00	0.02	7580.00	7580.00	0.00	3.01	7580.00	7580.00	0.00	2.47	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	7580.00	7580.00	0.00	26.70	4000	14	···
p11	8806.00	8806.00	0.00	0.02	8806.00	8806.00	0.00	2.83	8806.00	8806.00	0.00	2.57	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	8806.00	8806.00	0.00	29.97	3599	13	···
p12	9806.00	9806.00	0.00	0.03	9806.00	9806.00	0.00	2.97	9806.00	9806.00	0.00	2.62	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	9805.99	9806.00	0.00	35.99	3595	12	···
p13	8049.00	8049.00	0.00	0.05	8049.00	8049.00	0.00	3.29	8049.00	8049.00	0.00	8.64	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	8048.99	8049.00	0.00	123.72	6718	14	···
p14	7092.00	7092.00	0.00	0.04	7092.00	7092.00	0.00	3.28	7092.00	7092.00	0.00	10.44	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	7092.00	7092.00	0.00	102.71	6928	16	···
p15	8735.00	8735.00	0.00	0.05	8735.00	8735.00	0.00	3.12	8735.00	8735.00	0.00	9.34	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	8735.00	8735.00	0.00	176.96	6744	18	···
p16	10168.00	10168.00	0.00	0.05	10168.00	10168.00	0.00	3.19	10168.00	10168.00	0.00	9.15	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	10168.00	10168.00	0.00	270.86	6573	19	···
p17	8049.00	8049.00	0.00	0.05	8049.00	8049.00	0.00	3.27	8049.00	8049.00	0.00	8.49	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	8048.99	8049.00	0.00	117.54	6718	14	···
p18	7092.00	7092.00	0.00	0.05	7092.00	7092.00	0.00	3.22	7092.00	7092.00	0.00	10.35	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	7092.00	7092.00	0.00	113.00	6928	16	···
p19	8735.00	8735.00	0.00	0.05	8735.00	8735.00	0.00	3.27	8735.00	8735.00	0.00	9.26	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	8735.00	8735.00	0.00	179.07	6744	18	···
p2	7580.00	7580.00	0.00	0.02	7580.00	7580.00	0.00	3.55	7580.00	7580.00	0.00	2.47	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	7580.00	7580.00	0.00	29.11	4000	14	···
p20	10168.00	10168.00	0.00	0.05	10168.00	10168.00	0.00	3.55	10168.00	10168.00	0.00	9.07	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	10168.00	10168.00	0.00	271.55	6573	19	···
p21	8049.00	8049.00	0.00	0.05	8049.00	8049.00	0.00	3.11	8049.00	8049.00	0.00	8.70	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	8048.99	8049.00	0.00	112.47	6718	14	···
p22	7092.00	7092.00	0.00	0.05	7092.00	7092.00	0.00	3.16	7092.00	7092.00	0.00	10.45	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	7092.00	7092.00	0.00	99.02	6928	16	···
p23	8735.00	8735.00	0.00	0.05	8735.00	8735.00	0.00	3.16	8735.00	8735.00	0.00	9.51	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	8735.00	8735.00	0.00	180.26	6744	18	···
p24	10168.00	10168.00	0.00	0.05	10168.00	10168.00	0.00	3.39	10168.00	10168.00	0.00	9.27	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	10168.00	10168.00	0.00	266.98	6573	19	···
p25	11247.00	11247.00	0.00	0.25	11247.00	11247.00	0.00	7.17	11247.00	11247.00	0.00	315.25	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	11246.99	11247.00	0.00	1546.91	26508	21	···
p26	10543.00	10543.00	0.00	0.30	10543.00	10543.00	0.00	8.97	10543.00	10543.00	0.00	321.12	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	10542.99	10543.00	0.00	1268.52	27075	20	···
p27	11816.00	11816.00	0.00	0.22	11816.00	11816.00	0.00	8.44	11816.00	11816.00	0.00	340.57	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	11815.99	11816.00	0.00	1741.22	26475	20	···
p28	13016.00	13016.00	0.00	0.34	13016.00	13016.00	0.00	9.94	13016.00	13016.00	0.00	281.80	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	13015.99	13016.00	0.00	2087.73	26498	20	···
p29	11247.00	11247.00	0.00	0.24	11247.00	11247.00	0.00	6.93	11247.00	11247.00	0.00	316.23	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	11246.99	11247.00	0.00	1699.24	26508	21	···
p3	8806.00	8806.00	0.00	0.03	8806.00	8806.00	0.00	23.71	8806.00	8806.00	0.00	2.59	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	8806.00	8806.00	0.00	32.52	3599	13	···
p30	10543.00	10543.00	0.00	0.27	10543.00	10543.00	0.00	9.45	10543.00	10543.00	0.00	321.67	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	10542.99	10543.00	0.00	1550.41	27075	20	···
p31	11816.00	11816.00	0.00	0.26	11816.00	11816.00	0.00	8.17	11816.00	11816.00	0.00	342.37	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	11815.99	11816.00	0.00	1921.48	26475	20	···
p32	13016.00	13016.00	0.00	0.29	13016.00	13016.00	0.00	43.23	13016.00	13016.00	0.00	277.85	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	13015.99	13016.00	0.00	1802.58	26498	20	···
p33	11247.00	11247.00	0.00	0.34	11247.00	11247.00	0.00	9.40	11247.00	11247.00	0.00	312.95	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	11246.99	11247.00	0.00	1605.92	26508	21	···
p34	10543.00	10543.00	0.00	0.31	10543.00	10543.00	0.00	19.15	10543.00	10543.00	0.00	318.07	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	10542.99	10543.00	0.00	1257.40	27075	20	···
p35	11816.00	11816.00	0.00	0.27	11816.00	11816.00	0.00	9.08	11816.00	11816.00	0.00	343.63	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	11815.99	11816.00	0.00	1726.34	26475	20	···
p36	13016.00	13016.00	0.00	0.30	13016.00	13016.00	0.00	10.97	13016.00	13016.00	0.00	281.39	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	13015.99	13016.00	0.00	2108.09	26498	20	···
p37	11247.00	11247.00	0.00	0.28	11247.00	11247.00	0.00	7.56	11247.00	11247.00	0.00	317.65	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	11246.99	11247.00	0.00	1680.56	26508	21	···
p38	10543.00	10543.00	0.00	0.30	10543.00	10543.00	0.00	9.34	10543.00	10543.00	0.00	318.68	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	10542.99	10543.00	0.00	1404.77	27075	20	···
p39	11816.00	11816.00	0.00	0.38	11816.00	11816.00	0.00	9.15	11816.00	11816.00	0.00	340.18	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	11815.99	11816.00	0.00	1673.08	26475	20	···
p4	9806.00	9806.00	0.00	0.02	9806.00	9806.00	0.00	98.55	9806.00	9806.00	0.00	2.59	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	9805.99	9806.00	0.00	38.76	3595	12	···
p40	13016.00	13016.00	0.00	0.40	13016.00	13016.00	0.00	10.61	13016.00	13016.00	0.00	278.83	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	13015.99	13016.00	0.00	1955.21	26498	20	···
p41	6151.00	6151.00	0.00	0.05	6151.00	6151.00	0.00	3.42	6151.00	6151.00	0.00	11.77	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	6151.00	6151.00	0.00	248.72	6627	25	···
p42	5379.00	5379.00	0.00	0.07	5379.00	5379.00	0.00	3.87	5379.00	5379.00	0.00	47.07	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5379.00	5379.00	0.00	398.83	9485	15	···
p43	4865.00	4865.00	0.00	0.10	4865.00	4865.00	0.00	4.49	4865.00	4865.00	0.00	65.52	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4865.00	4865.00	0.00	888.25	12106	20	···
p44	6619.00	6619.00	0.00	0.04	6619.00	6619.00	0.00	3.43	6619.00	6619.00	0.00	10.84	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	6618.99	6619.00	0.00	146.70	7558	19	···
p45	5963.00	5963.00	0.00	0.09	5963.00	5963.00	0.00	8.98	5963.00	5963.00	0.00	43.38	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5963.00	5963.00	0.00	471.93	10105	22	···
p46	5255.00	5255.00	0.00	0.10	5255.00	5255.00	0.00	5.08	5255.00	5255.00	0.00	62.49	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5255.00	5255.00	0.00	872.18	12649	24	···
p47	5673.00	5673.00	0.00	0.04	5673.00	5673.00	0.00	3.60	5673.00	5673.00	0.00	0.19	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5673.00	5673.00	0.00	72.72	8278	11	···
p48	5358.00	5358.00	0.00	0.09	5358.00	5358.00	0.00	4.33	5358.00	5358.00	0.00	43.70	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5358.00	5358.00	0.00	313.65	10733	15	···
p49	5102.00	5102.00	0.00	0.13	5102.00	5102.00	0.00	5.65	5102.00	5102.00	0.00	67.77	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5102.00	5102.00	0.00	543.53	12879	14	···
p5	8332.00	8332.00	0.00	0.03	8332.00	8332.00	0.00	23.70	8332.00	8332.00	0.00	2.87	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	8332.00	8332.00	0.00	29.65	3603	14	···
p50	7307.00	7307.00	0.00	0.05	7307.00	7307.00	0.00	3.64	7307.00	7307.00	0.00	11.15	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	7306.99	7307.00	0.00	124.51	6797	12	···
p51	6705.00	6705.00	0.00	0.11	6705.00	6705.00	0.00	4.69	6705.00	6705.00	0.00	72.24	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	6705.00	6705.00	0.00	538.55	11805	19	···
p52	8166.00	8166.00	0.00	0.05	8166.00	8166.00	0.00	7.60	8166.00	8166.00	0.00	12.63	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	8165.99	8166.00	0.00	102.52	7605	18	···
p53	7731.00	7731.00	0.00	0.09	7731.00	7731.00	0.00	10.74	7731.00	7731.00	0.00	50.95	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	7730.99	7731.00	0.00	367.14	12603	18	···
p54	7728.00	7728.00	0.00	0.04	7728.00	7728.00	0.00	5.83	7728.00	7728.00	0.00	10.95	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	7728.00	7728.00	0.00	124.13	7250	19	···

Table 52 Detailed results for problem NLUFLP-A-holmberg, cost functions f_7

	GUROBI				**SCIP**				**COUENNE**				**NAIVE**					**CN24**						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	4507.16	4507.16	**0.00**	**0.45**	4507.16	4507.16	**0.00**	9.95	3192.24	4811.95	33.66	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4490.14	4517.70	0.61	3601.03	6250	13	···
p10	3939.69	3939.69	**0.00**	**0.58**	3939.69	3939.69	**0.00**	15.77	3022.56	4024.92	24.90	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3921.70	3949.22	0.70	3601.03	6589	14	···
p11	5078.86	5078.86	**0.00**	**0.87**	5078.86	5078.86	**0.00**	12.93	3870.50	5221.03	25.87	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5030.95	5097.99	1.31	3601.03	5846	8	···
p12	6021.03	6021.03	**0.00**	**1.16**	6021.03	6021.03	**0.00**	12.36	4252.49	6162.03	30.99	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5985.20	6036.73	0.85	3601.03	5635	9	···
p13	4394.04	4394.04	**0.00**	**1.37**	4394.04	4394.04	**0.00**	30.62	2204.43	5260.10	58.09	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4327.05	4427.47	2.27	3601.04	10764	5	···
p14	3971.43	3971.43	**0.00**	**32.42**	3971.43	3971.43	**0.00**	242.14	1512.66	4602.17	67.13	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3909.78	4010.88	2.52	3601.03	10992	7	···
p15	5177.00	5177.00	**0.00**	**23.68**	5177.00	5177.00	**0.00**	439.61	2428.14	6476.50	62.51	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5058.99	5229.47	3.26	3601.04	10489	4	···
p16	6177.00	6177.00	**0.00**	**17.65**	6177.00	6177.00	**0.00**	227.06	3321.23	6459.54	48.58	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	6085.71	6233.52	2.37	3601.04	10779	6	···
p17	4394.04	4394.04	**0.00**	**1.41**	4394.04	4394.04	**0.00**	30.70	2201.80	5260.10	58.14	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4327.05	4427.47	2.27	3601.04	10764	5	···
p18	3971.43	3971.43	**0.00**	**37.71**	3971.43	3971.43	**0.00**	266.51	1512.66	4602.17	67.13	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3904.55	4011.16	2.66	3601.03	10894	6	···
p19	5177.00	5177.00	**0.00**	**25.85**	5177.00	5177.00	**0.00**	443.25	2429.78	6476.50	62.48	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5058.99	5229.47	3.26	3601.03	10489	4	···
p2	3939.69	3939.69	**0.00**	**0.51**	3939.69	3939.69	**0.00**	17.39	3022.58	4024.92	24.90	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3921.70	3949.22	0.70	3601.03	6589	14	···
p20	6177.00	6177.00	**0.00**	**18.80**	6177.00	6177.00	**0.00**	220.85	3336.32	6459.54	48.35	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	6053.07	6233.52	2.89	3601.03	10632	5	···
p21	4394.04	4394.04	**0.00**	**1.33**	4394.04	4394.04	**0.00**	30.48	2201.80	5260.10	58.14	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4327.05	4427.47	2.27	3601.04	10764	5	···
p22	3971.43	3971.43	**0.00**	**36.45**	3971.43	3971.43	**0.00**	218.02	1512.79	4602.17	67.13	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3904.55	4011.16	2.66	3601.04	10894	6	···
p23	5177.00	5177.00	**0.00**	**26.04**	5177.00	5177.00	**0.00**	456.69	2428.14	6476.50	62.51	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5058.99	5229.47	3.26	3601.03	10489	4	···
p24	6177.00	6177.00	**0.00**	**20.37**	6177.00	6177.00	**0.00**	247.06	3321.23	6459.54	48.58	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	6085.71	6233.52	2.37	3601.03	10779	6	···
p25	6108.89	6108.89	**0.00**	**5.21**	6108.89	6108.89	**0.00**	474.18	2264.11	12556.62	81.97	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.15	45000	1	···
p26	5306.89	5306.89	**0.00**	**7.26**	5306.89	5306.89	**0.00**	612.16	2158.43	10363.81	79.17	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.18	45000	1	···
p27	6710.56	6710.56	**0.00**	**6.70**	6710.56	6710.56	**0.00**	800.41	2386.73	13750.36	82.64	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	6477.04	6790.83	4.62	3601.07	45311	2	···
p28	7910.56	7910.56	**0.00**	**10.79**	7910.56	7910.56	**0.00**	290.45	2649.61	16814.28	84.24	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	7677.04	7990.83	3.93	3601.13	45311	2	···
p29	6108.89	6108.89	**0.00**	**4.97**	6108.89	6108.89	**0.00**	436.99	2264.11	12556.62	81.97	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.16	45000	1	···
p3	5078.86	5078.86	**0.00**	**0.96**	5078.86	5078.86	**0.00**	87.72	3870.33	5221.03	25.87	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5030.95	5097.99	1.31	3601.03	5846	8	···
p30	5306.89	5306.89	**0.00**	**7.00**	5306.89	5306.89	**0.00**	572.70	2158.43	10363.81	79.17	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.25	45000	1	···
p31	6710.56	6710.56	**0.00**	**8.02**	6710.56	6710.56	**0.00**	751.05	2386.73	13750.36	82.64	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	6477.04	6790.83	4.62	3601.08	45311	2	···
p32	7910.56	7910.56	**0.00**	**11.32**	7910.56	7910.56	**0.00**	374.13	2649.61	16814.28	84.24	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	7677.04	7990.83	3.93	3601.13	45311	2	···
p33	6108.89	6108.89	**0.00**	**5.82**	6108.89	6108.89	**0.00**	538.57	2264.11	12556.62	81.97	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.17	45000	1	···
p34	5306.89	5306.89	**0.00**	**7.24**	5306.89	5306.89	**0.00**	640.93	2158.43	10363.81	79.17	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.19	45000	1	···
p35	6710.56	6710.56	**0.00**	**7.73**	6710.56	6710.56	**0.00**	872.75	2386.73	13750.36	82.64	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	6477.04	6790.83	4.62	3601.15	45311	2	···
p36	7910.56	7910.56	**0.00**	**11.07**	7910.56	7910.56	**0.00**	310.75	2649.61	16814.28	84.24	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	7677.04	7990.83	3.93	3601.14	45311	2	···
p37	6108.89	6108.89	**0.00**	**6.30**	6108.89	6108.89	**0.00**	466.49	2264.11	12556.62	81.97	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.18	45000	1	···
p38	5306.89	5306.89	**0.00**	**7.40**	5306.89	5306.89	**0.00**	631.72	2158.43	10363.81	79.17	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.18	45000	1	···
p39	6710.56	6710.56	**0.00**	**10.06**	6710.56	6710.56	**0.00**	851.27	2386.73	13750.36	82.64	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	6477.04	6790.83	4.62	3601.08	45311	2	···
p4	6021.03	6021.03	**0.00**	**1.13**	6021.03	6021.03	**0.00**	20.69	4247.86	6162.03	31.06	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	5985.20	6036.73	0.85	3601.03	5635	9	···
p40	7910.56	7910.56	**0.00**	**10.92**	7910.56	7910.56	**0.00**	294.47	2649.61	16814.28	84.24	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	7677.04	7990.83	3.93	3601.09	45311	2	···
p41	3450.00	3450.00	**0.00**	**1.69**	3450.00	3450.00	**0.00**	26.23	2031.51	4048.61	49.82	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3349.60	3498.59	4.26	3601.03	9348	2	···
p42	3170.88	3170.88	**0.00**	**3.06**	3170.88	3170.88	**0.00**	98.73	1399.77	4837.29	71.06	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3077.38	3211.18	4.17	3601.04	16304	2	···
p43	2800.94	2800.94	**0.00**	**4.66**	2800.94	2800.94	**0.00**	65.99	1065.90	4959.73	78.51	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2728.54	2836.40	3.80	3601.05	21231	2	···
p44	3856.63	3856.63	**0.00**	**1.38**	3856.63	3856.63	**0.00**	20.92	2713.41	4166.19	34.87	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3730.66	3921.14	4.86	3601.03	9351	2	···
p45	3557.77	3557.77	**0.00**	**6.13**	3557.77	3557.77	**0.00**	131.59	1559.80	5432.20	71.29	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3442.83	3604.95	4.50	3601.05	16274	2	···
p46	3224.52	3224.52	**0.00**	**10.08**	3224.52	3224.52	**0.00**	176.13	1139.69	5222.34	78.18	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.09	21000	1	···
p47	3671.58	3671.58	**0.00**	**1.49**	3671.58	3671.58	**0.00**	88.17	2569.37	3811.90	32.60	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3550.22	3706.84	4.23	3601.03	9526	3	···
p48	3168.15	3168.15	**0.00**	**6.45**	3168.15	3168.15	**0.00**	71.46	1431.19	5003.95	71.40	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3063.33	3197.18	4.19	3601.04	16262	2	···
p49	2989.64	2989.64	**0.00**	**23.68**	2989.64	2989.64	**0.00**	278.30	1061.64	5508.81	80.73	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2898.27	3019.86	4.03	3601.08	21419	3	···
p5	4507.16	4507.16	**0.00**	**0.45**	4507.16	4507.16	**0.00**	13.82	3193.65	4811.95	33.63	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4488.45	4517.70	0.65	3601.03	6170	12	···
p50	4116.12	4116.12	**0.00**	**2.26**	4116.12	4116.12	**0.00**	22.08	2403.52	4778.82	49.70	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3984.81	4170.82	4.46	3601.03	10503	3	···
p51	3908.24	3908.24	**0.00**	**5.92**	3908.24	3908.24	**0.00**	70.62	1573.58	6200.21	74.62	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.08	20000	1	···
p52	4626.02	4626.02	**0.00**	**1.93**	4626.02	4626.02	**0.00**	27.62	3176.00	5225.41	39.22	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4484.07	4705.26	4.70	3601.03	10309	2	···
p53	4428.87	4428.87	**0.00**	**4.58**	4428.87	4428.87	**0.00**	142.72	1861.36	7019.10	73.48	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4288.89	4488.78	4.45	3601.09	20371	2	···
p54	4405.98	4405.98	**0.00**	**1.83**	4405.98	4405.98	**0.00**	49.63	2652.18	4906.01	45.94	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4274.09	4466.10	4.30	3601.03	10331	2	···

Table 53 Detailed results for problem NLUFLP-A-holmberg, cost functions f_8

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	8332.00	8332.00	**0.00**	**0.05**	8332.00	8332.00	**0.00**	7.13	8332.00	8332.00	**0.00**	32.76	−∞	+∞	100.00	3600.01	+∞	8331.99	8332.00	**0.00**	47.94	5058	13	···
p10	7580.00	7580.00	**0.00**	**0.04**	7580.00	7580.00	**0.00**	6.29	7580.00	7580.00	**0.00**	109.95	−∞	+∞	100.00	3600.01	+∞	7580.00	7580.00	**0.00**	39.64	5470	12	···
p11	8806.00	8806.00	**0.00**	**0.04**	8806.00	8806.00	**0.00**	5.03	8806.00	8806.00	**0.00**	102.48	−∞	+∞	100.00	3600.01	+∞	8805.99	8806.00	**0.00**	59.21	5057	12	···
p12	9806.00	9806.00	**0.00**	**0.05**	9806.00	9806.00	**0.00**	4.60	9806.00	9806.00	**0.00**	28.09	−∞	+∞	100.00	3600.01	+∞	9806.00	9806.00	**0.00**	69.26	5076	13	···
p13	8049.00	8049.00	**0.00**	**0.12**	8049.00	8049.00	**0.00**	6.77	8049.00	8049.00	**0.00**	176.31	−∞	+∞	100.00	3600.01	+∞	8049.00	8049.00	**0.00**	173.79	9650	13	···
p14	7092.00	7092.00	**0.00**	**0.10**	7092.00	7092.00	**0.00**	5.83	7092.00	7092.00	**0.00**	486.79	−∞	+∞	100.00	3600.01	+∞	7092.00	7092.00	**0.00**	177.42	9888	14	···
p15	8735.00	8735.00	**0.00**	**0.10**	8735.00	8735.00	**0.00**	6.52	8735.00	8735.00	**0.00**	221.31	−∞	+∞	100.00	3600.01	+∞	8735.00	8735.00	**0.00**	238.21	9691	15	···
p16	10168.00	10168.00	**0.00**	**0.15**	10168.00	10168.00	**0.00**	11.05	10168.00	10168.00	**0.00**	80.47	−∞	+∞	100.00	3600.01	+∞	10168.00	10168.00	**0.00**	385.92	9473	16	···
p17	8049.00	8049.00	**0.00**	**0.12**	8049.00	8049.00	**0.00**	6.90	8049.00	8049.00	**0.00**	177.85	−∞	+∞	100.00	3600.01	+∞	8049.00	8049.00	**0.00**	184.31	9650	13	···
p18	7092.00	7092.00	**0.00**	**0.09**	7092.00	7092.00	**0.00**	5.78	7092.00	7092.00	**0.00**	489.82	−∞	+∞	100.00	3600.01	+∞	7092.00	7092.00	**0.00**	167.15	9888	14	···
p19	8735.00	8735.00	**0.00**	**0.12**	8735.00	8735.00	**0.00**	6.57	8735.00	8735.00	**0.00**	219.87	−∞	+∞	100.00	3600.01	+∞	8735.00	8735.00	**0.00**	272.58	9691	15	···
p2	7580.00	7580.00	**0.00**	**0.04**	7580.00	7580.00	**0.00**	6.96	7580.00	7580.00	**0.00**	110.62	−∞	+∞	100.00	3600.01	+∞	7580.00	7580.00	**0.00**	44.37	5470	12	···
p20	10168.00	10168.00	**0.00**	**0.14**	10168.00	10168.00	**0.00**	11.24	10168.00	10168.00	**0.00**	79.37	−∞	+∞	100.00	3600.01	+∞	10168.00	10168.00	**0.00**	380.79	9473	16	···
p21	8049.00	8049.00	**0.00**	**0.12**	8049.00	8049.00	**0.00**	6.81	8049.00	8049.00	**0.00**	179.34	−∞	+∞	100.00	3600.01	+∞	8049.00	8049.00	**0.00**	194.01	9650	13	···
p22	7092.00	7092.00	**0.00**	**0.12**	7092.00	7092.00	**0.00**	5.74	7092.00	7092.00	**0.00**	488.07	−∞	+∞	100.00	3600.01	+∞	7092.00	7092.00	**0.00**	188.52	9888	14	···
p23	8735.00	8735.00	**0.00**	**0.12**	8735.00	8735.00	**0.00**	6.53	8735.00	8735.00	**0.00**	222.20	−∞	+∞	100.00	3600.01	+∞	8735.00	8735.00	**0.00**	268.53	9691	15	···
p24	10168.00	10168.00	**0.00**	**0.15**	10168.00	10168.00	**0.00**	11.07	10168.00	10168.00	**0.00**	81.49	−∞	+∞	100.00	3600.01	+∞	10168.00	10168.00	**0.00**	390.17	9473	16	···
p25	11247.00	11247.00	**0.00**	**1.07**	11247.00	11247.00	**0.00**	84.76	11247.03	11247.03	**0.00**	1638.51	−∞	+∞	100.00	3600.01	+∞	11246.99	11247.00	**0.00**	2345.64	39814	17	···
p26	10543.00	10543.00	**0.00**	**1.29**	10543.00	10543.00	**0.00**	19.12	10543.03	10543.03	**0.00**	1659.61	−∞	+∞	100.00	3600.01	+∞	10542.99	10543.00	**0.00**	2405.75	40502	17	···
p27	11816.00	11816.00	**0.00**	**1.15**	11816.00	11816.00	**0.00**	74.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	11815.99	11816.00	**0.00**	1945.69	39849	17	···
p28	13016.00	13016.00	**0.00**	**1.96**	13016.00	13016.00	**0.00**	18.56	13016.03	13016.03	**0.00**	1648.65	−∞	+∞	100.00	3600.01	+∞	13015.99	13016.00	**0.00**	3271.71	39853	18	···
p29	11247.00	11247.00	**0.00**	**1.08**	11247.00	11247.00	**0.00**	85.46	11247.03	11247.03	**0.00**	1705.57	−∞	+∞	100.00	3600.01	+∞	11246.99	11247.00	**0.00**	2657.67	39814	17	···
p3	8806.00	8806.00	**0.00**	**0.05**	8806.00	8806.00	**0.00**	10.91	8806.00	8806.00	**0.00**	103.30	−∞	+∞	100.00	3600.01	+∞	8805.99	8806.00	**0.00**	67.16	5057	12	···
p30	10543.00	10543.00	**0.00**	**1.32**	10543.00	10543.00	**0.00**	19.47	10543.03	10543.03	**0.00**	1665.28	−∞	+∞	100.00	3600.01	+∞	10542.99	10543.00	**0.00**	2404.06	40502	17	···
p31	11816.00	11816.00	**0.00**	**1.52**	11816.00	11816.00	**0.00**	100.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	11815.99	11816.00	**0.00**	2776.34	39849	17	···
p32	13016.00	13016.00	**0.00**	**1.91**	13016.00	13016.00	**0.00**	35.20	13016.03	13016.03	**0.00**	1679.97	−∞	+∞	100.00	3600.01	+∞	13015.99	13016.00	**0.00**	2437.90	39853	18	···
p33	11247.00	11247.00	**0.00**	**1.32**	11247.00	11247.00	**0.00**	98.26	11247.03	11247.03	**0.00**	1710.88	−∞	+∞	100.00	3600.01	+∞	11246.99	11247.00	**0.00**	2559.42	39814	17	···
p34	10543.00	10543.00	**0.00**	**1.30**	10543.00	10543.00	**0.00**	33.91	10543.03	10543.03	**0.00**	1687.31	−∞	+∞	100.00	3600.01	+∞	10542.99	10543.00	**0.00**	2697.41	40502	17	···
p35	11816.00	11816.00	**0.00**	**1.38**	11816.00	11816.00	**0.00**	85.69	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	11815.99	11816.00	**0.00**	2494.60	39849	17	···
p36	13016.00	13016.00	**0.00**	**1.79**	13016.00	13016.00	**0.00**	19.94	13016.03	13016.03	**0.00**	1673.88	−∞	+∞	100.00	3600.01	+∞	13015.99	13016.00	**0.00**	2879.44	39853	18	···
p37	11247.00	11247.00	**0.00**	**1.16**	11247.00	11247.00	**0.00**	91.34	11247.03	11247.03	**0.00**	1639.00	−∞	+∞	100.00	3600.01	+∞	11246.99	11247.00	**0.00**	2558.02	39814	17	···
p38	10543.00	10543.00	**0.00**	**1.52**	10543.00	10543.00	**0.00**	18.88	10543.03	10543.03	**0.00**	1651.12	−∞	+∞	100.00	3600.01	+∞	10542.99	10543.00	**0.00**	2545.13	40502	17	···
p39	11816.00	11816.00	**0.00**	**1.85**	11816.00	11816.00	**0.00**	93.64	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	11815.99	11816.00	**0.00**	2373.23	39849	17	···
p4	9806.00	9806.00	**0.00**	**0.05**	9806.00	9806.00	**0.00**	10.80	9806.00	9806.00	**0.00**	30.08	−∞	+∞	100.00	3600.01	+∞	9806.00	9806.00	**0.00**	74.06	5076	13	···
p40	13016.00	13016.00	**0.00**	**1.90**	13016.00	13016.00	**0.00**	20.37	13016.03	13016.03	**0.00**	1666.15	−∞	+∞	100.00	3600.01	+∞	13015.99	13016.00	**0.00**	2937.05	39853	18	···
p41	6151.00	6151.00	**0.00**	**0.10**	6151.00	6151.00	**0.00**	7.06	6151.00	6151.00	**0.00**	134.38	−∞	+∞	100.00	3600.01	+∞	6151.00	6151.00	**0.00**	393.74	9145	19	···
p42	5379.00	5379.00	**0.00**	**0.22**	5379.00	5379.00	**0.00**	7.67	5379.00	5379.00	**0.00**	1706.85	−∞	+∞	100.00	3600.01	+∞	5379.00	5379.00	**0.00**	687.97	14206	14	···
p43	4865.00	4865.00	**0.00**	**0.52**	4865.00	4865.00	**0.00**	9.31	4865.00	4865.00	**0.00**	1810.17	−∞	+∞	100.00	3600.01	+∞	4865.00	4865.00	**0.00**	1160.62	18331	17	···
p44	6619.00	6619.00	**0.00**	**0.09**	6619.00	6619.01	**0.00**	11.47	6619.00	6619.00	**0.00**	44.37	−∞	+∞	100.00	3600.01	+∞	6619.00	6619.00	**0.00**	288.85	10198	17	···
p45	5963.00	5963.00	**0.00**	**0.27**	5963.00	5963.00	**0.00**	11.02	5963.00	5963.00	**0.00**	999.53	−∞	+∞	100.00	3600.01	+∞	5963.00	5963.00	**0.00**	749.03	14833	19	···
p46	5255.00	5255.00	**0.00**	**0.41**	5255.00	5255.00	**0.00**	10.85	5255.00	5255.00	**0.00**	417.84	−∞	+∞	100.00	3600.01	+∞	5255.00	5255.00	**0.00**	1004.61	18938	18	···
p47	5673.00	5673.00	**0.00**	**0.08**	5673.00	5673.00	**0.00**	41.59	5673.00	5673.00	**0.00**	145.90	−∞	+∞	100.00	3600.01	+∞	5673.00	5673.00	**0.00**	107.79	10859	10	···
p48	5358.00	5358.00	**0.00**	**0.22**	5358.00	5358.00	**0.00**	9.54	5358.00	5358.00	**0.00**	917.75	−∞	+∞	100.00	3600.01	+∞	5358.00	5358.00	**0.00**	518.34	15480	13	···
p49	5102.00	5102.00	**0.00**	**0.68**	5102.00	5102.00	**0.00**	10.29	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5102.00	5102.00	**0.00**	734.10	19146	14	···
p5	8332.00	8332.00	**0.00**	**0.04**	8332.00	8332.00	**0.00**	11.78	8332.00	8332.00	**0.00**	32.01	−∞	+∞	100.00	3600.01	+∞	8331.99	8332.00	**0.00**	47.41	5058	13	···
p50	7307.00	7307.00	**0.00**	**0.12**	7307.00	7307.00	**0.00**	9.49	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7307.00	7307.00	**0.00**	174.89	9748	12	···
p51	6705.00	6705.00	**0.00**	**0.34**	6705.00	6705.00	**0.00**	13.21	6705.00	6705.00	**0.00**	384.83	−∞	+∞	100.00	3600.01	+∞	6705.00	6705.00	**0.00**	713.52	17782	16	···
p52	8166.00	8166.00	**0.00**	**0.11**	8166.00	8166.00	**0.00**	24.48	8166.00	8166.00	**0.00**	212.80	−∞	+∞	100.00	3600.01	+∞	8165.99	8166.00	**0.00**	212.91	10516	16	···
p53	7731.00	7731.00	**0.00**	**0.35**	7731.00	7731.00	**0.00**	17.87	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7730.99	7731.00	**0.00**	560.91	18514	16	···
p54	7728.00	7728.00	**0.00**	**0.08**	7728.00	7728.00	**0.00**	14.96	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7728.00	7728.00	**0.00**	297.25	10219	17	···

Table 54 Detailed results for problem NLUFLP-A-holmberg, cost functions f_9

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	928.77	3382.70	72.54	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	525.57	6659.78	92.11	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3224.46	3317.49	**2.80**	3601.03	5070	4	···
p10	681.49	3043.58	77.61	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	443.27	5719.01	92.25	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2804.24	2840.95	**1.29**	3601.03	5467	6	···
p11	1067.61	4046.98	73.62	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	687.11	7319.01	90.61	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3832.16	3921.38	**2.28**	3601.03	5028	4	···
p12	1462.76	5091.08	71.27	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	911.66	8919.01	89.78	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4832.16	4921.38	**1.81**	3601.02	5028	4	···
p13	684.35	4001.94	82.90	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	7197.32	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3057.92	3108.20	**1.62**	3601.03	10010	6	···
p14	626.70	3407.05	81.61	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	330.97	5234.32	93.68	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2775.35	2819.51	**1.57**	3601.04	10017	6	···
p15	1025.13	5058.03	79.73	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	7834.32	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3831.44	3889.42	**1.49**	3601.03	9896	6	···
p16	1425.43	5609.39	74.59	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	10434.32	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4817.26	4889.42	**1.48**	3601.03	9767	5	···
p17	681.14	4040.12	83.14	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	7197.32	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3057.92	3108.20	**1.62**	3601.04	10010	6	···
p18	627.45	3407.05	81.58	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	330.97	5234.32	93.68	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2767.94	2819.51	**1.83**	3601.03	9874	5	···
p19	1022.56	5079.49	79.87	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	7834.32	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3831.44	3889.42	**1.49**	3601.03	9896	6	···
p2	679.99	3043.58	77.66	3600.01	644.02	3476.45	81.47	3603.12	443.27	5719.01	92.25	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2804.24	2840.95	**1.29**	3601.04	5468	6	···
p20	1425.00	5609.39	74.60	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	10434.32	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4817.25	4889.42	**1.48**	3601.04	9768	5	···
p21	681.21	4040.12	83.14	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	7197.32	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3057.92	3108.20	**1.62**	3601.05	10010	6	···
p22	625.69	3443.31	81.83	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	330.97	5234.32	93.68	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2775.35	2819.51	**1.57**	3601.04	10016	6	···
p23	1024.28	5058.03	79.75	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	7834.32	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3831.44	3889.42	**1.49**	3601.03	9896	6	···
p24	1424.92	5609.39	74.60	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	10434.32	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4817.26	4889.42	**1.48**	3601.03	9766	5	···
p25	442.70	14108.92	96.86	3600.07	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4356.41	4543.88	**4.13**	3601.06	41013	2	···
p26	385.50	10855.46	96.45	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3697.43	3850.05	**3.96**	3601.11	41239	2	···
p27	577.01	11650.92	95.05	3600.06	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4919.70	5101.37	**3.56**	3601.07	41088	2	···
p28	784.02	15446.68	**94.92**	3600.04	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.11	40500	1	···
p29	442.70	14101.16	96.86	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4356.41	4543.88	**4.13**	3601.13	41013	2	···
p3	1067.68	4046.97	73.62	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	687.11	7319.01	90.61	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3832.16	3921.38	**2.28**	3601.03	5029	4	···
p30	385.50	10759.34	96.42	3600.03	438.92	5739.73	92.35	3603.37	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3697.43	3850.05	**3.96**	3601.10	41239	2	···
p31	577.38	11646.51	95.04	3600.04	659.04	6539.73	**89.92**	3603.48	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.13	40500	1	···
p32	784.52	15385.48	94.90	3600.03	857.17	7339.73	**88.32**	3606.29	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.10	40500	1	···
p33	442.70	14076.41	96.86	3600.03	559.45	16501.70	96.61	3606.60	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4356.41	4543.88	**4.13**	3601.12	41013	2	···
p34	385.50	10869.53	96.45	3600.05	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3697.43	3850.05	**3.96**	3601.13	41239	2	···
p35	567.49	11696.47	**95.15**	3600.34	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.12	40500	1	···
p36	784.02	15462.81	**94.93**	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.10	40500	1	···
p37	442.70	14076.41	96.86	3600.04	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4356.41	4543.88	**4.13**	3601.13	41013	2	···
p38	385.50	10869.53	96.45	3600.04	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3697.43	3850.05	**3.96**	3601.09	41239	2	···
p39	567.49	11696.47	**95.15**	3600.32	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	$-\infty$	$+\infty$	100.00	3601.13	40500	1	···
p4	1468.29	5091.06	71.16	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	911.66	8919.01	89.78	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4832.16	4921.38	**1.81**	3601.03	5028	4	···
p40	784.02	15387.46	94.90	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	6041.87	6274.42	**3.71**	3601.12	40972	2	···
p41	656.38	3356.23	80.44	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	336.79	6157.21	94.53	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2593.99	2644.65	**1.92**	3601.03	9129	4	···
p42	577.58	3423.45	83.13	3600.02	634.68	2657.14	76.11	3603.23	325.69	5863.64	94.45	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2360.74	2409.91	**2.04**	3601.07	15338	4	···
p43	409.56	4619.00	91.13	3600.02	633.83	2333.23	72.83	3603.36	321.53	8430.03	96.19	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2171.99	2205.47	**1.52**	3601.07	19678	4	···
p44	673.80	3000.38	77.54	3600.02	670.05	3314.47	79.78	3603.13	341.64	6886.20	95.04	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2782.91	2846.97	**2.25**	3601.04	9088	4	···
p45	632.47	3664.82	82.74	3600.03	645.67	3054.86	78.86	3603.27	$-\infty$	6077.33	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2613.15	2667.82	**2.05**	3601.04	15253	4	···
p46	483.79	3675.29	86.84	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	327.49	8568.34	96.18	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2392.12	2434.41	**1.74**	3601.06	19707	4	···
p47	717.60	3429.21	79.07	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	341.56	6588.83	94.82	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2735.97	2790.11	**1.94**	3601.03	9112	4	···
p48	595.46	4077.68	85.40	3600.02	529.14	2672.09	80.20	3603.35	$-\infty$	6004.44	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2385.43	2428.89	**1.79**	3601.03	15292	4	···
p49	428.32	4553.24	90.59	3600.02	419.08	2570.57	83.70	3603.44	325.28	8287.33	96.07	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2283.39	2350.13	**2.84**	3601.05	19433	3	···
p5	917.75	3383.04	72.87	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	572.08	6659.78	91.41	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3224.45	3317.42	**2.80**	3601.03	5064	4	···
p50	665.05	3108.02	78.60	3600.02	673.98	3600.26	81.28	3603.24	345.93	6868.83	94.96	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2937.43	3004.60	**2.24**	3601.03	10063	4	···
p51	644.78	3169.41	79.66	3600.02	590.81	3453.90	82.89	3603.50	338.61	6244.32	94.58	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	2805.30	2912.83	**3.69**	3601.04	18745	3	···
p52	667.32	3561.53	81.26	3600.02	679.95	4327.77	84.29	3607.53	352.94	8427.43	95.81	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3301.15	3377.89	**2.27**	3601.04	10259	4	···
p53	593.59	5335.60	88.87	3600.01	616.95	3949.74	84.38	3603.32	345.19	6596.13	94.77	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3091.34	3200.96	**3.42**	3601.04	18857	3	···
p54	771.25	3225.75	76.09	3600.02	681.48	3843.55	82.27	3608.15	352.78	7517.82	95.31	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	3087.87	3214.30	**3.93**	3601.03	9788	3	···

Table 55 Detailed results for problem NLUFLP-A-holmberg, cost functions f_{10}

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	299.33	299.33	0.00	0.07	299.33	299.33	0.00	4.97	299.33	299.33	0.00	0.87	299.33	299.33	0.00	31.77	8280	299.33	299.33	0.00	3.15	90	5	···
10-20-10	302.53	302.54	0.00	0.04	302.54	302.54	0.00	2.80	302.54	302.54	0.00	0.31	302.54	302.54	0.00	46.48	8370	302.54	302.54	0.00	5.86	83	7	···
10-20-2	241.15	241.15	0.00	0.06	241.15	241.15	0.00	2.81	241.15	241.15	0.00	0.61	241.15	241.15	0.00	44.35	8470	241.15	241.15	0.00	6.59	108	7	···
10-20-3	277.48	277.48	0.00	0.05	277.48	277.48	0.00	4.71	277.48	277.48	0.00	4.11	277.48	277.48	0.00	35.36	8340	277.48	277.48	0.00	7.97	75	5	···
10-20-4	279.59	279.59	0.00	0.10	279.59	279.59	0.00	1056.20	279.59	279.59	0.00	3.44	279.59	279.59	0.00	39.92	8440	279.59	279.59	0.00	7.29	78	14	···
10-20-5	339.09	339.09	0.00	0.16	$-\infty$	$+\infty$	100.0	$+\infty$	339.09	339.09	0.00	10.99	339.09	339.09	0.00	38.92	8440	339.09	339.09	0.00	9.92	125	7	···
10-20-6	305.88	305.88	0.00	0.07	305.88	305.88	0.00	39.12	305.88	305.88	0.00	1.91	305.88	305.88	0.00	37.46	8440	305.88	305.88	0.00	7.88	92	9	···
10-20-7	236.77	236.77	0.00	0.04	236.77	236.77	0.00	3.03	236.77	236.77	0.00	3.84	236.77	236.77	0.00	39.52	8400	236.77	236.77	0.00	8.26	92	6	···
10-20-8	303.29	303.29	0.00	0.10	303.29	303.29	0.00	108.42	303.29	303.29	0.00	5.10	303.29	303.29	0.00	36.21	8530	303.29	303.29	0.00	6.17	93	13	···
10-20-9	304.67	304.67	0.00	0.08	304.67	304.67	0.00	12.09	304.67	304.67	0.00	4.65	304.67	304.67	0.00	49.65	8480	304.67	304.67	0.00	9.56	69	5	···
20-40-1	429.88	429.88	0.00	0.10	429.88	429.88	0.00	4.26	429.88	429.88	0.00	7.18	429.88	429.88	0.00	78.22	15740	429.88	429.88	0.00	4.35	172	9	···
20-40-10	438.85	438.85	0.00	0.08	438.85	438.85	0.00	4.71	438.85	438.85	0.00	12.89	438.85	438.85	0.00	71.36	16020	438.85	438.85	0.00	4.01	130	6	···
20-40-2	380.48	380.48	0.00	0.08	380.48	380.48	0.00	3.86	380.48	380.48	0.00	6.24	380.48	380.48	0.00	74.89	15900	380.48	380.48	0.00	3.77	117	6	···
20-40-3	409.42	409.42	0.00	0.14	409.42	409.42	0.00	89.32	409.42	409.42	0.00	9.38	409.42	409.42	0.00	88.67	15960	409.42	409.42	0.00	4.56	175	7	···
20-40-4	447.89	447.89	0.00	0.07	447.88	447.89	0.00	4.03	447.89	447.89	0.00	10.25	447.89	447.89	0.00	86.86	15800	447.89	447.89	0.00	4.12	153	5	···
20-40-5	390.04	390.04	0.00	0.07	390.04	390.04	0.00	4.05	390.04	390.04	0.00	19.71	390.04	390.04	0.00	84.68	16020	390.04	390.04	0.00	4.31	167	6	···
20-40-6	433.75	433.75	0.00	0.07	433.75	433.75	0.00	4.82	433.75	433.75	0.00	19.82	433.75	433.75	0.00	72.50	16000	433.75	433.75	0.00	4.90	154	7	···
20-40-7	442.71	442.71	0.00	0.06	442.71	442.71	0.00	3.75	442.71	442.71	0.00	6.83	442.71	442.71	0.00	70.34	15600	442.70	442.71	0.00	3.96	139	5	···
20-40-8	424.64	424.64	0.00	0.08	424.64	424.64	0.00	4.11	424.64	424.64	0.00	3.58	424.64	424.64	0.00	82.50	16000	424.64	424.64	0.00	4.23	173	6	···
20-40-9	403.39	403.39	0.00	0.07	403.39	403.39	0.00	4.52	403.39	403.39	0.00	21.56	403.39	403.39	0.00	70.63	16000	403.39	403.39	0.00	4.38	138	6	···
30-60-1	526.41	526.41	0.00	0.13	526.41	526.41	0.00	5.48	526.41	526.41	0.00	22.56	526.41	526.41	0.00	98.83	22470	526.41	526.41	0.00	6.60	219	9	···
30-60-10	398.31	398.31	0.00	0.14	398.31	398.31	0.00	5.36	398.31	398.31	0.00	64.02	398.31	398.31	0.00	119.91	23010	398.31	398.31	0.00	6.08	222	7	···
30-60-2	464.34	464.34	0.00	0.15	464.34	464.34	0.00	7.00	464.34	464.34	0.00	33.87	464.34	464.34	0.00	116.80	22620	464.34	464.34	0.00	5.86	232	7	···
30-60-3	492.00	492.00	0.00	0.21	492.00	492.00	0.00	5.03	530.34	530.34	0.00	7.69	492.00	492.00	0.00	130.49	23040	492.00	492.00	0.00	6.11	192	7	···
30-60-4	518.67	518.67	0.00	0.13	518.67	518.67	0.00	10.62	518.67	518.67	0.00	43.95	518.67	518.67	0.00	108.61	22650	518.67	518.67	0.00	6.49	223	7	···
30-60-5	487.68	487.68	0.00	0.13	487.68	487.68	0.00	4.48	487.68	487.68	0.00	65.07	487.68	487.68	0.00	120.88	22920	487.68	487.68	0.00	5.86	210	7	···
30-60-6	518.46	518.46	0.00	0.11	518.46	518.46	0.00	5.91	518.46	518.46	0.00	34.82	518.46	518.46	0.00	130.22	22890	518.46	518.46	0.00	5.54	185	7	···
30-60-7	581.97	581.97	0.00	0.13	581.97	581.97	0.00	4.49	581.97	581.97	0.00	69.84	581.97	581.97	0.00	90.15	22320	581.97	581.97	0.00	6.57	217	8	···
30-60-8	537.92	537.92	0.00	0.14	537.92	537.92	0.00	4.95	537.92	537.92	0.00	36.63	537.92	537.92	0.00	112.55	22590	537.92	537.92	0.00	6.38	213	7	···
30-60-9	502.96	502.96	0.00	0.12	502.96	502.96	0.00	4.22	502.96	502.96	0.00	36.38	502.96	502.96	0.00	128.41	22950	502.96	502.96	0.00	6.19	208	7	···
40-80-1	630.98	630.98	0.00	0.28	630.98	630.98	0.00	20.63	630.98	630.98	0.00	176.51	630.98	630.98	0.00	145.07	28920	630.98	630.98	0.00	8.75	273	7	···
40-80-10	622.32	622.32	0.00	0.23	622.32	622.32	0.00	30.41	622.32	622.32	0.00	59.60	622.32	622.32	0.00	129.46	29800	622.32	622.32	0.00	8.05	300	7	···
40-80-2	636.17	636.17	0.00	0.19	636.17	636.17	0.00	28.75	636.17	636.17	0.00	94.20	636.17	636.17	0.00	121.67	29120	636.17	636.17	0.00	7.99	253	8	···
40-80-3	567.04	567.04	0.00	0.23	567.04	567.04	0.00	33.45	567.04	567.04	0.00	93.95	567.04	567.04	0.00	133.43	29680	567.04	567.04	0.00	8.08	283	7	···
40-80-4	598.97	598.97	0.00	0.23	598.97	598.97	0.00	31.15	598.97	598.97	0.00	100.52	598.97	598.97	0.00	121.64	28960	598.97	598.97	0.00	8.07	255	7	···
40-80-5	612.72	612.72	0.00	0.21	612.72	612.72	0.00	46.99	612.72	612.72	0.00	97.80	612.72	612.72	0.00	120.82	29640	612.72	612.72	0.00	7.65	251	8	···
40-80-6	570.84	570.84	0.00	0.18	570.84	570.84	0.00	30.01	570.84	570.84	0.00	80.56	570.84	570.84	0.00	119.74	29200	570.84	570.84	0.00	7.92	258	7	···
40-80-7	655.43	655.43	0.00	0.25	655.43	655.43	0.00	47.00	655.43	655.43	0.00	169.90	655.43	655.43	0.00	126.23	28960	655.43	655.43	0.00	8.65	263	9	···
40-80-8	584.84	584.84	0.00	0.25	584.84	584.84	0.00	37.57	584.84	584.84	0.00	63.85	584.84	584.84	0.00	130.07	29160	584.84	584.84	0.00	8.35	295	9	···
40-80-9	589.11	589.11	0.00	0.24	589.11	589.11	0.00	32.32	589.11	589.11	0.00	180.16	589.11	589.11	0.00	127.05	29200	589.11	589.11	0.00	7.83	274	7	···
50-100-1	692.13	692.13	0.00	0.25	692.12	692.12	0.00	63.11	692.13	692.13	0.00	191.47	692.13	692.13	0.00	164.25	35200	692.12	692.13	0.00	9.75	315	7	···
50-100-10	722.11	722.11	0.00	0.26	722.11	722.11	0.00	32.60	722.11	722.11	0.00	190.07	722.11	722.11	0.00	201.35	36100	722.11	722.11	0.00	11.66	307	10	···
50-100-2	720.24	720.24	0.00	0.26	720.24	720.24	0.00	50.16	720.24	720.24	0.00	217.96	720.24	720.24	0.00	152.61	35500	720.24	720.24	0.00	10.93	327	7	···
50-100-3	689.91	689.91	0.00	0.32	689.91	689.91	0.00	44.62	689.91	689.91	0.00	309.68	689.91	689.91	0.00	205.77	35900	689.91	689.91	0.00	12.73	308	8	···
50-100-4	643.06	643.06	0.00	0.28	643.06	643.06	0.00	57.17	643.06	643.06	0.00	367.58	643.06	643.06	0.00	144.26	35150	643.06	643.06	0.00	9.98	309	7	···
50-100-5	632.44	632.44	0.00	0.31	632.44	632.44	0.00	53.64	632.44	632.44	0.00	180.45	632.44	632.44	0.00	158.08	35800	632.44	632.44	0.00	10.28	316	7	···
50-100-6	660.71	660.71	0.00	0.31	660.71	660.71	0.00	53.99	660.71	660.71	0.00	323.04	660.71	660.71	0.00	156.22	35350	660.71	660.71	0.00	11.42	301	8	···
50-100-7	762.64	762.64	0.00	0.29	762.64	762.64	0.00	301.58	762.64	762.64	0.00	333.03	762.64	762.64	0.00	224.36	35100	762.64	762.64	0.00	9.67	290	7	···
50-100-8	664.04	664.04	0.00	0.38	664.04	664.04	0.00	276.77	664.04	664.04	0.00	342.46	664.04	664.04	0.00	159.69	35200	664.04	664.04	0.00	11.64	343	9	···
50-100-9	722.86	722.86	0.00	0.30	722.86	722.86	0.00	56.43	722.86	722.86	0.00	319.16	722.86	722.86	0.00	266.93	35500	722.86	722.86	0.00	11.42	295	8	···

Table 56 Detailed results for problem NLUFLP-W-gunluk, cost functions f_1

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	279.41	279.41	**0.00**	**0.06**	279.41	279.41	**0.00**	11.29	279.41	279.41	**0.00**	5.07	279.41	279.41	**0.00**	196.22	9810	279.41	279.41	**0.00**	25.02	105	10	···
10-20-10	283.70	283.70	**0.00**	**0.06**	283.70	283.70	**0.00**	49.31	283.70	283.70	**0.00**	10.45	283.70	283.70	**0.00**	207.33	9810	283.70	283.70	**0.00**	45.98	91	9	···
10-20-2	228.08	228.08	**0.00**	**0.07**	228.08	228.08	**0.00**	10.63	228.08	228.08	**0.00**	6.28	228.08	228.08	**0.00**	195.02	9810	228.08	228.08	**0.00**	30.40	98	13	···
10-20-3	265.39	265.39	**0.00**	**0.07**	265.39	265.39	**0.00**	23.57	265.39	265.39	**0.00**	10.59	265.39	265.39	**0.00**	203.61	9810	265.39	265.39	**0.00**	32.38	91	8	···
10-20-4	252.46	252.46	**0.00**	**0.04**	252.46	252.46	**0.00**	6.73	252.46	252.46	**0.00**	5.91	252.46	252.46	**0.00**	197.59	9810	252.46	252.46	**0.00**	26.98	90	8	···
10-20-5	307.11	307.11	**0.00**	**0.04**	307.11	307.11	**0.00**	7.00	307.11	307.11	**0.00**	4.12	307.11	307.11	**0.00**	196.82	9810	307.11	307.11	**0.00**	15.28	93	9	···
10-20-6	266.19	266.19	**0.00**	**0.11**	266.19	266.19	**0.00**	10.48	266.19	266.19	**0.00**	10.78	266.19	266.19	**0.00**	197.26	9810	266.19	266.19	**0.00**	15.68	107	11	···
10-20-7	225.64	225.64	**0.00**	**0.05**	225.64	225.64	**0.00**	6.11	225.64	225.64	**0.00**	1.90	225.64	225.64	**0.00**	209.44	9810	225.64	225.64	**0.00**	47.13	88	8	···
10-20-8	271.78	271.78	**0.00**	**0.12**	271.78	271.78	**0.00**	11.11	271.78	271.78	**0.00**	12.90	271.78	271.78	**0.00**	205.89	9810	271.78	271.78	**0.00**	23.99	111	11	···
10-20-9	277.45	277.45	**0.00**	**0.09**	277.45	277.45	**0.00**	7.93	277.45	277.45	**0.00**	9.19	277.45	277.45	**0.00**	215.77	9810	277.45	277.45	**0.00**	42.38	112	14	···
20-40-1	401.24	401.24	**0.00**	**0.19**	401.24	401.24	**0.00**	1192.28	401.24	401.24	**0.00**	33.45	401.24	401.24	**0.00**	411.03	19620	401.24	401.24	**0.00**	16.93	172	10	···
20-40-10	414.82	414.82	**0.00**	**0.32**	414.79	414.82	0.01	3602.67	414.82	414.82	**0.00**	57.22	414.82	414.82	**0.00**	413.96	19620	414.82	414.82	**0.00**	23.24	179	11	···
20-40-2	345.40	345.40	**0.00**	**0.32**	345.39	345.40	0.00	3602.27	345.40	345.40	**0.00**	26.57	345.40	345.40	**0.00**	406.82	19620	345.40	345.40	**0.00**	17.82	196	12	···
20-40-3	366.77	366.77	**0.00**	**0.21**	366.77	366.77	**0.00**	109.18	366.77	366.77	**0.00**	200.31	366.77	366.77	**0.00**	397.49	19620	366.77	366.77	**0.00**	33.47	204	11	···
20-40-4	419.35	419.35	**0.00**	**0.34**	419.35	419.35	**0.00**	62.02	419.35	419.35	**0.00**	41.94	419.35	419.35	**0.00**	374.58	19620	419.35	419.35	**0.00**	16.29	179	10	···
20-40-5	366.34	366.34	**0.00**	**0.26**	366.32	366.34	0.00	3602.29	366.34	366.34	**0.00**	56.00	366.34	366.34	**0.00**	398.15	19620	366.34	366.34	**0.00**	17.93	193	14	···
20-40-6	401.78	401.78	**0.00**	**0.35**	401.78	401.78	**0.00**	38.49	401.78	401.78	**0.00**	91.05	401.78	401.78	**0.00**	496.72	19620	401.78	401.78	**0.00**	17.31	160	10	···
20-40-7	410.65	410.65	**0.00**	**0.30**	410.65	410.65	**0.00**	40.77	410.65	410.65	**0.00**	49.97	410.65	410.65	**0.00**	401.98	19620	410.65	410.65	**0.00**	33.34	182	9	···
20-40-8	388.99	388.99	**0.00**	**0.20**	388.98	388.98	**0.00**	38.55	388.99	388.99	**0.00**	16.00	388.99	388.99	**0.00**	395.87	19620	388.98	388.99	**0.00**	16.40	180	10	···
20-40-9	375.75	375.75	**0.00**	**0.23**	375.75	375.75	**0.00**	129.50	375.75	375.75	**0.00**	62.28	375.75	375.75	**0.00**	424.04	19620	375.75	375.75	**0.00**	16.46	186	11	···
30-60-1	472.01	472.01	**0.00**	**0.87**	472.01	472.01	**0.00**	3602.74	472.01	472.01	**0.00**	267.93	472.01	472.01	**0.00**	724.11	29430	472.01	472.01	**0.00**	24.36	270	11	···
30-60-10	370.09	370.09	**0.00**	**0.19**	370.09	370.09	**0.00**	16.42	370.09	370.09	**0.00**	62.86	370.09	370.09	**0.00**	582.14	29430	370.09	370.09	**0.00**	21.29	265	10	···
30-60-2	411.44	411.44	**0.00**	**0.79**	411.43	411.44	0.00	3602.69	411.44	411.44	**0.00**	145.26	411.44	411.44	**0.00**	573.23	29430	411.44	411.44	**0.00**	23.78	277	10	···
30-60-3	451.19	451.19	**0.00**	**0.39**	451.15	451.19	0.01	3602.27	451.19	451.19	**0.00**	58.90	451.19	451.19	**0.00**	653.89	29430	451.18	451.19	**0.00**	20.66	255	9	···
30-60-4	476.28	476.28	**0.00**	**1.32**	476.26	476.27	0.00	3602.51	476.28	476.28	**0.00**	1214.56	476.28	476.28	**0.00**	575.01	29430	476.27	476.28	**0.00**	24.33	275	12	···
30-60-5	458.95	458.95	**0.00**	**0.51**	458.95	458.95	**0.00**	21.59	458.95	458.95	**0.00**	124.92	458.95	458.95	**0.00**	556.80	29430	458.95	458.95	**0.00**	23.22	265	10	···
30-60-6	472.65	472.65	**0.00**	**0.49**	472.65	472.65	**0.00**	1283.25	472.65	472.65	**0.00**	166.67	472.65	472.65	**0.00**	734.91	29430	472.65	472.65	**0.00**	22.10	271	13	···
30-60-7	524.79	524.79	**0.00**	**1.89**	524.79	524.79	**0.00**	479.98	524.79	524.79	**0.00**	189.63	524.79	524.79	**0.00**	586.82	29430	524.79	524.79	**0.00**	23.80	299	11	···
30-60-8	490.55	490.55	**0.00**	**1.65**	490.55	490.55	**0.00**	3602.35	490.55	490.55	**0.00**	117.37	490.55	490.55	**0.00**	716.54	29430	490.55	490.55	**0.00**	25.66	286	11	···
30-60-9	464.16	464.16	**0.00**	**1.14**	464.16	464.16	**0.00**	424.42	464.16	464.16	**0.00**	548.01	464.16	464.16	**0.00**	626.40	29430	464.16	464.16	**0.00**	24.77	285	10	···
40-80-1	$-\infty$	$+\infty$	100.0	$+\infty$	571.01	571.01	**0.00**	2250.59	571.01	571.01	**0.00**	1227.94	571.01	571.01	**0.00**	1067.47	39240	571.01	571.01	**0.00**	**33.46**	345	12	···
40-80-10	$-\infty$	$+\infty$	100.0	$+\infty$	569.99	569.99	**0.00**	3602.34	569.99	569.99	**0.00**	291.72	569.99	569.99	**0.00**	766.35	39240	569.99	569.99	**0.00**	**31.04**	342	12	···
40-80-2	$-\infty$	$+\infty$	100.0	$+\infty$	564.71	564.73	0.00	3602.99	564.73	564.73	**0.00**	226.44	564.73	564.73	**0.00**	902.52	39240	564.73	564.73	**0.00**	**30.47**	340	10	···
40-80-3	518.10	518.10	**0.00**	**1.72**	518.09	518.10	0.00	3602.48	518.10	518.10	**0.00**	353.58	518.10	518.10	**0.00**	866.39	39240	518.10	518.10	**0.00**	31.30	354	13	···
40-80-4	$-\infty$	$+\infty$	100.0	$+\infty$	551.77	551.78	0.00	3602.81	551.78	551.78	**0.00**	797.09	551.78	551.78	**0.00**	951.29	39240	551.78	551.78	**0.00**	**31.49**	363	12	···
40-80-5	548.87	548.87	**0.00**	**1.64**	548.87	548.87	**0.00**	2330.48	548.87	548.87	**0.00**	813.02	548.87	548.87	**0.00**	909.76	39240	548.87	548.87	**0.00**	32.54	354	13	···
40-80-6	$-\infty$	$+\infty$	100.0	$+\infty$	539.24	539.25	**0.00**	3602.85	539.25	539.25	**0.00**	139.30	539.25	539.25	**0.00**	1054.78	39240	539.25	539.25	**0.00**	**28.95**	339	10	···
40-80-7	578.52	578.52	**0.00**	**7.58**	578.52	578.52	**0.00**	2976.82	578.52	578.52	**0.00**	567.59	578.52	578.52	**0.00**	1079.36	39240	578.52	578.52	**0.00**	32.29	350	12	···
40-80-8	528.50	528.50	**0.00**	**1.16**	528.49	528.50	**0.00**	3602.74	528.50	528.50	**0.00**	329.64	528.50	528.50	**0.00**	910.82	39240	528.50	528.50	**0.00**	31.19	369	12	···
40-80-9	$-\infty$	$+\infty$	100.0	$+\infty$	517.10	517.10	**0.00**	3602.59	517.10	517.10	**0.00**	228.63	517.10	517.10	**0.00**	935.03	39240	517.10	517.10	**0.00**	**28.18**	365	11	···
50-100-1	$-\infty$	$+\infty$	100.0	$+\infty$	616.13	616.43	0.05	3602.65	616.43	616.43	**0.00**	416.31	616.43	616.43	**0.00**	1185.43	49050	616.43	616.43	**0.00**	**33.54**	434	10	···
50-100-10	$-\infty$	$+\infty$	100.0	$+\infty$	649.84	649.97	0.02	3602.80	649.97	649.97	**0.00**	3572.63	649.97	649.97	**0.00**	2459.74	49050	649.97	649.97	**0.00**	**45.93**	445	13	···
50-100-2	$-\infty$	$+\infty$	100.0	$+\infty$	634.07	634.07	**0.00**	642.46	634.07	634.07	**0.00**	3210.86	634.07	634.07	**0.00**	1231.18	49050	634.07	634.07	**0.00**	**35.38**	463	10	···
50-100-3	$-\infty$	$+\infty$	100.0	$+\infty$	608.55	608.57	0.00	3602.57	608.57	608.57	**0.00**	2777.41	608.57	608.57	**0.00**	1606.51	49050	608.57	608.57	**0.00**	**46.45**	444	13	···
50-100-4	$-\infty$	$+\infty$	100.0	$+\infty$	576.29	576.29	**0.00**	3602.66	576.29	576.29	**0.00**	791.08	576.29	576.29	**0.00**	1091.57	49050	576.29	576.29	**0.00**	**36.02**	462	10	···
50-100-5	$-\infty$	$+\infty$	100.0	$+\infty$	571.76	571.76	**0.00**	91.28	571.76	571.76	**0.00**	1333.88	571.76	571.76	**0.00**	1159.81	49050	571.76	571.76	**0.00**	**38.04**	420	10	···
50-100-6	$-\infty$	$+\infty$	100.0	$+\infty$	607.65	607.65	**0.00**	3602.76	607.65	607.65	**0.00**	624.47	607.65	607.65	**0.00**	1240.55	49050	607.65	607.65	**0.00**	**44.67**	453	13	···
50-100-7	$-\infty$	$+\infty$	100.0	$+\infty$	701.79	701.79	**0.00**	1123.17	701.79	701.79	**0.00**	1854.08	701.79	701.79	**0.00**	1467.22	49050	701.79	701.79	**0.00**	**40.98**	406	13	···
50-100-8	$-\infty$	$+\infty$	100.0	$+\infty$	581.54	581.54	**0.00**	1331.37	581.54	581.54	**0.00**	1156.74	581.54	581.54	**0.00**	1710.28	49050	581.54	581.54	**0.00**	**40.46**	464	14	···
50-100-9	$-\infty$	$+\infty$	100.0	$+\infty$	639.98	639.98	**0.00**	3602.65	639.98	639.98	**0.00**	2261.73	639.98	639.98	**0.00**	1373.78	49050	639.98	639.98	**0.00**	**42.35**	450	11	···

Table 57 **Detailed results for problem NLUFLP-W-gunluk, cost functions f_2**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						...
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	...
10-20-1	221.18	221.18	0.00	0.03	221.18	221.18	0.00	2.85	221.18	221.18	0.00	3639.07	221.18	221.18	0.00	97.16	17010	221.18	221.18	0.00	8.93	152	13	...
10-20-10	211.03	211.03	0.00	0.04	211.03	211.03	0.00	3.09	211.03	211.03	0.00	3669.65	211.03	211.03	0.00	91.27	17010	211.03	211.03	0.00	8.53	181	17	...
10-20-2	173.67	173.67	0.00	0.05	173.67	173.67	0.00	3.13	$-\infty$	$+\infty$	100.0	3600.10	173.67	173.67	0.00	94.32	17010	173.67	173.67	0.00	7.97	203	17	...
10-20-3	201.33	201.33	0.00	0.04	201.33	201.33	0.00	3.16	$-\infty$	$+\infty$	100.0	3600.10	201.33	201.33	0.00	95.45	17010	201.33	201.33	0.00	9.16	203	18	...
10-20-4	217.86	217.86	0.00	0.05	217.86	217.86	0.00	2.98	$-\infty$	$+\infty$	100.0	3600.10	217.86	217.86	0.00	99.92	17010	217.86	217.86	0.00	9.06	164	13	...
10-20-5	283.01	283.01	0.00	0.05	283.01	283.01	0.00	3.70	283.01	283.01	0.00	3671.58	283.01	283.01	0.00	106.38	17010	283.01	283.01	0.00	9.50	200	19	...
10-20-6	155.77	155.77	0.00	0.04	155.77	155.77	0.00	3.33	155.77	155.77	0.00	3672.12	155.77	155.77	0.00	95.18	17010	155.77	155.77	0.00	6.98	222	15	...
10-20-7	166.21	166.21	0.00	0.03	166.21	166.21	0.00	2.60	166.21	166.21	0.00	3630.46	166.21	166.21	0.00	96.16	17010	166.21	166.21	0.00	7.28	195	13	...
10-20-8	177.57	177.57	0.00	0.05	177.57	177.57	0.00	3.23	$-\infty$	$+\infty$	100.0	3600.10	177.57	177.57	0.00	97.62	17010	177.57	177.57	0.00	8.37	196	19	...
10-20-9	184.98	184.98	0.00	0.03	184.98	184.98	0.00	2.84	184.98	184.98	0.00	3649.83	184.98	184.98	0.00	96.10	17010	184.98	184.98	0.00	7.11	215	17	...
20-40-1	264.57	264.57	0.00	0.08	264.57	264.57	0.00	3.92	264.57	264.57	0.00	3682.02	264.57	264.57	0.00	192.64	34020	264.57	264.57	0.00	12.07	339	12	...
20-40-10	271.54	271.54	0.00	0.10	271.54	271.54	0.00	3.03	271.54	271.54	0.00	3637.44	271.54	271.54	0.00	189.93	34020	271.54	271.54	0.00	10.75	325	13	...
20-40-2	236.03	236.03	0.00	0.10	236.03	236.03	0.00	2.76	236.03	236.03	0.00	3641.46	236.03	236.03	0.00	189.18	34020	236.03	236.03	0.00	13.16	334	17	...
20-40-3	223.15	223.15	0.00	0.09	223.15	223.15	0.00	3.30	223.15	223.15	0.00	3657.71	223.15	223.15	0.00	192.87	34020	223.15	223.15	0.00	12.30	326	11	...
20-40-4	316.79	316.79	0.00	0.11	316.79	316.79	0.00	3.44	316.79	316.79	0.00	3651.31	316.79	316.79	0.00	188.04	34020	316.79	316.79	0.00	11.86	317	13	...
20-40-5	224.73	224.73	0.00	0.06	224.73	224.73	0.00	2.89	224.73	224.73	0.00	159.10	224.73	224.73	0.00	189.65	34020	224.73	224.73	0.00	11.26	316	6	...
20-40-6	257.29	257.29	0.00	0.09	257.29	257.29	0.00	3.63	257.29	257.29	0.00	8.92	257.29	257.29	0.00	188.05	34020	257.29	257.29	0.00	11.18	296	6	...
20-40-7	285.88	285.88	0.00	0.09	285.88	285.88	0.00	3.54	285.88	285.88	0.00	3634.90	285.88	285.88	0.00	202.21	34020	285.88	285.88	0.00	12.14	367	8	...
20-40-8	240.17	240.17	0.00	0.07	240.17	240.17	0.00	4.00	240.17	240.17	0.00	3672.66	240.17	240.17	0.00	188.66	34020	240.17	240.17	0.00	11.72	351	11	...
20-40-9	237.41	237.41	0.00	0.08	237.41	237.41	0.00	3.37	237.41	237.41	0.00	3662.84	237.41	237.41	0.00	188.27	34020	237.41	237.41	0.00	13.02	400	11	...
30-60-1	273.68	273.68	0.00	0.12	273.68	273.68	0.00	4.25	273.68	273.68	0.00	3666.63	273.68	273.68	0.00	277.83	51030	273.68	273.68	0.00	20.93	580	13	...
30-60-10	236.83	236.83	0.00	0.13	236.83	236.83	0.00	3.51	236.83	236.83	0.00	3627.30	236.83	236.83	0.00	277.70	51030	236.83	236.83	0.00	17.67	589	11	...
30-60-2	270.70	270.70	0.00	0.12	270.70	270.70	0.00	4.78	270.70	270.70	0.00	3663.34	270.70	270.70	0.00	271.28	51030	270.70	270.70	0.00	19.93	505	13	...
30-60-3	271.30	271.30	0.00	0.14	271.30	271.30	0.00	4.69	271.30	271.30	0.00	3659.79	271.30	271.30	0.00	278.67	51030	271.30	271.30	0.00	26.23	523	14	...
30-60-4	304.65	304.65	0.00	0.14	304.65	304.65	0.00	4.03	304.65	304.65	0.00	3654.64	304.65	304.65	0.00	274.57	51030	304.65	304.65	0.00	17.42	478	10	...
30-60-5	300.09	300.09	0.00	0.11	300.09	300.09	0.00	3.91	300.09	300.09	0.00	3647.58	300.09	300.09	0.00	322.39	51030	300.09	300.09	0.00	23.41	496	8	...
30-60-6	284.38	284.39	0.00	0.12	284.38	284.38	0.00	4.54	284.38	284.39	0.00	3661.15	284.39	284.39	0.00	274.06	51030	284.39	284.39	0.00	21.61	486	17	...
30-60-7	340.10	340.10	0.00	0.16	340.10	340.10	0.00	5.27	340.10	340.10	0.00	3662.83	340.10	340.10	0.00	271.91	51030	340.10	340.10	0.00	20.51	519	15	...
30-60-8	309.69	309.69	0.00	0.11	309.69	309.69	0.00	3.34	309.69	309.69	0.00	4.92	309.69	309.69	0.00	274.28	51030	309.69	309.69	0.00	17.95	500	10	...
30-60-9	261.80	261.80	0.00	0.14	261.80	261.80	0.00	3.82	261.80	261.80	0.00	130.73	261.80	261.80	0.00	271.55	51030	261.80	261.80	0.00	17.33	471	9	...
40-80-1	335.34	335.34	0.00	0.17	335.34	335.34	0.00	4.31	335.34	335.34	0.00	171.71	335.34	335.34	0.00	368.51	68040	335.34	335.34	0.00	22.52	630	11	...
40-80-10	387.83	387.83	0.00	0.16	387.83	387.83	0.00	4.31	387.83	387.83	0.00	15.98	387.83	387.83	0.00	352.57	68040	387.83	387.83	0.00	21.21	567	6	...
40-80-2	344.78	344.78	0.00	0.17	344.78	344.78	0.00	5.50	344.78	344.78	0.00	3661.09	344.78	344.78	0.00	372.66	68040	344.78	344.78	0.00	26.89	637	9	...
40-80-3	318.48	318.48	0.00	0.15	318.47	318.47	0.00	4.87	318.48	318.48	0.00	3667.06	318.48	318.48	0.00	365.11	68040	318.48	318.48	0.00	30.92	696	11	...
40-80-4	365.14	365.14	0.00	0.18	365.14	365.14	0.00	5.35	365.14	365.14	0.00	3643.47	365.14	365.14	0.00	358.71	68040	365.14	365.14	0.00	26.73	597	13	...
40-80-5	335.74	335.74	0.00	0.18	335.74	335.74	0.00	4.97	335.74	335.74	0.00	307.50	335.74	335.74	0.00	357.93	68040	335.74	335.74	0.00	21.54	682	9	...
40-80-6	386.92	386.92	0.00	0.13	386.92	386.92	0.00	5.68	386.92	386.92	0.00	14.23	386.92	386.92	0.00	359.63	68040	386.92	386.92	0.00	19.62	575	9	...
40-80-7	350.08	350.08	0.00	0.23	350.08	350.08	0.00	4.35	350.08	350.08	0.00	3638.50	350.08	350.08	0.00	358.91	68040	350.08	350.08	0.00	23.98	592	7	...
40-80-8	340.24	340.24	0.00	0.16	340.23	340.23	0.00	4.11	340.24	340.24	0.00	15.81	340.24	340.24	0.00	369.56	68040	340.24	340.24	0.00	20.83	666	8	...
40-80-9	317.42	317.42	0.00	0.17	317.41	317.41	0.00	4.46	317.42	317.42	0.00	13.69	317.42	317.42	0.00	364.97	68040	317.42	317.42	0.00	30.15	596	9	...
50-100-1	383.93	383.93	0.00	0.95	383.93	383.93	0.00	6.15	383.93	383.93	0.00	3610.98	383.93	383.93	0.00	423.15	85050	383.93	383.93	0.00	28.28	765	12	...
50-100-10	**369.65**	**369.65**	**0.00**	**0.92**	**369.65**	**369.65**	**0.00**	**7.32**	**369.60**	**369.60**	**0.00**	**31.25**	**369.65**	**369.65**	**0.00**	**445.05**	**85050**	**369.65**	**369.65**	**0.00**	**31.69**	**763**	**14**	...
50-100-2	393.01	393.01	0.00	1.07	393.01	393.01	0.00	5.68	393.01	393.01	0.00	3613.94	393.01	393.01	0.00	415.99	85050	393.01	393.01	0.00	27.53	732	9	...
50-100-3	361.26	361.26	0.00	0.94	361.26	361.26	0.00	5.97	361.26	361.26	0.00	3645.64	361.26	361.26	0.00	432.31	85050	361.26	361.26	0.00	33.72	740	8	...
50-100-4	351.57	351.57	0.00	0.94	351.57	351.57	0.00	6.19	$-\infty$	$+\infty$	100.0	3600.10	351.57	351.57	0.00	430.33	85050	351.57	351.57	0.00	28.28	670	13	...
50-100-5	338.58	338.58	0.00	0.99	338.58	338.58	0.00	5.65	338.58	338.58	0.00	24.03	338.58	338.58	0.00	433.85	85050	338.58	338.58	0.00	26.45	709	7	...
50-100-6	381.53	381.53	0.00	0.92	381.53	381.53	0.00	6.74	381.53	381.53	0.00	3687.96	381.53	381.53	0.00	480.17	85050	381.53	381.53	0.00	30.47	713	10	...
50-100-7	394.13	394.13	0.00	1.01	394.13	394.13	0.00	6.13	394.13	394.13	0.00	3614.92	394.13	394.13	0.00	416.83	85050	394.13	394.13	0.00	31.26	752	12	...
50-100-8	326.52	326.52	0.00	1.06	326.52	326.52	0.00	6.28	326.52	326.52	0.00	28.55	326.52	326.52	0.00	406.73	85050	326.52	326.52	0.00	23.55	682	10	...
50-100-9	397.36	397.36	0.00	1.05	397.35	397.35	0.00	6.69	397.36	397.36	0.00	3636.80	397.36	397.36	0.00	416.64	85050	397.36	397.36	0.00	32.63	701	14	...

Table 58 **Detailed results for problem NLUFLP-W-gunluk, cost functions f_3**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	238.03	238.03	0.00	0.07	238.03	238.03	0.00	2.55	$-\infty$	$+\infty$	100.0	3600.10	238.03	238.03	0.00	150.33	22050	238.03	238.03	0.00	28.57	197	14	···
10-20-10	225.26	225.26	0.00	0.13	225.26	225.26	0.00	3.45	$-\infty$	$+\infty$	100.0	3600.10	225.26	225.26	0.00	160.99	22050	225.26	225.26	0.00	44.21	268	16	···
10-20-2	184.62	184.62	0.00	0.18	184.62	184.62	0.00	4.84	$-\infty$	$+\infty$	100.0	3600.10	184.62	184.62	0.00	171.76	22050	184.62	184.62	0.00	41.60	229	15	···
10-20-3	215.40	215.40	0.00	0.17	215.40	215.40	0.00	4.41	$-\infty$	$+\infty$	100.0	3600.10	215.40	215.40	0.00	156.05	22050	215.40	215.40	0.00	38.31	227	20	···
10-20-4	242.02	242.02	0.00	0.11	242.02	242.02	0.00	2.82	$-\infty$	$+\infty$	100.0	3600.10	242.02	242.02	0.00	174.87	22050	242.02	242.02	0.00	44.77	205	17	···
10-20-5	279.67	279.67	0.00	0.04	279.67	279.67	0.00	3.34	279.67	279.67	0.00	3640.65	279.67	279.67	0.00	145.32	22050	279.67	279.67	0.00	19.69	188	10	···
10-20-6	185.73	185.73	0.00	0.22	185.73	185.73	0.00	4.67	$-\infty$	$+\infty$	100.0	3600.10	185.73	185.73	0.00	149.79	22050	185.73	185.73	0.00	34.53	210	20	···
10-20-7	174.32	174.32	0.00	0.06	174.32	174.32	0.00	2.91	174.32	174.32	0.00	3666.09	174.32	174.32	0.00	144.96	22050	174.32	174.32	0.00	25.40	204	15	···
10-20-8	197.38	197.38	0.00	0.33	197.38	197.38	0.00	6.95	$-\infty$	$+\infty$	100.0	3600.10	197.38	197.38	0.00	152.82	22050	197.38	197.38	0.00	42.30	242	19	···
10-20-9	202.77	202.77	0.00	0.12	202.77	202.77	0.00	3.53	$-\infty$	$+\infty$	100.0	3600.10	202.77	202.77	0.00	165.30	22050	202.77	202.77	0.00	28.51	237	20	···
20-40-1	276.86	276.86	0.00	0.18	276.86	276.86	0.00	4.60	$-\infty$	$+\infty$	100.0	3600.10	276.86	276.86	0.00	245.63	44100	276.86	276.86	0.00	44.96	398	17	···
20-40-10	290.36	290.36	0.00	0.21	290.36	290.36	0.00	5.12	$-\infty$	$+\infty$	100.0	3600.10	290.36	290.36	0.00	305.68	44100	290.36	290.36	0.00	53.67	402	15	···
20-40-2	244.11	244.11	0.00	0.17	244.11	244.11	0.00	3.11	$-\infty$	$+\infty$	100.0	3600.10	244.11	244.11	0.00	289.41	44100	244.11	244.11	0.00	40.85	419	15	···
20-40-3	237.48	237.48	0.00	0.19	237.48	237.48	0.00	3.13	$-\infty$	$+\infty$	100.0	3600.10	237.48	237.48	0.00	265.82	44100	237.48	237.48	0.00	37.33	391	13	···
20-40-4	328.70	328.70	0.00	0.27	328.70	328.70	0.00	8.35	$-\infty$	$+\infty$	100.0	3600.10	328.70	328.70	0.00	261.53	44100	328.70	328.70	0.00	43.74	415	18	···
20-40-5	234.57	234.57	0.00	0.14	234.57	234.57	0.00	4.53	$-\infty$	$+\infty$	100.0	3600.10	234.57	234.57	0.00	266.93	44100	234.57	234.57	0.00	36.09	438	14	···
20-40-6	272.27	272.27	0.00	0.26	272.27	272.27	0.00	5.59	$-\infty$	$+\infty$	100.0	3600.10	272.27	272.27	0.00	289.07	44100	272.27	272.27	0.00	38.20	346	17	···
20-40-7	300.18	300.18	0.00	0.18	300.18	300.18	0.00	3.91	$-\infty$	$+\infty$	100.0	3600.10	300.18	300.18	0.00	289.81	44100	300.18	300.18	0.00	48.28	401	18	···
20-40-8	253.57	253.57	0.00	0.28	253.57	253.57	0.00	3.90	253.57	253.57	0.00	3668.62	253.57	253.57	0.00	252.36	44100	253.57	253.57	0.00	39.57	396	14	···
20-40-9	248.27	248.27	0.00	0.23	248.27	248.27	0.00	6.33	$-\infty$	$+\infty$	100.0	3600.10	248.27	248.27	0.00	255.99	44100	248.27	248.27	0.00	41.16	476	19	···
30-60-1	282.97	282.97	0.00	0.23	282.97	282.97	0.00	5.38	$-\infty$	$+\infty$	100.0	3600.10	282.97	282.97	0.00	367.36	66150	282.97	282.97	0.00	68.06	594	14	···
30-60-10	246.44	246.44	0.00	0.22	246.44	246.44	0.00	5.55	$-\infty$	$+\infty$	100.0	3600.10	246.44	246.44	0.00	375.94	66150	246.44	246.44	0.00	53.11	597	16	···
30-60-2	277.77	277.77	0.00	0.21	277.77	277.77	0.00	6.03	$-\infty$	$+\infty$	100.0	3600.10	277.77	277.77	0.00	402.13	66150	277.77	277.77	0.00	56.49	548	14	···
30-60-3	280.94	280.94	0.00	0.45	280.94	280.94	0.00	10.73	280.94	280.94	0.00	3676.81	280.94	280.94	0.00	342.50	66150	280.94	280.94	0.00	68.65	620	21	···
30-60-4	315.08	315.08	0.00	0.27	315.08	315.08	0.00	6.72	315.08	315.08	0.00	3659.61	315.08	315.08	0.00	363.53	66150	315.08	315.08	0.00	56.98	528	15	···
30-60-5	306.80	306.80	0.00	0.27	306.80	306.80	0.00	6.12	306.80	306.80	0.00	3683.52	306.80	306.80	0.00	369.14	66150	306.80	306.80	0.00	63.53	519	20	···
30-60-6	295.19	295.19	0.00	0.28	295.19	295.19	0.00	5.81	$-\infty$	$+\infty$	100.0	3600.10	295.19	295.19	0.00	381.66	66150	295.19	295.19	0.00	59.48	619	17	···
30-60-7	348.49	348.49	0.00	0.21	348.49	348.49	0.00	7.11	348.49	348.49	0.00	3644.71	348.49	348.49	0.00	344.36	66150	348.49	348.49	0.00	61.18	676	14	···
30-60-8	316.72	316.72	0.00	0.24	316.72	316.72	0.00	6.63	316.72	316.72	0.00	47.97	316.72	316.72	0.00	383.19	66150	316.72	316.72	0.00	53.44	586	14	···
30-60-9	271.04	271.04	0.00	0.25	271.04	271.04	0.00	9.04	271.04	271.04	0.00	3629.67	271.04	271.04	0.00	399.02	66150	271.04	271.04	0.00	52.74	562	13	···
40-80-1	342.68	342.68	0.00	0.48	342.68	342.68	0.00	8.22	$-\infty$	$+\infty$	100.0	3600.10	342.68	342.68	0.00	435.78	88200	342.68	342.68	0.00	80.29	721	13	···
40-80-10	394.59	394.59	0.00	0.28	394.59	394.59	0.00	7.99	394.59	394.59	0.00	3643.00	394.59	394.59	0.00	500.64	88200	394.59	394.59	0.00	76.10	674	11	···
40-80-2	352.30	352.30	0.00	0.35	352.30	352.30	0.00	6.42	352.30	352.30	0.00	3671.80	352.30	352.30	0.00	516.91	88200	352.30	352.30	0.00	77.87	709	12	···
40-80-3	324.92	324.92	0.00	0.49	324.91	324.91	0.00	11.48	$-\infty$	$+\infty$	100.0	3600.10	324.92	324.92	0.00	458.40	88200	324.92	324.92	0.00	80.81	754	18	···
40-80-4	372.76	372.76	0.00	0.36	372.76	372.76	0.00	8.54	372.76	372.76	0.00	3669.96	372.76	372.76	0.00	493.15	88200	372.76	372.76	0.00	76.86	702	15	···
40-80-5	343.89	343.89	0.00	0.34	343.89	343.89	0.00	7.87	343.89	343.89	0.00	260.20	343.89	343.89	0.00	511.20	88200	343.89	343.89	0.00	77.10	772	11	···
40-80-6	392.44	392.44	0.00	0.27	392.44	392.44	0.00	8.36	392.44	392.44	0.00	14.72	392.44	392.44	0.00	524.33	88200	392.44	392.44	0.00	75.33	776	6	···
40-80-7	358.46	358.46	0.00	0.45	358.46	358.46	0.00	16.55	358.46	358.46	0.00	3690.59	358.46	358.46	0.00	491.63	88200	358.46	358.46	0.00	78.05	862	17	···
40-80-8	345.73	345.73	0.00	0.34	345.73	345.73	0.00	8.91	345.73	345.73	0.00	74.50	345.73	345.73	0.00	530.23	88200	345.73	345.73	0.00	77.77	700	17	···
40-80-9	324.78	324.78	0.00	0.42	324.78	324.78	0.00	7.85	324.78	324.78	0.00	136.17	324.78	324.78	0.00	439.17	88200	324.78	324.78	0.00	75.75	739	17	···
50-100-1	389.51	389.51	0.00	1.40	389.51	389.51	0.00	8.46	389.51	389.51	0.00	55.80	389.51	389.51	0.00	642.63	110250	389.51	389.51	0.00	90.15	920	12	···
50-100-10	378.61	378.61	0.00	1.21	378.61	378.61	0.00	16.45	378.61	378.61	0.00	3632.81	378.61	378.61	0.00	589.55	110250	378.61	378.61	0.00	93.57	855	14	···
50-100-2	399.48	399.48	0.00	1.22	399.48	399.48	0.00	7.64	399.48	399.48	0.00	3614.37	399.48	399.48	0.00	647.13	110250	399.48	399.48	0.00	97.40	958	9	···
50-100-3	368.70	368.70	0.00	1.52	368.70	368.70	0.00	12.72	368.70	368.70	0.00	3675.49	368.70	368.70	0.00	622.51	110250	368.70	368.70	0.00	100.82	882	16	···
50-100-4	355.31	355.31	0.00	1.46	355.31	355.31	0.00	8.55	$-\infty$	$+\infty$	100.0	3600.10	355.31	355.31	0.00	546.61	110250	355.31	355.31	0.00	94.92	907	11	···
50-100-5	342.98	342.98	0.00	1.10	342.98	342.98	0.00	8.86	342.98	342.98	0.00	3655.32	342.98	342.98	0.00	568.59	110250	342.98	342.98	0.00	96.82	807	13	···
50-100-6	387.89	387.89	0.00	1.36	387.89	387.89	0.00	8.72	$-\infty$	$+\infty$	100.0	3600.10	387.89	387.89	0.00	536.81	110250	387.89	387.89	0.00	98.44	837	10	···
50-100-7	400.88	400.88	0.00	1.41	400.88	400.88	0.00	17.70	$-\infty$	$+\infty$	100.0	3600.10	400.88	400.88	0.00	564.07	110250	400.88	400.88	0.00	100.72	851	16	···
50-100-8	332.52	332.52	0.00	1.60	332.52	332.52	0.00	19.32	332.52	332.52	0.00	3659.78	332.52	332.52	0.00	533.66	110250	332.52	332.52	0.00	96.40	929	12	···
50-100-9	404.14	404.15	0.00	1.59	404.14	404.14	0.00	17.69	404.15	404.15	0.00	3645.71	404.15	404.15	0.00	566.66	110250	404.15	404.15	0.00	96.42	884	16	···

Table 59 Detailed results for problem NLUFLP-W-gunluk, cost functions f_4

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
10-20-1	224.74	224.74	0.00	0.14	224.74	224.74	0.00	3.09	$-\infty$	$+\infty$	100.0	3600.10	224.74	224.74	0.00	160.36	29618	224.74	224.74	0.00	36.05	207	12	⋯
10-20-10	210.52	210.52	0.00	0.11	210.52	210.52	0.00	3.07	210.52	210.52	0.00	3654.69	210.52	210.52	0.00	333.31	36650	210.52	210.52	0.00	19.58	226	8	⋯
10-20-2	177.85	177.85	0.00	0.12	177.85	177.85	0.00	3.12	177.85	177.85	0.00	684.57	177.84	177.85	0.00	278.99	36650	177.85	177.85	0.00	41.95	294	9	⋯
10-20-3	212.05	212.05	0.00	0.49	212.05	212.05	0.00	5.69	$-\infty$	$+\infty$	100.0	3600.10	212.04	212.05	0.00	369.28	36650	212.05	212.05	0.00	48.07	248	16	⋯
10-20-4	196.10	196.10	0.00	0.07	196.10	196.10	0.00	2.59	196.10	196.10	0.00	1.82	196.10	196.10	0.00	306.92	36650	196.10	196.10	0.00	25.65	210	7	⋯
10-20-5	275.01	275.01	0.00	0.23	275.01	275.01	0.00	3.38	$-\infty$	$+\infty$	100.0	3600.10	275.01	275.01	0.00	270.09	36650	275.01	275.01	0.00	35.85	247	21	⋯
10-20-6	172.68	172.68	0.00	0.90	172.68	172.68	0.00	6.99	$-\infty$	$+\infty$	100.0	3600.10	172.68	172.68	0.00	310.84	36650	172.68	172.68	0.00	43.33	274	17	⋯
10-20-7	174.33	174.33	0.00	0.14	174.33	174.33	0.00	3.03	$-\infty$	$+\infty$	100.0	3600.10	174.33	174.33	0.00	341.04	36650	174.33	174.33	0.00	44.88	267	9	⋯
10-20-8	193.59	193.59	0.00	1.43	193.59	193.59	0.00	5.77	$-\infty$	$+\infty$	100.0	3600.10	193.58	193.59	0.00	380.00	36650	193.59	193.59	0.00	99.59	233	16	⋯
10-20-9	199.85	199.85	0.00	0.37	199.85	199.85	0.00	4.73	$-\infty$	$+\infty$	100.0	3600.10	199.84	199.85	0.00	349.84	36650	199.85	199.85	0.00	59.90	267	20	⋯
20-40-1	281.31	281.31	0.00	0.92	281.31	281.31	0.00	9.80	$-\infty$	$+\infty$	100.0	3600.10	281.31	281.31	0.00	705.22	73300	281.31	281.31	0.00	65.07	478	15	⋯
20-40-10	291.64	291.64	0.00	1.10	291.64	291.64	0.00	13.97	$-\infty$	$+\infty$	100.0	3600.10	291.64	291.64	0.00	640.75	73300	291.64	291.64	0.00	42.18	464	19	⋯
20-40-2	247.53	247.53	0.00	0.94	247.53	247.53	0.00	7.04	$-\infty$	$+\infty$	100.0	3600.10	247.53	247.53	0.00	844.54	73300	247.53	247.53	0.00	63.67	432	13	⋯
20-40-3	241.58	241.58	0.00	1.95	241.58	241.58	0.00	25.19	$-\infty$	$+\infty$	100.0	3600.10	241.57	241.58	0.00	852.19	73300	241.57	241.58	0.00	104.75	512	23	⋯
20-40-4	331.03	331.03	0.00	4.93	331.03	331.03	0.00	36.22	$-\infty$	$+\infty$	100.0	3600.10	331.03	331.03	0.00	523.11	73300	331.03	331.03	0.00	51.00	445	15	⋯
20-40-5	238.93	238.93	0.00	0.92	238.93	238.93	0.00	12.26	$-\infty$	$+\infty$	100.0	3600.10	238.93	238.93	0.00	747.76	73300	238.93	238.93	0.00	68.67	565	15	⋯
20-40-6	275.85	275.85	0.00	1.33	275.85	275.85	0.00	10.69	$-\infty$	$+\infty$	100.0	3600.10	275.85	275.85	0.00	683.67	73300	275.85	275.85	0.00	69.73	448	16	⋯
20-40-7	299.88	299.88	0.00	0.80	299.88	299.88	0.00	9.33	$-\infty$	$+\infty$	100.0	3600.10	299.88	299.88	0.00	671.51	73300	299.88	299.88	0.00	70.20	475	13	⋯
20-40-8	258.20	258.20	0.00	1.79	258.20	258.20	0.00	23.66	$-\infty$	$+\infty$	100.0	3600.10	258.20	258.20	0.00	725.57	73300	258.20	258.20	0.00	68.66	462	17	⋯
20-40-9	252.19	252.19	0.00	1.33	252.19	252.19	0.00	11.61	$-\infty$	$+\infty$	100.0	3600.10	252.19	252.19	0.00	722.25	73300	252.19	252.19	0.00	83.47	625	18	⋯
30-60-1	287.96	287.96	0.00	1.41	287.96	287.96	0.00	55.92	$-\infty$	$+\infty$	100.0	3600.10	287.96	287.96	0.00	1277.39	109950	287.96	287.96	0.00	44.12	793	17	⋯
30-60-10	250.81	250.81	0.00	2.47	250.81	250.81	0.00	23.91	$-\infty$	$+\infty$	100.0	3600.10	250.81	250.81	0.00	1106.63	109950	250.81	250.81	0.00	64.19	691	16	⋯
30-60-2	281.42	281.42	0.00	1.81	281.42	281.42	0.00	13.99	$-\infty$	$+\infty$	100.0	3600.10	281.42	281.42	0.00	1168.17	109950	281.42	281.42	0.00	51.60	659	15	⋯
30-60-3	286.54	286.54	0.00	8.18	286.54	286.54	0.00	322.97	$-\infty$	$+\infty$	100.0	3600.10	286.54	286.54	0.00	1249.10	109950	286.54	286.54	0.00	51.95	771	17	⋯
30-60-4	319.95	319.95	0.00	2.05	319.95	319.95	0.00	26.41	$-\infty$	$+\infty$	100.0	3600.10	319.94	319.95	0.00	1246.86	109950	319.95	319.95	0.00	91.86	779	16	⋯
30-60-5	309.81	309.81	0.00	1.46	309.81	309.81	0.00	20.19	309.81	309.81	0.00	3656.17	309.81	309.81	0.00	1329.29	109950	309.81	309.81	0.00	40.25	633	17	⋯
30-60-6	300.56	300.56	0.00	1.90	300.56	300.56	0.00	53.39	$-\infty$	$+\infty$	100.0	3600.10	300.55	300.56	0.00	1123.76	109950	300.56	300.56	0.00	64.94	677	15	⋯
30-60-7	353.33	353.33	0.00	3.32	353.33	353.33	0.00	40.39	$-\infty$	$+\infty$	100.0	3600.10	353.33	353.33	0.00	1387.89	109950	353.33	353.33	0.00	45.43	662	14	⋯
30-60-8	320.75	320.75	0.00	1.11	320.75	320.75	0.00	21.06	320.75	320.75	0.00	50.24	320.75	320.75	0.00	1152.47	109950	320.75	320.75	0.00	51.73	693	16	⋯
30-60-9	275.99	275.99	0.00	1.34	275.99	275.99	0.00	29.25	275.99	275.99	0.00	3697.30	275.99	275.99	0.00	1060.06	109950	275.99	275.99	0.00	43.25	703	13	⋯
40-80-1	347.27	347.27	0.00	2.95	347.27	347.27	0.00	40.25	$-\infty$	$+\infty$	100.0	3600.10	347.27	347.27	0.00	2381.05	146600	347.27	347.27	0.00	46.60	988	17	⋯
40-80-10	398.48	398.48	0.00	2.21	398.48	398.48	0.00	24.11	$-\infty$	$+\infty$	100.0	3600.10	398.48	398.48	0.00	1639.11	146600	398.48	398.48	0.00	51.45	856	12	⋯
40-80-2	356.50	356.50	0.00	2.60	356.50	356.50	0.00	36.62	$-\infty$	$+\infty$	100.0	3600.10	356.50	356.50	0.00	2012.40	146600	356.50	356.50	0.00	64.30	873	14	⋯
40-80-3	328.93	328.93	0.00	2.32	328.93	328.93	0.00	91.58	$-\infty$	$+\infty$	100.0	3600.10	328.93	328.93	0.00	1831.71	146600	328.93	328.93	0.00	48.26	918	14	⋯
40-80-4	376.76	376.76	0.00	2.50	376.76	376.76	0.00	26.68	$-\infty$	$+\infty$	100.0	3600.10	376.76	376.76	0.00	1553.02	146600	376.76	376.76	0.00	75.05	870	12	⋯
40-80-5	348.29	348.29	0.00	1.58	348.29	348.29	0.00	21.35	348.29	348.29	0.00	3635.52	348.29	348.29	0.00	2078.35	146600	348.29	348.29	0.00	51.64	846	10	⋯
40-80-6	395.63	395.63	0.00	1.14	395.63	395.63	0.00	17.25	395.63	395.63	0.00	182.08	395.62	395.63	0.00	2755.83	146600	395.63	395.63	0.00	83.95	872	13	⋯
40-80-7	363.24	363.24	0.00	6.89	363.24	363.24	0.00	276.40	$-\infty$	$+\infty$	100.0	3600.10	363.24	363.24	0.00	1838.70	146600	363.24	363.24	0.00	51.91	875	14	⋯
40-80-8	349.17	349.17	0.00	1.69	349.17	349.17	0.00	96.06	$-\infty$	$+\infty$	100.0	3600.10	349.17	349.17	0.00	1915.87	146600	349.17	349.17	0.00	43.96	869	12	⋯
40-80-9	328.97	328.97	0.00	1.48	328.97	328.97	0.00	58.39	328.97	328.97	0.00	186.65	328.97	328.97	0.00	1660.86	146600	328.97	328.97	0.00	52.88	851	12	⋯
50-100-1	393.03	393.03	0.00	3.36	393.03	393.03	0.00	40.72	393.03	393.03	0.00	257.65	393.03	393.03	0.00	3270.55	183250	393.03	393.03	0.00	64.24	1048	16	⋯
50-100-10	383.89	383.90	0.00	3.79	383.89	383.89	0.00	234.99	$-\infty$	$+\infty$	100.0	3600.10	383.90	383.90	0.00	2122.89	183250	383.89	383.90	0.00	64.31	1122	16	⋯
50-100-2	403.04	403.04	0.00	2.49	403.04	403.04	0.00	14.08	403.04	403.04	0.00	3613.45	$-\infty$	$+\infty$	100.00	3602.51	183250	403.04	403.04	0.00	55.08	1003	9	⋯
50-100-3	372.48	372.48	0.00	4.03	372.48	372.48	0.00	54.22	$-\infty$	$+\infty$	100.0	3600.10	372.48	372.48	0.00	3570.55	183250	372.48	372.48	0.00	75.10	1053	15	⋯
50-100-4	357.73	357.73	0.00	4.67	357.73	357.73	0.00	25.72	$-\infty$	$+\infty$	100.0	3600.10	357.73	357.73	0.00	3274.10	183250	357.73	357.73	0.00	65.25	1083	17	⋯
50-100-5	345.82	345.82	0.00	3.34	345.82	345.82	0.00	44.81	345.82	345.82	0.00	3662.45	345.82	345.82	0.00	2397.62	183250	345.82	345.82	0.00	52.96	1108	11	⋯
50-100-6	391.72	391.72	0.00	3.44	391.72	391.72	0.00	47.01	$-\infty$	$+\infty$	100.0	3600.10	391.72	391.72	0.00	1652.30	183250	391.72	391.73	0.00	63.72	1019	10	⋯
50-100-7	405.13	405.13	0.00	9.72	405.13	405.13	0.00	194.72	405.13	405.13	0.00	3664.51	405.13	405.13	0.00	3242.72	183250	405.13	405.13	0.00	64.47	1106	15	⋯
50-100-8	336.06	336.06	0.00	2.92	336.06	336.06	0.00	141.99	336.06	336.06	0.00	3662.82	336.06	336.06	0.00	3296.45	183250	336.06	336.06	0.00	70.40	1052	17	⋯
50-100-9	407.99	407.99	0.00	4.06	407.99	407.99	0.00	458.91	407.99	407.99	0.00	3659.94	407.99	407.99	0.00	3259.88	183250	407.99	407.99	0.00	61.98	1137	15	⋯

Table 60 Detailed results for problem NLUFLP-W-gunluk, cost functions f_5

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	253.27	253.27	**0.00**	**0.49**	253.27	253.27	**0.00**	2.43	253.27	253.27	**0.00**	4.75	253.27	253.27	**0.00**	130.42	14430	253.27	253.27	**0.00**	81.51	128	10	···
10-20-10	245.44	245.44	**0.00**	**0.69**	245.44	245.44	**0.00**	13.09	245.44	245.44	**0.00**	1585.95	245.44	245.44	**0.00**	131.79	14430	245.44	245.44	**0.00**	72.94	155	19	···
10-20-2	197.28	197.28	**0.00**	**0.47**	197.28	197.28	**0.00**	4.43	197.28	197.28	**0.00**	37.25	197.28	197.28	**0.00**	154.69	14430	197.28	197.28	**0.00**	82.25	160	10	···
10-20-3	234.52	234.52	**0.00**	**0.76**	234.52	234.52	**0.00**	6.91	234.52	234.52	**0.00**	19.29	234.52	234.52	**0.00**	134.39	14430	234.52	234.52	**0.00**	86.80	175	14	···
10-20-4	250.65	250.65	**0.00**	**0.34**	250.65	250.65	**0.00**	4.36	250.65	250.65	**0.00**	18.31	250.65	250.65	**0.00**	111.20	14430	250.65	250.65	**0.00**	74.19	126	13	···
10-20-5	287.33	287.33	**0.00**	**0.24**	287.33	287.33	**0.00**	2.44	287.33	287.33	**0.00**	22.15	287.33	287.33	**0.00**	98.50	14430	287.33	287.33	**0.00**	65.02	157	9	···
10-20-6	218.44	218.44	**0.00**	**1.27**	218.44	218.44	**0.00**	3602.54	$-\infty$	$+\infty$	100.0	3600.10	218.44	218.44	**0.00**	117.28	14430	218.44	218.44	**0.00**	99.17	155	17	···
10-20-7	189.78	189.78	**0.00**	**0.72**	189.78	189.78	**0.00**	2.35	189.78	189.78	**0.00**	3.80	189.78	189.78	**0.00**	116.77	14430	189.78	189.78	**0.00**	86.32	153	8	···
10-20-8	224.32	224.32	**0.00**	**2.53**	224.32	224.32	**0.00**	3602.44	$-\infty$	$+\infty$	100.0	3600.10	224.32	224.32	**0.00**	128.07	14430	224.32	224.32	**0.00**	81.19	194	18	···
10-20-9	226.34	226.34	**0.00**	**0.87**	226.34	226.34	**0.00**	4.35	226.34	226.34	**0.00**	31.51	226.34	226.34	**0.00**	126.90	14430	226.34	226.34	**0.00**	64.76	156	15	···
20-40-1	310.39	310.39	**0.00**	**2.96**	310.39	310.39	**0.00**	12.32	310.39	310.39	**0.00**	255.81	310.39	310.39	**0.00**	198.04	28860	310.39	310.39	**0.00**	75.31	324	20	···
20-40-10	323.53	323.53	**0.00**	**2.83**	323.53	323.53	**0.00**	3602.09	$-\infty$	$+\infty$	100.0	3600.10	323.53	323.53	**0.00**	269.09	28860	323.53	323.53	**0.00**	82.49	301	18	···
20-40-2	265.70	265.70	**0.00**	**3.34**	265.70	265.70	**0.00**	7.56	265.70	265.70	**0.00**	54.07	265.70	265.70	**0.00**	227.09	28860	265.70	265.70	**0.00**	101.05	324	16	···
20-40-3	270.69	270.69	**0.00**	**2.58**	270.69	270.69	**0.00**	69.17	270.69	270.69	**0.00**	26.16	270.69	270.69	**0.00**	206.16	28860	270.69	270.69	**0.00**	63.76	345	15	···
20-40-4	353.96	353.96	**0.00**	**2.53**	353.96	353.96	**0.00**	592.79	353.96	353.96	**0.00**	2861.53	353.96	353.96	**0.00**	209.56	28860	353.96	353.97	**0.00**	99.42	247	18	···
20-40-5	260.10	260.10	**0.00**	**0.94**	260.10	260.10	**0.00**	31.54	260.10	260.10	**0.00**	39.86	260.10	260.10	**0.00**	221.91	28860	260.10	260.10	**0.00**	78.45	459	10	···
20-40-6	302.60	302.60	**0.00**	**1.71**	302.60	302.60	**0.00**	13.28	302.60	302.60	**0.00**	1018.47	302.60	302.60	**0.00**	205.51	28860	302.60	302.60	**0.00**	82.19	237	8	···
20-40-7	328.72	328.72	**0.00**	**1.84**	328.72	328.72	**0.00**	8.11	328.72	328.72	**0.00**	451.06	328.72	328.72	**0.00**	221.24	28860	328.72	328.72	**0.00**	71.88	269	14	···
20-40-8	285.88	285.88	**0.00**	**5.58**	285.88	285.88	**0.00**	50.16	285.88	285.88	**0.00**	160.48	285.88	285.88	**0.00**	241.11	28860	285.88	285.88	**0.00**	80.85	413	16	···
20-40-9	279.12	279.12	**0.00**	**1.76**	279.12	279.12	**0.00**	199.15	279.12	279.12	**0.00**	115.27	279.12	279.12	**0.00**	211.33	28860	279.12	279.12	**0.00**	91.05	362	13	···
30-60-1	321.44	321.44	**0.00**	**3.92**	321.44	321.44	**0.00**	3601.96	321.44	321.44	**0.00**	435.05	321.44	321.44	**0.00**	244.06	43290	321.44	321.44	**0.00**	79.11	589	9	···
30-60-10	278.28	278.28	**0.00**	**1.84**	278.28	278.28	**0.00**	3601.98	278.28	278.28	**0.00**	748.02	278.28	278.28	**0.00**	308.14	43290	278.28	278.28	**0.00**	82.99	459	19	···
30-60-2	302.91	302.91	**0.00**	**2.62**	302.90	302.91	**0.00**	3602.00	302.91	302.91	**0.00**	494.29	302.91	302.91	**0.00**	236.68	43290	302.91	302.91	**0.00**	81.30	544	19	···
30-60-3	318.10	318.10	**0.00**	**3.30**	318.10	318.10	**0.00**	32.63	318.10	318.10	**0.00**	480.19	318.10	318.10	**0.00**	242.31	43290	318.10	318.10	**0.00**	78.24	448	15	···
30-60-4	348.43	348.43	**0.00**	**3.11**	348.43	348.43	**0.00**	386.39	348.43	348.43	**0.00**	740.21	348.43	348.43	**0.00**	320.56	43290	348.43	348.43	**0.00**	82.00	535	16	···
30-60-5	331.70	331.70	**0.00**	**1.27**	331.70	331.70	**0.00**	3.07	331.70	331.70	**0.00**	319.85	331.70	331.70	**0.00**	313.78	43290	331.70	331.70	**0.00**	100.73	464	16	···
30-60-6	331.91	331.91	**0.00**	**5.59**	331.91	331.91	**0.00**	3601.96	331.92	331.92	**0.00**	337.94	331.91	331.92	**0.00**	313.75	43290	331.91	331.92	**0.00**	78.55	434	17	···
30-60-7	383.36	383.36	**0.00**	**3.05**	383.35	383.36	0.00	3601.96	382.26	383.36	0.29	3588.51	383.36	383.36	**0.00**	318.37	43290	383.36	383.36	**0.00**	83.01	453	15	···
30-60-8	346.53	346.53	**0.00**	**1.70**	346.53	346.53	**0.00**	2.89	346.53	346.53	**0.00**	256.00	346.53	346.53	**0.00**	321.22	43290	346.53	346.53	**0.00**	78.66	450	17	···
30-60-9	306.08	306.08	**0.00**	**1.61**	306.08	306.08	**0.00**	3601.99	306.08	306.08	**0.00**	441.62	306.08	306.08	**0.00**	294.45	43290	306.08	306.08	**0.00**	85.20	473	9	···
40-80-1	381.87	381.87	**0.00**	5.36	381.87	381.87	**0.00**	**3.80**	381.85	381.87	0.01	3542.58	381.87	381.87	**0.00**	428.51	57720	381.87	381.87	**0.00**	101.72	562	18	···
40-80-10	429.63	429.63	**0.00**	**7.33**	429.57	429.63	0.01	3602.04	429.63	429.63	**0.00**	3447.03	429.63	429.63	**0.00**	378.77	57720	429.63	429.63	**0.00**	95.10	530	9	···
40-80-2	386.37	386.37	**0.00**	**4.26**	386.34	386.37	0.01	3602.04	386.37	386.37	**0.00**	2162.41	386.37	386.37	**0.00**	421.86	57720	386.37	386.37	**0.00**	96.12	633	17	···
40-80-3	364.30	364.30	**0.00**	**9.15**	364.27	364.29	0.01	3601.99	362.67	364.30	0.45	3539.74	364.30	364.30	**0.00**	454.75	57720	364.30	364.30	**0.00**	106.13	634	20	···
40-80-4	405.40	405.40	**0.00**	**6.47**	405.40	405.40	**0.00**	102.99	405.40	405.40	**0.00**	2910.96	405.40	405.40	**0.00**	419.72	57720	405.40	405.40	**0.00**	99.77	524	18	···
40-80-5	378.99	378.99	**0.00**	**6.47**	378.99	378.99	**0.00**	35.41	378.99	378.99	**0.00**	613.29	378.99	378.99	**0.00**	374.51	57720	378.99	378.99	**0.00**	96.18	527	10	···
40-80-6	422.89	422.89	**0.00**	**3.86**	422.89	422.89	**0.00**	3602.65	422.89	422.89	**0.00**	3114.06	422.89	422.89	**0.00**	366.18	57720	422.89	422.89	**0.00**	101.06	544	13	···
40-80-7	398.46	398.46	**0.00**	**5.79**	398.46	398.46	**0.00**	507.95	395.54	398.46	0.73	3547.70	398.46	398.46	**0.00**	432.41	57720	398.46	398.46	**0.00**	101.06	511	19	···
40-80-8	381.45	381.45	**0.00**	**3.81**	381.45	381.45	**0.00**	3602.69	379.82	381.45	0.43	3553.07	381.45	381.45	**0.00**	398.34	57720	381.45	381.45	**0.00**	94.77	553	12	···
40-80-9	359.76	359.76	**0.00**	**3.04**	359.76	359.76	0.00	3602.04	359.76	359.76	**0.00**	2493.75	359.76	359.76	**0.00**	494.51	57720	359.76	359.76	**0.00**	95.99	632	19	···
50-100-1	429.63	429.63	**0.00**	9.57	429.63	429.63	**0.00**	**5.11**	428.44	429.65	0.28	3412.28	429.63	429.63	**0.00**	654.66	72150	429.63	429.63	**0.00**	116.16	817	11	···
50-100-10	427.67	427.67	**0.00**	**7.51**	427.55	427.67	0.03	3602.11	425.50	427.67	0.51	3350.59	427.67	427.67	**0.00**	498.25	72150	427.67	427.67	**0.00**	115.85	789	19	···
50-100-2	433.23	433.23	**0.00**	**4.73**	433.21	433.23	0.00	3602.47	432.71	433.23	0.12	3371.77	433.23	433.23	**0.00**	483.41	72150	433.23	433.23	**0.00**	107.94	749	12	···
50-100-3	408.28	408.28	**0.00**	**7.60**	408.20	408.28	0.02	3602.46	406.91	408.29	0.34	3416.91	408.28	408.28	**0.00**	493.45	72150	408.28	408.28	**0.00**	111.43	780	15	···
50-100-4	390.91	390.91	**0.00**	**6.88**	390.69	390.91	0.06	3602.06	389.96	390.91	0.25	3423.14	390.91	390.91	**0.00**	515.18	72150	390.91	390.91	**0.00**	123.58	846	12	···
50-100-5	376.19	376.19	**0.00**	**4.58**	376.06	376.19	0.03	3602.46	375.17	376.19	0.27	3343.79	376.19	376.19	**0.00**	399.16	72150	376.19	376.19	**0.00**	110.00	679	13	···
50-100-6	425.59	425.59	**0.00**	**9.59**	425.59	425.59	**0.00**	3602.57	423.93	425.59	0.39	3299.96	425.59	425.59	**0.00**	390.11	72150	425.59	425.59	**0.00**	116.44	737	13	···
50-100-7	444.76	444.76	**0.00**	7.19	444.76	444.76	**0.00**	**4.65**	442.88	444.77	0.42	3396.53	444.76	444.76	**0.00**	376.70	72150	444.76	444.76	**0.00**	110.61	747	18	···
50-100-8	371.92	371.92	**0.00**	**8.64**	371.83	371.92	0.02	3602.04	369.98	371.93	0.52	3398.07	371.94	371.94	**0.00**	521.33	72150	371.92	371.92	**0.00**	114.44	710	13	···
50-100-9	444.03	444.03	**0.00**	**6.18**	443.99	444.03	0.01	3602.05	442.16	444.03	0.42	3600.08	444.03	444.03	**0.00**	501.37	72150	444.03	444.03	**0.00**	116.54	714	16	···

Table 61 Detailed results for problem NLUFLP-W-gunluk, cost functions f_6

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	304.17	304.17	**0.00**	**0.11**	−∞	+∞	100.0	+∞	304.17	304.17	**0.00**	374.50	−∞	+∞	100.00	3600.01	+∞	304.17	304.17	**0.00**	6.25	136	12	···
10-20-10	287.50	287.50	**0.00**	**0.13**	−∞	+∞	100.0	+∞	287.50	287.50	**0.00**	3644.69	287.50	287.50	**0.00**	897.93	231725	287.50	287.50	**0.00**	6.65	138	12	···
10-20-2	235.28	235.28	**0.00**	**0.19**	−∞	+∞	100.0	+∞	235.28	235.28	**0.00**	2404.51	235.28	235.28	**0.00**	2578.97	763212	235.28	235.28	**0.00**	7.39	133	14	···
10-20-3	271.12	271.12	**0.00**	**0.20**	258.18	271.12	4.77	3602.30	271.12	271.12	**0.00**	2071.23	−∞	+∞	100.00	3600.01	+∞	271.12	271.12	**0.00**	7.69	138	12	···
10-20-4	289.17	289.17	**0.00**	**0.11**	284.94	289.17	1.46	3602.32	289.17	289.17	**0.00**	3.44	289.17	289.17	**0.00**	1987.33	763169	289.17	289.17	**0.00**	6.06	120	12	···
10-20-5	322.67	322.67	**0.00**	**0.07**	320.46	322.67	0.68	3602.92	322.67	322.67	**0.00**	20.20	322.67	322.67	**0.00**	2134.42	763013	322.67	322.67	**0.00**	5.94	126	12	···
10-20-6	304.84	304.84	**0.00**	**0.28**	−∞	+∞	100.0	+∞	304.84	304.84	**0.00**	1943.41	304.84	304.84	**0.00**	3441.32	762855	304.84	304.84	**0.00**	7.72	144	14	···
10-20-7	235.55	235.55	**0.00**	**0.13**	−∞	+∞	100.0	+∞	235.55	235.55	**0.00**	178.67	235.55	235.55	**0.00**	3112.81	763081	235.55	235.55	**0.00**	6.86	143	12	···
10-20-8	294.90	294.90	**0.00**	**0.50**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	294.90	294.90	**0.00**	2909.56	762839	294.90	294.90	**0.00**	10.24	145	13	···
10-20-9	302.88	302.88	**0.00**	**0.27**	−∞	+∞	100.0	+∞	302.88	302.88	**0.00**	3655.70	302.88	302.88	**0.00**	2603.18	762915	302.88	302.88	**0.00**	9.71	152	16	···
20-40-1	396.68	396.68	**0.00**	**7.80**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	396.67	396.68	**0.00**	1716.77	44824	396.67	396.68	**0.00**	11.36	285	12	···
20-40-10	398.46	398.46	**0.00**	**0.86**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	398.46	398.46	**0.00**	7.66	276	13	···
20-40-2	347.92	347.92	**0.00**	**10.92**	−∞	+∞	100.0	+∞	347.92	347.92	**0.00**	3686.25	−∞	+∞	100.00	3600.01	+∞	347.92	347.92	**0.00**	13.96	285	17	···
20-40-3	371.69	371.69	**0.00**	29.66	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	371.69	371.69	**0.00**	**14.49**	284	13	···
20-40-4	426.36	426.36	**0.00**	**2.94**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	426.36	426.36	**0.00**	2268.91	139936	426.36	426.36	**0.00**	13.32	245	15	···
20-40-5	347.72	347.72	**0.00**	**7.17**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	347.72	347.72	**0.00**	11.59	274	14	···
20-40-6	404.65	404.65	**0.00**	**10.51**	−∞	+∞	100.0	+∞	404.65	404.65	**0.00**	3677.66	−∞	+∞	100.00	3600.01	+∞	404.65	404.65	**0.00**	12.33	281	12	···
20-40-7	410.19	410.19	**0.00**	**5.57**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	410.19	410.19	**0.00**	14.99	307	13	···
20-40-8	377.01	377.01	**0.00**	**7.77**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	377.01	377.01	**0.00**	12.80	296	13	···
20-40-9	368.25	368.25	**0.00**	**12.02**	−∞	+∞	100.0	+∞	368.25	368.25	**0.00**	3615.57	−∞	+∞	100.00	3600.01	+∞	368.25	368.25	**0.00**	18.00	310	16	···
30-60-1	462.62	462.62	**0.00**	995.41	345.96	463.48	25.36	3602.84	432.74	463.07	6.55	3600.10	−∞	+∞	100.00	3600.01	+∞	462.62	462.62	**0.00**	**31.84**	448	15	···
30-60-10	346.59	346.59	**0.00**	33.33	288.75	346.59	16.69	3602.37	342.83	346.59	1.08	3600.10	−∞	+∞	100.00	3600.01	+∞	346.59	346.59	**0.00**	**14.92**	422	15	···
30-60-2	394.92	394.92	**0.00**	301.91	319.27	394.92	19.16	3602.43	381.00	394.92	3.52	3600.10	−∞	+∞	100.00	3600.01	+∞	394.92	394.92	**0.00**	**26.05**	439	16	···
30-60-3	416.69	416.69	**0.00**	301.51	333.42	417.00	20.04	3602.49	404.15	416.69	3.01	3600.10	−∞	+∞	100.00	3600.01	+∞	416.69	416.69	**0.00**	**18.66**	416	13	···
30-60-4	459.60	459.60	**0.00**	1330.26	365.56	459.60	20.46	3602.91	442.82	460.03	3.74	3600.10	−∞	+∞	100.00	3600.01	+∞	459.60	459.60	**0.00**	**33.11**	440	17	···
30-60-5	418.90	418.90	**0.00**	1989.14	342.10	419.17	18.39	3602.41	397.22	419.50	5.31	3600.10	−∞	+∞	100.00	3600.01	+∞	418.90	418.90	**0.00**	**23.81**	476	15	···
30-60-6	456.08	456.08	**0.00**	321.54	347.53	456.09	23.80	3602.42	438.07	456.08	3.95	3600.10	−∞	+∞	100.00	3600.01	+∞	456.08	456.08	**0.00**	**23.31**	410	15	···
30-60-7	513.44	513.44	**0.00**	735.46	403.82	513.49	21.36	3602.77	499.91	513.54	2.65	3600.10	−∞	+∞	100.00	3600.01	+∞	513.44	513.44	**0.00**	**38.92**	442	17	···
30-60-8	457.11	457.11	**0.00**	332.12	365.49	457.11	20.04	3602.31	440.98	457.11	3.53	3600.10	−∞	+∞	100.00	3600.01	+∞	457.11	457.11	**0.00**	**14.12**	437	13	···
30-60-9	425.27	427.97	0.63	3600.01	322.55	427.97	24.63	3602.39	408.97	428.24	4.50	3600.10	−∞	+∞	100.00	3600.01	+∞	427.97	427.97	**0.00**	**21.79**	444	13	···
40-80-1	516.86	535.44	3.47	3600.01	391.17	536.21	27.05	3602.38	488.84	536.48	8.88	3600.10	−∞	+∞	100.00	3600.01	+∞	535.44	535.44	**0.00**	**102.70**	592	16	···
40-80-10	536.02	536.03	**0.00**	2395.24	436.72	536.02	18.53	3602.37	513.36	536.20	4.26	3600.10	−∞	+∞	100.00	3600.01	+∞	536.03	536.03	**0.00**	**49.50**	573	14	···
40-80-2	508.51	517.08	**1.66**	3600.01	394.86	517.08	23.64	3602.87	489.69	517.31	5.34	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
40-80-3	489.63	493.97	0.88	3600.01	373.12	494.15	24.49	3602.48	469.70	493.98	4.92	3600.10	−∞	+∞	100.00	3600.01	+∞	493.97	493.97	**0.00**	**95.71**	624	19	···
40-80-4	506.97	511.65	0.91	3600.01	414.44	511.65	19.00	3602.80	495.24	511.65	3.21	3600.10	−∞	+∞	100.00	3600.01	+∞	511.65	511.65	**0.00**	**66.52**	572	14	···
40-80-5	497.23	502.17	0.98	3600.01	391.32	502.17	22.07	3602.63	481.07	502.17	4.20	3600.10	−∞	+∞	100.00	3600.01	+∞	502.17	502.17	**0.00**	**38.91**	587	15	···
40-80-6	509.14	509.14	**0.00**	292.16	429.72	509.14	15.60	3602.99	494.55	509.14	2.87	3600.10	−∞	+∞	100.00	3600.01	+∞	509.14	509.14	**0.00**	**23.83**	604	13	···
40-80-7	534.43	543.20	1.61	3600.01	407.58	543.20	24.97	3602.66	515.30	543.20	5.14	3600.10	−∞	+∞	100.00	3600.01	+∞	543.20	543.20	**0.00**	**73.90**	591	15	···
40-80-8	491.71	497.08	1.08	3600.01	389.16	497.08	21.71	3602.10	462.57	497.22	6.97	3600.10	−∞	+∞	100.00	3600.01	+∞	497.08	497.08	**0.00**	**41.40**	596	15	···
40-80-9	480.25	480.25	**0.00**	1796.71	375.77	480.25	21.76	3602.88	456.43	480.28	4.97	3600.10	−∞	+∞	100.00	3602.75	237402	480.25	480.25	**0.00**	**34.49**	567	15	···
50-100-1	545.12	552.45	1.33	3600.01	437.37	552.36	20.82	3602.56	516.61	552.64	6.52	3600.10	−∞	+∞	100.00	3600.01	+∞	552.21	552.21	**0.00**	**65.14**	747	14	···
50-100-10	565.16	584.19	3.26	3600.01	429.85	584.10	26.41	3604.54	535.45	584.40	8.38	3600.10	−∞	+∞	100.00	3600.01	+∞	583.93	583.93	**0.00**	**103.73**	728	16	···
50-100-2	557.34	566.79	1.67	3600.01	440.42	566.75	22.29	3602.64	532.40	566.76	6.06	3600.10	−∞	+∞	100.00	3600.01	+∞	566.76	566.76	**0.00**	**48.05**	691	14	···
50-100-3	528.31	533.38	0.95	3600.01	413.16	533.49	22.56	3603.04	497.95	533.49	6.66	3600.10	−∞	+∞	100.00	3600.01	+∞	533.28	533.28	**0.00**	**54.34**	692	15	···
50-100-4	501.93	508.95	**1.38**	3600.01	390.10	508.92	23.35	3604.13	467.30	509.72	8.32	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
50-100-5	487.74	496.17	1.70	3600.01	381.60	496.17	23.09	3603.24	466.24	496.17	6.03	3600.10	−∞	+∞	100.00	3600.01	+∞	496.17	496.17	**0.00**	**80.97**	704	14	···
50-100-6	544.30	552.55	1.49	3600.01	434.61	552.48	21.33	3602.81	522.20	552.48	5.48	3600.10	−∞	+∞	100.00	3600.01	+∞	552.43	552.43	**0.00**	**90.31**	715	17	···
50-100-7	595.37	618.94	3.81	3600.01	447.83	618.70	27.62	3604.25	562.58	619.59	9.20	3600.10	−∞	+∞	100.00	3600.01	+∞	618.70	618.70	**0.00**	**88.88**	758	14	···
50-100-8	500.34	510.34	1.96	3600.01	373.41	510.28	26.82	3602.89	472.05	510.44	7.52	3600.10	−∞	+∞	100.00	3600.01	+∞	510.28	510.28	**0.00**	**54.85**	730	13	···
50-100-9	557.42	567.74	1.82	3600.01	446.74	567.74	21.31	3602.58	530.46	568.06	6.62	3600.10	−∞	+∞	100.00	3600.01	+∞	567.71	567.71	**0.00**	**77.32**	713	15	···

Table 62 Detailed results for problem NLUFLP-W-gunluk, cost functions f_7

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	208.46	208.46	0.00	0.01	208.46	208.46	0.00	2.77	208.46	208.46	0.00	0.34	208.46	208.46	0.00	15.32	14970	208.46	208.46	0.00	0.89	167	6	···
10-20-10	198.71	198.71	0.00	0.01	198.71	198.71	0.00	2.86	198.71	198.71	0.00	0.22	198.71	198.71	0.00	15.44	14970	198.71	198.71	0.00	0.93	169	7	···
10-20-2	167.39	167.39	0.00	0.01	167.39	167.39	0.00	2.49	167.39	167.39	0.00	0.28	167.39	167.39	0.00	17.48	14970	167.39	167.39	0.00	1.24	196	11	···
10-20-3	193.34	193.34	0.00	0.01	193.34	193.34	0.00	2.65	193.34	193.34	0.00	0.51	193.34	193.34	0.00	15.67	14970	193.34	193.34	0.00	1.50	208	14	···
10-20-4	197.78	197.78	0.00	0.01	197.78	197.78	0.00	2.58	197.78	197.78	0.00	0.19	197.78	197.78	0.00	15.41	14970	197.78	197.78	0.00	0.71	140	4	···
10-20-5	268.71	268.71	0.00	0.01	268.70	268.71	0.00	3.17	268.71	268.71	0.00	0.28	268.71	268.71	0.00	15.49	14970	268.71	268.71	0.00	0.93	163	17	···
10-20-6	140.40	140.40	0.00	0.01	140.40	140.40	0.00	2.78	140.40	140.40	0.00	0.24	140.40	140.40	0.00	15.31	14970	140.40	140.40	0.00	1.05	175	10	···
10-20-7	162.19	162.19	0.00	0.01	162.19	162.19	0.00	2.76	162.19	162.19	0.00	0.27	162.19	162.19	0.00	15.53	14970	162.19	162.19	0.00	1.02	192	6	···
10-20-8	167.25	167.25	0.00	0.01	167.25	167.25	0.00	2.86	167.25	167.25	0.00	0.42	167.25	167.25	0.00	15.52	14970	167.25	167.25	0.00	9.18	260	14	···
10-20-9	175.08	175.08	0.00	0.01	175.08	175.08	0.00	2.89	175.08	175.08	0.00	0.28	175.08	175.08	0.00	15.09	14970	175.08	175.08	0.00	0.69	153	5	···
20-40-1	259.32	259.32	0.00	0.02	259.32	259.32	0.00	6.81	259.32	259.32	0.00	6.19	259.32	259.32	0.00	32.03	29940	259.32	259.32	0.00	5.04	305	16	···
20-40-10	262.16	262.16	0.00	0.02	262.16	262.16	0.00	6.19	262.16	262.16	0.00	1.32	262.16	262.16	0.00	31.18	29940	262.16	262.16	0.00	5.77	309	12	···
20-40-2	232.10	232.10	0.00	0.01	232.10	232.10	0.00	5.58	232.10	232.10	0.00	1.34	232.10	232.10	0.00	32.20	29940	232.10	232.10	0.00	9.27	307	12	···
20-40-3	215.85	215.85	0.00	0.02	215.85	215.85	0.00	6.25	215.85	215.85	0.00	1.40	215.85	215.85	0.00	32.00	29940	215.85	215.85	0.00	1.14	286	6	···
20-40-4	311.27	311.27	0.00	0.02	311.27	311.27	0.00	6.50	311.27	311.27	0.00	1.54	311.27	311.27	0.00	31.65	29940	311.27	311.27	0.00	5.60	259	9	···
20-40-5	220.32	220.32	0.00	0.02	220.32	220.32	0.00	5.89	220.32	220.32	0.00	1.21	220.32	220.32	0.00	31.77	29940	220.32	220.32	0.00	1.01	289	7	···
20-40-6	249.07	249.07	0.00	0.02	249.07	249.07	0.00	5.43	249.07	249.07	0.00	1.88	249.07	249.07	0.00	31.69	29940	249.07	249.07	0.00	1.49	304	15	···
20-40-7	278.97	278.97	0.00	0.02	278.97	278.97	0.00	5.62	278.97	278.97	0.00	1.77	278.97	278.97	0.00	31.46	29940	278.97	278.97	0.00	1.41	302	9	···
20-40-8	234.38	234.38	0.00	0.01	234.38	234.38	0.00	5.46	234.38	234.38	0.00	1.76	234.38	234.38	0.00	31.63	29940	234.38	234.38	0.00	6.21	324	9	···
20-40-9	232.66	232.66	0.00	0.01	232.66	232.66	0.00	6.49	232.66	232.66	0.00	5.40	232.66	232.66	0.00	31.64	29940	232.66	232.66	0.00	1.51	342	10	···
30-60-1	269.61	269.61	0.00	0.02	269.61	269.61	0.00	18.48	269.61	269.61	0.00	9.56	269.61	269.61	0.00	47.62	44910	269.61	269.61	0.00	5.99	476	11	···
30-60-10	232.64	232.64	0.00	0.02	232.64	232.64	0.00	18.53	232.64	232.64	0.00	6.27	232.64	232.64	0.00	46.73	44910	232.64	232.64	0.00	4.52	444	10	···
30-60-2	267.28	267.28	0.00	0.02	267.28	267.28	0.00	20.33	267.28	267.28	0.00	5.01	267.28	267.28	0.00	47.41	44910	267.28	267.28	0.00	5.04	381	10	···
30-60-3	267.11	267.11	0.00	0.02	267.11	267.11	0.00	18.97	267.11	267.11	0.00	7.64	267.11	267.11	0.00	47.44	44910	267.11	267.11	0.00	2.92	489	14	···
30-60-4	299.97	299.97	0.00	0.02	299.97	299.97	0.00	21.47	299.97	299.97	0.00	4.81	299.97	299.97	0.00	48.80	44910	299.97	299.97	0.00	5.74	459	8	···
30-60-5	296.80	296.80	0.00	0.02	296.80	296.80	0.00	15.41	296.80	296.80	0.00	4.73	296.80	296.80	0.00	47.44	44910	296.80	296.80	0.00	1.64	435	7	···
30-60-6	279.74	279.74	0.00	0.02	279.74	279.74	0.00	16.82	279.74	279.74	0.00	5.71	279.74	279.74	0.00	47.17	44910	279.74	279.74	0.00	4.75	518	9	···
30-60-7	336.09	336.09	0.00	0.02	336.09	336.09	0.00	19.36	336.09	336.09	0.00	6.32	336.09	336.09	0.00	47.45	44910	336.09	336.09	0.00	12.59	457	12	···
30-60-8	306.78	306.78	0.00	0.02	306.78	306.78	0.00	15.60	306.78	306.78	0.00	5.00	306.78	306.78	0.00	47.56	44910	306.78	306.78	0.00	5.29	434	10	···
30-60-9	257.98	257.98	0.00	0.02	257.98	257.98	0.00	18.23	257.98	257.98	0.00	4.86	257.98	257.98	0.00	47.16	44910	257.98	257.98	0.00	5.53	437	16	···
40-80-1	332.25	332.25	0.00	0.03	332.25	332.25	0.00	69.16	332.25	332.25	0.00	12.64	332.25	332.25	0.00	62.38	59880	332.25	332.25	0.00	13.43	549	11	···
40-80-10	384.97	384.97	0.00	0.04	384.97	384.97	0.00	53.80	384.97	384.97	0.00	13.77	384.97	384.97	0.00	70.97	59880	384.97	384.97	0.00	2.76	584	9	···
40-80-2	341.50	341.50	0.00	0.03	341.50	341.50	0.00	52.94	341.50	341.50	0.00	13.40	341.50	341.50	0.00	62.69	59880	341.50	341.50	0.00	5.35	564	7	···
40-80-3	315.70	315.70	0.00	0.03	315.70	315.70	0.00	53.75	315.70	315.70	0.00	12.77	315.70	315.70	0.00	61.50	59880	315.70	315.70	0.00	5.56	590	6	···
40-80-4	361.61	361.61	0.00	0.03	361.61	361.61	0.00	58.61	361.61	361.61	0.00	12.61	361.61	361.61	0.00	62.33	59880	361.61	361.61	0.00	5.15	520	6	···
40-80-5	332.27	332.27	0.00	0.03	332.27	332.27	0.00	51.31	332.27	332.27	0.00	12.27	332.27	332.27	0.00	60.82	59880	332.27	332.27	0.00	6.35	564	10	···
40-80-6	384.58	384.58	0.00	0.03	384.58	384.58	0.00	47.18	384.58	384.58	0.00	12.09	384.58	384.58	0.00	62.97	59880	384.58	384.58	0.00	2.36	559	13	···
40-80-7	346.35	346.35	0.00	0.03	346.35	346.35	0.00	58.06	346.35	346.35	0.00	13.19	346.35	346.35	0.00	62.65	59880	346.35	346.35	0.00	2.61	540	11	···
40-80-8	338.00	338.00	0.00	0.03	338.00	338.00	0.00	57.74	338.00	338.00	0.00	16.99	338.00	338.00	0.00	63.40	59880	338.00	338.00	0.00	1.82	530	8	···
40-80-9	314.36	314.36	0.00	0.03	314.36	314.36	0.00	55.59	314.36	314.36	0.00	13.26	314.36	314.36	0.00	62.05	59880	314.36	314.36	0.00	5.00	604	7	···
50-100-1	381.57	381.57	0.00	0.04	381.57	381.57	0.00	144.65	381.56	381.56	0.00	29.40	381.57	381.57	0.00	80.56	74850	381.57	381.57	0.00	10.45	646	6	···
50-100-10	365.87	365.87	0.00	0.04	365.87	365.87	0.00	140.42	365.82	365.82	0.00	27.24	365.87	365.87	0.00	82.77	74850	365.87	365.87	0.00	9.91	682	12	···
50-100-2	390.33	390.33	0.00	0.04	390.33	390.33	0.00	172.93	390.33	390.33	0.00	47.34	390.33	390.33	0.00	79.08	74850	390.33	390.33	0.00	11.23	701	6	···
50-100-3	358.15	358.15	0.00	0.04	358.15	358.15	0.00	140.08	358.15	358.15	0.00	30.74	358.15	358.15	0.00	79.09	74850	358.15	358.15	0.00	2.16	637	8	···
50-100-4	350.00	350.00	0.00	0.04	350.00	350.00	0.00	146.58	350.00	350.00	0.00	28.66	350.00	350.00	0.00	79.10	74850	350.00	350.00	0.00	5.50	773	6	···
50-100-5	336.76	336.76	0.00	0.04	336.76	336.76	0.00	169.86	336.76	336.76	0.00	27.83	336.76	336.76	0.00	79.28	74850	336.76	336.76	0.00	5.98	689	8	···
50-100-6	378.84	378.84	0.00	0.04	378.84	378.84	0.00	166.47	378.84	378.84	0.00	55.17	378.84	378.84	0.00	78.66	74850	378.84	378.84	0.00	6.83	731	10	···
50-100-7	391.28	391.28	0.00	0.04	391.28	391.28	0.00	183.97	391.28	391.28	0.00	25.06	391.28	391.28	0.00	79.61	74850	391.28	391.28	0.00	6.91	672	8	···
50-100-8	324.05	324.05	0.00	0.04	324.05	324.05	0.00	168.49	324.05	324.05	0.00	78.05	324.05	324.05	0.00	78.34	74850	324.05	324.05	0.00	6.03	688	8	···
50-100-9	394.41	394.41	0.00	0.04	394.41	394.41	0.00	170.05	394.41	394.41	0.00	107.87	394.41	394.41	0.00	75.44	74850	394.41	394.41	0.00	8.14	765	13	···

Table 63 **Detailed results for problem NLUFLP-W-gunluk, cost functions** f_8

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	295.32	295.32	**0.00**	**0.11**	292.23	295.32	1.05	3602.56	295.32	295.32	**0.00**	841.77	−∞	+∞	100.00	3608.63	1038950	295.32	295.32	**0.00**	3.66	197	18	···
10-20-10	289.70	289.70	**0.00**	**0.26**	280.14	289.72	3.31	3602.50	289.70	289.70	**0.00**	3021.70	−∞	+∞	100.00	3606.23	743464	289.70	289.70	**0.00**	6.25	201	19	···
10-20-2	231.99	231.99	**0.00**	**1.17**	221.78	231.99	4.40	3602.51	231.99	231.99	**0.00**	98.25	231.99	231.99	**0.00**	3474.55	713342	231.99	231.99	**0.00**	3.89	204	17	···
10-20-3	272.13	272.13	**0.00**	**0.64**	259.31	272.13	4.71	3602.53	272.12	272.14	0.01	3662.76	−∞	+∞	100.00	3608.92	1166780	272.13	272.13	**0.00**	5.78	187	16	···
10-20-4	280.23	280.23	**0.00**	**0.24**	280.18	280.23	0.02	3602.68	280.23	280.23	**0.00**	257.07	−∞	+∞	100.00	3607.10	780228	280.23	280.23	**0.00**	4.27	173	18	···
10-20-5	320.19	320.19	**0.00**	**0.09**	318.57	320.19	0.51	3602.41	320.19	320.19	**0.00**	136.35	−∞	+∞	100.00	3606.82	804357	320.19	320.19	**0.00**	4.07	204	17	···
10-20-6	287.63	287.63	**0.00**	**0.40**	−∞	+∞	100.0	+∞	287.63	287.63	**0.00**	1903.68	−∞	+∞	100.00	3606.81	791982	287.63	287.63	**0.00**	5.21	204	17	···
10-20-7	232.27	232.27	**0.00**	**0.28**	221.03	232.28	4.84	3602.56	232.27	232.27	**0.00**	148.05	−∞	+∞	100.00	3606.44	761697	232.27	232.27	**0.00**	5.42	200	20	···
10-20-8	284.25	284.25	**0.00**	**0.95**	−∞	+∞	100.0	+∞	284.25	284.25	**0.00**	3625.91	−∞	+∞	100.00	3606.25	708813	284.25	284.25	**0.00**	7.23	166	18	···
10-20-9	289.61	289.61	**0.00**	**0.29**	−∞	+∞	100.0	+∞	289.61	289.61	**0.00**	322.29	−∞	+∞	100.00	3606.22	709788	289.61	289.61	**0.00**	6.99	199	17	···
20-40-1	396.89	396.89	**0.00**	38.37	335.99	399.96	15.99	3602.82	383.50	397.32	3.48	3600.10	−∞	+∞	100.00	3600.01	+∞	396.89	396.89	**0.00**	**11.96**	404	17	···
20-40-10	403.22	403.22	**0.00**	13.37	355.00	405.15	12.38	3602.55	381.14	404.86	5.86	3600.10	−∞	+∞	100.00	3600.01	+∞	403.22	403.22	**0.00**	**11.67**	402	19	···
20-40-2	340.47	340.47	**0.00**	23.05	−∞	+∞	100.0	+∞	334.58	341.17	1.93	3600.10	−∞	+∞	100.00	3600.01	+∞	340.47	340.47	**0.00**	**12.36**	375	20	···
20-40-3	370.21	370.21	**0.00**	305.14	−∞	+∞	100.0	+∞	349.24	370.32	5.69	3600.10	−∞	+∞	100.00	3600.01	+∞	370.21	370.21	**0.00**	**16.80**	405	19	···
20-40-4	420.74	420.74	**0.00**	30.29	−∞	+∞	100.0	+∞	420.74	420.77	0.01	3564.04	−∞	+∞	100.00	3600.01	+∞	420.74	420.74	**0.00**	**13.20**	325	18	···
20-40-5	350.27	350.27	**0.00**	55.38	292.24	351.17	16.78	3602.78	323.28	350.84	7.85	3600.10	−∞	+∞	100.00	3600.01	+∞	350.27	350.27	**0.00**	**14.37**	373	20	···
20-40-6	402.94	402.94	**0.00**	132.23	−∞	+∞	100.0	+∞	366.10	407.08	10.07	3600.10	−∞	+∞	100.00	3600.01	+∞	402.94	402.94	**0.00**	**41.48**	414	24	···
20-40-7	407.11	407.11	**0.00**	30.92	356.61	409.79	12.98	3602.34	404.28	407.25	0.73	3586.53	−∞	+∞	100.00	3600.01	+∞	407.11	407.11	**0.00**	**23.76**	417	21	···
20-40-8	377.39	377.39	**0.00**	148.22	314.74	382.94	17.81	3602.70	362.41	377.94	4.11	3600.10	−∞	+∞	100.00	3600.01	+∞	377.39	377.39	**0.00**	**13.50**	397	18	···
20-40-9	366.85	366.85	**0.00**	63.84	305.47	368.40	17.08	3602.29	348.42	368.10	5.35	3600.10	−∞	+∞	100.00	3600.01	+∞	366.85	366.85	**0.00**	**17.91**	422	18	···
30-60-1	450.13	457.83	1.68	3600.01	345.99	468.18	26.10	3602.24	317.20	504.22	37.09	3600.10	−∞	+∞	100.00	3600.01	+∞	457.83	457.83	**0.00**	**50.08**	622	20	···
30-60-10	356.84	356.84	**0.00**	554.04	291.25	358.47	18.75	3602.35	−∞	420.04	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	356.84	356.84	**0.00**	**17.21**	580	18	···
30-60-2	394.04	394.04	**0.00**	2602.87	319.10	396.17	19.46	3602.25	−∞	414.65	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	394.04	394.04	**0.00**	**33.63**	576	19	···
30-60-3	424.20	424.20	**0.00**	2988.87	339.74	431.49	21.26	3602.32	−∞	490.57	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	424.20	424.20	**0.00**	**25.64**	584	20	···
30-60-4	453.19	459.73	1.42	3600.01	367.96	463.64	20.64	3602.70	−∞	496.69	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	459.73	459.73	**0.00**	**34.76**	573	19	···
30-60-5	415.03	429.45	3.36	3600.01	348.58	434.17	19.71	3602.44	369.77	431.33	14.27	3600.10	−∞	+∞	100.00	3600.01	+∞	429.45	429.45	**0.00**	**23.99**	579	19	···
30-60-6	459.01	459.01	**0.00**	2743.87	351.68	462.83	24.02	3602.33	−∞	509.11	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	459.01	459.01	**0.00**	**26.22**	591	19	···
30-60-7	503.25	508.91	1.11	3600.01	406.61	518.95	21.65	3602.31	−∞	558.99	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	508.91	508.91	**0.00**	**74.27**	637	19	···
30-60-8	453.22	462.02	1.90	3600.01	369.84	468.49	21.06	3602.89	−∞	503.57	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	462.02	462.02	**0.00**	**29.59**	569	22	···
30-60-9	417.78	435.97	4.17	3600.01	327.89	437.79	25.10	3602.54	−∞	473.21	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	435.86	435.86	**0.00**	**27.36**	566	18	···
40-80-1	500.79	545.71	8.23	3600.01	390.64	550.95	29.10	3603.40	380.67	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	544.65	544.65	**0.00**	**209.49**	827	21	···
40-80-10	530.24	549.04	3.42	3600.01	435.74	552.45	21.13	3602.73	427.35	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	549.04	549.04	**0.00**	**77.35**	784	19	···
40-80-2	491.06	532.62	7.80	3600.01	394.16	537.71	26.70	3602.42	−∞	560.84	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	532.40	532.40	**0.00**	**127.68**	835	23	···
40-80-3	477.84	497.18	3.89	3600.01	371.87	502.18	25.95	3602.89	361.28	572.07	36.85	3600.10	−∞	+∞	100.00	3600.01	+∞	497.18	497.18	**0.00**	**118.66**	829	19	···
40-80-4	500.46	523.66	4.43	3600.01	412.48	528.83	22.00	3603.01	402.89	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	522.77	522.77	**0.00**	**90.42**	781	19	···
40-80-5	489.38	512.95	4.59	3600.01	387.07	515.55	24.92	3602.81	375.45	557.41	32.64	3600.10	−∞	+∞	100.00	3600.01	+∞	512.95	512.95	**0.00**	**71.00**	782	20	···
40-80-6	513.46	519.70	1.20	3600.01	429.13	525.08	18.27	3603.15	421.04	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	519.70	519.70	**0.00**	**42.31**	835	21	···
40-80-7	518.32	546.46	5.15	3600.01	409.32	548.08	25.32	3603.14	396.81	611.07	35.06	3600.10	−∞	+∞	100.00	3600.01	+∞	546.46	546.46	**0.00**	**98.86**	807	18	···
40-80-8	483.05	505.45	4.43	3600.01	384.27	517.24	25.71	3602.52	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	505.44	505.45	**0.00**	**92.22**	860	19	···
40-80-9	466.65	488.51	4.47	3600.01	373.36	491.34	24.01	3602.64	357.55	535.62	33.25	3600.10	−∞	+∞	100.00	3600.01	+∞	488.51	488.51	**0.00**	**77.30**	739	25	···
50-100-1	527.68	571.19	7.62	3600.01	428.20	575.52	25.60	3602.41	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	569.37	569.37	**0.00**	**116.76**	1042	18	···
50-100-10	546.43	602.78	9.35	3600.01	427.09	605.74	29.49	3603.81	422.74	683.82	38.18	3600.10	−∞	+∞	100.00	3600.01	+∞	601.71	601.71	**0.00**	**242.54**	1004	21	···
50-100-2	533.83	588.75	9.33	3600.01	432.98	593.49	27.05	3602.71	428.93	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	586.82	586.82	**0.00**	**100.33**	954	20	···
50-100-3	508.59	555.46	8.44	3600.01	404.38	554.77	27.11	3602.82	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	552.16	552.16	**0.00**	**90.35**	964	21	···
50-100-4	489.39	537.67	8.98	3600.01	388.65	534.28	27.26	3603.19	386.72	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	531.97	531.97	**0.00**	**128.92**	1035	21	···
50-100-5	471.17	521.83	9.71	3600.01	375.83	524.47	28.34	3602.78	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	519.48	519.48	**0.00**	**107.10**	1005	20	···
50-100-6	530.58	568.95	6.74	3600.01	426.97	572.77	25.45	3606.51	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	567.52	567.52	**0.00**	**202.57**	1030	22	···
50-100-7	572.54	650.87	12.03	3600.01	443.15	654.56	32.30	3607.97	441.15	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	645.84	645.84	**0.00**	**280.50**	1100	25	···
50-100-8	482.44	532.49	9.40	3600.01	372.86	533.34	30.09	3603.72	367.81	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	529.54	529.54	**0.00**	**129.71**	1056	19	···
50-100-9	537.41	588.34	8.66	3600.01	440.41	593.49	25.79	3602.90	437.60	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	586.71	586.71	**0.00**	**176.75**	990	21	···

Table 64 Detailed results for problem NLUFLP-W-gunluk, cost functions f_9

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-20-1	206.75	206.75	0.00	0.51	206.12	206.75	0.30	3602.79	206.75	206.75	0.00	13.78	206.75	206.75	0.00	61.63	21350	206.75	206.75	0.00	6.75	161	13	···
10-20-10	194.55	194.55	0.00	0.47	192.74	194.55	0.93	3602.72	194.55	194.55	0.00	11.02	194.55	194.55	0.00	61.41	21350	194.55	194.55	0.00	7.39	169	15	···
10-20-2	164.43	164.43	0.00	0.10	164.43	164.43	0.00	184.83	164.43	164.43	0.00	2.08	164.43	164.43	0.00	62.08	21350	164.43	164.43	0.00	7.49	156	6	···
10-20-3	189.42	189.42	0.00	0.27	189.42	189.42	0.00	2183.82	189.42	189.42	0.00	6.35	189.42	189.42	0.00	59.78	21350	189.42	189.42	0.00	7.44	151	11	···
10-20-4	200.69	200.69	0.00	0.74	$-\infty$	$+\infty$	100.0	$+\infty$	200.69	200.69	0.00	17.32	200.69	200.69	0.00	63.05	21350	200.69	200.69	0.00	7.85	138	20	···
10-20-5	275.48	275.48	0.00	2.86	$-\infty$	$+\infty$	100.0	$+\infty$	275.48	275.48	0.00	51.64	275.48	275.48	0.00	72.27	21350	275.48	275.48	0.00	7.45	136	16	···
10-20-6	135.05	135.05	0.00	0.15	135.05	135.05	0.00	216.20	135.05	135.05	0.00	5.38	135.05	135.05	0.00	59.95	21350	135.05	135.05	0.00	7.27	161	7	···
10-20-7	161.18	161.18	0.00	0.10	161.18	161.18	0.00	5.47	161.18	161.18	0.00	0.44	161.18	161.18	0.00	65.39	21350	161.18	161.18	0.00	7.16	172	4	···
10-20-8	162.77	162.77	0.00	0.23	162.77	162.77	0.00	91.34	162.77	162.77	0.00	6.48	162.77	162.77	0.00	55.89	21350	162.77	162.77	0.00	7.39	150	12	···
10-20-9	170.74	170.74	0.00	0.09	170.74	170.74	0.00	16.09	170.74	170.74	0.00	2.11	170.74	170.74	0.00	60.63	21350	170.74	170.74	0.00	6.72	183	5	···
20-40-1	257.50	257.50	0.00	0.18	257.50	257.50	0.00	88.03	257.50	257.50	0.00	10.65	257.50	257.50	0.00	125.92	42700	257.50	257.50	0.00	12.07	255	6	···
20-40-10	258.24	258.24	0.00	0.24	258.04	258.24	0.08	3602.51	258.24	258.24	0.00	20.12	258.24	258.24	0.00	125.43	42700	258.24	258.24	0.00	11.91	278	9	···
20-40-2	230.39	230.39	0.00	0.12	230.39	230.39	0.00	22.01	230.39	230.39	0.00	14.44	230.39	230.39	0.00	125.92	42700	230.39	230.39	0.00	12.95	311	6	···
20-40-3	213.13	213.13	0.00	0.16	213.13	213.13	0.00	3602.73	213.13	213.13	0.00	8.54	213.13	213.13	0.00	128.45	42700	213.13	213.13	0.00	12.71	326	9	···
20-40-4	309.13	309.13	0.00	0.38	308.94	309.13	0.06	3602.75	309.13	309.13	0.00	19.42	309.13	309.13	0.00	128.48	42700	309.13	309.13	0.00	12.39	258	7	···
20-40-5	218.92	218.92	0.00	0.12	218.92	218.92	0.00	22.10	218.92	218.92	0.00	2.02	218.92	218.92	0.00	127.21	42700	218.92	218.92	0.00	12.07	243	7	···
20-40-6	245.58	245.58	0.00	0.17	245.58	245.58	0.00	888.86	245.58	245.58	0.00	10.31	245.58	245.58	0.00	126.17	42700	245.58	245.58	0.00	12.04	290	6	···
20-40-7	277.32	277.32	0.00	0.43	277.13	277.32	0.07	3602.29	277.32	277.32	0.00	17.47	277.32	277.32	0.00	126.78	42700	277.32	277.32	0.00	13.80	276	9	···
20-40-8	232.35	232.35	0.00	0.16	232.35	232.35	0.00	42.49	232.35	232.35	0.00	17.63	232.35	232.35	0.00	128.01	42700	232.35	232.35	0.00	11.77	274	6	···
20-40-9	231.43	231.43	0.00	0.16	231.43	231.43	0.00	875.95	231.43	231.43	0.00	2.21	231.43	231.43	0.00	141.40	42700	231.43	231.43	0.00	13.18	299	6	···
30-60-1	268.94	268.94	0.00	0.34	268.94	268.94	0.00	242.26	268.94	268.94	0.00	6.98	268.94	268.94	0.00	202.17	64050	268.94	268.94	0.00	17.76	437	7	···
30-60-10	231.84	231.84	0.00	0.34	231.84	231.84	0.00	183.29	231.84	231.84	0.00	51.94	231.84	231.84	0.00	197.01	64050	231.84	231.84	0.00	16.53	424	7	···
30-60-2	266.40	266.40	0.00	0.37	266.40	266.40	0.00	10.37	266.40	266.40	0.00	55.31	266.40	266.40	0.00	190.71	64050	266.40	266.40	0.00	15.58	375	4	···
30-60-3	266.37	266.37	0.00	1.10	266.34	266.37	0.01	3602.80	266.37	266.37	0.00	65.88	266.37	266.37	0.00	203.88	64050	266.37	266.37	0.00	17.64	415	8	···
30-60-4	298.90	298.90	0.00	0.43	298.90	298.90	0.00	371.13	298.90	298.90	0.00	72.31	298.90	298.90	0.00	197.65	64050	298.90	298.90	0.00	16.73	427	8	···
30-60-5	296.26	296.26	0.00	0.30	296.26	296.26	0.00	12.98	296.26	296.26	0.00	52.40	296.26	296.26	0.00	195.26	64050	296.26	296.26	0.00	16.16	401	5	···
30-60-6	278.76	278.76	0.00	0.34	278.76	278.76	0.00	280.66	278.76	278.76	0.00	48.26	278.76	278.76	0.00	201.31	64050	278.76	278.76	0.00	17.13	488	6	···
30-60-7	335.43	335.43	0.00	1.21	335.42	335.43	0.00	3602.03	335.43	335.43	0.00	105.38	335.43	335.43	0.00	200.92	64050	335.43	335.43	0.00	16.78	463	11	···
30-60-8	306.51	306.51	0.00	0.29	306.51	306.51	0.00	150.66	306.51	306.51	0.00	56.21	306.51	306.51	0.00	199.70	64050	306.51	306.51	0.00	16.06	453	5	···
30-60-9	257.38	257.38	0.00	0.32	257.38	257.38	0.00	217.27	257.38	257.38	0.00	47.64	257.38	257.38	0.00	200.32	64050	257.38	257.38	0.00	17.07	408	6	···
40-80-1	332.47	332.47	0.00	0.90	332.47	332.47	0.00	52.47	332.47	332.47	0.00	138.29	332.47	332.47	0.00	269.97	85400	332.47	332.47	0.00	20.70	604	6	···
40-80-10	385.22	385.22	0.00	1.41	385.21	385.21	0.00	3602.43	385.22	385.22	0.00	148.59	385.22	385.22	0.00	255.48	85400	385.21	385.22	0.00	21.38	549	6	···
40-80-2	341.45	341.45	0.00	0.59	341.45	341.45	0.00	37.61	341.45	341.45	0.00	184.47	341.45	341.45	0.00	261.83	85400	341.45	341.45	0.00	21.16	551	6	···
40-80-3	316.33	316.33	0.00	0.67	316.33	316.33	0.00	3602.45	316.33	316.33	0.00	185.32	316.33	316.33	0.00	267.32	85400	316.33	316.33	0.00	20.37	541	6	···
40-80-4	361.29	361.29	0.00	0.56	361.29	361.29	0.00	39.88	361.29	361.29	0.00	138.00	361.29	361.29	0.00	255.18	85400	361.29	361.29	0.00	22.59	574	6	···
40-80-5	331.83	331.83	0.00	0.54	331.83	331.83	0.00	1176.83	331.83	331.83	0.00	94.67	331.83	331.83	0.00	258.17	85400	331.83	331.83	0.00	20.53	556	5	···
40-80-6	384.95	384.95	0.00	1.10	384.95	384.95	0.00	64.13	384.95	384.95	0.00	127.37	384.95	384.95	0.00	269.91	85400	384.95	384.95	0.00	20.75	529	8	···
40-80-7	346.25	346.25	0.00	0.75	346.25	346.25	0.00	1343.28	346.25	346.25	0.00	137.58	346.25	346.25	0.00	258.20	85400	346.25	346.25	0.00	21.58	535	4	···
40-80-8	338.74	338.74	0.00	0.56	338.74	338.74	0.00	53.34	338.74	338.74	0.00	148.58	338.74	338.74	0.00	266.61	85400	338.74	338.74	0.00	22.14	677	5	···
40-80-9	314.41	314.41	0.00	0.64	314.40	314.40	0.00	40.99	314.41	314.41	0.00	199.41	314.41	314.41	0.00	259.09	85400	314.40	314.41	0.00	20.81	548	5	···
50-100-1	382.60	382.60	0.00	1.83	382.60	382.60	0.00	126.33	382.60	382.60	0.00	237.96	382.60	382.60	0.00	308.87	106750	382.60	382.60	0.00	26.99	760	6	···
50-100-10	366.33	366.33	0.00	1.06	366.31	366.33	0.00	3602.17	366.33	366.33	0.00	205.62	366.33	366.33	0.00	310.80	106750	366.33	366.33	0.00	25.34	752	8	···
50-100-2	390.71	390.71	0.00	0.76	390.71	390.71	0.00	108.37	390.71	390.71	0.00	235.93	390.71	390.71	0.00	304.62	106750	390.71	390.71	0.00	26.96	674	7	···
50-100-3	358.69	358.69	0.00	1.06	358.69	358.69	0.00	123.50	358.69	358.69	0.00	297.01	358.69	358.69	0.00	322.76	106750	358.69	358.69	0.00	29.58	666	6	···
50-100-4	351.31	351.31	0.00	0.97	351.31	351.31	0.00	131.92	351.31	351.31	0.00	242.20	351.31	351.31	0.00	311.56	106750	351.31	351.31	0.00	27.16	853	6	···
50-100-5	337.67	337.67	0.00	0.98	337.66	337.66	0.00	115.29	337.67	337.67	0.00	168.88	337.67	337.67	0.00	290.47	106750	337.67	337.67	0.00	27.20	722	8	···
50-100-6	379.43	379.43	0.00	2.70	379.43	379.43	0.00	226.01	379.43	379.43	0.00	329.18	379.43	379.43	0.00	301.60	106750	379.43	379.43	0.00	25.66	662	8	···
50-100-7	392.06	392.06	0.00	0.89	392.06	392.06	0.00	123.87	392.06	392.06	0.00	238.16	392.06	392.06	0.00	287.34	106750	392.06	392.06	0.00	23.77	728	5	···
50-100-8	324.88	324.88	0.00	1.10	324.88	324.88	0.00	108.36	324.88	324.88	0.00	235.34	324.88	324.88	0.00	302.23	106750	324.88	324.88	0.00	28.05	659	8	···
50-100-9	395.08	395.08	0.00	3.31	395.05	395.08	0.01	3602.81	395.08	395.08	0.00	600.38	395.08	395.08	0.00	295.03	106750	395.08	395.08	0.00	29.82	686	10	···

Table 65 **Detailed results for problem NLUFLP-W-gunluk, cost functions f_{10}**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	735715.27	735715.27	**0.00**	**0.85**	735715.25	735715.26	**0.00**	17.51	735715.30	735715.30	**0.00**	45.67	735715.25	735715.27	**0.00**	29.92	11881	735714.58	735715.27	**0.00**	2.12	202	7	···
cap102	778543.56	778543.56	**0.00**	**0.72**	778543.55	778543.55	**0.00**	25.66	778543.60	778543.60	**0.00**	40.24	778543.53	778543.56	**0.00**	37.68	11881	778542.90	778543.56	**0.00**	2.59	180	7	···
cap103	813769.10	813769.10	**0.00**	**0.59**	813769.07	813769.09	**0.00**	2.68	813769.10	813769.10	**0.00**	45.58	813769.06	813769.10	**0.00**	42.50	11881	813768.90	813769.10	**0.00**	2.53	168	9	···
cap104	854719.42	854719.52	**0.00**	**0.14**	854719.44	854719.51	**0.00**	3.02	854719.50	854719.50	**0.00**	10.66	854719.48	854719.52	**0.00**	43.76	11881	854719.00	854719.52	**0.00**	2.46	135	7	···
cap111	734721.49	734721.49	**0.00**	**1.44**	734721.46	734721.47	**0.00**	3.34	734721.50	734721.50	**0.00**	141.23	734721.46	734721.49	**0.00**	76.20	24256	734720.88	734721.49	**0.00**	3.11	271	7	···
cap112	775631.32	775631.32	**0.00**	**1.15**	775630.81	775631.30	**0.00**	3.08	775631.80	775631.80	**0.00**	125.92	775631.29	775631.32	**0.00**	83.29	24256	775630.79	775631.32	**0.00**	3.65	249	7	···
cap113	810560.23	810560.23	**0.00**	**0.89**	810559.51	810560.22	**0.00**	3.32	810560.20	810560.20	**0.00**	147.01	810560.19	810560.23	**0.00**	89.24	24256	810559.72	810560.23	**0.00**	3.83	240	10	···
cap114	851510.65	851510.65	**0.00**	**0.93**	851510.49	851510.64	**0.00**	3.08	851510.60	851510.60	**0.00**	136.26	851510.60	851510.65	**0.00**	96.43	24256	851510.12	851510.65	**0.00**	4.53	213	8	···
cap121	734721.49	734721.49	**0.00**	**1.52**	734721.46	734721.47	**0.00**	3.13	734721.50	734721.50	**0.00**	152.65	734721.46	734721.49	**0.00**	78.14	24256	734720.88	734721.49	**0.00**	3.14	271	7	···
cap122	775631.32	775631.32	**0.00**	**0.97**	775630.81	775631.30	**0.00**	3.08	775631.80	775631.80	**0.00**	124.70	775631.29	775631.32	**0.00**	81.56	24256	775630.79	775631.32	**0.00**	3.07	249	7	···
cap123	810560.23	810560.23	**0.00**	**0.80**	810559.51	810560.22	**0.00**	3.23	810560.20	810560.20	**0.00**	145.20	810560.19	810560.23	**0.00**	85.58	24256	810559.72	810560.23	**0.00**	4.01	240	10	···
cap124	851510.65	851510.65	**0.00**	**0.85**	851510.49	851510.64	**0.00**	6.84	851510.60	851510.60	**0.00**	147.95	851510.60	851510.65	**0.00**	97.65	24256	851510.12	851510.65	**0.00**	3.96	213	8	···
cap131	734721.49	734721.49	**0.00**	**1.55**	734721.46	734721.47	**0.00**	3.23	734721.50	734721.50	**0.00**	142.71	734721.46	734721.49	**0.00**	76.05	24256	734720.88	734721.49	**0.00**	3.51	271	7	···
cap132	775631.32	775631.32	**0.00**	**1.25**	775630.81	775631.30	**0.00**	3.28	775631.80	775631.80	**0.00**	120.78	775631.29	775631.32	**0.00**	83.73	24256	775630.79	775631.32	**0.00**	3.35	249	7	···
cap133	810560.23	810560.23	**0.00**	**0.71**	810559.51	810560.22	**0.00**	3.25	810560.20	810560.20	**0.00**	148.28	810560.19	810560.23	**0.00**	89.50	24256	810559.72	810560.23	**0.00**	4.21	240	10	···
cap134	851510.65	851510.65	**0.00**	**1.03**	851510.49	851510.64	**0.00**	3.06	851510.60	851510.60	**0.00**	134.21	851510.60	851510.65	**0.00**	93.01	24256	851510.12	851510.65	**0.00**	4.60	213	8	···
cap41	891641.18	891641.18	**0.00**	**0.38**	891641.15	891641.17	**0.00**	4.99	891641.20	891641.20	**0.00**	19.60	891641.16	891641.18	**0.00**	23.34	7426	891640.75	891641.18	**0.00**	1.59	122	7	···
cap42	919826.91	919826.91	**0.00**	**0.34**	919826.88	919826.90	**0.00**	14.81	919826.90	919826.90	**0.00**	19.53	919826.88	919826.91	**0.00**	28.45	7426	919826.48	919826.91	**0.00**	1.85	110	7	···
cap43	945126.60	945126.65	**0.00**	**0.12**	945126.51	945126.65	**0.00**	4.99	945126.60	945126.60	**0.00**	21.44	945126.62	945126.65	**0.00**	25.99	7426	945126.24	945126.65	**0.00**	1.76	104	7	···
cap44	977813.69	977813.69	**0.00**	**0.29**	977813.59	977813.69	**0.00**	4.99	977813.70	977813.70	**0.00**	17.70	977813.64	977813.69	**0.00**	26.61	7426	977813.30	977813.69	**0.00**	1.60	93	7	···
cap51	945126.60	945126.65	**0.00**	**0.12**	945126.51	945126.65	**0.00**	4.99	945126.60	945126.60	**0.00**	21.50	945126.62	945126.65	**0.00**	26.71	7426	945126.24	945126.65	**0.00**	1.60	104	7	···
cap61	891641.18	891641.18	**0.00**	**0.34**	891641.15	891641.17	**0.00**	5.84	891641.20	891641.20	**0.00**	20.29	891641.16	891641.18	**0.00**	23.25	7426	891640.75	891641.18	**0.00**	1.91	122	7	···
cap62	919826.91	919826.91	**0.00**	**0.33**	919826.88	919826.90	**0.00**	4.86	919826.90	919826.90	**0.00**	19.63	919826.88	919826.91	**0.00**	27.41	7426	919826.48	919826.91	**0.00**	1.68	110	7	···
cap63	945126.60	945126.65	**0.00**	**0.13**	945126.51	945126.65	**0.00**	5.63	945126.60	945126.60	**0.00**	21.44	945126.62	945126.65	**0.00**	26.87	7426	945126.24	945126.65	**0.00**	1.86	104	7	···
cap64	977813.69	977813.69	**0.00**	**0.31**	977813.59	977813.69	**0.00**	3.67	977813.70	977813.70	**0.00**	17.80	977813.64	977813.69	**0.00**	26.89	7426	977813.30	977813.69	**0.00**	1.60	93	7	···
cap71	891641.18	891641.18	**0.00**	**0.39**	891641.15	891641.17	**0.00**	7.77	891641.20	891641.20	**0.00**	20.55	891641.16	891641.18	**0.00**	23.62	7426	891640.75	891641.18	**0.00**	1.65	122	7	···
cap72	919826.91	919826.91	**0.00**	**0.34**	919826.88	919826.90	**0.00**	5.62	919826.90	919826.90	**0.00**	19.81	919826.88	919826.91	**0.00**	28.43	7426	919826.48	919826.91	**0.00**	1.74	110	7	···
cap73	945126.60	945126.65	**0.00**	**0.12**	945126.51	945126.65	**0.00**	6.41	945126.60	945126.60	**0.00**	21.46	945126.62	945126.65	**0.00**	26.76	7426	945126.24	945126.65	**0.00**	1.83	104	7	···
cap74	977813.69	977813.69	**0.00**	**0.25**	977813.59	977813.69	**0.00**	6.33	977813.70	977813.70	**0.00**	18.12	977813.64	977813.69	**0.00**	26.95	7426	977813.30	977813.69	**0.00**	1.55	93	7	···
cap81	735715.27	735715.27	**0.00**	**0.85**	735715.25	735715.26	**0.00**	45.75	735715.30	735715.30	**0.00**	45.48	735715.25	735715.27	**0.00**	29.54	11881	735714.58	735715.27	**0.00**	2.23	202	7	···
cap82	778543.56	778543.56	**0.00**	**0.71**	778543.55	778543.55	**0.00**	11.38	778543.60	778543.60	**0.00**	40.51	778543.53	778543.56	**0.00**	39.47	11881	778542.90	778543.56	**0.00**	2.40	180	7	···
cap83	813769.10	813769.10	**0.00**	**0.62**	813769.07	813769.09	**0.00**	16.48	813769.10	813769.10	**0.00**	44.72	813769.06	813769.10	**0.00**	43.01	11881	813768.90	813769.10	**0.00**	2.29	168	9	···
cap84	854719.42	854719.52	**0.00**	**0.15**	854719.44	854719.51	**0.00**	19.66	854719.50	854719.50	**0.00**	10.59	854719.48	854719.52	**0.00**	43.21	11881	854719.00	854719.52	**0.00**	2.21	135	7	···
cap91	735715.27	735715.27	**0.00**	**0.83**	735715.25	735715.26	**0.00**	19.67	735715.30	735715.30	**0.00**	49.72	735715.25	735715.27	**0.00**	30.82	11881	735714.58	735715.27	**0.00**	2.43	202	7	···
cap92	778543.56	778543.56	**0.00**	**0.77**	778543.55	778543.55	**0.00**	12.12	778543.60	778543.60	**0.00**	39.09	778543.53	778543.56	**0.00**	37.83	11881	778542.90	778543.56	**0.00**	2.10	180	7	···
cap93	813769.10	813769.10	**0.00**	**0.59**	813769.07	813769.09	**0.00**	11.51	813769.10	813769.10	**0.00**	45.41	813769.06	813769.10	**0.00**	42.53	11881	813768.90	813769.10	**0.00**	2.41	168	9	···
cap94	854719.42	854719.52	**0.00**	**0.14**	854719.44	854719.51	**0.00**	25.27	854719.50	854719.50	**0.00**	10.73	854719.48	854719.52	**0.00**	43.47	11881	854719.00	854719.52	**0.00**	2.00	135	7	···
#S			**37**				**37**				**37**				**37**					**37**				···
SGM			**0.00**	**0.63**			**0.00**	7.61			**0.00**	46.33			**0.00**	44.48	12697			**0.00**	2.54	162	7	···

Table 66 Detailed results for problem NLUFLP-W-orlib, cost functions f_1

	GUROBI				**SCIP**				**COUENNE**				**NAIVE**					**CN24**						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	339567.10	702801.00	51.68	3600.10	702801.06	702801.09	**0.00**	471.12	23545	702800.84	702801.09	**0.00**	**18.76**	341	9	···
cap102	732886.07	732886.81	**0.00**	221.62	$-\infty$	$+\infty$	100.0	$+\infty$	67397.84	732974.40	90.80	3600.10	732886.76	732886.81	**0.00**	487.73	23545	732886.43	732886.81	**0.00**	**20.46**	310	9	···
cap103	761104.15	761104.92	**0.00**	2687.49	$-\infty$	$+\infty$	100.0	$+\infty$	-196398.00	762470.20	125.76	3600.10	761104.86	761104.92	**0.00**	511.52	23545	761104.55	761104.92	**0.00**	**18.06**	304	9	···
cap104	797461.28	797462.87	**0.00**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-594488.70	798347.20	174.46	3600.10	797462.82	797462.90	**0.00**	527.32	23545	797462.42	797462.90	**0.00**	**20.53**	279	11	···
cap111	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	-238925.20	699433.90	134.16	3600.10	699433.85	699433.89	**0.00**	1037.34	48070	699433.56	699433.89	**0.00**	**34.92**	518	9	···
cap112	732487.50	732488.24	**0.00**	1072.65	$-\infty$	$+\infty$	100.0	$+\infty$	-863328.20	732488.20	184.84	3600.10	732488.20	732488.24	**0.00**	1060.32	48070	732487.95	732488.24	**0.00**	**33.79**	433	10	···
cap113	759434.38	759443.90	0.00	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-1484169.00	759443.90	151.17	3600.10	759443.84	759443.90	**0.00**	1072.98	48070	759443.42	759443.90	**0.00**	**34.05**	412	9	···
cap114	794460.84	794461.64	**0.00**	2769.33	$-\infty$	$+\infty$	100.0	$+\infty$	-2416402.00	795705.10	132.93	3600.10	794461.57	794461.64	**0.00**	1187.65	48070	794461.21	794461.64	**0.00**	**34.32**	400	11	···
cap121	$-\infty$	$+\infty$	100.0	$+\infty$	640533.37	711483.51	9.97	3602.90	-238663.90	699433.90	134.12	3600.10	699433.85	699433.89	**0.00**	943.30	48070	699433.56	699433.89	**0.00**	**34.02**	518	9	···
cap122	732487.50	732488.24	**0.00**	1120.67	645121.01	778317.85	17.11	3602.83	-863179.60	732488.20	184.86	3600.10	732488.20	732488.24	**0.00**	1084.09	48070	732487.95	732488.24	**0.00**	**31.55**	433	10	···
cap123	759434.35	759443.90	0.00	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-1483911.00	759443.90	151.18	3600.10	759443.84	759443.90	**0.00**	1008.46	48070	759443.42	759443.90	**0.00**	**35.12**	412	9	···
cap124	794460.84	794461.64	**0.00**	2836.25	$-\infty$	$+\infty$	100.0	$+\infty$	-2416318.00	795705.10	132.93	3600.10	794461.57	794461.64	**0.00**	1286.37	48070	794461.21	794461.64	**0.00**	**36.73**	400	11	···
cap131	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	-237605.70	699433.90	133.97	3600.10	699433.85	699433.89	**0.00**	1051.93	48070	699433.56	699433.89	**0.00**	**32.99**	518	9	···
cap132	732487.50	732488.24	**0.00**	968.29	$-\infty$	$+\infty$	100.0	$+\infty$	-863243.70	732488.20	184.85	3600.10	732488.20	732488.24	**0.00**	1018.54	48070	732487.95	732488.24	**0.00**	**32.87**	433	10	···
cap133	759434.74	759443.90	0.00	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-1483795.00	759443.90	151.18	3600.10	759443.84	759443.90	**0.00**	1048.43	48070	759443.42	759443.90	**0.00**	**34.00**	412	9	···
cap134	794460.84	794461.64	**0.00**	2770.22	$-\infty$	$+\infty$	100.0	$+\infty$	-2416402.00	795705.10	132.93	3600.10	794461.57	794461.64	**0.00**	1044.06	48070	794461.21	794461.64	**0.00**	**38.21**	400	11	···
cap41	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	722732.60	871088.30	17.03	3600.10	871088.44	871088.46	**0.00**	297.58	14716	871087.91	871088.46	**0.00**	**11.53**	186	8	···
cap42	891590.42	891591.24	**0.00**	**1.85**	$-\infty$	$+\infty$	100.0	$+\infty$	576745.90	891591.10	35.31	3600.10	891591.22	891591.24	**0.00**	288.74	14716	891590.38	891591.24	**0.00**	13.05	177	8	···
cap43	910419.74	910420.65	**0.00**	30.79	$-\infty$	$+\infty$	100.0	$+\infty$	418078.70	910453.40	54.08	3600.10	910420.61	910420.65	**0.00**	286.99	14716	910420.29	910420.65	**0.00**	**13.27**	179	9	···
cap44	935900.84	935901.78	**0.00**	1764.49	$-\infty$	$+\infty$	100.0	$+\infty$	192092.30	935903.20	79.48	3600.10	935901.74	935901.78	**0.00**	295.01	14716	935901.51	935901.78	**0.00**	**12.67**	177	10	···
cap51	910419.74	910420.65	**0.00**	32.64	904744.67	910420.49	0.62	3602.77	417968.00	910453.40	54.09	3600.10	910420.61	910420.65	**0.00**	299.05	14716	910420.29	910420.65	**0.00**	**13.52**	179	9	···
cap61	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	722040.40	871088.30	17.11	3600.10	871088.44	871088.46	**0.00**	297.75	14716	871087.91	871088.46	**0.00**	**13.65**	186	8	···
cap62	891590.42	891591.24	**0.00**	**1.76**	$-\infty$	$+\infty$	100.0	$+\infty$	576767.00	891591.10	35.31	3600.10	891591.22	891591.24	**0.00**	304.99	14716	891590.38	891591.24	**0.00**	12.18	177	8	···
cap63	910419.74	910420.65	**0.00**	35.03	$-\infty$	$+\infty$	100.0	$+\infty$	418278.90	910453.40	54.06	3600.10	910420.61	910420.65	**0.00**	293.78	14716	910420.29	910420.65	**0.00**	**13.66**	179	9	···
cap64	935900.84	935901.78	**0.00**	1785.90	$-\infty$	$+\infty$	100.0	$+\infty$	191621.10	935903.20	79.53	3600.10	935901.74	935901.78	**0.00**	286.89	14716	935901.51	935901.78	**0.00**	**13.70**	177	10	···
cap71	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	721952.70	871088.30	17.12	3600.10	871088.44	871088.46	**0.00**	306.19	14716	871087.91	871088.46	**0.00**	**12.50**	186	8	···
cap72	891590.42	891591.24	**0.00**	**1.74**	$-\infty$	$+\infty$	100.0	$+\infty$	576767.00	891591.10	35.31	3600.10	891591.22	891591.24	**0.00**	298.34	14716	891590.38	891591.24	**0.00**	11.64	177	8	···
cap73	910419.74	910420.65	**0.00**	34.12	$-\infty$	$+\infty$	100.0	$+\infty$	418299.60	910453.40	54.06	3600.10	910420.61	910420.65	**0.00**	303.13	14716	910420.29	910420.65	**0.00**	**13.60**	179	9	···
cap74	935900.84	935901.78	**0.00**	1671.96	$-\infty$	$+\infty$	100.0	$+\infty$	192423.40	935903.20	79.44	3600.10	935901.74	935901.78	**0.00**	302.28	14716	935901.51	935901.78	**0.00**	**13.72**	177	10	···
cap81	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	339568.00	702801.00	51.68	3600.10	702801.06	702801.09	**0.00**	487.92	23545	702800.84	702801.09	**0.00**	**17.92**	341	9	···
cap82	732886.07	732886.81	**0.00**	221.80	$-\infty$	$+\infty$	100.0	$+\infty$	67576.43	732974.40	90.78	3600.10	732886.76	732886.81	**0.00**	512.36	23545	732886.43	732886.81	**0.00**	**18.10**	310	9	···
cap83	761104.15	761104.92	**0.00**	2717.03	$-\infty$	$+\infty$	100.0	$+\infty$	-196668.10	762470.20	125.79	3600.10	761104.86	761104.92	**0.00**	504.58	23545	761104.55	761104.92	**0.00**	**19.36**	304	9	···
cap84	797461.19	797462.87	**0.00**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-594932.90	798347.20	174.52	3600.10	797462.82	797462.90	**0.00**	591.82	23545	797462.42	797462.90	**0.00**	**19.48**	279	11	···
cap91	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	339429.00	702801.00	51.70	3600.10	702801.06	702801.09	**0.00**	495.90	23545	702800.84	702801.09	**0.00**	**19.36**	341	9	···
cap92	732886.07	732886.81	**0.00**	220.00	$-\infty$	$+\infty$	100.0	$+\infty$	67424.35	732974.40	90.80	3600.10	732886.76	732886.81	**0.00**	489.73	23545	732886.43	732886.81	**0.00**	**18.50**	310	9	···
cap93	761104.15	761104.92	**0.00**	2759.29	$-\infty$	$+\infty$	100.0	$+\infty$	-196667.80	762470.20	125.79	3600.10	761104.86	761104.92	**0.00**	505.82	23545	761104.55	761104.92	**0.00**	**20.81**	304	9	···
cap94	797461.23	797462.87	**0.00**	3600.01	$-\infty$	$+\infty$	100.0	$+\infty$	-594137.00	798347.20	174.42	3600.10	797462.82	797462.90	**0.00**	546.56	23545	797462.42	797462.90	**0.00**	**20.42**	279	11	···
#S			22				0				0				**37**					**37**				···
SGM			2.07	1603.02			80.21	14400.00			68.98	14400.00			**0.00**	537.23	25162			**0.00**	**20.76**	286	9	···

Table 67 Detailed results for problem NLUFLP-W-orlib, cost functions f_2

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	653828.51	653828.51	**0.00**	**0.50**	653828.12	653828.12	**0.00**	19.46	653828.40	653828.40	**0.00**	3.61	653828.51	653828.51	**0.00**	229.56	40825	653828.25	653828.51	**0.00**	13.19	316	5	···
cap102	654853.42	654853.42	**0.00**	**0.47**	654852.46	654853.03	**0.00**	52.46	654853.30	654853.30	**0.00**	3.83	654853.42	654853.42	**0.00**	237.44	40825	654852.98	654853.42	**0.00**	14.76	316	5	···
cap103	655878.33	655878.33	**0.00**	**0.54**	655877.94	655877.94	**0.00**	3.74	655878.20	655878.20	**0.00**	16.24	655878.33	655878.33	**0.00**	230.16	40825	655877.71	655878.33	**0.00**	13.37	316	5	···
cap104	657415.70	657415.70	**0.00**	**0.49**	657415.31	657415.31	**0.00**	3.72	657415.60	657415.60	**0.00**	3.70	657415.70	657415.70	**0.00**	230.61	40825	657415.18	657415.70	**0.00**	14.14	334	6	···
cap111	625517.42	625517.42	**0.00**	**0.61**	625516.53	625516.53	**0.00**	4.15	625517.40	625517.40	**0.00**	47.94	625517.41	625517.42	**0.00**	459.42	83350	625517.10	625517.42	**0.00**	27.07	567	5	···
cap112	626481.41	626481.41	**0.00**	**0.65**	626480.52	626480.52	**0.00**	4.04	626481.40	626481.40	**0.00**	94.78	626481.38	626481.41	**0.00**	458.13	83350	626480.87	626481.41	**0.00**	30.91	567	5	···
cap113	627445.39	627445.39	**0.00**	**0.67**	627444.50	627444.50	**0.00**	4.88	627445.40	627445.40	**0.00**	142.35	627445.35	627445.39	**0.00**	462.05	83350	627445.15	627445.39	**0.00**	29.42	689	6	···
cap114	628891.36	628891.36	**0.00**	**0.53**	628890.47	628890.47	**0.00**	4.64	628891.40	628891.40	**0.00**	238.32	628891.31	628891.36	**0.00**	484.67	83350	628891.03	628891.36	**0.00**	27.95	689	6	···
cap121	625517.42	625517.42	**0.00**	**0.63**	625516.53	625516.53	**0.00**	4.09	625517.40	625517.40	**0.00**	48.20	625517.41	625517.42	**0.00**	504.06	83350	625517.10	625517.42	**0.00**	28.94	567	5	···
cap122	626481.41	626481.41	**0.00**	**0.55**	626480.52	626480.52	**0.00**	4.37	626481.40	626481.40	**0.00**	94.18	626481.38	626481.41	**0.00**	465.03	83350	626480.87	626481.41	**0.00**	28.84	567	5	···
cap123	627445.39	627445.39	**0.00**	**0.63**	627444.50	627444.50	**0.00**	4.74	627445.40	627445.40	**0.00**	143.09	627445.35	627445.39	**0.00**	448.05	83350	627445.15	627445.39	**0.00**	28.02	689	6	···
cap124	628891.36	628891.36	**0.00**	**0.64**	628890.47	628890.47	**0.00**	4.84	628891.40	628891.40	**0.00**	237.18	628891.31	628891.36	**0.00**	462.38	83350	628891.03	628891.36	**0.00**	29.90	689	6	···
cap131	625517.42	625517.42	**0.00**	**0.63**	625516.53	625516.53	**0.00**	6.76	625517.40	625517.40	**0.00**	47.68	625517.41	625517.42	**0.00**	456.46	83350	625517.10	625517.42	**0.00**	27.31	567	5	···
cap132	626481.41	626481.41	**0.00**	**0.67**	626480.52	626480.52	**0.00**	4.33	626481.40	626481.40	**0.00**	94.73	626481.38	626481.41	**0.00**	452.79	83350	626480.87	626481.41	**0.00**	28.38	567	5	···
cap133	627445.39	627445.39	**0.00**	**0.63**	627444.50	627444.50	**0.00**	8.56	627445.40	627445.40	**0.00**	140.81	627445.35	627445.39	**0.00**	455.89	83350	627445.15	627445.39	**0.00**	28.13	689	6	···
cap134	628891.36	628891.36	**0.00**	**0.62**	628890.47	628890.47	**0.00**	4.46	628891.40	628891.40	**0.00**	240.79	628891.31	628891.36	**0.00**	452.49	83350	628891.03	628891.36	**0.00**	26.97	689	6	···
cap41	840230.21	840230.21	**0.00**	**0.27**	840230.02	840230.02	**0.00**	5.46	840230.00	840230.00	**0.00**	1.71	840230.21	840230.21	**0.00**	144.35	25516	840229.64	840230.21	**0.00**	8.66	208	4	···
cap42	841736.89	841736.89	**0.00**	**0.31**	841736.70	841736.70	**0.00**	15.31	841736.70	841736.70	**0.00**	1.75	841736.89	841736.89	**0.00**	144.32	25516	841736.68	841736.89	**0.00**	8.82	219	5	···
cap43	843243.57	843243.57	**0.00**	**0.32**	843243.38	843243.38	**0.00**	5.69	843243.40	843243.40	**0.00**	1.80	843243.57	843243.57	**0.00**	146.51	25516	843243.28	843243.57	**0.00**	9.47	219	5	···
cap44	845503.43	845503.59	**0.00**	**0.20**	845503.40	845503.40	**0.00**	5.55	845503.40	845503.40	**0.00**	4.18	845503.59	845503.59	**0.00**	139.05	25516	845503.17	845503.59	**0.00**	9.69	219	5	···
cap51	843243.57	843243.57	**0.00**	**0.34**	843243.38	843243.38	**0.00**	5.66	843243.40	843243.40	**0.00**	1.75	843243.57	843243.57	**0.00**	143.18	25516	843243.28	843243.57	**0.00**	9.06	219	5	···
cap61	840230.21	840230.21	**0.00**	**0.32**	840230.02	840230.02	**0.00**	3.33	840230.00	840230.00	**0.00**	1.76	840230.21	840230.21	**0.00**	140.94	25516	840229.64	840230.21	**0.00**	9.16	208	4	···
cap62	841736.89	841736.89	**0.00**	**0.26**	841736.70	841736.70	**0.00**	8.75	841736.70	841736.70	**0.00**	1.78	841736.89	841736.89	**0.00**	140.92	25516	841736.68	841736.89	**0.00**	9.35	219	5	···
cap63	843243.57	843243.57	**0.00**	**0.29**	843243.38	843243.38	**0.00**	8.98	843243.40	843243.40	**0.00**	1.80	843243.57	843243.57	**0.00**	148.16	25516	843243.28	843243.57	**0.00**	9.65	219	5	···
cap64	845503.43	845503.59	**0.00**	**0.30**	845503.40	845503.40	**0.00**	7.38	845503.40	845503.40	**0.00**	4.25	845503.59	845503.59	**0.00**	144.68	25516	845503.17	845503.59	**0.00**	8.94	219	5	···
cap71	840230.21	840230.21	**0.00**	**0.31**	840230.02	840230.02	**0.00**	9.47	840230.00	840230.00	**0.00**	1.79	840230.21	840230.21	**0.00**	143.56	25516	840229.64	840230.21	**0.00**	9.66	208	4	···
cap72	841736.89	841736.89	**0.00**	**0.21**	841736.70	841736.70	**0.00**	9.51	841736.70	841736.70	**0.00**	1.74	841736.89	841736.89	**0.00**	143.86	25516	841736.68	841736.89	**0.00**	9.71	219	5	···
cap73	843243.57	843243.57	**0.00**	**0.29**	843243.38	843243.38	**0.00**	7.40	843243.40	843243.40	**0.00**	1.77	843243.57	843243.57	**0.00**	149.04	25516	843243.28	843243.57	**0.00**	8.72	219	5	···
cap74	845503.43	845503.59	**0.00**	**0.20**	845503.40	845503.40	**0.00**	18.55	845503.40	845503.40	**0.00**	4.03	845503.59	845503.59	**0.00**	141.99	25516	845503.17	845503.59	**0.00**	9.44	219	5	···
cap81	653828.51	653828.51	**0.00**	**0.36**	653828.12	653828.12	**0.00**	12.91	653828.40	653828.40	**0.00**	3.61	653828.51	653828.51	**0.00**	228.68	40825	653828.25	653828.51	**0.00**	12.57	316	5	···
cap82	654853.42	654853.42	**0.00**	**0.47**	654852.46	654853.03	**0.00**	14.66	654853.30	654853.30	**0.00**	3.83	654853.42	654853.42	**0.00**	233.35	40825	654852.98	654853.42	**0.00**	13.22	316	5	···
cap83	655878.33	655878.33	**0.00**	**0.43**	655877.94	655877.94	**0.00**	20.60	655878.20	655878.20	**0.00**	16.34	655878.33	655878.33	**0.00**	230.17	40825	655877.71	655878.33	**0.00**	12.98	316	5	···
cap84	657415.70	657415.70	**0.00**	**0.46**	657415.31	657415.31	**0.00**	36.38	657415.60	657415.60	**0.00**	3.74	657415.70	657415.70	**0.00**	230.86	40825	657415.18	657415.70	**0.00**	13.69	334	6	···
cap91	653828.51	653828.51	**0.00**	**0.57**	653828.12	653828.12	**0.00**	26.09	653828.40	653828.40	**0.00**	3.58	653828.51	653828.51	**0.00**	230.56	40825	653828.25	653828.51	**0.00**	12.61	316	5	···
cap92	654853.42	654853.42	**0.00**	**0.46**	654852.46	654853.03	**0.00**	20.41	654853.30	654853.30	**0.00**	3.81	654853.42	654853.42	**0.00**	229.87	40825	654852.98	654853.42	**0.00**	13.39	316	5	···
cap93	655878.33	655878.33	**0.00**	**0.68**	655877.94	655877.94	**0.00**	19.64	655878.20	655878.20	**0.00**	16.25	655878.33	655878.33	**0.00**	234.97	40825	655877.71	655878.33	**0.00**	14.66	316	5	···
cap94	657415.70	657415.70	**0.00**	**0.44**	657415.31	657415.31	**0.00**	19.82	657415.60	657415.60	**0.00**	3.68	657415.70	657415.70	**0.00**	233.85	40825	657415.18	657415.70	**0.00**	14.40	334	6	···
#S			**37**				**37**				**37**				**37**					**37**				···
SGM			**0.00**	**0.46**			**0.00**	9.69			**0.00**	18.33			**0.00**	246.43	43627			**0.00**	15.73	348	5	···

Table 68 **Detailed results for problem NLUFLP-W-orlib, cost functions** f_3

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	655762.97	655763.02	**0.00**	**0.31**	655762.63	655762.63	**0.00**	20.61	655762.90	655762.90	**0.00**	3.83	655763.02	655763.02	**0.00**	303.72	52921	655762.91	655763.02	**0.00**	49.71	375	6	···
cap102	658077.58	658077.60	**0.00**	**0.47**	658077.21	658077.21	**0.00**	53.18	658077.50	658077.50	**0.00**	6.30	658077.60	658077.60	**0.00**	318.98	52921	658077.42	658077.60	**0.00**	51.89	375	6	···
cap103	660392.15	660392.18	**0.00**	**0.46**	660391.27	660391.78	**0.00**	4.16	660392.00	660392.00	**0.00**	3.79	660392.17	660392.18	**0.00**	309.03	52921	660391.93	660392.18	**0.00**	52.34	375	6	···
cap104	663851.07	663851.07	**0.00**	**0.49**	663850.68	663850.68	**0.00**	3.89	663850.90	663850.90	**0.00**	4.04	663851.07	663851.07	**0.00**	304.63	52921	663850.86	663851.07	**0.00**	52.74	392	7	···
cap111	627320.20	627320.22	**0.00**	**0.67**	627319.32	627319.32	**0.00**	4.62	627320.20	627320.20	**0.00**	77.32	627320.21	627320.22	**0.00**	626.05	108046	627319.96	627320.22	**0.00**	110.66	680	6	···
cap112	629486.06	629486.06	**0.00**	**0.58**	629485.17	629485.17	**0.00**	4.56	629486.10	629486.10	**0.00**	136.91	629486.06	629486.06	**0.00**	647.48	108046	629485.63	629486.06	**0.00**	110.42	680	6	···
cap113	631651.90	631651.90	**0.00**	**0.55**	631651.01	631651.01	**0.00**	5.84	631651.90	631651.90	**0.00**	140.51	631651.90	631651.90	**0.00**	647.73	108046	631651.31	631651.90	**0.00**	110.08	680	6	···
cap114	634900.22	634900.67	**0.00**	**0.64**	634899.78	634899.78	**0.00**	5.96	634900.70	634900.70	**0.00**	937.98	634900.67	634900.67	**0.00**	638.07	108046	634900.44	634900.67	**0.00**	112.19	827	7	···
cap121	627320.20	627320.22	**0.00**	**0.65**	627319.32	627319.32	**0.00**	4.50	627320.20	627320.20	**0.00**	77.48	627320.21	627320.22	**0.00**	622.96	108046	627319.96	627320.22	**0.00**	104.21	680	6	···
cap122	629486.06	629486.06	**0.00**	**0.64**	629485.17	629485.17	**0.00**	4.64	629486.10	629486.10	**0.00**	132.34	629486.06	629486.06	**0.00**	642.85	108046	629485.63	629486.06	**0.00**	104.56	680	6	···
cap123	631651.90	631651.90	**0.00**	**0.60**	631651.01	631651.01	**0.00**	6.14	631651.90	631651.90	**0.00**	143.76	631651.90	631651.90	**0.00**	583.01	108046	631651.31	631651.90	**0.00**	101.28	680	6	···
cap124	634900.22	634900.67	**0.00**	**0.56**	634899.78	634899.78	**0.00**	6.41	634900.70	634900.70	**0.00**	949.67	634900.67	634900.67	**0.00**	610.14	108046	634900.44	634900.67	**0.00**	106.78	827	7	···
cap131	627320.20	627320.22	**0.00**	**0.56**	627319.32	627319.32	**0.00**	4.37	627320.20	627320.20	**0.00**	78.96	627320.21	627320.22	**0.00**	611.40	108046	627319.96	627320.22	**0.00**	99.16	680	6	···
cap132	629486.06	629486.06	**0.00**	**0.55**	629485.17	629485.17	**0.00**	4.65	629486.10	629486.10	**0.00**	134.84	629486.06	629486.06	**0.00**	596.97	108046	629485.63	629486.06	**0.00**	106.63	680	6	···
cap133	631651.90	631651.90	**0.00**	**0.62**	631651.01	631651.01	**0.00**	5.90	631651.90	631651.90	**0.00**	143.97	631651.90	631651.90	**0.00**	618.69	108046	631651.31	631651.90	**0.00**	104.90	680	6	···
cap134	634900.22	634900.67	**0.00**	**0.63**	634899.78	634899.78	**0.00**	6.04	634900.70	634900.70	**0.00**	952.85	634900.67	634900.67	**0.00**	614.32	108046	634900.44	634900.67	**0.00**	110.82	827	7	···
cap41	842961.99	842961.99	**0.00**	**0.27**	842961.45	842961.80	**0.00**	6.06	842961.80	842961.80	**0.00**	1.82	842961.89	842961.99	**0.00**	199.77	33076	842961.25	842961.99	**0.00**	48.33	227	6	···
cap42	846289.86	846289.86	**0.00**	**0.31**	846289.67	846289.67	**0.00**	15.92	846289.70	846289.70	**0.00**	1.80	846289.70	846289.86	**0.00**	204.78	33076	846289.35	846289.86	**0.00**	46.95	246	7	···
cap43	849617.73	849617.73	**0.00**	**0.31**	849617.55	849617.55	**0.00**	6.31	849617.60	849617.60	**0.00**	1.87	849617.50	849617.73	**0.00**	198.29	33076	849617.02	849617.73	**0.00**	49.53	246	7	···
cap44	854609.54	854609.54	**0.00**	**0.33**	854609.35	854609.35	**0.00**	5.71	854609.40	854609.40	**0.00**	8.36	854609.21	854609.54	**0.00**	200.67	33076	854609.20	854609.54	**0.00**	53.13	258	8	···
cap51	849617.73	849617.73	**0.00**	**0.30**	849617.55	849617.55	**0.00**	16.07	849617.60	849617.60	**0.00**	1.82	849617.50	849617.73	**0.00**	197.80	33076	849617.02	849617.73	**0.00**	49.36	246	7	···
cap61	842961.99	842961.99	**0.00**	**0.28**	842961.45	842961.80	**0.00**	8.17	842961.80	842961.80	**0.00**	1.77	842961.89	842961.99	**0.00**	202.45	33076	842961.25	842961.99	**0.00**	43.51	227	6	···
cap62	846289.86	846289.86	**0.00**	**0.22**	846289.67	846289.67	**0.00**	9.37	846289.70	846289.70	**0.00**	1.87	846289.70	846289.86	**0.00**	207.99	33076	846289.35	846289.86	**0.00**	48.31	246	7	···
cap63	849617.73	849617.73	**0.00**	**0.22**	849617.55	849617.55	**0.00**	19.27	849617.60	849617.60	**0.00**	1.87	849617.50	849617.73	**0.00**	211.47	33076	849617.02	849617.73	**0.00**	48.72	246	7	···
cap64	854609.54	854609.54	**0.00**	**0.30**	854609.35	854609.35	**0.00**	8.94	854609.40	854609.40	**0.00**	8.35	854609.21	854609.54	**0.00**	210.53	33076	854609.20	854609.54	**0.00**	54.72	258	8	···
cap71	842961.99	842961.99	**0.00**	**0.31**	842961.45	842961.80	**0.00**	15.64	842961.80	842961.80	**0.00**	1.81	842961.89	842961.99	**0.00**	192.90	33076	842961.25	842961.99	**0.00**	47.68	227	6	···
cap72	846289.86	846289.86	**0.00**	**0.26**	846289.67	846289.67	**0.00**	7.48	846289.70	846289.70	**0.00**	1.82	846289.70	846289.86	**0.00**	207.78	33076	846289.35	846289.86	**0.00**	50.24	246	7	···
cap73	849617.73	849617.73	**0.00**	**0.27**	849617.55	849617.55	**0.00**	7.47	849617.60	849617.60	**0.00**	1.89	849617.50	849617.73	**0.00**	209.29	33076	849617.02	849617.73	**0.00**	48.09	246	7	···
cap74	854609.54	854609.54	**0.00**	**0.31**	854609.35	854609.35	**0.00**	8.88	854609.40	854609.40	**0.00**	8.45	854609.21	854609.54	**0.00**	206.74	33076	854609.20	854609.54	**0.00**	53.82	258	8	···
cap81	655762.97	655763.02	**0.00**	**0.46**	655762.63	655762.63	**0.00**	14.87	655762.90	655762.90	**0.00**	3.80	655763.02	655763.02	**0.00**	312.93	52921	655762.91	655763.02	**0.00**	53.98	375	6	···
cap82	658077.58	658077.60	**0.00**	**0.51**	658077.21	658077.21	**0.00**	5.37	658077.50	658077.50	**0.00**	6.58	658077.60	658077.60	**0.00**	326.32	52921	658077.42	658077.60	**0.00**	51.62	375	6	···
cap83	660392.15	660392.18	**0.00**	**0.36**	660391.27	660391.78	**0.00**	21.29	660392.00	660392.00	**0.00**	3.85	660392.17	660392.18	**0.00**	303.08	52921	660391.93	660392.18	**0.00**	51.19	375	6	···
cap84	663851.07	663851.07	**0.00**	**0.51**	663850.68	663850.68	**0.00**	36.54	663850.90	663850.90	**0.00**	4.01	663851.07	663851.07	**0.00**	319.54	52921	663850.86	663851.07	**0.00**	52.76	392	7	···
cap91	655762.97	655763.02	**0.00**	**0.47**	655762.63	655762.63	**0.00**	20.60	655762.90	655762.90	**0.00**	3.69	655763.02	655763.02	**0.00**	318.27	52921	655762.91	655763.02	**0.00**	54.42	375	6	···
cap92	658077.58	658077.60	**0.00**	**0.35**	658077.21	658077.21	**0.00**	26.81	658077.50	658077.50	**0.00**	6.46	658077.60	658077.60	**0.00**	319.21	52921	658077.42	658077.60	**0.00**	50.45	375	6	···
cap93	660392.15	660392.18	**0.00**	**0.50**	660391.27	660391.78	**0.00**	20.38	660392.00	660392.00	**0.00**	3.87	660392.17	660392.18	**0.00**	297.24	52921	660391.93	660392.18	**0.00**	48.30	375	6	···
cap94	663851.07	663851.07	**0.00**	**0.51**	663850.68	663850.68	**0.00**	20.82	663850.90	663850.90	**0.00**	4.08	663851.07	663851.07	**0.00**	323.09	52921	663850.86	663851.07	**0.00**	51.45	392	7	···
#S			**37**				**37**				**37**				**37**					**37**				···
SGM			**0.00**	**0.44**			**0.00**	10.41			**0.00**	23.16			**0.00**	337.21	56553			**0.00**	64.85	400	7	···

Table 69 **Detailed results for problem NLUFLP-W-orlib, cost functions** f_4

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
cap101	655519.41	655519.41	0.00	0.51	655519.02	655519.02	0.00	48.26	655519.30	655519.30	0.00	3.89	655519.38	655519.41	0.00	769.41	87961	655519.08	655519.41	0.00	41.58	472	8	⋯
cap102	657671.59	657671.59	0.00	0.43	657671.20	657671.20	0.00	4.65	657671.50	657671.50	0.00	4.05	657671.54	657671.59	0.00	718.84	87961	657671.31	657671.59	0.00	44.71	511	9	⋯
cap103	659823.32	659823.77	0.00	0.50	659823.37	659823.37	0.00	4.33	659823.60	659823.60	0.00	4.01	659823.69	659823.77	0.00	860.48	87961	659823.30	659823.77	0.00	40.98	511	9	⋯
cap104	663051.70	663051.70	0.00	0.42	663050.73	663051.31	0.00	4.42	663051.60	663051.60	0.00	14.50	663051.70	663051.71	0.00	750.98	87961	663051.38	663051.79	0.00	42.39	536	11	⋯
cap111	627004.88	627004.92	0.00	0.66	627004.02	627004.02	0.00	4.69	627004.90	627004.90	0.00	151.05	627004.91	627004.92	0.00	1444.75	179586	627004.38	627004.92	0.00	65.84	879	6	⋯
cap112	628960.53	628960.56	0.00	0.72	628959.67	628959.67	0.00	4.89	628958.90	628958.90	0.00	14.63	628960.56	628960.56	0.00	1316.41	179586	628960.22	628960.56	0.00	67.90	948	7	⋯
cap113	630916.20	630916.20	0.00	0.65	630915.31	630915.31	0.00	7.14	630914.50	630914.50	0.00	14.45	630916.20	630916.20	0.00	1399.96	179586	630915.95	630916.20	0.00	62.61	1006	8	⋯
cap114	633849.67	633849.67	0.00	0.80	633848.78	633848.78	0.00	6.12	633848.00	633848.00	0.00	15.14	633849.66	633849.67	0.00	1312.03	179586	633849.31	633849.67	0.00	69.75	1006	8	⋯
cap121	627004.88	627004.92	0.00	0.62	627004.02	627004.02	0.00	4.54	627004.90	627004.90	0.00	151.81	627004.91	627004.92	0.00	1421.09	179586	627004.38	627004.92	0.00	68.73	879	6	⋯
cap122	628960.53	628960.56	0.00	0.78	628959.67	628959.67	0.00	4.92	628958.90	628958.90	0.00	14.62	628960.56	628960.56	0.00	1388.95	179586	628960.22	628960.56	0.00	68.05	948	7	⋯
cap123	630916.20	630916.20	0.00	0.75	630915.31	630915.31	0.00	6.89	630914.50	630914.50	0.00	13.80	630916.20	630916.20	0.00	1328.50	179586	630915.95	630916.20	0.00	65.62	1006	8	⋯
cap124	633849.67	633849.67	0.00	0.79	633848.78	633848.78	0.00	6.17	633848.00	633848.00	0.00	14.85	633849.66	633849.67	0.00	1282.90	179586	633849.31	633849.67	0.00	63.87	1006	8	⋯
cap131	627004.88	627004.92	0.00	0.62	627004.02	627004.02	0.00	4.37	627004.90	627004.90	0.00	150.72	627004.91	627004.92	0.00	1502.76	179586	627004.38	627004.92	0.00	62.66	879	6	⋯
cap132	628960.53	628960.56	0.00	0.73	628959.67	628959.67	0.00	4.67	628958.90	628958.90	0.00	14.45	628960.56	628960.56	0.00	1322.24	179586	628960.22	628960.56	0.00	63.80	948	7	⋯
cap133	630916.20	630916.20	0.00	0.73	630915.31	630915.31	0.00	7.11	630914.50	630914.50	0.00	14.52	630916.20	630916.20	0.00	1340.83	179586	630915.95	630916.20	0.00	63.58	1006	8	⋯
cap134	633849.67	633849.67	0.00	0.80	633848.78	633848.78	0.00	6.09	633848.00	633848.00	0.00	14.45	633849.66	633849.67	0.00	1266.38	179586	633849.31	633849.67	0.00	68.48	1006	8	⋯
cap41	842005.99	842005.99	0.00	0.22	842005.80	842005.80	0.00	6.29	842005.80	842005.80	0.00	1.87	842005.53	842005.99	0.00	558.44	54976	842005.22	842005.99	0.00	47.91	293	6	⋯
cap42	844696.52	844696.52	0.00	0.23	844695.70	844696.33	0.00	15.91	844696.40	844696.40	0.00	4.45	844695.76	844696.52	0.00	443.42	54976	844696.14	844696.52	0.00	47.31	307	7	⋯
cap43	847387.04	847387.04	0.00	0.32	847386.86	847386.86	0.00	6.58	847386.90	847386.90	0.00	6.77	847385.98	847387.05	0.00	593.01	54976	847386.52	847387.05	0.00	42.56	307	7	⋯
cap44	851422.85	851422.85	0.00	0.29	851422.66	851422.66	0.00	6.16	851422.70	851422.70	0.00	7.41	851421.33	851422.85	0.00	433.00	54976	851422.10	851422.85	0.00	44.52	308	7	⋯
cap51	847387.04	847387.04	0.00	0.28	847386.86	847386.86	0.00	4.59	847386.90	847386.90	0.00	6.63	847385.98	847387.05	0.00	612.96	54976	847386.52	847387.05	0.00	43.99	307	7	⋯
cap61	842005.99	842005.99	0.00	0.29	842005.80	842005.80	0.00	5.31	842005.80	842005.80	0.00	1.83	842005.53	842005.99	0.00	551.83	54976	842005.22	842005.99	0.00	41.91	293	6	⋯
cap62	844696.52	844696.52	0.00	0.29	844695.70	844696.33	0.00	9.47	844696.40	844696.40	0.00	4.62	844695.76	844696.52	0.00	433.95	54976	844696.14	844696.52	0.00	43.40	307	7	⋯
cap63	847387.04	847387.04	0.00	0.36	847386.86	847386.86	0.00	8.23	847386.90	847386.90	0.00	6.70	847385.98	847387.05	0.00	595.27	54976	847386.52	847387.05	0.00	46.08	307	7	⋯
cap64	851422.85	851422.85	0.00	0.43	851422.66	851422.66	0.00	9.33	851422.70	851422.70	0.00	7.37	851421.33	851422.85	0.00	457.16	54976	851422.10	851422.85	0.00	46.63	308	7	⋯
cap71	842005.99	842005.99	0.00	0.22	842005.80	842005.80	0.00	7.95	842005.80	842005.80	0.00	1.80	842005.53	842005.99	0.00	560.65	54976	842005.22	842005.99	0.00	42.48	293	6	⋯
cap72	844696.52	844696.52	0.00	0.24	844695.70	844696.33	0.00	7.78	844696.40	844696.40	0.00	4.47	844695.76	844696.52	0.00	437.84	54976	844696.14	844696.52	0.00	44.31	307	7	⋯
cap73	847387.04	847387.04	0.00	0.37	847386.86	847386.86	0.00	8.29	847386.90	847386.90	0.00	6.80	847385.98	847387.05	0.00	602.92	54976	847386.52	847387.05	0.00	45.56	307	7	⋯
cap74	851422.85	851422.85	0.00	0.44	851422.66	851422.66	0.00	10.09	851422.70	851422.70	0.00	7.34	851421.33	851422.85	0.00	524.57	54976	851422.10	851422.85	0.00	44.66	308	7	⋯
cap81	655519.41	655519.41	0.00	0.50	655519.02	655519.02	0.00	14.01	655519.30	655519.30	0.00	3.83	655519.38	655519.41	0.00	735.92	87961	655519.08	655519.41	0.00	43.60	472	8	⋯
cap82	657671.59	657671.59	0.00	0.56	657671.20	657671.20	0.00	20.82	657671.50	657671.50	0.00	3.97	657671.54	657671.59	0.00	731.09	87961	657671.31	657671.59	0.00	41.61	511	9	⋯
cap83	659823.32	659823.77	0.00	0.58	659823.37	659823.37	0.00	21.19	659823.60	659823.60	0.00	4.04	659823.69	659823.77	0.00	879.30	87961	659823.30	659823.77	0.00	41.70	511	9	⋯
cap84	663051.70	663051.70	0.00	0.60	663050.73	663051.31	0.00	21.34	663051.60	663051.60	0.00	14.34	663051.70	663051.71	0.00	741.33	87961	663051.38	663051.79	0.00	44.94	536	11	⋯
cap91	655519.41	655519.41	0.00	0.49	655519.02	655519.02	0.00	21.39	655519.30	655519.30	0.00	3.71	655519.38	655519.41	0.00	755.04	87961	655519.08	655519.41	0.00	41.94	472	8	⋯
cap92	657671.59	657671.59	0.00	0.52	657671.20	657671.20	0.00	20.70	657671.50	657671.50	0.00	3.93	657671.54	657671.59	0.00	732.55	87961	657671.31	657671.59	0.00	44.22	511	9	⋯
cap93	659823.32	659823.77	0.00	0.39	659823.37	659823.37	0.00	20.37	659823.60	659823.60	0.00	4.06	659823.69	659823.77	0.00	870.10	87961	659823.30	659823.77	0.00	43.46	511	9	⋯
cap94	663051.70	663051.70	0.00	0.57	663050.73	663051.31	0.00	21.45	663051.60	663051.60	0.00	14.52	663051.70	663051.71	0.00	755.33	87961	663051.38	663051.79	0.00	42.50	536	11	⋯
#S			37				37				37				37					37				⋯
SGM			0.00	0.50			0.00	9.46			0.00	10.99			0.00	807.70	93997			0.00	50.12	522	8	⋯

Table 70 Detailed results for problem NLUFLP-W-orlib, cost functions f_5

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	660688.45	660688.46	**0.00**	**0.05**	−∞	+∞	100.0	+∞	660688.30	660688.30	**0.00**	5.06	−∞	+∞	100.00	3600.01	+∞	660688.12	660688.46	**0.00**	104.76	319	6	···
cap102	666286.66	666286.67	**0.00**	**0.08**	−∞	+∞	100.0	+∞	666286.50	666286.50	**0.00**	5.36	−∞	+∞	100.00	3600.01	+∞	666286.10	666286.67	**0.00**	99.69	319	6	···
cap103	671884.88	671884.88	**0.00**	**0.21**	−∞	+∞	100.0	+∞	671884.70	671884.70	**0.00**	12.70	−∞	+∞	100.00	3600.01	+∞	671884.60	671884.88	**0.00**	107.42	359	7	···
cap104	680279.85	680279.85	**0.00**	**0.39**	−∞	+∞	100.0	+∞	680279.90	680279.90	**0.00**	58.34	−∞	+∞	100.00	3600.01	+∞	680279.23	680279.87	**0.00**	99.89	353	7	···
cap111	633935.25	633935.28	**0.00**	**0.07**	−∞	+∞	100.0	+∞	633935.30	633935.30	**0.00**	110.47	−∞	+∞	100.00	3600.01	+∞	633935.11	633935.28	**0.00**	178.69	814	8	···
cap112	640475.05	640475.05	**0.00**	**0.13**	−∞	+∞	100.0	+∞	640475.00	640475.00	**0.00**	106.21	−∞	+∞	100.00	3600.01	+∞	640474.76	640475.05	**0.00**	178.45	800	8	···
cap113	646928.41	646928.41	**0.00**	**0.74**	−∞	+∞	100.0	+∞	646928.40	646928.40	**0.00**	152.91	−∞	+∞	100.00	3600.01	+∞	646928.00	646928.41	**0.00**	178.03	792	8	···
cap114	656592.87	656592.87	**0.00**	**0.92**	−∞	+∞	100.0	+∞	656592.90	656592.90	**0.00**	161.09	−∞	+∞	100.00	3600.01	+∞	656592.58	656592.90	**0.00**	175.41	788	8	···
cap121	633935.25	633935.28	**0.00**	**0.08**	−∞	+∞	100.0	+∞	633935.30	633935.30	**0.00**	117.27	−∞	+∞	100.00	3600.01	+∞	633935.11	633935.28	**0.00**	180.73	814	8	···
cap122	640475.05	640475.05	**0.00**	**0.14**	−∞	+∞	100.0	+∞	640475.00	640475.00	**0.00**	106.38	−∞	+∞	100.00	3600.01	+∞	640474.76	640475.05	**0.00**	173.89	800	8	···
cap123	646928.41	646928.41	**0.00**	**0.77**	−∞	+∞	100.0	+∞	646928.40	646928.40	**0.00**	152.73	−∞	+∞	100.00	3600.01	+∞	646928.00	646928.41	**0.00**	182.83	792	8	···
cap124	656592.87	656592.87	**0.00**	**1.03**	−∞	+∞	100.0	+∞	656592.90	656592.90	**0.00**	157.36	−∞	+∞	100.00	3600.01	+∞	656592.58	656592.90	**0.00**	179.34	788	8	···
cap131	633935.25	633935.28	**0.00**	**0.08**	−∞	+∞	100.0	+∞	633935.30	633935.30	**0.00**	115.45	−∞	+∞	100.00	3600.01	+∞	633935.11	633935.28	**0.00**	176.78	814	8	···
cap132	640475.05	640475.05	**0.00**	**0.13**	−∞	+∞	100.0	+∞	640475.00	640475.00	**0.00**	106.29	−∞	+∞	100.00	3600.01	+∞	640474.76	640475.05	**0.00**	176.90	800	8	···
cap133	646928.41	646928.41	**0.00**	**0.68**	−∞	+∞	100.0	+∞	646928.40	646928.40	**0.00**	153.03	−∞	+∞	100.00	3600.01	+∞	646928.00	646928.41	**0.00**	162.36	792	8	···
cap134	656592.87	656592.87	**0.00**	**0.93**	−∞	+∞	100.0	+∞	656592.90	656592.90	**0.00**	157.39	−∞	+∞	100.00	3600.01	+∞	656592.58	656592.90	**0.00**	180.45	788	8	···
cap41	846858.86	846858.96	**0.00**	**0.05**	−∞	+∞	100.0	+∞	846859.00	846859.00	**0.00**	21.61	−∞	+∞	100.00	3600.01	+∞	846858.72	846858.96	**0.00**	109.24	207	7	···
cap42	852784.80	852784.80	**0.00**	**0.14**	−∞	+∞	100.0	+∞	852784.80	852784.80	**0.00**	21.80	−∞	+∞	100.00	3600.01	+∞	852784.41	852784.80	**0.00**	110.25	207	7	···
cap43	858710.65	858710.65	**0.00**	**0.17**	−∞	+∞	100.0	+∞	858710.60	858710.60	**0.00**	24.02	−∞	+∞	100.00	3600.01	+∞	858710.10	858710.65	**0.00**	107.47	207	7	···
cap44	867599.42	867599.42	**0.00**	**0.22**	−∞	+∞	100.0	+∞	867599.40	867599.40	**0.00**	32.37	−∞	+∞	100.00	3600.01	+∞	867599.04	867599.42	**0.00**	113.04	216	8	···
cap51	858710.65	858710.65	**0.00**	**0.20**	−∞	+∞	100.0	+∞	858710.60	858710.60	**0.00**	23.94	−∞	+∞	100.00	3600.01	+∞	858710.10	858710.65	**0.00**	101.23	207	7	···
cap61	846858.86	846858.96	**0.00**	**0.05**	−∞	+∞	100.0	+∞	846859.00	846859.00	**0.00**	21.54	−∞	+∞	100.00	3600.01	+∞	846858.72	846858.96	**0.00**	111.03	207	7	···
cap62	852784.80	852784.80	**0.00**	**0.16**	−∞	+∞	100.0	+∞	852784.80	852784.80	**0.00**	21.76	−∞	+∞	100.00	3600.01	+∞	852784.41	852784.80	**0.00**	100.52	207	7	···
cap63	858710.65	858710.65	**0.00**	**0.17**	−∞	+∞	100.0	+∞	858710.60	858710.60	**0.00**	23.96	−∞	+∞	100.00	3600.01	+∞	858710.10	858710.65	**0.00**	106.51	207	7	···
cap64	867599.42	867599.42	**0.00**	**0.22**	−∞	+∞	100.0	+∞	867599.40	867599.40	**0.00**	31.17	−∞	+∞	100.00	3600.01	+∞	867599.04	867599.42	**0.00**	115.28	216	8	···
cap71	846858.86	846858.96	**0.00**	**0.04**	−∞	+∞	100.0	+∞	846859.00	846859.00	**0.00**	21.58	−∞	+∞	100.00	3600.01	+∞	846858.72	846858.96	**0.00**	108.92	207	7	···
cap72	852784.80	852784.80	**0.00**	**0.17**	−∞	+∞	100.0	+∞	852784.80	852784.80	**0.00**	20.78	−∞	+∞	100.00	3600.01	+∞	852784.41	852784.80	**0.00**	108.17	207	7	···
cap73	858710.65	858710.65	**0.00**	**0.18**	−∞	+∞	100.0	+∞	858710.60	858710.60	**0.00**	23.01	−∞	+∞	100.00	3600.01	+∞	858710.10	858710.65	**0.00**	101.63	207	7	···
cap74	867599.42	867599.42	**0.00**	**0.20**	−∞	+∞	100.0	+∞	867599.40	867599.40	**0.00**	31.23	−∞	+∞	100.00	3600.01	+∞	867599.04	867599.42	**0.00**	105.80	216	8	···
cap81	660688.45	660688.46	**0.00**	**0.05**	−∞	+∞	100.0	+∞	660688.30	660688.30	**0.00**	5.00	−∞	+∞	100.00	3600.01	+∞	660688.12	660688.46	**0.00**	99.82	319	6	···
cap82	666286.66	666286.67	**0.00**	**0.09**	−∞	+∞	100.0	+∞	666286.50	666286.50	**0.00**	5.31	−∞	+∞	100.00	3600.01	+∞	666286.10	666286.67	**0.00**	100.51	319	6	···
cap83	671884.88	671884.88	**0.00**	**0.18**	−∞	+∞	100.0	+∞	671884.70	671884.70	**0.00**	12.59	−∞	+∞	100.00	3600.01	+∞	671884.60	671884.88	**0.00**	112.37	359	7	···
cap84	680279.85	680279.85	**0.00**	**0.35**	−∞	+∞	100.0	+∞	680279.90	680279.90	**0.00**	56.19	−∞	+∞	100.00	3600.01	+∞	680279.23	680279.87	**0.00**	103.43	353	7	···
cap91	660688.45	660688.46	**0.00**	**0.05**	−∞	+∞	100.0	+∞	660688.30	660688.30	**0.00**	5.15	−∞	+∞	100.00	3600.01	+∞	660688.12	660688.46	**0.00**	100.20	319	6	···
cap92	666286.66	666286.67	**0.00**	**0.09**	−∞	+∞	100.0	+∞	666286.50	666286.50	**0.00**	5.28	−∞	+∞	100.00	3600.01	+∞	666286.10	666286.67	**0.00**	111.48	319	6	···
cap93	671884.88	671884.88	**0.00**	**0.18**	−∞	+∞	100.0	+∞	671884.70	671884.70	**0.00**	12.56	−∞	+∞	100.00	3600.01	+∞	671884.60	671884.88	**0.00**	113.12	359	7	···
cap94	680279.85	680279.85	**0.00**	**0.42**	−∞	+∞	100.0	+∞	680279.90	680279.90	**0.00**	56.17	−∞	+∞	100.00	3600.01	+∞	680279.23	680279.87	**0.00**	115.43	353	7	···
#S			**37**				0				**37**				0					**37**				···
SGM			**0.00**	**0.26**			100.00	14400.00			**0.00**	38.54			100.00	14400.00	+∞			**0.00**	125.87	378	7	···

Table 71 Detailed results for problem NLUFLP-W-orlib, cost functions f_6

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	682610.20	682610.57	**0.00**	8.71	−∞	+∞	100.0	+∞	682610.50	682610.50	**0.00**	1670.30	682610.57	682610.58	**0.00**	1250.10	35065	682610.16	682610.58	**0.00**	**2.68**	327	9	···
cap102	702798.91	702798.91	**0.00**	11.95	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	702798.90	702798.92	**0.00**	1476.13	35065	702798.24	702798.92	**0.00**	**2.70**	326	9	···
cap103	722468.83	722468.83	**0.00**	27.62	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	722468.78	722468.84	**0.00**	3234.90	35065	722468.45	722468.85	**0.00**	**7.84**	343	10	···
cap104	751621.16	751621.16	**0.00**	126.08	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.58	35065	751620.65	751621.22	**0.00**	**7.11**	343	10	···
cap111	656390.69	656390.86	**0.00**	12.65	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.77	71590	656390.37	656390.88	**0.00**	**4.74**	638	9	···
cap112	677936.49	677937.14	**0.00**	358.17	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.95	71590	677936.80	677937.16	**0.00**	**7.30**	667	10	···
cap113	698992.44	698992.98	**0.00**	1090.92	−∞	+∞	100.0	+∞	697394.40	698993.30	0.23	3609.23	−∞	+∞	100.00	3601.89	71590	698992.37	698993.03	**0.00**	**12.16**	658	11	···
cap114	727770.23	730000.24	0.31	3600.01	−∞	+∞	100.0	+∞	720393.20	730000.80	1.32	3600.10	−∞	+∞	100.00	3601.69	71590	729999.68	730000.32	**0.00**	**14.67**	677	11	···
cap121	656390.69	656390.86	**0.00**	11.24	653921.99	656390.84	0.38	3602.94	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.68	71590	656390.37	656390.88	**0.00**	**4.94**	638	9	···
cap122	677936.49	677937.14	**0.00**	379.62	673666.26	677937.12	0.63	3610.64	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.74	71590	677936.80	677937.16	**0.00**	**6.79**	667	10	···
cap123	698992.44	698992.98	**0.00**	1214.78	−∞	+∞	100.0	+∞	697393.60	698993.30	0.23	3609.27	−∞	+∞	100.00	3601.91	71590	698992.37	698993.03	**0.00**	**13.34**	658	11	···
cap124	727869.26	730000.24	0.29	3600.01	−∞	+∞	100.0	+∞	720393.20	730000.80	1.32	3600.10	−∞	+∞	100.00	3601.73	71590	729999.68	730000.32	**0.00**	**13.06**	677	11	···
cap131	656390.69	656390.86	**0.00**	12.86	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.75	71590	656390.37	656390.88	**0.00**	**4.98**	638	9	···
cap132	677936.49	677937.14	**0.00**	319.53	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.93	71590	677936.80	677937.16	**0.00**	**6.76**	667	10	···
cap133	698992.44	698992.98	**0.00**	1284.76	−∞	+∞	100.0	+∞	697375.00	698993.30	0.23	3609.52	−∞	+∞	100.00	3601.81	71590	698992.37	698993.03	**0.00**	**11.81**	658	11	···
cap134	727869.05	730000.24	0.29	3600.01	−∞	+∞	100.0	+∞	720320.50	730000.80	1.33	3600.10	−∞	+∞	100.00	3601.78	71590	729999.68	730000.32	**0.00**	**14.46**	677	11	···
cap41	862781.65	862781.65	**0.00**	**0.68**	−∞	+∞	100.0	+∞	862781.50	862781.50	**0.00**	11.16	862781.64	862781.65	**0.00**	739.09	21916	862781.04	862781.65	**0.00**	1.21	187	8	···
cap42	879322.63	879322.63	**0.00**	1.61	−∞	+∞	100.0	+∞	879322.50	879322.50	**0.00**	20.51	879322.61	879322.63	**0.00**	894.38	21916	879322.09	879322.63	**0.00**	**1.50**	195	10	···
cap43	895313.45	895313.45	**0.00**	**2.13**	−∞	+∞	100.0	+∞	895313.30	895313.30	**0.00**	36.62	895313.42	895313.47	**0.00**	910.76	21916	895312.99	895313.48	**0.00**	4.66	203	9	···
cap44	918773.80	918773.80	**0.00**	13.62	−∞	+∞	100.0	+∞	918773.50	918773.60	**0.00**	3650.65	918773.76	918773.83	**0.00**	1070.06	21916	918773.17	918773.87	**0.00**	**5.67**	203	9	···
cap51	895313.45	895313.45	**0.00**	**1.89**	−∞	+∞	100.0	+∞	895313.30	895313.30	**0.00**	36.49	895313.42	895313.47	**0.00**	881.51	21916	895312.99	895313.48	**0.00**	4.53	203	9	···
cap61	862781.65	862781.65	**0.00**	**0.59**	−∞	+∞	100.0	+∞	862781.50	862781.50	**0.00**	11.23	862781.64	862781.65	**0.00**	739.81	21916	862781.04	862781.65	**0.00**	1.32	187	8	···
cap62	879322.63	879322.63	**0.00**	1.76	−∞	+∞	100.0	+∞	879322.50	879322.50	**0.00**	20.47	879322.61	879322.63	**0.00**	893.29	21916	879322.09	879322.63	**0.00**	**1.52**	195	10	···
cap63	895313.45	895313.45	**0.00**	**1.91**	−∞	+∞	100.0	+∞	895313.30	895313.30	**0.00**	36.87	895313.42	895313.47	**0.00**	887.02	21916	895312.99	895313.48	**0.00**	4.74	203	9	···
cap64	918773.80	918773.80	**0.00**	12.10	−∞	+∞	100.0	+∞	918773.50	918773.60	**0.00**	3656.13	918773.76	918773.83	**0.00**	1017.75	21916	918773.17	918773.87	**0.00**	**5.32**	203	9	···
cap71	862781.65	862781.65	**0.00**	**0.71**	−∞	+∞	100.0	+∞	862781.50	862781.50	**0.00**	11.24	862781.64	862781.65	**0.00**	740.40	21916	862781.04	862781.65	**0.00**	1.13	187	8	···
cap72	879322.63	879322.63	**0.00**	1.73	−∞	+∞	100.0	+∞	879322.50	879322.50	**0.00**	20.56	879322.61	879322.63	**0.00**	901.76	21916	879322.09	879322.63	**0.00**	**1.41**	195	10	···
cap73	895313.45	895313.45	**0.00**	**1.83**	−∞	+∞	100.0	+∞	895313.30	895313.30	**0.00**	36.59	895313.42	895313.47	**0.00**	872.17	21916	895312.99	895313.48	**0.00**	6.03	203	9	···
cap74	918773.80	918773.80	**0.00**	13.27	−∞	+∞	100.0	+∞	918773.50	918773.60	**0.00**	3652.31	918773.76	918773.83	**0.00**	1055.61	21916	918773.17	918773.87	**0.00**	**5.56**	203	9	···
cap81	682610.20	682610.57	**0.00**	8.45	−∞	+∞	100.0	+∞	682610.50	682610.50	**0.00**	1681.16	682610.57	682610.58	**0.00**	1290.79	35065	682610.16	682610.58	**0.00**	**2.27**	327	9	···
cap82	702798.91	702798.91	**0.00**	16.05	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	702798.90	702798.92	**0.00**	1571.65	35065	702798.24	702798.92	**0.00**	**2.65**	326	9	···
cap83	722468.83	722468.83	**0.00**	34.15	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	722468.78	722468.84	**0.00**	3188.53	35065	722468.45	722468.85	**0.00**	**7.71**	343	10	···
cap84	751621.16	751621.16	**0.00**	123.81	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.49	35065	751620.65	751621.22	**0.00**	**6.95**	343	10	···
cap91	682610.20	682610.57	**0.00**	5.30	−∞	+∞	100.0	+∞	682610.50	682610.50	**0.00**	1593.42	682610.57	682610.58	**0.00**	1274.47	35065	682610.16	682610.58	**0.00**	**2.53**	327	9	···
cap92	702798.91	702798.91	**0.00**	10.87	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	702798.90	702798.92	**0.00**	1621.07	35065	702798.24	702798.92	**0.00**	**2.61**	326	9	···
cap93	722468.83	722468.83	**0.00**	34.21	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	722468.78	722468.84	**0.00**	3245.50	35065	722468.45	722468.85	**0.00**	**7.78**	343	10	···
cap94	751621.16	751621.16	**0.00**	119.51	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.49	35065	751620.65	751621.22	**0.00**	**6.87**	343	10	···
#S			34				0				13				22					**37**				···
SGM			0.02	55.30			79.44	14400.00			6.07	2317.47			5.49	3292.69	37472			**0.00**	**5.50**	348	10	···

Table 72 Detailed results for problem NLUFLP-W-orlib, cost functions f_7

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	652930.71	652930.71	**0.00**	**0.02**	652930.69	652930.69	**0.00**	51.89	652930.60	652930.60	**0.00**	3.48	652930.71	652930.71	**0.00**	38.83	35929	652930.44	652930.71	**0.00**	1.01	329	3	···
cap102	653357.09	653357.09	**0.00**	**0.02**	653357.07	653357.07	**0.00**	3.60	653357.00	653357.00	**0.00**	3.60	653357.09	653357.09	**0.00**	39.42	35929	653356.64	653357.09	**0.00**	0.99	329	3	···
cap103	653783.46	653783.46	**0.00**	**0.02**	653783.44	653783.44	**0.00**	3.96	653783.30	653783.30	**0.00**	3.74	653783.46	653783.46	**0.00**	37.26	35929	653782.83	653783.46	**0.00**	1.10	329	3	···
cap104	654423.02	654423.02	**0.00**	**0.02**	654423.00	654423.00	**0.00**	3.71	654422.90	654422.90	**0.00**	4.41	654423.02	654423.02	**0.00**	39.10	35929	654422.67	654423.02	**0.00**	1.18	347	4	···
cap111	624673.83	624673.83	**0.00**	**0.02**	624673.79	624673.79	**0.00**	4.71	624673.80	624673.80	**0.00**	12.80	624673.82	624673.83	**0.00**	80.08	73354	624673.45	624673.83	**0.00**	4.41	569	3	···
cap112	625075.42	625075.42	**0.00**	**0.03**	625075.38	625075.38	**0.00**	4.77	625075.40	625075.40	**0.00**	13.28	625075.39	625075.42	**0.00**	82.04	73354	625075.19	625075.42	**0.00**	5.64	616	4	···
cap113	625477.01	625477.01	**0.00**	**0.03**	625476.97	625476.97	**0.00**	4.59	625477.00	625477.00	**0.00**	13.45	625476.97	625477.01	**0.00**	79.85	73354	625476.69	625477.01	**0.00**	4.98	616	4	···
cap114	626079.39	626079.39	**0.00**	**0.03**	626079.35	626079.35	**0.00**	4.38	626079.40	626079.40	**0.00**	14.21	626079.34	626079.39	**0.00**	81.35	73354	626078.93	626079.39	**0.00**	5.64	616	4	···
cap121	624673.83	624673.83	**0.00**	**0.02**	624673.79	624673.79	**0.00**	4.58	624673.80	624673.80	**0.00**	13.59	624673.82	624673.83	**0.00**	79.49	73354	624673.45	624673.83	**0.00**	4.27	569	3	···
cap122	625075.42	625075.42	**0.00**	**0.02**	625075.38	625075.38	**0.00**	11.45	625075.40	625075.40	**0.00**	13.68	625075.39	625075.42	**0.00**	80.73	73354	625075.19	625075.42	**0.00**	4.67	616	4	···
cap123	625477.01	625477.01	**0.00**	**0.03**	625476.97	625476.97	**0.00**	4.31	625477.00	625477.00	**0.00**	13.54	625476.97	625477.01	**0.00**	79.71	73354	625476.69	625477.01	**0.00**	4.45	616	4	···
cap124	626079.39	626079.39	**0.00**	**0.03**	626079.35	626079.35	**0.00**	4.62	626079.40	626079.40	**0.00**	14.12	626079.34	626079.39	**0.00**	78.80	73354	626078.93	626079.39	**0.00**	5.24	616	4	···
cap131	624673.83	624673.83	**0.00**	**0.03**	624673.79	624673.79	**0.00**	4.77	624673.80	624673.80	**0.00**	13.41	624673.82	624673.83	**0.00**	80.88	73354	624673.45	624673.83	**0.00**	5.09	569	3	···
cap132	625075.42	625075.42	**0.00**	**0.03**	625075.38	625075.38	**0.00**	4.52	625075.40	625075.40	**0.00**	13.46	625075.39	625075.42	**0.00**	80.37	73354	625075.19	625075.42	**0.00**	5.82	616	4	···
cap133	625477.01	625477.01	**0.00**	**0.03**	625476.97	625476.97	**0.00**	4.74	625477.00	625477.00	**0.00**	13.45	625476.97	625477.01	**0.00**	78.80	73354	625476.69	625477.01	**0.00**	5.02	616	4	···
cap134	626079.39	626079.39	**0.00**	**0.03**	626079.35	626079.35	**0.00**	4.55	626079.40	626079.40	**0.00**	13.51	626079.34	626079.39	**0.00**	75.44	73354	626078.93	626079.39	**0.00**	5.05	616	4	···
cap41	838916.05	838916.05	**0.00**	**0.02**	838916.04	838916.04	**0.00**	15.29	838915.90	838915.90	**0.00**	1.81	838916.05	838916.05	**0.00**	24.08	22456	838915.41	838916.05	**0.00**	0.72	179	3	···
cap42	839546.63	839546.63	**0.00**	**0.01**	839546.62	839546.62	**0.00**	5.50	839546.50	839546.50	**0.00**	2.09	839546.63	839546.63	**0.00**	23.88	22456	839546.34	839546.63	**0.00**	0.76	203	4	···
cap43	840177.20	840177.20	**0.00**	**0.02**	840177.20	840177.20	**0.00**	5.49	840177.00	840177.00	**0.00**	1.84	840177.20	840177.20	**0.00**	23.58	22456	840176.80	840177.20	**0.00**	0.75	203	4	···
cap44	841123.07	841123.07	**0.00**	**0.02**	841123.06	841123.06	**0.00**	5.53	841122.90	841122.90	**0.00**	1.83	841123.06	841123.07	**0.00**	23.95	22456	841122.49	841123.07	**0.00**	0.72	203	4	···
cap51	840177.20	840177.20	**0.00**	**0.01**	840177.20	840177.20	**0.00**	6.36	840177.00	840177.00	**0.00**	1.83	840177.20	840177.20	**0.00**	23.53	22456	840176.80	840177.20	**0.00**	0.80	203	4	···
cap61	838916.05	838916.05	**0.00**	**0.02**	838916.04	838916.04	**0.00**	7.62	838915.90	838915.90	**0.00**	1.82	838916.05	838916.05	**0.00**	24.15	22456	838915.41	838916.05	**0.00**	0.70	179	3	···
cap62	839546.63	839546.63	**0.00**	**0.02**	839546.62	839546.62	**0.00**	6.07	839546.50	839546.50	**0.00**	2.09	839546.63	839546.63	**0.00**	24.09	22456	839546.34	839546.63	**0.00**	0.87	203	4	···
cap63	840177.20	840177.20	**0.00**	**0.02**	840177.20	840177.20	**0.00**	6.00	840177.00	840177.00	**0.00**	1.81	840177.20	840177.20	**0.00**	23.65	22456	840176.80	840177.20	**0.00**	0.75	203	4	···
cap64	841123.07	841123.07	**0.00**	**0.02**	841123.06	841123.06	**0.00**	14.13	841122.90	841122.90	**0.00**	1.84	841123.06	841123.07	**0.00**	23.59	22456	841122.49	841123.07	**0.00**	0.78	203	4	···
cap71	838916.05	838916.05	**0.00**	**0.02**	838916.04	838916.04	**0.00**	4.04	838915.90	838915.90	**0.00**	1.81	838916.05	838916.05	**0.00**	22.88	22456	838915.41	838916.05	**0.00**	0.82	179	3	···
cap72	839546.63	839546.63	**0.00**	**0.02**	839546.62	839546.62	**0.00**	14.15	839546.50	839546.50	**0.00**	2.09	839546.63	839546.63	**0.00**	26.95	22456	839546.34	839546.63	**0.00**	0.75	203	4	···
cap73	840177.20	840177.20	**0.00**	**0.02**	840177.20	840177.20	**0.00**	5.97	840177.00	840177.00	**0.00**	1.76	840177.20	840177.20	**0.00**	23.30	22456	840176.80	840177.20	**0.00**	0.81	203	4	···
cap74	841123.07	841123.07	**0.00**	**0.02**	841123.06	841123.06	**0.00**	47.41	841122.90	841122.90	**0.00**	1.81	841123.06	841123.07	**0.00**	23.38	22456	841122.49	841123.07	**0.00**	0.80	203	4	···
cap81	652930.71	652930.71	**0.00**	**0.02**	652930.69	652930.69	**0.00**	36.35	652930.60	652930.60	**0.00**	3.51	652930.71	652930.71	**0.00**	40.21	35929	652930.44	652930.71	**0.00**	0.98	329	3	···
cap82	653357.09	653357.09	**0.00**	**0.02**	653357.07	653357.07	**0.00**	19.85	653357.00	653357.00	**0.00**	3.51	653357.09	653357.09	**0.00**	44.52	35929	653356.64	653357.09	**0.00**	1.02	329	3	···
cap83	653783.46	653783.46	**0.00**	**0.02**	653783.44	653783.44	**0.00**	11.17	653783.30	653783.30	**0.00**	3.69	653783.46	653783.46	**0.00**	39.04	35929	653782.83	653783.46	**0.00**	1.04	329	3	···
cap84	654423.02	654423.02	**0.00**	**0.02**	654423.00	654423.00	**0.00**	17.16	654422.90	654422.90	**0.00**	4.50	654423.02	654423.02	**0.00**	39.23	35929	654422.67	654423.02	**0.00**	1.09	347	4	···
cap91	652930.71	652930.71	**0.00**	**0.02**	652930.69	652930.69	**0.00**	11.14	652930.60	652930.60	**0.00**	3.53	652930.71	652930.71	**0.00**	39.88	35929	652930.44	652930.71	**0.00**	1.06	329	3	···
cap92	653357.09	653357.09	**0.00**	**0.02**	653357.07	653357.07	**0.00**	12.90	653357.00	653357.00	**0.00**	3.49	653357.09	653357.09	**0.00**	39.12	35929	653356.64	653357.09	**0.00**	1.00	329	3	···
cap93	653783.46	653783.46	**0.00**	**0.02**	653783.44	653783.44	**0.00**	12.12	653783.30	653783.30	**0.00**	3.72	653783.46	653783.46	**0.00**	43.58	35929	653782.83	653783.46	**0.00**	0.93	329	3	···
cap94	654423.02	654423.02	**0.00**	**0.02**	654423.00	654423.00	**0.00**	30.15	654422.90	654422.90	**0.00**	4.43	654423.02	654423.02	**0.00**	38.75	35929	654422.67	654423.02	**0.00**	1.09	347	4	···
#S			**37**				**37**				**37**				**37**					**37**				···
SGM			**0.00**	**0.02**			**0.00**	9.34			**0.00**	5.57			**0.00**	42.70	38395			**0.00**	2.09	337	4	···

Table 73 Detailed results for problem NLUFLP-W-orlib, cost functions f_8

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
cap101	685354.39	685354.52	0.00	37.07	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.66	218377	685354.22	685354.55	0.00	3.03	391	9	⋯
cap102	707119.45	707120.12	0.00	1121.48	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.70	218377	707119.89	707120.45	0.00	5.87	401	17	⋯
cap103	724723.98	728432.35	0.51	3600.01	−∞	+∞	100.0	+∞	722154.70	728435.40	0.86	3609.26	−∞	+∞	100.00	3602.77	218377	728432.47	728433.02	0.00	6.07	409	18	⋯
cap104	753260.03	760231.54	0.92	3600.01	−∞	+∞	100.0	+∞	740388.10	760234.90	2.61	3600.10	−∞	+∞	100.00	3602.92	218377	760231.23	760231.80	0.00	8.64	421	19	⋯
cap111	657345.37	666377.75	1.36	3600.01	−∞	+∞	100.0	+∞	655201.60	666389.10	1.68	3600.10	−∞	+∞	100.00	3604.75	445852	666377.73	666378.03	0.00	29.97	849	20	⋯
cap112	675474.24	693062.07	2.54	3600.01	−∞	+∞	100.0	+∞	670407.30	693154.00	3.28	3600.10	−∞	+∞	100.00	3604.75	445852	693061.72	693062.17	0.00	32.29	861	18	⋯
cap113	691521.23	718418.84	3.74	3600.01	−∞	+∞	100.0	+∞	684921.60	718878.40	4.72	3600.10	−∞	+∞	100.00	3604.35	445852	718418.76	718419.07	0.00	72.12	966	19	⋯
cap114	713494.99	754467.05	5.43	3600.01	−∞	+∞	100.0	+∞	703493.70	756178.00	6.97	3600.10	−∞	+∞	100.00	3604.44	445852	754466.62	754467.08	0.00	96.66	999	20	⋯
cap121	657230.46	666377.75	1.37	3600.01	655017.67	666426.77	1.71	3603.01	655213.70	666389.10	1.68	3600.10	−∞	+∞	100.00	3604.05	445852	666377.73	666378.03	0.00	30.19	849	20	⋯
cap122	675545.58	693062.07	2.53	3600.01	672720.92	693272.38	2.96	3602.86	670407.30	693154.00	3.28	3600.10	−∞	+∞	100.00	3604.19	445852	693061.72	693062.17	0.00	34.20	861	18	⋯
cap123	691447.04	718418.84	3.75	3600.01	−∞	+∞	100.0	+∞	684742.50	718878.40	4.75	3600.10	−∞	+∞	100.00	3604.35	445852	718418.76	718419.07	0.00	62.25	966	19	⋯
cap124	713661.87	754467.05	5.41	3600.01	−∞	+∞	100.0	+∞	703493.70	756178.00	6.97	3600.10	−∞	+∞	100.00	3604.65	445852	754466.62	754467.08	0.00	91.72	999	20	⋯
cap131	657271.55	666377.75	1.37	3600.01	−∞	+∞	100.0	+∞	655213.70	666389.10	1.68	3600.10	−∞	+∞	100.00	3604.06	445852	666377.73	666378.03	0.00	29.89	849	20	⋯
cap132	675464.43	693062.07	2.54	3600.01	−∞	+∞	100.0	+∞	670511.20	693154.00	3.27	3600.10	−∞	+∞	100.00	3604.83	445852	693061.72	693062.17	0.00	34.77	861	18	⋯
cap133	691654.12	718418.79	3.73	3600.01	−∞	+∞	100.0	+∞	684882.30	718878.40	4.73	3600.10	−∞	+∞	100.00	3604.29	445852	718418.76	718419.07	0.00	70.27	966	19	⋯
cap134	713451.38	754467.05	5.44	3600.01	−∞	+∞	100.0	+∞	703668.10	756178.00	6.94	3600.10	−∞	+∞	100.00	3604.28	445852	754466.62	754467.08	0.00	94.76	999	20	⋯
cap41	862939.80	862940.44	0.00	6.06	−∞	+∞	100.0	+∞	862940.30	862940.30	0.00	1378.71	863229.36	863229.38	0.00	2897.22	136486	862940.09	862940.45	0.00	1.42	244	9	⋯
cap42	879530.56	879530.56	0.00	28.89	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.75	136486	879530.35	879530.95	0.00	3.09	259	17	⋯
cap43	895356.99	895356.99	0.00	25.84	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.71	136486	895356.60	895357.44	0.00	2.45	257	16	⋯
cap44	918926.33	918926.33	0.00	46.15	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.90	136486	918926.20	918926.78	0.00	3.54	272	18	⋯
cap51	895356.99	895356.99	0.00	20.88	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.78	136486	895356.60	895357.44	0.00	2.61	257	16	⋯
cap61	862939.80	862940.44	0.00	5.22	−∞	+∞	100.0	+∞	862940.30	862940.30	0.00	1404.48	863229.36	863229.38	0.00	2801.06	136486	862940.09	862940.45	0.00	1.61	244	9	⋯
cap62	879530.56	879530.56	0.00	32.10	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.69	136486	879530.35	879530.95	0.00	2.98	259	17	⋯
cap63	895356.99	895356.99	0.00	25.59	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.68	136486	895356.60	895357.44	0.00	2.62	257	16	⋯
cap64	918926.33	918926.33	0.00	39.56	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.60	136486	918926.20	918926.78	0.00	3.34	272	18	⋯
cap71	862939.80	862940.44	0.00	6.36	−∞	+∞	100.0	+∞	862940.30	862940.30	0.00	1406.81	863229.36	863229.38	0.00	2663.57	136486	862940.09	862940.45	0.00	1.43	244	9	⋯
cap72	879530.56	879530.56	0.00	29.66	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.84	136486	879530.35	879530.95	0.00	3.09	259	17	⋯
cap73	895356.99	895356.99	0.00	29.00	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.71	136486	895356.60	895357.44	0.00	2.70	257	16	⋯
cap74	918926.33	918926.33	0.00	48.05	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.74	136486	918926.20	918926.78	0.00	3.49	272	18	⋯
cap81	685354.39	685354.52	0.00	48.97	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.62	218377	685354.22	685354.55	0.00	2.82	391	9	⋯
cap82	707119.45	707120.12	0.00	1182.97	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.71	218377	707119.89	707120.45	0.00	5.61	401	17	⋯
cap83	724678.75	728432.35	0.52	3600.01	−∞	+∞	100.0	+∞	722127.00	728435.40	0.87	3609.68	−∞	+∞	100.00	3602.68	218377	728432.47	728433.02	0.00	6.71	409	18	⋯
cap84	753337.25	760231.54	0.91	3600.01	−∞	+∞	100.0	+∞	740296.60	760234.90	2.62	3600.10	−∞	+∞	100.00	3602.95	218377	760231.23	760231.80	0.00	8.63	421	19	⋯
cap91	685354.39	685354.52	0.00	51.46	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.88	218377	685354.22	685354.55	0.00	3.06	391	9	⋯
cap92	707119.45	707120.12	0.00	1075.85	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.62	218377	707119.89	707120.45	0.00	5.38	401	17	⋯
cap93	724663.87	728432.35	0.52	3600.01	−∞	+∞	100.0	+∞	722120.80	728435.40	0.87	3609.40	−∞	+∞	100.00	3602.76	218377	728432.47	728433.02	0.00	6.57	409	18	⋯
cap94	753219.11	760231.54	0.92	3600.01	−∞	+∞	100.0	+∞	740557.40	760234.90	2.59	3600.10	−∞	+∞	100.00	3602.72	218377	760231.23	760231.80	0.00	8.80	421	19	⋯
#S			16				0				0				0					34				⋯
SGM			0.79	1254.15			81.56	14400.00			17.03	14400.00			100.00	14400.00	244670			0.00	14.09	477	17	⋯

Table 74 Detailed results for problem NLUFLP-W-orlib, cost functions f_9

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
cap101	652747.07	652747.07	**0.00**	**0.18**	652746.95	652747.05	**0.00**	17.86	652746.90	652746.90	**0.00**	4.98	652747.07	652747.07	**0.00**	157.28	51241	652746.45	652747.07	**0.00**	14.34	259	3	···
cap102	653051.02	653051.02	**0.00**	**0.18**	653050.97	653051.00	**0.00**	3.09	653050.90	653050.90	**0.00**	5.01	653051.02	653051.02	**0.00**	153.20	51241	653050.53	653051.02	**0.00**	12.76	279	4	···
cap103	653354.81	653354.97	**0.00**	**0.21**	653354.85	653354.95	**0.00**	3.32	653354.80	653354.80	**0.00**	5.11	653354.97	653354.97	**0.00**	159.96	51241	653354.71	653354.97	**0.00**	14.26	297	5	···
cap104	653810.89	653810.90	**0.00**	**0.20**	653810.49	653810.88	**0.00**	3.36	653810.80	653810.80	**0.00**	5.57	653810.90	653810.90	**0.00**	159.64	51241	653810.53	653810.90	**0.00**	13.73	297	5	···
cap111	624688.81	624688.81	**0.00**	**0.49**	624688.70	624688.77	**0.00**	3.86	624687.10	624687.10	**0.00**	20.62	624688.80	624688.81	**0.00**	318.12	104616	624688.43	624688.81	**0.00**	27.19	609	5	···
cap112	625100.36	625100.38	**0.00**	**0.50**	625099.86	625100.34	**0.00**	3.73	625100.40	625100.40	**0.00**	128.94	625100.37	625100.38	**0.00**	328.52	104616	625100.16	625100.38	**0.00**	28.00	700	6	···
cap113	625511.57	625511.95	**0.00**	**0.62**	625511.66	625511.91	**0.00**	3.80	625510.30	625510.30	**0.00**	20.86	625511.94	625511.95	**0.00**	327.04	104616	625511.64	625511.95	**0.00**	26.46	700	6	···
cap114	626129.28	626129.30	**0.00**	**0.57**	626128.79	626129.26	**0.00**	3.74	626127.60	626127.60	**0.00**	21.11	626129.30	626129.30	**0.00**	325.88	104616	626128.86	626129.30	**0.00**	26.65	700	6	···
cap121	624688.81	624688.81	**0.00**	**0.53**	624688.70	624688.77	**0.00**	3.71	624687.10	624687.10	**0.00**	20.74	624688.80	624688.81	**0.00**	320.17	104616	624688.43	624688.81	**0.00**	28.00	609	5	···
cap122	625100.36	625100.38	**0.00**	**0.55**	625099.86	625100.34	**0.00**	3.72	625100.40	625100.40	**0.00**	126.31	625100.37	625100.38	**0.00**	320.30	104616	625100.16	625100.38	**0.00**	26.38	700	6	···
cap123	625511.57	625511.95	**0.00**	**0.60**	625511.66	625511.91	**0.00**	3.61	625510.30	625510.30	**0.00**	20.61	625511.94	625511.95	**0.00**	306.76	104616	625511.64	625511.95	**0.00**	28.31	700	6	···
cap124	626129.28	626129.30	**0.00**	**0.49**	626128.79	626129.26	**0.00**	3.76	626127.60	626127.60	**0.00**	20.96	626129.30	626129.30	**0.00**	321.33	104616	626128.86	626129.30	**0.00**	27.54	700	6	···
cap131	624688.81	624688.81	**0.00**	**0.60**	624688.70	624688.77	**0.00**	6.31	624687.10	624687.10	**0.00**	20.41	624688.80	624688.81	**0.00**	314.10	104616	624688.43	624688.81	**0.00**	26.40	609	5	···
cap132	625100.36	625100.38	**0.00**	**0.48**	625099.86	625100.34	**0.00**	3.65	625100.40	625100.40	**0.00**	125.46	625100.37	625100.38	**0.00**	321.04	104616	625100.16	625100.38	**0.00**	29.92	700	6	···
cap133	625511.57	625511.95	**0.00**	**0.61**	625511.66	625511.91	**0.00**	6.30	625510.30	625510.30	**0.00**	20.62	625511.94	625511.95	**0.00**	296.64	104616	625511.64	625511.95	**0.00**	29.49	700	6	···
cap134	626129.28	626129.30	**0.00**	**0.56**	626128.79	626129.26	**0.00**	3.93	626127.60	626127.60	**0.00**	21.00	626129.30	626129.30	**0.00**	314.03	104616	626128.86	626129.30	**0.00**	24.92	700	6	···
cap41	838502.77	838502.83	**0.00**	**0.12**	838502.75	838502.82	**0.00**	5.18	838502.80	838502.80	**0.00**	24.16	838502.83	838502.83	**0.00**	101.14	32026	838502.19	838502.83	**0.00**	8.59	165	4	···
cap42	838857.75	838857.93	**0.00**	**0.13**	838857.76	838857.92	**0.00**	14.99	838857.90	838857.90	**0.00**	24.42	838857.92	838857.93	**0.00**	99.64	32026	838857.28	838857.93	**0.00**	10.11	166	5	···
cap43	839212.55	839213.02	**0.00**	**0.14**	839212.99	839213.01	**0.00**	5.27	839213.00	839213.00	**0.00**	24.62	839213.02	839213.02	**0.00**	99.43	32026	839212.58	839213.02	**0.00**	9.67	178	6	···
cap44	839745.62	839745.66	**0.00**	**0.13**	839745.58	839745.66	**0.00**	5.29	839745.70	839745.70	**0.00**	24.53	839745.66	839745.66	**0.00**	94.54	32026	839745.04	839745.66	**0.00**	9.01	178	6	···
cap51	839212.55	839213.02	**0.00**	**0.13**	839212.99	839213.01	**0.00**	3.16	839213.00	839213.00	**0.00**	24.37	839213.02	839213.02	**0.00**	101.68	32026	839212.58	839213.02	**0.00**	9.34	178	6	···
cap61	838502.77	838502.83	**0.00**	**0.12**	838502.75	838502.82	**0.00**	7.34	838502.80	838502.80	**0.00**	24.20	838502.83	838502.83	**0.00**	100.89	32026	838502.19	838502.83	**0.00**	9.68	165	4	···
cap62	838857.75	838857.93	**0.00**	**0.13**	838857.76	838857.92	**0.00**	5.76	838857.90	838857.90	**0.00**	24.87	838857.92	838857.93	**0.00**	99.42	32026	838857.28	838857.93	**0.00**	9.58	166	5	···
cap63	839212.55	839213.02	**0.00**	**0.13**	839212.99	839213.01	**0.00**	21.52	839213.00	839213.00	**0.00**	24.58	839213.02	839213.02	**0.00**	102.13	32026	839212.58	839213.02	**0.00**	11.13	178	6	···
cap64	839745.62	839745.66	**0.00**	**0.13**	839745.58	839745.66	**0.00**	7.32	839745.70	839745.70	**0.00**	23.62	839745.66	839745.66	**0.00**	98.63	32026	839745.04	839745.66	**0.00**	9.76	178	6	···
cap71	838502.77	838502.83	**0.00**	**0.12**	838502.75	838502.82	**0.00**	8.49	838502.80	838502.80	**0.00**	24.09	838502.83	838502.83	**0.00**	102.23	32026	838502.19	838502.83	**0.00**	9.74	165	4	···
cap72	838857.75	838857.93	**0.00**	**0.12**	838857.76	838857.92	**0.00**	5.81	838857.90	838857.90	**0.00**	24.58	838857.92	838857.93	**0.00**	98.79	32026	838857.28	838857.93	**0.00**	9.79	166	5	···
cap73	839212.55	839213.02	**0.00**	**0.13**	839212.99	839213.01	**0.00**	14.84	839213.00	839213.00	**0.00**	24.15	839213.02	839213.02	**0.00**	98.06	32026	839212.58	839213.02	**0.00**	9.27	178	6	···
cap74	839745.62	839745.66	**0.00**	**0.13**	839745.58	839745.66	**0.00**	7.32	839745.70	839745.70	**0.00**	24.58	839745.66	839745.66	**0.00**	94.23	32026	839745.04	839745.66	**0.00**	9.66	178	6	···
cap81	652747.07	652747.07	**0.00**	**0.18**	652746.95	652747.05	**0.00**	40.18	652746.90	652746.90	**0.00**	5.05	652747.07	652747.07	**0.00**	157.39	51241	652746.45	652747.07	**0.00**	14.96	259	3	···
cap82	653051.02	653051.02	**0.00**	**0.20**	653050.97	653051.00	**0.00**	19.37	653050.90	653050.90	**0.00**	4.99	653051.02	653051.02	**0.00**	156.58	51241	653050.53	653051.02	**0.00**	13.71	279	4	···
cap83	653354.81	653354.97	**0.00**	**0.19**	653354.85	653354.95	**0.00**	29.08	653354.80	653354.80	**0.00**	5.09	653354.97	653354.97	**0.00**	159.53	51241	653354.71	653354.97	**0.00**	13.51	297	5	···
cap84	653810.89	653810.90	**0.00**	**0.18**	653810.49	653810.88	**0.00**	19.52	653810.80	653810.80	**0.00**	5.55	653810.90	653810.90	**0.00**	159.74	51241	653810.53	653810.90	**0.00**	13.66	297	5	···
cap91	652747.07	652747.07	**0.00**	**0.21**	652746.95	652747.05	**0.00**	10.63	652746.90	652746.90	**0.00**	4.94	652747.07	652747.07	**0.00**	157.66	51241	652746.45	652747.07	**0.00**	14.27	259	3	···
cap92	653051.02	653051.02	**0.00**	**0.21**	653050.97	653051.00	**0.00**	11.82	653050.90	653050.90	**0.00**	5.01	653051.02	653051.02	**0.00**	161.13	51241	653050.53	653051.02	**0.00**	13.25	279	4	···
cap93	653354.81	653354.97	**0.00**	**0.21**	653354.85	653354.95	**0.00**	25.57	653354.80	653354.80	**0.00**	5.13	653354.97	653354.97	**0.00**	156.94	51241	653354.71	653354.97	**0.00**	13.90	297	5	···
cap94	653810.89	653810.90	**0.00**	**0.20**	653810.49	653810.88	**0.00**	25.50	653810.80	653810.80	**0.00**	5.44	653810.90	653810.90	**0.00**	159.35	51241	653810.53	653810.90	**0.00**	14.34	297	5	···
#S			**37**				**37**				**37**				**37**					**37**				···
SGM			**0.00**	**0.29**			**0.00**	8.57			**0.00**	18.69			**0.00**	169.45	54758			**0.00**	15.79	317	5	···

Table 75 **Detailed results for problem NLUFLP-W-orlib, cost functions f_{10}**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	6881.78	6881.78	0.00	0.06	6881.77	6881.78	0.00	3.76	6881.78	6881.78	0.00	1.57	6881.78	6881.78	0.00	34.25	8180	6881.78	6881.78	0.00	2.84	126	6	···
p10	6232.68	6232.69	0.00	0.07	6232.69	6232.69	0.00	3.94	6232.69	6232.69	0.00	1.35	6232.69	6232.69	0.00	32.47	8180	6232.68	6232.69	0.00	3.00	124	6	···
p11	7209.13	7209.13	0.00	0.06	7209.12	7209.13	0.00	3.62	7209.13	7209.13	0.00	1.46	7209.13	7209.13	0.00	30.21	8180	7209.12	7209.13	0.00	3.08	106	7	···
p12	7947.37	7947.37	0.00	0.05	7947.37	7947.37	0.00	5.55	7947.37	7947.37	0.00	1.61	7947.37	7947.37	0.00	38.88	8180	7947.37	7947.37	0.00	2.88	93	6	···
p13	6314.89	6314.89	0.00	0.09	6314.89	6314.89	0.00	5.07	6314.94	6314.94	0.00	21.71	6314.89	6314.89	0.00	64.02	15320	6314.89	6314.89	0.00	4.61	207	7	···
p14	5685.33	5685.33	0.00	0.09	5685.32	5685.32	0.00	5.08	5685.32	5685.32	0.00	18.94	5685.32	5685.33	0.00	59.16	15320	5685.32	5685.33	0.00	4.35	157	7	···
p15	6647.57	6647.57	0.00	0.08	6647.57	6647.57	0.00	5.12	6647.57	6647.57	0.00	16.35	6647.57	6647.57	0.00	67.02	15320	6647.57	6647.57	0.00	4.76	205	7	···
p16	7542.89	7542.90	0.00	0.09	7542.90	7542.90	0.00	5.39	7542.90	7542.90	0.00	15.99	7542.90	7542.90	0.00	69.98	15320	7542.90	7542.90	0.00	4.91	190	9	···
p17	6314.89	6314.89	0.00	0.09	6314.89	6314.89	0.00	5.12	6314.94	6314.94	0.00	21.84	6314.89	6314.89	0.00	65.62	15320	6314.89	6314.89	0.00	5.09	207	7	···
p18	5685.33	5685.33	0.00	0.09	5685.32	5685.32	0.00	5.70	5685.32	5685.32	0.00	19.05	5685.32	5685.33	0.00	60.07	15320	5685.32	5685.33	0.00	4.18	157	7	···
p19	6647.57	6647.57	0.00	0.09	6647.57	6647.57	0.00	5.35	6647.57	6647.57	0.00	16.51	6647.57	6647.57	0.00	67.64	15320	6647.57	6647.57	0.00	4.91	205	7	···
p2	6232.68	6232.69	0.00	0.07	6232.69	6232.69	0.00	3.86	6232.69	6232.69	0.00	1.32	6232.69	6232.69	0.00	29.61	8180	6232.68	6232.69	0.00	2.78	124	6	···
p20	7542.89	7542.90	0.00	0.09	7542.90	7542.90	0.00	5.75	7542.90	7542.90	0.00	16.10	7542.90	7542.90	0.00	70.09	15320	7542.90	7542.90	0.00	4.92	190	9	···
p21	6314.89	6314.89	0.00	0.08	6314.89	6314.89	0.00	4.95	6314.94	6314.94	0.00	22.01	6314.89	6314.89	0.00	64.43	15320	6314.89	6314.89	0.00	5.12	207	7	···
p22	5685.33	5685.33	0.00	0.09	5685.32	5685.32	0.00	10.24	5685.32	5685.32	0.00	18.91	5685.32	5685.33	0.00	59.46	15320	5685.32	5685.33	0.00	4.31	157	7	···
p23	6647.57	6647.57	0.00	0.09	6647.57	6647.57	0.00	4.91	6647.57	6647.57	0.00	16.33	6647.57	6647.57	0.00	67.04	15320	6647.57	6647.57	0.00	4.45	205	7	···
p24	7542.89	7542.90	0.00	0.09	7542.90	7542.90	0.00	8.27	7542.90	7542.90	0.00	15.92	7542.90	7542.90	0.00	69.87	15320	7542.90	7542.90	0.00	5.08	190	9	···
p25	9885.29	9885.29	0.00	0.19	9885.27	9885.28	0.00	266.98	9885.29	9885.29	0.00	267.86	9885.29	9885.29	0.00	117.89	22860	9885.28	9885.29	0.00	5.75	168	6	···
p26	9397.21	9397.22	0.00	0.22	9397.21	9397.21	0.00	61.66	9397.22	9397.22	0.00	343.77	9397.22	9397.22	0.00	86.86	22860	9397.22	9397.22	0.00	6.46	225	7	···
p27	10198.82	10198.83	0.00	0.24	10198.82	10198.83	0.00	1667.24	10198.83	10198.83	0.00	274.31	10198.83	10198.83	0.00	98.51	22860	10198.83	10198.83	0.00	6.45	208	7	···
p28	10901.16	10901.16	0.00	0.25	10901.13	10901.15	0.00	2705.80	10901.16	10901.16	0.00	309.26	10901.16	10901.16	0.00	99.43	22860	10901.16	10901.16	0.00	6.40	211	7	···
p29	9885.29	9885.29	0.00	0.19	9885.27	9885.28	0.00	178.42	9885.29	9885.29	0.00	270.25	9885.29	9885.29	0.00	106.19	22860	9885.28	9885.29	0.00	6.01	168	6	···
p3	7209.13	7209.13	0.00	0.06	7209.12	7209.13	0.00	3.62	7209.13	7209.13	0.00	1.42	7209.13	7209.13	0.00	28.01	8180	7209.12	7209.13	0.00	2.73	106	7	···
p30	9397.21	9397.22	0.00	0.21	9397.21	9397.21	0.00	44.64	9397.22	9397.22	0.00	340.84	9397.22	9397.22	0.00	86.55	22860	9397.22	9397.22	0.00	6.62	225	7	···
p31	10198.82	10198.83	0.00	0.24	10198.82	10198.83	0.00	1681.99	10198.83	10198.83	0.00	269.02	10198.83	10198.83	0.00	100.88	22860	10198.83	10198.83	0.00	6.99	208	7	···
p32	10901.16	10901.16	0.00	0.25	10901.13	10901.15	0.00	2658.10	10901.16	10901.16	0.00	305.25	10901.16	10901.16	0.00	103.74	22860	10901.16	10901.16	0.00	6.28	211	7	···
p33	9885.29	9885.29	0.00	0.20	9885.27	9885.28	0.00	181.74	9885.29	9885.29	0.00	268.29	9885.29	9885.29	0.00	108.56	22860	9885.28	9885.29	0.00	6.16	168	6	···
p34	9397.21	9397.22	0.00	0.21	9397.21	9397.21	0.00	39.15	9397.22	9397.22	0.00	346.12	9397.22	9397.22	0.00	90.10	22860	9397.22	9397.22	0.00	6.56	225	7	···
p35	10198.82	10198.83	0.00	0.25	10198.82	10198.83	0.00	1212.73	10198.83	10198.83	0.00	273.53	10198.83	10198.83	0.00	91.91	22860	10198.83	10198.83	0.00	6.96	208	7	···
p36	10901.16	10901.16	0.00	0.26	10901.13	10901.15	0.00	2836.57	10901.16	10901.16	0.00	305.44	10901.16	10901.16	0.00	104.64	22860	10901.16	10901.16	0.00	7.19	211	7	···
p37	9885.29	9885.29	0.00	0.19	9885.27	9885.28	0.00	206.09	9885.29	9885.29	0.00	265.53	9885.29	9885.29	0.00	105.91	22860	9885.28	9885.29	0.00	6.59	168	6	···
p38	9397.21	9397.22	0.00	0.20	9397.21	9397.21	0.00	46.90	9397.22	9397.22	0.00	339.62	9397.22	9397.22	0.00	80.34	22860	9397.22	9397.22	0.00	6.32	225	7	···
p39	10198.82	10198.83	0.00	0.25	10198.82	10198.83	0.00	1289.13	10198.83	10198.83	0.00	272.73	10198.83	10198.83	0.00	102.45	22860	10198.83	10198.83	0.00	7.13	208	7	···
p4	7947.37	7947.37	0.00	0.05	7947.37	7947.37	0.00	3.61	7947.37	7947.37	0.00	1.57	7947.37	7947.37	0.00	36.85	8180	7947.37	7947.37	0.00	2.86	93	6	···
p40	10901.16	10901.16	0.00	0.25	10901.13	10901.15	0.00	2515.35	10901.16	10901.16	0.00	308.68	10901.16	10901.16	0.00	103.36	22860	10901.16	10901.16	0.00	6.64	211	7	···
p41	4971.06	4971.06	0.00	0.22	4971.06	4971.06	0.00	172.08	4971.06	4971.06	0.00	6.71	4971.06	4971.06	0.00	31.40	8260	4971.06	4971.06	0.00	3.32	122	7	···
p42	4396.08	4396.08	0.00	0.33	4396.07	4396.08	0.00	566.17	4396.08	4396.08	0.00	35.45	4396.08	4396.08	0.00	88.02	16640	4396.08	4396.08	0.00	15.94	186	9	···
p43	4010.97	4010.97	0.00	0.37	4010.97	4010.97	0.00	2459.18	4010.97	4010.97	0.00	68.37	4010.97	4613.21	13.05	290.80	25110	4010.97	4010.97	0.00	26.64	206	9	···
p44	4963.17	4963.17	0.00	0.12	4963.17	4963.17	0.00	18.57	4963.17	4963.17	0.00	3.49	4963.17	4963.17	0.00	35.81	8130	4963.17	4963.17	0.00	3.48	193	7	···
p45	4572.22	4572.22	0.00	0.28	4572.21	4572.22	0.00	54.49	4572.22	4572.22	0.00	54.99	4572.22	4572.22	0.00	83.10	16420	4572.22	4572.22	0.00	6.63	186	10	···
p46	3970.76	3970.76	0.00	0.15	3970.76	3970.76	0.00	21.67	3970.76	3970.76	0.00	87.25	3970.76	3970.76	0.00	156.56	24840	3970.76	3970.76	0.00	6.54	233	8	···
p47	3741.91	3741.91	0.00	0.11	3741.91	3741.91	0.00	9.24	3741.91	3741.91	0.00	3.17	3741.91	3741.91	0.00	29.30	8130	3741.91	3741.91	0.00	3.52	179	7	···
p48	3560.38	3560.38	0.00	0.13	3560.38	3560.38	0.00	14.71	3560.38	3560.38	0.00	31.13	3560.38	3560.38	0.00	63.24	16380	3560.38	3560.38	0.00	5.23	223	10	···
p49	3338.19	3338.19	0.00	0.14	3338.19	3338.19	0.00	17.38	3338.19	3338.19	0.00	75.70	3338.19	3338.19	0.00	269.37	24840	3338.19	3338.19	0.00	6.51	234	8	···
p5	6881.78	6881.78	0.00	0.06	6881.77	6881.78	0.00	3.62	6881.78	6881.78	0.00	1.42	6881.78	6881.78	0.00	37.81	8180	6881.78	6881.78	0.00	2.69	126	6	···
p50	6432.37	6432.37	0.00	0.07	6432.36	6432.37	0.00	24.72	6432.37	6432.37	0.00	3.20	6432.37	6432.37	0.00	27.25	8180	6432.36	6432.37	0.00	2.93	99	5	···
p51	5874.71	5874.71	0.00	0.16	5874.71	5874.71	0.00	138.82	5874.71	5874.71	0.00	32.75	5874.71	5874.71	0.00	82.79	16360	5874.71	5874.71	0.00	5.20	229	6	···
p52	6968.57	6968.57	0.00	0.07	6968.57	6968.57	0.00	5.75	6968.57	6968.57	0.00	12.02	6968.57	6968.57	0.00	35.22	8070	6968.57	6968.57	0.00	3.54	176	5	···
p53	6569.17	6569.17	0.00	0.17	6569.17	6569.17	0.00	41.44	6569.17	6569.17	0.00	79.87	6569.17	6569.17	0.00	74.47	16140	6569.17	6569.18	0.00	5.20	163	8	···
p54	6771.44	6771.45	0.00	0.16	$-\infty$	$+\infty$	100.0	$+\infty$	6771.44	6771.44	0.00	22.04	6771.44	6771.45	0.00	29.74	8030	6771.44	6771.45	0.00	4.13	157	10	···

Table 76 Detailed results for problem NLUFLP-W-holmberg, cost functions f_1

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	6124.65	6124.65	**0.00**	**0.20**	6124.65	6124.65	**0.00**	7.54	6124.65	6124.65	**0.00**	6.50	6124.65	6124.65	**0.00**	181.66	9810	6124.65	6124.65	**0.00**	9.77	155	11	···
p10	5696.75	5696.75	**0.00**	**0.14**	5696.74	5696.75	**0.00**	6.24	5696.75	5696.75	**0.00**	3.68	5696.75	5696.75	**0.00**	209.36	9810	5696.75	5696.75	**0.00**	10.26	152	10	···
p11	6343.25	6343.25	**0.00**	**0.23**	6343.24	6343.25	**0.00**	10.24	6343.25	6343.25	**0.00**	9.18	6343.25	6343.25	**0.00**	213.94	9810	6343.24	6343.25	**0.00**	10.09	152	10	···
p12	6989.75	6989.75	**0.00**	**0.41**	6989.75	6989.75	**0.00**	14.85	6989.75	6989.75	**0.00**	8.15	6989.75	6989.75	**0.00**	213.79	9810	6989.75	6989.75	**0.00**	10.25	164	11	···
p13	5502.21	5502.21	**0.00**	**0.84**	5502.20	5502.21	**0.00**	125.09	5502.21	5502.21	**0.00**	37.98	5502.21	5502.21	**0.00**	423.07	19620	5502.21	5502.21	**0.00**	17.02	246	10	···
p14	4850.63	4850.63	**0.00**	**0.67**	4850.62	4850.63	**0.00**	311.12	4850.63	4850.63	**0.00**	11.91	4850.63	4850.63	**0.00**	439.53	19620	4850.63	4850.63	**0.00**	16.56	284	10	···
p15	5777.68	5777.68	**0.00**	**4.07**	5777.67	5777.68	**0.00**	143.84	5777.68	5777.68	**0.00**	73.64	5777.68	5777.68	**0.00**	438.23	19620	5777.68	5777.68	**0.00**	16.75	246	10	···
p16	6457.28	6457.28	**0.00**	**0.86**	6457.28	6457.28	**0.00**	58.02	6457.28	6457.28	**0.00**	83.89	6457.28	6457.28	**0.00**	477.24	19620	6457.28	6457.28	**0.00**	16.87	203	10	···
p17	5502.21	5502.21	**0.00**	**0.89**	5502.20	5502.21	**0.00**	90.81	5502.21	5502.21	**0.00**	38.20	5502.21	5502.21	**0.00**	444.11	19620	5502.21	5502.21	**0.00**	17.81	246	10	···
p18	4850.63	4850.63	**0.00**	**0.61**	4850.62	4850.63	**0.00**	272.54	4850.63	4850.63	**0.00**	11.96	4850.63	4850.63	**0.00**	446.69	19620	4850.63	4850.63	**0.00**	16.69	284	10	···
p19	5777.68	5777.68	**0.00**	**4.53**	5777.67	5777.68	**0.00**	193.92	5777.68	5777.68	**0.00**	74.30	5777.68	5777.68	**0.00**	447.89	19620	5777.68	5777.68	**0.00**	17.39	246	10	···
p2	5696.75	5696.75	**0.00**	**0.14**	5696.74	5696.75	**0.00**	5.86	5696.75	5696.75	**0.00**	3.62	5696.75	5696.75	**0.00**	189.85	9810	5696.75	5696.75	**0.00**	9.74	152	10	···
p20	6457.28	6457.28	**0.00**	**0.90**	6457.28	6457.28	**0.00**	58.21	6457.28	6457.28	**0.00**	84.26	6457.28	6457.28	**0.00**	468.34	19620	6457.28	6457.28	**0.00**	19.39	203	10	···
p21	5502.21	5502.21	**0.00**	**0.80**	5502.20	5502.21	**0.00**	240.03	5502.21	5502.21	**0.00**	38.30	5502.21	5502.21	**0.00**	437.43	19620	5502.21	5502.21	**0.00**	19.73	246	10	···
p22	4850.63	4850.63	**0.00**	**0.62**	4850.62	4850.63	**0.00**	265.58	4850.63	4850.63	**0.00**	11.87	4850.63	4850.63	**0.00**	443.15	19620	4850.63	4850.63	**0.00**	16.14	284	10	···
p23	5777.68	5777.68	**0.00**	**4.31**	5777.67	5777.68	**0.00**	281.74	5777.68	5777.68	**0.00**	73.62	5777.68	5777.68	**0.00**	444.99	19620	5777.68	5777.68	**0.00**	16.38	246	10	···
p24	6457.28	6457.28	**0.00**	**0.84**	6457.28	6457.28	**0.00**	97.57	6457.28	6457.28	**0.00**	83.56	6457.28	6457.28	**0.00**	481.40	19620	6457.28	6457.28	**0.00**	16.99	203	10	···
p25	9297.88	9297.88	**0.00**	**6.25**	$-\infty$	$+\infty$	100.0	$+\infty$	9297.88	9297.88	**0.00**	342.43	9297.88	9297.88	**0.00**	871.01	29430	9297.88	9297.88	**0.00**	27.11	231	9	···
p26	8943.77	8943.78	**0.00**	**5.03**	$-\infty$	$+\infty$	100.0	$+\infty$	8943.78	8943.78	**0.00**	239.96	8943.78	8943.78	**0.00**	690.67	29430	8943.77	8943.78	**0.00**	25.24	239	9	···
p27	9540.96	9540.96	**0.00**	**7.90**	$-\infty$	$+\infty$	100.0	$+\infty$	9540.96	9540.96	**0.00**	485.97	9540.96	9540.96	**0.00**	941.22	29430	9540.96	9540.96	**0.00**	31.81	249	11	···
p28	10084.28	10084.28	**0.00**	**7.89**	$-\infty$	$+\infty$	100.0	$+\infty$	10084.28	10084.28	**0.00**	439.63	10084.28	10084.28	**0.00**	944.67	29430	10084.27	10084.28	**0.00**	35.26	227	11	···
p29	9297.88	9297.88	**0.00**	**6.00**	$-\infty$	$+\infty$	100.0	$+\infty$	9297.88	9297.88	**0.00**	346.53	9297.88	9297.88	**0.00**	889.52	29430	9297.88	9297.88	**0.00**	26.66	231	9	···
p3	6343.25	6343.25	**0.00**	**0.22**	6343.24	6343.25	**0.00**	10.22	6343.25	6343.25	**0.00**	8.91	6343.25	6343.25	**0.00**	190.97	9810	6343.24	6343.25	**0.00**	9.60	152	10	···
p30	8943.77	8943.78	**0.00**	**4.60**	$-\infty$	$+\infty$	100.0	$+\infty$	8943.78	8943.78	**0.00**	238.92	8943.78	8943.78	**0.00**	671.12	29430	8943.77	8943.78	**0.00**	27.60	239	9	···
p31	9540.96	9540.96	**0.00**	**7.55**	$-\infty$	$+\infty$	100.0	$+\infty$	9540.96	9540.96	**0.00**	479.28	9540.96	9540.96	**0.00**	953.88	29430	9540.96	9540.96	**0.00**	32.13	249	11	···
p32	10084.28	10084.28	**0.00**	**7.63**	$-\infty$	$+\infty$	100.0	$+\infty$	10084.28	10084.28	**0.00**	434.00	10084.28	10084.28	**0.00**	949.47	29430	10084.27	10084.28	**0.00**	34.27	227	11	···
p33	9297.88	9297.88	**0.00**	**5.90**	$-\infty$	$+\infty$	100.0	$+\infty$	9297.88	9297.88	**0.00**	345.25	9297.88	9297.88	**0.00**	902.95	29430	9297.88	9297.88	**0.00**	26.74	231	9	···
p34	8943.77	8943.78	**0.00**	**4.51**	$-\infty$	$+\infty$	100.0	$+\infty$	8943.78	8943.78	**0.00**	241.80	8943.78	8943.78	**0.00**	687.83	29430	8943.77	8943.78	**0.00**	27.35	239	9	···
p35	9540.96	9540.96	**0.00**	**8.84**	$-\infty$	$+\infty$	100.0	$+\infty$	9540.96	9540.96	**0.00**	484.94	9540.96	9540.96	**0.00**	839.80	29430	9540.96	9540.96	**0.00**	32.78	249	11	···
p36	10084.28	10084.28	**0.00**	**7.74**	$-\infty$	$+\infty$	100.0	$+\infty$	10084.28	10084.28	**0.00**	434.19	10084.28	10084.28	**0.00**	951.85	29430	10084.27	10084.28	**0.00**	35.61	227	11	···
p37	9297.88	9297.88	**0.00**	**6.36**	$-\infty$	$+\infty$	100.0	$+\infty$	9297.88	9297.88	**0.00**	341.67	9297.88	9297.88	**0.00**	890.00	29430	9297.88	9297.88	**0.00**	26.79	231	9	···
p38	8943.77	8943.78	**0.00**	**5.10**	8943.74	8943.77	**0.00**	3602.63	8943.78	8943.78	**0.00**	242.09	8943.78	8943.78	**0.00**	617.12	29430	8943.77	8943.78	**0.00**	26.36	239	9	···
p39	9540.96	9540.96	**0.00**	**8.46**	9539.62	9540.96	0.01	3603.22	9540.96	9540.96	**0.00**	485.24	9540.96	9540.96	**0.00**	952.70	29430	9540.96	9540.96	**0.00**	32.33	249	11	···
p4	6989.75	6989.75	**0.00**	**0.37**	6989.75	6989.75	**0.00**	13.61	6989.75	6989.75	**0.00**	7.90	6989.75	6989.75	**0.00**	201.53	9810	6989.75	6989.75	**0.00**	10.45	164	11	···
p40	10084.28	10084.28	**0.00**	**7.22**	10082.68	10084.26	0.02	3602.90	10084.28	10084.28	**0.00**	439.35	10084.28	10084.28	**0.00**	960.28	29430	10084.27	10084.28	**0.00**	33.39	227	11	···
p41	4327.40	4327.41	**0.00**	**0.71**	4327.41	4327.41	**0.00**	23.95	4327.41	4327.41	**0.00**	23.85	4327.41	4327.41	**0.00**	256.79	9810	4327.41	4327.41	**0.00**	10.64	152	10	···
p42	3882.24	3882.24	**0.00**	**2.35**	3882.24	3882.24	**0.00**	571.65	3882.24	3882.24	**0.00**	1260.30	3882.24	3882.24	**0.00**	627.16	19620	3882.24	3882.24	**0.00**	39.43	206	11	···
p43	3540.89	3540.89	**0.00**	**14.04**	$-\infty$	$+\infty$	100.0	$+\infty$	3115.26	3540.89	12.02	3600.10	3540.89	3540.89	**0.00**	1767.88	29430	3540.89	3540.89	**0.00**	76.83	285	19	···
p44	4282.27	4282.27	**0.00**	**0.35**	4282.27	4282.27	**0.00**	15.57	4282.27	4282.27	**0.00**	11.55	4282.27	4282.27	**0.00**	240.31	9810	4282.27	4282.27	**0.00**	10.51	157	10	···
p45	3884.11	3884.11	**0.00**	**4.67**	3884.11	3884.11	**0.00**	93.46	3884.11	3884.11	**0.00**	638.20	3884.11	3884.11	**0.00**	579.61	19620	3884.11	3884.11	**0.00**	20.48	242	13	···
p46	3371.89	3371.89	**0.00**	**2.16**	3371.89	3371.89	**0.00**	92.29	3371.89	3371.89	**0.00**	888.46	3371.89	3371.89	**0.00**	855.57	29430	3371.89	3371.89	**0.00**	27.03	274	13	···
p47	3062.39	3062.39	**0.00**	**0.19**	3062.38	3062.38	**0.00**	15.43	3062.38	3062.38	**0.00**	8.80	3062.38	3062.39	**0.00**	189.93	9810	3062.38	3062.39	**0.00**	9.67	161	11	···
p48	2901.70	2901.70	**0.00**	**0.46**	2901.69	2901.69	**0.00**	42.23	2901.70	2901.70	**0.00**	88.66	2901.70	2901.70	**0.00**	542.67	19620	2901.70	2901.70	**0.00**	17.32	211	13	···
p49	2678.96	2678.96	**0.00**	**0.78**	2678.96	2678.96	**0.00**	57.02	2678.96	2678.96	**0.00**	1935.45	2678.96	2678.96	**0.00**	836.69	29430	2678.95	2678.96	**0.00**	21.39	267	10	···
p5	6124.65	6124.65	**0.00**	**0.20**	6124.65	6124.65	**0.00**	7.30	6124.65	6124.65	**0.00**	5.97	6124.65	6124.65	**0.00**	196.17	9810	6124.65	6124.65	**0.00**	10.32	155	11	···
p50	6144.15	6144.15	**0.00**	**0.23**	6144.15	6144.15	**0.00**	13.54	6144.15	6144.15	**0.00**	12.27	6144.15	6144.15	**0.00**	202.63	9810	6144.15	6144.15	**0.00**	28.24	94	9	···
p51	5454.00	5454.00	**0.00**	**1.61**	5454.00	5454.00	**0.00**	108.44	5454.00	5454.00	**0.00**	97.52	5454.00	5454.00	**0.00**	549.33	19620	5454.00	5454.00	**0.00**	25.42	180	10	···
p52	6594.08	6594.08	**0.00**	**0.27**	6594.08	6594.08	**0.00**	7.06	6594.08	6594.08	**0.00**	12.59	6594.08	6594.08	**0.00**	210.50	9810	6594.08	6594.08	**0.00**	11.43	118	9	···
p53	6134.86	6134.86	**0.00**	**0.71**	6134.86	6134.86	**0.00**	17.04	6134.86	6134.86	**0.00**	73.78	6134.86	6134.86	**0.00**	369.85	19620	6134.86	6134.86	**0.00**	17.98	168	11	···
p54	6440.15	6440.16	**0.00**	**0.36**	6440.16	6440.16	**0.00**	21.35	6440.16	6440.16	**0.00**	16.88	6440.16	6440.16	**0.00**	226.68	9810	6440.16	6440.16	**0.00**	12.35	115	10	···

Table 77 Detailed results for problem NLUFLP-W-holmberg, cost functions f_2

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	4850.50	4850.50	**0.00**	**0.04**	4850.49	4850.50	**0.00**	3.32	4850.50	4850.50	**0.00**	76.55	4850.50	4850.50	**0.00**	80.30	17010	4850.49	4850.50	**0.00**	6.52	163	10	···
p10	4810.33	4810.33	**0.00**	**0.03**	4810.33	4810.33	**0.00**	3.16	4810.33	4810.33	**0.00**	8.48	4810.33	4810.33	**0.00**	97.48	17010	4810.33	4810.33	**0.00**	6.54	158	6	···
p11	4866.56	4866.56	**0.00**	**0.04**	4866.55	4866.56	**0.00**	3.39	4866.56	4866.56	**0.00**	11.21	4866.56	4866.56	**0.00**	96.77	17010	4866.56	4866.56	**0.00**	6.63	203	7	···
p12	4922.75	4922.75	**0.00**	**0.05**	4922.75	4922.75	**0.00**	3.15	4922.75	4922.75	**0.00**	496.89	4922.75	4922.75	**0.00**	93.45	17010	4922.75	4922.75	**0.00**	6.71	168	9	···
p13	3255.98	3255.98	**0.00**	**0.11**	3255.98	3255.98	**0.00**	3.21	3255.98	3255.98	**0.00**	2.02	3255.98	3255.98	**0.00**	191.25	34020	3255.98	3255.98	**0.00**	11.87	365	8	···
p14	3230.07	3230.07	**0.00**	**0.10**	3230.06	3230.06	**0.00**	2.92	3230.06	3230.06	**0.00**	1.98	3230.07	3230.07	**0.00**	191.72	34020	3230.06	3230.07	**0.00**	11.64	386	8	···
p15	3263.44	3263.44	**0.00**	**0.09**	3263.44	3263.44	**0.00**	2.93	3263.44	3263.44	**0.00**	1.99	3263.44	3263.44	**0.00**	189.36	34020	3263.44	3263.44	**0.00**	11.73	386	8	···
p16	3296.82	3296.82	**0.00**	**0.11**	3296.82	3296.82	**0.00**	3.15	3296.82	3296.82	**0.00**	13.77	3296.82	3296.82	**0.00**	199.32	34020	3296.82	3296.82	**0.00**	11.25	387	8	···
p17	3255.98	3255.98	**0.00**	**0.09**	3255.98	3255.98	**0.00**	3.26	3255.98	3255.98	**0.00**	2.01	3255.98	3255.98	**0.00**	188.95	34020	3255.98	3255.98	**0.00**	11.95	365	8	···
p18	3230.07	3230.07	**0.00**	**0.10**	3230.06	3230.06	**0.00**	7.18	3230.06	3230.06	**0.00**	1.98	3230.07	3230.07	**0.00**	194.92	34020	3230.06	3230.07	**0.00**	11.16	386	8	···
p19	3263.44	3263.44	**0.00**	**0.09**	3263.44	3263.44	**0.00**	3.24	3263.44	3263.44	**0.00**	2.01	3263.44	3263.44	**0.00**	196.12	34020	3263.44	3263.44	**0.00**	12.23	386	8	···
p2	4810.33	4810.33	**0.00**	**0.03**	4810.33	4810.33	**0.00**	13.87	4810.33	4810.33	**0.00**	8.31	4810.33	4810.33	**0.00**	87.07	17010	4810.33	4810.33	**0.00**	6.28	158	6	···
p20	3296.82	3296.82	**0.00**	**0.10**	3296.82	3296.82	**0.00**	3.24	3296.82	3296.82	**0.00**	13.90	3296.82	3296.82	**0.00**	193.17	34020	3296.82	3296.82	**0.00**	11.24	387	8	···
p21	3255.98	3255.98	**0.00**	**0.08**	3255.98	3255.98	**0.00**	3.00	3255.98	3255.98	**0.00**	2.03	3255.98	3255.98	**0.00**	191.09	34020	3255.98	3255.98	**0.00**	12.30	365	8	···
p22	3230.07	3230.07	**0.00**	**0.08**	3230.06	3230.06	**0.00**	9.04	3230.06	3230.06	**0.00**	1.98	3230.07	3230.07	**0.00**	190.80	34020	3230.06	3230.07	**0.00**	12.05	386	8	···
p23	3263.44	3263.44	**0.00**	**0.10**	3263.44	3263.44	**0.00**	3.23	3263.44	3263.44	**0.00**	1.98	3263.44	3263.44	**0.00**	196.70	34020	3263.44	3263.44	**0.00**	11.21	386	8	···
p24	3296.82	3296.82	**0.00**	**0.09**	3296.82	3296.82	**0.00**	35.66	3296.82	3296.82	**0.00**	14.24	3296.82	3296.82	**0.00**	188.31	34020	3296.82	3296.82	**0.00**	11.25	387	8	···
p25	7952.92	7952.92	**0.00**	**0.71**	7952.90	7952.90	**0.00**	28.26	7952.92	7952.92	**0.00**	3626.64	7952.92	7952.92	**0.00**	287.34	51030	7952.92	7952.92	**0.00**	19.50	485	11	···
p26	7935.92	7935.92	**0.00**	**0.64**	7935.90	7935.90	**0.00**	4.98	7935.92	7935.92	**0.00**	3642.84	7935.92	7935.92	**0.00**	285.58	51030	7935.92	7935.92	**0.00**	18.49	495	11	···
p27	7965.88	7965.88	**0.00**	**0.64**	7965.86	7965.86	**0.00**	8.04	7965.88	7965.89	**0.00**	3662.04	7965.88	7965.88	**0.00**	283.31	51030	7965.88	7965.89	**0.00**	21.99	530	13	···
p28	7994.26	7994.26	**0.00**	**0.74**	7994.23	7994.24	**0.00**	5.89	7994.25	7994.26	**0.00**	3638.19	7994.26	7994.26	**0.00**	271.00	51030	7994.26	7994.26	**0.00**	21.12	505	10	···
p29	7952.92	7952.92	**0.00**	**0.68**	7952.90	7952.90	**0.00**	7.66	7952.92	7952.92	**0.00**	3625.98	7952.92	7952.92	**0.00**	278.16	51030	7952.92	7952.92	**0.00**	19.92	485	11	···
p3	4866.56	4866.56	**0.00**	**0.04**	4866.55	4866.56	**0.00**	3.55	4866.56	4866.56	**0.00**	10.91	4866.56	4866.56	**0.00**	89.10	17010	4866.56	4866.56	**0.00**	6.47	203	7	···
p30	7935.92	7935.92	**0.00**	**0.71**	7935.90	7935.90	**0.00**	5.17	7935.92	7935.92	**0.00**	3643.36	7935.92	7935.92	**0.00**	281.34	51030	7935.92	7935.92	**0.00**	18.18	495	11	···
p31	7965.88	7965.88	**0.00**	**0.66**	7965.86	7965.86	**0.00**	7.99	7965.88	7965.89	**0.00**	3661.68	7965.88	7965.88	**0.00**	289.03	51030	7965.88	7965.89	**0.00**	22.36	530	13	···
p32	7994.26	7994.26	**0.00**	**0.67**	7994.23	7994.24	**0.00**	6.38	7994.25	7994.26	**0.00**	3638.47	7994.26	7994.26	**0.00**	285.93	51030	7994.26	7994.26	**0.00**	21.30	505	10	···
p33	7952.92	7952.92	**0.00**	**0.69**	7952.90	7952.90	**0.00**	6.96	7952.92	7952.92	**0.00**	3626.76	7952.92	7952.92	**0.00**	283.97	51030	7952.92	7952.92	**0.00**	19.12	485	11	···
p34	7935.92	7935.92	**0.00**	**0.65**	7935.90	7935.90	**0.00**	4.91	7935.92	7935.92	**0.00**	3642.47	7935.92	7935.92	**0.00**	292.47	51030	7935.92	7935.92	**0.00**	19.46	495	11	···
p35	7965.88	7965.88	**0.00**	**0.70**	7965.86	7965.86	**0.00**	8.37	7965.88	7965.89	**0.00**	3661.80	7965.88	7965.88	**0.00**	286.66	51030	7965.88	7965.89	**0.00**	21.72	530	13	···
p36	7994.26	7994.26	**0.00**	**0.66**	7994.23	7994.24	**0.00**	6.32	7994.25	7994.26	**0.00**	3638.35	7994.26	7994.26	**0.00**	294.08	51030	7994.26	7994.26	**0.00**	21.36	505	10	···
p37	7952.92	7952.92	**0.00**	**0.61**	7952.90	7952.90	**0.00**	7.65	7952.92	7952.92	**0.00**	3626.37	7952.92	7952.92	**0.00**	274.73	51030	7952.92	7952.92	**0.00**	19.13	485	11	···
p38	7935.92	7935.92	**0.00**	**0.61**	7935.90	7935.90	**0.00**	5.28	7935.92	7935.92	**0.00**	3642.22	7935.92	7935.92	**0.00**	254.47	51030	7935.92	7935.92	**0.00**	18.61	495	11	···
p39	7965.88	7965.88	**0.00**	**0.71**	7965.86	7965.86	**0.00**	7.83	7965.88	7965.89	**0.00**	3661.98	7965.88	7965.88	**0.00**	296.40	51030	7965.88	7965.89	**0.00**	21.64	530	13	···
p4	4922.75	4922.75	**0.00**	**0.04**	4922.75	4922.75	**0.00**	3.29	4922.75	4922.75	**0.00**	492.10	4922.75	4922.75	**0.00**	90.17	17010	4922.75	4922.75	**0.00**	6.76	168	9	···
p40	7994.26	7994.26	**0.00**	**0.61**	7994.23	7994.24	**0.00**	6.39	7994.25	7994.26	**0.00**	3638.50	7994.26	7994.26	**0.00**	263.76	51030	7994.26	7994.26	**0.00**	18.68	505	10	···
p41	3173.62	3173.63	**0.00**	**0.06**	3173.62	3173.62	**0.00**	3.45	3173.62	3173.62	**0.00**	2586.02	3173.63	3173.63	**0.00**	84.38	17010	3173.62	3173.63	**0.00**	7.71	210	9	···
p42	2116.86	2116.86	**0.00**	**0.15**	2116.85	2116.86	**0.00**	3.89	2116.85	2116.85	**0.00**	4.27	2116.86	2116.86	**0.00**	195.08	34020	2116.86	2116.86	**0.00**	11.58	382	6	···
p43	1629.55	1629.55	**0.00**	**0.18**	1629.55	1629.55	**0.00**	12.50	1629.55	1629.55	**0.00**	16.56	1629.55	1629.55	**0.00**	287.68	51030	1629.55	1629.55	**0.00**	22.43	544	6	···
p44	3118.04	3118.05	**0.00**	**0.05**	3118.04	3118.04	**0.00**	5.46	3118.04	3118.04	**0.00**	19.62	3118.05	3118.05	**0.00**	96.28	17010	3118.04	3118.05	**0.00**	7.52	214	7	···
p45	2012.50	2012.50	**0.00**	**0.15**	2012.50	2012.50	**0.00**	8.23	2012.50	2012.50	**0.00**	4.26	2012.50	2012.50	**0.00**	188.95	34020	2012.50	2012.50	**0.00**	11.91	350	7	···
p46	1448.76	1448.76	**0.00**	**0.19**	1448.75	1448.75	**0.00**	11.81	1448.75	1448.75	**0.00**	43.52	1448.76	1448.76	**0.00**	288.82	51030	1448.76	1448.76	**0.00**	18.22	539	13	···
p47	1889.94	1889.94	**0.00**	**0.05**	1889.94	1889.94	**0.00**	7.27	1889.94	1889.94	**0.00**	1.78	1889.94	1889.94	**0.00**	98.32	17010	1889.94	1889.94	**0.00**	6.57	181	5	···
p48	1194.16	1194.16	**0.00**	**0.14**	1194.16	1194.16	**0.00**	11.69	1194.16	1194.16	**0.00**	10.65	1194.16	1194.16	**0.00**	183.25	34020	1194.16	1194.16	**0.00**	12.43	442	11	···
p49	887.82	887.82	**0.00**	**0.21**	887.82	887.82	**0.00**	12.00	887.81	887.82	**0.00**	3635.63	887.82	887.82	**0.00**	252.98	51030	887.82	887.82	**0.00**	20.04	562	17	···
p5	4850.50	4850.50	**0.00**	**0.04**	4850.49	4850.50	**0.00**	3.36	4850.50	4850.50	**0.00**	71.00	4850.50	4850.50	**0.00**	90.98	17010	4850.49	4850.50	**0.00**	6.98	163	10	···
p50	5358.21	5358.21	**0.00**	**0.09**	5358.21	5358.21	**0.00**	8.90	5358.21	5358.21	**0.00**	2845.40	5358.21	5358.21	**0.00**	86.89	17010	5358.21	5358.21	**0.00**	7.84	177	11	···
p51	4242.93	4242.93	**0.00**	**0.21**	4242.93	4242.93	**0.00**	4.38	4242.93	4242.93	**0.00**	3628.96	4242.93	4242.93	**0.00**	173.68	34020	4242.93	4242.93	**0.00**	12.72	382	11	···
p52	5757.86	5757.86	**0.00**	**0.07**	5757.85	5757.86	**0.00**	3.74	5757.86	5757.86	**0.00**	3616.71	5757.86	5757.86	**0.00**	93.81	17010	5757.85	5757.86	**0.00**	6.92	181	6	···
p53	5010.25	5010.25	**0.00**	**0.28**	5010.24	5010.25	**0.00**	4.18	5010.25	5010.25	**0.00**	3660.96	5010.25	5010.25	**0.00**	188.76	34020	5010.24	5010.25	**0.00**	12.71	328	8	···
p54	5480.01	5480.01	**0.00**	**0.07**	5480.01	5480.01	**0.00**	15.78	5480.01	5480.01	**0.00**	1764.47	5480.01	5480.01	**0.00**	95.36	17010	5480.01	5480.01	**0.00**	8.94	312	7	···

Table 78 Detailed results for problem NLUFLP-W-holmberg, cost functions f_3

Instance	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
p1	5015.45	5015.45	0.00	0.08	5015.45	5015.45	0.00	4.01	5015.45	5015.45	0.00	14.24	5015.45	5015.45	0.00	138.83	22050	5015.45	5015.45	0.00	23.75	202	6	···
p10	4923.50	4923.50	0.00	0.06	4923.50	4923.50	0.00	3.83	4923.50	4923.50	0.00	208.21	4923.50	4923.50	0.00	178.35	22050	4923.49	4923.50	0.00	24.38	184	6	···
p11	5054.96	5054.96	0.00	0.08	5054.96	5054.96	0.00	3.56	5054.96	5054.96	0.00	3633.58	5054.96	5054.96	0.00	161.77	22050	5054.96	5054.96	0.00	22.31	206	10	···
p12	5186.28	5186.28	0.00	0.12	5186.28	5186.28	0.00	3.47	5186.28	5186.28	0.00	3659.10	5186.28	5186.28	0.00	157.70	22050	5186.28	5186.28	0.00	23.28	207	9	···
p13	3363.92	3363.92	0.00	0.15	3363.92	3363.92	0.00	4.40	3363.92	3363.92	0.00	5.24	3363.92	3363.92	0.00	291.18	44100	3363.92	3363.92	0.00	38.82	437	10	···
p14	3301.25	3301.25	0.00	0.13	3301.24	3301.25	0.00	4.08	3301.25	3301.25	0.00	18.52	3301.25	3301.25	0.00	287.46	44100	3301.25	3301.25	0.00	38.93	454	10	···
p15	3382.08	3382.08	0.00	0.12	3382.08	3382.08	0.00	4.44	3382.08	3382.08	0.00	12.69	3382.08	3382.08	0.00	290.00	44100	3382.08	3382.08	0.00	38.23	438	10	···
p16	3462.91	3462.91	0.00	0.16	3462.91	3462.91	0.00	4.36	3462.91	3462.91	0.00	10.54	3462.91	3462.91	0.00	313.43	44100	3462.91	3462.91	0.00	39.55	440	10	···
p17	3363.92	3363.92	0.00	0.14	3363.92	3363.92	0.00	3.95	3363.92	3363.92	0.00	5.22	3363.92	3363.92	0.00	287.19	44100	3363.92	3363.92	0.00	39.88	437	10	···
p18	3301.25	3301.25	0.00	0.14	3301.24	3301.25	0.00	4.27	3301.25	3301.25	0.00	18.49	3301.25	3301.25	0.00	285.75	44100	3301.25	3301.25	0.00	38.84	454	10	···
p19	3382.08	3382.08	0.00	0.15	3382.08	3382.08	0.00	7.32	3382.08	3382.08	0.00	12.84	3382.08	3382.08	0.00	294.39	44100	3382.08	3382.08	0.00	39.56	438	10	···
p2	4923.50	4923.50	0.00	0.07	4923.50	4923.50	0.00	4.55	4923.50	4923.50	0.00	204.76	4923.50	4923.50	0.00	155.77	22050	4923.49	4923.50	0.00	22.10	184	6	···
p20	3462.91	3462.91	0.00	0.18	3462.91	3462.91	0.00	3.96	3462.91	3462.91	0.00	10.63	3462.91	3462.91	0.00	314.89	44100	3462.91	3462.91	0.00	42.53	440	10	···
p21	3363.92	3363.92	0.00	0.14	3363.92	3363.92	0.00	4.34	3363.92	3363.92	0.00	5.22	3363.92	3363.92	0.00	284.59	44100	3363.92	3363.92	0.00	39.41	437	10	···
p22	3301.25	3301.25	0.00	0.13	3301.24	3301.25	0.00	3.88	3301.25	3301.25	0.00	18.45	3301.25	3301.25	0.00	279.19	44100	3301.25	3301.25	0.00	41.40	454	10	···
p23	3382.08	3382.08	0.00	0.14	3382.08	3382.08	0.00	4.30	3382.08	3382.08	0.00	12.65	3382.08	3382.08	0.00	295.87	44100	3382.08	3382.08	0.00	39.10	438	10	···
p24	3462.91	3462.91	0.00	0.17	3462.91	3462.91	0.00	28.39	3462.91	3462.91	0.00	10.96	3462.91	3462.91	0.00	316.87	44100	3462.91	3462.91	0.00	38.15	440	10	···
p25	8039.07	8039.07	0.00	1.36	8039.05	8039.05	0.00	29.70	$-\infty$	$+\infty$	100.0	3600.10	8039.07	8039.07	0.00	512.23	66150	8039.07	8039.07	0.00	60.55	582	14	···
p26	7998.54	7998.54	0.00	0.99	7998.51	7998.51	0.00	16.95	7998.53	7998.54	0.00	3656.16	7998.54	7998.54	0.00	441.87	66150	7998.53	7998.54	0.00	57.60	564	13	···
p27	8066.71	8066.71	0.00	1.56	8066.68	8066.69	0.00	32.30	$-\infty$	$+\infty$	100.0	3600.10	8066.71	8066.71	0.00	516.36	66150	8066.71	8066.71	0.00	56.91	636	13	···
p28	8131.39	8131.40	0.00	2.21	8131.37	8131.37	0.00	46.00	$-\infty$	$+\infty$	100.0	3600.10	8131.40	8131.40	0.00	400.74	66150	8131.39	8131.40	0.00	62.66	600	17	···
p29	8039.07	8039.07	0.00	1.33	8039.05	8039.05	0.00	19.29	$-\infty$	$+\infty$	100.0	3600.10	8039.07	8039.07	0.00	490.61	66150	8039.07	8039.07	0.00	61.75	582	14	···
p3	5054.96	5054.96	0.00	0.09	5054.96	5054.96	0.00	3.65	5054.96	5054.96	0.00	3633.34	5054.96	5054.96	0.00	147.57	22050	5054.96	5054.96	0.00	23.61	206	10	···
p30	7998.54	7998.54	0.00	1.10	7998.51	7998.51	0.00	17.92	7998.53	7998.54	0.00	3656.77	7998.54	7998.54	0.00	431.99	66150	7998.53	7998.54	0.00	62.27	563	12	···
p31	8066.71	8066.71	0.00	1.52	8066.68	8066.69	0.00	31.64	$-\infty$	$+\infty$	100.0	3600.10	8066.71	8066.71	0.00	504.58	66150	8066.71	8066.71	0.00	58.98	636	13	···
p32	8131.39	8131.40	0.00	2.08	8131.37	8131.37	0.00	53.51	$-\infty$	$+\infty$	100.0	3600.10	8131.40	8131.40	0.00	420.69	66150	8131.39	8131.40	0.00	63.40	601	17	···
p33	8039.07	8039.07	0.00	1.31	8039.05	8039.05	0.00	18.63	$-\infty$	$+\infty$	100.0	3600.10	8039.07	8039.07	0.00	502.45	66150	8039.07	8039.07	0.00	60.40	582	14	···
p34	7998.54	7998.54	0.00	1.04	7998.51	7998.51	0.00	16.89	7998.53	7998.54	0.00	3656.40	7998.54	7998.54	0.00	427.42	66150	7998.53	7998.54	0.00	57.01	562	12	···
p35	8066.71	8066.71	0.00	1.53	8066.68	8066.69	0.00	31.60	$-\infty$	$+\infty$	100.0	3600.10	8066.71	8066.71	0.00	501.95	66150	8066.71	8066.71	0.00	59.84	636	13	···
p36	8131.39	8131.40	0.00	2.17	8131.37	8131.37	0.00	50.13	$-\infty$	$+\infty$	100.0	3600.10	8131.40	8131.40	0.00	421.33	66150	8131.39	8131.40	0.00	62.57	595	17	···
p37	8039.07	8039.07	0.00	1.31	8039.05	8039.05	0.00	18.16	$-\infty$	$+\infty$	100.0	3600.10	8039.07	8039.07	0.00	494.71	66150	8039.07	8039.07	0.00	60.18	582	14	···
p38	7998.54	7998.54	0.00	1.15	7998.51	7998.51	0.00	17.80	7998.53	7998.54	0.00	3656.45	7998.54	7998.54	0.00	396.98	66150	7998.53	7998.54	0.00	64.65	563	12	···
p39	8066.71	8066.71	0.00	1.61	8066.68	8066.69	0.00	31.69	$-\infty$	$+\infty$	100.0	3600.10	8066.71	8066.71	0.00	497.42	66150	8066.71	8066.71	0.00	58.34	636	13	···
p4	5186.28	5186.28	0.00	0.12	5186.28	5186.28	0.00	3.60	5186.28	5186.28	0.00	3661.05	5186.28	5186.28	0.00	144.08	22050	5186.28	5186.28	0.00	24.56	207	9	···
p40	8131.39	8131.40	0.00	1.87	8131.37	8131.37	0.00	51.78	$-\infty$	$+\infty$	100.0	3600.10	8131.40	8131.40	0.00	419.38	66150	8131.39	8131.40	0.00	69.45	578	19	···
p41	3308.71	3308.71	0.00	0.14	3308.71	3308.71	0.00	3.69	3308.71	3308.71	0.00	18.23	3308.71	3308.71	0.00	179.24	22050	3308.71	3308.71	0.00	21.54	252	10	···
p42	2205.58	2205.58	0.00	0.22	2205.58	2205.58	0.00	13.05	2205.58	2205.58	0.00	173.20	2205.58	2205.58	0.00	237.04	44100	2205.58	2205.58	0.00	37.92	491	9	···
p43	1696.12	1696.12	0.00	0.45	1696.12	1696.12	0.00	15.88	1696.12	1696.12	0.00	41.42	1696.12	1696.12	0.00	359.48	66150	1696.12	1696.12	0.00	60.42	631	23	···
p44	3257.01	3257.01	0.00	0.12	3257.01	3257.01	0.00	11.23	3257.01	3257.01	0.00	34.13	3257.01	3257.01	0.00	129.23	22050	3257.01	3257.01	0.00	22.11	380	7	···
p45	2106.17	2106.17	0.00	0.23	2106.17	2106.17	0.00	7.01	2106.17	2106.17	0.00	756.90	2106.17	2106.17	0.00	275.66	44100	2106.17	2106.17	0.00	41.71	403	8	···
p46	1510.23	1510.23	0.00	0.32	1510.23	1510.23	0.00	10.34	1510.23	1510.23	0.00	104.73	1510.23	1510.23	0.00	372.96	66150	1510.23	1510.23	0.00	63.71	622	15	···
p47	2021.22	2021.22	0.00	0.11	2021.22	2021.22	0.00	10.38	2021.22	2021.22	0.00	10.30	2021.22	2021.22	0.00	135.94	22050	2021.22	2021.22	0.00	19.26	183	5	···
p48	1269.81	1269.81	0.00	0.25	1269.81	1269.81	0.00	15.50	1269.81	1269.81	0.00	32.45	1269.81	1269.81	0.00	245.75	44100	1269.81	1269.81	0.00	40.38	391	14	···
p49	948.14	948.14	0.00	0.36	948.13	948.13	0.00	6.41	$-\infty$	$+\infty$	100.0	3600.10	948.14	948.14	0.00	340.36	66150	948.14	948.14	0.00	57.25	618	16	···
p5	5015.45	5015.45	0.00	0.08	5015.45	5015.45	0.00	3.83	5015.45	5015.45	0.00	13.10	5015.45	5015.45	0.00	149.69	22050	5015.45	5015.45	0.00	22.91	202	6	···
p50	5514.07	5514.07	0.00	0.17	5514.06	5514.07	0.00	8.12	5514.06	5514.06	0.00	3648.86	5514.07	5514.07	0.00	160.40	22050	5514.06	5514.07	0.00	44.41	234	14	···
p51	4345.99	4345.99	0.00	0.63	4345.98	4345.99	0.00	8.06	4345.99	4345.99	0.00	3654.86	4345.99	4345.99	0.00	290.69	44100	4345.99	4345.99	0.00	39.39	391	9	···
p52	5930.57	5930.57	0.00	0.21	5930.57	5930.57	0.00	4.56	5930.57	5930.57	0.00	3664.04	5930.57	5930.57	0.00	151.62	22050	5930.57	5930.57	0.00	25.74	260	11	···
p53	5122.51	5122.51	0.00	0.44	5122.51	5122.51	0.00	5.74	$-\infty$	$+\infty$	100.0	3600.10	5122.51	5122.51	0.00	294.72	44100	5122.51	5122.51	0.00	40.21	399	12	···
p54	5624.38	5624.38	0.00	0.30	5624.38	5624.38	0.00	16.48	5624.38	5624.38	0.00	3616.15	5624.38	5624.38	0.00	176.15	22050	5624.38	5624.38	0.00	22.42	215	13	···

Table 79 Detailed results for problem NLUFLP-W-holmberg, cost functions f_4

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
p1	5028.14	5028.14	0.00	0.33	5028.13	5028.13	0.00	4.39	5028.14	5028.14	0.00	8.59	5028.09	5028.14	0.00	265.68	36650	5028.13	5028.14	0.00	85.68	239	11	⋯
p10	4933.17	4933.17	0.00	0.31	4933.17	4933.17	0.00	4.14	4933.17	4933.17	0.00	3.72	4933.17	4933.17	0.00	283.00	36650	4933.17	4933.17	0.00	60.42	250	9	⋯
p11	5070.62	5070.62	0.00	0.37	5070.62	5070.62	0.00	4.30	5070.62	5070.62	0.00	20.33	5070.62	5070.62	0.00	283.19	36650	5070.62	5070.62	0.00	64.52	299	9	⋯
p12	5208.07	5208.07	0.00	0.45	5208.07	5208.07	0.00	4.64	5208.07	5208.07	0.00	161.75	5208.07	5208.07	0.00	306.43	36650	5208.07	5208.07	0.00	66.46	301	9	⋯
p13	3411.57	3411.57	0.00	1.18	3411.57	3411.57	0.00	4.69	3411.57	3411.57	0.00	11.95	3411.48	3411.57	0.00	919.27	73300	3411.57	3411.57	0.00	72.63	561	9	⋯
p14	3333.11	3333.11	0.00	0.87	3333.11	3333.11	0.00	4.05	3333.11	3333.11	0.00	3.83	3333.06	3333.11	0.00	840.74	73300	3333.11	3333.11	0.00	74.28	542	8	⋯
p15	3435.18	3435.18	0.00	1.13	3435.18	3435.18	0.00	5.75	3435.18	3435.18	0.00	279.36	3435.09	3435.18	0.00	762.78	73300	3435.18	3435.18	0.00	73.79	561	9	⋯
p16	3537.26	3537.26	0.00	1.18	3537.25	3537.25	0.00	5.63	3537.26	3537.26	0.00	59.65	3537.13	3537.26	0.00	574.27	73300	3537.25	3537.26	0.00	81.98	562	9	⋯
p17	3411.57	3411.57	0.00	1.20	3411.57	3411.57	0.00	4.34	3411.57	3411.57	0.00	11.99	3411.48	3411.57	0.00	979.37	73300	3411.57	3411.57	0.00	74.76	561	9	⋯
p18	3333.11	3333.11	0.00	1.01	3333.11	3333.11	0.00	4.27	3333.11	3333.11	0.00	3.84	3333.06	3333.11	0.00	889.45	73300	3333.11	3333.11	0.00	70.87	542	8	⋯
p19	3435.18	3435.18	0.00	1.57	3435.18	3435.18	0.00	11.88	3435.18	3435.18	0.00	283.47	3435.09	3435.18	0.00	860.76	73300	3435.18	3435.18	0.00	70.06	561	9	⋯
p2	4933.17	4933.17	0.00	0.32	4933.17	4933.17	0.00	3.97	4933.17	4933.17	0.00	3.68	4933.17	4933.17	0.00	255.23	36650	4933.17	4933.17	0.00	59.58	250	9	⋯
p20	3537.26	3537.26	0.00	1.55	3537.25	3537.25	0.00	5.62	3537.26	3537.26	0.00	59.71	3537.13	3537.26	0.00	581.38	73300	3537.25	3537.26	0.00	72.96	562	9	⋯
p21	3411.57	3411.57	0.00	0.84	3411.57	3411.57	0.00	4.55	3411.57	3411.57	0.00	11.93	3411.48	3411.57	0.00	920.86	73300	3411.57	3411.57	0.00	79.35	561	9	⋯
p22	3333.11	3333.11	0.00	0.97	3333.11	3333.11	0.00	9.81	3333.11	3333.11	0.00	3.80	3333.06	3333.11	0.00	896.01	73300	3333.11	3333.11	0.00	71.46	542	8	⋯
p23	3435.18	3435.18	0.00	1.35	3435.18	3435.18	0.00	5.67	3435.18	3435.18	0.00	279.78	3435.09	3435.18	0.00	798.73	73300	3435.18	3435.18	0.00	74.66	561	9	⋯
p24	3537.26	3537.26	0.00	1.62	3537.25	3537.25	0.00	29.92	3537.26	3537.26	0.00	59.18	3537.13	3537.26	0.00	581.48	73300	3537.25	3537.26	0.00	74.53	562	9	⋯
p25	8081.27	8081.27	0.00	23.38	8081.25	8081.25	0.00	758.25	8081.23	8081.28	0.00	3622.75	8081.27	8081.27	0.00	1136.82	109950	8081.27	8081.28	0.00	73.09	681	16	⋯
p26	8029.24	8029.24	0.00	10.30	8029.22	8029.22	0.00	163.91	$-\infty$	$+\infty$	100.0	3600.10	8029.24	8029.24	0.00	1101.76	109950	8029.24	8029.25	0.00	56.97	652	11	⋯
p27	8117.61	8117.62	0.00	32.04	8117.59	8117.60	0.00	1321.31	8116.48	8117.63	0.01	3606.31	8117.62	8117.62	0.00	1262.21	109950	8117.62	8117.62	0.00	67.88	675	15	⋯
p28	8202.42	8202.43	0.00	112.11	$-\infty$	$+\infty$	100.0	$+\infty$	8200.41	8202.44	0.02	3601.68	8202.43	8202.43	0.00	1269.46	109950	8202.42	8202.43	0.00	74.27	710	16	⋯
p29	8081.27	8081.27	0.00	24.37	8081.25	8081.25	0.00	881.90	8081.23	8081.28	0.00	3622.12	8081.27	8081.27	0.00	1161.67	109950	8081.27	8081.28	0.00	67.88	681	16	⋯
p3	5070.62	5070.62	0.00	0.37	5070.62	5070.62	0.00	4.01	5070.62	5070.62	0.00	19.79	5070.62	5070.62	0.00	263.31	36650	5070.62	5070.62	0.00	63.79	299	9	⋯
p30	8029.24	8029.24	0.00	12.67	8029.22	8029.22	0.00	145.24	$-\infty$	$+\infty$	100.0	3600.10	8029.24	8029.24	0.00	1117.34	109950	8029.24	8029.25	0.00	58.75	647	11	⋯
p31	8117.61	8117.62	0.00	34.19	8117.59	8117.60	0.00	1326.11	8116.51	8117.63	0.01	3606.20	8117.62	8117.62	0.00	1252.23	109950	8117.62	8117.62	0.00	71.05	675	15	⋯
p32	8202.42	8202.43	0.00	109.11	$-\infty$	$+\infty$	100.0	$+\infty$	8200.51	8202.44	0.02	3601.81	8202.43	8202.43	0.00	1253.96	109950	8202.42	8202.43	0.00	73.12	711	16	⋯
p33	8081.27	8081.27	0.00	21.21	8081.25	8081.25	0.00	632.03	8081.23	8081.28	0.00	3620.64	8081.27	8081.27	0.00	1051.74	109950	8081.27	8081.28	0.00	69.10	681	16	⋯
p34	8029.24	8029.24	0.00	11.09	8029.22	8029.22	0.00	155.07	$-\infty$	$+\infty$	100.0	3600.10	8029.24	8029.24	0.00	1086.57	109950	8029.24	8029.25	0.00	62.45	647	11	⋯
p35	8117.61	8117.62	0.00	38.60	8117.59	8117.60	0.00	1373.08	8116.49	8117.63	0.01	3607.92	8117.62	8117.62	0.00	1183.13	109950	8117.62	8117.62	0.00	68.50	676	15	⋯
p36	8202.42	8202.43	0.00	118.79	$-\infty$	$+\infty$	100.0	$+\infty$	8200.37	8202.44	0.03	3601.97	8202.43	8202.43	0.00	1312.75	109950	8202.42	8202.43	0.00	67.28	709	16	⋯
p37	8081.27	8081.27	0.00	19.98	8081.25	8081.25	0.00	785.07	8081.23	8081.28	0.00	3622.53	8081.27	8081.27	0.00	1142.37	109950	8081.27	8081.28	0.00	68.74	681	16	⋯
p38	8029.24	8029.24	0.00	11.41	8029.22	8029.22	0.00	173.64	$-\infty$	$+\infty$	100.0	3600.10	8029.24	8029.24	0.00	1037.32	109950	8029.24	8029.25	0.00	58.98	621	12	⋯
p39	8117.61	8117.62	0.00	37.29	8117.59	8117.60	0.00	1721.20	8116.47	8117.63	0.01	3606.32	8117.62	8117.62	0.00	1291.89	109950	8117.62	8117.62	0.00	69.98	668	15	⋯
p4	5208.07	5208.07	0.00	0.44	5208.07	5208.07	0.00	4.43	5208.07	5208.07	0.00	158.23	5208.07	5208.07	0.00	287.90	36650	5208.07	5208.07	0.00	65.14	301	9	⋯
p40	8202.42	8202.43	0.00	133.66	8202.40	8202.40	0.00	3350.14	8200.37	8202.44	0.03	3601.73	8202.43	8202.43	0.00	1119.77	109950	8202.42	8202.43	0.00	72.50	711	16	⋯
p41	3318.62	3318.62	0.00	0.77	3318.62	3318.62	0.00	6.22	3318.62	3318.62	0.00	75.40	3318.59	3318.62	0.00	355.27	36650	3318.62	3318.62	0.00	58.05	245	8	⋯
p42	2250.66	2250.66	0.00	6.07	2250.66	2250.66	0.00	48.19	2250.66	2250.66	0.00	700.11	2250.62	2250.66	0.00	734.82	73300	2250.66	2250.66	0.00	48.95	509	10	⋯
p43	1736.51	1736.51	0.00	8.54	1736.51	1736.51	0.00	153.15	1736.51	1736.51	0.00	188.41	1736.51	1736.51	0.00	1338.56	109950	1736.51	1736.51	0.00	44.85	787	16	⋯
p44	3289.56	3289.56	0.00	0.63	3289.56	3289.56	0.00	8.39	3289.56	3289.56	0.00	12.78	3289.53	3289.56	0.00	303.52	36650	3289.56	3289.56	0.00	66.65	240	9	⋯
p45	2143.17	2143.17	0.00	2.19	2143.16	2143.16	0.00	15.60	2143.16	2143.16	0.00	2472.12	2143.12	2143.17	0.00	847.55	73300	2143.17	2143.17	0.00	78.61	526	10	⋯
p46	1547.94	1547.94	0.00	4.24	1547.94	1547.94	0.00	55.99	1547.94	1547.94	0.00	3660.05	1547.94	1547.94	0.00	1295.28	109950	1547.94	1547.94	0.00	43.01	835	17	⋯
p47	2064.25	2064.25	0.00	0.55	2064.25	2064.25	0.00	4.64	2064.25	2064.25	0.00	9.70	2064.14	2064.25	0.01	369.57	36650	2064.25	2064.25	0.00	149.40	257	9	⋯
p48	1314.37	1314.37	0.00	7.06	1314.37	1314.37	0.00	67.72	1314.37	1314.37	0.00	3647.97	1314.37	1314.37	0.00	726.98	73300	1314.37	1314.37	0.00	37.95	516	15	⋯
p49	986.15	986.15	0.00	8.10	986.14	986.14	0.00	272.88	$-\infty$	$+\infty$	100.0	3600.10	986.15	986.15	0.00	1052.04	109950	986.15	986.15	0.00	38.61	876	19	⋯
p5	5028.14	5028.14	0.00	0.34	5028.13	5028.13	0.00	5.67	5028.14	5028.14	0.00	7.89	5028.09	5028.14	0.00	276.16	36650	5028.13	5028.14	0.00	81.56	239	11	⋯
p50	5460.48	5460.48	0.00	0.77	5460.48	5460.48	0.00	12.82	5460.48	5460.48	0.00	32.27	5460.45	5460.48	0.00	393.24	36650	5460.48	5460.48	0.00	70.09	217	9	⋯
p51	4353.54	4353.54	0.00	4.67	4353.54	4353.54	0.00	14.21	$-\infty$	$+\infty$	100.0	3600.10	4353.54	4353.54	0.00	758.76	73300	4353.53	4353.54	0.00	43.56	454	8	⋯
p52	5923.80	5923.80	0.00	1.06	5923.80	5923.80	0.00	6.03	5923.80	5923.80	0.00	3628.64	5923.80	5923.80	0.00	327.26	36650	5923.80	5923.80	0.00	52.35	225	9	⋯
p53	5152.93	5152.93	0.00	4.08	5152.93	5152.93	0.00	12.21	5152.93	5152.93	0.00	164.83	5152.92	5152.93	0.00	823.70	73300	5152.93	5152.93	0.00	75.43	432	10	⋯
p54	5643.91	5643.91	0.00	1.49	5643.91	5643.91	0.00	20.37	5643.91	5643.91	0.00	3622.95	5643.91	5643.91	0.00	282.74	36650	5643.91	5643.91	0.00	57.94	280	11	⋯

Table 80 Detailed results for problem NLUFLP-W-holmberg, cost functions f_5

	GUROBI				**SCIP**				**COUENNE**				**NAIVE**					**CN24**						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
p1	5252.04	5252.04	**0.00**	**0.38**	−∞	+∞	100.0	+∞	5252.04	5252.04	**0.00**	4.39	−∞	+∞	100.00	3600.01	+∞	5252.04	5252.04	**0.00**	92.10	179	8	⋯
p10	5088.42	5088.42	**0.00**	**0.26**	−∞	+∞	100.0	+∞	5088.42	5088.42	**0.00**	5.20	−∞	+∞	100.00	3600.01	+∞	5088.42	5088.42	**0.00**	88.95	174	8	⋯
p11	5330.03	5330.03	**0.00**	**0.44**	−∞	+∞	100.0	+∞	5330.03	5330.03	**0.00**	41.97	−∞	+∞	100.00	3600.01	+∞	5330.02	5330.03	**0.00**	86.15	169	9	⋯
p12	5571.52	5571.52	**0.00**	**0.69**	−∞	+∞	100.0	+∞	5571.52	5571.52	**0.00**	1285.46	−∞	+∞	100.00	3600.01	+∞	5571.52	5571.52	**0.00**	91.52	178	12	⋯
p13	3665.17	3665.17	**0.00**	**0.95**	−∞	+∞	100.0	+∞	3665.17	3665.17	**0.00**	9.75	−∞	+∞	100.00	3600.01	+∞	3665.17	3665.17	**0.00**	60.76	311	8	⋯
p14	3504.38	3504.38	**0.00**	**0.52**	−∞	+∞	100.0	+∞	3504.38	3504.38	**0.00**	6.99	−∞	+∞	100.00	3600.01	+∞	3504.38	3504.38	**0.00**	59.32	311	8	⋯
p15	3720.63	3720.63	**0.00**	**1.34**	−∞	+∞	100.0	+∞	3720.63	3720.63	**0.00**	14.21	−∞	+∞	100.00	3600.01	+∞	3720.63	3720.63	**0.00**	58.28	311	8	⋯
p16	3936.89	3936.89	**0.00**	**2.58**	−∞	+∞	100.0	+∞	3936.89	3936.89	**0.00**	12.95	−∞	+∞	100.00	3600.01	+∞	3936.89	3936.89	**0.00**	64.35	312	8	⋯
p17	3665.17	3665.17	**0.00**	**0.90**	−∞	+∞	100.0	+∞	3665.17	3665.17	**0.00**	9.35	−∞	+∞	100.00	3600.01	+∞	3665.17	3665.17	**0.00**	60.45	311	8	⋯
p18	3504.38	3504.38	**0.00**	**0.53**	−∞	+∞	100.0	+∞	3504.38	3504.38	**0.00**	7.01	−∞	+∞	100.00	3600.01	+∞	3504.38	3504.38	**0.00**	56.57	311	8	⋯
p19	3720.63	3720.63	**0.00**	**1.41**	−∞	+∞	100.0	+∞	3720.63	3720.63	**0.00**	13.54	−∞	+∞	100.00	3600.01	+∞	3720.63	3720.63	**0.00**	57.12	311	8	⋯
p2	5088.42	5088.42	**0.00**	**0.27**	−∞	+∞	100.0	+∞	5088.42	5088.42	**0.00**	5.23	−∞	+∞	100.00	3600.01	+∞	5088.42	5088.42	**0.00**	83.96	174	8	⋯
p20	3936.89	3936.89	**0.00**	**2.63**	−∞	+∞	100.0	+∞	3936.89	3936.89	**0.00**	12.92	−∞	+∞	100.00	3600.01	+∞	3936.89	3936.89	**0.00**	57.28	312	8	⋯
p21	3665.17	3665.17	**0.00**	**0.90**	−∞	+∞	100.0	+∞	3665.17	3665.17	**0.00**	9.56	−∞	+∞	100.00	3600.01	+∞	3665.17	3665.17	**0.00**	57.57	311	8	⋯
p22	3504.38	3504.38	**0.00**	**0.52**	−∞	+∞	100.0	+∞	3504.38	3504.38	**0.00**	7.00	−∞	+∞	100.00	3600.01	+∞	3504.38	3504.38	**0.00**	63.83	311	8	⋯
p23	3720.63	3720.63	**0.00**	**1.34**	−∞	+∞	100.0	+∞	3720.63	3720.63	**0.00**	13.48	−∞	+∞	100.00	3600.01	+∞	3720.63	3720.63	**0.00**	57.18	311	8	⋯
p24	3936.89	3936.89	**0.00**	**2.72**	−∞	+∞	100.0	+∞	3936.89	3936.89	**0.00**	12.92	−∞	+∞	100.00	3600.01	+∞	3936.89	3936.89	**0.00**	56.81	312	8	⋯
p25	8336.98	8336.99	**0.00**	**5.94**	−∞	+∞	100.0	+∞	8322.21	8336.99	0.18	2981.32	−∞	+∞	100.00	3600.01	+∞	8336.98	8336.99	**0.00**	81.37	507	12	⋯
p26	8216.18	8216.19	**0.00**	**5.93**	−∞	+∞	100.0	+∞	8211.26	8216.19	0.06	2977.61	−∞	+∞	100.00	3600.01	+∞	8216.18	8216.19	**0.00**	87.72	484	18	⋯
p27	8430.08	8430.08	**0.00**	**5.83**	−∞	+∞	100.0	+∞	8415.87	8430.09	0.17	3002.87	−∞	+∞	100.00	3600.01	+∞	8430.08	8430.09	**0.00**	82.89	506	15	⋯
p28	8642.33	8642.33	**0.00**	**7.02**	−∞	+∞	100.0	+∞	8584.71	8643.24	0.68	2993.27	−∞	+∞	100.00	3600.01	+∞	8642.33	8642.34	**0.00**	89.97	491	18	⋯
p29	8336.98	8336.99	**0.00**	**5.92**	−∞	+∞	100.0	+∞	8322.17	8336.99	0.18	2978.47	−∞	+∞	100.00	3600.01	+∞	8336.98	8336.99	**0.00**	85.05	507	12	⋯
p3	5330.03	5330.03	**0.00**	**0.51**	−∞	+∞	100.0	+∞	5330.03	5330.03	**0.00**	42.34	−∞	+∞	100.00	3600.01	+∞	5330.02	5330.03	**0.00**	84.70	169	9	⋯
p30	8216.18	8216.19	**0.00**	**6.77**	−∞	+∞	100.0	+∞	8211.08	8216.19	0.06	2971.22	−∞	+∞	100.00	3600.01	+∞	8216.18	8216.19	**0.00**	82.88	488	18	⋯
p31	8430.08	8430.08	**0.00**	**6.19**	−∞	+∞	100.0	+∞	8415.47	8430.09	0.17	2908.76	−∞	+∞	100.00	3600.01	+∞	8430.08	8430.09	**0.00**	88.64	501	15	⋯
p32	8642.33	8642.33	**0.00**	**6.08**	−∞	+∞	100.0	+∞	8581.07	8643.24	0.72	2914.82	−∞	+∞	100.00	3600.01	+∞	8642.33	8642.34	**0.00**	97.41	492	18	⋯
p33	8336.98	8336.99	**0.00**	**6.36**	−∞	+∞	100.0	+∞	8321.32	8336.99	0.19	2895.73	−∞	+∞	100.00	3600.01	+∞	8336.98	8336.99	**0.00**	79.55	507	12	⋯
p34	8216.18	8216.19	**0.00**	**5.77**	−∞	+∞	100.0	+∞	8211.56	8216.19	0.06	2935.06	−∞	+∞	100.00	3600.01	+∞	8216.18	8216.19	**0.00**	89.23	487	18	⋯
p35	8430.08	8430.08	**0.00**	**6.08**	−∞	+∞	100.0	+∞	8415.49	8430.09	0.17	2917.74	−∞	+∞	100.00	3600.01	+∞	8430.08	8430.09	**0.00**	91.20	501	15	⋯
p36	8642.33	8642.33	**0.00**	**6.79**	−∞	+∞	100.0	+∞	8579.12	8643.24	0.74	2895.53	−∞	+∞	100.00	3600.01	+∞	8642.33	8642.34	**0.00**	96.88	492	18	⋯
p37	8336.98	8336.99	**0.00**	**6.05**	−∞	+∞	100.0	+∞	8321.75	8336.99	0.18	2968.42	−∞	+∞	100.00	3600.01	+∞	8336.98	8336.99	**0.00**	83.04	507	12	⋯
p38	8216.18	8216.19	**0.00**	**6.46**	−∞	+∞	100.0	+∞	8210.97	8216.19	0.06	2887.32	−∞	+∞	100.00	3600.01	+∞	8216.18	8216.19	**0.00**	81.93	490	18	⋯
p39	8430.08	8430.08	**0.00**	**6.07**	−∞	+∞	100.0	+∞	8415.58	8430.09	0.17	2924.37	−∞	+∞	100.00	3600.01	+∞	8430.08	8430.09	**0.00**	84.88	501	15	⋯
p4	5571.52	5571.52	**0.00**	**0.63**	−∞	+∞	100.0	+∞	5571.52	5571.52	**0.00**	1268.07	−∞	+∞	100.00	3600.01	+∞	5571.52	5571.52	**0.00**	87.41	178	12	⋯
p40	8642.33	8642.33	**0.00**	**7.10**	−∞	+∞	100.0	+∞	8588.07	8643.24	0.64	2949.30	−∞	+∞	100.00	3600.01	+∞	8642.33	8642.34	**0.00**	93.24	506	17	⋯
p41	3519.99	3519.99	**0.00**	**0.92**	−∞	+∞	100.0	+∞	3519.99	3519.99	**0.00**	1146.18	−∞	+∞	100.00	3600.01	+∞	3519.99	3519.99	**0.00**	73.61	170	14	⋯
p42	2474.92	2474.92	**0.00**	**3.87**	−∞	+∞	100.0	+∞	2474.92	2474.92	**0.00**	49.22	−∞	+∞	100.00	3600.01	+∞	2474.92	2474.92	**0.00**	68.30	499	12	⋯
p43	1995.82	1995.82	**0.00**	**3.32**	−∞	+∞	100.0	+∞	1995.82	1995.82	**0.00**	2609.73	−∞	+∞	100.00	3600.01	+∞	1995.82	1995.82	**0.00**	77.05	554	16	⋯
p44	3476.58	3476.58	**0.00**	**0.66**	−∞	+∞	100.0	+∞	3476.58	3476.58	**0.00**	11.92	−∞	+∞	100.00	3600.01	+∞	3476.58	3476.58	**0.00**	62.11	159	6	⋯
p45	2374.25	2374.25	**0.00**	**1.53**	−∞	+∞	100.0	+∞	2374.24	2374.24	**0.00**	53.78	−∞	+∞	100.00	3600.01	+∞	2374.24	2374.25	**0.00**	69.87	366	10	⋯
p46	1811.40	1811.40	**0.00**	**4.55**	−∞	+∞	100.0	+∞	1801.11	1811.40	0.57	3549.49	−∞	+∞	100.00	3600.01	+∞	1811.40	1811.40	**0.00**	78.55	510	16	⋯
p47	2242.16	2242.16	**0.00**	**0.51**	−∞	+∞	100.0	+∞	2242.16	2242.16	**0.00**	6.65	−∞	+∞	100.00	3600.01	+∞	2242.15	2242.16	**0.00**	44.34	176	6	⋯
p48	1538.96	1538.96	**0.00**	**2.73**	−∞	+∞	100.0	+∞	1538.95	1538.95	**0.00**	702.07	−∞	+∞	100.00	3600.01	+∞	1538.96	1538.96	**0.00**	62.03	419	15	⋯
p49	1259.83	1259.83	**0.00**	**3.15**	−∞	+∞	100.0	+∞	1245.19	1259.89	1.17	3600.10	−∞	+∞	100.00	3600.01	+∞	1259.83	1259.83	**0.00**	77.43	520	18	⋯
p5	5252.04	5252.04	**0.00**	**0.43**	−∞	+∞	100.0	+∞	5252.04	5252.04	**0.00**	4.41	−∞	+∞	100.00	3600.01	+∞	5252.04	5252.04	**0.00**	83.80	179	8	⋯
p50	5706.64	5706.64	**0.00**	**0.89**	−∞	+∞	100.0	+∞	5706.64	5706.64	**0.00**	28.61	−∞	+∞	100.00	3600.01	+∞	5706.63	5706.64	**0.00**	79.54	132	7	⋯
p51	4605.77	4605.78	**0.00**	**3.89**	−∞	+∞	100.0	+∞	4605.78	4605.78	**0.00**	3570.42	−∞	+∞	100.00	3600.01	+∞	4605.77	4605.78	**0.00**	98.12	324	12	⋯
p52	6139.02	6139.02	**0.00**	**0.77**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	6139.02	6139.02	**0.00**	98.11	202	10	⋯
p53	5379.14	5379.14	**0.00**	**7.26**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5379.14	5379.14	**0.00**	72.08	335	14	⋯
p54	5838.81	5838.81	**0.00**	**0.58**	−∞	+∞	100.0	+∞	5838.81	5838.81	**0.00**	22.35	−∞	+∞	100.00	3600.01	+∞	5838.81	5838.82	**0.00**	67.89	191	6	⋯

Table 81 **Detailed results for problem NLUFLP-W-holmberg, cost functions f_6**

Instance	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
p1	6207.39	6207.39	**0.00**	**1.10**	−∞	+∞	100.0	+∞	6207.39	6207.39	**0.00**	3662.96	−∞	+∞	100.00	3609.97	1253119	6207.39	6207.39	**0.00**	2.06	152	12	⋯
p10	5747.97	5747.97	**0.00**	**1.10**	−∞	+∞	100.0	+∞	5747.97	5747.97	**0.00**	13.20	−∞	+∞	100.00	3610.11	1253210	5747.97	5747.97	**0.00**	1.51	161	9	⋯
p11	6428.62	6428.62	**0.00**	**1.20**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3609.78	1253100	6428.62	6428.62	**0.00**	2.08	173	10	⋯
p12	7104.98	7104.98	**0.00**	**1.75**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3609.48	1253570	7104.97	7104.98	**0.00**	2.66	175	11	⋯
p13	5039.85	5039.85	**0.00**	448.97	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5039.85	5039.85	**0.00**	**3.62**	321	12	⋯
p14	4438.19	4438.19	**0.00**	79.07	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	4438.19	4438.19	**0.00**	**3.04**	326	11	⋯
p15	5272.31	5272.31	**0.00**	555.99	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5272.31	5272.31	**0.00**	**3.38**	311	11	⋯
p16	6097.15	6097.16	**0.00**	2375.72	−∞	+∞	100.0	+∞	6058.23	6097.16	0.64	3605.00	−∞	+∞	100.00	3600.01	+∞	6097.15	6097.16	**0.00**	**9.31**	311	11	⋯
p17	5039.85	5039.85	**0.00**	454.07	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5039.85	5039.85	**0.00**	**3.61**	321	12	⋯
p18	4438.19	4438.19	**0.00**	88.20	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	4438.19	4438.19	**0.00**	**3.20**	326	11	⋯
p19	5272.31	5272.31	**0.00**	589.94	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5272.31	5272.31	**0.00**	**3.44**	311	11	⋯
p2	5747.97	5747.97	**0.00**	**1.10**	−∞	+∞	100.0	+∞	5747.97	5747.97	**0.00**	12.78	−∞	+∞	100.00	3609.68	1253210	5747.97	5747.97	**0.00**	1.45	161	9	⋯
p20	6097.15	6097.16	**0.00**	2161.52	4264.32	6097.15	30.06	3602.72	6056.81	6097.16	0.66	3605.30	−∞	+∞	100.00	3600.01	+∞	6097.15	6097.16	**0.00**	**9.06**	311	11	⋯
p21	5039.85	5039.85	**0.00**	416.81	3963.61	5039.85	21.35	3602.67	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5039.85	5039.85	**0.00**	**3.71**	321	12	⋯
p22	4438.19	4438.19	**0.00**	81.03	3711.53	4438.19	16.37	3602.89	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	4438.19	4438.19	**0.00**	**3.17**	326	11	⋯
p23	5272.31	5272.31	**0.00**	533.83	3986.92	5272.31	24.38	3602.71	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5272.31	5272.31	**0.00**	**3.43**	311	11	⋯
p24	6097.15	6097.16	**0.00**	2289.57	−∞	+∞	100.0	+∞	6057.43	6097.16	0.65	3605.05	−∞	+∞	100.00	3600.01	+∞	6097.15	6097.16	**0.00**	**10.49**	311	11	⋯
p25	9370.69	9370.70	**0.00**	2021.49	8333.33	9370.67	11.07	3606.88	9273.09	9370.90	1.04	3600.10	−∞	+∞	100.00	3600.01	+∞	9370.69	9370.70	**0.00**	**135.19**	438	17	⋯
p26	8990.46	8990.47	**0.00**	805.72	−∞	+∞	100.0	+∞	8951.17	8990.44	0.44	3593.29	−∞	+∞	100.00	3600.01	+∞	8990.46	8990.47	**0.00**	**107.00**	415	16	⋯
p27	9681.02	9681.03	**0.00**	1019.31	−∞	+∞	100.0	+∞	9585.59	9681.01	0.99	3594.12	−∞	+∞	100.00	3600.01	+∞	9681.03	9681.04	**0.00**	**190.16**	419	17	⋯
p28	10346.45	10346.45	**0.00**	794.06	−∞	+∞	100.0	+∞	10188.77	10346.99	1.53	3600.10	−∞	+∞	100.00	3600.01	+∞	10346.45	10346.46	**0.00**	**221.09**	422	16	⋯
p29	9370.69	9370.70	**0.00**	2485.51	−∞	+∞	100.0	+∞	9273.27	9370.90	1.04	3600.10	−∞	+∞	100.00	3600.01	+∞	9370.69	9370.70	**0.00**	**138.04**	438	17	⋯
p3	6428.62	6428.62	**0.00**	**1.11**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3609.90	1253100	6428.62	6428.62	**0.00**	2.15	173	10	⋯
p30	8990.46	8990.47	**0.00**	752.60	−∞	+∞	100.0	+∞	8951.55	8990.44	0.43	3593.38	−∞	+∞	100.00	3600.01	+∞	8990.46	8990.47	**0.00**	**107.17**	415	16	⋯
p31	9681.02	9681.03	**0.00**	1144.90	−∞	+∞	100.0	+∞	9585.90	9681.01	0.98	3594.25	−∞	+∞	100.00	3600.01	+∞	9681.03	9681.04	**0.00**	**174.79**	419	17	⋯
p32	10346.45	10346.45	**0.00**	798.26	−∞	+∞	100.0	+∞	10188.77	10346.99	1.53	3600.10	−∞	+∞	100.00	3600.01	+∞	10346.45	10346.46	**0.00**	**215.15**	422	16	⋯
p33	9370.69	9370.70	**0.00**	1920.25	−∞	+∞	100.0	+∞	9272.85	9370.90	1.05	3600.10	−∞	+∞	100.00	3600.01	+∞	9370.69	9370.70	**0.00**	**139.34**	438	17	⋯
p34	8990.46	8990.47	**0.00**	923.26	−∞	+∞	100.0	+∞	8951.00	8990.44	0.44	3593.23	−∞	+∞	100.00	3600.01	+∞	8990.46	8990.47	**0.00**	**116.24**	415	16	⋯
p35	9681.02	9681.03	**0.00**	947.21	−∞	+∞	100.0	+∞	9585.90	9681.01	0.98	3594.25	−∞	+∞	100.00	3600.01	+∞	9681.03	9681.04	**0.00**	**185.60**	419	17	⋯
p36	10346.45	10346.45	**0.00**	755.42	−∞	+∞	100.0	+∞	10190.44	10346.99	1.51	3600.10	−∞	+∞	100.00	3600.01	+∞	10346.45	10346.46	**0.00**	**214.18**	422	16	⋯
p37	9370.69	9370.70	**0.00**	2101.38	−∞	+∞	100.0	+∞	9273.09	9370.90	1.04	3600.10	−∞	+∞	100.00	3600.01	+∞	9370.69	9370.70	**0.00**	**139.28**	438	17	⋯
p38	8990.46	8990.47	**0.00**	892.06	8214.01	8990.44	8.64	3603.20	8951.00	8990.44	0.44	3593.34	−∞	+∞	100.00	3600.01	+∞	8990.46	8990.47	**0.00**	**114.04**	415	16	⋯
p39	9681.02	9681.03	**0.00**	1064.29	8414.14	9681.01	13.09	3603.99	9585.06	9681.01	0.99	3594.16	−∞	+∞	100.00	3600.01	+∞	9681.03	9681.04	**0.00**	**184.07**	419	17	⋯
p4	7104.98	7104.98	**0.00**	**1.55**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3609.94	1253570	7104.97	7104.98	**0.00**	2.74	175	11	⋯
p40	10346.45	10346.45	**0.00**	805.91	8598.75	10346.43	16.89	3602.97	10190.44	10346.99	1.51	3600.10	−∞	+∞	100.00	3600.01	+∞	10346.45	10346.46	**0.00**	**207.00**	422	16	⋯
p41	4388.64	4388.64	**0.00**	4.28	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3609.29	1328755	4388.64	4388.64	**0.00**	**2.98**	160	11	⋯
p42	3599.00	3673.91	2.04	3600.01	−∞	+∞	100.0	+∞	3527.47	3674.73	4.01	3600.10	−∞	+∞	100.00	3600.01	+∞	3673.91	3673.91	**0.00**	**36.60**	340	15	⋯
p43	3090.90	3291.61	6.10	3600.01	−∞	+∞	100.0	+∞	3000.72	3291.61	8.84	3600.10	−∞	+∞	100.00	3605.32	694839	3290.42	3290.42	**0.00**	**107.26**	461	17	⋯
p44	4381.18	4381.18	**0.00**	1.61	−∞	+∞	100.0	+∞	4381.18	4381.18	**0.00**	2732.35	−∞	+∞	100.00	3608.30	1253039	4381.18	4381.18	**0.00**	**1.50**	163	9	⋯
p45	3570.90	3607.74	1.02	3600.01	−∞	+∞	100.0	+∞	3511.31	3607.74	2.67	3600.10	−∞	+∞	100.00	3600.01	+∞	3607.74	3607.74	**0.00**	**6.31**	318	11	⋯
p46	2967.64	3166.33	6.27	3600.01	−∞	+∞	100.0	+∞	2859.08	3167.22	9.73	3600.10	−∞	+∞	100.00	3600.01	+∞	3166.33	3166.33	**0.00**	**68.07**	478	17	⋯
p47	3162.39	3162.39	**0.00**	**1.17**	−∞	+∞	100.0	+∞	3162.39	3162.39	**0.00**	30.93	−∞	+∞	100.00	3608.50	1256852	3162.39	3162.39	**0.00**	1.23	159	10	⋯
p48	2749.80	2805.11	1.97	3600.01	−∞	+∞	100.0	+∞	2668.81	2805.11	4.86	3600.10	−∞	+∞	100.00	3600.01	+∞	2805.11	2805.11	**0.00**	**18.96**	355	19	⋯
p49	2395.91	2629.83	8.90	3600.01	1345.32	2632.19	48.89	3625.41	2308.78	2632.80	12.31	3600.10	−∞	+∞	100.00	3600.01	+∞	2629.74	2629.74	**0.00**	**70.57**	494	17	⋯
p5	6207.39	6207.39	**0.00**	**1.28**	5557.64	6207.39	10.47	3602.96	6207.39	6207.39	**0.00**	3661.48	−∞	+∞	100.00	3610.06	1253119	6207.39	6207.39	**0.00**	2.27	152	12	⋯
p50	6359.91	6359.91	**0.00**	**2.32**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3608.85	1256867	6359.90	6359.91	**0.00**	7.18	147	11	⋯
p51	5503.16	5503.16	**0.00**	340.96	−∞	+∞	100.0	+∞	5465.53	5503.34	0.69	3605.49	−∞	+∞	100.00	3600.01	+∞	5503.16	5503.16	**0.00**	**63.13**	312	18	⋯
p52	6828.84	6828.84	**0.00**	**0.87**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3608.52	1252969	6828.84	6828.84	**0.00**	7.97	144	11	⋯
p53	6272.53	6272.53	**0.00**	**22.00**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	6272.53	6272.53	**0.00**	47.95	288	17	⋯
p54	6662.44	6662.44	**0.00**	**2.91**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3608.48	1253004	6662.43	6662.44	**0.00**	9.15	161	10	⋯

Table 82 **Detailed results for problem NLUFLP-W-holmberg, cost functions** f_7

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
p1	4777.53	4777.53	**0.00**	**0.01**	4777.52	4777.52	**0.00**	17.68	4777.52	4777.52	**0.00**	0.71	4777.53	4777.53	**0.00**	16.17	14970	4777.52	4777.53	**0.00**	0.85	170	7	⋯
p10	4760.70	4760.70	**0.00**	**0.01**	4760.70	4760.70	**0.00**	9.43	4760.70	4760.70	**0.00**	0.63	4760.70	4760.70	**0.00**	15.85	14970	4760.70	4760.70	**0.00**	0.88	166	7	⋯
p11	4783.84	4783.84	**0.00**	**0.01**	4783.84	4783.84	**0.00**	4.66	4783.84	4783.84	**0.00**	0.68	4783.84	4783.84	**0.00**	15.63	14970	4783.84	4783.84	**0.00**	0.95	166	7	⋯
p12	4806.98	4806.98	**0.00**	**0.01**	4806.98	4806.98	**0.00**	4.12	4806.98	4806.98	**0.00**	0.75	4806.98	4806.98	**0.00**	15.39	14970	4806.97	4806.98	**0.00**	0.93	166	7	⋯
p13	3210.98	3210.98	**0.00**	**0.02**	3210.98	3210.98	**0.00**	8.89	3210.98	3210.98	**0.00**	2.08	3210.98	3210.98	**0.00**	36.41	29940	3210.98	3210.98	**0.00**	1.29	330	7	⋯
p14	3200.41	3200.41	**0.00**	**0.02**	3200.41	3200.41	**0.00**	7.29	3200.41	3200.41	**0.00**	1.89	3200.41	3200.41	**0.00**	31.84	29940	3200.41	3200.41	**0.00**	1.26	314	6	⋯
p15	3214.02	3214.02	**0.00**	**0.02**	3214.02	3214.02	**0.00**	7.36	3214.02	3214.02	**0.00**	1.95	3214.02	3214.02	**0.00**	31.84	29940	3214.02	3214.02	**0.00**	1.49	352	7	⋯
p16	3227.63	3227.63	**0.00**	**0.02**	3227.62	3227.62	**0.00**	7.57	3227.63	3227.63	**0.00**	2.11	3227.63	3227.63	**0.00**	31.71	29940	3227.62	3227.63	**0.00**	1.55	352	7	⋯
p17	3210.98	3210.98	**0.00**	**0.02**	3210.98	3210.98	**0.00**	7.27	3210.98	3210.98	**0.00**	2.07	3210.98	3210.98	**0.00**	32.02	29940	3210.98	3210.98	**0.00**	1.28	330	7	⋯
p18	3200.41	3200.41	**0.00**	**0.02**	3200.41	3200.41	**0.00**	7.26	3200.41	3200.41	**0.00**	1.92	3200.41	3200.41	**0.00**	32.34	29940	3200.41	3200.41	**0.00**	1.40	314	6	⋯
p19	3214.02	3214.02	**0.00**	**0.02**	3214.02	3214.02	**0.00**	7.26	3214.02	3214.02	**0.00**	1.98	3214.02	3214.02	**0.00**	31.87	29940	3214.02	3214.02	**0.00**	1.45	352	7	⋯
p2	4760.70	4760.70	**0.00**	**0.01**	4760.70	4760.70	**0.00**	4.59	4760.70	4760.70	**0.00**	0.65	4760.70	4760.70	**0.00**	14.13	14970	4760.70	4760.70	**0.00**	0.85	166	7	⋯
p20	3227.63	3227.63	**0.00**	**0.02**	3227.62	3227.62	**0.00**	7.65	3227.63	3227.63	**0.00**	2.13	3227.63	3227.63	**0.00**	31.25	29940	3227.62	3227.63	**0.00**	1.41	352	7	⋯
p21	3210.98	3210.98	**0.00**	**0.02**	3210.98	3210.98	**0.00**	7.23	3210.98	3210.98	**0.00**	2.08	3210.98	3210.98	**0.00**	32.13	29940	3210.98	3210.98	**0.00**	1.37	330	7	⋯
p22	3200.41	3200.41	**0.00**	**0.02**	3200.41	3200.41	**0.00**	7.05	3200.41	3200.41	**0.00**	1.89	3200.41	3200.41	**0.00**	31.58	29940	3200.41	3200.41	**0.00**	1.16	314	6	⋯
p23	3214.02	3214.02	**0.00**	**0.02**	3214.02	3214.02	**0.00**	7.22	3214.02	3214.02	**0.00**	1.95	3214.02	3214.02	**0.00**	31.92	29940	3214.02	3214.02	**0.00**	1.70	352	7	⋯
p24	3227.63	3227.63	**0.00**	**0.02**	3227.62	3227.62	**0.00**	20.02	3227.63	3227.63	**0.00**	2.20	3227.63	3227.63	**0.00**	31.71	29940	3227.62	3227.63	**0.00**	1.43	352	7	⋯
p25	7915.35	7915.35	**0.00**	**0.04**	7915.32	7915.32	**0.00**	131.50	7915.35	7915.35	**0.00**	91.14	7915.35	7915.35	**0.00**	47.16	44910	7915.34	7915.35	**0.00**	3.33	427	10	⋯
p26	7908.22	7908.22	**0.00**	**0.04**	7908.20	7908.20	**0.00**	126.64	7908.22	7908.22	**0.00**	119.20	7908.22	7908.22	**0.00**	47.15	44910	7908.22	7908.23	**0.00**	3.59	425	11	⋯
p27	7921.04	7921.04	**0.00**	**0.04**	7921.01	7921.01	**0.00**	116.58	7921.04	7921.04	**0.00**	87.62	7921.04	7921.04	**0.00**	46.08	44910	7921.04	7921.04	**0.00**	4.74	452	13	⋯
p28	7933.66	7933.66	**0.00**	**0.04**	7933.63	7933.63	**0.00**	128.27	7933.66	7933.66	**0.00**	90.69	7933.66	7933.66	**0.00**	45.44	44910	7933.65	7933.66	**0.00**	4.39	438	13	⋯
p29	7915.35	7915.35	**0.00**	**0.04**	7915.32	7915.32	**0.00**	127.05	7915.35	7915.35	**0.00**	90.40	7915.35	7915.35	**0.00**	46.30	44910	7915.34	7915.35	**0.00**	3.35	427	10	⋯
p3	4783.84	4783.84	**0.00**	**0.01**	4783.84	4783.84	**0.00**	4.51	4783.84	4783.84	**0.00**	0.67	4783.84	4783.84	**0.00**	14.76	14970	4783.84	4783.84	**0.00**	0.85	166	7	⋯
p30	7908.22	7908.22	**0.00**	**0.04**	7908.20	7908.20	**0.00**	111.80	7908.22	7908.22	**0.00**	118.29	7908.22	7908.22	**0.00**	48.30	44910	7908.22	7908.23	**0.00**	3.54	425	11	⋯
p31	7921.04	7921.04	**0.00**	**0.04**	7921.01	7921.01	**0.00**	123.14	7921.04	7921.04	**0.00**	86.64	7921.04	7921.04	**0.00**	48.09	44910	7921.04	7921.04	**0.00**	4.84	452	13	⋯
p32	7933.66	7933.66	**0.00**	**0.04**	7933.63	7933.63	**0.00**	136.77	7933.66	7933.66	**0.00**	90.70	7933.66	7933.66	**0.00**	47.92	44910	7933.65	7933.66	**0.00**	4.49	438	13	⋯
p33	7915.35	7915.35	**0.00**	**0.04**	7915.32	7915.32	**0.00**	117.23	7915.35	7915.35	**0.00**	90.92	7915.35	7915.35	**0.00**	44.46	44910	7915.34	7915.35	**0.00**	3.49	427	10	⋯
p34	7908.22	7908.22	**0.00**	**0.04**	7908.20	7908.20	**0.00**	134.00	7908.22	7908.22	**0.00**	119.81	7908.22	7908.22	**0.00**	47.38	44910	7908.22	7908.23	**0.00**	3.73	425	11	⋯
p35	7921.04	7921.04	**0.00**	**0.04**	7921.01	7921.01	**0.00**	119.41	7921.04	7921.04	**0.00**	86.37	7921.04	7921.04	**0.00**	47.61	44910	7921.04	7921.04	**0.00**	4.84	452	13	⋯
p36	7933.66	7933.66	**0.00**	**0.04**	7933.63	7933.63	**0.00**	141.30	7933.66	7933.66	**0.00**	89.78	7933.66	7933.66	**0.00**	48.17	44910	7933.65	7933.66	**0.00**	4.41	438	13	⋯
p37	7915.35	7915.35	**0.00**	**0.04**	7915.32	7915.32	**0.00**	113.42	7915.35	7915.35	**0.00**	90.44	7915.35	7915.35	**0.00**	40.29	44910	7915.34	7915.35	**0.00**	3.35	427	10	⋯
p38	7908.22	7908.22	**0.00**	**0.04**	7908.20	7908.20	**0.00**	105.41	7908.22	7908.22	**0.00**	119.75	7908.22	7908.22	**0.00**	47.96	44910	7908.22	7908.23	**0.00**	3.58	425	11	⋯
p39	7921.04	7921.04	**0.00**	**0.04**	7921.01	7921.01	**0.00**	104.44	7921.04	7921.04	**0.00**	87.25	7921.04	7921.04	**0.00**	48.82	44910	7921.04	7921.04	**0.00**	4.74	452	13	⋯
p4	4806.98	4806.98	**0.00**	**0.01**	4806.98	4806.98	**0.00**	4.27	4806.98	4806.98	**0.00**	0.74	4806.98	4806.98	**0.00**	14.55	14970	4806.97	4806.98	**0.00**	0.93	166	7	⋯
p40	7933.66	7933.66	**0.00**	**0.04**	7933.63	7933.63	**0.00**	127.82	7933.66	7933.66	**0.00**	90.42	7933.66	7933.66	**0.00**	40.61	44910	7933.65	7933.66	**0.00**	4.70	438	13	⋯
p41	3113.86	3113.86	**0.00**	**0.02**	3113.86	3113.86	**0.00**	5.76	3113.86	3113.86	**0.00**	1.54	3113.86	3113.86	**0.00**	15.87	14970	3113.86	3113.86	**0.00**	0.87	156	6	⋯
p42	2080.18	2080.18	**0.00**	**0.02**	2080.18	2080.18	**0.00**	27.95	2080.18	2080.18	**0.00**	4.12	2080.18	2080.18	**0.00**	32.33	29940	2080.18	2080.18	**0.00**	13.83	363	6	⋯
p43	1601.91	1601.91	**0.00**	**0.02**	1601.91	1601.91	**0.00**	25.23	1601.90	1601.90	**0.00**	5.90	1601.91	1601.91	**0.00**	54.45	44910	1601.91	1601.91	**0.00**	5.09	496	7	⋯
p44	3058.42	3058.42	**0.00**	**0.02**	3058.41	3058.41	**0.00**	10.93	3058.41	3058.41	**0.00**	1.57	3058.42	3058.42	**0.00**	15.55	14970	3058.42	3058.42	**0.00**	5.57	209	5	⋯
p45	1973.15	1973.15	**0.00**	**0.02**	1973.15	1973.15	**0.00**	15.92	1973.15	1973.15	**0.00**	4.30	1973.15	1973.15	**0.00**	32.03	29940	1973.15	1973.15	**0.00**	1.68	379	6	⋯
p46	1423.37	1423.37	**0.00**	**0.02**	1423.36	1423.37	**0.00**	24.43	1423.37	1423.37	**0.00**	7.22	1423.37	1423.37	**0.00**	47.95	44910	1423.37	1423.37	**0.00**	7.01	497	12	⋯
p47	1834.41	1834.41	**0.00**	**0.02**	1834.41	1834.41	**0.00**	11.44	1834.40	1834.40	**0.00**	1.80	1834.41	1834.41	**0.00**	13.67	14970	1834.40	1834.41	**0.00**	0.93	179	3	⋯
p48	1163.20	1163.20	**0.00**	**0.02**	1163.19	1163.19	**0.00**	21.72	1163.19	1163.19	**0.00**	4.23	1163.20	1163.20	**0.00**	32.12	29940	1163.19	1163.20	**0.00**	2.13	352	10	⋯
p49	863.19	863.19	**0.00**	**0.02**	863.18	863.18	**0.00**	23.07	863.19	863.19	**0.00**	9.82	863.19	863.19	**0.00**	54.43	44910	863.19	863.19	**0.00**	6.32	477	13	⋯
p5	4777.53	4777.53	**0.00**	**0.01**	4777.52	4777.52	**0.00**	5.63	4777.52	4777.52	**0.00**	0.65	4777.53	4777.53	**0.00**	13.97	14970	4777.52	4777.53	**0.00**	0.92	170	7	⋯
p50	5283.56	5283.56	**0.00**	**0.02**	5283.56	5283.56	**0.00**	7.38	5283.56	5283.56	**0.00**	1.80	5283.56	5283.56	**0.00**	15.72	14970	5283.56	5283.56	**0.00**	0.81	146	5	⋯
p51	4197.23	4197.23	**0.00**	**0.02**	4197.22	4197.22	**0.00**	25.81	4197.22	4197.22	**0.00**	6.72	4197.23	4197.23	**0.00**	32.36	29940	4197.22	4197.23	**0.00**	1.96	310	6	⋯
p52	5679.15	5679.15	**0.00**	**0.02**	5679.15	5679.15	**0.00**	7.46	5679.15	5679.15	**0.00**	1.92	5679.15	5679.15	**0.00**	15.67	14970	5679.14	5679.15	**0.00**	5.27	142	8	⋯
p53	4962.28	4962.28	**0.00**	**0.02**	4962.27	4962.27	**0.00**	35.50	4962.28	4962.28	**0.00**	7.33	4962.28	4962.28	**0.00**	28.38	29940	4962.27	4962.28	**0.00**	1.03	262	5	⋯
p54	5416.17	5416.17	**0.00**	**0.01**	5416.17	5416.17	**0.00**	7.15	5416.17	5416.17	**0.00**	1.87	5416.17	5416.17	**0.00**	14.12	14970	5416.17	5416.17	**0.00**	0.87	157	5	⋯

Table 83 **Detailed results for problem NLUFLP-W-holmberg, cost functions f_8**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
p1	6062.45	6062.45	**0.00**	**1.42**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3607.68	1066144	6062.45	6062.45	**0.00**	1.61	176	12	⋯
p10	5648.89	5648.89	**0.00**	**1.13**	−∞	+∞	100.0	+∞	5648.88	5648.89	**0.00**	3646.54	−∞	+∞	100.00	3608.96	1066150	5648.88	5648.89	**0.00**	1.50	179	10	⋯
p11	6263.48	6263.48	**0.00**	2.85	5754.62	6263.48	8.12	3603.09	6263.45	6263.48	**0.00**	3689.72	−∞	+∞	100.00	3608.84	1066130	6263.47	6263.48	**0.00**	**1.76**	183	11	⋯
p12	6878.06	6878.06	**0.00**	4.81	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3608.64	1066150	6878.07	6878.07	**0.00**	**2.14**	193	11	⋯
p13	4908.10	5050.88	2.83	3600.01	−∞	+∞	100.0	+∞	4831.57	5051.19	4.35	3600.10	−∞	+∞	100.00	3600.01	+∞	5050.88	5050.88	**0.00**	**4.31**	374	17	⋯
p14	4413.75	4460.78	1.05	3600.01	−∞	+∞	100.0	+∞	4366.89	4460.78	2.10	3600.10	−∞	+∞	100.00	3600.01	+∞	4460.78	4460.78	**0.00**	**2.81**	369	10	⋯
p15	5093.39	5296.67	3.84	3600.01	−∞	+∞	100.0	+∞	4973.16	5297.20	6.12	3600.10	−∞	+∞	100.00	3600.01	+∞	5296.68	5296.68	**0.00**	**5.19**	367	17	⋯
p16	5776.89	6090.44	5.15	3600.01	−∞	+∞	100.0	+∞	5608.08	6115.16	8.29	3600.10	−∞	+∞	100.00	3600.01	+∞	6090.44	6090.44	**0.00**	**11.84**	395	18	⋯
p17	4911.82	5050.88	2.75	3600.01	−∞	+∞	100.0	+∞	4831.57	5051.19	4.35	3600.10	−∞	+∞	100.00	3600.01	+∞	5050.88	5050.88	**0.00**	**4.17**	374	17	⋯
p18	4407.56	4460.78	1.19	3600.01	−∞	+∞	100.0	+∞	4365.92	4460.78	2.13	3600.10	−∞	+∞	100.00	3600.01	+∞	4460.78	4460.78	**0.00**	**2.82**	369	10	⋯
p19	5088.01	5296.67	3.94	3600.01	−∞	+∞	100.0	+∞	4973.10	5297.20	6.12	3600.10	−∞	+∞	100.00	3600.01	+∞	5296.68	5296.68	**0.00**	**5.65**	367	17	⋯
p2	5648.89	5648.89	**0.00**	**1.07**	−∞	+∞	100.0	+∞	5648.89	5648.89	**0.00**	3645.97	−∞	+∞	100.00	3608.61	1066150	5648.88	5648.89	**0.00**	1.44	179	10	⋯
p20	5767.62	6090.44	5.30	3600.01	4341.15	6113.96	29.00	3606.09	5605.30	6115.16	8.34	3600.10	−∞	+∞	100.00	3600.01	+∞	6090.44	6090.44	**0.00**	**10.72**	395	18	⋯
p21	4914.85	5050.88	2.69	3600.01	3993.93	5050.88	20.93	3602.76	4831.57	5051.19	4.35	3600.10	−∞	+∞	100.00	3600.01	+∞	5050.88	5050.88	**0.00**	**4.48**	374	17	⋯
p22	4414.33	4460.78	1.04	3600.01	3767.51	4460.78	15.54	3602.89	4366.99	4460.78	2.10	3600.10	−∞	+∞	100.00	3600.01	+∞	4460.78	4460.78	**0.00**	**2.61**	369	10	⋯
p23	5090.81	5296.67	3.89	3600.01	4038.46	5296.69	23.76	3602.46	4976.75	5297.20	6.05	3600.10	−∞	+∞	100.00	3600.01	+∞	5296.68	5296.68	**0.00**	**5.93**	367	17	⋯
p24	5765.93	6090.44	5.33	3600.01	−∞	+∞	100.0	+∞	5607.73	6115.16	8.30	3600.10	−∞	+∞	100.00	3600.01	+∞	6090.44	6090.44	**0.00**	**10.79**	395	18	⋯
p25	9255.09	9255.09	**0.00**	1623.14	8356.54	9263.81	9.79	3611.34	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9255.09	9255.10	**0.00**	**122.59**	641	18	⋯
p26	8906.22	8906.23	**0.00**	745.05	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8906.22	8906.23	**0.00**	**114.24**	650	16	⋯
p27	9530.24	9530.24	**0.00**	836.63	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9530.24	9530.24	**0.00**	**123.16**	654	17	⋯
p28	10117.90	10117.91	**0.00**	540.33	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	10117.90	10117.91	**0.00**	**168.77**	635	19	⋯
p29	9255.09	9255.09	**0.00**	1664.47	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9255.09	9255.10	**0.00**	**127.16**	641	18	⋯
p3	6263.48	6263.48	**0.00**	2.69	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3608.69	1066130	6263.47	6263.48	**0.00**	**1.77**	183	11	⋯
p30	8906.22	8906.23	**0.00**	813.47	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8906.22	8906.23	**0.00**	**112.55**	650	16	⋯
p31	9530.24	9530.24	**0.00**	817.24	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9530.24	9530.24	**0.00**	**125.64**	654	17	⋯
p32	10117.90	10117.91	**0.00**	496.79	−∞	+∞	100.0	+∞	8589.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	10117.90	10117.91	**0.00**	**169.73**	635	19	⋯
p33	9255.09	9255.09	**0.00**	1577.81	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9255.09	9255.10	**0.00**	**117.52**	641	18	⋯
p34	8906.22	8906.23	**0.00**	760.26	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8906.22	8906.23	**0.00**	**115.55**	650	16	⋯
p35	9530.24	9530.24	**0.00**	920.95	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9530.24	9530.24	**0.00**	**125.77**	654	17	⋯
p36	10117.90	10117.91	**0.00**	558.00	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	10117.90	10117.91	**0.00**	**187.25**	635	19	⋯
p37	9255.09	9255.09	**0.00**	1422.37	8356.54	9263.81	9.79	3607.79	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9255.09	9255.10	**0.00**	**120.96**	641	18	⋯
p38	8906.22	8906.23	**0.00**	687.23	8261.48	8906.92	7.25	3632.75	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8906.22	8906.23	**0.00**	**106.87**	650	16	⋯
p39	9530.24	9530.24	**0.00**	843.91	8432.62	9530.50	11.52	3636.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9530.24	9530.24	**0.00**	**127.57**	654	17	⋯
p4	6878.06	6878.06	**0.00**	4.50	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3608.54	1066150	6878.07	6878.07	**0.00**	**2.06**	193	11	⋯
p40	10117.90	10117.91	**0.00**	564.18	8659.07	10120.31	14.44	3636.10	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	10117.90	10117.91	**0.00**	**169.65**	635	19	⋯
p41	4265.39	4265.39	**0.00**	6.27	3653.91	4265.39	14.34	3602.84	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3607.53	1035466	4265.39	4265.39	**0.00**	**3.13**	180	11	⋯
p42	3441.09	3683.73	6.59	3600.01	−∞	+∞	100.0	+∞	3121.75	3692.81	15.46	3600.10	−∞	+∞	100.00	3600.01	+∞	3680.70	3680.71	**0.00**	**46.02**	404	18	⋯
p43	2844.79	3315.25	14.19	3600.01	−∞	+∞	100.0	+∞	1985.61	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	3315.25	3315.25	**0.00**	**115.02**	590	20	⋯
p44	4234.71	4234.71	**0.00**	2.24	−∞	+∞	100.0	+∞	4234.70	4234.71	**0.00**	3635.21	−∞	+∞	100.00	3607.49	1067253	4234.71	4234.71	**0.00**	**1.47**	176	10	⋯
p45	3410.92	3634.05	6.14	3600.01	−∞	+∞	100.0	+∞	3366.25	3635.46	7.41	3600.10	−∞	+∞	100.00	3600.01	+∞	3634.05	3634.05	**0.00**	**15.01**	410	18	⋯
p46	2733.03	3207.42	14.79	3600.01	−∞	+∞	100.0	+∞	2416.65	3258.78	25.84	3600.10	−∞	+∞	100.00	3600.01	+∞	3193.02	3193.02	**0.00**	**117.85**	644	21	⋯
p47	3012.98	3012.98	**0.00**	**1.14**	−∞	+∞	100.0	+∞	3012.98	3012.98	**0.00**	63.60	−∞	+∞	100.00	3607.47	1065152	3012.98	3012.98	**0.00**	1.58	203	10	⋯
p48	2592.75	2763.91	6.19	3600.01	−∞	+∞	100.0	+∞	2295.89	2817.21	18.51	3600.10	−∞	+∞	100.00	3600.01	+∞	2763.91	2763.92	**0.00**	**12.43**	404	19	⋯
p49	2195.56	2602.41	15.63	3600.01	1364.92	2715.26	49.73	3608.95	1923.31	2660.38	27.71	3600.10	−∞	+∞	100.00	3600.01	+∞	2593.11	2593.11	**0.00**	**59.17**	611	17	⋯
p5	6062.45	6062.45	**0.00**	**1.57**	5649.74	6062.45	6.81	3602.91	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3608.57	1066144	6062.45	6062.45	**0.00**	1.65	176	12	⋯
p50	6251.47	6251.47	**0.00**	**5.87**	−∞	+∞	100.0	+∞	6251.43	6251.50	0.00	3618.79	−∞	+∞	100.00	3609.17	1066483	6251.46	6251.47	**0.00**	7.13	196	16	⋯
p51	5418.15	5418.15	**0.00**	467.14	−∞	+∞	100.0	+∞	5220.02	5429.78	3.86	3600.10	−∞	+∞	100.00	3600.01	+∞	5418.15	5418.15	**0.00**	**26.05**	410	17	⋯
p52	6675.49	6675.49	**0.00**	**1.23**	−∞	+∞	100.0	+∞	6675.47	6675.50	**0.00**	3637.37	−∞	+∞	100.00	3600.01	+∞	6675.49	6675.50	**0.00**	6.44	196	17	⋯
p53	6154.66	6154.66	**0.00**	73.50	−∞	+∞	100.0	+∞	6045.74	6163.64	1.91	3600.10	−∞	+∞	100.00	3600.01	+∞	6154.66	6154.66	**0.00**	**23.23**	427	17	⋯
p54	6516.20	6516.20	**0.00**	**3.73**	−∞	+∞	100.0	+∞	6516.20	6516.21	**0.00**	3661.71	−∞	+∞	100.00	3600.01	+∞	6516.20	6516.21	**0.00**	7.15	189	18	⋯

Table 84 **Detailed results for problem NLUFLP-W-holmberg, cost functions f_9**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
p1	4747.63	4747.63	**0.00**	**0.07**	4747.62	4747.63	**0.00**	7.33	4747.63	4747.63	**0.00**	0.93	4747.63	4747.63	**0.00**	52.42	21350	4747.63	4747.63	**0.00**	6.54	138	4	⋯
p10	4740.53	4740.53	**0.00**	**0.07**	4740.52	4740.53	**0.00**	4.15	4740.53	4740.53	**0.00**	0.85	4740.53	4740.53	**0.00**	61.03	21350	4740.53	4740.53	**0.00**	6.94	138	4	⋯
p11	4750.21	4750.21	**0.00**	**0.08**	4750.21	4750.21	**0.00**	8.90	4750.21	4750.21	**0.00**	0.87	4750.21	4750.21	**0.00**	59.67	21350	4750.21	4750.21	**0.00**	6.89	138	4	⋯
p12	4759.90	4759.90	**0.00**	**0.10**	4759.89	4759.90	**0.00**	21.77	4759.90	4759.90	**0.00**	0.88	4759.90	4759.90	**0.00**	60.27	21350	4759.89	4759.90	**0.00**	6.93	138	4	⋯
p13	3198.90	3198.90	**0.00**	**0.15**	3198.89	3198.89	**0.00**	7.02	3198.90	3198.90	**0.00**	2.67	3198.90	3198.90	**0.00**	130.48	42700	3198.90	3198.90	**0.00**	12.69	366	5	⋯
p14	3192.66	3192.66	**0.00**	**0.13**	3192.66	3192.66	**0.00**	5.95	3192.66	3192.66	**0.00**	2.60	3192.66	3192.66	**0.00**	130.56	42700	3192.66	3192.66	**0.00**	12.25	328	4	⋯
p15	3201.10	3201.10	**0.00**	**0.14**	3201.09	3201.09	**0.00**	7.45	3201.10	3201.10	**0.00**	2.65	3201.10	3201.10	**0.00**	130.83	42700	3201.09	3201.10	**0.00**	12.81	383	5	⋯
p16	3209.53	3209.54	**0.00**	**0.15**	3209.53	3209.53	**0.00**	10.76	3209.53	3209.53	**0.00**	2.92	3209.54	3209.54	**0.00**	131.60	42700	3209.53	3209.54	**0.00**	12.69	383	5	⋯
p17	3198.90	3198.90	**0.00**	**0.14**	3198.89	3198.89	**0.00**	6.61	3198.90	3198.90	**0.00**	2.66	3198.90	3198.90	**0.00**	131.96	42700	3198.90	3198.90	**0.00**	12.19	366	5	⋯
p18	3192.66	3192.66	**0.00**	**0.14**	3192.66	3192.66	**0.00**	5.39	3192.66	3192.66	**0.00**	2.64	3192.66	3192.66	**0.00**	133.09	42700	3192.66	3192.66	**0.00**	11.97	328	4	⋯
p19	3201.10	3201.10	**0.00**	**0.15**	3201.09	3201.09	**0.00**	11.31	3201.10	3201.10	**0.00**	2.68	3201.10	3201.10	**0.00**	130.47	42700	3201.09	3201.10	**0.00**	13.24	383	5	⋯
p2	4740.53	4740.53	**0.00**	**0.07**	4740.52	4740.53	**0.00**	4.19	4740.53	4740.53	**0.00**	0.85	4740.53	4740.53	**0.00**	56.08	21350	4740.53	4740.53	**0.00**	7.15	138	4	⋯
p20	3209.53	3209.54	**0.00**	**0.15**	3209.53	3209.53	**0.00**	16.03	3209.53	3209.53	**0.00**	2.96	3209.54	3209.54	**0.00**	129.52	42700	3209.53	3209.54	**0.00**	13.55	383	5	⋯
p21	3198.90	3198.90	**0.00**	**0.14**	3198.89	3198.89	**0.00**	6.79	3198.90	3198.90	**0.00**	2.66	3198.90	3198.90	**0.00**	131.95	42700	3198.90	3198.90	**0.00**	13.77	366	5	⋯
p22	3192.66	3192.66	**0.00**	**0.14**	3192.66	3192.66	**0.00**	10.39	3192.66	3192.66	**0.00**	2.59	3192.66	3192.66	**0.00**	129.36	42700	3192.66	3192.66	**0.00**	12.34	328	4	⋯
p23	3201.10	3201.10	**0.00**	**0.14**	3201.09	3201.09	**0.00**	7.08	3201.10	3201.10	**0.00**	2.64	3201.10	3201.10	**0.00**	130.34	42700	3201.09	3201.10	**0.00**	12.79	383	5	⋯
p24	3209.53	3209.54	**0.00**	**0.15**	3209.53	3209.53	**0.00**	30.97	3209.53	3209.53	**0.00**	3.09	3209.54	3209.54	**0.00**	130.01	42700	3209.53	3209.54	**0.00**	12.58	383	5	⋯
p25	7907.35	7907.36	**0.00**	**8.18**	$-\infty$	$+\infty$	100.0	$+\infty$	7907.36	7907.36	**0.00**	388.12	7907.36	7908.33	0.01	197.83	64050	7907.36	7907.36	**0.00**	18.60	398	7	⋯
p26	7902.32	7902.33	**0.00**	**6.47**	$-\infty$	$+\infty$	100.0	$+\infty$	7902.33	7902.33	**0.00**	489.11	7902.33	7902.33	**0.00**	196.13	64050	7902.32	7902.33	**0.00**	18.49	388	6	⋯
p27	7911.21	7911.21	**0.00**	**8.12**	$-\infty$	$+\infty$	100.0	$+\infty$	7911.21	7911.21	**0.00**	503.48	7911.21	7911.21	**0.00**	195.37	64050	7911.21	7911.21	**0.00**	19.48	398	7	⋯
p28	7920.09	7920.10	**0.00**	50.79	$-\infty$	$+\infty$	100.0	$+\infty$	7920.10	7920.10	**0.00**	463.04	7920.10	7920.10	**0.00**	192.40	64050	7920.10	7920.10	**0.00**	**17.80**	402	7	⋯
p29	7907.35	7907.36	**0.00**	**8.07**	$-\infty$	$+\infty$	100.0	$+\infty$	7907.36	7907.36	**0.00**	388.90	7907.36	7908.33	0.01	191.40	64050	7907.36	7907.36	**0.00**	17.78	398	7	⋯
p3	4750.21	4750.21	**0.00**	**0.08**	4750.21	4750.21	**0.00**	8.99	4750.21	4750.21	**0.00**	0.85	4750.21	4750.21	**0.00**	57.39	21350	4750.21	4750.21	**0.00**	6.94	138	4	⋯
p30	7902.32	7902.33	**0.00**	**7.96**	$-\infty$	$+\infty$	100.0	$+\infty$	7902.33	7902.33	**0.00**	487.64	7902.33	7902.33	**0.00**	224.50	64050	7902.32	7902.33	**0.00**	18.19	391	6	⋯
p31	7911.21	7911.21	**0.00**	**7.80**	$-\infty$	$+\infty$	100.0	$+\infty$	7911.21	7911.21	**0.00**	495.28	7911.21	7911.21	**0.00**	198.40	64050	7911.21	7911.21	**0.00**	19.07	402	7	⋯
p32	7920.09	7920.10	**0.00**	52.53	$-\infty$	$+\infty$	100.0	$+\infty$	7920.10	7920.10	**0.00**	464.83	7920.10	7920.10	**0.00**	199.81	64050	7920.10	7920.10	**0.00**	**18.17**	402	7	⋯
p33	7907.35	7907.36	**0.00**	**8.40**	$-\infty$	$+\infty$	100.0	$+\infty$	7907.36	7907.36	**0.00**	390.24	7907.36	7908.33	0.01	178.52	64050	7907.36	7907.36	**0.00**	19.03	398	7	⋯
p34	7902.32	7902.33	**0.00**	**7.51**	$-\infty$	$+\infty$	100.0	$+\infty$	7902.33	7902.33	**0.00**	496.44	7902.33	7902.33	**0.00**	167.97	64050	7902.32	7902.33	**0.00**	17.44	393	6	⋯
p35	7911.21	7911.21	**0.00**	**7.32**	$-\infty$	$+\infty$	100.0	$+\infty$	7911.21	7911.21	**0.00**	497.44	7911.21	7911.21	**0.00**	198.04	64050	7911.21	7911.21	**0.00**	17.64	398	7	⋯
p36	7920.09	7920.10	**0.00**	54.84	$-\infty$	$+\infty$	100.0	$+\infty$	7920.10	7920.10	**0.00**	458.19	7920.10	7920.10	**0.00**	185.20	64050	7920.10	7920.10	**0.00**	**18.27**	401	7	⋯
p37	7907.35	7907.36	**0.00**	**8.63**	7903.80	7907.35	0.04	3604.10	7907.36	7907.36	**0.00**	389.00	7907.36	7908.33	0.01	200.85	64050	7907.36	7907.36	**0.00**	18.79	398	7	⋯
p38	7902.32	7902.33	**0.00**	**6.79**	7900.17	$+\infty$	100.00	$+\infty$	7902.33	7902.33	**0.00**	491.05	7902.33	7902.33	**0.00**	200.62	64050	7902.32	7902.33	**0.00**	18.36	387	6	⋯
p39	7911.21	7911.21	**0.00**	**8.13**	7905.75	7911.20	0.07	3602.76	7911.21	7911.21	**0.00**	503.17	7911.21	7911.21	**0.00**	200.53	64050	7911.21	7911.21	**0.00**	18.48	398	7	⋯
p4	4759.90	4759.90	**0.00**	**0.10**	4759.89	4759.90	**0.00**	21.38	4759.90	4759.90	**0.00**	0.86	4759.90	4759.90	**0.00**	57.39	21350	4759.89	4759.90	**0.00**	6.97	138	4	⋯
p40	7920.09	7920.10	**0.00**	53.11	7911.93	$+\infty$	100.00	$+\infty$	7920.10	7920.10	**0.00**	462.98	7920.10	7920.10	**0.00**	172.27	64050	7920.10	7920.10	**0.00**	**17.60**	402	7	⋯
p41	3090.40	3090.40	**0.00**	**0.17**	3089.58	$+\infty$	100.00	$+\infty$	3090.40	3090.40	**0.00**	3.24	3090.40	3090.40	**0.00**	55.11	21350	3090.40	3090.40	**0.00**	7.42	216	4	⋯
p42	2071.15	2071.15	**0.00**	**0.47**	2071.15	2071.15	**0.00**	2478.90	2071.15	2071.15	**0.00**	5.18	2071.15	2071.15	**0.00**	133.65	42700	2071.15	2071.15	**0.00**	13.08	298	7	⋯
p43	1600.60	1600.60	**0.00**	**0.48**	1600.60	1600.60	**0.00**	1182.29	1600.60	1600.60	**0.00**	8.80	1600.60	1600.60	**0.00**	193.52	64050	1600.60	1600.60	**0.00**	18.10	455	7	⋯
p44	3033.90	3033.90	**0.00**	**0.10**	3033.89	3033.90	**0.00**	40.06	3033.90	3033.90	**0.00**	1.84	3033.90	3033.90	**0.00**	57.29	21350	3033.90	3033.90	**0.00**	7.42	195	5	⋯
p45	1963.03	1963.03	**0.00**	**0.23**	1963.02	1963.02	**0.00**	112.21	1963.02	1963.02	**0.00**	9.24	1963.03	1963.03	**0.00**	117.69	42700	1963.02	1963.03	**0.00**	12.46	344	8	⋯
p46	1423.63	1423.63	**0.00**	**1.20**	1423.63	1423.63	**0.00**	863.34	1423.63	1423.63	**0.00**	47.26	1423.63	1423.63	**0.00**	202.07	64050	1423.63	1423.63	**0.00**	19.56	586	11	⋯
p47	1811.59	1811.59	**0.00**	**0.08**	1811.59	1811.59	**0.00**	9.61	1811.59	1811.59	**0.00**	1.79	1811.59	1811.59	**0.00**	50.21	21350	1811.59	1811.59	**0.00**	7.12	148	4	⋯
p48	1157.19	1157.19	**0.00**	**0.96**	1157.19	1157.19	**0.00**	3189.30	1157.19	1157.19	**0.00**	53.16	1157.19	1157.19	**0.00**	140.97	42700	1157.19	1157.19	**0.00**	14.81	408	15	⋯
p49	864.08	864.08	**0.00**	**1.36**	864.08	864.08	**0.00**	3616.76	864.08	864.08	**0.00**	111.28	864.08	864.08	**0.00**	179.00	64050	864.08	864.08	**0.00**	19.00	566	10	⋯
p5	4747.63	4747.63	**0.00**	**0.08**	4747.62	4747.63	**0.00**	7.40	4747.63	4747.63	**0.00**	0.83	4747.63	4747.63	**0.00**	54.34	21350	4747.63	4747.63	**0.00**	7.13	138	4	⋯
p50	5257.22	5257.22	**0.00**	**0.22**	5246.98	5257.22	0.19	3603.65	5257.22	5257.22	**0.00**	8.28	5257.22	5257.22	**0.00**	73.03	21350	5257.22	5257.22	**0.00**	7.16	160	6	⋯
p51	4183.27	4183.27	**0.00**	**1.50**	$-\infty$	$+\infty$	100.0	$+\infty$	4183.27	4183.27	**0.00**	53.85	4183.27	4183.27	**0.00**	129.28	42700	4183.26	4183.27	**0.00**	12.21	288	10	⋯
p52	5645.91	5645.91	**0.00**	**0.15**	$-\infty$	$+\infty$	100.0	$+\infty$	5645.91	5645.91	**0.00**	6.03	5645.91	5645.91	**0.00**	62.16	21350	5645.91	5645.91	**0.00**	7.85	164	4	⋯
p53	4947.07	4947.08	**0.00**	**1.16**	$-\infty$	$+\infty$	100.0	$+\infty$	4947.07	4947.07	**0.00**	12.68	4947.08	4947.08	**0.00**	142.22	42700	4947.07	4947.08	**0.00**	13.35	278	7	⋯
p54	5389.63	5389.63	**0.00**	**0.13**	$-\infty$	$+\infty$	100.0	$+\infty$	5389.62	5389.62	**0.00**	2.23	5389.63	5389.63	**0.00**	61.32	21350	5389.62	5389.63	**0.00**	7.01	129	5	⋯

Table 85 **Detailed results for problem NLUFLP-W-holmberg, cost functions f_{10}**

Instance	GUROBI DB	PB	GAP	CPU	SCIP DB	PB	GAP	CPU	COUENNE DB	PB	GAP	CPU	NAIVE DB	PB	GAP	CPU	NP	CN24 DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	1921.43	2677.26	28.23	3600.01	1871.70	2660.55	29.65	3603.22	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.15	72132	2508.44	2600.35	**3.53**	**3591.85**	410	2	⋯
c100_400_10_F_T_10	9960.89	10439.25	4.58	3600.01	9139.54	10559.02	13.44	3602.95	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.18	85606	10427.79	10427.80	**0.00**	3601.02	1377	14	⋯
c100_400_10_V_L_10	2929.91	4051.99	27.69	3600.01	2886.35	4196.70	31.22	3603.50	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.17	72169	3979.66	4060.19	**1.98**	3601.02	424	4	⋯
c100_400_30_F_L_10	6059.81	9348.21	35.18	3600.01	6179.57	9344.57	**33.87**	3602.51	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.24	76473	$-\infty$	$+\infty$	100.00	3601.21	400	1	⋯
c100_400_30_F_T_10	11077.33	22130.98	49.95	3600.01	11049.46	19452.02	43.20	3602.42	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.20	84032	17401.05	18520.15	**6.04**	3601.02	470	2	⋯
c100_400_30_V_T_10	32525.61	63179.40	48.52	3600.01	33098.38	59416.14	**44.29**	3602.46	19520.98	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.29	82847	$-\infty$	$+\infty$	100.00	3601.09	395	1	⋯
c25_100_10_F_L_5	2809.07	3695.96	24.00	3600.01	2951.12	3714.09	20.54	3602.25	1478.84	3905.35	62.13	3600.10	3682.93	3682.93	**0.00**	521.22	19668	3682.92	3682.93	**0.00**	**463.67**	321	22	⋯
c25_100_10_F_T_5	3618.75	3777.81	4.21	3600.01	3675.86	3775.99	2.65	3602.27	2163.36	3806.18	43.16	3600.10	3775.99	3775.99	**0.00**	159.15	21398	3775.99	3775.99	**0.00**	**114.77**	398	21	⋯
c25_100_10_V_L_5	1950.50	1950.51	**0.00**	594.72	1950.51	1950.51	**0.00**	370.42	1205.77	1975.73	38.97	3600.10	1950.51	1950.51	**0.00**	150.75	16465	1950.51	1950.51	**0.00**	**84.20**	277	12	⋯
c25_100_30_F_L_5	5038.56	7311.18	31.08	3600.01	5588.37	7231.49	22.72	3602.23	$-\infty$	$+\infty$	100.0	3600.10	7231.48	7231.49	**0.00**	**835.36**	20717	7231.47	7231.49	**0.00**	3601.02	458	22	⋯
c25_100_30_F_T_5	6667.97	10685.91	37.60	3600.01	7707.79	10569.53	27.08	3602.23	3670.82	11511.70	68.11	3600.10	10569.53	10569.53	**0.00**	**469.47**	21351	10569.52	10569.53	**0.00**	3325.09	490	20	⋯
c25_100_30_V_T_5	31348.52	43351.78	27.69	3600.01	34329.62	42967.78	20.10	3602.19	18180.33	51092.35	64.42	3600.10	42967.76	42967.78	**0.00**	**184.29**	21391	42967.76	42967.80	**0.00**	1393.98	509	25	⋯
c33	26367.15	26912.31	2.03	3600.01	26832.11	26891.28	0.22	3602.35	$-\infty$	37764.88	100.00	3600.10	26891.26	26891.28	**0.00**	107.99	14918	26891.27	26891.28	**0.00**	**52.15**	355	26	⋯
c35	25090.96	25219.81	0.51	3600.01	25202.60	25219.81	0.07	3602.16	20916.80	26883.30	22.19	3600.10	25219.79	25219.81	**0.00**	84.73	14691	25219.80	25219.82	**0.00**	**42.28**	430	21	⋯
c36	44654.02	44699.57	0.10	3600.02	44698.94	44699.57	0.00	3602.54	27978.84	44699.57	37.41	3600.10	44699.55	44699.58	**0.00**	78.84	14667	44699.55	44699.59	**0.00**	**41.63**	323	18	⋯
c37	8842.44	8915.44	0.82	3600.02	8881.24	8915.44	0.38	3603.13	$-\infty$	$+\infty$	100.0	3600.10	8915.43	8915.44	**0.00**	88.96	15292	8915.43	8915.44	**0.00**	**75.75**	420	19	⋯
c38	13052.41	13110.11	0.44	3600.02	13085.26	13110.11	0.19	3603.25	$-\infty$	$+\infty$	100.0	3600.10	13110.10	13110.11	**0.00**	133.67	15095	13110.10	13110.11	**0.00**	**117.50**	604	21	⋯
c39	8840.19	8917.63	0.87	3600.03	8872.95	8917.63	0.50	3602.89	$-\infty$	$+\infty$	100.0	3600.10	8917.63	8917.64	**0.00**	121.44	15233	8917.63	8917.64	**0.00**	**94.69**	578	21	⋯
c40	11799.06	11910.36	0.93	3600.02	11849.95	11910.36	0.51	3602.83	$-\infty$	$+\infty$	100.0	3600.10	11910.35	11910.36	**0.00**	**89.66**	15351	11910.35	11910.36	**0.00**	102.36	425	19	⋯
c41	36883.01	36907.22	0.07	3600.02	36907.18	36907.22	**0.00**	3602.13	13658.75	44943.19	69.61	3600.10	36907.20	36907.22	**0.00**	150.87	14801	36907.20	36907.23	**0.00**	**45.51**	381	21	⋯
c42	37179.03	37194.33	0.04	3600.02	37194.29	37194.33	**0.00**	2888.91	13655.37	41380.66	67.00	3600.10	37194.31	37194.34	**0.00**	76.38	14450	37194.31	37194.34	**0.00**	**43.09**	360	19	⋯
c43	32966.23	33036.19	0.21	3600.01	33031.76	33036.19	0.01	3602.35	11965.46	35218.85	66.03	3600.10	33036.17	33036.19	**0.00**	94.48	14743	33036.16	33036.19	**0.00**	**38.77**	378	20	⋯
c44	52524.24	52527.87	0.01	3600.02	52527.82	52527.87	**0.00**	3060.66	19065.06	57651.25	66.93	3600.10	52527.84	52527.88	**0.00**	77.47	14590	52527.84	52527.89	**0.00**	**45.58**	407	20	⋯
c45	6016.99	6044.91	0.46	3600.02	6034.02	6044.91	0.18	3602.61	$-\infty$	$+\infty$	100.0	3600.10	6044.90	6044.91	**0.00**	192.47	15554	6044.90	6044.91	**0.00**	**179.68**	613	21	⋯
c46	9603.76	9633.15	0.31	3600.04	9622.22	9633.15	0.11	3602.63	$-\infty$	$+\infty$	100.0	3600.10	9633.14	9633.15	**0.00**	286.01	15297	9633.14	9633.15	**0.00**	**130.19**	416	20	⋯
c47	6337.55	6360.38	0.36	3600.02	6354.69	6360.38	0.09	3602.86	$-\infty$	$+\infty$	100.0	3600.10	6360.38	6360.39	**0.00**	**143.24**	15589	6360.38	6360.39	**0.00**	177.23	380	20	⋯
c48	8478.69	8505.16	0.31	3600.03	8497.25	8505.16	0.09	3602.86	$-\infty$	$+\infty$	100.0	3600.10	8505.15	8505.16	**0.00**	**127.57**	15416	8505.15	8505.16	**0.00**	145.42	398	19	⋯
c49	5380.57	5604.86	4.00	3600.02	5459.45	5589.59	2.33	3602.78	1617.22	$+\infty$	100.00	3600.10	5589.43	5589.43	**0.00**	**248.29**	25275	5589.43	5589.44	**0.00**	377.14	728	31	⋯
c50	8023.73	8228.79	2.49	3600.02	8099.98	8228.79	1.57	3602.71	2353.05	$+\infty$	100.00	3600.10	8228.79	8228.80	**0.00**	**176.57**	24987	8228.79	8228.80	**0.00**	236.73	681	26	⋯
c51	4619.07	4889.49	5.53	3600.02	4569.88	4869.93	6.16	3602.66	1476.34	$+\infty$	100.00	3600.10	4869.93	4869.93	**0.00**	334.04	25210	4869.93	4869.93	**0.00**	**292.86**	603	28	⋯
c52	8310.98	8629.04	3.69	3600.02	8259.36	8629.04	4.28	3602.65	2442.63	$+\infty$	100.00	3600.10	8629.04	8629.04	**0.00**	**228.99**	25027	8629.04	8629.04	**0.00**	276.78	589	25	⋯
c53	8817.10	8935.15	1.32	3600.04	7668.55	13307.85	42.38	3605.53	$-\infty$	$+\infty$	100.0	3600.10	8935.14	8935.15	**0.00**	**224.71**	22521	8935.14	8935.15	**0.00**	1226.97	668	24	⋯
c54	11465.07	11583.11	1.02	3600.06	11398.83	11583.11	1.59	3607.69	$-\infty$	$+\infty$	100.0	3600.10	11583.11	11583.12	**0.00**	**196.67**	22489	11583.11	11583.12	**0.00**	877.97	615	21	⋯
c55	7383.38	7700.90	4.12	3600.06	6865.51	11681.41	41.23	3607.53	$-\infty$	$+\infty$	100.0	3600.10	7685.64	7685.65	**0.00**	**365.40**	22853	7685.64	7685.65	**0.00**	1325.54	652	29	⋯
c56	10779.35	10941.78	1.48	3600.04	8846.33	$+\infty$	100.00	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	10941.78	10941.79	**0.00**	**222.66**	22535	10941.78	10941.79	**0.00**	1170.88	649	24	⋯
c57	4351.96	4433.88	1.85	3600.04	4394.32	4432.87	0.87	3603.80	1310.68	$+\infty$	100.00	3600.10	4432.87	4432.87	**0.00**	**239.25**	25142	4432.87	4432.87	**0.00**	418.09	698	28	⋯
c58	8968.98	8990.00	0.23	3600.03	8952.43	8990.00	0.42	3604.69	2174.19	$+\infty$	100.00	3600.10	8989.99	8990.00	**0.00**	**236.48**	24996	8989.99	8990.00	**0.00**	240.51	552	19	⋯
c59	5269.80	5299.13	0.55	3600.03	5255.66	5299.13	0.82	3605.81	1481.87	$+\infty$	100.00	3600.10	5299.13	5299.13	**0.00**	**272.33**	25276	5299.13	5299.13	**0.00**	374.11	616	23	⋯
c60	5350.26	5449.62	1.82	3600.02	5389.38	5445.87	1.04	3604.68	1588.64	$+\infty$	100.00	3600.10	5445.86	5445.87	**0.00**	**198.27**	25208	5445.86	5445.87	**0.00**	373.99	544	26	⋯
c61	7736.22	7769.61	0.43	3600.07	7534.32	12528.69	39.86	3614.75	$-\infty$	$+\infty$	100.0	3600.10	7769.60	7769.61	**0.00**	**215.78**	22733	7769.60	7769.61	**0.00**	1374.34	864	22	⋯
c62	10404.71	10437.86	0.32	3600.08	10218.80	15185.31	32.71	3613.53	$-\infty$	$+\infty$	100.0	3600.10	10437.86	10437.87	**0.00**	**177.23**	22401	10437.86	10437.87	**0.00**	1336.81	593	22	⋯
c63	6550.96	6589.79	0.59	3600.08	5800.54	9648.61	39.88	3610.20	$-\infty$	$+\infty$	100.0	3600.10	6589.78	6589.79	**0.00**	**256.18**	22842	6589.79	6589.79	**0.00**	956.44	593	19	⋯
c64	9264.39	9317.10	0.57	3600.09	8274.69	14339.85	42.30	3607.58	$-\infty$	$+\infty$	100.0	3600.10	9317.10	9317.10	**0.00**	**201.01**	22521	9317.10	9317.11	**0.00**	714.91	666	20	⋯
#S			1				3				0				**37**					36				⋯
SGM			2.62	13375.61			4.40	12296.83			84.75	14400.00			0.90	**344.60**	23417			**0.38**	475.92	503	18	⋯

Table 86 Detailed results for problem NLMCFP-N, cost functions f_1

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	711.62	712.41	0.11	3600.01	-1070.88	1344.84	179.63	3603.19	-1934.90	1301.06	167.24	3600.10	−∞	+∞	100.00	3601.25	95400	712.24	712.47	**0.03**	3601.02	844	15	⋯
c100_400_10_F_T_10	5930.58	6662.20	10.98	3600.01	-17891.94	19261.12	192.89	3602.94	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.27	95400	6065.66	6687.05	**9.29**	3601.04	543	2	⋯
c100_400_10_V_L_10	1075.84	1079.53	0.34	3600.01	-1430.00	1757.44	181.37	3602.92	-3143.26	1825.00	158.06	3600.10	−∞	+∞	100.00	3601.25	95400	1079.05	1079.46	**0.04**	3601.02	930	29	⋯
c100_400_30_F_L_10	3055.92	3103.30	1.53	3600.01	-2642.90	4428.67	159.68	3602.48	-6016.21	+∞	100.00	3600.10	−∞	+∞	100.00	3601.28	95400	3077.37	3118.55	**1.32**	3601.02	612	6	⋯
c100_400_30_F_T_10	7604.85	7744.92	**1.81**	3600.01	-7769.69	12786.55	160.76	3602.54	-15877.02	+∞	100.00	3600.10	−∞	+∞	100.00	3601.26	95400	7568.95	7824.64	3.27	3601.02	583	3	⋯
c100_400_30_V_T_10	18827.18	19211.13	**2.00**	3600.01	-10860.53	24940.39	143.55	3603.33	-28844.56	+∞	100.00	3600.10	−∞	+∞	100.00	3601.28	95400	18542.85	19514.32	4.98	3601.02	541	2	⋯
c25_100_10_F_L_5	1058.30	1058.30	**0.00**	**2.93**	535.19	1071.79	50.07	3602.25	-1910.49	1374.19	171.93	3600.10	1058.30	1058.30	**0.00**	740.34	23850	1058.30	1058.30	**0.00**	162.96	298	19	⋯
c25_100_10_F_T_5	2058.87	2058.87	**0.00**	**34.84**	-519.96	2088.43	124.90	3602.22	-2366.98	2481.67	195.38	3600.10	2058.87	2058.87	**0.00**	1364.82	23850	2058.87	2058.87	**0.00**	630.18	385	22	⋯
c25_100_10_V_L_5	492.39	492.39	**0.00**	**48.86**	334.63	494.64	32.35	3602.67	-253.65	513.34	149.41	3600.10	492.39	492.39	**0.00**	693.36	23850	492.39	492.39	**0.00**	157.78	397	30	⋯
c25_100_30_F_L_5	2498.77	2498.77	**0.00**	**148.16**	725.34	2509.95	71.10	3602.27	-3115.29	2696.62	186.56	3600.10	2498.77	2498.77	**0.00**	762.16	23850	2498.77	2498.77	**0.00**	661.04	463	25	⋯
c25_100_30_F_T_5	3834.94	3834.95	**0.00**	625.92	1799.75	3834.95	53.07	3602.24	-2974.41	3990.49	174.54	3600.10	3834.94	3834.95	**0.00**	722.26	23850	3834.94	3834.95	**0.00**	**393.07**	485	28	⋯
c25_100_30_V_T_5	18920.98	18921.82	0.00	3600.01	7207.79	19109.79	62.28	3602.17	-19338.15	19900.48	197.17	3600.10	18921.79	18921.81	**0.00**	721.83	23850	18921.79	18921.81	**0.00**	**575.29**	491	22	⋯
c33	16986.62	17640.18	3.70	3600.02	-47421.31	17556.63	137.02	3602.20	-54815.22	21689.98	139.57	3600.10	17475.08	17475.10	**0.00**	537.17	19080	17475.08	17475.10	**0.00**	**161.20**	682	28	⋯
c35	17007.07	17404.67	2.28	3600.01	-65172.07	17555.41	126.94	3602.14	−∞	+∞	100.0	3600.10	17209.94	17209.96	**0.00**	565.57	19080	17209.94	17209.96	**0.00**	**262.23**	493	27	⋯
c36	30772.36	31041.23	0.87	3600.01	-106219.86	31538.96	129.69	3602.12	-105437.90	35036.08	133.23	3600.10	30997.20	30997.23	**0.00**	544.52	19080	30997.20	30997.23	**0.00**	**242.48**	442	21	⋯
c37	6371.46	6374.13	0.04	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	6374.07	6374.08	**0.00**	**876.93**	19080	6374.07	6374.08	**0.00**	2923.25	413	22	⋯
c38	9319.23	9344.42	0.27	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	9331.95	9331.96	**0.00**	**688.97**	19080	9331.95	9331.96	**0.00**	2331.02	423	31	⋯
c39	6360.45	6360.75	0.00	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	6360.74	6360.75	**0.00**	**630.22**	19080	6360.74	6360.75	**0.00**	2381.57	406	22	⋯
c40	8499.63	8508.53	0.10	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	8505.60	8505.61	**0.00**	**734.37**	19080	8505.60	8505.61	**0.00**	3084.08	424	26	⋯
c41	24599.65	24800.68	0.81	3600.01	-72449.49	24848.74	134.30	3602.13	-74858.51	27170.85	136.30	3600.10	24681.18	24681.20	**0.00**	559.74	19080	24681.17	24681.20	**0.00**	**231.34**	546	21	⋯
c42	25499.40	25687.69	0.73	3600.01	-90092.66	26260.60	129.15	3602.10	-90194.76	26540.79	129.43	3600.10	25587.62	25587.64	**0.00**	545.24	19080	25587.62	25587.64	**0.00**	**237.24**	548	22	⋯
c43	22403.52	22534.74	0.58	3600.01	-70249.45	23060.80	132.83	3602.09	-67512.84	23426.16	134.70	3600.10	22495.78	22495.80	**0.00**	549.11	19080	22495.78	22495.80	**0.00**	**215.89**	516	21	⋯
c44	35897.99	35960.26	0.17	3600.01	-92801.14	36347.63	139.17	3602.49	-106612.50	37361.30	135.04	3600.10	35941.56	35941.60	**0.00**	538.39	19080	35941.57	35941.60	**0.00**	**286.80**	636	25	⋯
c45	4325.27	4325.32	0.00	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	4325.32	4325.32	**0.00**	**625.47**	19080	4325.32	4325.32	**0.00**	2102.25	439	27	⋯
c46	6952.82	6953.06	0.00	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	6953.02	6953.03	**0.00**	**827.62**	19080	6953.02	6953.03	**0.00**	3302.75	415	28	⋯
c47	4504.27	4504.28	**0.00**	882.51	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	4504.27	4504.28	**0.00**	**701.28**	19080	4504.27	4504.28	**0.00**	3000.49	403	23	⋯
c48	6083.39	6083.40	**0.00**	**697.89**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	6083.39	6083.40	**0.00**	720.98	19080	6083.39	6083.40	**0.00**	3593.69	407	26	⋯
c49	3216.22	3216.58	0.01	3600.03	625.46	3256.60	80.79	3602.73	-4152.77	3257.18	178.43	3600.10	3216.58	3216.58	**0.00**	**876.86**	28620	3216.58	3216.58	**0.00**	3014.78	639	23	⋯
c50	5443.10	5443.59	0.01	3600.02	-255.55	5689.33	104.49	3687.30	−∞	+∞	100.0	3600.10	5443.59	5443.60	**0.00**	**952.62**	28620	5443.59	5443.59	**0.00**	3396.15	624	21	⋯
c51	2874.02	2874.53	0.02	3600.02	346.61	2940.71	88.21	3602.61	-3767.88	2941.19	178.06	3600.10	2874.53	2874.53	**0.00**	**880.96**	28620	2874.53	2874.53	**0.00**	3147.92	705	23	⋯
c52	5295.94	5296.99	0.02	3600.03	-37.73	5538.74	100.68	3602.57	−∞	+∞	100.0	3600.10	5296.99	5296.99	**0.00**	**917.26**	28620	5296.99	5296.99	**0.00**	1799.27	617	25	⋯
c53	6317.70	6326.18	0.13	3600.11	-27047.38	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	6322.76	6322.76	**0.00**	**1207.42**	28620	5991.84	6521.36	8.12	3601.04	177	2	⋯
c54	8231.03	8280.10	0.59	3600.04	-29953.16	8393.98	128.02	3606.78	−∞	+∞	100.0	3600.10	8277.63	8277.64	**0.00**	**2501.00**	28620	8016.57	8371.28	4.24	3601.06	214	3	⋯
c55	5288.04	5290.77	0.05	3600.05	-22209.72	14119.00	163.57	3606.89	−∞	+∞	100.0	3600.10	5290.69	5290.70	**0.00**	**1217.20**	28620	5109.93	5531.58	7.62	3601.06	179	2	⋯
c56	7737.04	7751.11	0.18	3600.05	-32823.63	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	7745.06	7745.06	**0.00**	**1940.58**	28620	7347.29	8014.28	8.32	3601.06	179	2	⋯
c57	2802.07	2802.40	0.01	3600.03	350.62	2839.48	87.65	3604.50	-3761.57	+∞	100.00	3600.10	2802.39	2802.40	**0.00**	**964.08**	28620	2802.39	2802.40	**0.00**	2726.04	619	27	⋯
c58	5952.06	5952.07	**0.00**	3404.33	−∞	+∞	100.0	+∞	-8208.78	+∞	100.00	3600.10	5952.06	5952.07	**0.00**	**937.29**	28620	5952.06	5952.07	**0.00**	2451.67	552	24	⋯
c59	3025.93	3025.94	**0.00**	3600.03	554.92	3076.54	81.96	3701.36	-4055.50	+∞	100.00	3600.10	3025.93	3025.94	**0.00**	**901.30**	28620	3025.93	3025.94	**0.00**	1928.18	728	23	⋯
c60	3303.74	3304.08	0.01	3600.03	567.34	3368.02	83.16	3604.60	-4329.99	+∞	100.00	3600.10	3304.08	3304.08	**0.00**	**930.27**	28620	3304.05	3304.09	0.00	3601.02	539	27	⋯
c61	5567.59	5568.02	0.01	3600.05	-21499.12	5636.40	126.22	3610.43	−∞	+∞	100.0	3600.10	5567.98	5567.99	**0.00**	**1903.24**	28620	5199.08	5664.88	8.22	3601.09	180	2	⋯
c62	7520.39	7521.29	**0.01**	3600.16	-31097.45	7645.12	124.58	3613.79	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	6976.90	7653.28	8.84	3601.05	180	2	⋯
c63	4652.63	4653.81	0.03	3600.07	-19044.43	4695.66	124.66	3611.79	−∞	+∞	100.0	3600.10	4653.81	4653.81	**0.00**	**2776.13**	28620	4541.66	4696.40	3.29	3601.07	213	3	⋯
c64	6738.22	6738.65	0.01	3600.07	-27750.53	6819.01	124.57	3617.91	−∞	+∞	100.0	3600.10	6738.64	6738.65	**0.00**	**3063.10**	28620	6675.34	6789.32	1.68	3601.05	242	4	⋯
#S			8				0				0				**36**					28				⋯
SGM			**0.33**	6654.75			90.89	14400.00			100.00	14400.00			1.12	**1410.47**	47771			0.66	2474.20	435	16	⋯

Table 87 Detailed results for problem NLMCFP-N, cost functions f_2

Instance	GUROBI DB	PB	GAP	CPU	SCIP DB	PB	GAP	CPU	COUENNE DB	PB	GAP	CPU	NAIVE DB	PB	GAP	CPU	NP	CN24 DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	538.73	598.19	9.94	3600.01	540.57	598.19	9.63	3603.17	405.18	600.99	32.58	3600.10	598.19	598.19	**0.00**	**842.33**	115000	598.16	598.20	0.01	3601.03	1821	22	···
c100_400_10_F_T_10	537.18	540.76	0.66	3600.01	516.30	540.76	4.52	3602.48	266.31	708.60	62.42	3600.10	540.22	541.24	0.19	**331.51**	115000	540.76	540.76	**0.00**	3601.02	2298	35	···
c100_400_10_V_L_10	839.33	940.69	10.77	3600.01	836.35	940.39	11.06	3602.96	486.63	941.31	48.30	3600.10	$-\infty$	$+\infty$	100.00	3601.22	115000	940.22	940.49	**0.03**	3601.03	1548	17	···
c100_400_30_F_L_10	1621.15	1967.23	17.59	3600.01	1653.03	1966.29	15.93	3602.46	895.83	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.30	115000	1962.04	1967.84	**0.29**	3601.02	1226	11	···
c100_400_30_F_T_10	4445.63	6419.84	30.75	3600.01	4494.47	6415.77	29.95	3602.40	2818.83	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.25	115000	6023.82	6467.95	**6.87**	3601.02	971	2	···
c100_400_30_V_T_10	14320.19	19376.96	26.10	3600.01	14371.45	19341.01	25.69	3603.36	10675.56	19752.90	45.95	3600.10	$-\infty$	$+\infty$	100.00	3601.26	115000	18790.38	19523.86	**3.76**	3601.02	998	3	···
c25_100_10_F_L_5	808.91	808.91	**0.00**	2267.23	807.92	808.91	0.12	3602.27	654.79	808.91	19.05	3600.10	808.91	808.91	**0.00**	126.07	28750	808.91	808.91	**0.00**	**117.64**	611	28	···
c25_100_10_F_T_5	859.66	859.66	**0.00**	248.49	859.66	859.66	**0.00**	3602.25	543.34	859.66	36.80	3600.10	859.66	859.66	**0.00**	146.46	28750	859.66	859.66	**0.00**	**95.04**	585	21	···
c25_100_10_V_L_5	516.20	516.20	**0.00**	**83.29**	516.20	516.20	**0.00**	280.08	476.70	516.20	7.65	3600.10	516.20	516.20	**0.00**	116.47	28750	516.20	516.20	**0.00**	226.44	560	19	···
c25_100_30_F_L_5	2097.51	2166.11	3.17	3600.01	2108.67	2166.11	2.65	3602.20	1126.80	2166.99	48.00	3600.10	2166.10	2166.11	**0.00**	**155.05**	28750	2166.10	2166.11	**0.00**	819.14	636	22	···
c25_100_30_F_T_5	4705.83	5108.68	7.89	3600.01	4840.69	5108.68	5.25	3602.22	3085.11	5120.99	39.76	3600.10	5108.68	5108.68	**0.00**	**2200.91**	28750	5108.60	5108.69	0.00	3601.02	468	13	···
c25_100_30_V_T_5	19508.88	20547.70	5.06	3600.01	19856.56	20547.70	3.36	3602.19	11601.66	20715.93	44.00	3600.10	20547.69	20547.70	**0.00**	**612.62**	28750	20547.68	20547.70	**0.00**	1855.29	584	13	···
c33	20905.50	20905.52	**0.00**	**21.13**	20905.52	20905.52	**0.00**	87.34	$-\infty$	22045.75	100.00	3600.10	20905.51	20905.52	**0.00**	107.86	23000	20905.51	20905.52	**0.00**	182.91	439	22	···
c35	14561.98	14561.99	**0.00**	**62.20**	14561.99	14561.99	**0.00**	178.63	7306.85	15989.83	54.30	3600.10	14561.98	14561.99	**0.00**	102.24	23000	14561.98	14562.00	**0.00**	117.80	418	22	···
c36	30786.56	30786.59	**0.00**	297.47	30786.58	30786.60	**0.00**	1619.47	14545.13	52211.23	72.14	3600.10	30786.58	30786.60	**0.00**	**110.24**	23000	30786.57	30786.60	**0.00**	184.04	421	26	···
c37	3551.53	3565.18	0.38	3600.05	3509.97	3564.31	1.52	3602.92	$-\infty$	$+\infty$	100.0	3600.10	3564.31	3564.31	**0.00**	**125.01**	23000	3564.31	3564.31	**0.00**	563.02	436	22	···
c38	4769.34	4810.35	0.85	3600.02	4783.04	4810.34	0.57	3603.24	$-\infty$	$+\infty$	100.0	3600.10	4810.34	4810.34	**0.00**	**110.84**	23000	4810.34	4810.34	**0.00**	376.19	493	26	···
c39	3640.40	3650.15	0.27	3600.04	3630.61	3650.15	0.54	3602.75	$-\infty$	$+\infty$	100.0	3600.10	3650.14	3650.15	**0.00**	**113.62**	23000	3650.14	3650.15	**0.00**	653.92	520	26	···
c40	4962.49	5123.92	3.15	3600.02	5094.98	5123.85	0.56	3602.92	$-\infty$	$+\infty$	100.0	3600.10	5123.85	5123.85	**0.00**	**128.01**	23000	5123.85	5123.85	**0.00**	579.77	479	24	···
c41	24002.94	24002.96	**0.00**	**16.25**	24002.95	24002.95	**0.00**	59.59	11847.24	32459.12	63.50	3600.10	24002.94	24002.95	**0.00**	103.77	23000	24002.93	24002.95	**0.00**	144.76	451	14	···
c42	22901.85	22901.87	**0.00**	**41.75**	22901.87	22901.88	**0.00**	355.13	10924.87	32795.44	66.69	3600.10	22901.87	22901.88	**0.00**	100.85	23000	22901.86	22901.88	**0.00**	165.58	405	21	···
c43	21394.16	21394.18	**0.00**	**89.30**	21394.16	21394.18	**0.00**	235.86	10839.20	29678.15	63.48	3600.10	21394.17	21394.18	**0.00**	101.03	23000	21394.17	21394.18	**0.00**	176.42	538	25	···
c44	32156.01	32156.04	**0.00**	**77.63**	32156.05	32156.05	**0.00**	141.67	16513.54	44053.55	62.51	3600.10	32156.03	32156.05	**0.00**	151.18	23000	32156.02	32156.05	**0.00**	217.78	486	23	···
c45	1915.07	2610.84	26.65	3600.02	2573.91	2574.01	0.00	3602.62	1112.29	3950.65	71.85	3600.10	2574.00	2574.01	**0.00**	**127.77**	23000	2574.00	2574.01	**0.00**	628.02	490	22	···
c46	3654.71	3654.75	0.00	3600.02	3654.74	3654.74	**0.00**	1491.89	$-\infty$	$+\infty$	100.0	3600.10	3654.74	3654.74	**0.00**	**112.31**	23000	3654.74	3654.74	**0.00**	457.61	462	22	···
c47	2351.60	2968.71	20.79	3600.02	2965.54	2965.79	0.01	3602.98	$-\infty$	$+\infty$	100.0	3600.10	2965.78	2965.79	**0.00**	**106.44**	23000	2965.78	2965.79	**0.00**	772.21	463	21	···
c48	2591.08	3410.82	24.03	3600.02	3375.02	3375.63	0.02	3602.83	$-\infty$	$+\infty$	100.0	3600.10	3375.63	3375.63	**0.00**	**116.85**	23000	3375.63	3375.63	**0.00**	993.26	449	26	···
c49	2573.72	2573.73	**0.00**	2967.12	2573.73	2573.73	**0.00**	2104.40	1262.42	$+\infty$	100.00	3600.10	2573.72	2573.73	**0.00**	**157.32**	34500	2573.72	2573.73	**0.00**	981.70	663	27	···
c50	3325.26	3390.03	1.91	3600.02	3375.00	3375.23	0.01	3602.95	1535.18	$+\infty$	100.00	3600.10	3375.23	3375.23	**0.00**	**175.69**	34500	3375.23	3375.23	**0.00**	589.71	798	26	···
c51	2353.09	2470.01	4.73	3600.02	2424.47	2470.79	1.87	3602.62	1170.08	$+\infty$	100.00	3600.10	2466.31	2466.31	**0.00**	**171.03**	34500	2466.31	2466.31	**0.00**	2288.04	663	25	···
c52	3522.38	4036.99	12.75	3600.02	3850.36	3909.11	1.50	3602.60	1819.41	$+\infty$	100.00	3600.10	3909.11	3909.11	**0.00**	**200.35**	34500	3909.11	3909.12	**0.00**	2114.77	714	31	···
c53	0.00	$+\infty$	100.00	$+\infty$	4247.62	4507.55	5.77	3606.70	$-\infty$	$+\infty$	100.0	3600.10	4483.48	4483.48	**0.00**	**324.25**	34500	4481.27	4484.28	0.07	3601.05	426	10	···
c54	5.46	$+\infty$	100.00	$+\infty$	4927.08	5491.75	10.28	3609.12	$-\infty$	$+\infty$	100.0	3600.10	5454.26	5454.26	**0.00**	**294.54**	34500	5447.20	5456.13	0.16	3601.04	383	12	···
c55	2.55	$+\infty$	100.00	$+\infty$	4357.05	4889.77	10.89	3614.54	$-\infty$	$+\infty$	100.0	3600.10	4842.68	4842.68	**0.00**	**728.28**	34500	4792.65	4854.30	1.27	3601.04	300	5	···
c56	0.00	$+\infty$	100.00	$+\infty$	5269.94	5714.60	7.78	3608.13	$-\infty$	$+\infty$	100.0	3600.10	5653.05	5653.05	**0.00**	**263.38**	34500	5646.26	5654.72	0.15	3601.05	401	12	···
c57	1664.81	1995.06	16.55	3600.03	1995.03	1995.03	**0.00**	531.80	923.82	$+\infty$	100.00	3600.10	1995.03	1995.03	**0.00**	**166.10**	34500	1995.03	1995.03	**0.00**	406.20	761	17	···
c58	2546.38	2938.46	13.34	3600.03	2926.85	2926.85	**0.00**	1682.79	1210.12	$+\infty$	100.00	3600.10	2926.84	2926.85	**0.00**	**166.33**	34500	2926.84	2926.85	**0.00**	619.79	667	23	···
c59	2198.52	2429.69	9.51	3600.07	2422.79	2422.79	**0.00**	1424.41	1128.60	$+\infty$	100.00	3600.10	2422.79	2422.79	**0.00**	**161.96**	34500	2422.79	2422.79	**0.00**	1083.09	756	29	···
c60	2115.37	2570.85	17.72	3600.02	2534.25	2534.27	**0.00**	3605.04	1161.63	$+\infty$	100.00	3600.10	2534.27	2534.27	**0.00**	**169.20**	34500	2534.27	2534.27	**0.00**	1101.86	656	23	···
c61	10.77	$+\infty$	100.00	$+\infty$	3419.36	3439.89	0.60	3609.77	$-\infty$	$+\infty$	100.0	3600.10	3439.89	3439.89	**0.00**	**207.96**	34500	3439.89	3439.89	**0.00**	3503.68	703	23	···
c62	2.59	$+\infty$	100.00	$+\infty$	4047.35	4246.25	4.68	3612.84	$-\infty$	$+\infty$	100.0	3600.10	4193.42	4193.42	**0.00**	**281.60**	34500	4193.39	4193.43	**0.00**	3601.05	636	18	···
c63	16.78	$+\infty$	100.00	$+\infty$	3324.80	3437.67	3.28	3608.51	$-\infty$	$+\infty$	100.0	3600.10	3415.63	3415.64	**0.00**	**266.24**	34500	3415.63	3415.64	**0.00**	3601.04	695	21	···
c64	7.41	$+\infty$	100.00	$+\infty$	3788.69	3930.88	3.62	3610.26	$-\infty$	$+\infty$	100.0	3600.10	3928.81	3928.81	**0.00**	**453.40**	34500	3928.55	3928.86	0.01	3601.11	550	17	···
#S			11				13				0				**38**					29				···
SGM			6.36	4517.74			1.49	5054.73			72.68	14400.00			0.54	**290.67**	34540			**0.13**	1482.06	609	19	···

Table 88 Detailed results for problem NLMCFP-N, cost functions f_3

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	602.97	786.83	23.37	3600.01	611.90	778.29	21.38	3603.05	251.60	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.28	163500	761.51	784.42	**2.92**	3601.02	992	4	···
c100_400_10_F_T_10	920.25	1255.37	26.69	3600.01	897.52	1255.37	28.51	3602.45	114.24	1996.25	94.28	3600.10	$-\infty$	$+\infty$	100.00	3601.35	163500	1255.26	1255.45	**0.01**	3601.02	1781	21	···
c100_400_10_V_L_10	982.34	1283.19	23.45	3600.01	1026.27	1241.86	17.36	3602.97	442.35	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.34	163500	1230.59	1245.91	**1.23**	3601.02	1035	7	···
c100_400_30_F_L_10	2192.71	3057.23	28.28	3600.01	2141.77	3015.94	28.99	3602.58	917.39	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.44	163500	2799.95	3034.29	**7.72**	3601.02	957	2	···
c100_400_30_F_T_10	5510.81	9540.76	42.24	3600.01	5652.98	9171.43	**38.36**	3602.40	2579.08	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.34	163500	$-\infty$	$+\infty$	100.00	3601.09	900	1	···
c100_400_30_V_T_10	16196.58	24677.51	34.37	3600.01	16475.88	23686.91	**30.44**	3603.32	8843.75	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.35	163500	$-\infty$	$+\infty$	100.00	3601.22	900	1	···
c25_100_10_F_L_5	1194.91	1194.91	**0.00**	2805.37	1194.76	1194.91	0.01	3602.45	681.02	1267.57	46.27	3600.10	1194.91	1194.91	**0.00**	**505.97**	40875	1194.91	1194.91	**0.00**	987.57	794	32	···
c25_100_10_F_T_5	1569.41	1593.71	1.52	3600.01	1563.40	1593.21	1.87	3602.25	955.73	1601.38	40.32	3600.10	1593.21	1593.21	**0.00**	**1276.21**	40875	1593.21	1593.21	**0.00**	1516.03	704	23	···
c25_100_10_V_L_5	601.23	601.23	**0.00**	3146.31	601.23	601.23	**0.00**	3602.24	399.78	619.09	35.42	3600.10	601.23	601.23	**0.00**	390.03	40875	601.23	601.23	**0.00**	**317.72**	580	23	···
c25_100_30_F_L_5	2452.15	2971.85	17.49	3600.01	2716.16	2963.26	8.34	3602.23	826.48	3444.67	76.01	3600.10	$-\infty$	$+\infty$	100.00	3601.16	40875	2947.34	2965.67	**0.62**	3601.02	348	7	···
c25_100_30_F_T_5	4887.12	5114.11	4.44	3600.01	5014.95	5107.14	1.81	3602.23	$-\infty$	$+\infty$	100.0	3600.10	5107.14	5107.15	**0.00**	**2304.53**	40875	5107.13	5107.15	**0.00**	3601.02	571	21	···
c25_100_30_V_T_5	22057.79	22927.16	3.79	3600.01	22787.99	22926.42	0.60	3602.20	$-\infty$	28263.36	100.00	3600.10	22926.41	22926.42	**0.00**	**2263.68**	40875	22926.26	22926.44	**0.00**	3601.02	593	16	···
c33	21671.53	21671.55	**0.00**	**108.93**	21671.52	21671.54	**0.00**	1046.06	$-\infty$	$+\infty$	100.0	3600.10	21671.53	21671.55	**0.00**	256.83	32700	21671.53	21671.55	**0.00**	330.89	497	28	···
c35	16565.38	16565.39	**0.00**	**87.50**	4802.74	30792.33	84.40	3605.92	$-\infty$	$+\infty$	100.0	3600.10	16565.38	16565.40	**0.00**	251.62	32700	16565.38	16565.40	**0.00**	366.08	491	32	···
c36	37132.41	37132.44	**0.00**	**74.06**	9406.63	54888.08	82.86	3675.33	$-\infty$	$+\infty$	100.0	3600.10	37132.42	37132.46	**0.00**	259.84	32700	37132.42	37132.46	**0.00**	256.61	450	24	···
c37	5131.42	5290.57	3.01	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	5253.66	5253.66	**0.00**	**530.45**	32700	5253.66	5253.66	**0.00**	2437.97	747	23	···
c38	7275.33	7277.67	0.03	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	7277.67	7277.67	**0.00**	**349.76**	32700	7277.67	7277.67	**0.00**	1882.83	471	23	···
c39	5123.09	5223.21	1.92	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	5216.56	5216.56	**0.00**	**616.92**	32700	5216.56	5216.56	**0.00**	2512.41	471	20	···
c40	7365.10	7656.05	3.80	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	7617.52	7617.52	**0.00**	**595.97**	32700	7617.52	7617.52	**0.00**	2847.95	466	20	···
c41	28055.35	28055.38	**0.00**	**42.27**	7688.69	41534.59	81.49	3655.32	$-\infty$	$+\infty$	100.0	3600.10	28055.36	28055.38	**0.00**	255.50	32700	28055.36	28055.39	**0.00**	253.69	449	23	···
c42	27846.65	27846.68	**0.00**	**28.72**	6845.81	44178.62	84.50	3618.06	$-\infty$	$+\infty$	100.0	3600.10	27846.67	27846.69	**0.00**	301.01	32700	27846.66	27846.69	**0.00**	236.60	526	20	···
c43	25646.86	25646.89	**0.00**	**90.76**	6611.84	33459.07	80.24	3646.32	$-\infty$	$+\infty$	100.0	3600.10	25646.87	25646.89	**0.00**	259.68	32700	25646.87	25646.89	**0.00**	235.16	487	20	···
c44	38114.66	38114.69	**0.00**	**66.27**	10981.14	45696.65	75.97	3612.33	13913.14	$+\infty$	100.00	3600.10	38114.67	38114.71	**0.00**	256.89	32700	38114.67	38114.70	**0.00**	326.96	586	22	···
c45	3882.71	3882.71	**0.00**	2813.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	3882.71	3882.71	**0.00**	**408.95**	32700	3882.71	3882.71	**0.00**	1584.81	432	19	···
c46	5900.84	6115.07	3.50	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	6006.95	6006.95	**0.00**	**316.18**	32700	6006.95	6006.95	**0.00**	1453.99	476	16	···
c47	4378.67	4406.91	0.64	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	4406.91	4406.91	**0.00**	**484.09**	32700	4406.91	4406.91	**0.00**	2030.39	453	21	···
c48	5204.23	5257.69	1.02	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	5257.49	5257.49	**0.00**	**425.28**	32700	5257.49	5257.49	**0.00**	1759.99	465	14	···
c49	3359.05	3359.06	**0.00**	2474.26	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	3359.05	3359.06	**0.00**	**488.09**	49050	3359.05	3359.06	**0.00**	3008.54	839	28	···
c50	4754.31	4754.31	**0.00**	3125.36	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	4754.31	4754.32	**0.00**	**498.64**	49050	4754.31	4754.32	**0.00**	3050.20	1125	28	···
c51	2957.05	2958.91	0.06	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	2958.79	2958.79	**0.00**	**499.47**	49050	2958.79	2958.79	**0.00**	2973.66	746	31	···
c52	2724.64	4930.29	44.74	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	4789.45	4789.45	**0.00**	**463.42**	49050	4789.45	4789.45	**0.00**	2578.98	702	25	···
c53	0.00	$+\infty$	100.00	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	6102.87	6102.88	**0.00**	**2148.58**	49050	5785.17	6243.23	7.34	3601.04	310	2	···
c54	0.00	$+\infty$	100.00	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	7660.83	7660.84	**0.00**	**2336.49**	49050	7531.33	7801.91	3.47	3601.05	326	3	···
c55	0.00	$+\infty$	100.00	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	5930.68	5930.68	**0.00**	**2541.01**	49050	5849.16	6094.47	4.03	3601.04	324	3	···
c56	0.00	$+\infty$	100.00	$+\infty$	7564.41	7617.72	0.70	3611.65	$-\infty$	$+\infty$	100.0	3600.10	7615.47	7615.48	**0.00**	**2078.81**	49050	7534.70	7667.03	1.73	3579.89	356	5	···
c57	1311.05	2927.45	55.22	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	2879.60	2879.61	**0.00**	**489.57**	49050	2879.60	2879.61	**0.00**	3314.27	749	25	···
c58	2305.33	4761.61	51.59	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	4706.51	4706.51	**0.00**	**406.60**	49050	4706.51	4706.51	**0.00**	1870.32	770	21	···
c59	1717.14	3226.57	46.78	3600.03	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	3193.10	3193.11	**0.00**	**460.77**	49050	3193.10	3193.11	**0.00**	1658.64	759	21	···
c60	1821.90	3460.39	47.35	3600.02	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	3406.06	3406.07	**0.00**	**467.46**	49050	3406.06	3406.07	**0.00**	2437.88	693	24	···
c61	0.00	$+\infty$	100.00	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	5330.94	5330.94	**0.00**	**1391.50**	49050	5234.55	5423.51	3.48	3601.06	334	3	···
c62	0.00	$+\infty$	100.00	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	6794.89	6794.89	**0.00**	**2047.77**	49050	6720.82	6816.08	1.40	3601.04	364	4	···
c63	0.00	$+\infty$	100.00	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	4829.09	4829.10	**0.00**	**1415.23**	49050	$-\infty$	$+\infty$	100.00	3601.93	270	1	···
c64	0.00	$+\infty$	100.00	$+\infty$	$-\infty$	$+\infty$	100.0	$+\infty$	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.76	49050	6194.85	6278.68	**1.34**	**3587.95**	337	3	···
#S			12				1				0				**35**					26				···
SGM			7.27	5080.37			38.32	13550.30			93.19	14400.00			1.36	**1121.15**	49106			**0.92**	2933.82	568	14	···

Table 89 Detailed results for problem NLMCFP-N, cost functions f_4

	GUROBI				SCIP				COUENNE				NAIVE					CN24						
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	207.85	239.44	13.19	3600.01	193.04	232.20	16.86	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.46	292700	227.46	232.91	**2.34**	3601.02	1369	5	⋯
c100_400_10_F_T_10	352.28	879.36	59.94	3600.01	378.99	863.96	**56.13**	3602.94	122.26	1097.50	88.86	3600.10	−∞	+∞	100.00	3601.73	292700	−∞	+∞	100.00	3601.06	1300	1	⋯
c100_400_10_V_L_10	269.16	318.94	15.61	3600.01	242.05	314.13	22.95	3602.76	137.97	+∞	100.00	3600.10	−∞	+∞	100.00	3601.56	292700	303.41	306.88	**1.13**	3601.02	1497	9	⋯
c100_400_30_F_L_10	502.70	656.19	23.39	3600.01	487.45	617.39	21.05	3602.65	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.47	292700	598.33	616.95	**3.02**	3601.02	1481	5	⋯
c100_400_30_F_T_10	1130.73	1538.16	26.49	3600.01	1105.29	1450.26	23.79	3602.40	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.50	292700	1327.09	1450.74	**8.52**	3601.02	1371	2	⋯
c100_400_30_V_T_10	3958.99	4520.05	**12.41**	3600.01	3836.52	4639.29	17.30	3602.44	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.68	292700	−∞	+∞	100.00	3601.19	1300	1	⋯
c25_100_10_F_L_5	192.85	192.85	**0.00**	3600.01	192.62	192.85	0.12	3602.24	143.84	192.87	25.42	3600.10	192.82	192.87	0.03	389.09	73175	192.85	192.85	**0.00**	**186.39**	743	36	⋯
c25_100_10_F_T_5	334.44	334.46	0.00	3600.01	334.46	334.46	**0.00**	920.33	188.86	342.90	44.92	3600.10	334.46	334.48	0.01	515.56	73175	334.46	334.46	**0.00**	**421.96**	762	33	⋯
c25_100_10_V_L_5	130.96	130.96	**0.00**	474.58	130.93	130.96	0.03	3602.26	120.96	130.96	7.64	3600.10	130.96	130.96	**0.00**	265.11	73175	130.96	130.96	**0.00**	**110.30**	737	31	⋯
c25_100_30_F_L_5	452.35	452.36	0.00	3600.01	452.36	452.36	**0.00**	**121.74**	372.24	462.30	19.48	3600.10	452.31	452.41	0.02	373.12	73175	452.36	452.36	**0.00**	219.17	741	32	⋯
c25_100_30_F_T_5	1213.62	1213.62	**0.00**	3302.56	1213.59	1213.62	0.00	3602.18	1084.83	1213.64	10.61	3600.10	1213.62	1213.62	**0.00**	270.62	73175	1213.62	1213.62	**0.00**	**103.02**	719	29	⋯
c25_100_30_V_T_5	3830.73	3830.78	0.00	3600.01	3830.77	3830.79	**0.00**	3602.20	−∞	4150.95	100.00	3600.10	3830.78	3830.79	**0.00**	331.38	73175	3830.78	3830.79	**0.00**	**101.59**	775	32	⋯
c33	7582.97	7582.98	**0.00**	**74.99**	7582.98	7582.98	**0.00**	442.75	−∞	+∞	100.0	3600.10	7582.83	7583.00	0.00	235.39	58540	7582.98	7582.98	**0.00**	107.07	571	25	⋯
c35	4791.39	4791.40	**0.00**	**51.05**	4791.40	4791.40	**0.00**	509.02	−∞	+∞	100.0	3600.10	4791.39	4791.88	0.01	208.61	58540	4791.40	4791.40	**0.00**	118.89	541	23	⋯
c36	6973.16	6973.16	**0.00**	**74.45**	6970.84	6973.17	0.03	3602.41	0.00	+∞	100.00	3600.10	6973.08	6973.29	0.00	272.72	58540	6973.16	6973.17	**0.00**	120.25	555	29	⋯
c37	1027.28	1027.30	0.00	3600.07	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	1027.30	1027.30	**0.00**	**352.24**	58540	1027.30	1027.30	**0.00**	424.86	600	22	⋯
c38	1515.42	2000.46	24.25	3600.03	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	1515.49	1515.50	**0.00**	**405.87**	58540	1515.50	1515.50	**0.00**	483.75	600	22	⋯
c39	1026.63	1026.65	0.00	3600.03	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	1026.65	1026.65	**0.00**	**354.84**	58540	1026.65	1026.65	**0.00**	465.07	600	22	⋯
c40	1363.93	1363.96	0.00	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	1363.96	1363.96	**0.00**	**381.55**	58540	1363.96	1363.97	**0.00**	549.45	600	22	⋯
c41	9052.04	9052.05	**0.00**	**21.56**	9052.05	9052.05	**0.00**	537.27	−∞	+∞	100.0	3600.10	9052.04	9052.46	0.00	306.38	58540	9052.05	9052.05	**0.00**	112.29	538	24	⋯
c42	6722.91	6722.92	**0.00**	**21.72**	6722.92	6722.92	**0.00**	1520.13	−∞	+∞	100.0	3600.10	6722.91	6723.77	0.01	368.79	58540	6722.92	6722.92	**0.00**	99.77	534	22	⋯
c43	6568.34	6568.35	**0.00**	131.27	6568.35	6568.35	**0.00**	982.36	−∞	+∞	100.0	3600.10	6568.35	6568.37	**0.00**	381.71	58540	6568.35	6568.35	**0.00**	**124.67**	748	23	⋯
c44	11279.97	11279.98	**0.00**	184.80	11279.98	11279.98	**0.00**	134.80	−∞	+∞	100.0	3600.10	11279.97	11280.16	0.00	252.90	58540	11279.98	11279.99	**0.00**	**124.78**	550	23	⋯
c45	424.88	791.09	46.29	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	689.69	689.69	**0.00**	394.17	58540	689.69	689.69	**0.00**	**349.22**	600	22	⋯
c46	579.67	1284.72	54.88	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	1111.45	1111.45	**0.00**	**305.73**	58540	1111.45	1111.45	**0.00**	373.87	600	22	⋯
c47	721.81	721.83	0.00	3600.04	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	721.83	721.83	**0.00**	**298.00**	58540	721.83	721.83	**0.00**	421.48	590	22	⋯
c48	978.52	978.54	0.00	3600.04	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	978.54	978.54	**0.00**	424.92	58540	978.54	978.54	**0.00**	**356.35**	593	22	⋯
c49	587.32	723.48	18.82	3600.04	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	699.43	699.43	**0.00**	**391.31**	87810	699.43	699.43	**0.00**	589.86	893	22	⋯
c50	867.32	867.35	0.00	3600.04	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	867.35	867.38	0.00	**421.57**	87810	867.35	867.35	**0.00**	604.76	870	22	⋯
c51	443.84	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	508.49	508.49	**0.00**	**352.39**	87810	508.49	508.49	**0.00**	860.25	940	23	⋯
c52	698.69	936.15	25.37	3600.04	−∞	+∞	100.0	+∞	545.04	+∞	100.00	3600.10	861.79	861.79	**0.00**	**709.85**	87810	861.79	861.79	**0.00**	821.63	1272	27	⋯
c53	0.00	+∞	100.00	+∞	1037.13	1037.13	**0.00**	2521.70	−∞	+∞	100.0	3600.10	1037.13	1037.13	**0.00**	**1951.68**	87810	1036.40	1037.16	0.07	3601.04	660	13	⋯
c54	0.00	+∞	100.00	+∞	1327.42	1327.42	**0.00**	2902.04	−∞	+∞	100.0	3600.10	1327.42	1327.42	**0.00**	**1187.51**	87810	1327.24	1327.45	0.02	3601.04	703	16	⋯
c55	0.00	+∞	100.00	+∞	-552.58	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	914.24	914.24	**0.00**	**1050.47**	87810	911.87	914.37	0.27	3601.11	581	11	⋯
c56	0.00	+∞	100.00	+∞	-818.28	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	1278.00	1278.00	**0.00**	**1003.35**	87810	1277.13	1278.03	0.07	3601.11	654	13	⋯
c57	458.83	458.84	0.00	3600.05	−∞	+∞	100.0	+∞	272.36	+∞	100.00	3600.10	458.84	458.86	0.00	**406.68**	87810	458.84	458.84	**0.00**	556.63	964	22	⋯
c58	669.62	1137.71	41.14	3600.02	−∞	+∞	100.0	+∞	367.46	+∞	100.00	3600.10	954.73	954.77	0.00	**360.64**	87810	954.73	954.73	**0.00**	617.12	868	24	⋯
c59	570.49	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	727.67	727.71	0.01	**402.77**	87810	727.67	727.67	**0.00**	469.96	877	24	⋯
c60	536.22	+∞	100.00	+∞	−∞	+∞	100.0	+∞	371.52	+∞	100.00	3600.10	641.78	641.79	0.00	**444.10**	87810	641.78	641.78	**0.00**	603.11	938	22	⋯
c61	0.00	+∞	100.00	+∞	864.12	888.82	2.78	3613.98	−∞	+∞	100.0	3600.10	888.82	888.82	**0.00**	**2109.77**	87810	888.82	888.82	**0.00**	2695.39	897	22	⋯
c62	0.00	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	1205.58	1205.58	**0.00**	**734.94**	87810	1205.42	1205.61	0.02	3600.26	690	15	⋯
c63	0.00	+∞	100.00	+∞	719.32	770.37	6.63	3614.58	−∞	+∞	100.0	3600.10	770.37	770.37	**0.00**	**687.90**	87810	770.37	770.37	**0.00**	1798.31	988	22	⋯
c64	0.00	+∞	100.00	+∞	1075.31	1075.31	**0.00**	2313.93	−∞	+∞	100.0	3600.10	1075.31	1075.31	**0.00**	**711.06**	87810	1075.31	1075.31	**0.00**	2491.56	900	22	⋯
#S			9				11				0				**37**					32				⋯
SGM			7.22	5444.91			12.07	6809.31			81.96	14400.00			0.91	**736.28**	87909			**0.43**	913.98	782	19	⋯

Table 90 Detailed results for problem NLMCFP-N, cost functions f_5

	GUROBI				**SCIP**				**COUENNE**				**NAIVE**					**CN24**						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	453.75	862.12	**47.37**	3600.01	328.63	853.35	61.49	3602.49	71.11	+∞	100.00	3600.10	−∞	+∞	100.00	3601.23	144300	−∞	+∞	100.00	3600.01	+∞	0	···
c100_400_10_F_T_10	-154.33	2838.76	105.44	3600.01	-962.52	2992.75	132.16	3602.47	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.25	144300	2794.62	2803.78	**0.33**	3601.03	875	7	···
c100_400_10_V_L_10	715.62	1394.26	48.67	3600.01	547.15	1356.28	59.66	3602.52	212.88	2203.30	90.34	3600.10	−∞	+∞	100.00	3601.31	144300	1314.81	1327.66	**0.97**	3601.02	770	7	···
c100_400_30_F_L_10	1821.05	3581.16	49.15	3600.01	1495.86	3499.19	57.25	3602.61	372.82	+∞	100.00	3600.10	−∞	+∞	100.00	3601.29	144300	3261.02	3383.77	**3.63**	3601.03	672	3	···
c100_400_30_F_T_10	4095.91	9932.80	58.76	3600.01	4043.05	9361.86	**56.81**	3602.68	1807.96	+∞	100.00	3600.10	−∞	+∞	100.00	3601.28	144300	−∞	+∞	100.00	3601.10	600	1	···
c100_400_30_V_T_10	11825.09	24967.26	52.64	3600.01	11627.69	24028.37	**51.61**	3602.67	7796.94	+∞	100.00	3600.10	−∞	+∞	100.00	3601.39	144300	−∞	+∞	100.00	3601.10	600	1	···
c25_100_10_F_L_5	1026.67	1282.25	19.93	3600.01	969.46	1282.26	24.39	3602.68	320.80	1332.99	75.93	3600.10	1279.11	1279.11	**0.00**	**572.37**	36075	1279.11	1279.11	**0.00**	715.52	550	25	···
c25_100_10_F_T_5	1454.65	1873.89	22.37	3600.01	1523.69	1873.11	18.65	3602.33	389.70	1908.63	79.58	3600.10	1872.87	1872.87	**0.00**	**1146.06**	36075	1872.87	1872.87	**0.00**	2068.00	460	15	···
c25_100_10_V_L_5	563.32	596.13	5.50	3600.01	561.05	595.88	5.84	3602.36	363.42	603.29	39.76	3600.10	595.82	595.82	**0.00**	**267.52**	36075	595.82	595.82	**0.00**	330.00	417	20	···
c25_100_30_F_L_5	2274.85	2983.22	23.75	3600.01	2263.29	2969.61	23.79	3602.34	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.13	36075	2965.30	2970.95	**0.19**	3601.02	334	13	···
c25_100_30_F_T_5	3849.25	4709.35	18.26	3600.01	3873.65	4706.80	17.70	3602.30	2621.29	4726.45	44.54	3600.10	4706.80	4706.80	**0.00**	**334.68**	36075	4706.80	4706.80	**0.00**	1178.80	584	24	···
c25_100_30_V_T_5	18552.24	22528.02	17.65	3600.01	17616.94	22464.81	21.58	3602.30	10498.64	22711.49	53.77	3600.10	22456.98	22457.00	**0.00**	**721.70**	36075	22456.96	22457.00	**0.00**	3601.03	513	26	···
c33	20487.14	21573.51	5.04	3600.01	20848.88	21510.09	3.07	3602.41	−∞	+∞	100.0	3600.10	21495.37	21495.39	**0.00**	**185.35**	28860	21495.38	21495.39	**0.00**	287.89	358	22	···
c35	16171.21	17679.80	8.53	3600.01	16238.45	17657.03	8.03	3603.96	5462.88	+∞	100.00	3600.10	17655.44	17655.45	**0.00**	**179.75**	28860	17655.44	17655.45	**0.00**	302.68	400	26	···
c36	33142.17	35803.50	7.43	3600.02	35301.84	35803.50	1.40	3602.39	−∞	+∞	100.0	3600.10	35803.48	35803.51	**0.00**	**195.69**	28860	35803.48	35803.52	**0.00**	311.05	382	21	···
c37	5453.43	6125.20	10.97	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	6042.26	6042.27	**0.00**	**514.96**	28860	6042.26	6042.27	**0.00**	3136.96	412	26	···
c38	7512.13	8325.12	9.77	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	8292.05	8292.06	**0.00**	**632.93**	28860	8292.05	8292.06	**0.00**	2480.06	420	25	···
c39	5456.40	6033.96	9.57	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	5968.54	5968.55	**0.00**	**466.92**	28860	5968.54	5968.55	**0.00**	2566.43	407	21	···
c40	7651.75	8520.22	10.19	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	8484.44	8484.45	**0.00**	**435.36**	28860	8484.37	8484.47	0.00	3601.03	343	20	···
c41	28233.17	28233.19	**0.00**	2013.69	27894.18	28233.20	1.20	3602.43	−∞	+∞	100.0	3600.10	28233.18	28233.20	**0.00**	**230.07**	28860	28233.18	28233.20	**0.00**	309.90	364	19	···
c42	27862.40	27862.43	**0.00**	2669.98	27862.44	27862.44	**0.00**	1732.75	−∞	+∞	100.0	3600.10	27862.42	27862.44	**0.00**	**199.57**	28860	27862.42	27862.44	**0.00**	270.14	346	21	···
c43	24157.83	25053.38	3.57	3600.01	24813.51	25071.92	1.03	3602.31	−∞	+∞	100.0	3600.10	25052.70	25052.72	**0.00**	**256.51**	28860	25052.70	25052.72	**0.00**	298.43	458	22	···
c44	39187.29	39187.32	**0.00**	1054.71	38845.33	39187.33	0.87	3602.32	−∞	+∞	100.0	3600.10	39187.31	39187.34	**0.00**	**216.69**	28860	39187.30	39187.34	**0.00**	341.81	398	24	···
c45	4198.67	4389.19	4.34	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	4389.12	4389.13	**0.00**	**326.50**	28860	4389.12	4389.13	**0.00**	2713.46	389	24	···
c46	6640.85	6917.52	4.00	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	6888.53	6888.53	**0.00**	**257.96**	28860	6888.53	6888.53	**0.00**	1714.60	359	16	···
c47	4433.83	4758.79	6.83	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	4758.54	4758.54	**0.00**	**280.47**	28860	4758.54	4758.54	**0.00**	3185.41	382	23	···
c48	5689.60	5964.36	4.61	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	5961.39	5961.39	**0.00**	**272.12**	28860	5961.39	5961.39	**0.00**	2386.52	367	19	···
c49	3142.53	3495.50	10.10	3600.02	3113.12	3508.84	11.28	3602.94	−∞	+∞	100.0	3600.10	3486.96	3486.96	**0.00**	**338.83**	43290	3486.96	3486.96	**0.00**	2405.50	630	28	···
c50	4684.58	5360.60	12.61	3600.02	4707.35	5359.22	12.16	3602.88	−∞	+∞	100.0	3600.10	5305.82	5305.83	**0.00**	**330.50**	43290	5305.82	5305.83	**0.00**	2509.11	572	21	···
c51	2578.06	3048.83	15.44	3600.03	2479.39	3125.19	20.66	3602.89	−∞	+∞	100.0	3600.10	3033.70	3033.71	**0.00**	**438.17**	43290	3033.70	3033.71	**0.00**	2939.90	610	26	···
c52	4378.34	5055.22	13.39	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	5028.96	5028.96	**0.00**	**356.95**	43290	5028.96	5028.96	**0.00**	2074.54	826	23	···
c53	5562.57	6668.46	16.58	3600.06	-1353.36	7007.07	119.31	3606.70	−∞	+∞	100.0	3600.10	6566.58	6566.58	**0.00**	**1207.31**	43290	6218.58	6789.77	8.41	3601.28	212	2	···
c54	7282.82	8557.61	14.90	3600.05	-1770.18	9051.49	119.56	3606.16	−∞	+∞	100.0	3600.10	8453.78	8453.79	**0.00**	**1199.61**	43290	7993.26	8719.09	8.32	3601.06	211	2	···
c55	5053.74	6178.46	18.20	3600.05	-1119.80	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	6120.90	6120.90	**0.00**	**887.40**	43290	5855.27	6417.44	8.76	3601.06	210	2	···
c56	6904.06	8303.22	16.85	3600.05	-1647.84	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	8181.39	8181.39	**0.00**	**3419.83**	43290	7742.80	8445.33	8.32	3601.05	213	2	···
c57	2694.46	3036.53	11.26	3600.03	−∞	+∞	100.0	+∞	695.74	+∞	100.00	3600.10	3031.07	3031.07	**0.00**	**383.40**	43290	3031.07	3031.07	**0.00**	3367.39	622	27	···
c58	5031.01	5361.68	6.17	3600.04	1448.81	5365.49	73.00	3602.60	−∞	+∞	100.0	3600.10	5350.29	5350.30	**0.00**	**387.99**	43290	5350.29	5350.30	**0.00**	2063.75	527	22	···
c59	3016.14	3248.68	7.16	3600.02	2783.90	3280.40	15.14	3605.21	−∞	+∞	100.0	3600.10	3246.01	3246.01	**0.00**	**392.26**	43290	3246.01	3246.01	**0.00**	2540.64	590	27	···
c60	3164.66	3525.37	10.23	3600.05	−∞	+∞	100.0	+∞	911.53	+∞	100.00	3600.10	3520.96	3520.96	**0.00**	**369.76**	43290	3520.96	3520.96	**0.00**	2347.24	698	21	···
c61	5037.98	5911.85	14.78	3600.06	-1191.66	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	5809.23	5809.23	**0.00**	**1042.18**	43290	5424.12	5966.04	9.08	3601.06	211	2	···
c62	6665.06	7651.49	12.89	3600.06	-1610.37	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	7612.59	7612.59	**0.00**	**915.74**	43290	7118.66	7783.96	8.55	3601.07	209	2	···
c63	4427.23	5110.57	13.37	3600.13	-990.84	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	5097.93	5097.93	**0.00**	**1255.34**	43290	4819.59	5265.71	8.47	3601.06	214	2	···
c64	6249.25	7031.38	11.12	3600.06	-1439.73	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	7009.42	7009.43	**0.00**	**1187.61**	43290	6547.02	7148.51	8.41	3601.06	208	2	···
#S			3				1				0				**36**					26				···
SGM			11.17	12450.53			33.19	13709.19			93.38	14400.00			**1.12**	**813.25**	43339			1.23	3238.82	774	14	···

Table 91 Detailed results for problem NLMCFP-N, cost functions f_6

Instance	GUROBI DB	PB	GAP	CPU	SCIP DB	PB	GAP	CPU	COUENNE DB	PB	GAP	CPU	NAIVE DB	PB	GAP	CPU	NP	CN24 DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	1233.61	1313.60	6.09	3600.01	1216.47	1314.27	7.44	3602.43	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1274.27	1315.29	**3.12**	3601.02	520	4	···
c100_400_10_F_T_10	5129.73	8010.52	**35.96**	3600.01	4401.50	7996.29	44.96	3602.96	2890.56	8172.63	64.63	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	500	1	···
c100_400_10_V_L_10	1929.97	1974.80	2.27	3600.01	1918.82	1974.80	2.83	3602.38	−∞	2430.66	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	1966.21	1977.62	**0.58**	3601.02	571	6	···
c100_400_30_F_L_10	4973.05	5564.92	10.64	3600.01	4989.44	5542.57	9.98	3602.41	3492.05	6128.40	43.02	3600.10	−∞	+∞	100.00	3600.01	+∞	5091.86	5484.20	**7.15**	3601.02	527	2	···
c100_400_30_F_T_10	12340.55	15042.08	17.96	3600.01	12421.92	14977.82	**17.06**	3602.43	9412.17	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.13	500	1	···
c100_400_30_V_T_10	32211.54	38131.18	**15.52**	3600.01	28868.48	38210.06	24.45	3602.38	25402.97	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	500	1	···
c25_100_10_F_L_5	2061.46	2061.46	**0.00**	**50.68**	2061.46	2061.46	**0.00**	329.82	1883.28	2070.35	9.04	3600.10	−∞	+∞	100.00	3600.01	+∞	2061.46	2061.46	**0.00**	975.12	410	19	···
c25_100_10_F_T_5	3336.09	3336.09	**0.00**	**623.92**	3336.09	3336.09	**0.00**	1375.36	2887.40	3341.50	13.59	3600.10	−∞	+∞	100.00	3600.01	+∞	3336.08	3336.09	**0.00**	3601.02	424	15	···
c25_100_10_V_L_5	921.85	921.85	**0.00**	**1.39**	921.85	921.85	**0.00**	4.21	921.85	921.85	**0.00**	3649.26	−∞	+∞	100.00	3600.01	+∞	921.85	921.85	**0.00**	34.42	369	14	···
c25_100_30_F_L_5	4718.67	4749.62	0.65	3600.01	4696.00	4749.62	1.13	3602.23	3088.36	4790.67	35.53	3600.10	−∞	+∞	100.00	3600.01	+∞	4738.82	4750.27	**0.24**	3601.02	213	6	···
c25_100_30_F_T_5	7543.96	7595.78	0.68	3600.01	7225.92	7595.42	4.86	3602.22	5868.45	7603.98	22.82	3600.10	−∞	+∞	100.00	3600.01	+∞	7594.15	7595.62	**0.02**	3601.02	282	10	···
c25_100_30_V_T_5	36208.16	36685.73	1.30	3600.01	33922.82	36685.73	7.53	3602.19	26024.47	36719.14	29.13	3600.10	−∞	+∞	100.00	3600.01	+∞	36574.09	36719.73	**0.40**	3601.02	221	8	···
c33	33608.72	33608.73	**0.00**	**5.11**	33608.73	33608.73	**0.00**	47.14	−∞	33916.63	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	33608.71	33608.74	**0.00**	56.32	303	17	···
c35	29531.71	29531.72	**0.00**	**2.72**	29531.72	29531.72	**0.00**	44.63	17724.96	31025.24	42.87	3600.10	−∞	+∞	100.00	3600.01	+∞	29531.70	29531.73	**0.00**	47.04	320	15	···
c36	58087.61	58087.63	**0.00**	**11.55**	58087.63	58087.63	**0.00**	70.99	−∞	64594.63	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	58087.60	58087.65	**0.00**	193.21	329	17	···
c37	10437.26	10437.27	**0.00**	**509.17**	7896.14	10466.60	24.56	3603.24	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	10296.45	10438.32	1.36	3601.03	161	7	···
c38	14973.62	14973.62	**0.00**	**339.98**	10643.85	14978.03	28.94	3602.81	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	14802.40	14989.36	1.25	3601.03	151	6	···
c39	10385.12	10385.12	**0.00**	**249.39**	10385.12	10385.12	**0.00**	860.63	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	10359.78	10387.85	0.27	3601.03	188	11	···
c40	14223.90	14223.92	**0.00**	**197.66**	10715.97	14224.49	24.67	3602.52	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	14164.76	14223.92	0.42	3601.02	184	10	···
c41	45338.10	45338.10	**0.00**	**1.62**	45338.10	45338.10	**0.00**	155.74	27236.41	49534.97	45.02	3600.10	−∞	+∞	100.00	3600.01	+∞	45338.07	45338.11	**0.00**	76.26	304	14	···
c42	46280.78	46280.79	**0.00**	**2.74**	46280.79	46280.79	**0.00**	74.06	27256.57	52080.01	47.66	3600.10	−∞	+∞	100.00	3600.01	+∞	46280.77	46280.80	**0.00**	43.66	314	18	···
c43	40646.97	40646.97	**0.00**	**3.27**	40646.97	40646.97	**0.00**	65.15	23873.89	44703.31	46.59	3600.10	−∞	+∞	100.00	3600.01	+∞	40646.95	40646.98	**0.00**	60.69	325	14	···
c44	64299.47	64299.47	**0.00**	**4.02**	64299.47	64299.47	**0.00**	35.03	36777.84	70158.02	47.58	3600.10	−∞	+∞	100.00	3600.01	+∞	64299.43	64299.49	**0.00**	49.29	312	15	···
c45	7230.20	7230.21	**0.00**	**49.62**	7230.20	7230.20	**0.00**	265.83	3591.09	8066.55	55.48	3600.10	−∞	+∞	100.00	3600.01	+∞	7230.20	7230.20	**0.00**	2675.62	320	16	···
c46	11393.77	11393.78	**0.00**	**63.99**	11393.78	11393.78	**0.00**	340.55	5352.01	11433.72	53.19	3600.10	−∞	+∞	100.00	3600.01	+∞	11393.75	11393.78	**0.00**	3601.03	330	20	···
c47	7768.24	7768.24	**0.00**	**112.29**	5956.97	7787.04	23.50	3602.95	3933.71	7856.77	49.93	3600.10	−∞	+∞	100.00	3600.01	+∞	7761.01	7768.46	0.10	3601.04	199	13	···
c48	10018.97	10018.97	**0.00**	**103.17**	7482.10	10019.95	25.33	3602.87	4778.91	10038.22	52.39	3600.10	−∞	+∞	100.00	3600.01	+∞	10012.30	10020.16	0.08	3601.03	212	18	···
c49	5683.46	5683.46	**0.00**	**86.71**	5683.46	5683.46	**0.00**	549.35	3172.87	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	5683.45	5683.46	**0.00**	1291.63	457	16	···
c50	8936.23	8936.24	**0.00**	**77.45**	8936.24	8936.24	**0.00**	396.12	4648.28	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	8936.24	8936.24	**0.00**	1747.66	463	15	···
c51	5088.82	5088.82	**0.00**	**91.53**	5088.82	5088.82	**0.00**	756.71	2915.26	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	5086.71	5088.82	0.04	3601.03	356	15	···
c52	8761.78	8761.79	**0.00**	**417.54**	6488.41	8771.58	26.03	3603.12	4829.39	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	8757.54	8762.14	0.05	3601.03	317	13	···
c53	10900.99	10901.00	**0.00**	**3396.60**	6678.37	11011.75	39.35	3607.90	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	10579.72	10921.63	3.13	3601.04	193	3	···
c54	14056.87	14056.87	**0.00**	**1490.38**	14040.80	14073.32	0.23	3607.84	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	13665.49	14083.23	2.97	3601.06	201	3	···
c55	9757.83	9772.62	**0.15**	3600.04	6437.91	9844.38	34.60	3609.75	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9131.87	9841.79	7.21	3601.06	173	2	···
c56	13409.18	13409.18	**0.00**	**1037.53**	8194.42	13414.13	38.91	3608.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	12482.11	13441.29	7.14	3601.05	181	2	···
c57	4900.71	4900.71	**0.00**	**58.88**	4900.71	4900.71	**0.00**	660.91	2586.56	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	4900.71	4900.71	**0.00**	1313.07	491	19	···
c58	9231.99	9231.99	**0.00**	**96.21**	6599.28	9239.86	28.58	3604.36	4307.18	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	9231.98	9231.99	**0.00**	1416.52	501	18	···
c59	5342.90	5342.90	**0.00**	**65.02**	5342.90	5342.90	**0.00**	507.80	2922.24	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	5342.90	5342.91	**0.00**	1487.11	472	19	···
c60	5766.96	5766.96	**0.00**	**41.81**	4363.61	5767.73	24.34	3605.87	3137.57	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	5766.96	5766.96	**0.00**	1960.75	476	21	···
c61	9412.35	9412.35	**0.00**	**391.81**	5275.74	9420.28	44.00	3611.59	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8866.60	9415.57	5.83	3601.09	176	2	···
c62	12439.24	12439.24	**0.00**	**614.03**	7681.68	12459.04	38.34	3607.48	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	12279.53	12459.91	1.45	3601.06	204	4	···
c63	8199.16	8199.16	**0.00**	**355.51**	8199.16	8199.16	**0.00**	3554.44	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8102.36	8231.98	1.57	3601.04	207	4	···
c64	11315.80	11315.81	**0.00**	**992.92**	11305.49	11317.89	0.11	3610.59	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	11171.22	11323.88	1.35	3601.05	209	5	···
#S			**33**				19				0				0					16				···
SGM			**0.48**	**347.15**			3.59	2341.60			62.21	14400.00			100.00	14400.00	+∞			1.13	3534.88	303	10	···

Table 92 Detailed results for problem NLMCFP-N, cost functions f_7

	GUROBI				SCIP				COUENNE				NAIVE					CN24						
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	300.04	300.04	0.00	0.25	300.04	300.04	0.00	6.65	300.04	300.04	0.00	3429.95	300.04	300.05	0.00	121.81	92000	300.04	300.04	0.00	3601.03	2333	38	···
c100_400_10_F_T_10	222.64	222.64	0.00	0.25	222.64	222.64	0.00	8.16	222.64	296.75	24.97	3600.10	222.00	223.22	0.55	132.15	92000	222.64	222.64	0.00	3601.03	2496	44	···
c100_400_10_V_L_10	454.91	454.91	0.00	0.29	454.91	454.91	0.00	6.57	454.91	455.61	0.15	3523.00	454.91	454.91	0.00	157.56	92000	454.91	454.91	0.00	1849.64	2399	38	···
c100_400_30_F_L_10	930.65	930.65	0.00	0.80	930.65	930.65	0.00	30.50	823.91	+∞	100.00	3600.10	930.65	930.66	0.00	128.54	92000	930.64	930.65	0.00	3601.03	2396	35	···
c100_400_30_F_T_10	2964.63	2964.63	0.00	0.71	2964.63	2964.63	0.00	27.70	2696.23	+∞	100.00	3600.10	2964.63	2964.63	0.00	126.61	92000	2963.66	2964.99	0.04	3601.02	1693	20	···
c100_400_30_V_T_10	9291.61	9291.61	0.00	1.23	9291.60	9291.61	0.00	32.51	8798.35	+∞	100.00	3600.10	9291.60	9291.61	0.00	127.31	92000	9290.83	9291.96	0.01	3601.03	1784	25	···
c25_100_10_F_L_5	364.43	364.43	0.00	0.07	364.43	364.43	0.00	2.52	364.57	364.57	0.00	86.44	364.43	364.43	0.00	37.07	23000	364.43	364.43	0.00	33.06	612	25	···
c25_100_10_F_T_5	377.21	377.21	0.00	0.06	377.21	377.21	0.00	3.01	377.21	377.21	0.00	167.14	377.21	377.21	0.00	38.05	23000	377.21	377.21	0.00	44.40	596	24	···
c25_100_10_V_L_5	274.29	274.29	0.00	0.03	274.29	274.29	0.00	2.90	274.29	274.29	0.00	49.36	274.29	274.29	0.00	37.50	23000	274.29	274.29	0.00	39.25	597	35	···
c25_100_30_F_L_5	1012.03	1012.03	0.00	0.11	1012.03	1012.03	0.00	3.34	1012.03	1012.03	0.00	244.99	1012.03	1012.03	0.00	37.85	23000	1012.03	1012.03	0.00	72.53	637	27	···
c25_100_30_F_T_5	2663.72	2663.72	0.00	0.09	2663.72	2663.72	0.00	3.17	2663.72	2663.72	0.00	777.84	2663.72	2663.72	0.00	38.51	23000	2663.72	2663.72	0.00	49.79	619	20	···
c25_100_30_V_T_5	10304.11	10304.11	0.00	0.11	10304.11	10304.11	0.00	3.32	10304.11	10304.11	0.00	1317.30	10304.10	10304.11	0.00	38.11	23000	10304.11	10304.11	0.00	102.07	676	28	···
c33	11619.18	11619.18	0.00	0.16	11619.18	11619.18	0.00	7.83	11619.18	11619.18	0.00	3376.83	11619.18	11619.19	0.00	32.42	18400	11619.18	11619.19	0.00	33.11	456	23	···
c35	7760.86	7760.86	0.00	0.31	7760.86	7760.86	0.00	7.35	−∞	12297.01	100.00	3600.10	7760.86	7760.87	0.00	32.76	18400	7760.86	7760.87	0.00	32.72	474	28	···
c36	14945.85	14945.85	0.00	0.32	14945.84	14945.85	0.00	8.00	−∞	15014.98	100.00	3600.10	14945.84	14945.85	0.00	32.61	18400	14945.84	14945.85	0.00	38.81	483	26	···
c37	1622.34	1622.34	0.00	1.82	1622.34	1622.34	0.00	36.83	−∞	+∞	100.0	3600.10	1622.34	1622.34	0.00	33.52	18400	1622.34	1622.34	0.00	119.89	557	31	···
c38	2135.34	2135.34	0.00	2.00	2135.34	2135.34	0.00	30.57	1969.57	2179.70	9.64	3600.10	2135.34	2135.34	0.00	33.86	18400	2135.34	2135.34	0.00	137.47	527	27	···
c39	1671.80	1671.80	0.00	1.72	1671.80	1671.80	0.00	34.53	1545.28	2547.17	39.33	3600.10	1671.80	1671.80	0.00	34.11	18400	1671.80	1671.80	0.00	106.95	515	27	···
c40	2385.52	2385.52	0.00	1.78	2385.52	2385.52	0.00	38.50	−∞	+∞	100.0	3600.10	2385.52	2385.53	0.00	34.02	18400	2385.52	2385.52	0.00	117.76	504	26	···
c41	12507.39	12507.39	0.00	0.36	12507.38	12507.39	0.00	8.21	−∞	20035.00	100.00	3600.10	12507.38	12507.39	0.00	32.79	18400	12507.38	12507.39	0.00	38.79	466	24	···
c42	11357.60	11357.60	0.00	0.43	11357.59	11357.60	0.00	8.69	10916.76	19372.48	43.65	3600.10	11357.59	11357.60	0.00	32.77	18400	11357.59	11357.60	0.00	41.69	485	22	···
c43	11246.33	11246.33	0.00	0.48	11246.33	11246.33	0.00	10.15	−∞	15937.04	100.00	3600.10	11246.33	11246.33	0.00	32.63	18400	11246.33	11246.34	0.00	38.05	603	23	···
c44	16586.49	16586.49	0.00	0.44	16586.48	16586.49	0.00	8.68	16586.49	16586.49	0.00	2914.95	16586.47	16586.49	0.00	32.63	18400	16586.48	16586.49	0.00	43.17	434	24	···
c45	1178.13	1178.13	0.00	2.23	1178.12	1178.13	0.00	44.94	1106.92	2039.34	45.72	3600.10	1178.12	1178.13	0.00	34.03	18400	1178.12	1178.13	0.00	199.37	523	25	···
c46	1645.80	1645.80	0.00	2.82	1645.79	1645.80	0.00	48.99	−∞	+∞	100.0	3600.10	1645.79	1645.80	0.00	33.37	18400	1645.79	1645.80	0.00	219.98	493	27	···
c47	1348.31	1348.31	0.00	2.78	1348.31	1348.31	0.00	39.97	−∞	+∞	100.0	3600.10	1348.31	1348.31	0.00	33.36	18400	1348.31	1348.31	0.00	215.57	521	26	···
c48	1529.88	1529.88	0.00	2.57	1529.88	1529.88	0.00	48.71	1449.63	2510.86	42.27	3600.10	1529.88	1529.88	0.00	33.02	18400	1529.88	1529.88	0.00	201.81	583	29	···
c49	1341.01	1341.01	0.00	3.60	1341.01	1341.01	0.00	60.87	1257.26	+∞	100.00	3600.10	1341.01	1341.01	0.00	44.02	27600	1341.01	1341.01	0.00	215.06	777	33	···
c50	1644.89	1644.89	0.00	3.44	1644.89	1644.89	0.00	65.93	1535.52	+∞	100.00	3600.10	1644.89	1644.89	0.00	44.50	27600	1644.89	1644.89	0.00	238.02	797	31	···
c51	1222.96	1222.96	0.00	3.49	1222.96	1222.96	0.00	64.82	1176.84	+∞	100.00	3600.10	1222.96	1222.96	0.00	45.41	27600	1222.96	1222.96	0.00	351.25	705	28	···
c52	1966.09	1966.09	0.00	3.30	1966.09	1966.09	0.00	58.12	1830.74	+∞	100.00	3600.10	1966.09	1966.09	0.00	45.34	27600	1966.09	1966.09	0.00	202.66	773	35	···
c53	2249.46	2249.46	0.00	17.29	2249.46	2249.46	0.00	312.86	−∞	+∞	100.0	3600.10	2249.46	2249.46	0.00	51.39	27600	2249.46	2249.46	0.00	1261.40	840	34	···
c54	2646.94	2646.94	0.00	19.78	2646.94	2646.94	0.00	233.89	−∞	+∞	100.0	3600.10	2646.94	2646.94	0.00	50.99	27600	2646.94	2646.94	0.00	1665.80	776	34	···
c55	2565.65	2565.65	0.00	18.64	2565.65	2565.65	0.00	376.77	−∞	+∞	100.0	3600.10	2565.65	2565.65	0.00	51.06	27600	2565.65	2565.65	0.00	1571.84	781	33	···
c56	2869.83	2869.83	0.00	16.30	2869.82	2869.83	0.00	334.49	−∞	+∞	100.0	3600.10	2869.82	2869.83	0.00	51.02	27600	2869.82	2869.83	0.00	1707.87	792	31	···
c57	975.53	975.53	0.00	5.29	975.53	975.53	0.00	106.07	927.29	+∞	100.00	3600.10	975.53	975.53	0.00	44.42	27600	975.53	975.53	0.00	254.20	720	30	···
c58	1319.37	1319.37	0.00	5.01	1319.37	1319.37	0.00	111.80	1215.76	+∞	100.00	3600.10	1319.37	1319.37	0.00	44.52	27600	1319.37	1319.37	0.00	238.62	838	28	···
c59	1183.03	1183.03	0.00	5.69	1183.03	1183.03	0.00	101.99	1132.08	+∞	100.00	3600.10	1183.03	1183.03	0.00	43.85	27600	1183.03	1183.03	0.00	279.39	913	25	···
c60	1231.80	1231.80	0.00	5.64	1231.80	1231.80	0.00	111.51	1166.94	+∞	100.00	3600.10	1231.80	1231.80	0.00	44.13	27600	1231.80	1231.80	0.00	395.60	791	34	···
c61	1645.08	1645.08	0.00	30.88	1645.08	1645.08	0.00	413.23	−∞	+∞	100.0	3600.10	1645.07	1645.08	0.00	50.87	27600	1645.08	1645.08	0.00	1146.25	1003	26	···
c62	1907.28	1907.28	0.00	29.60	1907.28	1907.28	0.00	329.45	−∞	+∞	100.0	3600.10	1907.28	1907.29	0.00	52.57	27600	1907.28	1907.29	0.00	2067.24	803	27	···
c63	1639.85	1639.85	0.00	29.49	1639.85	1639.85	0.00	313.02	−∞	+∞	100.0	3600.10	1639.85	1639.85	0.00	53.02	27600	1639.85	1639.85	0.00	1287.74	731	28	···
c64	1782.78	1782.78	0.00	17.27	1782.77	1782.77	0.00	404.59	−∞	+∞	100.0	3600.10	1782.77	1782.78	0.00	42.48	27600	1782.77	1782.78	0.00	1417.51	831	29	···
#S			42				42				8				41					37				···
SGM			0.00	4.09			0.00	43.46			30.94	7917.73			0.01	47.75	27753			0.00	357.59	762	28	···

Table 93 Detailed results for problem NLMCFP-N, cost functions f_8

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	1140.22	1213.10	6.01	3600.01	1007.94	1225.24	17.74	3602.38	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1201.15	1226.27	**2.05**	3601.02	802	6	···
c100_400_10_F_T_10	4624.83	7542.17	**38.68**	3600.01	3599.53	7413.03	51.44	3602.87	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	600	1	···
c100_400_10_V_L_10	1798.74	1828.29	1.62	3600.01	1557.25	1852.65	15.94	3602.32	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1826.67	1836.46	**0.53**	3601.02	1022	8	···
c100_400_30_F_L_10	4529.94	4951.85	8.52	3600.02	4264.82	4974.68	14.27	3602.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	4809.40	4968.62	**3.20**	3601.03	712	3	···
c100_400_30_F_T_10	11564.18	13679.97	15.47	3600.01	11477.89	13609.18	15.66	3602.33	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	12299.33	13638.00	**9.82**	3601.05	678	2	···
c100_400_30_V_T_10	30187.58	35266.22	**14.40**	3600.01	26094.20	35310.22	26.10	3602.45	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	600	1	···
c25_100_10_F_L_5	1840.97	1840.97	**0.00**	**41.46**	1840.97	1840.97	**0.00**	366.24	1426.37	1913.59	25.46	3600.10	−∞	+∞	100.00	3600.01	+∞	1840.97	1840.97	**0.00**	395.50	434	17	···
c25_100_10_F_T_5	2936.07	2936.07	**0.00**	**329.05**	2771.24	2936.10	5.61	3602.66	2030.24	2936.51	30.86	3600.10	−∞	+∞	100.00	3600.01	+∞	2936.07	2936.07	**0.00**	2580.69	486	18	···
c25_100_10_V_L_5	835.84	835.84	**0.00**	**1.88**	835.84	835.84	**0.00**	17.82	751.39	848.58	11.45	3600.10	−∞	+∞	100.00	3600.01	+∞	835.84	835.84	**0.00**	28.41	381	16	···
c25_100_30_F_L_5	4225.90	4225.90	**0.00**	**1047.93**	3854.27	4250.96	9.33	3602.25	−∞	4337.92	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	4225.71	4225.98	0.01	3601.02	346	14	···
c25_100_30_F_T_5	6988.43	7013.25	0.35	3600.01	6793.81	7016.91	3.18	3602.18	−∞	7109.35	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	7012.36	7014.44	**0.03**	3601.02	330	12	···
c25_100_30_V_T_5	33001.05	33176.86	0.53	3600.01	31392.27	33180.59	5.39	3602.23	24528.93	33349.99	26.45	3600.10	−∞	+∞	100.00	3600.01	+∞	33171.88	33181.00	**0.03**	3601.02	302	9	···
c33	30803.21	30803.21	**0.00**	**2.75**	30803.22	30803.22	**0.00**	53.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	30803.20	30803.22	**0.00**	54.71	354	22	···
c35	26803.44	26803.44	**0.00**	**1.48**	26803.44	26803.44	**0.00**	1544.26	17984.21	27766.56	35.23	3600.10	−∞	+∞	100.00	3600.01	+∞	26803.43	26803.44	**0.00**	71.49	352	22	···
c36	52703.81	52703.83	**0.00**	**10.92**	52703.84	52703.84	**0.00**	1271.79	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	52703.79	52703.84	**0.00**	281.65	382	22	···
c37	9147.88	9147.88	**0.00**	**899.31**	9147.88	9147.88	**0.00**	2762.95	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9138.41	9156.23	0.19	3601.03	225	9	···
c38	13126.64	13126.65	**0.00**	**216.98**	10076.75	13126.65	23.23	3603.15	7086.15	13232.33	46.45	3600.10	−∞	+∞	100.00	3600.01	+∞	13122.98	13126.65	0.03	3601.03	250	9	···
c39	9136.51	9136.51	**0.00**	**403.31**	7252.23	9137.20	20.63	3603.32	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9132.66	9137.06	0.05	3601.03	237	10	···
c40	12471.48	12471.49	**0.00**	**357.02**	10374.02	12472.37	16.82	3602.80	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	12470.20	12471.49	0.01	3601.03	258	11	···
c41	40703.08	40703.08	**0.00**	**1.09**	40703.08	40703.08	**0.00**	600.75	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	40703.05	40703.08	**0.00**	107.35	335	21	···
c42	41704.04	41704.04	**0.00**	**3.62**	41704.04	41704.04	**0.00**	1500.62	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	41704.01	41704.04	**0.00**	81.25	408	22	···
c43	36538.98	36538.98	**0.00**	**2.84**	36538.98	36538.98	**0.00**	534.84	23873.89	36541.75	34.67	3600.10	−∞	+∞	100.00	3600.01	+∞	36538.96	36538.98	**0.00**	47.17	312	15	···
c44	57375.33	57375.35	**0.00**	**2.01**	57375.35	57375.35	**0.00**	1372.79	37091.23	57543.92	35.54	3600.10	−∞	+∞	100.00	3600.01	+∞	57375.33	57375.35	**0.00**	45.90	358	20	···
c45	6327.34	6327.34	**0.00**	**48.34**	6327.34	6327.34	**0.00**	272.84	3591.09	7240.31	50.40	3600.10	−∞	+∞	100.00	3600.01	+∞	6327.33	6327.34	**0.00**	1215.60	330	12	···
c46	9882.69	9882.69	**0.00**	**57.31**	9882.69	9882.69	**0.00**	382.20	5352.01	9897.95	45.93	3600.10	−∞	+∞	100.00	3600.01	+∞	9882.69	9882.69	**0.00**	1042.53	367	15	···
c47	6826.74	6826.74	**0.00**	**364.19**	5499.04	6829.35	19.48	3602.90	3933.71	7715.00	49.01	3600.10	−∞	+∞	100.00	3600.01	+∞	6825.91	6826.74	0.01	3601.10	290	12	···
c48	8709.35	8709.35	**0.00**	**56.56**	8709.35	8709.35	**0.00**	329.15	4778.91	8717.00	45.18	3600.10	−∞	+∞	100.00	3600.01	+∞	8709.35	8709.35	**0.00**	1384.07	364	13	···
c49	5066.30	5066.30	**0.00**	**94.34**	5066.30	5066.30	**0.00**	581.35	3172.87	5091.70	37.69	3600.10	−∞	+∞	100.00	3600.01	+∞	5066.30	5066.30	**0.00**	1455.00	502	17	···
c50	7924.44	7924.44	**0.00**	**43.94**	5772.61	7926.29	27.17	3602.59	4648.28	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	7924.44	7924.44	**0.00**	1222.41	492	18	···
c51	4595.12	4595.12	**0.00**	**195.37**	3720.08	4595.12	19.04	3602.91	2915.26	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	4595.04	4595.12	0.00	3601.02	447	14	···
c52	7905.99	7906.00	**0.00**	**645.41**	7906.00	7906.00	**0.00**	1481.07	4829.39	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	7905.93	7906.00	**0.00**	3601.03	494	16	···
c53	9658.60	9658.60	**0.00**	**3359.00**	6868.82	9658.98	28.89	3605.43	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9498.87	9679.46	1.87	3601.04	247	3	···
c54	12399.68	12419.54	**0.16**	3600.14	12369.69	12420.15	0.41	3608.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	12169.89	12443.82	2.20	3601.04	240	3	···
c55	8757.22	8757.23	**0.00**	**2312.40**	6581.35	8793.26	25.15	3606.37	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8622.27	8757.50	1.54	3601.08	252	3	···
c56	11905.30	11905.31	**0.00**	**1328.07**	8141.16	11961.73	31.94	3607.94	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	11824.61	11918.93	0.79	3601.03	276	4	···
c57	4331.88	4331.88	**0.00**	**34.70**	4331.88	4331.88	**0.00**	672.59	2586.56	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	4331.88	4331.88	**0.00**	1237.68	535	20	···
c58	8109.99	8109.99	**0.00**	**59.31**	5682.66	8115.45	29.98	3604.73	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8109.98	8109.99	**0.00**	1096.06	516	20	···
c59	4738.89	4738.89	**0.00**	**67.12**	−∞	+∞	100.0	+∞	2922.24	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	4738.89	4738.89	**0.00**	1543.11	537	19	···
c60	5120.72	5120.72	**0.00**	**34.11**	−∞	+∞	100.0	+∞	3137.57	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	5120.72	5120.72	**0.00**	1721.97	524	19	···
c61	8237.07	8237.07	**0.00**	**334.11**	8237.07	8237.07	**0.00**	2208.88	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8218.77	8237.07	0.22	3601.05	308	5	···
c62	10858.88	10858.88	**0.00**	**859.99**	7314.14	10858.88	32.64	3608.19	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	10829.10	10860.98	0.29	3601.05	311	5	···
c63	7234.09	7234.09	**0.00**	**1791.77**	7231.82	7234.57	0.04	3613.11	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7207.72	7237.36	0.41	3601.05	302	5	···
c64	9861.26	9861.27	**0.00**	**538.24**	9861.27	9861.27	**0.00**	3172.85	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9803.48	9866.78	0.64	3601.10	286	5	···
#S			**34**				18				0				0					19				···
SGM			**0.43**	**342.00**			4.25	3924.59			72.59	14400.00			100.00	14400.00	+∞			0.59	2942.18	387	11	···

Table 94 Detailed results for problem NLMCFP-N, cost functions f_9

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	419.41	509.66	17.71	3600.01	417.53	587.85	28.97	3603.11	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	481.50	481.50	**0.00**	**1201.32**	2148	37	···
c100_400_10_F_T_10	64.19	125.33	48.78	3600.01	60.00	159.03	62.27	3602.99	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	120.77	120.77	**0.00**	3601.02	1866	29	···
c100_400_10_V_L_10	547.83	730.31	24.99	3600.01	546.31	967.56	43.54	3602.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	666.29	666.29	**0.00**	3601.02	2165	36	···
c100_400_30_F_L_10	512.80	1088.21	52.88	3600.01	501.36	1361.75	63.18	3602.45	91.02	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	984.99	985.20	**0.02**	3601.02	1632	23	···
c100_400_30_F_T_10	675.72	4365.97	84.52	3600.01	669.93	4363.54	84.65	3602.53	59.90	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	3450.30	3480.42	**0.87**	3601.03	1234	8	···
c100_400_30_V_T_10	6896.67	17189.16	59.88	3600.02	6852.17	16493.96	58.46	3602.46	4019.26	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	14290.72	15136.12	**5.59**	3601.02	946	3	···
c25_100_10_F_L_5	209.89	420.32	50.06	3600.01	212.51	387.54	45.16	3602.30	103.71	600.54	82.73	3600.10	−∞	+∞	100.00	3600.01	+∞	387.54	387.54	**0.00**	**56.00**	628	31	···
c25_100_10_F_T_5	317.21	351.47	9.75	3600.01	313.34	351.47	10.85	3602.23	144.53	355.13	59.30	3600.10	−∞	+∞	100.00	3600.01	+∞	351.47	351.47	**0.00**	**53.49**	611	27	···
c25_100_10_V_L_5	356.07	511.67	30.41	3600.01	408.71	505.35	19.12	3602.22	150.76	705.58	78.63	3600.10	−∞	+∞	100.00	3600.01	+∞	505.35	505.35	**0.00**	**98.87**	521	26	···
c25_100_30_F_L_5	370.33	1651.42	77.57	3600.01	520.48	1550.80	66.44	3602.24	106.18	2135.28	95.03	3600.10	−∞	+∞	100.00	3600.01	+∞	1550.80	1550.80	**0.00**	3601.02	603	26	···
c25_100_30_F_T_5	1769.55	5718.10	69.05	3600.01	2425.15	5537.60	56.21	3602.17	180.14	6801.23	97.35	3600.10	−∞	+∞	100.00	3600.01	+∞	5537.37	5537.63	**0.00**	3601.02	455	15	···
c25_100_30_V_T_5	5144.73	21977.27	76.59	3600.01	6400.19	19869.00	67.79	3602.22	1129.55	22645.81	95.01	3600.10	−∞	+∞	100.00	3600.01	+∞	19868.83	19869.02	**0.00**	3601.02	512	22	···
c33	18725.56	20136.35	7.01	3600.02	19610.26	20105.66	2.46	3602.15	61.02	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	20105.64	20105.66	**0.00**	**92.33**	703	23	···
c35	9362.93	10913.45	14.21	3600.01	10098.21	10774.49	6.28	3602.37	44.05	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	10774.48	10774.49	**0.00**	**57.82**	449	25	···
c36	25506.23	28219.15	9.61	3600.01	26388.66	27547.53	4.21	3602.42	84.73	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	27547.51	27547.54	**0.00**	**96.43**	474	24	···
c37	635.74	1587.76	59.96	3600.02	362.37	1634.53	77.83	3602.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1587.76	1587.76	**0.00**	**265.43**	537	31	···
c38	1250.29	2007.18	37.71	3600.04	1509.89	2007.18	24.78	3603.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	2007.18	2007.18	**0.00**	**119.06**	592	32	···
c39	708.75	1687.21	57.99	3600.03	484.81	1799.12	73.05	3602.90	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1687.21	1687.21	**0.00**	**382.58**	531	29	···
c40	1206.03	2680.96	55.02	3600.02	25.89	2918.90	99.11	3602.90	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	2550.08	2550.08	**0.00**	**279.55**	509	31	···
c41	22151.26	22571.99	1.86	3600.01	22533.89	22570.08	0.16	3602.25	68.27	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	22570.07	22570.09	**0.00**	**89.82**	413	25	···
c42	20847.57	21099.29	1.19	3600.02	21083.00	21095.33	0.06	3602.07	65.90	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	21095.32	21095.33	**0.00**	**139.17**	451	21	···
c43	19849.08	20506.91	3.21	3600.01	20289.78	20503.51	1.04	3602.24	61.59	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	20503.50	20503.52	**0.00**	**95.11**	422	23	···
c44	29369.78	29573.32	0.69	3600.02	29435.53	29563.53	0.43	3602.42	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	29563.51	29563.54	**0.00**	**92.61**	476	25	···
c45	857.23	1260.64	32.00	3600.02	192.98	1308.10	85.25	3602.70	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1260.64	1260.64	**0.00**	**422.90**	817	26	···
c46	1036.87	1510.59	31.36	3600.03	16.89	1869.76	99.10	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1510.59	1510.59	**0.00**	**299.59**	634	30	···
c47	1369.50	1707.63	19.80	3600.02	466.80	1709.50	72.69	3603.00	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1692.69	1692.70	**0.00**	**472.33**	517	28	···
c48	1315.32	1662.85	20.90	3600.02	15.48	2009.75	99.23	3602.85	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1662.85	1662.85	**0.00**	**257.82**	656	28	···
c49	1563.20	1924.54	18.78	3600.03	1546.12	1953.58	20.86	3602.81	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1896.70	1896.70	**0.00**	**429.69**	703	24	···
c50	1142.54	2058.95	44.51	3600.02	1119.45	2002.16	44.09	3602.62	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1863.05	1863.05	**0.00**	**232.75**	743	33	···
c51	1488.85	2094.85	28.93	3600.03	1470.83	2126.58	30.84	3602.61	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1980.03	1980.03	**0.00**	**1098.24**	771	29	···
c52	1783.06	2847.61	37.38	3600.04	1879.61	3191.79	41.11	3603.57	11.57	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	2735.36	2735.36	**0.00**	**845.61**	835	27	···
c53	1610.75	3321.32	51.50	3600.05	12.14	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	2902.05	2903.48	**0.05**	3601.04	518	18	···
c54	1826.04	3497.07	47.78	3600.05	16.86	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	3180.42	3180.45	**0.00**	3601.04	705	21	···
c55	2000.16	3790.86	47.24	3600.05	10.37	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	3631.84	3678.88	**1.28**	3601.04	348	8	···
c56	2114.13	3866.33	45.32	3600.05	15.59	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	3658.95	3658.98	**0.00**	3601.04	672	23	···
c57	1178.18	1441.52	18.27	3600.05	1232.33	1438.81	14.35	3604.27	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1416.22	1416.22	**0.00**	**425.51**	683	24	···
c58	1423.71	1498.24	4.97	3600.05	1433.14	1498.24	4.35	3604.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1498.24	1498.24	**0.00**	**299.69**	821	25	···
c59	2082.02	2125.96	2.07	3600.03	2117.48	2124.53	0.33	3604.57	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	2124.53	2124.53	**0.00**	**466.81**	629	25	···
c60	1688.68	1935.00	12.73	3600.08	1673.12	2036.01	17.82	3604.80	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1907.52	1907.52	**0.00**	**540.99**	894	24	···
c61	1472.47	1958.35	24.81	3600.06	1447.54	1958.35	26.08	3611.50	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1910.43	1910.43	**0.00**	**2587.97**	1003	26	···
c62	1510.81	1966.24	23.16	3600.07	18.73	1966.24	99.05	3612.31	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1966.23	1966.24	**0.00**	3601.04	713	28	···
c63	1884.61	2477.64	23.94	3600.07	9.23	2572.24	99.64	3607.04	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	2391.47	2391.48	**0.00**	3601.04	774	25	···
c64	1263.71	1817.12	30.46	3600.06	14.56	1884.59	99.23	3610.58	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	1817.11	1817.12	**0.00**	3601.06	770	31	···
#S			0				0				0				0					**28**				···
SGM			22.79	14400.00			27.34	14400.00			97.52	14400.00			100.00	14400.00	+∞			**0.08**	**1045.61**	695	24	···

Table 95 Detailed results for problem NLMCFP-N, cost functions f_{10}

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	23005.10	37303.95	38.33	3600.01	23818.22	37816.40	37.02	3602.71	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.60	287297	37006.95	37303.96	**0.80**	3601.02	1614	4	···
c100_400_10_F_T_10	69328.67	152183.45	54.44	3600.02	77140.93	140759.04	45.20	3602.21	38313.95	+∞	100.00	3600.10	−∞	+∞	100.00	3601.47	295013	111285.09	118742.07	**6.28**	3601.02	1554	2	···
c100_400_10_V_L_10	44545.41	85030.10	47.61	3600.02	46897.28	86281.13	**45.65**	3603.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.99	286141	−∞	+∞	100.00	3601.14	1508	1	···
c100_400_30_F_L_10	51108.69	87161.84	**41.36**	3600.02	52798.46	94537.08	44.15	3602.87	25714.76	+∞	100.00	3600.10	−∞	+∞	100.00	3601.52	319675	−∞	+∞	100.00	3601.09	1578	1	···
c100_400_30_F_T_10	**241022.50**	**436240.62**	**44.75**	**3600.01**	**1297914.42**	**1297914.42**	**0.00**	**118.46**	**152532.90**	**+∞**	**100.00**	**3600.10**	**−∞**	**+∞**	**100.00**	**3601.54**	**314348**	**−∞**	**+∞**	**100.00**	**3601.09**	**1581**	**1**	···
c100_400_30_V_T_10	1103678.60	1692592.18	**34.79**	3600.01	1116427.49	8003554.78	86.05	3602.70	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.54	313879	−∞	+∞	100.00	3601.08	1573	1	···
c25_100_10_F_L_5	15823.04	24534.60	35.51	3600.01	16440.40	25037.69	34.34	3602.50	7263.32	33278.27	78.17	3600.10	−∞	+∞	100.00	3601.15	71487	24534.60	24534.62	**0.00**	**610.84**	556	13	···
c25_100_10_F_T_5	74947.83	91157.76	17.78	3600.01	77762.45	90284.03	13.87	3602.89	42499.42	97484.06	56.40	3600.10	90283.98	90284.03	**0.00**	2591.42	72587	90283.98	90284.03	**0.00**	**386.71**	1084	12	···
c25_100_10_V_L_5	39566.18	49024.45	19.29	3600.01	37062.58	49083.53	24.49	3602.69	24102.08	74297.42	67.56	3600.10	−∞	+∞	100.00	3601.15	70520	48083.63	48083.66	**0.00**	**760.30**	629	15	···
c25_100_30_F_L_5	54277.86	68771.20	21.07	3600.01	56162.91	67281.49	16.53	3602.75	21361.85	86175.74	75.21	3600.10	−∞	+∞	100.00	3601.15	70777	65438.13	65438.16	**0.00**	**2040.14**	845	13	···
c25_100_30_F_T_5	194338.87	235161.70	17.36	3600.02	203256.05	233912.79	13.11	3602.57	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.17	78860	231016.29	231016.88	**0.00**	3601.03	914	13	···
c25_100_30_V_T_5	1203057.97	1452708.46	17.19	3600.01	1218883.18	1450163.09	15.95	3602.48	570121.60	2293339.00	75.14	3600.10	−∞	+∞	100.00	3601.15	80107	1412034.11	1415841.95	**0.27**	3601.02	499	5	···
c33	424151.11	691531.47	38.66	3600.02	417948.12	711206.36	41.23	3602.76	189758.70	+∞	100.00	3600.10	−∞	+∞	100.00	3601.29	166739	581823.73	597272.26	**2.59**	3601.02	968	4	···
c35	227647.62	630517.11	63.90	3600.01	238456.28	616960.81	61.35	3602.80	181717.50	+∞	100.00	3600.10	−∞	+∞	100.00	3601.32	187186	505034.16	539800.43	**6.44**	3601.02	1002	3	···
c36	381514.60	1102389.12	65.39	3600.02	398044.46	1140067.48	**65.09**	3602.77	294913.40	+∞	100.00	3600.10	−∞	+∞	100.00	3601.33	185765	−∞	+∞	100.00	3601.07	920	1	···
c37	47097.30	151927.55	69.00	3600.03	49805.67	134526.81	**62.98**	3603.40	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.39	191275	−∞	+∞	100.00	3601.15	912	1	···
c38	67355.02	217749.02	**69.07**	3600.04	70935.58	647231.20	89.04	3603.06	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.44	192700	−∞	+∞	100.00	3601.29	920	1	···
c39	49179.34	155823.89	68.44	3600.03	52954.19	146359.58	**63.82**	3603.21	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.41	193311	−∞	+∞	100.00	3601.14	916	1	···
c40	67522.67	234975.92	**71.26**	3600.02	73846.48	468442.01	84.24	3603.57	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.44	193152	−∞	+∞	100.00	3601.24	912	1	···
c41	351497.24	679587.81	**48.28**	3600.01	346774.97	9115010.32	96.20	3602.28	193829.30	+∞	100.00	3600.10	−∞	+∞	100.00	3601.34	216778	−∞	+∞	100.00	3601.07	1152	1	···
c42	357214.91	922238.77	**61.27**	3600.01	357983.31	1133632.31	68.42	3602.21	247593.10	+∞	100.00	3600.10	−∞	+∞	100.00	3601.46	224931	−∞	+∞	100.00	3601.07	1176	1	···
c43	286523.89	947551.56	69.76	3600.02	302641.22	880447.60	**65.63**	3602.49	226586.80	+∞	100.00	3600.10	−∞	+∞	100.00	3601.41	235810	−∞	+∞	100.00	3601.08	1176	1	···
c44	353213.55	979263.60	**63.93**	3600.01	367060.47	1027039.14	64.26	3602.55	275875.90	+∞	100.00	3600.10	−∞	+∞	100.00	3601.44	234694	−∞	+∞	100.00	3601.07	1176	1	···
c45	37693.91	134626.20	**72.00**	3600.02	41450.74	386014.20	89.26	3602.80	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.51	249930	−∞	+∞	100.00	3601.17	1176	1	···
c46	56743.92	190945.18	**70.28**	3600.03	62209.25	543952.35	88.56	3603.32	44697.18	243981.40	81.68	3600.10	−∞	+∞	100.00	3601.54	247712	−∞	+∞	100.00	3601.27	1168	1	···
c47	38743.66	140906.78	72.50	3600.04	43045.18	132094.16	**67.41**	3603.56	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.49	247748	−∞	+∞	100.00	3601.25	1164	1	···
c48	53615.72	191033.99	**71.93**	3600.02	59147.65	482232.24	87.73	3603.13	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.63	247635	−∞	+∞	100.00	3601.22	1164	1	···
c49	31050.14	107388.22	**71.09**	3600.03	33579.53	669463.17	94.98	3603.70	22877.35	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.26	2072	1	···
c50	48484.65	166661.36	**70.91**	3600.02	53111.45	927398.17	94.27	3603.23	34832.39	+∞	100.00	3600.10	−∞	+∞	100.00	3601.70	440944	−∞	+∞	100.00	3601.16	2064	1	···
c51	29490.48	113541.81	**74.03**	3600.02	33047.58	490067.79	93.26	3603.23	24869.61	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.24	2076	1	···
c52	50249.04	181965.72	**72.39**	3600.02	54969.72	707518.99	92.23	3602.74	40065.26	+∞	100.00	3600.10	−∞	+∞	100.00	3601.80	441307	−∞	+∞	100.00	3601.22	2068	1	···
c53	58042.05	205209.02	**71.72**	3600.05	-0.00	+∞	100.00	+∞	49383.76	+∞	100.00	3600.10	−∞	+∞	100.00	3602.26	439089	−∞	+∞	100.00	3601.58	2080	1	···
c54	75428.32	272800.49	**72.35**	3600.27	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.23	439155	−∞	+∞	100.00	3601.56	2080	1	···
c55	59698.37	223980.45	**73.35**	3600.07	-0.00	+∞	100.00	+∞	51737.03	+∞	100.00	3600.10	−∞	+∞	100.00	3602.17	438304	−∞	+∞	100.00	3601.69	2064	1	···
c56	77140.89	285981.68	**73.03**	3600.06	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.08	439851	−∞	+∞	100.00	3601.71	2072	1	···
c57	27778.12	93923.86	**70.42**	3600.04	30205.64	952512.16	96.83	3604.12	20214.15	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.26	2720	1	···
c58	31305.77	112724.60	**72.23**	3600.03	34304.09	1011962.17	96.61	3603.42	23521.37	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.29	2720	1	···
c59	25255.58	98175.36	**74.28**	3600.03	28229.69	632241.20	95.53	3602.94	21844.28	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.24	2748	1	···
c60	29958.82	106107.08	**71.77**	3600.03	32988.48	701238.89	95.30	3604.00	23634.93	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.29	2744	1	···
c61	49875.57	178201.34	**72.01**	3600.08	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.90	2740	1	···
c62	66901.87	230203.54	**70.94**	3600.08	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.67	2716	1	···
c63	48797.36	175267.94	**72.16**	3600.08	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.71	2712	1	···
c64	65746.12	241438.34	**72.77**	3600.06	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.54	2732	1	···
#S			0				0				0				1					**4**				···
SGM			54.52	14400.00			63.66	14400.00			95.39	14400.00			89.49	13824.48	24537522			**37.91**	10918.25	1429	2	···

Table 96 Detailed results for problem NLMCFP-A, cost functions f_1

Instance	GUROBI DB	PB	GAP	CPU	SCIP DB	PB	GAP	CPU	COUENNE DB	PB	GAP	CPU	NAIVE DB	PB	GAP	CPU	NP	CN24 DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	13879.13	14256.05	2.64	3600.01	-19799.30	32570.19	160.79	3602.65	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	14233.72	14262.31	**0.20**	3601.02	2142	8	⋯
c100_400_10_F_T_10	48943.48	56596.61	**13.52**	3600.01	-214324.07	377378.19	156.79	3603.06	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	2000	1	⋯
c100_400_10_V_L_10	26519.47	26830.56	1.16	3600.01	-26727.87	48311.15	155.32	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	26534.67	26760.85	**0.85**	3601.02	2089	6	⋯
c100_400_30_F_L_10	31669.67	33622.91	**5.81**	3600.02	-37386.27	235090.68	115.90	3604.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.14	2000	1	⋯
c100_400_30_F_T_10	178529.53	192462.77	**7.24**	3600.02	-61898.04	785981.62	107.88	3602.59	-320613.60	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.15	2000	1	⋯
c100_400_30_V_T_10	745230.53	800021.76	**6.85**	3600.01	-16809.25	4804430.06	100.35	3602.58	-1043383.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.15	2000	1	⋯
c25_100_10_F_L_5	9319.65	9319.66	**0.00**	**166.35**	-9031.47	12148.90	174.34	3602.48	-24708.99	24034.57	197.27	3600.10	−∞	+∞	100.00	3601.20	95400	9319.45	9319.46	**0.00**	2172.16	889	29	⋯
c25_100_10_F_T_5	51072.05	51170.83	0.19	3600.01	-918.03	61341.37	101.50	3602.72	-68385.56	73691.47	192.80	3600.10	−∞	+∞	100.00	3601.20	95400	51138.41	51192.01	**0.10**	3601.02	805	13	⋯
c25_100_10_V_L_5	**18243.80**	**18243.80**	**0.00**	**8.62**	**10035.64**	**18483.83**	**45.71**	**3602.37**	**-6014.40**	**27293.24**	**122.04**	**3600.10**	**−∞**	**+∞**	**100.00**	**3601.20**	**95400**	**16903.98**	**16903.99**	**0.00**	**387.30**	**874**	**23**	⋯
c25_100_30_F_L_5	34785.37	35023.95	**0.68**	3600.02	10872.19	39243.79	72.30	3602.72	-34983.50	51574.91	167.83	3600.10	−∞	+∞	100.00	3601.21	95400	34781.06	35085.84	0.87	3601.02	677	9	⋯
c25_100_30_F_T_5	138555.90	140297.71	**1.24**	3600.01	109591.94	142490.25	23.09	3602.95	-33856.08	171543.00	119.74	3600.10	−∞	+∞	100.00	3601.20	95400	137951.29	141090.05	2.22	3601.02	576	6	⋯
c25_100_30_V_T_5	807355.36	846893.69	4.67	3600.01	603474.82	845416.74	28.62	3602.47	-178816.20	1137481.00	115.72	3600.10	−∞	+∞	100.00	3601.20	95400	831208.12	846222.85	**1.77**	3601.02	581	9	⋯
c33	343039.97	411600.75	16.66	3600.02	-3326062.85	2193476.77	165.95	3602.57	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	360145.27	380661.70	**5.39**	3601.03	1159	2	⋯
c35	217110.57	302784.87	28.30	3600.02	-1874245.23	1005926.89	153.67	3602.50	-1779691.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	258937.17	274780.29	**5.77**	3601.02	1181	2	⋯
c36	368125.20	506141.47	**27.27**	3600.01	-2650691.36	1219885.47	146.02	3602.58	-2412099.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	1150	1	⋯
c37	49297.41	67371.16	**26.83**	3600.02	-520181.79	455555.80	187.58	3603.13	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.18	1140	1	⋯
c38	67824.41	101312.40	**33.05**	3600.03	-768722.46	677256.79	188.10	3602.61	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.24	1150	1	⋯
c39	51665.78	72375.52	**28.61**	3600.05	-497018.43	429558.40	186.43	3602.57	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.15	1145	1	⋯
c40	71232.13	99267.45	**28.24**	3600.02	-563846.27	498329.60	188.38	3602.61	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.16	1140	1	⋯
c41	313719.13	401047.54	**21.78**	3600.02	-3505880.86	1777983.65	150.71	3602.60	-2787001.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	1440	1	⋯
c42	340589.21	476745.75	**28.56**	3600.02	-3833872.48	1840604.31	148.01	3602.14	-3558571.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.13	1470	1	⋯
c43	278389.73	424852.52	**34.47**	3600.02	-3499355.06	1361959.76	138.92	3602.10	-2459286.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.13	1470	1	⋯
c44	340190.16	463999.73	**26.68**	3600.01	-3161126.75	1531961.84	148.46	3602.27	-2849287.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.13	1470	1	⋯
c45	43282.87	51281.46	**15.60**	3600.02	-379672.30	366299.46	196.48	3602.63	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.23	1470	1	⋯
c46	61222.16	97207.75	**37.02**	3600.02	-477628.65	545045.94	187.63	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.20	1460	1	⋯
c47	45339.26	53549.41	**15.33**	3600.03	-334603.04	348019.92	196.14	3603.27	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.16	1455	1	⋯
c48	59724.38	76524.81	**21.95**	3600.02	-395167.67	482173.17	181.96	3603.27	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	1455	1	⋯
c49	34212.54	36969.17	**7.46**	3600.02	-246014.49	417540.93	158.92	3603.20	-281364.70	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.16	2590	1	⋯
c50	53684.57	64411.69	**16.65**	3600.02	-476771.73	601901.43	179.21	3603.20	-468400.70	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.20	2580	1	⋯
c51	34344.75	36076.10	**4.80**	3600.02	-277981.72	308679.42	190.06	3603.14	-273512.80	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	2595	1	⋯
c52	57698.10	62504.23	7.69	3600.02	-290612.04	449040.46	164.72	3602.94	-407939.30	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	58020.76	62653.55	**7.39**	3601.03	2673	2	⋯
c53	67453.25	87804.59	**23.18**	3600.10	-2116012.94	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.55	2600	1	⋯
c54	83327.07	128822.11	**35.32**	3600.06	-2409374.41	811276.53	133.67	3606.04	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.54	2600	1	⋯
c55	71860.33	82532.15	**12.93**	3600.06	-1665170.06	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.56	2580	1	⋯
c56	89499.88	113438.48	**21.10**	3600.06	-1966106.63	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.53	2590	1	⋯
c57	30577.83	32622.98	**6.27**	3600.02	-296562.97	585042.11	150.69	3603.63	-354229.50	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.19	3400	1	⋯
c58	35338.20	38653.30	**8.58**	3600.03	-426474.08	635225.37	167.14	3602.82	-416667.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.24	3400	1	⋯
c59	29341.88	31831.75	**7.82**	3600.05	-193288.74	383662.46	150.38	3602.77	-299770.90	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.20	3435	1	⋯
c60	33890.29	36505.65	**7.16**	3600.04	-337963.24	433586.85	177.95	3603.52	-333459.80	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	34372.95	37070.28	7.28	3601.02	3503	2	⋯
c61	57265.11	67534.61	**15.21**	3600.07	-2643583.69	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.75	3425	1	⋯
c62	74664.14	108590.64	**31.24**	3600.06	-3232951.10	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.72	3395	1	⋯
c63	58577.37	64917.38	**9.77**	3600.07	-2089676.97	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.73	3390	1	⋯
c64	77210.32	92380.94	**16.42**	3600.16	-2612200.72	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.62	3415	1	⋯
#S			**1**				0				0				0					**1**				⋯
SGM			**11.04**	12965.77			93.08	14400.00			100.00	14400.00			100.00	14400.00	8436107989667			38.86	13766.71	1771	2	⋯

Table 97 Detailed results for problem NLMCFP-A, cost functions f_2

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	2541.14	3202.22	20.64	3600.02	2584.58	3202.17	19.29	3603.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	3157.55	3240.42	**2.56**	3601.02	3948	8	⋯
c100_400_10_F_T_10	8902.14	14596.73	39.01	3600.02	9175.07	14372.79	36.16	3602.94	7420.62	14357.11	48.31	3600.10	−∞	+∞	100.00	3601.74	460000	14094.92	14453.81	**2.48**	3601.02	3797	6	⋯
c100_400_10_V_L_10	7237.71	9881.63	26.76	3600.01	7291.60	9856.46	26.02	3603.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.67	460000	9711.93	9945.36	**2.35**	3601.02	4049	8	⋯
c100_400_30_F_L_10	2552.65	3512.31	27.32	3600.02	2721.34	3511.04	22.49	3603.44	205.18	5868.28	96.50	3600.10	3299.28	3314.39	**0.46**	**2936.67**	460000	3219.82	3347.44	3.81	3601.02	3887	5	⋯
c100_400_30_F_T_10	20630.58	29634.66	30.38	3600.02	21184.02	29270.80	27.63	3602.45	4727.42	+∞	100.00	3600.10	−∞	+∞	100.00	3601.74	460000	27259.92	29417.98	**7.34**	3601.02	3622	2	⋯
c100_400_30_V_T_10	85863.72	119857.11	28.36	3600.02	88377.41	119504.65	**26.05**	3603.06	25528.89	+∞	100.00	3600.10	−∞	+∞	100.00	3601.78	460000	−∞	+∞	100.00	3601.18	3600	1	⋯
c25_100_10_F_L_5	3077.87	3507.05	12.24	3600.01	3076.93	3507.05	12.26	3602.62	2088.65	+∞	100.00	3600.10	3506.70	3507.31	0.02	**358.60**	115000	3507.05	3507.06	**0.00**	3601.02	2500	43	⋯
c25_100_10_F_T_5	16868.45	18273.54	7.69	3600.01	16498.76	18270.51	9.70	3603.23	11946.26	18599.99	35.77	3600.10	18270.43	18271.38	0.01	**425.06**	115000	18270.50	18270.52	**0.00**	3601.02	2295	40	⋯
c25_100_10_V_L_5	6564.93	6666.09	1.52	3600.01	6516.74	6666.09	2.24	3602.40	5110.72	6705.63	23.78	3600.10	6656.93	6662.94	0.09	**397.02**	115000	6658.70	6658.71	**0.00**	3601.02	2433	42	⋯
c25_100_30_F_L_5	8373.13	9719.10	13.85	3600.02	8508.66	9719.10	12.45	3602.43	5287.62	9719.10	45.60	3600.10	9712.31	9712.96	0.01	**1028.00**	115000	9712.28	9712.74	**0.00**	3601.01	1870	27	⋯
c25_100_30_F_T_5	27129.60	30032.36	9.67	3600.02	27363.15	30028.91	8.88	3603.28	−∞	+∞	100.0	3600.10	30017.83	30020.26	0.01	**1006.72**	115000	30017.94	30019.52	**0.01**	3601.02	1757	26	⋯
c25_100_30_V_T_5	142760.66	156194.50	8.60	3600.01	144836.34	156190.54	7.27	3602.60	94183.39	158969.70	40.75	3600.10	156190.39	156192.62	**0.00**	**1046.83**	115000	156181.42	156193.00	0.01	3601.02	1639	24	⋯
c33	76160.83	104843.34	27.36	3600.01	83422.40	104827.88	20.42	3602.45	43558.05	221676.20	80.35	3600.10	104542.15	104912.48	0.35	**1610.86**	262200	104554.73	104775.86	**0.21**	3601.02	3386	21	⋯
c35	110824.98	202106.25	45.16	3600.02	113544.82	197349.90	42.47	3602.66	39180.20	363965.50	89.24	3600.10	196912.30	196943.47	**0.02**	**2531.74**	264500	196132.15	197398.47	0.64	3601.02	2921	16	⋯
c36	180537.01	316951.77	43.04	3600.01	179021.24	315011.47	43.17	3602.58	69026.95	694708.80	90.06	3600.10	314530.23	314609.60	**0.03**	**3070.69**	264500	314348.43	314673.68	0.10	3601.02	3538	22	⋯
c37	10168.55	+∞	100.00	+∞	10048.42	23911.94	57.98	3602.50	−∞	+∞	100.0	3600.10	23055.70	23057.05	**0.01**	**1639.39**	262200	22945.65	23110.46	0.71	3601.03	3018	12	⋯
c38	14890.77	+∞	100.00	+∞	15144.08	33783.47	55.17	3602.51	−∞	+∞	100.0	3600.10	33782.79	33783.61	**0.00**	**1067.52**	264500	33229.39	34078.22	2.49	3601.03	2668	6	⋯
c39	12671.96	+∞	100.00	+∞	13248.67	29388.40	54.92	3602.49	3377.58	76444.32	95.58	3600.10	−∞	+∞	100.00	3601.49	263350	28959.37	29703.67	**2.51**	3601.02	2535	8	⋯
c40	20498.25	+∞	100.00	+∞	22359.88	45800.31	51.18	3602.62	9418.99	107525.80	91.24	3600.10	−∞	+∞	100.00	3601.45	262200	45020.92	46040.40	**2.21**	3601.03	2625	8	⋯
c41	61444.92	107190.63	42.68	3600.01	67280.71	107176.50	37.22	3602.24	0.00	287043.70	100.00	3600.10	106430.41	106768.31	**0.32**	**2560.44**	331200	106391.56	106733.57	0.32	3601.02	4159	21	⋯
c42	88514.27	161905.31	45.33	3600.01	96836.17	161858.60	40.17	3602.02	0.00	365082.20	100.00	3600.10	−∞	+∞	100.00	3601.53	338100	159941.63	160964.52	**0.64**	3601.02	3872	16	⋯
c43	114203.75	223277.59	48.85	3600.01	121168.38	222193.09	45.47	3602.11	20511.15	482731.80	95.75	3600.10	−∞	+∞	100.00	3601.62	338100	219541.10	222371.72	**1.27**	3601.03	3554	13	⋯
c44	114344.88	233508.97	51.03	3600.01	114227.02	233249.77	51.03	3602.48	2723.67	449574.30	99.39	3600.10	−∞	+∞	100.00	3601.52	338100	230435.98	233066.82	**1.13**	3601.02	3713	13	⋯
c45	10050.53	+∞	100.00	+∞	10225.87	23779.42	57.00	3602.61	−∞	+∞	100.0	3600.10	23262.29	23263.49	**0.01**	**3265.79**	338100	22462.84	23599.67	4.82	3601.04	3086	5	⋯
c46	14272.96	+∞	100.00	+∞	14452.37	33492.83	56.85	3602.78	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.63	335800	32969.45	33700.18	**2.17**	3601.03	3415	8	⋯
c47	0.00	+∞	100.00	+∞	11675.66	26799.25	56.43	3603.10	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.61	334650	25126.70	26631.61	**5.65**	3601.03	3065	4	⋯
c48	16090.93	+∞	100.00	+∞	16613.66	37238.38	55.39	3603.26	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.59	334650	35464.45	37804.24	**6.19**	3601.02	2851	4	⋯
c49	6042.99	+∞	100.00	+∞	5617.00	15119.63	62.85	3603.22	79.42	+∞	100.00	3600.10	−∞	+∞	100.00	3602.05	595700	12942.27	13765.01	**5.98**	3601.03	5093	4	⋯
c50	0.00	+∞	100.00	+∞	7843.91	20799.34	62.29	3603.46	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3601.91	593400	18529.11	19487.22	**4.92**	3601.03	5323	6	⋯
c51	9628.00	+∞	100.00	+∞	9618.26	21872.54	56.03	3603.12	1163.50	+∞	100.00	3600.10	−∞	+∞	100.00	3601.97	596850	20366.21	22284.40	**8.61**	3601.02	4742	2	⋯
c52	0.00	+∞	100.00	+∞	13629.26	31730.92	57.05	3602.38	1080.06	+∞	100.00	3600.10	−∞	+∞	100.00	3602.09	594550	29740.50	31527.57	**5.67**	3601.03	5048	4	⋯
c53	0.00	+∞	100.00	+∞	13805.59	40229.74	65.68	3605.82	430.72	+∞	100.00	3600.10	−∞	+∞	100.00	3602.59	598000	35633.24	38777.19	**8.11**	3601.06	4869	2	⋯
c54	0.00	+∞	100.00	+∞	18005.79	45959.59	60.82	3606.12	850.28	+∞	100.00	3600.10	−∞	+∞	100.00	3603.28	598000	41021.37	44613.66	**8.05**	3601.04	4942	2	⋯
c55	0.00	+∞	100.00	+∞	21438.41	52437.43	**59.12**	3605.05	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.53	593400	−∞	+∞	100.00	3601.56	4644	1	⋯
c56	0.00	+∞	100.00	+∞	21790.23	+∞	100.00	+∞	6974.48	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.72	4662	1	⋯
c57	4760.84	+∞	100.00	+∞	4436.52	12280.28	63.87	3603.75	0.00	+∞	100.00	3600.10	10875.38	10899.79	**0.22**	**3151.94**	782000	10564.38	11061.22	4.49	3601.02	6946	6	⋯
c58	0.00	+∞	100.00	+∞	4406.82	10840.10	59.35	3603.14	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3604.55	782000	10489.90	11023.64	**4.84**	3601.04	7049	6	⋯
c59	0.00	+∞	100.00	+∞	5813.89	14119.02	58.82	3602.76	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3602.42	790050	13419.22	14251.46	**5.84**	3601.05	6979	5	⋯
c60	0.00	+∞	100.00	+∞	6551.06	17435.68	62.43	3602.59	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	15180.98	16074.93	**5.56**	3601.02	7075	6	⋯
c61	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3602.82	787750	22653.93	24581.06	**7.84**	3601.09	6813	3	⋯
c62	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	29049.47	31668.71	**8.27**	3601.06	6481	2	⋯
c63	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	825.81	+∞	100.00	3600.10	−∞	+∞	100.00	3602.88	779700	29237.82	31923.02	**8.41**	3601.07	6288	2	⋯
c64	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3603.86	785450	35396.18	38522.68	**8.12**	3601.06	6789	3	⋯
#S			0				0				0				**11**					0				⋯
SGM			51.67	14400.00			39.68	14400.00			87.99	14400.00			19.83	6351.34	2188510			**3.41**	14400.00	3839	9	⋯

Table 98 **Detailed results for problem NLMCFP-A, cost functions f_3**

Instance	GUROBI DB	PB	GAP	CPU	SCIP DB	PB	GAP	CPU	COUENNE DB	PB	GAP	CPU	NAIVE DB	PB	GAP	CPU	NP	CN24 DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	2792.85	7103.38	60.68	3600.02	3462.03	6192.88	44.10	3603.07	289.09	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	6102.36	6218.29	**1.86**	3601.02	3898	9	···
c100_400_10_F_T_10	9542.12	24672.43	61.32	3600.02	8586.17	22321.16	61.53	3603.51	4242.97	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	20409.43	21894.12	**6.78**	3601.02	3624	2	···
c100_400_10_V_L_10	8913.67	17507.76	**49.09**	3600.01	8616.63	16951.66	49.17	3602.46	1280.73	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	3600	1	···
c100_400_30_F_L_10	3633.18	8182.07	55.60	3600.02	4013.13	9505.77	57.78	3602.56	63.24	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	7333.69	7868.88	**6.80**	3601.04	3688	2	···
c100_400_30_F_T_10	27960.84	+∞	100.00	+∞	28641.45	62953.89	**54.50**	3602.67	5746.98	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	3600	1	···
c100_400_30_V_T_10	128988.87	+∞	100.00	+∞	131483.12	252280.47	**47.88**	3602.36	43035.43	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	3600	1	···
c25_100_10_F_L_5	4145.80	6326.28	34.47	3600.01	4212.48	6242.23	32.52	3602.80	2185.33	6632.45	67.05	3600.10	−∞	+∞	100.00	3601.23	163500	6239.33	6243.39	**0.07**	3601.02	1685	29	···
c25_100_10_F_T_5	24203.25	27482.79	11.93	3600.01	21843.19	27489.70	20.54	3602.74	12053.72	30967.46	61.08	3600.10	27277.76	27278.78	0.00	**2092.26**	163500	27277.75	27278.28	**0.00**	3601.02	1998	28	···
c25_100_10_V_L_5	9810.19	10552.42	7.03	3600.01	9880.07	10552.38	6.37	3602.40	6346.84	10890.53	41.72	3600.10	10275.03	10281.18	0.06	**2041.08**	163500	10275.70	10275.95	**0.00**	3601.02	1846	28	···
c25_100_30_F_L_5	11269.38	14015.21	19.59	3600.01	11232.37	14169.86	20.73	3602.42	−∞	19870.57	100.00	3600.10	13964.11	13964.81	**0.01**	**2978.85**	163500	13963.77	13964.73	0.01	3601.02	1756	25	···
c25_100_30_F_T_5	39175.50	45243.20	13.41	3600.01	39880.43	45343.45	12.05	3603.85	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.27	163500	45075.21	45098.05	**0.05**	3601.02	1578	21	···
c25_100_30_V_T_5	186252.04	224214.67	16.93	3600.01	192345.71	223834.60	14.07	3602.55	89916.70	226768.30	60.35	3600.10	−∞	+∞	100.00	3601.33	163500	222443.95	222461.76	**0.01**	3601.02	1759	23	···
c33	85451.57	224864.45	62.00	3600.02	98897.40	219654.20	54.98	3602.66	−∞	459471.00	100.00	3600.10	−∞	+∞	100.00	3601.63	372780	214322.86	217670.01	**1.54**	3601.02	2672	10	···
c35	102505.28	329070.98	68.85	3600.01	107760.88	286532.79	62.39	3602.66	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.59	376050	258901.74	281027.42	**7.87**	3601.02	2075	2	···
c36	158486.51	520580.05	69.56	3600.02	170903.52	471325.56	63.74	3602.55	87206.20	+∞	100.00	3600.10	−∞	+∞	100.00	3601.64	376050	421676.86	459590.41	**8.25**	3601.02	2086	2	···
c37	10333.63	+∞	100.00	+∞	9871.97	52655.44	**81.25**	3602.51	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.73	372780	−∞	+∞	100.00	3601.25	2052	1	···
c38	14893.05	+∞	100.00	+∞	14833.98	68429.86	**78.32**	3602.47	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.73	376050	−∞	+∞	100.00	3601.26	2070	1	···
c39	12419.19	+∞	100.00	+∞	12423.47	62415.25	**80.10**	3602.58	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.73	374415	−∞	+∞	100.00	3601.21	2061	1	···
c40	0.00	+∞	100.00	+∞	20499.84	76236.36	**73.11**	3602.63	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.60	372780	−∞	+∞	100.00	3601.24	2052	1	···
c41	58675.11	230124.38	74.50	3600.01	65716.77	224691.61	70.75	3602.26	66.34	602660.20	99.99	3600.10	−∞	+∞	100.00	3601.72	470880	215527.44	223149.96	**3.42**	**3597.55**	2930	9	···
c42	85458.95	355549.60	75.96	3600.01	93234.60	335564.79	72.22	3601.99	2771.60	671296.80	99.59	3600.10	−∞	+∞	100.00	3601.69	480690	295925.06	315043.52	**6.07**	3601.02	2722	5	···
c43	106953.30	388961.35	72.50	3600.01	107301.52	339481.80	68.39	3602.09	40753.22	560876.80	92.73	3600.10	−∞	+∞	100.00	3601.73	480690	302349.70	332556.40	**9.08**	3601.02	2654	2	···
c44	103670.48	469448.06	77.92	3600.01	108782.91	415996.03	**73.85**	3602.13	27177.00	621810.90	95.63	3600.10	−∞	+∞	100.00	3601.70	480690	−∞	+∞	100.00	3601.07	2646	1	···
c45	9658.60	+∞	100.00	+∞	9705.90	43240.85	**77.55**	3602.64	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.83	480690	−∞	+∞	100.00	3601.27	2646	1	···
c46	14254.88	+∞	100.00	+∞	13606.62	70065.36	80.58	3602.88	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.82	477420	60189.98	63784.22	**5.63**	3601.02	2707	3	···
c47	0.00	+∞	100.00	+∞	10909.39	46842.03	**76.71**	3603.11	5921.38	+∞	100.00	3600.10	−∞	+∞	100.00	3602.13	475785	−∞	+∞	100.00	3601.26	2619	1	···
c48	0.00	+∞	100.00	+∞	15539.14	65404.33	**76.24**	3603.14	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.02	475785	−∞	+∞	100.00	3601.23	2619	1	···
c49	5702.99	+∞	100.00	+∞	5220.67	36553.28	**85.72**	3603.46	345.44	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.16	4662	1	···
c50	8021.95	+∞	100.00	+∞	7635.47	41292.44	**81.51**	3602.96	342.71	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	4644	1	···
c51	9242.59	+∞	100.00	+∞	9080.11	35039.72	**74.09**	3603.25	3263.83	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.25	4671	1	···
c52	0.00	+∞	100.00	+∞	12782.11	58900.36	78.30	3602.35	4503.68	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	51704.16	56321.65	**8.20**	3601.02	4700	2	···
c53	0.00	+∞	100.00	+∞	14717.48	100389.20	**85.34**	3605.83	5377.17	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.55	4680	1	···
c54	0.00	+∞	100.00	+∞	16719.61	83696.08	**80.02**	3606.09	6014.64	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.54	4680	1	···
c55	0.00	+∞	100.00	+∞	19912.56	86941.32	77.10	3606.05	12273.58	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	70221.79	76475.24	**8.18**	3601.05	4704	2	···
c56	0.00	+∞	100.00	+∞	21882.90	95051.93	**76.98**	3605.55	13468.86	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.58	4662	1	···
c57	0.00	+∞	100.00	+∞	4120.45	30626.18	**86.55**	3603.74	35.28	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.36	6120	1	···
c58	0.00	+∞	100.00	+∞	4090.39	37312.07	**89.04**	3603.19	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.26	6120	1	···
c59	0.00	+∞	100.00	+∞	5338.80	38188.94	**86.02**	3603.31	76.65	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.29	6183	1	···
c60	0.00	+∞	100.00	+∞	6079.93	47298.73	**87.15**	3603.03	271.13	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.19	6174	1	···
c61	0.00	+∞	100.00	+∞	2295.77	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.65	6165	1	···
c62	0.00	+∞	100.00	+∞	4250.40	+∞	100.00	+∞	267.23	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.69	6111	1	···
c63	0.00	+∞	100.00	+∞	7243.12	+∞	100.00	+∞	3404.58	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.54	6102	1	···
c64	0.00	+∞	100.00	+∞	9217.38	+∞	100.00	+∞	3172.85	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.55	6147	1	···
#S			0				0				0				**3**					0				···
SGM			69.46	14400.00			59.92	14400.00			94.60	14400.00			72.31	12685.36	7147626789			**23.81**	13943.29	3300	4	···

Table 99 Detailed results for problem NLMCFP-A, cost functions f_4

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	1321.26	4365.21	69.73	3600.02	1269.22	3816.11	**66.74**	3602.43	55.05	+∞	100.00	3600.10	−∞	+∞	100.00	3602.89	1170800	−∞	+∞	100.00	3601.07	5200	1	···
c100_400_10_F_T_10	3916.96	16349.32	**76.04**	3600.02	3017.20	22764.69	86.75	3603.47	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.62	1170800	−∞	+∞	100.00	3601.19	5200	1	···
c100_400_10_V_L_10	4003.00	10160.49	60.60	3600.02	3915.36	9853.92	**60.27**	3602.44	983.92	+∞	100.00	3600.10	−∞	+∞	100.00	3602.69	1170800	−∞	+∞	100.00	3601.08	5200	1	···
c100_400_30_F_L_10	1563.53	+∞	100.00	+∞	1757.67	8396.99	**79.07**	3602.94	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.78	1170800	−∞	+∞	100.00	3601.09	5200	1	···
c100_400_30_F_T_10	16094.73	+∞	100.00	+∞	16476.76	36376.28	**54.70**	3602.72	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.97	1170800	−∞	+∞	100.00	3601.11	5200	1	···
c100_400_30_V_T_10	92345.64	+∞	100.00	+∞	91046.91	159563.55	**42.94**	3603.46	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.94	1170800	−∞	+∞	100.00	3601.09	5200	1	···
c25_100_10_F_L_5	1812.21	2521.20	28.12	3600.01	1539.81	2514.42	38.76	3602.37	746.31	3134.39	76.19	3600.10	−∞	+∞	100.00	3601.43	292700	1940.76	1955.26	**0.74**	3601.02	1614	16	···
c25_100_10_F_T_5	**10902.74**	**12569.77**	**13.26**	**3600.01**	**9374.35**	**13591.09**	**31.03**	**3602.77**	**4736.21**	**17868.32**	**73.49**	**3600.10**	**−∞**	**+∞**	**100.00**	**3601.45**	**292700**	**9986.32**	**9986.33**	**0.00**	**1885.10**	**2720**	**33**	···
c25_100_10_V_L_5	**3798.45**	**4048.93**	**6.19**	**3600.01**	**3539.00**	**4048.17**	**12.58**	**3603.24**	**2359.35**	**5069.07**	**53.46**	**3600.10**	**−∞**	**+∞**	**100.00**	**3601.47**	**292700**	**3176.57**	**3176.71**	**0.00**	**3601.02**	**2326**	**30**	···
c25_100_30_F_L_5	**5390.23**	**9367.13**	**42.46**	**3600.01**	**6521.29**	**9251.52**	**29.51**	**3603.23**	**−∞**	**+∞**	**100.0**	**3600.10**	**−∞**	**+∞**	**100.00**	**3601.49**	**292700**	**6441.07**	**6454.31**	**0.21**	**3601.02**	**1711**	**14**	···
c25_100_30_F_T_5	**28917.26**	**36277.28**	**20.29**	**3600.01**	**25837.77**	**36758.47**	**29.71**	**3602.66**	**5519.39**	**+∞**	**100.00**	**3600.10**	**−∞**	**+∞**	**100.00**	**3601.43**	**292700**	**25766.18**	**25916.20**	**0.58**	**3601.01**	**1724**	**12**	···
c25_100_30_V_T_5	118834.54	179713.68	33.88	3600.01	130658.17	178177.26	26.67	3602.63	26880.86	+∞	100.00	3600.10	−∞	+∞	100.00	3601.47	292700	146406.18	146872.52	**0.32**	3601.02	1697	15	···
c33	21711.57	158124.22	86.27	3600.01	19226.24	99622.41	**80.70**	3602.51	0.00	630722.70	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.18	2964	1	···
c35	30874.20	99650.78	69.02	3600.01	29479.20	81834.78	**63.98**	3602.62	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.97	673210	−∞	+∞	100.00	3601.07	2990	1	···
c36	49545.43	158465.76	68.73	3600.02	45517.57	126509.80	64.02	3602.64	20416.59	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	92451.89	102270.75	**9.60**	3601.08	3047	2	···
c37	3149.80	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.13	2964	1	···
c38	4616.90	+∞	100.00	+∞	4136.50	26655.62	**84.48**	3602.51	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.33	2990	1	···
c39	4025.88	+∞	100.00	+∞	3747.93	14084.55	**73.39**	3602.55	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.16	670283	−∞	+∞	100.00	3601.15	2977	1	···
c40	6375.15	+∞	100.00	+∞	6008.90	61289.59	**90.20**	3602.67	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.28	667356	−∞	+∞	100.00	3601.21	2964	1	···
c41	16282.05	175968.58	90.75	3600.02	15447.60	100994.29	**84.70**	3602.26	0.00	760849.10	100.00	3600.10	−∞	+∞	100.00	3602.33	842976	−∞	+∞	100.00	3601.17	3744	1	···
c42	24213.82	221003.23	89.04	3600.01	21776.90	137262.87	**84.13**	3602.04	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.55	860538	−∞	+∞	100.00	3601.07	3822	1	···
c43	31670.60	+∞	100.00	+∞	30184.94	108077.38	**72.07**	3602.12	5934.55	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	3822	1	···
c44	31997.54	202775.75	84.22	3600.02	30144.37	117104.14	**74.26**	3602.56	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	3822	1	···
c45	3149.61	+∞	100.00	+∞	1378.67	15139.87	**90.89**	3644.32	427.16	+∞	100.00	3600.10	−∞	+∞	100.00	3602.52	860538	−∞	+∞	100.00	3601.25	3822	1	···
c46	4485.36	+∞	100.00	+∞	4193.64	17706.98	**76.32**	3602.77	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.41	854684	−∞	+∞	100.00	3601.17	3796	1	···
c47	3578.51	+∞	100.00	+∞	3309.49	13777.18	**75.98**	3603.46	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.77	851757	−∞	+∞	100.00	3601.24	3783	1	···
c48	4977.24	+∞	100.00	+∞	4725.72	16262.74	**70.94**	3603.11	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.36	851757	−∞	+∞	100.00	3601.17	3783	1	···
c49	1842.01	+∞	100.00	+∞	1672.46	10035.16	**83.33**	3603.25	20.57	+∞	100.00	3600.10	−∞	+∞	100.00	3603.24	1516186	−∞	+∞	100.00	3601.25	6734	1	···
c50	2609.35	+∞	100.00	+∞	2345.97	13980.55	**83.22**	3603.22	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	6708	1	···
c51	3024.11	+∞	100.00	+∞	2815.49	15911.89	**82.31**	3603.19	329.13	+∞	100.00	3600.10	−∞	+∞	100.00	3603.26	1519113	−∞	+∞	100.00	3601.34	6747	1	···
c52	4196.50	+∞	100.00	+∞	3918.60	22368.32	82.48	3602.44	301.39	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	12009.06	13204.76	**9.06**	3601.02	6850	2	···
c53	0.00	+∞	100.00	+∞	−∞	+∞	100.0	+∞	99.71	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.55	6760	1	···
c54	0.00	+∞	100.00	+∞	−∞	+∞	100.0	+∞	249.96	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.59	6760	1	···
c55	0.00	+∞	100.00	+∞	6465.20	22432.64	**71.18**	3606.00	2370.79	+∞	100.00	3600.10	−∞	+∞	100.00	3603.42	1510332	−∞	+∞	100.00	3601.63	6708	1	···
c56	0.00	+∞	100.00	+∞	5068.89	26073.68	**80.56**	3621.58	2001.32	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.55	6734	1	···
c57	0.00	+∞	100.00	+∞	1298.06	7293.49	**82.20**	3602.93	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	8840	1	···
c58	0.00	+∞	100.00	+∞	1284.34	8751.31	**85.32**	3603.70	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.27	8840	1	···
c59	0.00	+∞	100.00	+∞	1711.91	9424.51	**81.84**	3602.98	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.19	8931	1	···
c60	0.00	+∞	100.00	+∞	1947.12	9792.87	**80.12**	3603.23	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.47	8918	1	···
c61	0.00	+∞	100.00	+∞	-2261.91	+∞	100.00	+∞	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.91	8905	1	···
c62	0.00	+∞	100.00	+∞	-2053.98	+∞	100.00	+∞	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.12	8827	1	···
c63	0.00	+∞	100.00	+∞	325.08	+∞	100.00	+∞	241.87	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.79	8814	1	···
c64	0.00	+∞	100.00	+∞	691.43	+∞	100.00	+∞	0.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.78	8879	1	···
#S			0				0				0				0					0				···
SGM			88.55	14400.00			76.00	14400.00			99.31	14400.00			100.00	14400.00	7422661175			**71.44**	14400.00	4900	2	···

Table 100 **Detailed results for problem NLMCFP-A, cost functions** f_5

Instance	GUROBI DB	PB	GAP	CPU	SCIP DB	PB	GAP	CPU	COUENNE DB	PB	GAP	CPU	NAIVE DB	PB	GAP	CPU	NP	CN24 DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	369.96	14032.71	97.36	3600.02	-131.85	13019.53	101.01	3602.30	-15062.68	+∞	100.00	3600.10	−∞	+∞	100.00	3601.85	577200	10836.75	11125.56	**2.60**	3601.02	2480	4	···
c100_400_10_F_T_10	-12345.32	52086.97	123.70	3600.02	-14293.67	66390.22	121.53	3602.31	-24723.26	+∞	100.00	3600.10	−∞	+∞	100.00	3601.76	577200	−∞	+∞	100.00	3601.19	2400	1	···
c100_400_10_V_L_10	5805.76	35577.95	83.68	3600.02	3849.01	30508.00	87.38	3602.26	-59648.39	+∞	100.00	3600.10	−∞	+∞	100.00	3601.70	577200	25825.91	27717.81	**6.83**	3601.02	2429	2	···
c100_400_30_F_L_10	3477.64	23964.94	85.49	3600.01	4114.43	24809.54	**83.42**	3602.40	-19140.03	+∞	100.00	3600.10	−∞	+∞	100.00	3601.77	577200	−∞	+∞	100.00	3601.22	2400	1	···
c100_400_30_F_T_10	54010.61	94178.92	**42.65**	3600.02	50937.74	113016.69	54.93	3602.38	-68439.79	+∞	100.00	3600.10	−∞	+∞	100.00	3601.77	577200	−∞	+∞	100.00	3601.19	2400	1	···
c100_400_30_V_T_10	256635.56	429944.43	**40.31**	3600.01	248260.02	454136.03	45.33	3602.33	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.96	577200	−∞	+∞	100.00	3601.10	2400	1	···
c25_100_10_F_L_5	2830.38	9438.99	70.01	3600.01	910.19	9316.28	90.23	3602.08	-1533.83	10535.51	114.56	3600.10	−∞	+∞	100.00	3601.22	144300	8758.24	8769.52	**0.13**	3601.03	880	14	···
c25_100_10_F_T_5	25798.68	37682.09	31.54	3600.01	15779.23	37437.32	57.85	3602.09	3276.20	41570.55	92.12	3600.10	−∞	+∞	100.00	3601.23	144300	36973.23	36993.86	**0.06**	3601.02	985	15	···
c25_100_10_V_L_5	12553.46	15241.32	17.64	3600.01	9326.76	15444.03	39.61	3602.13	3043.99	17275.33	82.38	3600.10	−∞	+∞	100.00	3601.22	144300	14837.14	14837.30	**0.00**	3601.02	1096	28	···
c25_100_30_F_L_5	13933.35	20917.99	33.39	3600.01	12666.76	21023.75	39.75	3602.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.23	144300	20801.49	20805.97	**0.02**	3601.03	1008	15	···
c25_100_30_F_T_5	58658.05	68369.17	14.20	3600.02	56656.27	68084.57	16.79	3602.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.26	144300	67855.56	67871.06	**0.02**	3601.03	1026	15	···
c25_100_30_V_T_5	312447.72	372472.89	16.12	3600.01	289902.76	372485.61	22.17	3602.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.25	144300	368122.00	369652.45	**0.41**	3601.03	839	8	···
c33	-215168.06	553610.68	138.87	3600.01	-202286.48	598645.13	133.79	3602.22	-570795.80	+∞	100.00	3600.10	−∞	+∞	100.00	3601.48	329004	389282.98	397020.58	**1.95**	3601.02	1476	5	···
c35	-90729.32	378409.29	123.98	3600.02	-131385.89	404411.13	132.49	3602.21	-191922.60	+∞	100.00	3600.10	−∞	+∞	100.00	3601.50	331890	323223.19	347694.55	**7.04**	3601.03	1421	2	···
c36	-94194.76	633783.61	114.86	3600.02	-163291.32	642958.48	125.40	3602.25	-235430.10	+∞	100.00	3600.10	−∞	+∞	100.00	3601.46	331890	−∞	+∞	100.00	3601.08	1380	1	···
c37	-23353.07	99624.21	123.44	3600.02	-28100.34	166641.06	116.86	3602.74	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.53	329004	−∞	+∞	100.00	3601.16	1368	1	···
c38	-32831.49	143478.93	122.88	3600.02	-39578.97	233048.14	116.98	3602.99	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.53	331890	−∞	+∞	100.00	3601.25	1380	1	···
c39	-18617.45	105885.32	117.58	3600.02	-23540.21	142007.61	116.58	3602.69	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.54	330447	−∞	+∞	100.00	3601.17	1374	1	···
c40	-16767.90	133004.01	112.61	3600.02	-24067.79	178041.25	113.52	3602.74	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.54	329004	99529.03	106326.82	**6.39**	3601.03	1468	2	···
c41	-339795.03	569142.49	159.70	3600.01	-354977.62	647108.83	154.86	3602.40	-593111.30	+∞	100.00	3600.10	−∞	+∞	100.00	3601.56	415584	−∞	+∞	100.00	3601.09	1728	1	···
c42	-375097.63	653085.85	157.43	3600.02	-421250.57	2269539.87	118.56	3602.30	-624024.80	+∞	100.00	3600.10	−∞	+∞	100.00	3602.22	424242	−∞	+∞	100.00	3601.09	1764	1	···
c43	-199146.59	538883.55	136.96	3600.02	-257884.13	1396776.86	118.46	3603.40	-340399.40	+∞	100.00	3600.10	−∞	+∞	100.00	3601.58	424242	413837.28	446076.97	**7.23**	3601.02	1806	2	···
c44	-274886.48	657073.62	141.83	3600.02	-327168.46	1780334.13	118.38	3602.72	-428737.60	+∞	100.00	3600.10	−∞	+∞	100.00	3601.58	424242	−∞	+∞	100.00	3601.09	1764	1	···
c45	-18709.92	89030.77	121.02	3600.03	-23055.40	138513.17	116.64	3603.34	-29232.69	+∞	100.00	3600.10	−∞	+∞	100.00	3601.67	424242	−∞	+∞	100.00	3601.26	1764	1	···
c46	-26468.70	128560.88	120.59	3600.03	-32784.19	207223.40	115.82	3603.29	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.67	421356	−∞	+∞	100.00	3601.30	1752	1	···
c47	-16928.56	95983.10	117.64	3600.05	-21716.73	132039.98	116.45	3603.32	-27421.08	+∞	100.00	3600.10	−∞	+∞	100.00	3602.42	419913	−∞	+∞	100.00	3601.19	1746	1	···
c48	-20656.47	121370.81	117.02	3600.02	-27724.08	182701.39	115.17	3603.52	-35475.73	+∞	100.00	3600.10	−∞	+∞	100.00	3601.94	419913	−∞	+∞	100.00	3601.33	1746	1	···
c49	-30257.29	138162.28	121.90	3600.03	-34558.45	177044.17	119.52	3604.09	-46387.94	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.18	3108	1	···
c50	-47390.89	217750.53	121.76	3600.02	-65778.46	317558.10	120.71	3604.95	-68396.88	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	3096	1	···
c51	-22495.79	123878.01	118.16	3600.03	-68163.01	162980.69	141.82	3605.17	-34635.08	+∞	100.00	3600.10	−∞	+∞	100.00	3602.08	748917	−∞	+∞	100.00	3601.17	3114	1	···
c52	-32688.88	173084.02	118.89	3600.03	-98069.35	248287.52	139.50	3605.25	-50011.79	+∞	100.00	3600.10	−∞	+∞	100.00	3602.19	746031	−∞	+∞	100.00	3601.29	3102	1	···
c53	-38544.86	168339.76	122.90	3600.06	-107233.78	+∞	100.00	+∞	-56871.60	+∞	100.00	3600.10	−∞	+∞	100.00	3602.55	750360	−∞	+∞	100.00	3601.64	3120	1	···
c54	-43782.94	194001.76	122.57	3600.05	-122100.55	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.27	750360	−∞	+∞	100.00	3601.65	3120	1	···
c55	-24826.54	198924.07	112.48	3600.06	-84386.29	+∞	100.00	+∞	-43195.55	+∞	100.00	3600.10	−∞	+∞	100.00	3602.38	744588	−∞	+∞	100.00	3601.59	3096	1	···
c56	-31692.62	196335.40	116.14	3600.06	-99636.94	+∞	100.00	+∞	-52519.62	+∞	100.00	3600.10	−∞	+∞	100.00	3602.29	747474	−∞	+∞	100.00	3601.65	3108	1	···
c57	-43480.87	186659.15	123.29	3600.03	-141249.56	232432.19	160.77	3606.17	-65110.81	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.33	4080	1	···
c58	-48868.75	207933.86	123.50	3600.03	-148703.78	275815.51	153.91	3603.61	-71266.91	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	4080	1	···
c59	-31425.96	146876.17	121.40	3600.03	-89894.86	212740.17	142.26	3604.83	-45816.23	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	4122	1	···
c60	-33993.38	160983.37	121.12	3600.03	-100071.81	231049.66	143.31	3603.60	-50250.17	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.32	4116	1	···
c61	-49512.63	170099.45	129.11	3600.07	-133969.64	+∞	100.00	+∞	-71056.12	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.69	4110	1	···
c62	-59186.09	222265.53	126.63	3600.10	-163837.17	+∞	100.00	+∞	-86563.50	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.69	4074	1	···
c63	-37166.55	149975.72	124.78	3600.09	-105899.15	+∞	100.00	+∞	-55661.21	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.70	4068	1	···
c64	-48588.93	197218.47	124.64	3600.08	-132379.23	+∞	100.00	+∞	-70793.96	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.00	4098	1	···
#S			0				0				0				0					0				···
SGM			79.02	14400.00			84.13	14400.00			99.36	14400.00			100.00	14400.00	35011380			**35.13**	14400.00	2124	3	···

Table 101 Detailed results for problem NLMCFP-A, cost functions f_6

	GUROBI				SCIP				COUENNE				NAIVE					CN24						...
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	...
c100_400_10_F_L_10	17475.57	20483.43	**14.68**	3600.01	16674.99	20818.87	19.90	3602.34	−∞	21656.33	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	2000	1	...
c100_400_10_F_T_10	49655.67	71279.45	**30.34**	3600.02	45981.36	70575.03	34.85	3604.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	2000	1	...
c100_400_10_V_L_10	36173.43	41558.16	**12.96**	3600.01	34808.42	41711.34	16.55	3602.56	−∞	52376.96	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	2000	1	...
c100_400_30_F_L_10	29741.79	45168.12	34.15	3600.01	30014.48	44607.42	32.71	3602.53	21403.07	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	43112.69	44902.64	**3.99**	3601.02	2133	3	...
c100_400_30_F_T_10	150359.02	204927.23	26.63	3600.01	150731.59	199646.88	**24.50**	3602.86	106389.70	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	2000	1	...
c100_400_30_V_T_10	663528.86	849014.42	**21.85**	3600.02	594019.56	833776.43	28.76	3602.86	525621.60	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	2000	1	...
c25_100_10_F_L_5	14012.14	14012.14	**0.00**	**1426.24**	−∞	+∞	100.0	+∞	10623.62	15253.35	30.35	3600.10	−∞	+∞	100.00	3600.01	+∞	14011.83	14014.01	0.02	3601.02	1101	15	...
c25_100_10_F_T_5	65254.14	66688.54	2.15	3600.01	63891.14	66842.33	4.42	3602.74	46458.85	67296.51	30.96	3600.10	−∞	+∞	100.00	3600.01	+∞	66049.07	66752.21	**1.05**	3601.02	583	4	...
c25_100_10_V_L_5	**26746.74**	**26746.74**	**0.00**	**72.71**	**26746.74**	**26746.74**	**0.00**	**247.07**	**23077.07**	**27008.06**	**14.55**	**3600.10**	**−∞**	**+∞**	**100.00**	**3600.01**	**+∞**	**26125.43**	**26144.14**	**0.07**	**3601.02**	**942**	**12**	...
c25_100_30_F_L_5	38913.17	42100.39	**7.57**	3600.01	38311.23	42795.46	10.48	3602.79	22070.07	46360.04	52.39	3600.10	−∞	+∞	100.00	3600.01	+∞	38879.70	42508.31	8.54	3601.02	516	2	...
c25_100_30_F_T_5	147229.92	156797.24	6.10	3600.01	147453.12	156199.37	**5.60**	3602.45	83910.48	162757.40	48.44	3600.10	−∞	+∞	100.00	3600.01	+∞	144993.03	157273.75	7.81	3601.02	528	2	...
c25_100_30_V_T_5	826309.43	855009.85	3.36	3600.01	784477.72	853051.37	8.04	3602.78	464618.40	940446.70	50.60	3600.10	−∞	+∞	100.00	3600.01	+∞	847924.89	853765.05	**0.68**	3601.02	559	5	...
c33	536056.56	712867.84	24.80	3600.01	529681.22	718673.68	26.30	3602.54	−∞	895831.50	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	579213.19	641703.91	**9.74**	3601.02	1148	2	...
c35	393580.64	463144.32	15.02	3600.01	400848.43	467920.87	**14.33**	3602.75	−∞	648932.50	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	1150	1	...
c36	649589.02	799845.22	**18.79**	3600.02	656943.92	811520.61	19.05	3602.48	584306.40	1036117.00	43.61	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	1150	1	...
c37	73220.74	106425.22	**31.20**	3600.02	73639.03	108827.61	32.33	3603.21	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	1140	1	...
c38	105037.73	153807.48	31.71	3600.03	106640.70	147157.58	**27.53**	3602.81	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.15	1150	1	...
c39	78679.50	107467.90	**26.79**	3600.02	79413.44	108947.07	27.11	3602.88	73975.88	124877.10	40.76	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	1145	1	...
c40	106829.96	145129.17	**26.39**	3600.02	106798.33	149348.16	28.49	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.15	1140	1	...
c41	477793.33	696095.42	31.36	3600.02	477062.45	680236.93	**29.87**	3601.98	386461.70	879614.30	56.06	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	1440	1	...
c42	589440.09	748547.83	21.26	3600.01	591145.25	750019.55	**21.18**	3602.07	492690.10	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	1470	1	...
c43	498030.88	607354.81	18.00	3600.01	508474.66	604663.53	**15.91**	3602.07	448170.40	1009619.00	55.61	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	1470	1	...
c44	598874.20	731141.84	18.09	3600.01	608197.94	733324.24	**17.06**	3602.16	546666.40	1129370.00	51.60	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	1470	1	...
c45	61314.94	81868.07	25.11	3600.02	62318.12	82206.42	**24.19**	3602.67	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.16	1470	1	...
c46	91521.52	123953.79	26.16	3600.03	92772.19	123825.79	**25.08**	3603.44	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.18	1460	1	...
c47	63714.03	81542.03	**21.86**	3600.04	64081.25	86800.21	26.17	3603.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.20	1455	1	...
c48	88444.71	113412.33	**22.01**	3600.02	88167.47	114444.45	22.96	3603.05	84733.46	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.18	1455	1	...
c49	46101.68	66025.83	30.18	3600.02	44285.53	70984.26	37.61	3602.80	44156.46	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	57545.45	63717.31	**9.69**	3601.02	2601	2	...
c50	70479.81	112605.72	**37.41**	3600.02	67559.76	110841.59	39.05	3603.29	67559.76	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.16	2580	1	...
c51	47647.96	58885.98	**19.08**	3600.02	47347.78	64200.66	26.25	3602.44	46616.36	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	2595	1	...
c52	78344.49	110417.91	**29.05**	3600.02	77801.63	111660.44	30.32	3603.28	76709.71	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.20	2585	1	...
c53	97755.94	+∞	100.00	+∞	96690.87	142358.72	**32.08**	3606.58	96690.87	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.53	2600	1	...
c54	123963.06	+∞	100.00	+∞	122679.60	190248.75	**35.52**	3605.74	122679.60	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.64	2600	1	...
c55	101628.79	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	100648.60	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.68	2580	1	...
c56	127690.71	+∞	100.00	+∞	126356.53	253480.80	**50.15**	3605.44	126356.50	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.55	2590	1	...
c57	40439.20	59374.43	**31.89**	3600.02	39055.20	61231.55	36.22	3602.68	39055.20	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.24	3400	1	...
c58	47026.95	71849.35	**34.55**	3600.02	45708.84	73238.27	37.59	3602.90	45708.84	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.20	3400	1	...
c59	41603.53	53956.61	**22.89**	3600.02	41977.03	59885.75	29.90	3603.19	40991.78	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	3435	1	...
c60	45587.59	64013.78	**28.78**	3600.07	44810.50	68558.88	34.64	3602.90	44810.50	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.22	3430	1	...
c61	80094.64	+∞	100.00	+∞	79059.22	123696.39	**36.09**	3608.73	79059.22	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.79	3425	1	...
c62	107685.63	+∞	100.00	+∞	106564.04	+∞	100.00	+∞	106564.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.69	3395	1	...
c63	81885.74	+∞	100.00	+∞	81216.58	+∞	100.00	+∞	81216.58	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.91	3390	1	...
c64	110194.12	+∞	100.00	+∞	109364.24	+∞	100.00	+∞	109364.20	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.02	3415	1	...
#S			**1**				0				0				0					0				...
SGM			**25.42**	13630.18			28.24	14400.00			82.79	14400.00			100.00	14400.00	+∞			54.85	14400.00	1744	2	...

Table 102 Detailed results for problem NLMCFP-A, cost functions f_7

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	1367.40	1367.40	0.00	0.26	1367.40	1367.40	0.00	20.31	$-\infty$	1390.94	100.00	3600.10	1362.78	1377.92	1.10	1086.40	368000	1354.17	1380.14	1.88	3601.03	4143	11	···
c100_400_10_F_T_10	7156.05	7156.05	0.00	0.32	7156.04	7156.05	0.00	16.66	$-\infty$	8995.30	100.00	3600.10	7151.90	7162.96	0.15	1162.92	368000	7120.60	7182.52	0.86	3601.02	4552	13	···
c100_400_10_V_L_10	4322.70	4322.70	0.00	0.31	4322.70	4322.70	0.00	20.37	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	4258.41	4359.34	2.32	3601.02	4212	8	···
c100_400_30_F_L_10	1439.14	1439.14	0.00	0.95	1439.13	1439.14	0.00	77.34	87.19	$+\infty$	100.00	3600.10	1341.48	1364.20	1.67	2468.62	368000	1281.37	1379.27	7.10	3601.03	3807	3	···
c100_400_30_F_T_10	13871.21	13871.21	0.00	1.18	13871.21	13871.21	0.00	72.41	4396.46	$+\infty$	100.00	3600.10	13645.30	13662.82	0.13	2878.68	368000	13395.56	13792.59	2.88	3601.02	4009	6	···
c100_400_30_V_T_10	54385.87	54385.87	0.00	1.16	54385.83	54385.87	0.00	75.28	18868.80	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.61	368000	51712.43	53605.12	3.53	3601.03	4028	6	···
c25_100_10_F_L_5	1473.83	1473.83	0.00	0.07	1473.83	1473.83	0.00	4.24	1473.83	1473.83	0.00	289.50	1473.13	1474.17	0.07	120.17	92000	1473.83	1473.83	0.00	1019.55	2515	41	···
c25_100_10_F_T_5	9638.42	9638.42	0.00	0.05	9638.42	9638.42	0.00	3.42	9638.42	9638.42	0.00	309.67	9638.11	9639.13	0.01	119.88	92000	9638.42	9638.43	0.00	537.84	2363	38	···
c25_100_10_V_L_5	2846.54	2846.54	0.00	0.04	2846.53	2846.54	0.00	3.54	2846.54	2846.54	0.00	211.61	2838.67	2846.08	0.26	131.96	92000	2841.19	2841.20	0.00	1307.61	2358	39	···
c25_100_30_F_L_5	4537.73	4537.73	0.00	0.14	4537.72	4537.72	0.00	5.63	4537.73	4537.73	0.00	2239.03	4503.86	4504.68	0.02	118.40	92000	4504.24	4504.25	0.00	3601.02	2421	37	···
c25_100_30_F_T_5	15936.95	15936.95	0.00	0.13	15936.94	15936.95	0.00	5.01	15936.95	15936.95	0.00	1545.99	15724.09	15724.72	0.00	142.72	92000	15724.18	15724.19	0.00	3231.13	2675	43	···
c25_100_30_V_T_5	85389.99	85389.99	0.00	0.08	85389.94	85389.98	0.00	6.08	85389.99	85389.99	0.00	1832.60	85389.62	85392.45	0.00	117.44	92000	85389.93	85390.04	0.00	3601.02	2340	37	···
c33	43314.18	43314.18	0.00	0.33	43314.15	43314.18	0.00	26.05	$-\infty$	44061.71	100.00	3600.10	42969.01	43571.73	1.38	322.83	209760	43224.50	43250.07	0.06	3601.02	3973	27	···
c35	92864.14	92864.14	0.00	0.34	92864.06	92864.14	0.00	27.86	92848.19	93304.04	0.49	3434.01	92134.98	92192.32	0.06	297.83	211600	92149.74	92151.42	0.00	3601.02	4988	47	···
c36	141243.11	141243.11	0.00	0.32	141243.03	141243.11	0.00	26.21	$-\infty$	384000.80	100.00	3600.10	141218.30	141324.46	0.08	310.70	211600	141240.68	141244.70	0.00	3601.02	4888	44	···
c37	9520.56	9520.56	0.00	1.91	9520.55	9520.56	0.00	140.44	$-\infty$	$+\infty$	100.0	3600.10	9518.45	9521.03	0.03	367.64	209760	9445.44	9570.16	1.30	3601.03	2881	10	···
c38	13951.76	13951.76	0.00	1.21	13951.75	13951.76	0.00	142.68	1648.27	40827.25	95.96	3600.10	13950.17	13952.82	0.02	406.72	211600	13752.71	14110.03	2.53	3601.02	2630	7	···
c39	12363.09	12363.09	0.00	1.42	12363.08	12363.09	0.00	126.10	$-\infty$	$+\infty$	100.0	3600.10	12362.97	12363.47	0.00	431.46	210680	12295.75	12407.38	0.90	3601.02	2927	11	···
c40	19670.81	19670.81	0.00	1.94	19670.80	19670.81	0.00	115.27	$-\infty$	$+\infty$	100.0	3600.10	19670.62	19671.72	0.01	412.97	209760	19429.13	19832.33	2.03	3601.02	2594	9	···
c41	44164.03	44164.03	0.00	0.46	44164.02	44164.03	0.00	34.37	0.00	125023.00	100.00	3600.10	43560.74	44187.01	1.42	414.13	264960	43869.92	43949.60	0.18	3601.02	4561	25	···
c42	66772.25	66772.25	0.00	0.46	66772.22	66772.25	0.00	35.37	0.00	172886.40	100.00	3600.10	66201.00	66708.66	0.76	417.64	270480	66335.54	66437.45	0.15	3601.03	4725	26	···
c43	96742.19	96742.19	0.00	0.51	96742.09	96742.19	0.00	34.35	24453.07	126726.10	80.70	3600.10	96345.65	96455.79	0.11	434.69	270480	96330.87	96389.88	0.06	3601.02	4995	31	···
c44	97236.45	97236.45	0.00	0.28	97236.38	97236.45	0.00	35.77	5648.84	135530.70	95.83	3600.10	96966.83	97046.40	0.08	375.29	270480	96994.21	97004.18	0.01	3601.02	5653	37	···
c45	9735.05	9735.05	0.00	2.76	9735.04	9735.05	0.00	197.40	$-\infty$	$+\infty$	100.0	3600.10	9733.34	9735.75	0.02	550.43	270480	9613.19	9806.51	1.97	3601.30	3506	7	···
c46	13899.34	13899.34	0.00	2.81	13899.33	13899.34	0.00	191.88	2923.57	43645.11	93.30	3600.10	13897.51	13900.69	0.02	507.89	268640	13637.16	14074.59	3.11	3601.03	3313	6	···
c47	11118.60	11118.60	0.00	1.77	11118.59	11118.60	0.00	190.08	$-\infty$	$+\infty$	100.0	3600.10	11118.56	11118.62	0.00	387.75	267720	10960.44	11218.83	2.30	3601.03	3289	9	···
c48	15629.95	15629.95	0.00	1.71	15629.94	15629.95	0.00	182.96	5967.41	43449.39	86.27	3600.10	15628.57	15630.75	0.01	396.27	267720	15426.66	15742.57	2.01	3601.03	3436	8	···
c49	5642.24	5642.24	0.00	2.86	5642.24	5642.24	0.00	293.91	124.82	10107.88	98.77	3600.10	5558.88	5571.93	0.23	1408.54	476560	5400.10	5667.11	4.71	3601.03	5275	6	···
c50	7881.43	7881.43	0.00	3.35	7881.43	7881.43	0.00	323.07	$-\infty$	$+\infty$	100.0	3600.10	7875.10	7884.83	0.12	1272.63	474720	7656.72	8004.91	4.35	3601.02	5335	6	···
c51	9385.24	9385.24	0.00	3.14	9385.24	9385.24	0.00	238.76	1502.77	$+\infty$	100.00	3600.10	9381.50	9388.08	0.07	1723.42	477480	9208.32	9509.85	3.17	3601.03	5575	7	···
c52	13112.25	13112.25	0.00	2.12	13112.25	13112.25	0.00	276.19	$-\infty$	$+\infty$	100.0	3600.10	13108.51	13114.94	0.05	821.65	475640	12657.08	13300.26	4.84	3601.02	5280	5	···
c53	15956.73	15956.73	0.00	10.84	15956.72	15956.73	0.00	1729.88	$-\infty$	$+\infty$	100.0	3600.10	15956.04	15956.88	0.01	1159.16	478400	15055.97	16336.22	7.84	3601.05	5042	2	···
c54	18088.42	18088.42	0.00	17.44	18088.41	18088.42	0.00	1733.81	$-\infty$	$+\infty$	100.0	3600.10	18087.89	18088.55	0.00	1168.16	478400	17092.92	18559.37	7.90	3601.04	5193	3	···
c55	21539.61	21539.61	0.00	17.81	21539.59	21539.61	0.00	1921.26	$-\infty$	$+\infty$	100.0	3600.10	21539.54	21539.87	0.00	1239.91	474720	20270.35	22083.00	8.21	3601.04	5139	3	···
c56	23677.59	23677.59	0.00	12.67	23677.58	23677.59	0.00	1531.86	$-\infty$	$+\infty$	100.0	3600.10	23676.57	23678.07	0.01	1448.54	476560	22369.94	24237.81	7.71	3601.05	5019	2	···
c57	4456.80	4456.80	0.00	5.60	4456.80	4456.80	0.00	592.45	$-\infty$	$+\infty$	100.0	3600.10	4431.08	4473.55	0.95	3003.81	625600	4314.92	4531.75	4.78	3601.02	7133	7	···
c58	4429.52	4429.52	0.00	5.31	4429.52	4429.52	0.00	799.75	$-\infty$	$+\infty$	100.0	3600.10	4409.03	4440.48	0.71	2460.05	625600	4240.97	4519.71	6.17	3601.02	6866	5	···
c59	5780.69	5780.69	0.00	4.91	5780.69	5780.69	0.00	654.35	$-\infty$	$+\infty$	100.0	3600.10	5776.80	5782.83	0.10	1742.97	632040	5607.05	5872.42	4.52	3601.06	7159	6	···
c60	6582.58	6582.58	0.00	4.65	6582.58	6582.58	0.00	644.55	0.00	$+\infty$	100.00	3600.10	6547.98	6555.77	0.12	1830.75	631120	6363.32	6648.28	4.29	3601.05	7329	7	···
c61	9857.47	9857.47	0.00	30.17	9857.46	9857.47	0.00	2980.73	0.00	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	9544.13	10016.49	4.72	3601.06	7211	5	···
c62	12696.85	12696.85	0.00	30.69	12696.84	12696.85	0.00	3322.54	$-\infty$	$+\infty$	100.0	3600.10	12694.88	12697.96	0.02	1805.98	624680	12251.24	12921.68	5.19	3601.19	6946	4	···
c63	13028.72	13028.72	0.00	25.80	13028.70	13028.72	0.00	3446.72	971.73	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3602.63	623760	12560.05	13249.63	5.20	3601.06	6849	4	···
c64	15601.49	15601.49	0.00	19.05	15601.47	15601.48	0.00	3021.19	6.81	$+\infty$	100.00	3600.10	15600.73	15601.67	0.01	1737.35	628360	14714.96	16028.66	8.20	3601.07	6904	3	···
#S			29				29				3				20					2				···
SGM			0.00	5.28			0.00	251.32			61.11	10289.61			0.77	978.00	1257646			2.73	11743.85	4364	9	···

Table 103 Detailed results for problem NLMCFP-A, cost functions f_8

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	16202.20	18802.61	**13.83**	3600.01	14308.29	19723.14	27.45	3602.77	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	17566.13	22713.92	22.66	3601.02	2649	2	···
c100_400_10_F_T_10	47238.77	70569.78	**33.06**	3600.01	39682.55	97398.72	59.26	3602.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	2400	1	···
c100_400_10_V_L_10	34484.20	38729.04	**10.96**	3600.01	31764.05	42154.03	24.65	3602.69	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	35749.46	55703.53	35.82	3601.03	2525	2	···
c100_400_30_F_L_10	28591.33	45952.24	37.78	3600.01	27890.15	42922.10	**35.02**	3602.92	21990.49	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	2400	1	···
c100_400_30_F_T_10	147073.00	232410.10	**36.72**	3600.02	137393.61	233869.88	41.25	3602.80	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	2400	1	···
c100_400_30_V_T_10	637100.10	958555.49	33.54	3600.01	650550.81	884758.44	**26.47**	3602.45	514375.30	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	2400	1	···
c25_100_10_F_L_5	12934.43	12934.43	**0.00**	**778.87**	12393.22	13271.94	6.62	3603.03	−∞	16551.52	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	12934.42	12934.44	**0.00**	1579.36	1645	18	···
c25_100_10_F_T_5	59094.90	60850.83	2.89	3600.01	58129.21	60912.26	4.57	3602.51	40297.91	66183.23	39.11	3600.10	−∞	+∞	100.00	3600.01	+∞	60669.76	61050.99	**0.62**	3601.02	937	6	···
c25_100_10_V_L_5	**24708.58**	**24708.58**	**0.00**	**97.53**	**22068.88**	**25194.64**	**12.41**	**3602.37**	**−∞**	**26917.09**	**100.00**	**3600.10**	**−∞**	**+∞**	**100.00**	**3600.01**	**+∞**	**24105.37**	**24117.27**	**0.05**	**3601.01**	**1568**	**20**	···
c25_100_30_F_L_5	36506.26	39647.07	7.92	3600.01	32916.68	40039.72	17.79	3602.77	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	37205.32	40391.45	**7.89**	3601.03	659	2	···
c25_100_30_F_T_5	136189.00	144914.20	6.02	3600.01	134226.36	144881.02	7.35	3602.63	−∞	154339.30	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	140782.72	146630.95	**3.99**	3601.02	664	3	···
c25_100_30_V_T_5	783020.03	820435.90	4.56	3600.01	759668.29	819309.36	7.28	3602.75	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	810470.70	823445.97	**1.58**	3601.02	797	6	···
c33	510826.01	659942.69	22.60	3600.02	485097.19	702191.85	30.92	3602.52	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	579538.03	720586.73	**19.57**	3601.02	1784	3	···
c35	390543.37	447691.71	**12.77**	3600.02	372366.18	457253.35	18.56	3603.36	359174.60	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	409188.92	484860.39	15.61	3601.03	1570	2	···
c36	643664.28	797655.39	**19.31**	3600.01	594494.29	839532.32	29.19	3602.49	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	1380	1	···
c37	70152.29	115154.06	**39.08**	3600.02	71478.32	138751.29	48.48	3602.76	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	1368	1	···
c38	101194.21	166986.94	**39.40**	3600.02	101106.12	194957.81	48.14	3602.47	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.16	1380	1	···
c39	76099.79	117652.18	**35.32**	3600.02	74974.05	141159.86	46.89	3602.93	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.15	1374	1	···
c40	103266.00	157996.50	**34.64**	3600.02	102086.19	193153.66	47.15	3602.69	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.14	1368	1	···
c41	463005.19	660810.97	29.93	3600.01	432073.84	706379.00	38.83	3602.34	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	578851.72	787390.56	**26.48**	3601.02	2005	2	···
c42	574011.40	727880.20	**21.14**	3600.01	526933.50	824022.14	36.05	3602.00	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	1764	1	···
c43	491052.88	587688.64	**16.44**	3600.01	463945.45	642728.61	27.82	3602.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	528433.68	646817.38	18.30	3601.02	2002	2	···
c44	596304.13	719069.13	**17.07**	3600.01	556356.23	729894.67	23.78	3602.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	642293.92	786410.01	18.33	3601.02	1925	2	···
c45	60907.30	84994.95	28.34	3600.02	61264.22	82293.69	**25.55**	3603.11	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.18	1764	1	···
c46	89675.04	134070.29	**33.11**	3600.03	88421.75	160126.19	44.78	3602.67	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	1752	1	···
c47	61833.18	87684.75	29.48	3600.02	63564.50	87504.45	**27.36**	3602.73	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.18	1746	1	···
c48	85930.58	118967.51	**27.77**	3600.02	85104.54	151762.94	43.92	3603.10	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	1746	1	···
c49	44918.36	+∞	100.00	+∞	44588.22	+∞	100.00	+∞	44156.47	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.14	3108	1	···
c50	68809.82	+∞	100.00	+∞	67559.76	+∞	100.00	+∞	67559.78	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	3096	1	···
c51	47295.07	+∞	100.00	+∞	46616.36	+∞	100.00	+∞	46616.37	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	3114	1	···
c52	77640.88	+∞	100.00	+∞	76822.92	+∞	100.00	+∞	76709.73	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	3102	1	···
c53	97213.01	+∞	100.00	+∞	96692.81	168389.96	**42.58**	3605.54	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.51	3120	1	···
c54	123642.72	+∞	100.00	+∞	122679.60	+∞	100.00	+∞	122679.60	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.53	3120	1	···
c55	101011.81	+∞	100.00	+∞	100770.02	165936.98	**39.27**	3605.57	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.64	3096	1	···
c56	127059.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.56	3108	1	···
c57	39857.27	+∞	100.00	+∞	39147.42	+∞	100.00	+∞	39055.21	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.26	4080	1	···
c58	46347.83	+∞	100.00	+∞	45708.84	+∞	100.00	+∞	45708.85	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.19	4080	1	···
c59	41483.33	+∞	100.00	+∞	40991.78	+∞	100.00	+∞	40991.79	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.20	4122	1	···
c60	45431.94	+∞	100.00	+∞	44810.50	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.23	4116	1	···
c61	79364.56	+∞	100.00	+∞	79059.22	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.88	4110	1	···
c62	107123.49	+∞	100.00	+∞	106564.04	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.33	4074	1	···
c63	81559.95	+∞	100.00	+∞	81220.99	137645.36	**40.99**	3606.54	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.03	4068	1	···
c64	110133.56	+∞	100.00	+∞	109364.24	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.62	4098	1	···
#S			**1**				0				0				0					**1**				···
SGM			**34.49**	13436.97			40.98	14400.00			97.80	14400.00			100.00	14400.00	+∞			50.46	13663.12	2175	2	···

Table 104 Detailed results for problem NLMCFP-A, cost functions f_9

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	69.05	3304.73	97.91	3600.01	76.45	1404.80	94.56	3602.76	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	717.72	756.50	**5.13**	3601.02	3686	2	···
c100_400_10_F_T_10	218.45	41523.59	99.47	3600.02	282.54	5472.07	94.84	3603.00	150.78	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	4089.93	4275.52	**4.34**	3601.03	3895	5	···
c100_400_10_V_L_10	187.78	18070.65	98.96	3600.01	241.38	5016.86	95.19	3603.31	134.22	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	2571.13	2746.04	**6.37**	3601.02	3674	2	···
c100_400_30_F_L_10	183.63	16089.14	98.86	3600.01	163.01	8075.93	**97.98**	3603.81	74.97	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	3600	1	···
c100_400_30_F_T_10	829.53	105326.88	99.21	3600.02	932.27	22238.54	95.81	3603.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	8179.47	8710.27	**6.09**	3601.02	3789	2	···
c100_400_30_V_T_10	4571.17	183340.52	97.51	3600.01	5681.47	75880.27	**92.51**	3602.55	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	3600	1	···
c25_100_10_F_L_5	89.61	908.24	90.13	3600.01	56.41	912.19	93.82	3602.55	26.70	3131.24	99.15	3600.10	−∞	+∞	100.00	3600.01	+∞	672.71	672.71	**0.00**	3601.02	2552	41	···
c25_100_10_F_T_5	608.10	12899.81	95.29	3600.01	557.33	8156.81	93.17	3602.40	294.86	8996.74	96.72	3600.10	−∞	+∞	100.00	3600.01	+∞	7078.66	7078.67	**0.00**	**600.94**	2391	42	···
c25_100_10_V_L_5	279.31	1657.75	83.15	3600.01	256.06	1798.52	85.76	3602.77	190.21	3695.07	94.85	3600.10	−∞	+∞	100.00	3600.01	+∞	1451.42	1451.43	**0.00**	**1912.51**	1833	40	···
c25_100_30_F_L_5	177.98	7359.82	97.58	3600.01	203.31	5199.64	96.09	3602.78	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	2926.52	2926.52	**0.00**	**1241.62**	2438	40	···
c25_100_30_F_T_5	766.23	19052.17	95.98	3600.01	1105.70	16138.29	93.15	3602.63	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	10683.33	10683.35	**0.00**	3601.02	2211	36	···
c25_100_30_V_T_5	4709.50	91858.69	94.87	3600.01	5600.99	75947.96	92.63	3602.83	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	58949.84	58950.78	**0.00**	3601.03	1976	32	···
c33	1431.94	29747.73	95.19	3600.01	1836.19	42650.70	95.69	3602.46	780.79	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	22892.59	24671.20	**7.21**	3601.02	2213	3	···
c35	920.20	128534.77	99.28	3600.02	860.94	80927.20	98.94	3602.79	880.51	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	48359.23	48514.70	**0.32**	3601.02	3266	18	···
c36	1365.19	276062.75	99.51	3600.02	1315.07	120176.57	98.91	3602.91	1350.19	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	71784.32	71790.55	**0.01**	3600.16	4041	38	···
c37	163.95	31051.54	99.47	3600.02	148.55	51898.12	99.71	3602.83	134.76	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	3920.91	4037.73	**2.89**	3601.02	2711	6	···
c38	235.11	40893.20	99.43	3600.02	223.49	82806.57	99.73	3602.54	193.63	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	5742.75	5872.27	**2.21**	3601.02	2748	7	···
c39	180.53	34355.78	99.47	3600.03	144.55	50173.29	99.71	3602.81	147.30	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	5397.13	5562.31	**2.97**	3601.03	2641	6	···
c40	234.22	54315.02	99.57	3600.02	193.15	60942.05	99.68	3602.66	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	11225.44	11520.06	**2.56**	3601.03	2484	10	···
c41	1163.18	42244.37	97.25	3600.02	1212.20	54194.29	97.76	3602.02	783.68	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	23740.19	24487.08	**3.05**	3601.02	3066	6	···
c42	1269.18	247047.29	99.49	3600.02	1321.22	70240.31	98.12	3602.10	1060.72	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	32186.19	33969.03	**5.25**	3601.02	3001	4	···
c43	1066.68	117382.76	99.09	3600.01	881.00	102138.37	99.14	3602.14	1089.83	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	45722.68	45881.46	**0.35**	3601.03	4039	20	···
c44	1231.10	384237.54	99.68	3600.02	1038.89	106575.39	99.03	3602.24	1234.88	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	42155.54	43627.56	**3.37**	3601.02	3354	6	···
c45	143.84	35946.26	99.60	3600.02	96.44	45500.18	99.79	3602.66	121.47	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	4025.52	4222.91	**4.67**	3601.04	3195	4	···
c46	210.87	46227.41	99.54	3600.03	124.79	64955.59	99.81	3602.74	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5704.88	5857.20	**2.60**	3601.02	3411	7	···
c47	144.48	32885.78	99.56	3600.05	115.49	47369.21	99.76	3602.79	130.43	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	4776.27	4928.02	**3.08**	3601.03	3429	6	···
c48	185.18	47189.46	99.61	3600.05	-0.00	+∞	100.00	+∞	175.74	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	6832.44	7068.12	**3.33**	3601.02	3389	6	···
c49	108.15	31060.03	**99.65**	3600.03	91.92	37806.53	99.76	3602.55	93.78	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.25	4662	1	···
c50	165.08	35805.95	**99.54**	3600.02	132.27	64083.90	99.79	3603.31	137.60	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.22	4644	1	···
c51	104.63	41971.02	99.75	3600.02	85.09	40973.12	99.79	3603.43	109.62	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	4158.72	4392.71	**5.33**	3601.03	5356	4	···
c52	169.62	57809.55	99.71	3600.03	125.30	71326.58	99.82	3603.22	166.40	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	5500.71	6041.20	**8.95**	3601.03	4963	2	···
c53	201.34	68589.67	99.71	3600.12	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	6422.16	7053.58	**8.95**	3601.04	5120	2	···
c54	261.80	71461.88	99.63	3600.07	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	7202.59	7893.59	**8.75**	3601.04	5001	2	···
c55	206.09	75948.84	99.73	3600.06	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9369.88	10292.71	**8.97**	3601.05	5277	3	···
c56	267.27	80265.01	99.67	3600.07	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	9610.08	10528.97	**8.73**	3601.04	5087	2	···
c57	92.81	29828.28	**99.69**	3600.03	81.52	41025.80	99.80	3602.87	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.22	6120	1	···
c58	106.33	30865.55	**99.66**	3600.03	69.67	52034.81	99.87	3602.43	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	6120	1	···
c59	83.93	32212.43	**99.74**	3600.03	-0.00	41190.14	100.00	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.24	6183	1	···
c60	100.00	32275.88	**99.69**	3600.02	-0.00	47709.73	100.00	3602.52	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.29	6174	1	···
c61	173.40	49374.31	**99.65**	3600.07	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.95	6165	1	···
c62	232.44	59123.85	**99.61**	3600.07	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.67	6111	1	···
c63	168.83	59858.38	99.72	3600.07	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	5441.11	5960.16	**8.71**	3601.07	6545	2	···
c64	227.71	73770.47	99.69	3600.07	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	6327.46	6921.93	**8.59**	3601.07	6716	2	···
#S			0				0				0				0					**3**				···
SGM			98.31	14400.00			97.86	14400.00			99.78	14400.00			100.00	14400.00	+∞			**7.09**	12060.31	3777	7	···

Table 105 Detailed results for problem NLMCFP-A, cost functions f_{10}

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	22622.96	24515.36	7.72	3600.01	21920.92	24653.56	11.08	3603.11	17365.40	+∞	100.00	3600.10	−∞	+∞	100.00	3601.17	72132	24384.46	24515.37	**0.53**	3601.02	424	3	⋯
c100_400_10_F_T_10	53027.21	59442.30	10.79	3600.01	51119.50	70574.67	27.57	3603.09	36288.70	+∞	100.00	3600.10	−∞	+∞	100.00	3601.16	85606	58950.99	59274.77	**0.55**	3601.02	453	2	⋯
c100_400_10_V_L_10	5450.08	7509.62	**27.43**	3600.02	5470.92	8072.72	32.23	3603.12	3286.76	+∞	100.00	3600.10	−∞	+∞	100.00	3601.16	72169	−∞	+∞	100.00	3601.20	400	1	⋯
c100_400_30_F_L_10	44086.37	56645.75	**22.17**	3600.02	41415.56	149651.16	72.33	3603.29	37303.70	+∞	100.00	3600.10	−∞	+∞	100.00	3601.23	76472	−∞	+∞	100.00	3601.12	400	1	⋯
c100_400_30_F_T_10	75109.15	103239.79	**27.25**	3600.02	73497.23	107931.43	31.90	3603.28	63312.22	+∞	100.00	3600.10	−∞	+∞	100.00	3601.21	84032	−∞	+∞	100.00	**458.70**	396	1	⋯
c100_400_30_V_T_10	39830.25	75248.09	**47.07**	3600.02	33721.78	76656.25	56.01	3603.30	24399.48	+∞	100.00	3600.10	−∞	+∞	100.00	3601.20	82845	−∞	+∞	100.00	3601.10	395	1	⋯
c25_100_10_F_L_5	15742.40	16933.08	7.03	3600.02	15213.98	16933.09	10.15	3602.97	12163.66	18751.82	35.13	3600.10	16933.09	16933.09	**0.00**	2776.76	19668	16933.08	16933.09	**0.00**	**1354.49**	220	9	⋯
c25_100_10_F_T_5	30872.18	32608.74	5.33	3600.01	30619.27	32608.75	6.10	3602.97	27676.50	33153.68	16.52	3600.10	32608.75	32608.75	**0.00**	786.97	21398	32608.73	32608.75	**0.00**	**265.41**	440	8	⋯
c25_100_10_V_L_5	3364.43	3364.43	**0.00**	572.04	3088.73	3364.43	8.19	3602.97	2363.32	3613.98	34.61	3600.10	3364.43	3364.43	**0.00**	445.69	16465	3364.43	3364.43	**0.00**	**152.02**	215	12	⋯
c25_100_30_F_L_5	31004.60	35048.64	11.54	3600.01	30953.47	35001.46	11.57	3603.10	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.08	20717	34994.12	35001.46	**0.02**	3601.02	220	6	⋯
c25_100_30_F_T_5	37971.15	46031.89	17.51	3600.01	38284.56	45642.39	16.12	3603.04	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.09	21351	45460.09	45642.39	**0.40**	3601.01	164	3	⋯
c25_100_30_V_T_5	33945.85	48291.62	29.71	3600.01	33195.55	48756.29	31.92	3603.03	−∞	+∞	100.0	3600.10	47520.90	47520.92	**0.00**	**818.37**	21391	47520.88	47520.92	**0.00**	1719.82	458	22	⋯
c33	51626.88	69487.87	25.70	3600.02	51924.02	71431.55	27.31	3618.78	38838.63	+∞	100.00	3600.10	66998.82	66998.84	**0.00**	**2378.81**	14918	66847.02	67007.52	0.24	3597.20	122	5	⋯
c35	35819.81	40167.89	10.82	3600.02	35678.43	41952.13	14.95	3603.11	22579.95	+∞	100.00	3600.10	39531.23	39531.25	**0.00**	**1395.93**	14691	38682.88	40084.27	3.50	3601.02	96	2	⋯
c36	93481.59	114005.50	**18.00**	3600.02	94128.12	120559.06	21.92	3603.08	72986.39	+∞	100.00	3600.10	−∞	+∞	100.00	3601.13	14667	−∞	+∞	100.00	3601.16	80	1	⋯
c37	30400.90	40766.45	**25.43**	3600.03	30219.57	305900.46	90.12	3603.26	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.29	15292	−∞	+∞	100.00	3601.34	80	1	⋯
c38	63337.60	81603.89	**22.38**	3600.04	63200.53	143622.20	56.00	3603.19	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.27	15095	−∞	+∞	100.00	3601.32	80	1	⋯
c39	32495.95	42424.70	**23.40**	3600.02	32522.99	62171.07	47.69	3603.29	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	15233	−∞	+∞	100.00	3601.31	80	1	⋯
c40	65173.75	84048.92	**22.46**	3600.03	65260.60	133752.28	51.21	3603.23	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	15351	−∞	+∞	100.00	3601.31	80	1	⋯
c41	57195.81	73459.12	**22.14**	3600.01	56794.01	88169.94	35.59	3603.24	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.11	14801	−∞	+∞	100.00	3601.10	80	1	⋯
c42	101327.45	127988.19	**20.83**	3600.02	100032.99	142653.36	29.88	3603.23	81817.34	+∞	100.00	3600.10	−∞	+∞	100.00	3601.10	14450	−∞	+∞	100.00	3601.10	80	1	⋯
c43	44852.91	54284.93	17.38	3600.01	44791.06	58573.25	23.53	3603.43	29581.38	+∞	100.00	3600.10	−∞	+∞	100.00	3601.11	14743	50997.73	53282.98	**4.29**	3601.02	93	2	⋯
c44	98740.24	125719.41	21.46	3600.01	99284.68	134039.13	25.93	3603.26	74908.14	+∞	100.00	3600.10	−∞	+∞	100.00	3601.13	14590	116628.15	119539.66	**2.44**	3601.02	95	2	⋯
c45	23673.95	30598.31	**22.63**	3600.02	23603.50	44057.46	46.43	3602.96	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.34	15554	−∞	+∞	100.00	3601.38	80	1	⋯
c46	53636.49	63624.76	**15.70**	3600.03	53417.30	528873.90	89.90	3602.97	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	15297	−∞	+∞	100.00	3601.42	80	1	⋯
c47	22951.83	30434.84	**24.59**	3600.03	22903.90	220227.92	89.60	3602.95	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.27	15589	−∞	+∞	100.00	3601.38	80	1	⋯
c48	49119.67	58602.90	**16.18**	3600.03	49199.13	88649.01	44.50	3603.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.30	15416	−∞	+∞	100.00	3601.47	80	1	⋯
c49	10329.20	20250.17	**48.99**	3600.04	10183.05	35465.89	71.29	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.29	25275	−∞	+∞	100.00	3601.36	120	1	⋯
c50	31311.15	47830.96	**34.54**	3600.03	30457.39	593478.29	94.87	3603.37	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.30	24987	−∞	+∞	100.00	3601.45	120	1	⋯
c51	7962.62	15043.77	**47.07**	3600.03	7640.98	25807.89	70.39	3603.26	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.30	25210	−∞	+∞	100.00	3601.38	120	1	⋯
c52	30696.81	45809.17	**32.99**	3600.03	29901.18	130861.53	77.15	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	25027	−∞	+∞	100.00	3601.38	120	1	⋯
c53	8566.56	+∞	100.00	+∞	7387.82	222555.69	**96.68**	3606.46	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.65	22521	−∞	+∞	100.00	3602.32	120	1	⋯
c54	11218.17	+∞	100.00	+∞	9450.30	434231.73	**97.82**	3606.38	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.57	22489	−∞	+∞	100.00	3602.31	120	1	⋯
c55	7067.70	+∞	100.00	+∞	5974.69	173350.51	**96.55**	3606.56	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.08	22853	−∞	+∞	100.00	3602.29	120	1	⋯
c56	10464.49	+∞	100.00	+∞	8697.28	350259.36	**97.52**	3606.57	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.00	22535	−∞	+∞	100.00	3602.31	120	1	⋯
c57	8491.80	17255.21	**50.79**	3600.05	3169.75	161456.39	98.04	3603.19	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.38	25142	−∞	+∞	100.00	3601.46	120	1	⋯
c58	17215.73	31110.67	**44.66**	3600.05	7077.62	330593.03	97.86	3603.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.37	24996	−∞	+∞	100.00	3601.53	120	1	⋯
c59	7974.02	16315.58	**51.13**	3600.04	3573.87	107764.57	96.68	3603.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.62	25276	−∞	+∞	100.00	3601.50	120	1	⋯
c60	12710.36	23454.14	**45.81**	3600.04	3697.71	218386.95	98.31	3603.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.43	25208	−∞	+∞	100.00	3601.50	120	1	⋯
c61	7534.50	+∞	100.00	+∞	6259.38	283372.53	**97.79**	3609.12	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.85	22733	−∞	+∞	100.00	3603.27	120	1	⋯
c62	10217.60	+∞	100.00	+∞	8426.12	578603.20	**98.54**	3607.87	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.75	22401	−∞	+∞	100.00	3602.52	120	1	⋯
c63	6272.56	+∞	100.00	+∞	5314.05	219797.34	**97.58**	3607.77	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.24	22842	−∞	+∞	100.00	3602.60	120	1	⋯
c64	9114.21	+∞	100.00	+∞	7608.68	451858.68	**98.32**	3606.26	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.92	22521	−∞	+∞	100.00	3602.61	120	1	⋯
#S			1				0				0				**6**					4				⋯
SGM			**27.66**	13363.72			45.46	14400.00			91.41	14400.00			52.05	10162.91	23417			30.91	9530.30	143	2	⋯

Table 106 Detailed results for problem NLMCNDP-N, cost functions f_1

	GUROBI				**SCIP**				**COUENNE**				**NAIVE**					**CN24**						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	20990.24	20990.26	**0.00**	**214.65**	18170.30	21792.45	16.62	3603.11	14062.99	+∞	100.00	3600.10	−∞	+∞	100.00	3601.20	95400	20990.27	20990.31	**0.00**	3601.01	750	13	···
c100_400_10_F_T_10	49436.83	54966.34	**10.06**	3600.01	20579.78	71861.76	71.36	3603.13	3339.69	+∞	100.00	3600.10	−∞	+∞	100.00	3601.23	95400	−∞	+∞	100.00	3601.07	500	1	···
c100_400_10_V_L_10	3580.02	3580.02	**0.00**	**1187.66**	422.86	4609.96	90.83	3603.10	-1442.13	+∞	100.00	3600.10	−∞	+∞	100.00	3601.21	95400	3579.02	3580.15	0.03	3601.01	686	5	···
c100_400_30_F_L_10	40703.80	43974.38	**7.44**	3600.02	31415.94	138366.31	77.30	3603.21	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	95400	−∞	+∞	100.00	3601.11	500	1	···
c100_400_30_F_T_10	72118.92	83182.85	**13.30**	3600.02	49870.17	193890.45	74.28	3603.19	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.26	95400	−∞	+∞	100.00	3601.11	500	1	···
c100_400_30_V_T_10	26330.64	28098.82	**6.29**	3600.02	-13176.99	44928.31	129.33	3603.25	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.26	95400	−∞	+∞	100.00	3601.10	500	1	···
c25_100_10_F_L_5	12971.09	12971.11	**0.00**	**5.48**	−∞	+∞	100.0	+∞	8530.56	15479.13	44.89	3600.10	12971.11	12971.11	**0.00**	2100.36	23850	12971.10	12971.11	**0.00**	181.98	350	8	···
c25_100_10_F_T_5	28183.67	28183.68	**0.00**	**6.43**	25724.60	28183.68	8.73	3602.98	23162.67	28674.39	19.22	3600.10	28183.68	28183.68	**0.00**	1103.25	23850	28183.67	28183.68	**0.00**	114.25	472	7	···
c25_100_10_V_L_5	1574.77	1574.77	**0.00**	**0.15**	1290.67	1574.77	18.04	3602.97	789.11	1611.74	51.04	3600.10	1574.77	1574.77	**0.00**	744.56	23850	1574.77	1574.77	**0.00**	40.77	346	7	···
c25_100_30_F_L_5	28476.02	28476.04	**0.00**	**56.75**	24689.87	28524.73	13.44	3603.12	−∞	+∞	100.0	3600.10	28476.09	28476.09	**0.00**	2417.97	23850	28476.07	28476.09	**0.00**	1021.27	399	8	···
c25_100_30_F_T_5	35733.80	35733.84	**0.00**	**37.34**	32019.12	35733.84	10.40	3603.03	−∞	+∞	100.0	3600.10	35733.83	35733.84	**0.00**	1374.19	23850	35733.82	35733.84	**0.00**	932.75	441	17	···
c25_100_30_V_T_5	23102.27	23118.04	0.07	3600.01	-10.45	24024.08	100.04	3603.07	−∞	+∞	100.0	3600.10	23117.78	23117.79	**0.00**	**1804.82**	23850	23117.62	23117.82	**0.00**	3601.01	404	25	···
c33	40661.03	56093.51	27.51	3600.02	-32900.34	56134.58	158.61	3603.03	-35760.80	+∞	100.00	3600.10	−∞	+∞	100.00	3601.14	19080	50314.95	51543.81	**2.38**	3601.01	113	2	···
c35	27581.50	33048.19	**16.54**	3600.01	-59829.52	34066.26	156.94	3603.11	-48352.01	+∞	100.00	3600.10	−∞	+∞	100.00	3601.15	19080	−∞	+∞	100.00	3601.07	100	1	···
c36	79066.65	91582.31	13.67	3600.02	-74892.93	98728.89	175.86	3602.57	-64043.34	+∞	100.00	3600.10	−∞	+∞	100.00	3601.15	19080	89031.19	91219.45	**2.40**	3601.01	115	2	···
c37	28716.61	238957.00	**87.98**	3600.03	3527.42	60567.00	94.18	3603.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.35	19080	−∞	+∞	100.00	3601.27	100	1	···
c38	60688.26	74608.58	**18.66**	3600.03	24869.26	141570.31	82.43	3686.81	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.34	19080	−∞	+∞	100.00	3601.30	100	1	···
c39	30704.62	37648.75	**18.44**	3600.02	358.00	67533.55	99.47	3603.23	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	19080	−∞	+∞	100.00	3601.31	100	1	···
c40	62698.84	74661.12	**16.02**	3600.03	24978.75	141060.02	82.29	3603.23	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.29	19080	−∞	+∞	100.00	3601.28	100	1	···
c41	41574.35	57199.85	**27.32**	3600.02	-60603.50	69152.94	187.64	3603.29	-60345.68	+∞	100.00	3600.10	−∞	+∞	100.00	3601.13	19080	−∞	+∞	100.00	3601.09	100	1	···
c42	85186.44	111490.81	**23.59**	3600.02	-35882.57	131089.47	127.37	3603.20	-34651.32	+∞	100.00	3600.10	−∞	+∞	100.00	3601.14	19080	−∞	+∞	100.00	3601.08	100	1	···
c43	33724.24	44490.12	**24.20**	3600.01	-61501.87	45692.21	174.29	3603.32	-57282.52	+∞	100.00	3600.10	−∞	+∞	100.00	3601.13	19080	−∞	+∞	100.00	3601.09	100	1	···
c44	79367.50	105190.39	**24.55**	3600.01	-83295.86	112836.03	173.82	3603.22	-66682.47	+∞	100.00	3600.10	−∞	+∞	100.00	3601.14	19080	−∞	+∞	100.00	3601.08	100	1	···
c45	22797.28	26561.82	**14.17**	3600.03	8859.25	43222.26	79.50	3602.96	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.31	19080	−∞	+∞	100.00	3601.33	100	1	···
c46	51715.09	59318.73	**12.82**	3600.03	19991.27	106890.00	81.30	3602.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.30	19080	−∞	+∞	100.00	3601.32	100	1	···
c47	22072.82	24709.01	**10.67**	3600.03	4199.66	40842.73	89.72	3602.99	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.30	19080	−∞	+∞	100.00	3601.33	100	1	···
c48	47315.64	53769.05	**12.00**	3600.03	20833.28	89312.26	76.67	3603.00	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	19080	−∞	+∞	100.00	3601.34	100	1	···
c49	10418.68	12573.52	**17.14**	3600.02	-5408.52	122831.80	104.40	3603.24	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.50	28620	−∞	+∞	100.00	3601.35	150	1	···
c50	30984.86	38360.22	**19.23**	3600.03	9772.97	78476.00	87.55	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.30	28620	−∞	+∞	100.00	3601.38	150	1	···
c51	7691.85	8747.30	**12.07**	3600.03	-6441.13	15577.00	141.35	3603.32	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.26	28620	−∞	+∞	100.00	3601.37	150	1	···
c52	29870.53	35508.20	**15.88**	3600.04	2610.16	64850.46	95.98	3603.36	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.26	28620	−∞	+∞	100.00	3601.41	150	1	···
c53	4279.00	+∞	100.00	+∞	-26797.63	227121.00	111.80	3606.43	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.45	28620	−∞	+∞	100.00	3602.25	150	1	···
c54	5637.60	+∞	100.00	+∞	-35202.66	439598.00	108.01	3606.48	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.40	28620	−∞	+∞	100.00	3602.23	150	1	···
c55	3550.58	+∞	100.00	+∞	-22165.47	176263.00	112.58	3606.54	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.67	28620	−∞	+∞	100.00	3602.30	150	1	···
c56	5227.00	+∞	100.00	+∞	-32898.13	354524.00	109.28	3606.58	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.39	28620	−∞	+∞	100.00	3602.31	150	1	···
c57	8686.37	10664.32	**18.55**	3600.03	-12185.47	164500.11	107.41	3603.15	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.35	28620	−∞	+∞	100.00	3601.44	150	1	···
c58	16192.02	19556.26	**17.20**	3600.03	-26601.88	337718.20	107.88	3603.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.32	28620	−∞	+∞	100.00	3601.46	150	1	···
c59	7779.34	8965.69	**13.23**	3600.03	-13301.25	110572.00	112.03	3603.11	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.42	28620	−∞	+∞	100.00	3601.44	150	1	···
c60	12878.35	15189.65	15.22	3600.03	-14232.22	221701.10	106.42	3603.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.36	28620	14615.60	14906.51	**1.95**	3601.04	173	2	···
c61	3840.42	+∞	100.00	+∞	-23683.34	285930.00	108.28	3609.27	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.55	28620	−∞	+∞	100.00	3602.58	150	1	···
c62	5123.59	+∞	100.00	+∞	-32155.22	583215.00	105.51	3607.83	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.98	28620	−∞	+∞	100.00	3602.65	150	1	···
c63	3248.72	+∞	100.00	+∞	-19661.50	222551.00	108.83	3607.83	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.47	28620	−∞	+∞	100.00	3602.74	150	1	···
c64	4617.06	+∞	100.00	+∞	-28634.28	455145.00	106.29	3606.19	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.95	28620	−∞	+∞	100.00	3602.74	150	1	···
#S			**7**				0				0				6					5				···
SGM			**13.28**	5860.58			74.91	14400.00			93.06	14400.00			52.05	10486.26	28653			32.69	9022.72	183	2	···

Table 107 Detailed results for problem NLMCNDP-N, cost functions f_2

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	20767.38	21279.68	2.41	3600.02	20562.98	21279.68	3.37	3603.12	16939.84	+∞	100.00	3600.10	−∞	+∞	100.00	3601.24	115000	21279.65	21279.68	**0.00**	3601.02	1210	8	···
c100_400_10_F_T_10	42242.61	48464.32	**12.84**	3600.02	41090.09	50157.52	18.08	3603.12	33411.16	+∞	100.00	3600.10	−∞	+∞	100.00	3601.23	115000	−∞	+∞	100.00	3601.07	900	1	···
c100_400_10_V_L_10	3193.08	3774.60	15.41	3600.01	3131.80	3776.57	17.07	3603.12	2615.24	+∞	100.00	3600.10	−∞	+∞	100.00	3601.27	115000	3648.28	3774.60	**3.35**	3601.01	921	2	···
c100_400_30_F_L_10	39890.88	46412.13	**14.05**	3600.02	38404.63	120256.69	68.06	3603.27	36198.18	+∞	100.00	3600.10	−∞	+∞	100.00	3601.27	115000	−∞	+∞	100.00	3601.11	900	1	···
c100_400_30_F_T_10	69012.47	94073.18	**26.64**	3600.02	67433.09	128118.05	47.37	3603.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.26	115000	−∞	+∞	100.00	3601.11	900	1	···
c100_400_30_V_T_10	22028.26	31557.38	**30.20**	3600.02	21673.98	34080.84	36.40	3603.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.29	115000	−∞	+∞	100.00	3601.20	900	1	···
c25_100_10_F_L_5	13335.83	13335.83	**0.00**	**104.32**	13335.83	13335.83	**0.00**	167.17	11388.67	16903.34	32.62	3600.10	13335.83	13335.83	**0.00**	3590.80	28750	13335.82	13335.83	**0.00**	445.91	335	7	···
c25_100_10_F_T_5	29075.08	29075.11	**0.00**	3175.35	28928.60	29075.11	0.50	3602.94	27001.16	29762.53	9.28	3600.10	29075.11	29075.11	**0.00**	634.60	28750	29075.09	29075.11	**0.00**	**396.44**	515	10	···
c25_100_10_V_L_5	1845.52	1845.52	**0.00**	**15.10**	1845.52	1845.52	**0.00**	77.70	1664.66	1941.40	14.26	3600.10	1845.52	1845.52	**0.00**	234.02	28750	1845.52	1845.52	**0.00**	233.10	345	9	···
c25_100_30_F_L_5	28574.30	29122.94	1.88	3600.01	28342.34	29122.94	2.68	3603.05	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.10	28750	29122.89	29122.94	**0.00**	3601.02	463	9	···
c25_100_30_F_T_5	36860.87	38052.15	3.13	3600.01	36591.51	38052.15	3.84	3603.12	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.10	28750	38024.58	38052.15	**0.07**	3601.02	298	4	···
c25_100_30_V_T_5	21607.99	25076.28	13.83	3600.01	18930.85	25071.64	24.49	3602.99	−∞	+∞	100.0	3600.10	25041.16	25041.17	**0.00**	**2023.08**	28750	25039.91	25041.17	0.01	3601.02	475	9	···
c33	54632.96	57482.34	4.96	3600.01	52286.77	57482.35	9.04	3612.45	45695.23	+∞	100.00	3600.10	−∞	+∞	100.00	3601.16	23000	55022.28	57552.61	**4.40**	3601.02	190	2	···
c35	30374.04	32524.26	6.61	3600.02	28204.52	32899.13	14.27	3609.70	22276.60	+∞	100.00	3600.10	−∞	+∞	100.00	3601.12	23000	31169.60	32587.27	**4.35**	3601.02	191	2	···
c36	**96331.19**	**101439.16**	**5.04**	**3600.01**	**92556.91**	**105641.19**	**12.39**	**3604.38**	**78863.28**	+∞	**100.00**	**3600.10**	**103533.45**	**103533.47**	**0.00**	**650.29**	**23000**	**97942.46**	**101795.42**	**3.79**	**3601.01**	**188**	**2**	···
c37	0.00	+∞	100.00	+∞	29332.07	60246.34	**51.31**	3603.19	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.30	23000	−∞	+∞	100.00	3601.35	180	1	···
c38	0.00	+∞	100.00	+∞	59240.38	158281.40	**62.57**	3603.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.35	23000	−∞	+∞	100.00	3601.34	180	1	···
c39	0.00	+∞	100.00	+∞	30833.53	66449.69	**53.60**	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.30	23000	−∞	+∞	100.00	3601.25	180	1	···
c40	62214.00	76826.82	**19.02**	3600.02	63855.56	132657.67	51.86	3603.25	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.31	23000	−∞	+∞	100.00	3601.34	180	1	···
c41	59028.26	62160.80	5.04	3600.02	57004.10	62718.63	9.11	3614.65	46163.68	+∞	100.00	3600.10	−∞	+∞	100.00	3601.11	23000	58816.42	61812.64	**4.85**	3601.02	190	2	···
c42	102165.61	116192.05	**12.07**	3600.01	101094.77	121283.61	16.65	3603.11	92162.96	+∞	100.00	3600.10	−∞	+∞	100.00	3601.17	23000	−∞	+∞	100.00	3601.09	180	1	···
c43	42909.12	45176.07	**5.02**	3600.01	41614.26	46677.24	10.85	3604.26	31737.51	+∞	100.00	3600.10	−∞	+∞	100.00	3601.17	23000	42860.32	45148.93	5.07	3601.02	194	2	···
c44	97562.44	103700.78	5.92	3600.02	95388.73	103700.80	8.02	3603.20	80937.83	+∞	100.00	3600.10	−∞	+∞	100.00	3601.16	23000	99877.27	103700.80	**3.69**	3601.02	191	2	···
c45	0.00	+∞	100.00	+∞	22502.45	49726.22	**54.75**	3603.33	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.37	23000	−∞	+∞	100.00	3601.42	180	1	···
c46	0.00	+∞	100.00	+∞	50731.83	108281.67	**53.15**	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.37	23000	−∞	+∞	100.00	3601.35	180	1	···
c47	0.00	+∞	100.00	+∞	21730.40	42085.96	**48.37**	3603.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.34	23000	−∞	+∞	100.00	3601.35	180	1	···
c48	0.00	+∞	100.00	+∞	47502.62	86608.65	**45.15**	3602.94	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.35	23000	−∞	+∞	100.00	3601.45	180	1	···
c49	233.01	+∞	100.00	+∞	11242.59	31664.23	**64.49**	3603.25	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.36	34500	−∞	+∞	100.00	3601.40	270	1	···
c50	199.03	+∞	100.00	+∞	31457.37	82073.12	**61.67**	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.33	34500	−∞	+∞	100.00	3601.36	270	1	···
c51	189.06	+∞	100.00	+∞	8191.15	16835.67	**51.35**	3603.34	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.36	34500	−∞	+∞	100.00	3601.39	270	1	···
c52	368.38	+∞	100.00	+∞	30475.94	69462.80	**56.13**	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.32	34500	−∞	+∞	100.00	3601.39	270	1	···
c53	0.00	+∞	100.00	+∞	0.00	214608.32	100.00	3606.59	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.04	34500	−∞	+∞	100.00	3602.33	270	1	···
c54	5.46	+∞	100.00	+∞	0.00	422707.06	100.00	3606.58	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.58	34500	−∞	+∞	100.00	3602.41	270	1	···
c55	2.55	+∞	100.00	+∞	0.00	167136.98	100.00	3606.56	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.75	34500	−∞	+∞	100.00	3602.42	270	1	···
c56	0.00	+∞	100.00	+∞	0.00	339388.00	100.00	3606.31	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.29	34500	−∞	+∞	100.00	3602.43	270	1	···
c57	132.74	+∞	100.00	+∞	9384.84	31184.47	**69.91**	3603.18	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.39	34500	−∞	+∞	100.00	3601.53	270	1	···
c58	264.05	+∞	100.00	+∞	16423.78	54881.16	**70.07**	3603.15	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.38	34500	−∞	+∞	100.00	3601.46	270	1	···
c59	191.60	+∞	100.00	+∞	7690.19	18909.78	59.33	3603.15	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.41	34500	9938.68	10351.52	**3.99**	3601.04	288	2	···
c60	233.52	+∞	100.00	+∞	12927.48	31290.13	**58.69**	3603.15	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.44	34500	−∞	+∞	100.00	3601.58	270	1	···
c61	10.77	+∞	100.00	+∞	0.00	274365.29	100.00	3608.20	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.10	34500	−∞	+∞	100.00	3603.55	270	1	···
c62	2.59	+∞	100.00	+∞	0.00	567073.04	100.00	3607.80	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.54	34500	−∞	+∞	100.00	3603.60	270	1	···
c63	16.78	+∞	100.00	+∞	0.00	213520.67	100.00	3607.77	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.91	34500	−∞	+∞	100.00	3603.67	270	1	···
c64	7.41	+∞	100.00	+∞	0.00	440873.71	100.00	3606.10	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.21	34500	−∞	+∞	100.00	3603.63	270	1	···
#S			3				2				0				**4**					3				···
SGM			28.65	10640.10			28.57	11482.84			88.08	14400.00			64.08	11204.34	34876			**27.61**	11052.33	302	2	···

Table 108 Detailed results for problem NLMCNDP-N, cost functions f_3

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	20558.04	21444.41	4.13	3600.01	20410.54	21480.45	4.98	3603.08	16859.70	+∞	100.00	3600.10	−∞	+∞	100.00	3601.29	163500	21440.86	21444.41	**0.02**	3601.02	1034	5	···
c100_400_10_F_T_10	42546.52	49512.45	**14.07**	3600.01	41764.54	49862.54	16.24	3603.09	33541.70	+∞	100.00	3600.10	−∞	+∞	100.00	3601.31	163500	−∞	+∞	100.00	3601.07	900	1	···
c100_400_10_V_L_10	3235.60	3970.22	**18.50**	3600.01	3183.44	4005.13	20.52	3603.12	2591.78	+∞	100.00	3600.10	−∞	+∞	100.00	3601.28	163500	−∞	+∞	100.00	3601.07	900	1	···
c100_400_30_F_L_10	39359.45	+∞	100.00	+∞	38844.85	97986.13	**60.36**	3603.23	35853.42	+∞	100.00	3600.10	−∞	+∞	100.00	3601.39	163500	−∞	+∞	100.00	3601.11	900	1	···
c100_400_30_F_T_10	69314.19	106277.86	**34.78**	3600.02	68422.19	130824.78	47.70	3603.18	61186.34	+∞	100.00	3600.10	−∞	+∞	100.00	3601.31	163500	−∞	+∞	100.00	3601.11	900	1	···
c100_400_30_V_T_10	22728.88	37062.04	**38.67**	3600.03	23127.25	42525.15	45.62	3603.13	16081.21	+∞	100.00	3600.10	−∞	+∞	100.00	3601.37	163500	−∞	+∞	100.00	3601.20	900	1	···
c25_100_10_F_L_5	13468.94	13468.96	**0.00**	425.40	13468.96	13468.96	**0.00**	**221.84**	11351.95	16807.29	32.46	3600.10	−∞	+∞	100.00	3601.14	40875	13468.95	13468.96	**0.00**	640.94	429	9	···
c25_100_10_F_T_5	29121.25	29488.04	1.24	3600.01	29488.04	29488.04	**0.00**	1059.53	27496.26	29833.49	7.83	3600.10	29488.04	29488.04	**0.00**	2074.01	40875	29488.03	29488.04	**0.00**	**437.95**	478	6	···
c25_100_10_V_L_5	1769.77	1769.77	**0.00**	**17.92**	1769.77	1769.77	**0.00**	228.94	1714.43	1769.78	3.13	3600.10	1769.77	1769.77	**0.00**	540.59	40875	1769.77	1769.77	**0.00**	218.74	373	10	···
c25_100_30_F_L_5	28496.19	29349.87	2.91	3600.01	28499.08	29349.87	2.90	3603.10	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.15	40875	29347.17	29349.87	**0.01**	3601.02	366	6	···
c25_100_30_F_T_5	36987.16	36987.21	**0.00**	3600.02	36987.21	36987.21	**0.00**	**2472.87**	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.11	40875	36987.19	36987.22	**0.00**	3002.09	501	11	···
c25_100_30_V_T_5	24975.72	27578.72	9.44	3600.01	22735.34	27943.45	18.64	3603.00	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.14	40875	27458.28	27481.64	**0.09**	3601.02	375	10	···
c33	52120.55	55333.02	**5.81**	3600.02	50644.97	55207.38	8.26	3603.46	43305.34	+∞	100.00	3600.10	−∞	+∞	100.00	3601.16	32700	−∞	+∞	100.00	3601.08	180	1	···
c35	31101.53	34737.64	**10.47**	3600.01	30441.53	36951.47	17.62	3604.80	20628.97	+∞	100.00	3600.10	−∞	+∞	100.00	3601.13	32700	−∞	+∞	100.00	3601.08	180	1	···
c36	93066.79	99760.57	6.71	3600.02	92462.37	100196.30	7.72	3603.05	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.12	32700	97213.26	99190.27	**1.99**	3601.02	203	2	···
c37	0.00	+∞	100.00	+∞	29949.16	59385.81	**49.57**	3603.22	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.30	32700	−∞	+∞	100.00	3601.31	180	1	···
c38	0.00	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.37	32700	−∞	+∞	100.00	3601.34	180	1	···
c39	29228.18	38495.06	**24.07**	3600.03	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.32	32700	−∞	+∞	100.00	3601.33	180	1	···
c40	61591.74	77511.24	**20.54**	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.30	32700	−∞	+∞	100.00	3601.35	180	1	···
c41	56814.20	62434.58	**9.00**	3600.01	56314.38	62836.60	10.38	3612.25	44005.57	+∞	100.00	3600.10	−∞	+∞	100.00	3601.16	32700	−∞	+∞	100.00	3601.09	180	1	···
c42	100220.78	112205.72	**10.68**	3600.02	100456.26	117048.68	14.18	3603.17	89675.93	+∞	100.00	3600.10	−∞	+∞	100.00	3601.16	32700	−∞	+∞	100.00	3601.21	180	1	···
c43	43107.38	46341.01	**6.98**	3600.01	43008.05	47254.42	8.99	3603.13	29997.18	+∞	100.00	3600.10	−∞	+∞	100.00	3601.14	32700	−∞	+∞	100.00	3601.20	180	1	···
c44	96672.28	104572.29	**7.55**	3600.01	95630.29	110020.73	13.08	3603.20	77828.00	+∞	100.00	3600.10	−∞	+∞	100.00	3601.15	32700	−∞	+∞	100.00	3601.09	180	1	···
c45	0.00	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.36	32700	−∞	+∞	100.00	3601.34	180	1	···
c46	0.00	+∞	100.00	+∞	50389.19	104231.17	**51.66**	3603.33	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.34	32700	−∞	+∞	100.00	3601.41	180	1	···
c47	0.00	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	32700	−∞	+∞	100.00	3601.43	180	1	···
c48	0.00	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.29	32700	−∞	+∞	100.00	3601.40	180	1	···
c49	5.16	+∞	100.00	+∞	10172.48	32476.19	**68.68**	3636.05	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.34	49050	−∞	+∞	100.00	3601.37	270	1	···
c50	0.00	+∞	100.00	+∞	29982.69	83817.38	**64.23**	3607.66	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.32	49050	−∞	+∞	100.00	3601.37	270	1	···
c51	0.25	+∞	100.00	+∞	7274.60	17180.44	**57.66**	3605.91	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.43	49050	−∞	+∞	100.00	3601.36	270	1	···
c52	32.11	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.34	49050	−∞	+∞	100.00	3601.43	270	1	···
c53	0.00	+∞	100.00	+∞	-1048.11	216794.53	100.48	3606.44	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.06	49050	−∞	+∞	100.00	3602.34	270	1	···
c54	0.00	+∞	100.00	+∞	-1372.48	425812.69	100.32	3606.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.05	49050	−∞	+∞	100.00	3602.48	270	1	···
c55	0.00	+∞	100.00	+∞	-864.59	168773.74	100.51	3606.42	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.11	49050	−∞	+∞	100.00	3602.45	270	1	···
c56	0.00	+∞	100.00	+∞	-1280.32	342049.01	100.37	3606.79	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.03	49050	−∞	+∞	100.00	3602.50	270	1	···
c57	5.19	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.43	49050	−∞	+∞	100.00	3601.47	270	1	···
c58	6.45	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.38	49050	−∞	+∞	100.00	3601.51	270	1	···
c59	3.63	+∞	100.00	+∞	7367.74	105957.62	**93.05**	3603.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.39	49050	−∞	+∞	100.00	3601.54	270	1	···
c60	3.89	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.43	49050	−∞	+∞	100.00	3601.48	270	1	···
c61	0.00	+∞	100.00	+∞	-921.66	276633.35	100.33	3608.23	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.31	49050	−∞	+∞	100.00	3603.57	270	1	···
c62	0.00	+∞	100.00	+∞	-1250.13	570161.94	100.22	3607.81	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.39	49050	−∞	+∞	100.00	3603.55	270	1	···
c63	0.00	+∞	100.00	+∞	-767.16	215175.45	100.36	3607.86	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.26	49050	−∞	+∞	100.00	3602.64	270	1	···
c64	0.00	+∞	100.00	+∞	-1115.04	443490.65	100.25	3606.14	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.21	49050	−∞	+∞	100.00	3603.56	270	1	···
#S			2				**4**				0				2					**4**				···
SGM			**29.47**	11477.70			31.20	10742.85			85.35	14400.00			80.49	12759.14	49106			43.01	10821.08	298	2	···

Table 109 Detailed results for problem NLMCNDP-N, cost functions f_4

Instance	GUROBI DB	PB	GAP	CPU	SCIP DB	PB	GAP	CPU	COUENNE DB	PB	GAP	CPU	NAIVE DB	PB	GAP	CPU	NP	CN24 DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	20371.46	20371.48	0.00	3381.00	20371.48	20371.48	0.00	1972.13	16537.10	+∞	100.00	3600.10	−∞	+∞	100.00	3601.47	292700	20391.57	20391.62	0.00	3601.03	1516	10	···
c100_400_10_F_T_10	41798.62	47379.18	11.78	3600.02	40830.47	50321.00	18.86	3603.08	33101.32	+∞	100.00	3600.10	−∞	+∞	100.00	3601.49	292700	−∞	+∞	100.00	3601.20	1300	1	···
c100_400_10_V_L_10	2530.76	2811.30	9.98	3600.02	2476.44	2850.49	13.12	3603.09	2161.38	+∞	100.00	3600.10	−∞	+∞	100.00	3601.57	292700	2838.07	2858.76	0.72	3601.01	1320	2	···
c100_400_30_F_L_10	37465.96	43646.21	14.16	3600.03	36699.89	120743.38	69.61	3603.23	34406.90	+∞	100.00	3600.10	−∞	+∞	100.00	3601.62	292700	−∞	+∞	100.00	1446.56	1300	1	···
c100_400_30_F_T_10	64542.02	92903.12	30.53	3600.02	62926.83	169565.73	62.89	3603.22	57824.72	+∞	100.00	3600.10	−∞	+∞	100.00	3601.54	292700	−∞	+∞	100.00	3601.10	1300	1	···
c100_400_30_V_T_10	11657.42	16159.54	27.86	3600.02	11060.61	22466.61	50.77	3603.07	9596.84	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	1300	1	···
c25_100_10_F_L_5	12220.94	12220.95	0.00	26.42	12220.95	12220.95	0.00	67.78	10917.79	14806.58	26.26	3600.10	−∞	+∞	100.00	3601.20	73175	12240.92	12240.93	0.00	721.82	449	17	···
c25_100_10_F_T_5	26652.71	26652.73	0.00	4.52	26652.71	26652.73	0.00	118.18	26652.73	26652.73	0.00	3439.48	26652.73	26652.73	0.00	2649.47	73175	26652.71	26652.73	0.00	89.91	568	9	···
c25_100_10_V_L_5	1406.68	1406.68	0.00	4.75	1406.68	1406.68	0.00	39.50	1406.68	1406.68	0.00	1424.67	1406.68	1406.68	0.00	639.84	73175	1406.68	1406.68	0.00	60.63	608	8	···
c25_100_30_F_L_5	27107.30	27107.33	0.00	1092.74	27107.33	27107.33	0.00	3180.94	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.23	73175	27226.75	27226.82	0.00	3601.01	533	14	···
c25_100_30_F_T_5	34123.19	35457.54	3.76	3600.01	33515.63	35453.06	5.46	3603.00	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.18	73175	35663.88	35771.17	0.30	3601.02	363	4	···
c25_100_30_V_T_5	8261.08	8261.15	0.00	3600.02	8211.36	8261.15	0.60	3602.97	−∞	+∞	100.0	3600.10	8280.71	8280.72	0.00	3305.12	73175	8280.65	8280.75	0.00	3601.02	630	26	···
c33	40068.65	45337.13	11.62	3600.02	38285.74	48825.74	21.59	3609.08	35483.48	+∞	100.00	3600.10	−∞	+∞	100.00	3601.24	58540	−∞	+∞	100.00	3601.08	260	1	···
c35	16861.45	18843.25	10.52	3600.02	17297.40	19658.16	12.01	3603.06	14918.80	+∞	100.00	3600.10	−∞	+∞	100.00	3601.23	58540	−∞	+∞	100.00	3601.08	260	1	···
c36	68741.65	76682.55	10.36	3600.02	65916.21	83938.18	21.47	3603.01	62559.42	+∞	100.00	3600.10	−∞	+∞	100.00	3601.20	58540	−∞	+∞	100.00	3601.08	260	1	···
c37	0.00	+∞	100.00	+∞	24811.51	56114.82	55.78	3603.21	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.35	58540	−∞	+∞	100.00	3601.31	260	1	···
c38	0.00	+∞	100.00	+∞	54503.87	139925.68	61.05	3603.15	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.35	58540	−∞	+∞	100.00	3601.33	260	1	···
c39	26833.29	33198.56	19.17	3600.03	26576.03	60461.09	56.04	3603.27	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.32	58540	−∞	+∞	100.00	3601.33	260	1	···
c40	58342.96	70939.55	17.76	3600.03	57946.76	127759.63	54.64	3603.24	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.34	58540	−∞	+∞	100.00	3601.30	260	1	···
c41	38168.67	42428.64	10.04	3600.02	37001.51	45887.35	19.36	3607.64	33684.62	+∞	100.00	3600.10	−∞	+∞	100.00	3601.18	58540	−∞	+∞	100.00	3601.10	260	1	···
c42	81891.06	98622.88	16.97	3600.01	79604.01	106700.27	25.39	3603.18	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.15	58540	−∞	+∞	100.00	3601.09	260	1	···
c43	25115.19	27082.37	7.26	3600.01	24551.65	27565.59	10.93	3610.81	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.22	58540	27050.59	27653.63	2.18	3601.02	278	2	···
c44	70516.38	78696.85	10.39	3600.01	69001.37	87125.08	20.80	3603.11	63946.57	+∞	100.00	3600.10	−∞	+∞	100.00	3601.19	58540	81022.36	82148.86	1.37	3601.02	275	2	···
c45	0.00	+∞	100.00	+∞	20144.36	45374.28	55.60	3602.96	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.41	58540	−∞	+∞	100.00	3601.35	260	1	···
c46	0.00	+∞	100.00	+∞	47276.02	99036.31	52.26	3603.00	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.38	58540	−∞	+∞	100.00	3601.39	260	1	···
c47	0.00	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.34	58540	−∞	+∞	100.00	3601.38	260	1	···
c48	0.00	+∞	100.00	+∞	43092.34	80865.53	46.71	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.37	58540	−∞	+∞	100.00	3601.43	260	1	···
c49	3.30	+∞	100.00	+∞	8416.13	28503.25	70.47	3603.23	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.44	87810	−∞	+∞	100.00	3601.41	390	1	···
c50	0.00	+∞	100.00	+∞	27101.60	79956.20	66.10	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.33	87810	−∞	+∞	100.00	3601.38	390	1	···
c51	0.16	+∞	100.00	+∞	5883.80	13955.83	57.84	3603.38	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.38	87810	−∞	+∞	100.00	3601.39	390	1	···
c52	20.67	+∞	100.00	+∞	26316.72	63444.25	58.52	3603.32	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.36	87810	−∞	+∞	100.00	3601.38	390	1	···
c53	0.00	+∞	100.00	+∞	-669.87	211726.52	100.32	3606.60	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.23	87810	−∞	+∞	100.00	3602.29	390	1	···
c54	0.00	+∞	100.00	+∞	-877.18	419358.31	100.21	3606.55	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.13	87810	−∞	+∞	100.00	3602.33	390	1	···
c55	0.00	+∞	100.00	+∞	-552.58	163189.81	100.34	3606.49	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.17	87810	−∞	+∞	100.00	3602.33	390	1	···
c56	0.00	+∞	100.00	+∞	-818.28	335383.50	100.24	3606.80	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.99	87810	−∞	+∞	100.00	3602.25	390	1	···
c57	3.31	+∞	100.00	+∞	6803.32	157688.21	95.69	3603.22	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.51	87810	−∞	+∞	100.00	3601.54	390	1	···
c58	4.12	+∞	100.00	+∞	11864.11	323079.05	96.33	3603.23	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.45	87810	−∞	+∞	100.00	3601.48	390	1	···
c59	2.32	+∞	100.00	+∞	5836.88	15530.86	62.42	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.47	87810	−∞	+∞	100.00	3601.50	390	1	···
c60	2.49	+∞	100.00	+∞	10588.51	213799.12	95.05	3603.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.51	87810	−∞	+∞	100.00	3601.54	390	1	···
c61	0.00	+∞	100.00	+∞	-589.05	272317.27	100.22	3608.18	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.60	87810	−∞	+∞	100.00	3603.47	390	1	···
c62	0.00	+∞	100.00	+∞	-798.99	564987.57	100.14	3607.77	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.28	87810	−∞	+∞	100.00	3603.57	390	1	···
c63	0.00	+∞	100.00	+∞	-490.31	211152.34	100.23	3607.90	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.67	87810	−∞	+∞	100.00	3603.51	390	1	···
c64	0.00	+∞	100.00	+∞	-712.65	438834.58	100.16	3606.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.29	87810	−∞	+∞	100.00	3603.64	390	1	···
#S			2				2				2				2					2				···
SGM			40.77	9815.86			44.02	10785.68			77.16	12979.30			77.16	12605.19	146498			70.00	10149.80	391	1	···

Table 110 Detailed results for problem NLMCNDP-N, cost functions f_5

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	20781.37	21314.30	2.50	3600.01	20228.12	21336.96	5.20	3603.07	16739.42	+∞	100.00	3600.10	−∞	+∞	100.00	3601.26	144300	21312.62	21314.30	**0.01**	3601.02	701	9	⋯
c100_400_10_F_T_10	41375.23	52420.66	**21.07**	3600.01	40126.47	53910.34	25.57	3603.11	30797.42	+∞	100.00	3600.10	−∞	+∞	100.00	3601.26	144300	−∞	+∞	100.00	**340.24**	600	1	⋯
c100_400_10_V_L_10	3066.14	3888.41	21.15	3600.01	2934.37	4016.29	26.94	3602.98	2307.20	+∞	100.00	3600.10	−∞	+∞	100.00	3601.26	144300	3862.74	3888.41	**0.66**	3601.02	632	3	⋯
c100_400_30_F_L_10	39519.17	45712.74	**13.55**	3600.02	38367.63	149815.82	74.39	3603.31	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	144300	−∞	+∞	100.00	3601.14	600	1	⋯
c100_400_30_F_T_10	69259.57	87219.88	**20.59**	3600.02	67156.73	135417.95	50.41	3603.58	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	144300	−∞	+∞	100.00	3601.13	600	1	⋯
c100_400_30_V_T_10	19878.59	35397.21	**43.84**	3600.02	19214.02	43780.31	56.11	3603.28	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	144300	−∞	+∞	100.00	3601.13	600	1	⋯
c25_100_10_F_L_5	13284.18	13284.19	**0.00**	721.36	13284.20	13284.20	**0.00**	3168.15	11508.63	14322.30	19.65	3600.10	13284.19	13284.20	**0.00**	1448.38	36075	13284.19	13284.20	**0.00**	**327.25**	279	9	⋯
c25_100_10_F_T_5	28675.60	29024.67	1.20	3600.01	28404.00	29024.67	2.14	3604.22	27131.67	29336.05	7.51	3600.10	29024.67	29024.67	**0.00**	1166.12	36075	29024.65	29024.67	**0.00**	**161.84**	758	5	⋯
c25_100_10_V_L_5	1721.85	1721.85	**0.00**	393.50	1721.85	1721.85	**0.00**	3150.06	1520.52	1723.25	11.76	3600.10	1721.85	1721.85	**0.00**	268.38	36075	1721.85	1721.85	**0.00**	**176.71**	412	10	⋯
c25_100_30_F_L_5	28275.34	29238.70	3.29	3600.01	27854.99	29241.45	4.74	3603.82	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.15	36075	29238.42	29238.70	**0.00**	3601.02	375	11	⋯
c25_100_30_F_T_5	36000.61	37002.53	2.71	3600.01	35523.66	37002.53	4.00	3604.27	−∞	+∞	100.0	3600.10	37002.53	37002.53	**0.00**	2526.22	36075	37002.51	37002.53	**0.00**	**1496.89**	405	11	⋯
c25_100_30_V_T_5	21170.07	27029.66	21.68	3600.02	18798.96	27072.32	30.56	3603.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.10	36075	26789.02	26791.65	**0.01**	3601.02	374	12	⋯
c33	49119.05	57494.77	14.57	3600.01	46690.68	59030.61	20.90	3615.29	41936.62	+∞	100.00	3600.10	−∞	+∞	100.00	3601.12	28860	54784.61	56647.54	**3.29**	3601.02	132	2	⋯
c35	28178.90	34747.13	18.90	3600.01	26496.15	36315.84	27.04	3606.76	20646.31	+∞	100.00	3600.10	−∞	+∞	100.00	3601.16	28860	33545.20	34135.78	**1.73**	3601.02	140	2	⋯
c36	85971.21	99681.66	13.75	3600.02	82302.80	100257.26	17.91	3605.95	74634.80	+∞	100.00	3600.10	−∞	+∞	100.00	3601.12	28860	97318.34	99863.20	**2.55**	3601.02	130	2	⋯
c37	27524.47	36585.38	**24.77**	3600.03	27278.96	65432.85	58.31	3602.77	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	28860	−∞	+∞	100.00	3601.35	120	1	⋯
c38	58918.70	74258.11	**20.66**	3600.03	58199.38	171050.50	65.98	3611.84	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.29	28860	−∞	+∞	100.00	3601.24	120	1	⋯
c39	28917.20	37744.50	**23.39**	3600.03	28827.02	66428.44	56.60	3603.40	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.30	28860	−∞	+∞	100.00	3601.37	120	1	⋯
c40	61468.91	74648.39	**17.66**	3600.03	61283.36	136200.30	55.00	3714.65	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.30	28860	−∞	+∞	100.00	3601.36	120	1	⋯
c41	50864.50	61227.79	**16.93**	3600.02	48369.68	62929.41	23.14	3609.48	42830.56	+∞	100.00	3600.10	−∞	+∞	100.00	3601.11	28860	−∞	+∞	100.00	3601.11	120	1	⋯
c42	94661.88	111998.90	**15.48**	3600.01	93583.17	117231.60	20.17	3603.59	87657.54	+∞	100.00	3600.10	−∞	+∞	100.00	3601.17	28860	−∞	+∞	100.00	3601.11	120	1	⋯
c43	37141.13	45258.14	17.93	3600.01	35161.29	46154.70	23.82	3603.59	29241.93	+∞	100.00	3600.10	−∞	+∞	100.00	3601.18	28860	43594.21	45038.94	**3.21**	3601.02	141	2	⋯
c44	88206.63	105040.88	**16.03**	3600.01	87285.49	110909.51	21.30	3603.59	76104.30	+∞	100.00	3600.10	−∞	+∞	100.00	3601.18	28860	−∞	+∞	100.00	3601.11	120	1	⋯
c45	21989.70	28076.31	**21.68**	3600.02	21874.22	50277.54	56.49	3603.84	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.37	28860	−∞	+∞	100.00	3601.41	120	1	⋯
c46	50212.87	60666.06	**17.23**	3600.03	49924.07	115358.03	56.72	3631.23	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.34	28860	−∞	+∞	100.00	3601.35	120	1	⋯
c47	21150.84	27231.89	**22.33**	3600.03	21611.08	39622.99	45.46	3603.53	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.36	28860	−∞	+∞	100.00	3601.40	120	1	⋯
c48	45659.39	54990.10	**16.97**	3600.03	45546.96	84626.99	46.18	3603.48	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.43	28860	−∞	+∞	100.00	3601.37	120	1	⋯
c49	9952.07	13856.66	**28.18**	3600.03	9866.70	31344.89	68.52	3603.28	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.37	43290	−∞	+∞	100.00	3601.42	180	1	⋯
c50	29848.10	40602.60	**26.49**	3600.03	29496.89	88672.98	66.74	3603.26	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.34	43290	−∞	+∞	100.00	3601.44	180	1	⋯
c51	7216.58	15212.30	**52.56**	3600.03	7286.80	16251.72	55.16	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.41	43290	−∞	+∞	100.00	3601.36	180	1	⋯
c52	28808.99	38783.90	**25.72**	3600.03	28638.42	68329.36	58.09	3603.25	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.36	43290	−∞	+∞	100.00	3601.37	180	1	⋯
c53	-1328.19	+∞	100.00	+∞	-1348.81	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.92	43290	−∞	+∞	100.00	3602.39	180	1	⋯
c54	-1739.24	+∞	100.00	+∞	-1770.16	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.09	43290	−∞	+∞	100.00	3602.26	180	1	⋯
c55	-1095.63	+∞	100.00	+∞	-1114.74	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.70	43290	−∞	+∞	100.00	3602.29	180	1	⋯
c56	-1622.45	+∞	100.00	+∞	-1653.37	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.61	43290	−∞	+∞	100.00	3602.29	180	1	⋯
c57	8262.99	11723.79	**29.52**	3600.03	-612.57	160174.03	100.38	3603.53	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.42	43290	−∞	+∞	100.00	3601.46	180	1	⋯
c58	14719.71	22080.36	**33.34**	3600.04	14371.61	327104.11	95.61	3603.39	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.40	43290	−∞	+∞	100.00	3601.40	180	1	⋯
c59	7217.22	9997.55	**27.81**	3600.04	-670.08	105882.41	100.63	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.38	43290	−∞	+∞	100.00	3601.48	180	1	⋯
c60	12411.40	16599.20	**25.23**	3600.04	12146.01	216627.96	94.39	3603.27	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.50	43290	−∞	+∞	100.00	3601.47	180	1	⋯
c61	-1167.95	+∞	100.00	+∞	-1190.24	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.60	43290	−∞	+∞	100.00	3603.29	180	1	⋯
c62	-1584.20	+∞	100.00	+∞	-1615.53	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.28	43290	−∞	+∞	100.00	3603.27	180	1	⋯
c63	-972.17	+∞	100.00	+∞	-988.91	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.16	43290	−∞	+∞	100.00	3603.44	180	1	⋯
c64	-1413.01	+∞	100.00	+∞	-1439.34	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.21	43290	−∞	+∞	100.00	3603.45	180	1	⋯
#S			2				2				0				**4**					**4**				⋯
SGM			**20.81**	12362.19			35.44	13419.87			86.58	14400.00			64.75	11283.15	43339			30.80	9360.56	212	2	⋯

Table 111 Detailed results for problem NLMCNDP-N, cost functions f_6

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	22155.74	22155.76	**0.00**	**1653.93**	21150.40	22181.29	4.65	3602.96	17691.75	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	22154.73	22155.76	0.00	3601.02	640	7	⋯
c100_400_10_F_T_10	48227.12	55817.11	**13.60**	3600.01	45906.44	56558.57	18.83	3603.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	500	1	⋯
c100_400_10_V_L_10	4377.46	4782.43	**8.47**	3600.01	4190.61	4901.16	14.50	3603.11	3584.85	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	500	1	⋯
c100_400_30_F_L_10	43316.07	47209.43	**8.25**	3600.02	41266.67	104184.40	60.39	3603.24	39094.50	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.12	500	1	⋯
c100_400_30_F_T_10	78199.17	90120.66	**13.23**	3600.02	75990.11	99294.27	23.47	3603.19	68927.34	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	500	1	⋯
c100_400_30_V_T_10	38750.06	46854.66	**17.30**	3600.02	38298.69	52297.07	26.77	3603.14	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	500	1	⋯
c25_100_10_F_L_5	14248.71	14248.71	**0.00**	**26.07**	14248.71	14248.71	**0.00**	109.17	12827.30	16071.91	20.19	3600.10	−∞	+∞	100.00	3600.01	+∞	14248.71	14248.71	**0.00**	340.99	259	10	⋯
c25_100_10_F_T_5	31381.56	31381.59	**0.00**	981.64	31305.77	31381.59	0.24	3602.94	29061.40	31649.94	8.18	3600.10	−∞	+∞	100.00	3600.01	+∞	31381.57	31381.59	**0.00**	**519.94**	421	16	⋯
c25_100_10_V_L_5	2047.46	2047.46	**0.00**	**0.90**	2047.46	2047.46	**0.00**	10.05	2047.46	2047.46	**0.00**	3587.15	−∞	+∞	100.00	3600.01	+∞	2047.46	2047.46	**0.00**	33.10	256	11	⋯
c25_100_30_F_L_5	30928.87	30928.89	**0.00**	**834.44**	30928.89	30928.89	**0.00**	2109.14	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	30928.83	30928.89	**0.00**	3601.02	329	10	⋯
c25_100_30_F_T_5	39399.81	39399.85	**0.00**	**553.50**	39399.85	39399.85	**0.00**	2139.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	39399.84	39399.85	**0.00**	3456.64	362	11	⋯
c25_100_30_V_T_5	39979.27	40988.04	2.46	3600.01	37618.43	41012.72	8.28	3602.97	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	40381.09	40989.46	**1.48**	3601.02	172	4	⋯
c33	65294.14	66975.45	**2.51**	3600.02	63108.33	67446.19	6.43	3619.13	55800.61	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	100	1	⋯
c35	46044.61	46896.67	**1.82**	3600.02	44321.06	48162.58	7.98	3614.66	33766.24	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	44032.69	47004.71	6.32	3601.02	106	2	⋯
c36	117468.60	121563.87	**3.37**	3600.02	115860.39	121999.04	5.03	3612.98	100359.30	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	115033.05	121563.88	5.37	3601.03	104	2	⋯
c37	32561.53	39405.40	**17.37**	3600.03	31852.58	66794.27	52.31	3603.41	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.26	100	1	⋯
c38	65969.53	81919.20	**19.47**	3600.03	64988.26	165548.07	60.74	3603.29	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.26	100	1	⋯
c39	33833.62	42425.53	**20.25**	3600.03	36692.86	71818.53	48.91	3603.32	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.31	100	1	⋯
c40	67128.84	82363.02	**18.50**	3600.03	71052.30	139386.02	49.02	3603.31	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.31	100	1	⋯
c41	75194.50	77078.53	**2.44**	3600.02	73957.90	77949.35	5.12	3608.25	62091.14	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.12	100	1	⋯
c42	120334.05	128200.48	**6.14**	3600.01	117787.66	133819.02	11.98	3603.23	107268.20	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	100	1	⋯
c43	59561.82	61008.06	**2.37**	3600.01	58836.42	61823.06	4.83	3604.73	45352.80	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.12	100	1	⋯
c44	123674.69	128141.44	**3.49**	3600.01	122231.28	130388.91	6.26	3603.13	103842.90	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	100	1	⋯
c45	25226.91	30936.50	**18.46**	3600.03	27014.93	48339.42	44.11	3603.01	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.33	100	1	⋯
c46	55338.95	65156.73	**15.07**	3600.03	54956.26	107248.47	48.76	3602.96	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.34	100	1	⋯
c47	24623.86	29310.69	**15.99**	3600.03	26528.84	41293.42	35.76	3603.01	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.31	100	1	⋯
c48	50353.35	60110.74	**16.23**	3600.03	49964.76	89768.12	44.34	3602.94	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.34	100	1	⋯
c49	12683.87	15684.90	**19.13**	3600.03	13229.97	31741.56	58.32	3603.36	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.27	150	1	⋯
c50	34489.67	43216.18	**20.19**	3600.03	35635.16	83573.25	57.36	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	150	1	⋯
c51	9764.66	11306.15	**13.63**	3600.03	10271.67	16684.63	38.44	3603.32	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.30	150	1	⋯
c52	33625.38	40252.84	**16.46**	3600.04	34653.27	67285.86	48.50	3603.64	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	150	1	⋯
c53	0.00	+∞	100.00	+∞	0.00	221016.75	100.00	3606.53	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.02	150	1	⋯
c54	168.24	+∞	100.00	+∞	0.00	431400.00	100.00	3606.58	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.99	150	1	⋯
c55	105.07	+∞	100.00	+∞	0.00	172004.62	100.00	3606.31	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.00	150	1	⋯
c56	0.00	+∞	100.00	+∞	0.00	347171.92	100.00	3606.71	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.02	150	1	⋯
c57	10640.39	13344.84	**20.27**	3600.04	10298.77	31109.05	66.89	3603.12	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	150	1	⋯
c58	19922.73	25787.40	**22.74**	3600.04	21631.33	52541.38	58.83	3603.13	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.33	150	1	⋯
c59	10131.24	12039.00	**15.85**	3600.04	9534.32	18985.61	49.78	3603.13	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.34	150	1	⋯
c60	15194.19	18092.72	**16.02**	3600.03	14783.36	32373.79	54.34	3603.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.34	150	1	⋯
c61	386.75	+∞	100.00	+∞	0.00	280304.30	100.00	3608.61	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.89	150	1	⋯
c62	99.37	+∞	100.00	+∞	0.00	575369.63	100.00	3607.87	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.88	150	1	⋯
c63	578.63	+∞	100.00	+∞	0.00	218268.97	100.00	3607.85	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.73	150	1	⋯
c64	329.50	+∞	100.00	+∞	0.00	448321.60	100.00	3606.12	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.24	150	1	⋯
#S			**6**				4				1				0					4				⋯
SGM			**11.16**	8219.07			21.89	10110.98			81.74	13942.36			100.00	14400.00	+∞			41.94	10334.12	171	2	⋯

Table 112 **Detailed results for problem NLMCNDP-N, cost functions f_7**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	20614.31	20614.31	**0.00**	**842.79**	20614.31	20614.31	**0.00**	2483.12	16931.78	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.19	92000	20614.30	20614.31	**0.00**	2574.95	1164	9	⋯
c100_400_10_F_T_10	42383.55	46774.87	**9.39**	3600.01	40883.91	54298.52	24.71	3603.06	33411.49	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.23	92000	$-\infty$	$+\infty$	100.00	3601.07	900	1	⋯
c100_400_10_V_L_10	3032.72	3032.72	**0.00**	**898.39**	2956.75	3032.72	2.50	3603.19	2585.92	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.19	92000	3032.72	3032.72	**0.00**	3068.28	1265	9	⋯
c100_400_30_F_L_10	39448.08	42391.21	**6.94**	3600.02	38433.31	101983.52	62.31	3603.19	36126.09	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.25	92000	$-\infty$	$+\infty$	100.00	**683.71**	900	1	⋯
c100_400_30_F_T_10	68690.03	77022.71	**10.82**	3600.02	67558.23	101619.83	33.52	3603.24	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.26	92000	$-\infty$	$+\infty$	100.00	3601.12	900	1	⋯
c100_400_30_V_T_10	18421.52	19531.60	**5.68**	3600.02	18176.07	21815.86	16.68	3603.13	16239.20	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.28	92000	$-\infty$	$+\infty$	100.00	3601.11	900	1	⋯
c25_100_10_F_L_5	12582.51	12582.51	**0.00**	**9.97**	12582.51	12582.51	**0.00**	45.45	11684.56	13643.51	14.36	3600.10	12582.51	12582.51	**0.00**	1127.45	23000	12582.50	12582.51	**0.00**	175.58	347	6	⋯
c25_100_10_F_T_5	27481.40	27481.40	**0.00**	**1.35**	27481.40	27481.40	**0.00**	8.81	27481.40	27481.40	**0.00**	2133.26	27481.40	27481.40	**0.00**	136.35	23000	27481.37	27481.40	**0.00**	45.51	478	9	⋯
c25_100_10_V_L_5	1514.65	1514.65	**0.00**	**0.29**	1514.65	1514.65	**0.00**	3.89	1514.65	1514.65	**0.00**	892.36	1514.65	1514.65	**0.00**	62.86	23000	1514.65	1514.65	**0.00**	30.45	436	6	⋯
c25_100_30_F_L_5	27606.64	27606.64	**0.00**	**33.80**	27606.64	27606.64	**0.00**	131.55	$-\infty$	$+\infty$	100.0	3600.10	27606.64	27606.64	**0.00**	919.29	23000	27606.63	27606.64	**0.00**	708.07	559	10	⋯
c25_100_30_F_T_5	35650.82	35650.82	**0.00**	**18.50**	35650.82	35650.82	**0.00**	81.81	$-\infty$	$+\infty$	100.0	3600.10	35650.81	35650.82	**0.00**	539.71	23000	35650.80	35650.82	**0.00**	383.82	471	8	⋯
c25_100_30_V_T_5	14665.90	14665.90	**0.00**	**4.12**	14665.90	14665.90	**0.00**	51.23	$-\infty$	$+\infty$	100.0	3600.10	14665.89	14665.90	**0.00**	152.49	23000	14665.89	14665.90	**0.00**	567.48	821	14	⋯
c33	48369.06	48369.06	**0.00**	**170.73**	48369.06	48369.06	**0.00**	2813.98	45736.25	$+\infty$	100.00	3600.10	48369.05	48369.06	**0.00**	826.76	18400	48369.05	48369.06	**0.00**	3273.16	560	9	⋯
c35	24048.68	24048.68	**0.00**	**1562.82**	23285.49	28095.63	17.12	3603.03	22329.45	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.12	18400	23744.40	24147.23	1.67	3601.02	209	3	⋯
c36	84629.07	84629.07	**0.00**	**1711.78**	81183.48	90194.51	9.99	3603.00	79065.97	$+\infty$	100.00	3600.10	84629.06	84629.07	**0.00**	2038.08	18400	84417.09	84629.07	0.25	3601.02	220	5	⋯
c37	28148.58	32540.31	**13.50**	3600.03	28148.43	34153.37	17.58	3603.46	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.30	18400	$-\infty$	$+\infty$	100.00	3601.25	180	1	⋯
c38	59216.96	136895.96	**56.74**	3600.03	59251.44	153740.76	61.46	3603.30	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.30	18400	$-\infty$	$+\infty$	100.00	3601.25	180	1	⋯
c39	29657.36	34108.47	**13.05**	3600.03	29592.43	63448.18	53.36	3603.58	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.30	18400	$-\infty$	$+\infty$	100.00	3601.26	180	1	⋯
c40	61931.95	72448.31	**14.52**	3600.05	62000.43	75115.78	17.46	3603.66	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.30	18400	$-\infty$	$+\infty$	100.00	3601.27	180	1	⋯
c41	48547.37	48547.37	**0.00**	**201.89**	48547.37	48547.37	**0.00**	1908.97	46317.55	$+\infty$	100.00	3600.10	48547.36	48547.37	**0.00**	1090.26	18400	48519.84	48547.37	0.06	3601.02	252	7	⋯
c42	98330.56	101220.69	**2.86**	3600.02	94956.49	108471.11	12.46	3603.19	92132.97	$+\infty$	100.00	3600.10	$-\infty$	$+\infty$	100.00	3601.12	18400	$-\infty$	$+\infty$	100.00	3601.10	180	1	⋯
c43	33100.89	33100.89	**0.00**	**141.98**	32870.87	33100.89	0.69	3603.24	31744.14	$+\infty$	100.00	3600.10	33100.88	33100.89	**0.00**	222.16	18400	33100.86	33100.89	**0.00**	2946.00	489	16	⋯
c44	85429.58	85429.58	**0.00**	**394.58**	83162.31	85796.95	3.07	3603.15	81012.32	$+\infty$	100.00	3600.10	85429.57	85429.58	**0.00**	1421.13	18400	85424.88	85429.58	0.01	3601.02	280	8	⋯
c45	22510.14	25891.94	**13.06**	3600.03	22508.08	47994.89	53.10	3603.01	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.34	18400	$-\infty$	$+\infty$	100.00	3601.29	180	1	⋯
c46	0.00	$+\infty$	100.00	$+\infty$	50734.89	102856.10	**50.67**	3602.99	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.40	18400	$-\infty$	$+\infty$	100.00	3601.30	180	1	⋯
c47	21733.66	25002.29	**13.07**	3600.04	21735.67	40029.04	45.70	3603.03	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.36	18400	$-\infty$	$+\infty$	100.00	3601.29	180	1	⋯
c48	46062.57	51240.23	**10.10**	3600.07	46246.11	81940.11	43.56	3603.05	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.36	18400	$-\infty$	$+\infty$	100.00	3601.29	180	1	⋯
c49	10510.65	12207.88	**13.90**	3600.02	10458.99	30456.60	65.66	3603.26	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.33	27600	$-\infty$	$+\infty$	100.00	3601.28	270	1	⋯
c50	30417.29	38775.77	**21.56**	3600.03	30414.68	81577.92	62.72	3603.34	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.34	27600	$-\infty$	$+\infty$	100.00	3601.27	270	1	⋯
c51	7519.74	8325.08	**9.67**	3600.02	7485.64	16252.74	53.94	3603.28	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.33	27600	$-\infty$	$+\infty$	100.00	3601.27	270	1	⋯
c52	29387.60	35966.16	**18.29**	3600.04	29502.42	68145.07	56.71	3603.76	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.35	27600	$-\infty$	$+\infty$	100.00	3601.27	270	1	⋯
c53	0.00	$+\infty$	100.00	$+\infty$	0.00	212547.80	100.00	3606.43	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3603.16	27600	$-\infty$	$+\infty$	100.00	3602.51	270	1	⋯
c54	2.22	$+\infty$	100.00	$+\infty$	0.00	420027.23	100.00	3606.54	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3602.02	27600	$-\infty$	$+\infty$	100.00	3602.46	270	1	⋯
c55	1.03	$+\infty$	100.00	$+\infty$	0.00	165281.62	100.00	3606.59	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3603.18	27600	$-\infty$	$+\infty$	100.00	3602.51	270	1	⋯
c56	0.00	$+\infty$	100.00	$+\infty$	0.00	336840.19	100.00	3606.58	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3602.63	27600	$-\infty$	$+\infty$	100.00	3602.53	270	1	⋯
c57	56.39	$+\infty$	100.00	$+\infty$	8711.98	30113.58	**71.07**	3603.18	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.44	27600	$-\infty$	$+\infty$	100.00	3601.37	270	1	⋯
c58	110.95	$+\infty$	100.00	$+\infty$	15169.06	52343.58	**71.02**	3603.17	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.47	27600	$-\infty$	$+\infty$	100.00	3601.39	270	1	⋯
c59	7686.42	8438.63	**8.91**	3600.03	7693.08	18202.00	57.73	3603.11	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.41	27600	$-\infty$	$+\infty$	100.00	3601.38	270	1	⋯
c60	97.50	$+\infty$	100.00	$+\infty$	12912.45	32356.37	**60.09**	3603.10	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3601.42	27600	$-\infty$	$+\infty$	100.00	3601.38	270	1	⋯
c61	4.37	$+\infty$	100.00	$+\infty$	0.00	272702.38	100.00	3608.51	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3603.38	27600	$-\infty$	$+\infty$	100.00	3602.72	270	1	⋯
c62	1.05	$+\infty$	100.00	$+\infty$	0.00	564750.83	100.00	3607.84	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3603.33	27600	$-\infty$	$+\infty$	100.00	3602.73	270	1	⋯
c63	6.81	$+\infty$	100.00	$+\infty$	0.00	211816.28	100.00	3607.81	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3603.37	27600	$-\infty$	$+\infty$	100.00	3602.73	270	1	⋯
c64	3.01	$+\infty$	100.00	$+\infty$	0.00	438839.54	100.00	3606.16	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3603.47	27600	$-\infty$	$+\infty$	100.00	3602.74	270	1	⋯
#S			**14**				9				2				11					10				⋯
SGM			**8.94**	**3053.75**			17.09	5715.82			77.00	12915.11			30.02	6125.34	27632			22.15	6380.85	332	3	⋯

Table 113 Detailed results for problem NLMCNDP-N, cost functions f_8

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	21931.48	21931.48	**0.00**	**787.93**	21152.19	21931.48	3.55	3603.10	17694.72	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	21931.42	21931.48	**0.00**	3601.02	809	9	⋯
c100_400_10_F_T_10	47932.16	54805.41	**12.54**	3600.01	46271.47	55321.72	16.36	3603.10	37156.88	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	600	1	⋯
c100_400_10_V_L_10	4234.12	4589.30	**7.74**	3600.02	4108.26	4616.16	11.00	3603.07	3572.66	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	600	1	⋯
c100_400_30_F_L_10	43119.50	46499.83	**7.27**	3600.02	41867.78	78943.41	46.96	3603.26	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.12	600	1	⋯
c100_400_30_F_T_10	77072.51	90463.03	**14.80**	3600.02	75689.39	115939.39	34.72	3603.22	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.12	600	1	⋯
c100_400_30_V_T_10	37471.22	43927.94	**14.70**	3600.02	36787.44	44877.85	18.03	3624.20	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	600	1	⋯
c25_100_10_F_L_5	14017.62	14017.62	**0.00**	**20.07**	13998.96	14017.62	0.13	3602.91	12411.05	23140.97	46.37	3600.10	−∞	+∞	100.00	3600.01	+∞	14017.61	14017.62	**0.00**	231.23	281	9	⋯
c25_100_10_F_T_5	30830.43	30830.45	**0.00**	315.26	30830.45	30830.45	**0.00**	279.29	27720.44	31880.11	13.05	3600.10	−∞	+∞	100.00	3600.01	+∞	30830.43	30830.45	**0.00**	**176.26**	357	9	⋯
c25_100_10_V_L_5	1977.91	1977.91	**0.00**	**1.00**	1977.66	1977.91	0.01	3603.04	1977.91	2051.68	3.60	3600.10	−∞	+∞	100.00	3600.01	+∞	1977.91	1977.91	**0.00**	28.07	274	10	⋯
c25_100_30_F_L_5	30467.55	30467.58	**0.00**	**255.37**	30380.48	30467.58	0.29	3602.61	26457.74	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	30467.56	30467.58	**0.00**	3143.53	345	10	⋯
c25_100_30_F_T_5	38912.53	38912.56	**0.00**	**167.65**	38912.57	38912.57	**0.00**	678.71	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	38912.54	38912.57	**0.00**	1599.48	367	10	⋯
c25_100_30_V_T_5	37019.79	37472.48	1.21	3600.01	35185.32	37472.50	6.10	3602.94	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	37425.89	37472.50	**0.12**	3601.02	251	5	⋯
c33	64310.12	64761.12	**0.70**	3600.02	61624.40	64872.44	5.01	3605.89	55800.61	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	64101.44	64775.32	1.04	3601.02	159	3	⋯
c35	43558.70	44120.40	**1.27**	3600.02	−∞	+∞	100.0	+∞	33766.24	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	43414.64	44226.09	1.83	3601.01	148	3	⋯
c36	114179.07	117127.72	**2.52**	3600.02	112870.81	117245.82	3.73	3603.01	100359.30	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	112444.62	117279.53	4.12	3601.02	137	2	⋯
c37	32561.74	38575.10	**15.59**	3600.03	34096.21	64937.61	47.49	3603.21	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.25	120	1	⋯
c38	66357.11	98159.57	**32.40**	3600.04	64988.26	163836.87	60.33	3603.13	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	120	1	⋯
c39	34099.68	41311.89	**17.46**	3600.03	35924.65	70671.68	49.17	3603.24	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	120	1	⋯
c40	67532.05	80425.56	**16.03**	3600.03	69983.15	137917.12	49.26	3603.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	120	1	⋯
c41	73132.94	74093.50	**1.30**	3600.02	71525.04	75618.09	5.41	3623.91	62091.14	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	70818.86	74537.52	4.99	3601.02	138	2	⋯
c42	118955.72	124281.31	**4.29**	3600.02	116128.90	128368.99	9.54	3603.22	107268.20	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	120	1	⋯
c43	56953.49	57489.80	**0.93**	3600.01	56104.73	57623.06	2.63	3603.14	45352.80	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	56631.40	57507.71	1.52	3601.02	158	3	⋯
c44	119857.02	122501.87	**2.16**	3600.02	118136.90	122756.10	3.76	3603.10	102333.70	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	117241.86	122394.64	4.21	3601.02	142	2	⋯
c45	25411.72	29950.76	**15.16**	3600.02	26527.41	50390.69	47.36	3602.96	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.30	120	1	⋯
c46	55615.69	63681.31	**12.67**	3600.03	57659.46	104977.96	45.07	3603.01	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.31	120	1	⋯
c47	24784.73	28773.50	**13.86**	3600.03	26025.09	42722.31	39.08	3603.05	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.31	120	1	⋯
c48	50654.84	58986.54	**14.12**	3600.04	49963.70	88618.38	43.62	3602.99	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.31	120	1	⋯
c49	12665.14	15098.26	**16.12**	3600.04	12990.23	31697.60	59.02	3603.29	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.27	180	1	⋯
c50	34371.68	42320.27	**18.78**	3600.03	35131.58	82664.00	57.50	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.27	180	1	⋯
c51	9673.66	10926.08	**11.46**	3600.03	10017.47	16319.60	38.62	3603.31	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	180	1	⋯
c52	33525.08	40089.48	**16.37**	3600.02	0.00	422086.44	100.00	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	180	1	⋯
c53	0.00	+∞	100.00	+∞	0.00	219704.27	100.00	3606.58	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.01	180	1	⋯
c54	163.79	+∞	100.00	+∞	0.00	429640.90	100.00	3606.44	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.98	180	1	⋯
c55	107.75	+∞	100.00	+∞	0.00	170939.62	100.00	3606.75	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.56	180	1	⋯
c56	0.00	+∞	100.00	+∞	0.00	345577.73	100.00	3606.68	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.02	180	1	⋯
c57	10692.91	12809.66	**16.52**	3600.03	10221.55	30896.80	66.92	3603.21	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.36	180	1	⋯
c58	19952.70	24598.78	**18.89**	3600.04	21067.83	52214.37	59.65	3603.11	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	180	1	⋯
c59	10118.86	11540.47	**12.32**	3600.04	9533.85	18123.56	47.40	3603.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.34	180	1	⋯
c60	15140.79	17780.07	**14.84**	3600.04	14622.25	32283.61	54.71	3603.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	180	1	⋯
c61	387.66	+∞	100.00	+∞	0.00	279117.68	100.00	3608.44	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.87	180	1	⋯
c62	100.55	+∞	100.00	+∞	0.00	573711.98	100.00	3607.78	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.77	180	1	⋯
c63	575.18	+∞	100.00	+∞	0.00	217267.59	100.00	3607.75	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.78	180	1	⋯
c64	343.21	+∞	100.00	+∞	0.00	446821.77	100.00	3606.12	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.78	180	1	⋯
#S			**6**				2				0				0					5				⋯
SGM			**9.80**	7430.03			22.23	12249.67			87.21	14400.00			100.00	14400.00	+∞			29.05	9452.27	208	2	⋯

Table 114 Detailed results for problem NLMCNDP-N, cost functions f_9

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	20475.60	21535.99	4.92	3600.01	19746.27	21652.95	8.81	3603.08	16291.22	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	21528.92	21535.99	**0.03**	3601.02	986	4	···
c100_400_10_F_T_10	42200.37	48015.45	**12.11**	3600.01	40201.49	53745.61	25.20	3603.11	32911.67	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	900	1	···
c100_400_10_V_L_10	2665.29	3928.13	**32.15**	3600.01	2565.64	4700.00	45.41	3603.12	1844.72	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	900	1	···
c100_400_30_F_L_10	36572.68	46526.38	**21.39**	3600.02	34768.74	148489.93	76.59	3603.19	32413.27	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	900	1	···
c100_400_30_F_T_10	62669.19	88468.09	**29.16**	3600.02	59957.37	194546.91	69.18	3603.22	53845.66	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.19	900	1	···
c100_400_30_V_T_10	13172.07	40101.74	67.15	3600.02	13010.20	33832.50	**61.55**	3603.18	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	900	1	···
c25_100_10_F_L_5	12844.37	13545.95	5.18	3600.01	12602.42	13545.96	6.97	3602.96	10439.11	14537.39	28.19	3600.10	−∞	+∞	100.00	3600.01	+∞	13545.95	13545.96	**0.00**	**1087.68**	339	8	···
c25_100_10_F_T_5	27151.28	29103.47	6.71	3600.01	26827.86	29103.48	7.82	3602.94	25138.63	30207.78	16.78	3600.10	−∞	+∞	100.00	3600.01	+∞	29103.48	29103.48	**0.00**	**240.51**	446	9	···
c25_100_10_V_L_5	1745.48	1937.13	9.89	3600.01	1660.72	1937.13	14.27	3603.07	1093.65	2367.30	53.80	3600.10	−∞	+∞	100.00	3600.01	+∞	1937.13	1937.13	**0.00**	**134.85**	356	9	···
c25_100_30_F_L_5	26081.65	29503.88	11.60	3600.02	25801.23	29590.71	12.81	3603.13	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	29438.53	29503.90	**0.22**	3601.02	266	3	···
c25_100_30_F_T_5	32159.02	39399.74	18.38	3600.02	32552.26	39513.43	17.62	3603.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	38426.80	39171.59	**1.90**	3601.02	252	2	···
c25_100_30_V_T_5	7588.47	27772.95	72.68	3600.01	7634.19	26573.58	71.27	3602.96	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	24555.81	24610.40	**0.22**	3601.02	337	8	···
c33	42842.09	58672.19	**26.98**	3600.02	42070.59	63611.79	33.86	3604.63	18198.38	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	180	1	···
c35	20470.47	31866.46	**35.76**	3600.01	20133.54	38290.39	47.42	3603.07	8892.64	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	180	1	···
c36	77759.70	106166.60	**26.76**	3600.02	77467.37	118523.91	34.64	3603.05	41622.25	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	180	1	···
c37	21622.07	35379.49	**38.89**	3600.03	22901.04	50452.62	54.61	3603.23	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.33	180	1	···
c38	50762.40	78635.93	**35.45**	3600.02	50397.33	138532.67	63.62	3603.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.34	180	1	···
c39	23812.51	39572.64	**39.83**	3600.02	24940.25	56636.69	55.96	3603.26	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.36	180	1	···
c40	53614.43	76514.66	**29.93**	3600.02	56026.83	128910.29	56.54	3603.21	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	180	1	···
c41	42248.93	62185.61	**32.06**	3600.05	42164.70	72250.62	41.64	3603.33	13758.80	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.19	180	1	···
c42	85627.12	122530.25	**30.12**	3600.01	84747.42	132712.32	36.14	3603.22	54142.16	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.19	180	1	···
c43	33599.49	46083.78	**27.09**	3600.01	33843.44	55713.63	39.25	3603.24	9921.88	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	180	1	···
c44	78124.24	109186.34	**28.45**	3600.01	76945.72	127323.80	39.57	3603.16	39747.34	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.21	180	1	···
c45	17862.72	29691.17	**39.84**	3600.07	17774.36	40793.67	56.43	3602.93	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.43	180	1	···
c46	44260.24	62445.12	**29.12**	3600.03	44211.24	100131.01	55.85	3603.06	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.47	180	1	···
c47	16868.07	28336.79	**40.47**	3600.04	16831.96	39111.40	56.96	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.36	180	1	···
c48	40979.39	57672.13	**28.94**	3600.03	40776.48	84101.11	51.51	3603.04	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.46	180	1	···
c49	5943.00	15048.15	**60.51**	3600.03	5758.66	34065.15	83.10	3603.25	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.36	270	1	···
c50	23773.72	41522.40	**42.74**	3600.04	23346.24	587239.06	96.02	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	270	1	···
c51	4091.17	11427.91	**64.20**	3600.03	3891.09	23672.57	83.56	3603.36	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	270	1	···
c52	23251.73	40223.04	**42.19**	3600.03	22833.27	135968.95	83.21	3603.37	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.40	270	1	···
c53	15.10	+∞	100.00	+∞	13.02	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.30	270	1	···
c54	20.61	+∞	100.00	+∞	16.66	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.41	270	1	···
c55	12.90	+∞	100.00	+∞	10.53	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.35	270	1	···
c56	18.45	+∞	100.00	+∞	15.33	338279.48	100.00	3606.67	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.34	270	1	···
c57	4587.20	13244.58	**65.37**	3600.04	5.57	158467.18	100.00	3603.18	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.43	270	1	···
c58	8704.15	24825.10	**64.94**	3600.04	12.44	323401.88	100.00	3603.47	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.40	270	1	···
c59	3728.15	12034.11	**69.02**	3600.04	6.27	104756.19	99.99	3603.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.44	270	1	···
c60	8204.64	18173.71	**54.85**	3600.04	6.49	214892.01	100.00	3603.10	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.45	270	1	···
c61	15.03	+∞	100.00	+∞	11.03	272845.77	100.00	3608.59	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3603.45	270	1	···
c62	18.44	+∞	100.00	+∞	14.85	565096.79	100.00	3607.81	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3603.61	270	1	···
c63	13.73	+∞	100.00	+∞	9.37	212595.24	100.00	3607.85	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3603.57	270	1	···
c64	17.40	+∞	100.00	+∞	13.41	438828.41	100.00	3606.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3603.54	270	1	···
#S			0				0				0				0					**3**				···
SGM			**37.19**	14400.00			51.55	14400.00			91.91	14400.00			100.00	14400.00	+∞			48.34	11088.75	286	2	···

Table 115 Detailed results for problem NLMCNDP-N, cost functions f_{10}

	GUROBI				**SCIP**				**COUENNE**				**NAIVE**					**CN24**						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	43601.58	58067.82	24.91	3600.01	42754.65	58458.40	26.86	3603.97	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.64	287299	55621.05	58007.48	**4.11**	3601.02	1533	2	⋯
c100_400_10_F_T_10	114269.80	186783.95	**38.82**	3600.01	119794.77	218444.62	45.16	3604.04	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.65	295012	−∞	+∞	100.00	3601.07	1529	1	⋯
c100_400_10_V_L_10	47253.97	88401.11	**46.55**	3600.01	48422.15	90990.75	46.78	3603.05	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.63	286141	−∞	+∞	100.00	3601.07	1508	1	⋯
c100_400_30_F_L_10	89123.61	161083.91	**44.67**	3600.02	23172.63	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.65	319676	−∞	+∞	100.00	3601.11	1578	1	⋯
c100_400_30_F_T_10	326320.51	554198.09	**41.12**	3600.02	119988.47	+∞	100.00	+∞	293385.70	+∞	100.00	3600.10	−∞	+∞	100.00	3601.64	314346	−∞	+∞	100.00	3601.10	1581	1	⋯
c100_400_30_V_T_10	1125350.15	1803011.78	**37.58**	3600.02	516567.88	+∞	100.00	+∞	765427.00	+∞	100.00	3600.10	−∞	+∞	100.00	3601.55	313877	−∞	+∞	100.00	3601.19	1573	1	⋯
c25_100_10_F_L_5	29103.45	36591.55	20.46	3600.01	28795.70	36591.56	21.31	3602.91	23232.89	45012.18	48.39	3600.10	−∞	+∞	100.00	3601.19	71488	36591.54	36591.56	**0.00**	**1093.15**	540	11	⋯
c25_100_10_F_T_5	105976.64	117963.01	10.16	3600.01	108149.97	117963.03	8.32	3602.47	88300.58	123991.90	28.79	3600.10	117962.98	117963.03	**0.00**	2175.38	72588	117962.97	117963.03	**0.00**	**378.24**	1099	11	⋯
c25_100_10_V_L_5	39922.98	50260.44	20.57	3600.01	36109.80	50260.47	28.15	3602.86	29221.05	53804.87	45.69	3600.10	−∞	+∞	100.00	3601.16	70521	49537.62	49537.66	**0.00**	**962.47**	634	16	⋯
c25_100_30_F_L_5	80937.04	92397.03	12.40	3600.01	82177.25	91505.77	10.19	3602.91	65349.11	105409.00	38.00	3600.10	−∞	+∞	100.00	3601.13	70778	90256.81	90256.90	**0.00**	**2775.75**	746	18	⋯
c25_100_30_F_T_5	231571.36	268717.82	13.82	3600.01	239056.72	268985.61	11.13	3602.92	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.15	78859	260795.33	260799.44	**0.00**	3601.02	740	13	⋯
c25_100_30_V_T_5	1205984.78	1454340.79	17.08	3600.01	1217030.12	1454888.05	16.35	3602.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.20	80107	1407647.36	1419490.95	**0.83**	3601.01	477	4	⋯
c33	453864.14	781416.53	41.92	3600.02	444916.16	910598.54	51.14	3603.13	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.35	166739	619649.75	647316.26	**4.27**	3601.02	956	3	⋯
c35	243868.23	701729.50	65.25	3600.02	256518.43	703332.00	63.53	3603.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.40	187186	534173.27	568104.43	**5.97**	3601.02	998	3	⋯
c36	460621.43	1151071.09	**59.98**	3600.02	473668.25	1249577.67	62.09	3603.47	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.41	185765	−∞	+∞	100.00	3601.17	920	1	⋯
c37	78956.24	288379.28	**72.62**	3600.03	81156.88	681939.51	88.10	3603.87	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.53	191274	−∞	+∞	100.00	3601.25	912	1	⋯
c38	130046.89	517057.05	**74.85**	3600.03	129253.03	1255964.04	89.71	3604.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.53	192700	−∞	+∞	100.00	3601.30	920	1	⋯
c39	80844.41	314619.13	**74.30**	3600.07	81096.71	642326.39	87.37	3603.36	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.58	193311	−∞	+∞	100.00	3601.27	916	1	⋯
c40	130542.35	484984.82	**73.08**	3600.03	136711.88	814587.89	83.22	3603.15	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.58	193152	−∞	+∞	100.00	3601.27	912	1	⋯
c41	374362.60	785667.83	**52.35**	3600.02	374343.12	886027.30	57.75	3603.26	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.87	216778	−∞	+∞	100.00	3601.10	1152	1	⋯
c42	466048.38	1045524.97	**55.42**	3600.01	463741.61	1224932.88	62.14	3603.23	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.43	224931	−∞	+∞	100.00	3601.18	1176	1	⋯
c43	311294.35	838459.32	**62.87**	3600.01	323945.38	979458.82	66.93	3603.24	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.38	235810	−∞	+∞	100.00	3601.10	1176	1	⋯
c44	450126.61	1222517.03	**63.18**	3600.01	462128.27	1316977.56	64.91	3603.18	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.38	234695	−∞	+∞	100.00	3601.09	1176	1	⋯
c45	63595.95	186554.22	**65.91**	3600.03	64140.10	621358.13	89.68	3603.89	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.68	249931	−∞	+∞	100.00	3601.31	1176	1	⋯
c46	110968.49	299166.40	**62.91**	3600.03	111424.89	1060491.37	89.49	3603.31	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.60	247712	−∞	+∞	100.00	3601.34	1168	1	⋯
c47	63552.32	183671.91	**65.40**	3600.03	64262.44	574775.82	88.82	3603.60	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.57	247748	−∞	+∞	100.00	3601.39	1164	1	⋯
c48	100972.63	275883.50	**63.40**	3600.04	102329.72	919854.93	88.88	3603.09	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.47	247637	−∞	+∞	100.00	3601.38	1164	1	⋯
c49	42651.60	206001.05	**79.30**	3600.03	41987.11	785217.75	94.65	3603.73	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.37	2072	1	⋯
c50	83058.61	405681.49	**79.53**	3600.02	82862.03	1514039.69	94.53	3603.42	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.83	440944	−∞	+∞	100.00	3601.29	2064	1	⋯
c51	39314.26	185093.90	**78.76**	3600.03	-0.00	567086.11	100.00	3603.33	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.91	442493	−∞	+∞	100.00	3601.29	2076	1	⋯
c52	83441.34	374137.34	**77.70**	3600.03	84145.14	1120504.43	92.49	3603.18	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.87	441308	−∞	+∞	100.00	3601.33	2068	1	⋯
c53	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.74	439092	−∞	+∞	100.00	3602.10	2080	1	⋯
c54	-0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.66	439156	−∞	+∞	100.00	3602.06	2080	1	⋯
c55	-0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.59	438304	−∞	+∞	100.00	3601.79	2064	1	⋯
c56	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.60	439853	−∞	+∞	100.00	3601.81	2072	1	⋯
c57	38076.24	117881.30	**67.70**	3600.04	-0.00	1115125.26	100.00	3603.72	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.29	2720	1	⋯
c58	48942.79	+∞	100.00	+∞	-0.00	1333473.62	100.00	3603.43	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.37	2720	1	⋯
c59	34074.05	117104.61	**70.90**	3600.04	-0.00	739552.13	100.00	3603.82	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.36	2748	1	⋯
c60	45090.97	205553.15	**78.06**	3600.04	-0.00	920675.31	100.00	3603.82	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.36	2744	1	⋯
c61	-0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.99	2740	1	⋯
c62	-0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.64	2716	1	⋯
c63	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.34	2712	1	⋯
c64	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.56	2732	1	⋯
#S			0				0				0				1					**4**				⋯
SGM			54.95	14400.00			64.48	14400.00			91.76	14400.00			89.72	13781.59	14175577			**43.04**	11271.91	1417	2	⋯

Table 116 Detailed results for problem NLMCNDP-A, cost functions f_1

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	34146.50	34655.19	1.47	3600.01	7587.12	41907.53	81.90	3604.10	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	34618.47	34662.86	**0.13**	3601.02	2115	6	⋯
c100_400_10_F_T_10	98050.66	106668.89	**8.08**	3600.02	-65374.46	292275.59	122.37	3604.18	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.15	2000	1	⋯
c100_400_10_V_L_10	29460.66	29740.10	**0.94**	3600.01	-542680.47	509499.74	193.89	3603.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	29130.39	29846.18	2.40	3601.02	2023	4	⋯
c100_400_30_F_L_10	71679.01	76949.84	**6.85**	3600.02	-65576.53	888054.61	107.38	3603.20	-268132.10	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	2000	1	⋯
c100_400_30_F_T_10	256922.95	300442.43	**14.49**	3600.02	−∞	+∞	100.0	+∞	-1019989.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	2000	1	⋯
c100_400_30_V_T_10	714934.82	863337.26	**17.19**	3600.02	−∞	+∞	100.0	+∞	-7426142.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.16	2000	1	⋯
c25_100_10_F_L_5	21643.63	21643.65	**0.00**	**95.99**	5871.13	23490.32	75.01	3602.86	-19055.23	30816.96	161.83	3600.10	−∞	+∞	100.00	3601.23	95400	21643.64	21643.66	**0.00**	1278.47	748	23	⋯
c25_100_10_F_T_5	79080.37	80427.68	1.68	3600.02	-14691.22	88485.29	116.60	3602.93	-46637.07	101693.90	145.86	3600.10	−∞	+∞	100.00	3601.25	95400	80427.61	80427.69	**0.00**	**2749.98**	885	20	⋯
c25_100_10_V_L_5	**19527.24**	**19527.25**	**0.00**	**70.75**	**-48785.60**	**20552.58**	**142.13**	**3602.90**	**-97470.83**	**55284.88**	**156.72**	**3600.10**	−∞	+∞	**100.00**	**3601.24**	**95400**	**18074.02**	**18074.04**	**0.00**	**366.22**	**809**	**19**	⋯
c25_100_30_F_L_5	60514.12	63576.52	4.82	3600.02	-11367.27	69191.46	116.43	3602.84	-42167.05	82437.09	151.15	3600.10	−∞	+∞	100.00	3601.22	95400	61964.70	62177.59	**0.34**	3601.02	628	7	⋯
c25_100_30_F_T_5	168777.94	177376.16	4.85	3600.01	-170160.61	190039.08	189.54	3602.77	-175951.40	225799.80	177.92	3600.10	−∞	+∞	100.00	3601.23	95400	174379.64	176830.32	**1.39**	3601.02	600	7	⋯
c25_100_30_V_T_5	772592.12	851787.56	9.30	3600.01	-1916507.19	1113629.55	158.11	3603.06	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.47	95400	835035.20	850005.85	**1.76**	3601.02	582	9	⋯
c33	385317.34	482954.22	**20.22**	3600.02	-25749292.28	2208413.80	108.58	3660.73	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	1140	1	⋯
c35	238300.65	349510.74	**31.82**	3600.02	-3128543.95	2165252.18	169.21	3636.39	-2576910.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.12	1150	1	⋯
c36	449013.86	659878.82	**31.96**	3600.01	-3754743.33	1557860.48	141.49	3650.70	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.12	1150	1	⋯
c37	82131.46	1154685.99	**92.89**	3600.03	-217970.24	1161194.57	118.77	3603.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.26	1140	1	⋯
c38	130288.40	241867.34	**46.13**	3600.03	-166574.03	1843168.00	109.04	3604.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.32	1150	1	⋯
c39	85116.46	116448.23	**26.91**	3600.03	-176641.24	980671.66	118.01	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.23	1145	1	⋯
c40	136685.55	181299.23	**24.61**	3600.03	-100799.23	1277046.13	107.89	3603.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.25	1140	1	⋯
c41	348283.41	496709.49	**29.88**	3600.01	-23538796.15	1531544.24	106.51	3609.13	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	1440	1	⋯
c42	457974.26	623583.27	**26.56**	3600.01	-21404206.84	7841314.76	136.63	3608.18	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.14	1470	1	⋯
c43	304519.40	475915.38	**36.01**	3600.02	-6694036.72	3527316.87	152.69	3647.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.15	1470	1	⋯
c44	445179.65	611080.85	**27.15**	3600.01	-7584538.55	4773680.74	162.94	3613.35	-5252799.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.17	1470	1	⋯
c45	72404.25	88191.78	**17.90**	3600.03	-147037.00	919423.19	115.99	3603.83	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.33	1470	1	⋯
c46	118581.62	156596.35	**24.28**	3600.03	-92706.90	1486017.79	106.24	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	1460	1	⋯
c47	73345.50	88410.92	**17.04**	3600.06	-115383.79	837090.04	113.78	3603.49	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.30	1455	1	⋯
c48	110555.81	144387.55	**23.43**	3600.04	-77631.99	1268335.83	106.12	3603.00	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	1455	1	⋯
c49	49101.01	54414.22	**9.76**	3600.03	-301008.68	1178469.70	125.54	3603.59	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.27	2590	1	⋯
c50	93908.75	118757.62	**20.92**	3600.03	-338637.02	2005104.69	116.89	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.31	2580	1	⋯
c51	46342.44	49717.97	**6.79**	3600.03	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.34	2595	1	⋯
c52	100298.73	110748.63	**9.44**	3600.03	-97836.04	1269782.99	107.70	3603.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	2585	1	⋯
c53	0.00	+∞	100.00	+∞	-1471796.17	1572310.52	193.61	3606.31	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.11	2600	1	⋯
c54	0.00	+∞	100.00	+∞	-1532951.47	1963180.92	178.09	3606.63	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.07	2600	1	⋯
c55	0.00	+∞	100.00	+∞	-1154700.35	1209305.27	195.48	3605.46	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.90	2580	1	⋯
c56	0.00	+∞	100.00	+∞	-1233740.46	1572400.37	178.46	3606.68	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.79	2590	1	⋯
c57	43742.13	49357.06	**11.38**	3600.07	-449273.61	1732433.51	125.93	3603.69	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.38	3400	1	⋯
c58	55739.88	64629.63	**13.75**	3600.04	-336645.12	1991162.02	116.91	3603.26	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.38	3400	1	⋯
c59	**40921.74**	**45497.06**	**10.06**	**3600.03**	**-289529.31**	**870290.10**	**133.27**	**3603.80**	**-3258127.00**	**-3258127.00**	**0.00**	**0.51**	−∞	+∞	**100.00**	**3600.01**	+∞	−∞	+∞	**100.00**	**3601.38**	**3435**	**1**	⋯
c60	53744.24	59248.61	**9.29**	3600.03	-230481.49	1086181.35	121.22	3603.70	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.36	3430	1	⋯
c61	0.00	+∞	100.00	+∞	-2020707.45	1967204.24	197.35	3609.28	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.47	3425	1	⋯
c62	0.00	+∞	100.00	+∞	-2116476.33	2640598.91	180.15	3606.71	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.52	3395	1	⋯
c63	0.00	+∞	100.00	+∞	-1419637.34	1513727.94	193.78	3608.77	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.98	3390	1	⋯
c64	0.00	+∞	100.00	+∞	-1630855.70	2076188.35	178.55	3610.83	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.15	3415	1	⋯
#S			1				0				0				0					**2**				⋯
SGM			**19.38**	12772.91			98.82	14400.00			100.00	14400.00			100.00	14400.00	7942376584386			49.05	13039.15	1731	2	⋯

Table 117 **Detailed results for problem NLMCNDP-A, cost functions f_2**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	24583.02	29714.46	17.27	3600.02	26473.09	29714.46	10.91	3603.95	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.73	460000	29419.46	29463.36	**0.15**	3601.02	3715	5	⋯
c100_400_10_F_T_10	54764.64	75587.09	**27.55**	3600.02	63072.51	88530.62	28.76	3604.00	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.68	460000	−∞	+∞	100.00	3601.07	3600	1	⋯
c100_400_10_V_L_10	8397.64	19722.80	57.42	3600.01	12388.82	20821.79	**40.50**	3603.11	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.79	460000	−∞	+∞	100.00	3601.07	3600	1	⋯
c100_400_30_F_L_10	44283.96	169492.72	**73.87**	3600.02	6069.90	+∞	100.00	+∞	38681.20	+∞	100.00	3600.10	−∞	+∞	100.00	3601.87	460000	−∞	+∞	100.00	3601.10	3600	1	⋯
c100_400_30_F_T_10	87450.22	132857.16	**34.18**	3600.02	10324.05	+∞	100.00	+∞	72667.52	+∞	100.00	3600.10	−∞	+∞	100.00	3601.99	460000	−∞	+∞	100.00	3601.21	3600	1	⋯
c100_400_30_V_T_10	72572.16	+∞	100.00	+∞	10734.85	+∞	100.00	+∞	34223.33	+∞	100.00	3600.10	−∞	+∞	100.00	3601.90	460000	−∞	+∞	100.00	3601.21	3600	1	⋯
c25_100_10_F_L_5	19034.87	20507.46	7.18	3600.01	19172.04	20507.46	6.51	3602.89	15554.05	26342.19	40.95	3600.10	−∞	+∞	100.00	3601.21	115000	20507.45	20507.49	**0.00**	3601.02	1920	21	⋯
c25_100_10_F_T_5	49761.77	50738.35	1.92	3600.01	46833.96	50754.97	7.73	3602.89	38745.71	52903.66	26.76	3600.10	−∞	+∞	100.00	3601.26	115000	50738.35	50738.37	**0.00**	**514.27**	1367	13	⋯
c25_100_10_V_L_5	7891.34	9796.16	19.44	3600.02	6352.18	9796.16	35.16	3602.89	5297.94	11131.56	52.41	3600.10	−∞	+∞	100.00	3601.21	115000	9796.14	9796.20	**0.00**	3601.02	2023	25	⋯
c25_100_30_F_L_5	35901.59	38486.23	6.72	3600.01	34907.05	38486.23	9.30	3602.94	30468.39	+∞	100.00	3600.10	−∞	+∞	100.00	3601.23	115000	38362.28	38362.30	**0.00**	**2455.72**	1705	14	⋯
c25_100_30_F_T_5	61061.23	66833.69	8.64	3600.01	54752.23	66756.45	17.98	3603.23	49697.11	71130.65	30.13	3600.10	−∞	+∞	100.00	3601.28	115000	66471.49	66489.28	**0.03**	3601.02	1111	8	⋯
c25_100_30_V_T_5	134554.85	161865.28	16.87	3600.02	123675.90	162238.78	23.77	3602.58	93502.40	168280.30	44.44	3600.10	−∞	+∞	100.00	3601.22	115000	161829.45	161873.16	**0.03**	3601.02	1477	19	⋯
c33	92580.29	244734.69	62.17	3600.02	138217.73	246188.01	43.86	3603.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.46	262200	230470.96	242611.23	**5.00**	3601.02	2099	2	⋯
c35	120436.93	297469.26	59.51	3600.02	166626.05	288412.88	42.23	3603.27	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.49	264500	272538.14	287043.36	**5.05**	3601.02	2151	3	⋯
c36	241969.91	597983.86	59.54	3600.03	347795.82	580407.26	40.08	3603.28	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.47	264500	536872.40	571699.65	**6.09**	3601.02	2115	2	⋯
c37	0.00	+∞	100.00	+∞	80583.00	153369.35	**47.46**	3604.67	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.82	262200	−∞	+∞	100.00	3601.26	2052	1	⋯
c38	0.00	+∞	100.00	+∞	145610.98	293698.23	**50.42**	3604.44	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.81	264500	−∞	+∞	100.00	3601.30	2070	1	⋯
c39	0.00	+∞	100.00	+∞	86666.57	171793.72	**49.55**	3603.37	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.65	263350	−∞	+∞	100.00	3601.27	2061	1	⋯
c40	0.00	+∞	100.00	+∞	152468.79	284229.33	**46.36**	3603.38	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.66	262200	−∞	+∞	100.00	3601.30	2052	1	⋯
c41	87504.66	304161.34	71.23	3600.02	143701.28	250634.94	42.67	3603.40	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.62	331200	240545.89	250747.29	**4.07**	3601.02	2711	3	⋯
c42	194785.25	575228.48	66.14	3600.02	314336.81	538267.31	**41.60**	3603.22	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.63	338100	−∞	+∞	100.00	3601.07	2646	1	⋯
c43	130258.84	369040.82	64.70	3600.03	192535.79	349161.12	44.86	3603.44	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.66	338100	322761.63	345541.07	**6.59**	3601.02	2689	2	⋯
c44	200819.84	586856.96	65.78	3600.02	339340.04	592468.41	42.72	3603.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.68	338100	542903.45	571319.21	**4.97**	3601.02	2697	2	⋯
c45	0.00	+∞	100.00	+∞	69547.28	122218.56	**43.10**	3603.64	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.91	338100	−∞	+∞	100.00	3601.34	2646	1	⋯
c46	0.00	+∞	100.00	+∞	129974.59	252547.81	**48.53**	3603.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.77	335800	−∞	+∞	100.00	3601.32	2628	1	⋯
c47	0.00	+∞	100.00	+∞	68717.50	123401.83	**44.31**	3603.05	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.75	334650	−∞	+∞	100.00	3601.38	2619	1	⋯
c48	0.00	+∞	100.00	+∞	121486.60	189474.87	**35.88**	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.86	334650	−∞	+∞	100.00	3601.34	2619	1	⋯
c49	0.00	+∞	100.00	+∞	31820.80	66238.57	**51.96**	3603.57	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.00	595700	−∞	+∞	100.00	3601.34	4662	1	⋯
c50	0.00	+∞	100.00	+∞	64487.72	266693.45	**75.82**	3603.82	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.93	593400	−∞	+∞	100.00	3601.27	4644	1	⋯
c51	0.00	+∞	100.00	+∞	32066.02	58001.21	**44.71**	3602.64	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.08	596850	−∞	+∞	100.00	3601.34	4671	1	⋯
c52	0.00	+∞	100.00	+∞	85752.06	193356.83	**55.65**	3602.89	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.99	594550	−∞	+∞	100.00	3601.31	4653	1	⋯
c53	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.65	598000	−∞	+∞	100.00	3602.10	4680	1	⋯
c54	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.89	598000	−∞	+∞	100.00	3601.94	4680	1	⋯
c55	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.81	593400	−∞	+∞	100.00	3601.81	4644	1	⋯
c56	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.91	595700	−∞	+∞	100.00	3601.89	4662	1	⋯
c57	**0.00**	**+∞**	**100.00**	**+∞**	**0.00**	**170057.07**	**100.00**	**3603.76**	**−∞**	**+∞**	**100.0**	**3600.10**	**−∞**	**+∞**	**100.00**	**3602.65**	**782000**	**139727.72**	**135193.16**	**-3.35**	**2296.61**	**6338**	**2**	⋯
c58	0.00	+∞	100.00	+∞	0.00	334608.51	100.00	3603.40	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.48	782000	263858.18	264996.39	**0.43**	3601.04	6243	2	⋯
c59	0.00	+∞	100.00	+∞	0.00	117844.54	100.00	3603.69	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.68	790050	−∞	+∞	100.00	3601.35	6183	1	⋯
c60	0.00	+∞	100.00	+∞	0.00	230297.34	100.00	3603.48	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.45	788900	−∞	+∞	100.00	3601.47	6174	1	⋯
c61	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.56	787750	−∞	+∞	100.00	3603.53	6165	1	⋯
c62	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3604.26	780850	−∞	+∞	100.00	3603.45	6111	1	⋯
c63	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3605.45	779700	−∞	+∞	100.00	3602.04	6102	1	⋯
c64	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3605.64	785450	−∞	+∞	100.00	3602.15	6147	1	⋯
#S			0				0				0				0					**2**				⋯
SGM			58.45	14400.00			48.36	14400.00			89.13	14400.00			100.00	14400.00	371946			**27.55**	12758.63	3168	3	⋯

Table 118 Detailed results for problem NLMCNDP-A, cost functions f_3

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	24248.10	31506.70	23.04	3600.01	25941.38	31506.71	17.66	3604.01	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	31080.75	31272.88	**0.61**	3601.03	3665	4	···
c100_400_10_F_T_10	54429.70	81611.28	33.31	3600.02	63596.98	86635.84	**26.59**	3604.05	44118.87	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	3600	1	···
c100_400_10_V_L_10	8715.91	27470.50	68.27	3600.02	12452.74	27755.33	**55.13**	3603.05	7911.39	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	3600	1	···
c100_400_30_F_L_10	43779.60	346153.51	**87.35**	3600.02	-41990.20	+∞	100.00	+∞	38797.96	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	3600	1	···
c100_400_30_F_T_10	86901.19	1483887.35	**94.14**	3600.02	-139808.76	+∞	100.00	+∞	72839.66	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.19	3600	1	···
c100_400_30_V_T_10	85834.41	8872864.71	**99.03**	3600.03	-951504.05	+∞	100.00	+∞	50727.39	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	3600	1	···
c25_100_10_F_L_5	19234.17	20748.65	7.30	3600.01	18711.05	20748.65	9.82	3602.06	15494.65	25905.25	40.19	3600.10	−∞	+∞	100.00	3601.26	163500	20748.64	20748.70	**0.00**	3601.01	1926	28	···
c25_100_10_F_T_5	51869.34	62019.02	16.37	3600.01	48842.01	62110.56	21.36	3602.51	40125.38	68712.36	41.60	3600.10	−∞	+∞	100.00	3601.39	163500	61727.58	61727.73	**0.00**	3601.02	2146	25	···
c25_100_10_V_L_5	10016.26	13315.92	24.78	3600.02	7101.54	13541.24	47.56	3602.89	6328.00	15926.49	60.27	3600.10	−∞	+∞	100.00	3601.31	163500	12960.04	12960.06	**0.00**	3601.02	2164	24	···
c25_100_30_F_L_5	37342.71	43712.71	14.57	3600.01	34688.44	43645.19	20.52	3602.50	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.23	163500	43467.92	43493.06	**0.06**	3601.03	1181	12	···
c25_100_30_F_T_5	63662.22	85801.72	25.80	3600.01	57072.38	87712.95	34.93	3603.01	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.25	163500	84382.53	84867.51	**0.57**	**3582.52**	1048	5	···
c25_100_30_V_T_5	127457.22	234982.12	45.76	3600.01	104248.15	232189.10	55.10	3603.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.28	163500	227952.54	228252.79	**0.13**	3601.02	1432	17	···
c33	89334.24	527055.40	83.05	3600.02	-1106385.16	647955.18	158.57	3603.12	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.58	372780	355517.37	371659.75	**4.34**	3601.02	2082	3	···
c35	113803.95	402780.81	**71.75**	3600.02	-351319.23	+∞	100.00	+∞	76999.53	+∞	100.00	3600.10	−∞	+∞	100.00	3601.61	376050	−∞	+∞	100.00	3601.08	2070	1	···
c36	235360.24	741085.37	**68.24**	3600.01	-443195.35	+∞	100.00	+∞	180430.20	+∞	100.00	3600.10	−∞	+∞	100.00	3601.69	376050	−∞	+∞	100.00	3601.09	2070	1	···
c37	0.00	+∞	100.00	+∞	79003.04	172533.99	**54.21**	3604.38	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.85	372780	−∞	+∞	100.00	3601.28	2052	1	···
c38	0.00	+∞	100.00	+∞	139878.50	328187.66	**57.38**	3605.65	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.83	376050	−∞	+∞	100.00	3601.29	2070	1	···
c39	0.00	+∞	100.00	+∞	84496.59	218419.95	**61.31**	3603.39	52043.69	+∞	100.00	3600.10	−∞	+∞	100.00	3601.76	374415	−∞	+∞	100.00	3601.34	2061	1	···
c40	0.00	+∞	100.00	+∞	147558.80	337823.50	**56.32**	3603.38	97477.03	+∞	100.00	3600.10	−∞	+∞	100.00	3601.69	372780	−∞	+∞	100.00	3601.31	2052	1	···
c41	84418.95	521198.55	**83.80**	3600.02	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.33	470880	−∞	+∞	100.00	3601.09	2592	1	···
c42	190257.47	829728.29	77.07	3600.02	-1094534.48	+∞	100.00	+∞	113280.90	+∞	100.00	3600.10	−∞	+∞	100.00	3601.83	480690	600738.97	637585.82	**5.78**	3601.02	2680	2	···
c43	123927.85	493822.15	74.90	3600.01	−∞	+∞	100.0	+∞	69978.67	+∞	100.00	3600.10	−∞	+∞	100.00	3601.73	480690	412024.86	443004.32	**6.99**	3601.02	2670	2	···
c44	195682.21	827391.70	**76.35**	3600.02	-671532.65	+∞	100.00	+∞	126288.20	+∞	100.00	3600.10	−∞	+∞	100.00	3601.68	480690	−∞	+∞	100.00	3601.10	2646	1	···
c45	0.00	+∞	100.00	+∞	67068.79	163488.86	**58.98**	3603.26	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.94	480690	−∞	+∞	100.00	3601.30	2646	1	···
c46	0.00	+∞	100.00	+∞	123709.18	297688.22	**58.44**	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.93	477420	−∞	+∞	100.00	3601.37	2628	1	···
c47	0.00	+∞	100.00	+∞	66097.65	156265.66	**57.70**	3603.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.88	475785	−∞	+∞	100.00	3601.32	2619	1	···
c48	0.00	+∞	100.00	+∞	116561.57	222934.97	**47.71**	3603.06	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.93	475785	−∞	+∞	100.00	3601.31	2619	1	···
c49	0.00	+∞	100.00	+∞	31423.79	95697.10	**67.16**	3603.61	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.30	4662	1	···
c50	0.00	+∞	100.00	+∞	78958.01	258672.87	**69.48**	3603.80	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.31	4644	1	···
c51	0.00	+∞	100.00	+∞	25101.67	107316.48	**76.61**	3602.94	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.34	4671	1	···
c52	0.00	+∞	100.00	+∞	78479.20	296492.84	**73.53**	3602.87	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.31	4653	1	···
c53	0.00	+∞	100.00	+∞	-84621.02	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.01	4680	1	···
c54	0.00	+∞	100.00	+∞	-96352.77	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.98	4680	1	···
c55	0.00	+∞	100.00	+∞	-66591.46	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.85	4644	1	···
c56	0.00	+∞	100.00	+∞	-78626.15	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.87	4662	1	···
c57	0.00	+∞	100.00	+∞	-111463.77	192588.89	157.88	3603.73	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.38	6120	1	···
c58	0.00	+∞	100.00	+∞	-117346.09	358549.07	132.73	3603.41	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.40	6120	1	···
c59	0.00	+∞	100.00	+∞	-70938.41	145484.60	148.76	3603.70	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.44	6183	1	···
c60	0.00	+∞	100.00	+∞	-78969.31	260157.56	130.35	3603.48	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.44	6174	1	···
c61	0.00	+∞	100.00	+∞	-105718.98	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.69	6165	1	···
c62	0.00	+∞	100.00	+∞	-129288.25	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.64	6111	1	···
c63	0.00	+∞	100.00	+∞	-83567.82	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.54	6102	1	···
c64	0.00	+∞	100.00	+∞	-104463.95	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.13	6147	1	···
#S			0				0				0				0					0				···
SGM			70.99	14400.00			66.92	14400.00			94.83	14400.00			100.00	14400.00	7147626789			**39.43**	13941.94	3210	3	···

Table 119 Detailed results for problem NLMCNDP-A, cost functions f_4

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	22653.82	25988.54	12.83	3600.01	23271.22	26110.08	10.87	3603.95	18704.72	+∞	100.00	3600.10	−∞	+∞	100.00	3603.08	1170800	25782.34	26019.09	**0.91**	3601.02	5252	3	···
c100_400_10_F_T_10	50003.05	71594.65	30.16	3600.01	53143.14	75918.53	**30.00**	3603.98	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.44	1170800	−∞	+∞	100.00	3601.07	5200	1	···
c100_400_10_V_L_10	6144.33	15737.45	60.96	3600.01	7621.06	16276.19	53.18	3603.06	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.80	1170800	13410.37	14219.06	**5.69**	3601.02	5227	2	···
c100_400_30_F_L_10	41028.88	149851.24	72.62	3600.02	42617.32	138417.47	**69.21**	3603.24	35846.69	+∞	100.00	3600.10	−∞	+∞	100.00	3602.98	1170800	−∞	+∞	100.00	3601.11	5200	1	···
c100_400_30_F_T_10	80834.86	141232.29	**42.76**	3600.02	-85745.99	+∞	100.00	+∞	64502.39	+∞	100.00	3600.10	−∞	+∞	100.00	3603.38	1170800	−∞	+∞	100.00	3601.10	5200	1	···
c100_400_30_V_T_10	55473.24	+∞	100.00	+∞	-607841.51	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.15	1170800	−∞	+∞	100.00	3601.10	5200	1	···
c25_100_10_F_L_5	16770.83	17351.79	3.35	3600.01	16002.63	17351.79	7.78	3602.88	13328.57	23120.83	42.35	3600.10	−∞	+∞	100.00	3601.45	292700	17227.41	17227.48	**0.00**	3601.02	2148	22	···
c25_100_10_F_T_5	36413.64	36413.71	**0.00**	3600.02	34637.52	36413.72	4.88	3602.90	30816.64	38620.65	20.21	3600.10	−∞	+∞	100.00	3601.45	292700	36413.71	36413.75	**0.00**	**579.49**	2024	22	···
c25_100_10_V_L_5	4351.00	5285.95	17.69	3600.01	3625.31	6065.00	40.23	3602.46	3064.15	6681.31	54.14	3600.10	−∞	+∞	100.00	3601.55	292700	4996.98	5016.47	**0.39**	3601.02	1477	10	···
c25_100_30_F_L_5	31787.11	35694.32	10.95	3600.01	30518.84	37857.91	19.39	3602.81	25298.77	+∞	100.00	3600.10	−∞	+∞	100.00	3601.47	292700	32476.56	32476.59	**0.00**	**1118.68**	2085	20	···
c25_100_30_F_T_5	49034.14	66708.41	26.49	3600.02	44156.93	76001.63	41.90	3603.03	37111.33	+∞	100.00	3600.10	−∞	+∞	100.00	3601.56	292700	56651.70	56651.75	**0.00**	**2087.86**	2359	27	···
c25_100_30_V_T_5	65848.91	175386.95	62.46	3600.01	52570.29	174662.52	69.90	3602.55	30766.52	+∞	100.00	3600.10	−∞	+∞	100.00	3601.46	292700	150834.11	153290.32	**1.60**	3601.02	1507	9	···
c33	55689.54	211830.60	**73.71**	3600.01	-707114.43	731123.83	196.72	3603.15	46385.10	+∞	100.00	3600.10	−∞	+∞	100.00	3602.30	667356	−∞	+∞	100.00	3601.16	2964	1	···
c35	61801.78	131092.17	**52.86**	3600.03	-224535.64	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.05	673210	−∞	+∞	100.00	3601.08	2990	1	···
c36	155668.27	234112.04	33.51	3600.02	-283255.63	+∞	100.00	+∞	102112.00	+∞	100.00	3600.10	−∞	+∞	100.00	3602.28	673210	222635.77	230102.73	**3.25**	3601.02	3112	3	···
c37	0.00	+∞	100.00	+∞	56370.68	123213.25	**54.25**	3603.99	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.08	667356	−∞	+∞	100.00	3601.29	2964	1	···
c38	0.00	+∞	100.00	+∞	108741.65	233730.32	**53.48**	3604.33	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.31	673210	−∞	+∞	100.00	3601.29	2990	1	···
c39	0.00	+∞	100.00	+∞	61021.84	123500.97	**50.59**	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.43	670283	−∞	+∞	100.00	3601.34	2977	1	···
c40	0.00	+∞	100.00	+∞	114943.43	231880.94	**50.43**	3603.36	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.35	667356	−∞	+∞	100.00	3601.25	2964	1	···
c41	53681.30	182758.32	70.63	3600.02	76918.95	146703.40	**47.57**	3603.31	20297.49	+∞	100.00	3600.10	−∞	+∞	100.00	3602.16	842976	−∞	+∞	100.00	3601.09	3744	1	···
c42	143578.03	303223.25	52.65	3600.01	184963.13	290040.00	**36.23**	3603.21	85362.50	+∞	100.00	3600.10	−∞	+∞	100.00	3602.20	860538	−∞	+∞	100.00	3601.10	3822	1	···
c43	68068.40	209889.01	67.57	3600.02	92809.44	143964.64	**35.53**	3603.19	30055.00	+∞	100.00	3600.10	−∞	+∞	100.00	3602.22	860538	−∞	+∞	100.00	3601.17	3822	1	···
c44	139915.34	251124.79	44.28	3600.02	186800.19	244986.42	**23.75**	3603.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.37	860538	−∞	+∞	100.00	3601.11	3822	1	···
c45	0.00	+∞	100.00	+∞	48008.23	94793.08	**49.35**	3603.24	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.49	860538	−∞	+∞	100.00	3601.45	3822	1	···
c46	0.00	+∞	100.00	+∞	96910.30	196168.92	**50.60**	3603.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.56	854684	−∞	+∞	100.00	3601.46	3796	1	···
c47	0.00	+∞	100.00	+∞	45989.73	77309.11	**40.51**	3603.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.72	851757	−∞	+∞	100.00	3601.35	3783	1	···
c48	0.00	+∞	100.00	+∞	89940.95	158233.02	**43.16**	3603.00	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.53	851757	−∞	+∞	100.00	3601.34	3783	1	···
c49	0.00	+∞	100.00	+∞	15966.17	82314.08	**80.60**	3603.40	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.77	1516186	−∞	+∞	100.00	3601.35	6734	1	···
c50	0.00	+∞	100.00	+∞	52407.84	200428.84	**73.85**	3603.81	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.27	6708	1	···
c51	0.00	+∞	100.00	+∞	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.54	1519113	−∞	+∞	100.00	3601.40	6747	1	···
c52	0.00	+∞	100.00	+∞	59266.87	173398.80	**65.82**	3602.88	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.69	1513259	−∞	+∞	100.00	3601.31	6721	1	···
c53	0.00	+∞	100.00	+∞	-54083.10	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.20	6760	1	···
c54	0.00	+∞	100.00	+∞	-61581.12	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.04	6760	1	···
c55	0.00	+∞	100.00	+∞	-42560.03	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.80	6708	1	···
c56	0.00	+∞	100.00	+∞	-50251.65	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3604.63	1516186	−∞	+∞	100.00	3601.86	6734	1	···
c57	0.00	+∞	100.00	+∞	-71238.88	178626.13	139.88	3603.43	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.38	8840	1	···
c58	0.00	+∞	100.00	+∞	-74998.40	345881.59	121.68	3603.37	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.47	8840	1	···
c59	0.00	+∞	100.00	+∞	-45338.26	126912.75	135.72	3603.75	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	8931	1	···
c60	0.00	+∞	100.00	+∞	-50470.98	239627.61	121.06	3603.39	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.37	8918	1	···
c61	0.00	+∞	100.00	+∞	-67567.27	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.76	8905	1	···
c62	0.00	+∞	100.00	+∞	-82630.89	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3603.46	8827	1	···
c63	0.00	+∞	100.00	+∞	-53409.98	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.60	8814	1	···
c64	0.00	+∞	100.00	+∞	-66765.15	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.13	8879	1	···
#S			0				0				0				0					**3**				···
SGM			59.10	14400.00			56.76	14400.00			93.17	14400.00			100.00	14400.00	139179698			**42.46**	12045.77	4519	3	···

Table 120 Detailed results for problem NLMCNDP-A, cost functions f_5

	GUROBI				**SCIP**				**COUENNE**				**NAIVE**					**CN24**						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	25029.22	32898.33	23.92	3600.01	24595.02	32972.72	25.41	3602.92	10384.44	+∞	100.00	3600.10	−∞	+∞	100.00	3601.70	577200	32688.23	32693.87	**0.02**	**3588.48**	2568	8	···
c100_400_10_F_T_10	58977.99	91803.89	**35.76**	3600.02	55332.32	112043.95	50.62	3602.96	20766.50	+∞	100.00	3600.10	−∞	+∞	100.00	3601.69	577200	−∞	+∞	100.00	3601.22	2400	1	···
c100_400_10_V_L_10	-26845.12	38795.79	169.20	3600.01	-41327.86	61387.42	167.32	3602.99	-67784.63	+∞	100.00	3600.10	−∞	+∞	100.00	3601.74	577200	28873.93	30523.58	**5.40**	3601.02	2428	2	···
c100_400_30_F_L_10	45744.56	69511.22	**34.19**	3600.02	-54037.33	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.75	577200	−∞	+∞	100.00	**857.54**	2400	1	···
c100_400_30_F_T_10	64960.88	241976.19	**73.15**	3600.02	-179841.26	+∞	100.00	+∞	-20690.37	+∞	100.00	3600.10	−∞	+∞	100.00	3601.89	577200	−∞	+∞	100.00	3601.22	2400	1	···
c100_400_30_V_T_10	-335149.56	760046.66	144.10	3600.02	-1205911.85	+∞	100.00	+∞	-559202.70	+∞	100.00	3600.10	−∞	+∞	100.00	3601.87	577200	−∞	+∞	100.00	3601.11	2400	1	···
c25_100_10_F_L_5	20195.58	21620.01	6.59	3600.01	17617.04	21620.01	18.52	3602.84	11641.93	26667.11	56.34	3600.10	−∞	+∞	100.00	3601.23	144300	21620.00	21620.01	**0.00**	**1051.69**	1142	13	···
c25_100_10_F_T_5	53775.12	71467.32	24.76	3600.02	44317.98	71302.52	37.85	3602.83	33670.07	78101.00	56.89	3600.10	−∞	+∞	100.00	3601.28	144300	70329.55	70329.83	**0.00**	3601.02	992	12	···
c25_100_10_V_L_5	9608.20	17877.45	46.26	3600.02	791.98	20097.96	96.06	3602.86	-3644.91	24708.47	114.75	3600.10	−∞	+∞	100.00	3601.32	144300	16911.54	16912.43	**0.01**	3601.02	874	19	···
c25_100_30_F_L_5	37251.49	49967.92	25.45	3600.01	29613.26	50826.02	41.74	3602.86	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.23	144300	49553.62	49607.93	**0.11**	3601.10	706	6	···
c25_100_30_F_T_5	68529.36	110336.18	37.89	3600.01	33862.50	118743.45	71.48	3602.94	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.25	144300	108423.93	108628.19	**0.19**	3601.02	761	7	···
c25_100_30_V_T_5	54464.70	390620.01	86.06	3600.01	-74341.20	400925.19	118.54	3602.89	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.32	144300	369979.15	376313.56	**1.68**	3601.02	737	5	···
c33	-200005.49	681571.07	129.34	3600.02	-306229.97	960046.76	131.90	3603.21	-706180.20	+∞	100.00	3600.10	−∞	+∞	100.00	3601.62	329004	444974.30	472843.26	**5.89**	3601.03	1398	2	···
c35	-7464.27	436739.55	101.71	3600.01	-75315.42	1400790.41	105.38	3603.24	-202393.50	+∞	100.00	3600.10	−∞	+∞	100.00	3601.52	331890	351881.60	376446.94	**6.53**	3601.02	1419	2	···
c36	208844.65	870793.93	**76.02**	3600.02	148636.43	942123.83	84.22	3603.12	-168712.80	+∞	100.00	3600.10	−∞	+∞	100.00	3601.54	331890	−∞	+∞	100.00	3601.09	1380	1	···
c37	72614.36	142913.91	**49.19**	3600.03	74889.54	508899.75	85.28	3602.57	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.68	329004	−∞	+∞	100.00	3601.33	1368	1	···
c38	135488.17	214070.45	**36.71**	3600.03	138170.98	913374.86	84.87	3602.59	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.67	331890	−∞	+∞	100.00	3601.38	1380	1	···
c39	81346.98	139009.46	**41.48**	3600.04	82057.40	165159.17	50.32	3602.89	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.73	330447	−∞	+∞	100.00	3601.34	1374	1	···
c40	143653.89	205683.06	**30.16**	3600.03	143469.30	265764.91	46.02	3602.91	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.75	329004	−∞	+∞	100.00	3601.27	1368	1	···
c41	-213640.04	636326.83	133.57	3600.01	-315786.58	1045696.00	130.20	3602.73	-701452.10	+∞	100.00	3600.10	−∞	+∞	100.00	3601.57	415584	−∞	+∞	100.00	3601.26	1728	1	···
c42	119569.09	918889.30	**86.99**	3600.01	20741.58	1314601.00	98.42	3602.75	-581760.20	+∞	100.00	3600.10	−∞	+∞	100.00	3601.64	424242	−∞	+∞	100.00	3601.22	1764	1	···
c43	-78307.02	617560.42	112.68	3600.01	-169199.00	1445708.69	111.70	3602.86	-367923.20	+∞	100.00	3600.10	−∞	+∞	100.00	3601.59	424242	−∞	+∞	100.00	3601.11	1764	1	···
c44	140869.61	1088733.93	**87.06**	3600.02	54680.17	1008791.62	94.58	3603.03	-352886.70	+∞	100.00	3600.10	−∞	+∞	100.00	3601.57	424242	−∞	+∞	100.00	3601.10	1764	1	···
c45	64268.50	123584.08	**48.00**	3600.04	63581.28	414057.61	84.64	3603.14	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.88	424242	−∞	+∞	100.00	3601.35	1764	1	···
c46	122199.59	190720.66	**35.93**	3600.03	123015.13	243321.33	49.44	3603.37	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.90	421356	−∞	+∞	100.00	3601.38	1752	1	···
c47	62936.66	124964.91	**49.64**	3600.03	62474.74	378945.23	83.51	3603.39	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.90	419913	−∞	+∞	100.00	3601.40	1746	1	···
c48	113311.34	168408.72	**32.72**	3600.04	113492.18	182101.70	37.68	3603.19	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.80	419913	−∞	+∞	100.00	3601.37	1746	1	···
c49	15759.90	116663.44	**86.49**	3600.03	-98222.45	426624.84	123.02	3603.27	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.36	3108	1	···
c50	72406.99	268595.60	**73.04**	3600.05	-134152.07	1011814.38	113.26	3603.25	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.13	744588	−∞	+∞	100.00	3601.36	3096	1	···
c51	17142.17	104932.73	**83.66**	3600.03	-68163.01	292424.15	123.31	3603.30	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.11	748917	−∞	+∞	100.00	3601.36	3114	1	···
c52	80881.32	245772.91	**67.09**	3600.03	25924.67	726200.17	96.43	3603.71	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.20	746031	−∞	+∞	100.00	3601.36	3102	1	···
c53	-119333.55	+∞	100.00	+∞	-107233.78	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.81	750360	−∞	+∞	100.00	3602.16	3120	1	···
c54	-135877.80	+∞	100.00	+∞	-122100.55	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.07	3120	1	···
c55	-93908.05	+∞	100.00	+∞	-84386.29	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.89	744588	−∞	+∞	100.00	3602.10	3096	1	···
c56	-110879.51	+∞	100.00	+∞	-99636.94	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.76	3108	1	···
c57	9765.28	121713.07	**91.98**	3600.04	-141249.56	606213.83	123.30	3603.69	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.43	4080	1	···
c58	27556.39	129620.80	**78.74**	3600.04	-148703.78	794186.84	118.72	3603.87	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.42	4080	1	···
c59	12739.68	109725.22	**88.39**	3600.04	-89894.86	388349.05	123.15	3603.63	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.51	4122	1	···
c60	31625.82	176885.48	**82.12**	3600.04	-100071.81	531132.68	118.84	3603.63	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.44	4116	1	···
c61	-149086.15	+∞	100.00	+∞	-133969.64	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.33	4110	1	···
c62	-182323.80	+∞	100.00	+∞	-163837.17	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3603.35	4074	1	···
c63	-117848.32	+∞	100.00	+∞	-105899.15	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3603.23	4068	1	···
c64	-147316.29	+∞	100.00	+∞	-132379.23	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3603.30	4098	1	···
#S			0				0				0				0					**2**				···
SGM			61.87	14400.00			79.27	14400.00			97.40	14400.00			100.00	14400.00	54090037			**39.72**	12290.69	2082	2	···

Table 121 Detailed results for problem NLMCNDP-A, cost functions f_6

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	37020.68	41522.60	**10.84**	3600.01	31910.50	41522.61	23.15	3604.01	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	2000	1	···
c100_400_10_F_T_10	92063.23	122675.04	**24.95**	3600.01	87932.51	125758.28	30.08	3603.06	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.06	2000	1	···
c100_400_10_V_L_10	37360.32	44452.30	**15.95**	3600.01	33873.72	46867.64	27.72	3603.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	2000	1	···
c100_400_30_F_L_10	68574.13	86157.58	**20.41**	3600.01	10303.18	+∞	100.00	+∞	64320.65	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	2000	1	···
c100_400_30_F_T_10	189560.36	301556.34	**37.14**	3600.02	183005.24	352183.04	48.04	3603.03	172714.70	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	**156.06**	2000	1	···
c100_400_30_V_T_10	542762.80	868369.22	**37.50**	3600.02	202712.49	+∞	100.00	+∞	533089.00	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	2000	1	···
c25_100_10_F_L_5	26081.08	26081.08	**0.00**	**426.55**	23456.06	26081.08	10.06	3602.87	21704.85	26736.39	18.82	3600.10	−∞	+∞	100.00	3600.01	+∞	26081.06	26081.08	**0.00**	3205.56	721	12	···
c25_100_10_F_T_5	93338.82	99679.47	6.36	3600.01	80344.63	100101.12	19.74	3602.86	74460.20	103961.60	28.38	3600.10	−∞	+∞	100.00	3600.01	+∞	98109.11	99749.87	**1.64**	3601.02	530	3	···
c25_100_10_V_L_5	**27697.88**	**28180.05**	**1.71**	**3600.01**	**22396.97**	**28180.05**	**20.52**	**3602.87**	**20663.79**	**30611.11**	**32.50**	**3600.10**	**−∞**	**+∞**	**100.00**	**3600.01**	**+∞**	**27492.19**	**27492.55**	**0.00**	**3601.02**	**1012**	**12**	···
c25_100_30_F_L_5	61442.16	71215.50	**13.72**	3600.02	50861.07	71215.51	28.58	3602.88	47785.52	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	500	1	···
c25_100_30_F_T_5	165978.77	196303.39	15.45	3600.01	128792.26	197740.98	34.87	3603.04	116159.40	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	185490.13	196191.20	**5.45**	3601.01	526	2	···
c25_100_30_V_T_5	796145.72	860363.53	7.46	3600.02	502654.16	876704.79	42.67	3603.03	467873.30	1007134.00	53.54	3600.10	−∞	+∞	100.00	3600.01	+∞	843238.84	858918.72	**1.83**	3601.02	550	4	···
c33	507068.38	754601.38	**32.80**	3600.02	444279.37	841870.00	47.23	3603.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	1140	1	···
c35	412150.38	474654.83	**13.17**	3600.02	388908.38	499050.22	22.07	3609.42	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	1150	1	···
c36	748816.32	878542.60	**14.77**	3600.02	699196.24	963229.16	27.41	3608.24	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.08	1150	1	···
c37	123518.80	145883.93	**15.33**	3600.02	122948.48	194308.65	36.73	3603.95	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.22	1140	1	···
c38	194857.82	232513.53	**16.20**	3600.03	193878.38	306258.78	36.69	3603.42	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	1150	1	···
c39	128084.04	141996.32	**9.80**	3600.03	127642.74	173023.58	26.23	3603.38	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.23	1145	1	···
c40	194674.22	218509.56	**10.91**	3600.03	194205.10	305120.48	36.35	3603.33	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.22	1140	1	···
c41	**501644.46**	**715819.58**	**29.92**	**3600.01**	**445654.25**	**811256.97**	**45.07**	**3603.22**	**−∞**	**+∞**	**100.0**	**3600.10**	**−∞**	**+∞**	**100.00**	**3600.01**	**+∞**	**1154237.23**	**1212109.58**	**4.77**	**3601.02**	**1484**	**2**	···
c42	692023.67	884128.29	**21.73**	3600.02	660766.32	945959.26	30.15	3603.19	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	1470	1	···
c43	514332.36	632743.77	**18.71**	3600.01	485448.73	658855.20	26.32	3603.17	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	1470	1	···
c44	725790.62	826751.20	**12.21**	3600.01	686274.16	900745.48	23.81	3611.96	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	1470	1	···
c45	99130.92	115247.95	**13.98**	3600.03	98923.82	130235.54	24.04	3603.19	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.30	1470	1	···
c46	166904.53	196736.07	**15.16**	3600.03	166071.27	240801.55	31.03	3603.47	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.29	1460	1	···
c47	99621.41	118385.53	**15.85**	3600.03	99395.77	145510.12	31.69	3603.05	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.30	1455	1	···
c48	151753.94	172819.14	**12.19**	3600.03	151236.40	219434.10	31.08	3603.71	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.30	1455	1	···
c49	63349.90	86229.53	**26.53**	3600.02	63243.00	90544.24	30.15	3603.36	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.26	2590	1	···
c50	123086.21	158028.17	**22.11**	3600.02	0.00	730460.79	100.00	3603.04	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.26	2580	1	···
c51	59858.66	75557.72	**20.78**	3600.02	0.00	150657.21	100.00	3603.60	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.25	2595	1	···
c52	126949.73	158744.39	**20.03**	3600.03	126636.57	189336.93	33.12	3602.86	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.27	2585	1	···
c53	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.49	2600	1	···
c54	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.37	2600	1	···
c55	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.11	2580	1	···
c56	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.74	2590	1	···
c57	0.00	+∞	100.00	+∞	0.00	230520.73	100.00	3603.41	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	3400	1	···
c58	0.00	+∞	100.00	+∞	0.00	413697.69	100.00	3603.63	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.37	3400	1	···
c59	53564.29	68625.79	**21.95**	3600.04	0.00	174032.03	100.00	3603.72	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	3435	1	···
c60	68383.63	83504.15	**18.11**	3600.04	0.00	293412.42	100.00	3603.45	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	3430	1	···
c61	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.99	3425	1	···
c62	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.46	3395	1	···
c63	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.41	3390	1	···
c64	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.47	3415	1	···
#S			**1**				0				0				0					**1**				···
SGM			**24.23**	13221.96			47.34	14400.00			91.78	14400.00			100.00	14400.00	+∞			69.77	12449.43	1725	1	···

Table 122 Detailed results for problem NLMCNDP-A, cost functions f_7

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
c100_400_10_F_L_10	**26015.50**	**26015.50**	**0.00**	**376.65**	**26015.50**	**26015.50**	**0.00**	**322.62**	**−∞**	**+∞**	**100.0**	**3600.10**	**−∞**	**+∞**	**100.00**	**3601.67**	**368000**	**25618.74**	**25619.02**	**0.00**	**3601.02**	**4885**	**9**	···
c100_400_10_F_T_10	64970.11	67225.98	**3.36**	3600.01	64088.40	69246.72	7.45	3603.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.63	368000	−∞	+∞	100.00	3601.07	3600	1	···
c100_400_10_V_L_10	12434.67	12556.37	**0.97**	3600.01	12255.77	12525.62	2.15	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.64	368000	−∞	+∞	100.00	3601.07	3600	1	···
c100_400_30_F_L_10	43180.94	170035.46	**74.60**	3600.02	6028.34	+∞	100.00	+∞	38747.11	+∞	100.00	3600.10	−∞	+∞	100.00	3601.75	368000	−∞	+∞	100.00	3601.16	3600	1	···
c100_400_30_F_T_10	105069.25	118236.96	**11.14**	3600.02	10131.39	+∞	100.00	+∞	72261.80	+∞	100.00	3600.10	−∞	+∞	100.00	3601.61	368000	−∞	+∞	100.00	3601.10	3600	1	···
c100_400_30_V_T_10	64426.66	79483.48	**18.94**	3600.02	4847.89	+∞	100.00	+∞	27575.19	+∞	100.00	3600.10	−∞	+∞	100.00	3601.82	368000	−∞	+∞	100.00	3601.11	3600	1	···
c25_100_10_F_L_5	17976.85	17976.85	**0.00**	**23.68**	17976.86	17976.86	**0.00**	45.21	16098.92	19227.55	16.27	3600.10	−∞	+∞	100.00	3601.17	92000	17976.85	17976.86	**0.00**	441.38	1174	9	···
c25_100_10_F_T_5	40590.07	40590.07	**0.00**	9.04	40590.08	40590.08	**0.00**	**6.90**	40590.08	40590.08	**0.00**	2791.34	40590.07	40593.61	0.01	811.52	92000	40590.06	40590.09	**0.00**	235.15	1453	19	···
c25_100_10_V_L_5	5451.40	5451.40	**0.00**	**6.42**	5451.39	5451.39	**0.00**	16.24	4968.69	5539.60	10.31	3600.10	5451.39	5465.78	0.26	1193.42	92000	5451.39	5451.40	**0.00**	1232.41	2544	33	···
c25_100_30_F_L_5	**34243.23**	**34243.25**	**0.00**	**49.52**	**34243.27**	**34243.27**	**0.00**	**66.15**	**30402.75**	**36992.99**	**17.81**	**3600.10**	**−∞**	**+∞**	**100.00**	**3601.21**	**92000**	**33123.73**	**33123.76**	**0.00**	**2141.58**	**1916**	**27**	···
c25_100_30_F_T_5	**52130.60**	**52130.56**	**0.00**	**182.54**	**52130.56**	**52130.56**	**0.00**	**71.19**	**48726.12**	**54744.32**	**10.99**	**3600.10**	**51293.43**	**51296.52**	**0.01**	**1229.51**	**92000**	**51293.44**	**51293.47**	**0.00**	**2526.93**	**2060**	**25**	···
c25_100_30_V_T_5	91126.14	91126.14	**0.00**	36.39	91126.14	91126.14	**0.00**	**26.31**	91126.14	91126.14	**0.00**	2953.70	91125.87	91127.56	0.00	215.51	92000	91125.70	91126.47	**0.00**	3601.01	2135	31	···
c33	140532.99	140532.99	**0.00**	**3291.77**	139743.63	141640.29	1.34	3603.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.37	209760	138478.53	142726.84	2.98	3601.03	2136	4	···
c35	167352.60	167968.47	**0.37**	3600.10	166598.64	168464.87	1.11	3604.29	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.37	211600	161301.04	170595.65	5.45	3601.02	2126	2	···
c36	351414.21	354972.54	**1.00**	3600.02	348583.80	352537.10	1.12	3603.51	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.39	211600	342360.14	357685.26	4.28	3601.02	2157	3	···
c37	0.00	+∞	100.00	+∞	82783.23	170585.38	**51.47**	3603.81	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.65	209760	−∞	+∞	100.00	3601.23	2052	1	···
c38	0.00	+∞	100.00	+∞	148481.70	234798.35	**36.76**	3603.68	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.63	211600	−∞	+∞	100.00	3601.24	2070	1	···
c39	60069.60	241757.23	75.15	3600.03	88347.54	105544.87	**16.29**	3603.53	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.77	210680	−∞	+∞	100.00	3601.23	2061	1	···
c40	0.00	+∞	100.00	+∞	155261.66	228778.50	**32.13**	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.63	209760	−∞	+∞	100.00	3601.24	2052	1	···
c41	145745.49	147493.92	**1.19**	3600.02	144489.68	147492.29	2.04	3603.19	21073.71	+∞	100.00	3600.10	−∞	+∞	100.00	3601.48	264960	142986.65	150125.22	4.76	3601.02	2645	2	···
c42	**322536.34**	**326139.93**	**1.10**	**3600.01**	**317736.01**	**337698.65**	**5.91**	**3603.17**	**−∞**	**641932.50**	**100.00**	**3600.10**	**−∞**	**+∞**	**100.00**	**3601.54**	**270480**	**318243.75**	**321984.88**	**1.16**	**3601.02**	**2752**	**5**	···
c43	194026.02	195472.23	**0.74**	3600.02	193071.71	196343.33	1.67	3603.19	52784.35	+∞	100.00	3600.10	−∞	+∞	100.00	3601.49	270480	186723.47	197786.41	5.59	3601.02	2718	2	···
c44	347577.45	349960.08	**0.68**	3600.01	344092.40	349247.66	1.48	3603.24	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.46	270480	338548.34	353024.06	4.10	3601.02	2710	2	···
c45	0.00	+∞	100.00	+∞	71002.86	117342.95	**39.49**	3603.11	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.91	270480	−∞	+∞	100.00	3601.29	2646	1	···
c46	0.00	+∞	100.00	+∞	132297.76	222260.94	**40.48**	3603.77	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.97	268640	−∞	+∞	100.00	3601.30	2628	1	···
c47	0.00	+∞	100.00	+∞	69410.63	102101.74	**32.02**	3603.53	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.55	267720	−∞	+∞	100.00	3601.29	2619	1	···
c48	0.00	+∞	100.00	+∞	122406.56	168644.44	**27.42**	3603.77	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.68	267720	−∞	+∞	100.00	3601.29	2619	1	···
c49	0.00	+∞	100.00	+∞	31724.19	50019.78	**36.58**	3603.31	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.96	476560	−∞	+∞	100.00	3601.28	4662	1	···
c50	0.00	+∞	100.00	+∞	75492.91	229472.72	**67.10**	3602.86	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.90	474720	−∞	+∞	100.00	3601.30	4644	1	···
c51	0.00	+∞	100.00	+∞	32368.71	39855.51	**18.78**	3603.44	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.87	477480	−∞	+∞	100.00	3601.28	4671	1	···
c52	0.00	+∞	100.00	+∞	89647.79	128000.63	**29.96**	3602.83	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3601.99	475640	−∞	+∞	100.00	3601.28	4653	1	···
c53	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.67	478400	−∞	+∞	100.00	3602.57	4680	1	···
c54	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.87	478400	−∞	+∞	100.00	3602.04	4680	1	···
c55	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.60	474720	−∞	+∞	100.00	3602.41	4644	1	···
c56	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.69	476560	−∞	+∞	100.00	3601.79	4662	1	···
c57	0.00	+∞	100.00	+∞	0.00	163118.92	100.00	3603.41	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.15	625600	−∞	+∞	100.00	3601.39	6120	1	···
c58	0.00	+∞	100.00	+∞	0.00	327572.52	100.00	3603.38	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.78	625600	−∞	+∞	100.00	3601.39	6120	1	···
c59	0.00	+∞	100.00	+∞	0.00	109442.46	100.00	3603.74	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.11	632040	−∞	+∞	100.00	3601.31	6183	1	···
c60	0.00	+∞	100.00	+∞	0.00	220873.89	100.00	3603.40	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.20	631120	−∞	+∞	100.00	3601.39	6174	1	···
c61	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.36	630200	−∞	+∞	100.00	3603.16	6165	1	···
c62	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.59	624680	−∞	+∞	100.00	3603.21	6111	1	···
c63	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3602.71	623760	−∞	+∞	100.00	3602.40	6102	1	···
c64	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3603.15	628360	−∞	+∞	100.00	3602.47	6147	1	···
#S			**5**				4				2				2					3				···
SGM			23.85	7260.99			**20.88**	7657.79			71.03	13258.28			70.26	11283.36	321133			39.15	11144.23	3379	2	···

Table 123 Detailed results for problem NLMCNDP-A, cost functions f_8

Instance	GUROBI DB	PB	GAP	CPU	SCIP DB	PB	GAP	CPU	COUENNE DB	PB	GAP	CPU	NAIVE DB	PB	GAP	CPU	NP	CN24 DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	35895.90	39575.79	9.30	3600.01	32001.20	39603.05	19.20	3603.97	29064.27	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	39148.13	42875.83	**8.69**	3601.02	2860	3	⋯
c100_400_10_F_T_10	89705.95	115807.90	**22.54**	3600.01	87741.25	136837.10	35.88	3604.01	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	2400	1	⋯
c100_400_10_V_L_10	35858.65	41471.62	**13.53**	3600.01	33216.14	45868.57	27.58	3603.05	32360.46	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	38642.55	57577.07	32.89	3601.02	2518	2	⋯
c100_400_30_F_L_10	69734.57	83406.36	**16.39**	3600.01	11014.24	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	**1662.20**	2400	1	⋯
c100_400_30_F_T_10	196760.33	313206.27	**37.18**	3600.01	191327.69	360645.68	46.95	3603.12	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	**135.30**	2400	1	⋯
c100_400_30_V_T_10	541566.09	983416.91	**44.93**	3600.02	191380.09	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	2400	1	⋯
c25_100_10_F_L_5	25063.61	25063.61	**0.00**	**549.44**	21987.67	25063.62	12.27	3602.54	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	25063.44	25063.62	**0.00**	3601.02	1638	21	⋯
c25_100_10_F_T_5	**86664.23**	**92600.89**	**6.41**	**3600.02**	**78143.49**	**92600.93**	**15.61**	**3602.88**	**69253.97**	**76301.28**	**9.24**	**3600.10**	**−∞**	**+∞**	**100.00**	**3600.01**	**+∞**	**92248.02**	**93157.99**	**0.98**	**3601.02**	**812**	**5**	⋯
c25_100_10_V_L_5	**25504.58**	**26073.58**	**2.18**	**3600.01**	**20874.70**	**26073.60**	**19.94**	**3602.50**	**−∞**	**+∞**	**100.0**	**3600.10**	**−∞**	**+∞**	**100.00**	**3600.01**	**+∞**	**25421.91**	**25421.94**	**0.00**	**2949.29**	**1509**	**18**	⋯
c25_100_30_F_L_5	58791.79	67366.74	12.73	3600.01	51024.06	67591.89	24.51	3602.87	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	64866.66	67903.41	**4.47**	3601.03	644	2	⋯
c25_100_30_F_T_5	157453.79	183898.60	14.38	3600.01	130632.01	185371.63	29.53	3602.96	115420.80	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	173218.47	185349.49	**6.54**	3601.03	628	2	⋯
c25_100_30_V_T_5	715752.17	828826.02	13.64	3600.01	511919.85	836131.96	38.78	3603.02	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	814333.79	831712.97	**2.09**	3601.02	752	5	⋯
c33	507478.44	688360.56	**26.28**	3600.01	442118.52	763329.97	42.08	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	600213.13	815192.25	26.37	3601.03	1559	2	⋯
c35	412590.89	457222.31	**9.76**	3600.01	388196.00	474407.04	18.17	3603.89	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	433752.44	509403.32	14.85	3601.03	1570	2	⋯
c36	745221.13	853216.71	**12.66**	3600.02	698396.66	896339.55	22.08	3602.99	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	1380	1	⋯
c37	123492.02	141101.91	**12.48**	3600.04	122946.63	148535.20	17.23	3604.57	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.25	1368	1	⋯
c38	194857.13	235044.38	**17.10**	3600.03	193890.96	338219.85	42.67	3603.74	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.22	1380	1	⋯
c39	127836.56	143577.11	**10.96**	3600.03	127647.52	144643.43	11.75	3603.60	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.24	1374	1	⋯
c40	194879.01	214082.13	**8.97**	3600.02	194202.07	213579.95	9.07	3603.68	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.22	1368	1	⋯
c41	494730.74	665155.31	**25.62**	3600.01	447609.62	720740.99	37.90	3603.20	445654.40	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.13	1728	1	⋯
c42	681894.95	841658.86	**18.98**	3600.01	661905.77	897104.87	26.22	3603.20	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	1764	1	⋯
c43	512823.33	603097.26	**14.97**	3600.01	485039.67	631260.26	23.16	3603.20	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	1764	1	⋯
c44	718802.43	810153.57	**11.28**	3600.02	686017.12	853608.54	19.63	3603.27	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.10	1764	1	⋯
c45	99167.15	114080.14	**13.07**	3600.03	98918.11	197405.88	49.89	3603.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	1764	1	⋯
c46	166886.53	191458.38	**12.83**	3600.03	166063.47	380507.72	56.36	3603.79	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.29	1752	1	⋯
c47	99606.54	113812.59	**12.48**	3600.03	99382.20	197212.58	49.61	3603.10	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	1746	1	⋯
c48	151594.72	167493.79	**9.49**	3600.03	151209.95	206654.74	26.83	3604.01	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.30	1746	1	⋯
c49	63370.12	80298.34	**21.08**	3600.03	63042.74	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.27	3108	1	⋯
c50	123053.60	149997.40	**17.96**	3600.05	122369.98	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.29	3096	1	⋯
c51	59843.38	75994.97	**21.25**	3600.03	59465.55	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.26	3114	1	⋯
c52	127047.63	151661.65	**16.23**	3600.03	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.27	3102	1	⋯
c53	0.00	+∞	100.00	+∞	0.00	381057.64	100.00	3606.54	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.55	3120	1	⋯
c54	0.00	+∞	100.00	+∞	0.00	659747.04	100.00	3606.25	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.63	3120	1	⋯
c55	0.00	+∞	100.00	+∞	0.00	329727.82	100.00	3606.58	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.47	3096	1	⋯
c56	0.00	+∞	100.00	+∞	0.00	560812.87	100.00	3606.63	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.22	3108	1	⋯
c57	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	0.02	0.02	**0.00**	**0.50**	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.36	4080	1	⋯
c58	**74038.66**	**89557.18**	**17.33**	**3600.04**	**73862.71**	**+∞**	**100.00**	**+∞**	**0.02**	**0.02**	**0.00**	**0.49**	**−∞**	**+∞**	**100.00**	**3600.01**	**+∞**	**−∞**	**+∞**	**100.00**	**3601.36**	**4080**	**1**	⋯
c59	**53523.58**	**64468.28**	**16.98**	**3600.03**	**-0.00**	**+∞**	**100.00**	**+∞**	**0.01**	**0.01**	**0.00**	**0.48**	**−∞**	**+∞**	**100.00**	**3600.01**	**+∞**	**−∞**	**+∞**	**100.00**	**3601.38**	**4122**	**1**	⋯
c60	**68381.07**	**84156.96**	**18.75**	**3600.04**	**-0.00**	**+∞**	**100.00**	**+∞**	**0.01**	**0.01**	**0.00**	**0.49**	**−∞**	**+∞**	**100.00**	**3600.01**	**+∞**	**−∞**	**+∞**	**100.00**	**3601.36**	**4116**	**1**	⋯
c61	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3603.01	4110	1	⋯
c62	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3603.01	4074	1	⋯
c63	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.41	4068	1	⋯
c64	0.00	+∞	100.00	+∞	0.00	636430.93	100.00	3606.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.40	4098	1	⋯
#S			**1**				0				**1**				0					0				⋯
SGM			**22.77**	13219.24			45.36	14400.00			88.45	11905.15			100.00	14400.00	+∞			58.35	12054.57	2083	2	⋯

Table 124 Detailed results for problem NLMCNDP-A, cost functions f_9

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
c100_400_10_F_L_10	20336.78	29053.22	30.00	3600.01	19708.35	29522.60	33.24	3604.01	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	28237.09	28911.09	**2.33**	3601.02	3621	2	⋯
c100_400_10_F_T_10	41775.37	73789.56	**43.39**	3600.01	40514.70	83080.79	51.23	3603.94	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	3600	1	⋯
c100_400_10_V_L_10	2527.54	19948.09	**87.33**	3600.01	2551.26	35745.79	92.86	3603.03	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.07	3600	1	⋯
c100_400_30_F_L_10	36828.20	85062.70	**56.70**	3600.02	6820.56	+∞	100.00	+∞	32398.63	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.11	3600	1	⋯
c100_400_30_F_T_10	64872.70	154213.90	**57.93**	3600.02	10272.80	+∞	100.00	+∞	54654.89	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.13	3600	1	⋯
c100_400_30_V_T_10	12619.55	255113.31	**95.05**	3600.02	3070.95	+∞	100.00	+∞	9704.04	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.12	3600	1	⋯
c25_100_10_F_L_5	13741.63	19694.97	30.23	3600.02	13271.07	19695.00	32.62	3602.86	10620.01	25702.67	58.68	3600.10	−∞	+∞	100.00	3600.01	+∞	19404.83	19404.84	**0.00**	**2187.71**	1139	13	⋯
c25_100_10_F_T_5	29403.47	39023.63	24.65	3600.02	28626.63	38872.44	26.36	3602.89	25173.09	42790.64	41.17	3600.10	−∞	+∞	100.00	3600.01	+∞	38786.25	38786.25	**0.00**	**186.18**	1330	12	⋯
c25_100_10_V_L_5	1818.20	4302.45	57.74	3600.02	1648.06	6261.18	73.68	3602.43	1266.81	8147.91	84.45	3600.10	−∞	+∞	100.00	3600.01	+∞	4110.19	4110.19	**0.00**	**1426.51**	1367	26	⋯
c25_100_30_F_L_5	25540.97	35214.21	27.47	3600.01	25390.47	34524.61	26.46	3602.92	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	32433.60	32433.62	**0.00**	**1446.85**	1239	15	⋯
c25_100_30_F_T_5	31444.12	50130.11	37.27	3600.02	31356.42	48368.02	35.17	3603.04	28349.79	54345.57	47.83	3600.10	−∞	+∞	100.00	3600.01	+∞	45158.80	45158.84	**0.00**	**1510.80**	1388	20	⋯
c25_100_30_V_T_5	8238.34	94053.12	91.24	3600.01	8528.74	71390.67	88.05	3603.76	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	64831.98	64833.22	**0.00**	3601.02	1921	36	⋯
c33	22675.90	145602.75	84.43	3600.01	22617.94	484455.22	95.33	3603.12	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	96385.86	98536.10	**2.18**	3601.02	2123	3	⋯
c35	11449.91	339254.81	96.62	3600.02	11883.62	265102.08	**95.52**	3603.25	11518.89	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	2070	1	⋯
c36	46553.56	727908.96	93.60	3600.02	50391.29	737933.39	**93.17**	3603.51	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	2070	1	⋯
c37	21972.63	197411.95	**88.87**	3600.04	22484.31	316763.49	92.90	3604.90	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.30	2052	1	⋯
c38	51187.58	246477.17	**79.23**	3600.03	51209.40	689622.35	92.57	3603.22	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.25	2070	1	⋯
c39	24032.47	181784.27	**86.78**	3600.03	24621.39	293339.85	91.61	3603.35	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.33	2061	1	⋯
c40	53877.82	240255.86	**77.57**	3600.02	54527.27	485876.68	88.78	3603.37	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.28	2052	1	⋯
c41	16728.33	293395.25	**94.30**	3600.01	17072.48	492589.02	96.53	3603.36	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	2592	1	⋯
c42	58680.55	659775.23	91.11	3600.02	60291.85	364268.28	**83.45**	3603.15	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	2646	1	⋯
c43	12396.66	746220.70	98.34	3600.02	13953.14	462997.18	**96.99**	3603.16	13306.79	+∞	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	2646	1	⋯
c44	43592.75	859230.66	94.93	3600.02	48339.14	410905.31	**88.24**	3603.18	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.09	2646	1	⋯
c45	18064.38	139578.83	**87.06**	3600.03	18562.51	283710.16	93.46	3603.16	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.34	2646	1	⋯
c46	44494.53	204926.83	**78.29**	3600.03	45135.80	585147.36	92.29	3603.79	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.44	2628	1	⋯
c47	17064.47	131226.15	**87.00**	3600.03	17571.40	258569.48	93.20	3603.11	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.32	2619	1	⋯
c48	41094.81	186210.90	**77.93**	3600.04	41731.21	501790.49	91.68	3603.75	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	2619	1	⋯
c49	5944.43	111999.09	**94.69**	3600.03	5912.53	151293.12	96.09	3603.39	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.30	4662	1	⋯
c50	23889.90	201274.94	**88.13**	3600.03	23927.22	655545.86	96.35	3603.37	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.31	4644	1	⋯
c51	4105.83	140701.00	**97.08**	3600.03	-0.00	121092.23	100.00	3603.68	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.44	4671	1	⋯
c52	23138.15	205452.29	**88.74**	3600.03	-0.00	482325.54	100.00	3602.88	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.34	4653	1	⋯
c53	-0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.03	4680	1	⋯
c54	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.98	4680	1	⋯
c55	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.10	4644	1	⋯
c56	0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.86	4662	1	⋯
c57	4703.29	+∞	100.00	+∞	-0.00	209381.98	100.00	3603.50	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.46	6120	1	⋯
c58	8709.33	+∞	100.00	+∞	-0.00	377877.98	100.00	3603.43	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.49	6120	1	⋯
c59	3823.65	102175.16	**96.26**	3600.04	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.44	6183	1	⋯
c60	8213.46	111563.77	**92.64**	3600.04	-0.00	259685.91	100.00	3603.44	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3601.35	6174	1	⋯
c61	-0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3603.25	6165	1	⋯
c62	-0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3603.40	6111	1	⋯
c63	-0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.09	6102	1	⋯
c64	-0.00	+∞	100.00	+∞	-0.00	+∞	100.00	+∞	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3602.12	6147	1	⋯
#S			0				0				0				0					**5**				⋯
SGM			76.97	14400.00			82.39	14400.00			94.75	14400.00			100.00	14400.00	+∞			**44.22**	10634.65	3147	3	⋯

Table 125 Detailed results for problem NLMCNDP-A, cost functions f_{10}

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-1	-878.52	-878.52	**0.00**	**23.47**	-878.51	-878.51	**0.00**	65.21	-878.51	-878.51	**0.00**	309.89	-878.51	-878.51	**0.00**	51.85	8510	-878.51	-878.51	**0.00**	66.47	186	28	···
10-10	-505.81	-505.81	**0.00**	**0.44**	-505.81	-505.81	**0.00**	3.66	-505.81	-505.81	**0.00**	2.98	-505.81	-505.81	**0.00**	45.03	8510	-505.81	-505.81	**0.00**	42.34	121	19	···
10-2	-583.05	-583.05	**0.00**	**4.63**	-583.04	-583.04	**0.00**	14.01	-583.04	-583.04	**0.00**	38.21	-583.04	-583.04	**0.00**	51.33	8510	-583.04	-583.04	**0.00**	54.50	163	26	···
10-3	-476.31	-476.31	**0.00**	**8.24**	-476.31	-476.31	**0.00**	16.81	-476.31	-476.31	**0.00**	49.84	-476.31	-476.31	**0.00**	57.61	8510	-476.31	-476.31	**0.00**	55.38	170	29	···
10-4	-774.36	-774.36	**0.00**	**0.31**	-774.36	-774.36	**0.00**	3.54	-774.36	-774.36	**0.00**	0.93	-774.36	-774.36	**0.00**	54.69	8510	-774.36	-774.36	**0.00**	32.22	81	13	···
10-5	-797.61	-797.61	**0.00**	**0.35**	-797.60	-797.60	**0.00**	3.72	-797.60	-797.60	**0.00**	3.75	-797.60	-797.60	**0.00**	52.01	8510	-797.60	-797.60	**0.00**	29.63	106	17	···
10-6	-432.44	-432.44	**0.00**	**0.66**	-432.43	-432.43	**0.00**	4.97	-432.43	-432.43	**0.00**	7.23	-432.43	-432.43	**0.00**	39.76	8510	-432.43	-432.43	**0.00**	38.76	138	25	···
10-7	-410.57	-410.57	**0.00**	**1.26**	-410.57	-410.57	**0.00**	5.57	-410.57	-410.57	**0.00**	17.64	-410.57	-410.57	**0.00**	51.01	8510	-410.57	-410.57	**0.00**	37.82	127	22	···
10-8	-552.51	-552.51	**0.00**	**21.36**	-552.50	-552.50	**0.00**	63.30	-552.50	-552.50	**0.00**	3144.47	-552.50	-552.50	**0.00**	54.78	8510	-552.50	-552.50	**0.00**	75.22	184	30	···
10-9	-458.39	-458.39	**0.00**	**1.67**	-458.38	-458.38	**0.00**	5.19	-458.38	-458.38	**0.00**	33.66	-458.38	-458.38	**0.00**	55.80	8510	-458.38	-458.38	**0.00**	39.29	123	20	···
100-1	-7176.46	-6733.11	6.18	3600.31	-7168.62	-6745.01	5.91	3603.11	-7182.12	-6524.23	9.16	3600.10	-6757.99	-6757.98	**0.00**	544.33	85100	-6757.99	-6757.98	**0.00**	**434.96**	1453	35	···
100-10	-7246.46	-6953.00	4.05	3600.20	-7251.38	-6960.39	4.01	3602.92	-7264.91	-6808.37	6.28	3600.10	-6961.57	-6961.56	**0.00**	**376.64**	85100	-6961.57	-6961.56	**0.00**	400.20	1341	27	···
100-2	-7015.36	-6543.47	6.73	3600.30	-7004.14	-6545.87	6.54	3603.12	-7019.67	-6419.84	8.54	3600.10	-6563.05	-6563.05	**0.00**	509.51	85100	-6563.05	-6563.05	**0.00**	**436.47**	1531	36	···
100-3	-7291.85	-6902.45	5.34	3600.29	-7289.47	-6891.60	5.46	3603.17	-7305.45	-6770.45	7.32	3600.10	-6905.73	-6905.73	**0.00**	424.44	85100	-6905.73	-6905.73	**0.00**	**409.32**	1365	30	···
100-4	-7218.68	-6772.63	6.18	3600.31	-7220.86	-6760.68	6.37	3603.32	-7236.28	-6640.73	8.23	3600.10	-6774.25	-6774.25	**0.00**	440.82	85100	-6774.25	-6774.25	**0.00**	**369.28**	1411	36	···
100-5	-6643.11	-6327.10	4.76	3600.18	-6637.00	-6327.16	4.67	3603.07	-6648.16	-6180.99	7.03	3600.10	-6328.88	-6328.88	**0.00**	510.25	85100	-6328.88	-6328.88	**0.00**	**382.51**	1487	42	···
100-6	-6974.90	-6469.18	7.25	3600.29	-6978.09	-6470.06	7.28	3603.22	-6995.46	-6355.84	9.14	3600.10	-6476.94	-6476.94	**0.00**	437.78	85100	-6476.94	-6476.94	**0.00**	**397.35**	1473	30	···
100-7	-7823.19	-7278.22	6.97	3600.20	-7812.20	-7274.18	6.89	3603.21	-7829.96	-7055.25	9.89	3600.10	-7279.22	-7279.22	**0.00**	389.95	85100	-7279.22	-7279.22	**0.00**	**350.20**	1365	27	···
100-8	-6469.34	-6157.10	4.83	3600.31	-6461.74	-6150.45	4.82	3603.32	-6475.22	-6047.91	6.60	3600.10	-6157.42	-6157.42	**0.00**	426.08	85100	-6157.42	-6157.41	**0.00**	**374.93**	1254	28	···
100-9	-7790.43	-7225.02	7.26	3600.19	-7781.48	-7184.92	7.67	3602.82	-7799.11	-7110.01	8.84	3600.10	-7228.61	-7228.61	**0.00**	438.56	85100	-7228.61	-7228.60	**0.00**	**397.64**	1446	27	···
1000-1	-71344.25	-66898.14	**6.23**	3600.47	-71338.95	-66830.93	6.32	3603.13	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
1000-10	-72483.95	-68008.28	6.17	3600.46	-72479.22	-67918.30	6.29	3603.01	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-68317.58	-68078.07	**0.35**	**3586.62**	5408	5	···
1000-2	-73343.39	-69011.32	5.91	3600.33	-73341.58	-68924.99	6.02	3602.95	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-69316.73	-68987.43	**0.48**	**3431.87**	5012	4	···
1000-3	-73381.86	-68625.60	**6.48**	3600.47	-73375.35	-68430.44	6.74	3602.99	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
1000-4	-74653.97	-69729.64	6.60	3600.46	-74649.11	-69473.49	6.93	3603.31	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-69994.66	-69646.96	**0.50**	**3457.38**	5016	4	···
1000-5	-74264.18	-69628.76	**6.24**	3600.50	-74258.14	-69485.63	6.43	3602.96	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
1000-6	-76998.41	-72112.25	**6.35**	3600.30	-76993.73	-71952.63	6.55	3603.26	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
1000-7	-72565.90	-68089.65	**6.17**	3600.45	-72561.53	-67963.89	6.34	3603.03	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
1000-8	-72859.03	-68424.04	6.09	3600.32	-72855.70	-68242.78	6.33	3603.31	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-69378.53	-68048.04	**1.96**	**3374.72**	4360	3	···
1000-9	-75197.18	-70348.07	**6.45**	3600.46	-75193.89	-70052.99	6.84	3603.01	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	···
20-1	-1556.54	-1556.54	**0.00**	331.08	-1556.52	-1556.52	**0.00**	662.38	-1600.75	-1554.70	2.88	3600.10	-1556.52	-1556.52	**0.00**	91.23	17020	-1556.52	-1556.52	**0.00**	**85.82**	281	23	···
20-10	-1698.82	-1698.72	0.01	3600.30	-1698.71	-1698.71	**0.00**	3583.22	-1807.06	-1645.55	8.94	3600.10	-1698.71	-1698.71	**0.00**	108.88	17020	-1698.71	-1698.70	**0.00**	**83.53**	299	25	···
20-2	-967.19	-967.19	**0.00**	3298.39	-967.20	-967.17	0.00	3603.35	-1026.97	-938.85	8.58	3600.10	-967.18	-967.17	**0.00**	89.62	17020	-967.18	-967.17	**0.00**	**82.50**	314	27	···
20-3	-1741.80	-1741.80	**0.00**	1610.77	-1741.79	-1741.79	**0.00**	197.68	-1840.34	-1696.62	7.81	3600.10	-1741.79	-1741.79	**0.00**	108.21	17020	-1741.79	-1741.79	**0.00**	**82.22**	262	20	···
20-4	-1115.26	-1115.26	**0.00**	493.68	-1115.25	-1115.25	**0.00**	77.72	-1147.20	-1110.97	3.16	3600.10	-1115.25	-1115.25	**0.00**	92.90	17020	-1115.25	-1115.25	**0.00**	**70.90**	266	21	···
20-5	-1782.59	-1782.59	**0.00**	458.94	-1782.58	-1782.58	**0.00**	232.59	-1896.11	-1763.58	6.99	3600.10	-1782.58	-1782.58	**0.00**	90.94	17020	-1782.58	-1782.58	**0.00**	**75.69**	277	27	···
20-6	-1250.74	-1247.60	0.25	3600.29	-1247.59	-1247.59	**0.00**	569.68	-1315.55	-1238.62	5.85	3600.10	-1247.59	-1247.59	**0.00**	79.38	17020	-1247.59	-1247.59	**0.00**	**71.81**	263	26	···
20-7	-1116.22	-1116.22	**0.00**	2506.29	-1116.21	-1116.21	**0.00**	445.97	-1197.56	-1098.88	8.24	3600.10	-1116.21	-1116.21	**0.00**	98.57	17020	-1116.21	-1116.21	**0.00**	**77.44**	278	23	···
20-8	-1269.30	-1269.30	**0.00**	1125.80	-1269.28	-1269.28	**0.00**	1212.91	-1328.85	-1269.28	4.48	3600.10	-1269.28	-1269.28	**0.00**	101.57	17020	-1269.28	-1269.28	**0.00**	**92.69**	291	25	···
20-9	-1355.42	-1355.42	**0.00**	823.53	-1355.41	-1355.41	**0.00**	722.44	-1362.98	-1355.41	0.56	3602.09	-1355.41	-1355.41	**0.00**	87.18	17020	-1355.41	-1355.41	**0.00**	**85.24**	289	22	···
200-1	-14249.86	-13306.69	6.62	3600.19	-14246.24	-13310.60	6.57	3603.07	-14256.51	-13060.02	8.39	3600.10	-13356.08	-13356.07	**0.00**	934.14	170200	-13356.08	-13356.07	**0.00**	**790.42**	2671	32	···
200-10	-14801.14	-13827.09	6.58	3600.21	-14799.63	-13808.36	6.70	3603.05	-14813.66	-13622.73	8.04	3600.10	-13836.17	-13836.16	**0.00**	893.41	170200	-13836.18	-13836.16	**0.00**	**738.24**	2807	34	···
200-2	-14666.08	-13710.23	6.52	3600.32	-14657.45	-13696.65	6.56	3603.15	-14670.30	-13262.65	9.60	3600.10	-13719.33	-13719.32	**0.00**	849.22	170200	-13719.33	-13719.32	**0.00**	**801.64**	2835	42	···
200-3	-15901.11	-14972.92	5.84	3600.31	-15898.86	-14960.77	5.90	3603.12	-15912.71	-14532.40	8.67	3600.10	-14989.44	-14989.43	**0.00**	982.88	170200	-14989.44	-14989.42	**0.00**	**738.70**	2815	35	···
200-4	-14792.17	-13871.46	6.22	3600.31	-14794.25	-13881.87	6.17	3602.83	-14805.78	-13433.62	9.27	3600.10	-13890.34	-13890.33	**0.00**	1209.90	170200	-13890.34	-13890.33	**0.00**	**879.96**	2889	44	···
200-5	-14446.85	-13576.26	6.03	3600.34	-14441.01	-13598.80	5.83	3603.18	-14453.00	-13522.76	6.44	3600.10	-13622.71	-13622.70	**0.00**	1030.56	170200	-13622.71	-13622.70	**0.00**	**741.70**	2824	41	···
200-6	-13987.51	-13013.08	6.97	3600.33	-13979.16	-13001.73	6.99	3603.02	-13992.66	-12604.96	9.92	3600.10	-13030.25	-13030.24	**0.00**	1037.05	170200	-13030.25	-13030.24	**0.00**	**864.77**	2968	55	···
200-7	-14608.08	-13788.91	5.61	3600.32	-14604.23	-13774.13	5.68	3603.01	-14616.03	-13386.79	8.41	3600.10	-13794.62	-13794.61	**0.00**	812.12	170200	-13794.62	-13794.61	**0.00**	**721.80**	2921	43	···
200-8	-14843.23	-13850.32	6.69	3600.30	-14839.11	-13849.68	6.67	3603.09	-14850.14	-13394.20	9.80	3600.10	-13873.58	-13873.57	**0.00**	**775.40**	170200	-13873.58	-13873.57	**0.00**	858.74	2829	35	···
200-9	-15957.63	-14759.83	7.51	3600.32	-15984.24	-14743.15	7.76	3602.99	-15999.52	-14707.20	8.08	3600.10	-14773.03	-14773.02	**0.00**	1544.63	170200	-14773.03	-14773.02	**0.00**	**844.75**	2913	37	···

Table 126 Detailed results for problem NLCKP, cost functions f_1

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
10-1	-410.89	-410.89	**0.00**	**0.37**	-410.89	-410.89	**0.00**	6.88	-410.89	-410.89	**0.00**	19.46	-410.89	-410.89	**0.00**	280.62	9530	-410.89	-410.89	**0.00**	8.65	125	12	⋯
10-10	-247.70	-247.70	**0.00**	**0.29**	-247.70	-247.70	**0.00**	10.51	-247.70	-247.70	**0.00**	7.80	-247.70	-247.70	**0.00**	307.61	9530	-247.70	-247.70	**0.00**	8.30	128	19	⋯
10-2	-291.03	-291.03	**0.00**	**0.48**	-291.03	-291.03	**0.00**	8.02	-291.03	-291.03	**0.00**	12.77	-291.03	-291.03	**0.00**	282.95	9530	-291.03	-291.03	**0.00**	7.98	131	16	⋯
10-3	-224.94	-224.94	**0.00**	**0.29**	-224.94	-224.94	**0.00**	12.98	-224.94	-224.94	**0.00**	28.29	-224.94	-224.94	**0.00**	285.42	9530	-224.94	-224.94	**0.00**	9.42	131	15	⋯
10-4	-389.09	-389.09	**0.00**	**0.29**	-389.09	-389.09	**0.00**	5.15	-389.09	-389.09	**0.00**	3.89	-389.09	-389.09	**0.00**	268.77	9530	-389.09	-389.09	**0.00**	7.93	114	13	⋯
10-5	-375.74	-375.74	**0.00**	**0.29**	-375.74	-375.74	**0.00**	6.90	-375.74	-375.74	**0.00**	9.52	-375.74	-375.74	**0.00**	282.57	9530	-375.74	-375.74	**0.00**	10.24	125	12	⋯
10-6	-206.26	-206.26	**0.00**	**0.34**	-206.26	-206.26	**0.00**	9.77	-206.26	-206.26	**0.00**	20.78	-206.26	-206.26	**0.00**	261.67	9530	-206.26	-206.26	**0.00**	8.96	138	20	⋯
10-7	-195.77	-195.77	**0.00**	**0.32**	-195.77	-195.77	**0.00**	12.76	-195.77	-195.77	**0.00**	15.54	-195.77	-195.77	**0.00**	258.74	9530	-195.77	-195.77	**0.00**	9.05	139	19	⋯
10-8	-250.94	-250.94	**0.00**	**0.29**	-250.94	-250.94	**0.00**	72.59	-250.94	-250.94	**0.00**	142.26	-250.94	-250.94	**0.00**	302.94	9530	-250.94	-250.94	**0.00**	8.87	130	14	⋯
10-9	-215.41	-215.41	**0.00**	**0.29**	-215.41	-215.41	**0.00**	11.65	-215.41	-215.41	**0.00**	18.55	-215.41	-215.41	**0.00**	289.68	9530	-215.41	-215.41	**0.00**	9.18	127	12	⋯
100-1	-3270.88	-3270.88	**0.00**	**15.91**	-4950.21	-3129.23	36.79	3603.24	-4858.25	-3181.11	34.52	3600.10	-3270.88	-3270.88	**0.00**	2833.73	95300	-3270.88	-3270.88	**0.00**	94.02	1243	23	⋯
100-10	-3409.65	-3409.65	**0.00**	**0.26**	-5011.29	-3242.86	35.29	3603.20	-4949.71	-3185.52	35.64	3600.10	-3409.65	-3409.65	**0.00**	2456.67	95300	-3409.65	-3409.65	**0.00**	88.95	1166	17	⋯
100-2	-3185.82	-3185.82	**0.00**	**38.68**	-4837.75	-3059.60	36.76	3602.86	-4714.90	-3122.48	33.77	3600.10	-3185.82	-3185.82	**0.00**	2854.07	95300	-3185.82	-3185.82	**0.00**	89.82	1254	23	⋯
100-3	-3385.09	-3385.09	**0.00**	**2.09**	-5033.03	-3138.80	37.64	3603.17	-4916.98	-3345.25	31.97	3600.10	-3385.09	-3385.09	**0.00**	2844.36	95300	-3385.09	-3385.09	**0.00**	86.95	1235	20	⋯
100-4	-3278.42	-3278.42	**0.00**	**0.75**	-4991.23	-3076.40	38.36	3603.19	-4876.75	-3158.70	35.23	3600.10	-3278.42	-3278.42	**0.00**	2912.34	95300	-3278.42	-3278.42	**0.00**	79.94	1208	18	⋯
100-5	-3084.12	-3084.12	**0.00**	**1.29**	-4593.16	-2995.90	34.77	3602.94	-4514.89	-2890.19	35.99	3600.10	-3084.12	-3084.12	**0.00**	2818.81	95300	-3084.12	-3084.12	**0.00**	91.48	1209	23	⋯
100-6	-3126.86	-3126.85	**0.00**	**1.00**	-4806.13	-2896.57	39.73	3603.22	-4618.01	-3097.61	32.92	3600.10	-3126.85	-3126.85	**0.00**	2454.65	95300	-3126.86	-3126.85	**0.00**	78.29	1242	22	⋯
100-7	-3504.58	-3504.58	**0.00**	**0.63**	-5386.79	-3316.76	38.43	3603.30	-5256.56	-3430.58	34.74	3600.10	-3504.58	-3504.58	**0.00**	2812.33	95300	-3504.58	-3504.58	**0.00**	77.28	1221	18	⋯
100-8	-3024.77	-3024.76	**0.00**	**2.33**	-4459.30	-2748.72	38.36	3603.27	-4354.33	-2995.68	31.20	3600.10	-3024.76	-3024.76	**0.00**	2831.82	95300	-3024.77	-3024.76	**0.00**	81.87	1220	21	⋯
100-9	-3480.60	-3480.59	**0.00**	**29.39**	-5368.06	-3384.84	36.94	3603.20	-5236.03	-3429.90	34.49	3600.10	-3480.59	-3480.59	**0.00**	2747.11	95300	-3480.60	-3480.59	**0.00**	79.16	1259	25	⋯
1000-1	-32625.28	-32625.04	**0.00**	3600.37	-49193.09	-19348.96	60.67	3603.03	-49170.35	-14800.89	69.90	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32625.08	-32625.04	**0.00**	**924.56**	12027	28	⋯
1000-10	-33169.13	-33168.77	**0.00**	3600.36	-49978.71	-20573.63	58.84	3603.19	-49956.62	-15037.76	69.90	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33168.81	-33168.78	**0.00**	**890.63**	11970	37	⋯
1000-2	$-\infty$	$+\infty$	100.0	$+\infty$	-50583.21	-20807.08	58.87	3603.11	-50557.65	-15218.93	69.90	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33662.62	-33662.59	**0.00**	**791.29**	12061	28	⋯
1000-3	-33222.63	-33222.27	0.00	3600.31	-50594.01	-20294.87	59.89	3602.90	-50575.90	-15222.78	69.90	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33222.29	-33222.26	**0.00**	**962.95**	12314	29	⋯
1000-4	-33806.77	-33806.44	**0.00**	3600.39	-51500.84	-21241.63	58.75	3603.01	-51474.05	-15494.71	69.90	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33806.48	-33806.44	**0.00**	**850.57**	12126	33	⋯
1000-5	-33722.06	-33721.75	**0.00**	3600.38	-51215.44	-20716.09	59.55	3603.02	-51204.62	-15412.52	69.90	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33721.79	-33721.75	**0.00**	**796.21**	12066	32	⋯
1000-6	-35003.45	-35003.05	0.00	3600.39	-53119.27	-21826.65	58.91	3602.99	-53098.91	-15979.34	69.91	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-35003.08	-35003.05	**0.00**	**852.92**	12271	32	⋯
1000-7	-33186.05	-33185.82	**0.00**	3600.35	-50058.29	-20817.38	58.41	3602.85	-50043.99	-15061.40	69.90	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33185.85	-33185.82	**0.00**	**802.41**	12215	34	⋯
1000-8	-33298.53	-33298.26	**0.00**	3600.37	-50246.78	-21376.69	57.46	3603.17	-50222.91	-15118.75	69.90	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33298.30	-33298.27	**0.00**	**796.45**	12123	31	⋯
1000-9	-34140.76	-34140.41	**0.00**	3600.26	-51817.18	-20869.92	59.72	3602.88	-51786.34	-15592.50	69.89	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-34140.46	-34140.43	**0.00**	**963.37**	12212	42	⋯
20-1	-750.11	-750.11	**0.00**	**0.33**	-750.11	-750.11	**0.00**	3283.31	-829.74	-743.85	10.35	3600.10	-750.11	-750.11	**0.00**	511.97	19060	-750.11	-750.11	**0.00**	16.72	262	21	⋯
20-10	-820.54	-820.54	**0.00**	**0.23**	$-\infty$	$+\infty$	100.0	$+\infty$	-883.65	-814.18	7.86	3600.10	-820.54	-820.54	**0.00**	589.63	19060	-820.54	-820.54	**0.00**	16.93	239	12	⋯
20-2	-453.83	-453.83	**0.00**	**0.31**	-454.75	-453.83	0.20	3603.49	-518.59	-446.90	13.82	3600.10	-453.83	-453.83	**0.00**	531.80	19060	-453.83	-453.83	**0.00**	16.37	263	19	⋯
20-3	-836.87	-836.87	**0.00**	**0.31**	-910.42	-836.87	8.08	3603.18	-899.53	-832.73	7.43	3600.10	-836.87	-836.87	**0.00**	526.65	19060	-836.87	-836.87	**0.00**	19.51	247	14	⋯
20-4	-549.20	-549.20	**0.00**	**0.31**	-549.20	-549.20	**0.00**	3238.80	-549.20	-549.20	**0.00**	3482.73	-549.20	-549.20	**0.00**	572.77	19060	-549.20	-549.20	**0.00**	16.35	235	12	⋯
20-5	-853.24	-853.24	**0.00**	**0.29**	$-\infty$	$+\infty$	100.0	$+\infty$	-942.19	-847.92	10.01	3600.10	-853.24	-853.24	**0.00**	515.89	19060	-853.24	-853.24	**0.00**	17.93	246	13	⋯
20-6	-602.90	-602.90	**0.00**	**0.32**	-649.90	$+\infty$	100.00	$+\infty$	-725.94	-594.90	18.05	3600.10	-602.90	-602.90	**0.00**	618.92	19060	-602.90	-602.90	**0.00**	16.40	254	13	⋯
20-7	-523.79	-523.79	**0.00**	**0.33**	-533.51	-523.79	1.82	3603.32	-593.22	-513.36	13.46	3600.10	-523.79	-523.79	**0.00**	619.73	19060	-523.79	-523.79	**0.00**	18.09	260	18	⋯
20-8	-604.94	-604.94	**0.00**	**0.31**	-618.13	-604.77	2.16	3603.18	-669.98	-602.95	10.00	3600.10	-604.94	-604.94	**0.00**	518.30	19060	-604.94	-604.94	**0.00**	18.96	252	20	⋯
20-9	-656.93	-656.93	**0.00**	**0.32**	-656.93	-656.93	**0.00**	2042.39	-707.41	-650.25	8.08	3600.10	-656.93	-656.93	**0.00**	572.68	19060	-656.93	-656.93	**0.00**	17.68	254	20	⋯
200-1	-6500.92	-6500.92	**0.00**	**0.59**	-9818.84	-6138.94	37.48	3603.04	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6500.92	-6500.92	**0.00**	168.89	2393	17	⋯
200-10	-6689.72	-6689.71	**0.00**	**6.48**	-10204.41	-6326.25	38.00	3602.86	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6689.72	-6689.71	**0.00**	161.28	2447	24	⋯
200-2	-6655.61	-6655.61	**0.00**	440.64	-10107.37	-6400.35	36.68	3603.09	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6655.61	-6655.61	**0.00**	**167.63**	2478	26	⋯
200-3	-7272.87	-7272.87	**0.00**	**81.07**	-10975.51	-6810.56	37.95	3602.95	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-7272.87	-7272.87	**0.00**	171.40	2463	23	⋯
200-4	-6728.41	-6728.41	**0.00**	**29.28**	-10212.64	-6422.41	37.11	3602.85	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6728.41	-6728.41	**0.00**	154.08	2481	23	⋯
200-5	-6638.59	-6638.58	**0.00**	623.82	-9971.67	-6013.25	39.70	3603.07	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6638.59	-6638.58	**0.00**	**151.98**	2440	21	⋯
200-6	-6309.71	-6309.70	**0.00**	**139.79**	-9629.87	-6024.45	37.44	3602.83	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6309.71	-6309.70	**0.00**	173.34	2519	24	⋯
200-7	-6667.11	-6667.11	**0.00**	303.44	-10091.45	-6331.91	37.25	3602.94	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6667.11	-6667.11	**0.00**	**168.78**	2392	25	⋯
200-8	-6706.68	-6706.67	**0.00**	**5.20**	-10230.27	-6378.42	37.65	3603.04	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6706.68	-6706.67	**0.00**	172.42	2431	21	⋯
200-9	$-\infty$	$+\infty$	100.0	$+\infty$	-10990.45	-6785.05	38.26	3603.16	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-7131.67	-7131.66	**0.00**	**161.21**	2604	29	⋯

Table 127 Detailed results for problem NLCKP, cost functions f_2

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
10-1	-405.57	-405.57	**0.00**	**0.56**	-405.57	-405.57	**0.00**	7.14	$-\infty$	$+\infty$	100.0	3600.10	-405.57	-405.57	**0.00**	36.55	11730	-405.57	-405.57	**0.00**	1.41	190	28	⋯
10-10	-243.75	-243.75	**0.00**	**0.36**	-243.75	-243.75	**0.00**	3.43	$-\infty$	$+\infty$	100.0	3600.10	-243.75	-243.75	**0.00**	46.23	11730	-243.75	-243.75	**0.00**	2.43	142	20	⋯
10-2	-286.56	-286.56	**0.00**	**0.37**	-286.56	-286.56	**0.00**	4.44	$-\infty$	$+\infty$	100.0	3600.10	-286.56	-286.56	**0.00**	35.84	11730	-286.56	-286.56	**0.00**	1.34	164	23	⋯
10-3	-219.04	-219.04	**0.00**	**0.39**	-219.04	-219.04	**0.00**	4.94	$-\infty$	$+\infty$	100.0	3600.10	-219.04	-219.04	**0.00**	49.52	11730	-219.04	-219.04	**0.00**	2.50	183	25	⋯
10-4	-385.86	-385.86	**0.00**	**0.29**	-385.86	-385.86	**0.00**	3.66	$-\infty$	$+\infty$	100.0	3600.10	-385.86	-385.86	**0.00**	67.75	11730	-385.86	-385.86	**0.00**	0.55	77	13	⋯
10-5	-386.21	-386.21	**0.00**	**0.36**	-386.21	-386.21	**0.00**	3.57	$-\infty$	$+\infty$	100.0	3600.10	-386.21	-386.21	**0.00**	34.81	11730	-386.21	-386.21	**0.00**	0.91	156	22	⋯
10-6	-203.52	-203.52	**0.00**	**0.39**	-203.52	-203.52	**0.00**	3.96	$-\infty$	$+\infty$	100.0	3600.10	-203.52	-203.52	**0.00**	46.79	11730	-203.52	-203.52	**0.00**	1.32	189	23	⋯
10-7	-188.51	-188.51	**0.00**	**0.39**	-188.51	-188.51	**0.00**	3.79	$-\infty$	$+\infty$	100.0	3600.10	-188.51	-188.51	**0.00**	42.16	11730	-188.51	-188.51	**0.00**	2.83	183	21	⋯
10-8	-230.59	-230.59	**0.00**	**1.09**	-230.59	-230.59	**0.00**	13.69	$-\infty$	$+\infty$	100.0	3600.10	-230.59	-230.58	**0.00**	57.82	11730	-230.59	-230.58	**0.00**	3.06	242	29	⋯
10-9	-207.71	-207.71	**0.00**	**0.43**	-207.71	-207.71	**0.00**	5.28	$-\infty$	$+\infty$	100.0	3600.10	-207.71	-207.71	**0.00**	53.78	11730	-207.71	-207.71	**0.00**	2.99	205	25	⋯
100-1	-3204.65	-3198.15	0.20	3600.32	-3280.30	-3197.39	2.53	3603.16	-3456.60	-3193.18	7.62	3600.10	-3198.15	-3198.15	**0.00**	400.29	117300	-3198.15	-3198.15	**0.00**	**12.38**	1865	37	⋯
100-10	-3362.33	-3362.22	0.00	3600.20	-3413.17	-3362.22	1.49	3603.13	-3589.07	-3360.77	6.36	3600.10	-3362.22	-3362.22	**0.00**	551.99	117300	-3362.22	-3362.22	**0.00**	**11.38**	1660	38	⋯
100-2	-3108.24	-3101.05	0.23	3600.29	-3183.84	-3099.81	2.64	3603.15	-3353.33	-3098.45	7.60	3600.10	-3101.06	-3101.06	**0.00**	363.95	117300	-3101.06	-3101.06	**0.00**	**10.62**	1906	37	⋯
100-3	-3316.29	-3315.40	0.03	3600.19	-3383.66	-3315.40	2.02	3603.07	-3557.49	-3284.20	7.68	3600.10	-3315.40	-3315.40	**0.00**	348.26	117300	-3315.40	-3315.40	**0.00**	**11.20**	1770	39	⋯
100-4	-3205.93	-3201.29	0.14	3600.30	-3279.07	-3201.29	2.37	3603.12	$-\infty$	$+\infty$	100.0	3600.10	-3201.29	-3201.29	**0.00**	525.36	117300	-3201.29	-3201.29	**0.00**	**17.95**	1986	50	⋯
100-5	-3034.29	-3031.68	0.09	3600.32	-3098.25	-3030.81	2.18	3603.12	-3259.32	-3029.24	7.06	3600.10	-3031.68	-3031.68	**0.00**	342.57	117300	-3031.68	-3031.68	**0.00**	**12.75**	1769	41	⋯
100-6	-3052.52	-3041.25	0.37	3600.32	-3122.58	-3039.86	2.65	3603.02	-3299.43	-3030.68	8.15	3600.10	-3041.25	-3041.25	**0.00**	334.74	117300	-3041.25	-3041.25	**0.00**	**9.15**	1775	41	⋯
100-7	-3429.89	-3427.40	0.07	3600.19	-3498.22	-3427.40	2.02	3603.32	-3681.78	-3421.53	7.07	3600.10	-3427.40	-3427.40	**0.00**	399.11	117300	-3427.40	-3427.40	**0.00**	**19.91**	2017	50	⋯
100-8	-2966.63	-2966.12	0.02	3600.30	-3025.19	-2966.12	1.95	3603.18	-3167.34	-2944.46	7.04	3600.10	-2966.12	-2966.12	**0.00**	365.25	117300	-2966.12	-2966.12	**0.00**	**10.94**	1756	36	⋯
100-9	-3395.18	-3382.27	0.38	3600.19	-3481.22	-3381.53	2.86	3603.23	$-\infty$	$+\infty$	100.0	3600.10	-3382.28	-3382.27	**0.00**	402.00	117300	-3382.28	-3382.27	**0.00**	**15.32**	1923	47	⋯
1000-1	-32319.65	-31914.74	1.25	3600.37	-32963.46	-31892.05	3.25	3602.93	-34684.29	-30699.76	11.49	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-31918.88	-31918.85	**0.00**	**158.34**	18281	80	⋯
1000-10	-32852.53	-32459.29	1.20	3600.29	-34126.01	-32432.73	4.96	3602.98	-35237.62	-31219.40	11.40	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32460.50	-32460.47	**0.00**	**140.78**	17976	74	⋯
1000-2	-33340.49	-32956.83	1.15	3600.31	-34023.92	-32915.40	3.26	3603.25	-35753.87	-23669.40	33.80	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32958.35	-32958.33	**0.00**	**155.21**	18067	76	⋯
1000-3	-32920.13	-32451.16	1.42	3600.49	-33584.83	-32416.89	3.48	3602.83	-35466.70	-31017.48	12.54	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32458.28	-32458.26	**0.00**	**206.97**	18742	106	⋯
1000-4	-33459.75	-32993.38	1.39	3600.37	-34133.20	-32947.62	3.47	3603.00	-36117.46	-31449.76	12.92	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33004.11	-33004.08	**0.00**	**151.83**	18453	78	⋯
1000-5	-33460.82	-32989.08	1.41	3600.35	-34379.98	-32931.21	4.21	3602.99	-36003.67	-31575.29	12.30	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32991.91	-32991.88	**0.00**	**249.32**	18275	89	⋯
1000-6	-34693.36	-34215.81	1.38	3600.41	-35383.48	-34167.28	3.44	3602.97	-37312.72	-32836.98	12.00	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-34219.68	-34219.66	**0.00**	**195.65**	18649	85	⋯
1000-7	-32860.47	-32470.06	1.19	3600.24	-33526.75	-32431.83	3.27	3602.94	-35272.42	-22116.96	37.30	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32470.49	-32470.45	**0.00**	**158.84**	17987	86	⋯
1000-8	-33006.84	-32587.62	1.27	3600.47	-33668.51	-32551.43	3.32	3602.93	-35409.33	-21689.88	38.75	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32590.84	-32590.81	**0.00**	**109.14**	18085	86	⋯
1000-9	-33795.76	-33351.02	1.32	3600.39	-34471.24	-33304.99	3.38	3602.81	-36313.53	-31958.34	11.99	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33355.38	-33355.35	**0.00**	**229.00**	18621	104	⋯
20-1	-738.37	-738.37	**0.00**	5.26	-738.37	-738.36	**0.00**	110.00	$-\infty$	$+\infty$	100.0	3600.10	-738.37	-738.36	**0.00**	98.45	23460	-738.37	-738.36	**0.00**	**4.07**	353	29	⋯
20-10	-790.71	-790.71	**0.00**	17.74	-790.71	-790.71	**0.00**	223.01	-790.80	-790.71	0.01	3613.51	-790.71	-790.71	**0.00**	85.33	23460	-790.71	-790.71	**0.00**	**5.26**	370	28	⋯
20-2	-442.50	-442.50	**0.00**	25.59	-442.50	-442.50	**0.00**	709.36	-442.93	-442.50	0.10	3623.15	-442.50	-442.50	**0.00**	95.66	23460	-442.50	-442.50	**0.00**	**4.25**	401	30	⋯
20-3	-821.64	-821.64	**0.00**	13.81	-821.64	-821.64	**0.00**	131.05	-821.70	-821.64	0.01	3615.19	-821.64	-821.64	**0.00**	86.19	23460	-821.64	-821.64	**0.00**	**6.12**	380	31	⋯
20-4	-536.67	-536.67	**0.00**	12.56	-536.67	-536.67	**0.00**	86.35	$-\infty$	$+\infty$	100.0	3600.10	-536.67	-536.67	**0.00**	85.21	23460	-536.67	-536.67	**0.00**	**5.44**	327	29	⋯
20-5	-840.23	-840.23	**0.00**	4.48	-840.23	-840.23	**0.00**	152.97	-840.29	-840.23	0.01	3623.49	-840.23	-840.23	**0.00**	69.78	23460	-840.23	-840.23	**0.00**	**2.26**	356	25	⋯
20-6	-587.20	-587.20	**0.00**	7.93	-587.20	-587.20	**0.00**	122.77	$-\infty$	$+\infty$	100.0	3600.10	-587.20	-587.20	**0.00**	70.91	23460	-587.20	-587.20	**0.00**	**3.02**	395	30	⋯
20-7	-515.08	-515.08	**0.00**	3.51	-515.08	-515.08	**0.00**	48.17	$-\infty$	$+\infty$	100.0	3600.10	-515.08	-515.08	**0.00**	75.88	23460	-515.08	-515.08	**0.00**	**2.39**	396	30	⋯
20-8	-592.30	-592.30	**0.00**	**3.76**	-592.30	-592.30	**0.00**	87.17	$-\infty$	$+\infty$	100.0	3600.10	-592.30	-592.30	**0.00**	76.54	23460	-592.30	-592.30	**0.00**	5.06	380	29	⋯
20-9	-644.95	-644.95	**0.00**	7.93	-644.95	-644.95	**0.00**	166.48	-644.96	-644.95	0.00	3631.50	-644.95	-644.95	**0.00**	104.79	23460	-644.95	-644.95	**0.00**	**3.17**	355	26	⋯
200-1	-6391.84	-6342.33	0.77	3600.19	-6532.54	-6340.25	2.94	3603.29	$-\infty$	$+\infty$	100.0	3600.10	-6342.50	-6342.49	**0.00**	1059.22	234600	-6342.50	-6342.49	**0.00**	**35.52**	3786	54	⋯
200-10	-6591.86	-6544.61	0.72	3600.32	-6748.34	-6542.80	3.05	3602.99	$-\infty$	$+\infty$	100.0	3600.10	-6544.62	-6544.61	**0.00**	936.42	234600	-6544.62	-6544.62	**0.00**	**40.33**	3826	68	⋯
200-2	-6551.22	-6489.88	0.94	3600.31	-6694.87	-6487.89	3.09	3602.89	$-\infty$	$+\infty$	100.0	3600.10	-6491.51	-6491.51	**0.00**	666.43	234600	-6491.51	-6491.51	**0.00**	**17.36**	3600	46	⋯
200-3	-7178.44	-7131.79	0.65	3600.33	-7351.36	-7130.02	3.01	3603.04	-7707.15	-7129.46	7.50	3600.10	-7132.33	-7132.33	**0.00**	689.90	234600	-7132.33	-7132.33	**0.00**	**21.05**	3551	46	⋯
200-4	-6630.53	-6578.29	0.79	3600.21	-6787.47	-6575.90	3.12	3603.35	$-\infty$	$+\infty$	100.0	3600.10	-6578.30	-6578.30	**0.00**	855.28	234600	-6578.31	-6578.30	**0.00**	**26.30**	3674	52	⋯
200-5	-6540.15	-6488.32	0.79	3600.33	-6689.00	-6484.22	3.06	3602.86	$-\infty$	$+\infty$	100.0	3600.10	-6488.35	-6488.35	**0.00**	1066.18	234600	-6488.36	-6488.35	**0.00**	**28.66**	3569	50	⋯
200-6	-6202.69	-6143.89	0.95	3600.31	-6344.23	-6141.53	3.19	3603.16	-6719.17	-6134.88	8.70	3600.10	-6144.09	-6144.09	**0.00**	759.14	234600	-6144.10	-6144.09	**0.00**	**36.64**	3900	58	⋯
200-7	-6590.70	-6542.09	0.74	3600.32	-6754.46	-6541.51	3.15	3603.07	$-\infty$	$+\infty$	100.0	3600.10	-6542.40	-6542.39	**0.00**	697.62	234600	-6542.40	-6542.39	**0.00**	**23.01**	3822	50	⋯
200-8	-6605.90	-6548.95	0.86	3600.30	-6756.82	-6543.08	3.16	3602.89	$-\infty$	$+\infty$	100.0	3600.10	-6549.21	-6549.21	**0.00**	639.81	234600	-6549.21	-6549.21	**0.00**	**21.27**	3639	48	⋯
200-9	-7000.32	-6923.57	1.10	3600.34	-7156.82	-6916.40	3.36	3603.10	$-\infty$	$+\infty$	100.0	3600.10	-6923.92	-6923.91	**0.00**	2076.37	234600	-6923.92	-6923.91	**0.00**	**36.60**	3846	64	⋯

Table 128 **Detailed results for problem NLCKP, cost functions f_3**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
10-1	-401.86	-401.86	**0.00**	**0.47**	-401.86	-401.86	**0.00**	7.59	$-\infty$	$+\infty$	100.0	3600.10	-401.86	-401.86	**0.00**	56.54	16650	-401.86	-401.86	**0.00**	2.07	211	30	⋯
10-10	-243.66	-243.66	**0.00**	**0.32**	-243.66	-243.66	**0.00**	3.91	-243.66	-243.66	**0.00**	1.58	-243.66	-243.66	**0.00**	58.27	16650	-243.66	-243.66	**0.00**	1.17	201	12	⋯
10-2	-286.19	-286.19	**0.00**	**0.31**	-286.19	-286.19	**0.00**	4.22	-286.19	-286.19	**0.00**	0.40	-286.19	-286.19	**0.00**	88.54	16650	-286.19	-286.19	**0.00**	1.12	207	14	⋯
10-3	-220.35	-220.35	**0.00**	**0.39**	-220.35	-220.35	**0.00**	6.31	$-\infty$	$+\infty$	100.0	3600.10	-220.35	-220.35	**0.00**	75.04	16650	-220.35	-220.35	**0.00**	1.98	221	27	⋯
10-4	-385.89	-385.89	**0.00**	0.30	-385.89	-385.89	**0.00**	3.90	-385.89	-385.89	**0.00**	**0.23**	-385.89	-385.89	**0.00**	99.33	16650	-385.89	-385.89	**0.00**	1.17	201	14	⋯
10-5	-374.96	-374.96	**0.00**	**0.32**	-374.96	-374.96	**0.00**	4.25	-374.96	-374.96	**0.00**	36.26	-374.96	-374.96	**0.00**	88.38	16650	-374.96	-374.96	**0.00**	4.17	206	21	⋯
10-6	-203.50	-203.50	**0.00**	**0.31**	-203.50	-203.50	**0.00**	4.35	-203.50	-203.50	**0.00**	14.61	-203.50	-203.50	**0.00**	64.96	16650	-203.50	-203.50	**0.00**	1.60	206	17	⋯
10-7	-189.73	-189.73	**0.00**	**0.39**	-189.73	-189.73	**0.00**	4.74	$-\infty$	$+\infty$	100.0	3600.10	-189.73	-189.73	**0.00**	59.61	16650	-189.73	-189.73	**0.00**	3.26	219	25	⋯
10-8	-236.24	-236.24	**0.00**	**1.50**	-236.24	-236.24	**0.00**	46.84	$-\infty$	$+\infty$	100.0	3600.10	-236.24	-236.24	**0.00**	49.93	16650	-236.24	-236.24	**0.00**	1.84	235	28	⋯
10-9	-203.01	-203.01	**0.00**	**0.35**	-203.01	-203.01	**0.00**	5.11	-203.01	-203.01	**0.00**	62.73	-203.01	-203.01	**0.00**	73.33	16650	-203.01	-203.01	**0.00**	3.49	220	23	⋯
100-1	-3210.61	-3177.59	1.03	3600.29	-3788.86	-2871.72	24.21	3603.10	-5245.99	-3177.78	39.42	3600.10	-3178.47	-3178.47	**0.00**	669.65	166500	-3178.47	-3178.47	**0.00**	**14.35**	2160	36	⋯
100-10	-3339.03	-3339.02	**0.00**	596.31	-3854.76	-3057.91	20.67	3603.08	-5404.52	-3338.40	38.23	3600.10	-3339.03	-3339.02	**0.00**	748.14	166500	-3339.02	-3339.02	**0.00**	**11.45**	2112	31	⋯
100-2	-3127.06	-3091.85	1.13	3600.32	-3682.83	-2815.47	23.55	3603.46	-5146.07	-3090.44	39.95	3600.10	-3092.16	-3092.16	**0.00**	736.05	166500	-3092.16	-3092.16	**0.00**	**11.83**	2181	37	⋯
100-3	-3315.58	-3303.22	0.37	3600.19	-3827.26	-3008.64	21.39	3603.15	-5437.46	-3303.07	39.25	3600.10	-3303.25	-3303.25	**0.00**	748.81	166500	-3303.25	-3303.25	**0.00**	**15.98**	2203	40	⋯
100-4	-3201.60	-3176.16	0.79	3600.18	-3806.16	-2871.58	24.55	3603.13	-5263.39	-3171.27	39.75	3600.10	-3176.20	-3176.20	**0.00**	619.20	166500	-3176.20	-3176.20	**0.00**	**13.09**	2159	41	⋯
100-5	-3028.65	-3017.57	0.37	3600.20	-3521.78	-2728.49	22.53	3603.08	-4896.33	-3017.01	38.38	3600.10	-3017.77	-3017.77	**0.00**	929.11	166500	-3017.77	-3017.77	**0.00**	**13.68**	2132	33	⋯
100-6	-3078.61	-3025.95	1.71	3600.32	-3637.86	-2730.26	24.95	3602.92	-5059.79	-3026.66	40.18	3600.10	-3027.34	-3027.33	**0.00**	529.41	166500	-3027.33	-3027.33	**0.00**	**13.64**	2176	39	⋯
100-7	-3401.39	-3383.02	0.54	3600.19	-4115.85	-3013.67	26.78	3603.19	-5631.38	-3381.39	39.95	3600.10	-3383.03	-3383.03	**0.00**	858.76	166500	-3383.03	-3383.03	**0.00**	**20.09**	2232	48	⋯
100-8	-2964.87	-2957.18	0.26	3600.33	-3375.46	-2715.50	19.55	3603.07	-4836.03	-2957.03	38.85	3600.10	-2957.19	-2957.19	**0.00**	610.59	166500	-2957.19	-2957.19	**0.00**	**10.60**	2115	34	⋯
100-9	-3428.59	-3368.51	1.75	3600.30	-4096.94	-3068.62	25.10	3602.92	-5639.38	-3366.92	40.30	3600.10	-3368.62	-3368.61	**0.00**	827.18	166500	-3368.62	-3368.61	**0.00**	**11.24**	2137	42	⋯
1000-1	-32659.07	-31715.88	2.89	3600.45	-39682.18	-27971.46	29.51	3603.18	-53937.76	-31211.26	42.13	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-31726.99	-31726.96	**0.00**	**180.90**	21585	86	⋯
1000-10	-33185.66	-32238.64	2.85	3600.38	-40312.73	-28562.57	29.15	3602.97	-54799.29	-31688.59	42.17	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32255.19	-32255.15	**0.00**	**201.35**	21128	76	⋯
1000-2	-33707.43	-32735.60	2.88	3600.47	-40785.65	-29409.06	27.89	3602.96	-55557.04	-32251.93	41.95	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32746.42	-32746.39	**0.00**	**165.32**	21392	70	⋯
1000-3	-33377.73	-32190.24	3.56	3600.29	-40735.06	-28967.73	28.89	3603.05	-55179.47	-31687.15	42.57	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32201.94	-32201.91	**0.00**	**198.45**	21497	84	⋯
1000-4	-33991.97	-32781.15	3.56	3600.28	-41477.21	-29451.27	28.99	3602.99	-56255.16	-32290.48	42.60	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32804.57	-32804.54	**0.00**	**170.85**	21554	87	⋯
1000-5	-33858.73	-32734.83	3.32	3600.36	-41261.07	-29098.59	29.48	3603.00	-55927.76	-32203.71	42.42	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32748.22	-32748.18	**0.00**	**185.78**	21148	87	⋯
1000-6	-35084.68	-33955.53	3.22	3600.32	-45080.56	-29238.73	35.14	3602.93	-57986.27	-33395.04	42.41	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33970.52	-33970.48	**0.00**	**182.87**	21308	82	⋯
1000-7	-33232.98	-32237.22	3.00	3600.33	-40375.32	-28817.59	28.63	3603.13	-54878.78	-31708.21	42.22	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32253.46	-32253.43	**0.00**	**153.70**	21345	67	⋯
1000-8	-33335.34	-32357.86	2.93	3600.38	-40524.03	-28925.26	28.62	3602.98	-55049.40	-31810.15	42.22	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32375.81	-32375.78	**0.00**	**203.89**	21497	82	⋯
1000-9	-34172.70	-33109.61	3.11	3600.39	-41713.04	-29384.57	29.56	3602.90	-56564.42	-32466.33	42.60	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33126.85	-33126.82	**0.00**	**277.63**	21820	117	⋯
20-1	-734.66	-734.66	**0.00**	**1.53**	-734.66	-734.66	**0.00**	145.77	-734.66	-734.66	**0.00**	3652.97	-734.66	-734.66	**0.00**	140.59	33300	-734.66	-734.66	**0.00**	4.14	438	28	⋯
20-10	-790.20	-790.20	**0.00**	**1.02**	-790.20	-790.20	**0.00**	144.96	$-\infty$	$+\infty$	100.0	3600.10	-790.20	-790.20	**0.00**	132.75	33300	-790.20	-790.20	**0.00**	3.98	434	23	⋯
20-2	-435.69	-435.69	**0.00**	**3.71**	-435.69	-435.69	**0.00**	1004.92	$-\infty$	$+\infty$	100.0	3600.10	-435.69	-435.69	**0.00**	149.51	33300	-435.69	-435.69	**0.00**	4.01	456	31	⋯
20-3	-822.76	-822.76	**0.00**	**0.45**	-822.76	-822.76	**0.00**	107.19	-822.76	-822.76	**0.00**	80.83	-822.76	-822.76	**0.00**	107.63	33300	-822.76	-822.76	**0.00**	3.16	441	25	⋯
20-4	-535.04	-535.04	**0.00**	**0.57**	-535.04	-535.04	**0.00**	54.80	$-\infty$	$+\infty$	100.0	3600.10	-535.04	-535.04	**0.00**	119.95	33300	-535.04	-535.04	**0.00**	4.53	422	27	⋯
20-5	-828.12	-828.12	**0.00**	**0.47**	-828.12	-828.12	**0.00**	84.41	-828.12	-828.12	**0.00**	47.97	-828.12	-828.12	**0.00**	117.82	33300	-828.12	-828.12	**0.00**	3.74	434	21	⋯
20-6	-582.67	-582.67	**0.00**	**1.45**	-582.67	-582.67	**0.00**	334.71	$-\infty$	$+\infty$	100.0	3600.10	-582.67	-582.67	**0.00**	132.54	33300	-582.67	-582.67	**0.00**	4.09	441	25	⋯
20-7	-504.63	-504.63	**0.00**	**2.30**	-504.63	-504.63	**0.00**	651.19	$-\infty$	$+\infty$	100.0	3600.10	-504.63	-504.63	**0.00**	120.18	33300	-504.63	-504.63	**0.00**	4.50	440	31	⋯
20-8	-592.05	-592.04	**0.00**	**1.39**	-592.04	-592.04	**0.00**	173.86	$-\infty$	$+\infty$	100.0	3600.10	-592.05	-592.04	**0.00**	104.22	33300	-592.04	-592.04	**0.00**	2.73	442	29	⋯
20-9	-642.19	-642.19	**0.00**	**0.49**	-642.19	-642.19	**0.00**	56.87	-642.19	-642.19	**0.00**	33.58	-642.19	-642.19	**0.00**	111.06	33300	-642.19	-642.19	**0.00**	2.62	420	21	⋯
200-1	-6417.31	-6299.11	1.84	3600.19	-7767.83	-5499.22	29.21	3602.94	-10617.23	-6296.85	40.69	3600.10	-6300.02	-6300.01	**0.00**	1080.96	333000	-6300.02	-6300.01	**0.00**	**30.00**	4303	48	⋯
200-10	-6608.91	-6475.00	2.03	3600.32	-8043.61	-5699.81	29.14	3603.01	-11037.34	-6470.38	41.38	3600.10	-6475.47	-6475.46	**0.00**	1498.99	333000	-6475.46	-6475.46	**0.00**	**25.55**	4407	46	⋯
200-2	-6597.53	-6455.59	2.15	3600.31	-7974.72	-5117.30	35.83	3602.58	-10944.89	-6451.94	41.05	3600.10	-6456.79	-6456.79	**0.00**	1295.28	333000	-6456.79	-6456.79	**0.00**	**24.58**	4302	50	⋯
200-3	-7173.92	-7067.49	1.48	3600.31	-8704.40	-6085.75	30.08	3603.31	-11814.89	-7066.11	40.19	3600.10	-7069.28	-7069.28	**0.00**	1627.31	333000	-7069.28	-7069.27	**0.00**	**25.24**	4298	46	⋯
200-4	-6704.38	-6543.05	2.41	3600.32	-8096.70	-5726.06	29.28	3602.95	-11069.33	-6540.93	40.91	3600.10	-6544.83	-6544.82	**0.00**	1456.92	333000	-6544.83	-6544.82	**0.00**	**22.72**	4244	41	⋯
200-5	-6568.24	-6456.17	1.71	3600.31	-7910.73	-5097.65	35.56	3603.04	-10788.61	-6451.62	40.20	3600.10	-6457.01	-6457.00	**0.00**	1368.42	333000	-6457.01	-6457.00	**0.00**	**22.69**	4341	48	⋯
200-6	-6272.88	-6111.70	2.57	3600.31	-7585.96	-4915.48	35.20	3603.23	-10393.33	-6106.43	41.25	3600.10	-6113.44	-6113.43	**0.00**	2081.81	333000	-6113.44	-6113.43	**0.00**	**27.92**	4389	53	⋯
200-7	-6616.76	-6488.75	1.93	3600.33	-7979.30	-5794.14	27.39	3603.07	-10901.22	-6485.72	40.50	3600.10	-6488.84	-6488.84	**0.00**	2489.72	333000	-6488.84	-6488.84	**0.00**	**26.51**	4351	54	⋯
200-8	-6637.84	-6483.46	2.33	3600.32	-8068.68	-5737.44	28.89	3602.99	-11012.21	-6480.66	41.15	3600.10	-6486.12	-6486.11	**0.00**	1060.60	333000	-6486.12	-6486.11	**0.00**	**25.30**	4345	43	⋯
200-9	-7059.87	-6868.07	2.72	3600.36	-8630.38	-5898.86	31.65	3603.01	$-\infty$	$+\infty$	100.0	3600.10	-6870.53	-6870.53	**0.00**	3150.81	333000	-6870.53	-6870.52	**0.00**	**38.71**	4547	67	⋯

Table 129 **Detailed results for problem NLCKP, cost functions f_4**

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
10-1	-429.12	-429.12	**0.00**	**2.25**	-429.13	-429.12	0.00	3603.18	−∞	+∞	100.0	3600.10	-429.12	-429.12	**0.00**	115.17	29880	-429.12	-429.12	**0.00**	2.81	263	26	⋯
10-10	-252.63	-252.63	**0.00**	**0.50**	-252.63	-252.63	**0.00**	525.61	−∞	+∞	100.0	3600.10	-252.63	-252.63	**0.00**	101.42	29880	-252.63	-252.63	**0.00**	2.15	266	25	⋯
10-2	-291.70	-291.70	**0.00**	**0.70**	-291.70	-291.70	**0.00**	24.35	−∞	+∞	100.0	3600.10	-291.70	-291.70	**0.00**	115.42	29880	-291.70	-291.70	**0.00**	3.53	276	27	⋯
10-3	-228.86	-228.86	**0.00**	**3.54**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	-228.86	-228.86	**0.00**	112.57	29880	-228.86	-228.86	**0.00**	5.23	310	31	⋯
10-4	-387.55	-387.55	**0.00**	**0.62**	-387.55	-387.55	**0.00**	186.11	−∞	+∞	100.0	3600.10	-387.55	-387.55	**0.00**	111.68	29880	-387.55	-387.55	**0.00**	3.45	257	25	⋯
10-5	-399.23	-399.23	**0.00**	**0.82**	-399.23	-399.23	**0.00**	61.71	−∞	+∞	100.0	3600.10	-399.23	-399.23	**0.00**	109.33	29880	-399.23	-399.23	**0.00**	2.27	253	24	⋯
10-6	-213.18	-213.17	**0.00**	**0.67**	-213.18	-213.17	**0.00**	3603.18	−∞	+∞	100.0	3600.10	-213.17	-213.17	**0.00**	107.32	29880	-213.18	-213.17	**0.00**	3.93	265	24	⋯
10-7	-198.72	-198.72	**0.00**	**0.56**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	-198.72	-198.72	**0.00**	106.41	29880	-198.72	-198.72	**0.00**	4.05	285	25	⋯
10-8	-250.47	-250.47	**0.00**	**0.75**	-250.47	-250.47	**0.00**	3603.31	−∞	+∞	100.0	3600.10	-250.47	-250.47	**0.00**	121.12	29880	-250.47	-250.47	**0.00**	5.09	306	28	⋯
10-9	-225.29	-225.29	**0.00**	**0.55**	−∞	+∞	100.0	+∞	−∞	+∞	100.0	3600.10	-225.29	-225.29	**0.00**	97.78	29880	-225.29	-225.29	**0.00**	1.56	274	22	⋯
100-1	-3376.16	-3376.06	0.00	3600.30	-3977.85	-3370.02	15.28	3603.00	-4241.04	-3366.57	20.62	3600.10	-3376.06	-3376.06	**0.00**	1844.47	298800	-3376.06	-3376.06	**0.00**	**26.84**	2747	38	⋯
100-10	-3505.92	-3505.89	**0.00**	3600.20	-4103.26	-3505.45	14.57	3603.05	-4392.33	-3502.05	20.27	3600.10	-3505.90	-3505.89	**0.00**	1092.74	298800	-3505.90	-3505.89	**0.00**	**21.95**	2706	35	⋯
100-2	-3266.88	-3266.74	0.00	3600.18	-3849.30	-3257.82	15.37	3602.85	-4125.68	-3255.13	21.10	3600.10	-3266.74	-3266.74	**0.00**	1814.41	298798	-3266.74	-3266.74	**0.00**	**24.89**	2789	47	⋯
100-3	-3451.97	-3451.92	0.00	3600.19	-4069.07	-3448.99	15.24	3603.04	−∞	+∞	100.0	3600.10	-3451.92	-3451.92	**0.00**	1584.42	298800	-3451.92	-3451.92	**0.00**	**33.51**	2812	50	⋯
100-4	-3375.69	-3375.56	0.00	3600.32	-3975.38	-3366.53	15.32	3603.14	-4262.82	-3368.41	20.98	3600.10	-3375.56	-3375.56	**0.00**	1591.55	298800	-3375.56	-3375.56	**0.00**	**28.94**	2823	44	⋯
100-5	-3177.92	-3177.89	0.00	3600.19	-3743.85	-3172.03	15.27	3603.43	−∞	+∞	100.0	3600.10	-3177.89	-3177.88	**0.00**	1396.22	298800	-3177.89	-3177.88	**0.00**	**22.89**	2705	33	⋯
100-6	-3201.91	-3201.81	0.00	3600.30	-3764.58	-3195.20	15.12	3603.03	-4041.48	-3191.17	21.04	3600.10	-3201.81	-3201.81	**0.00**	1403.72	298800	-3201.81	-3201.81	**0.00**	**25.90**	2792	44	⋯
100-7	-3627.14	-3627.08	0.00	3600.19	-4247.46	-3617.34	14.84	3603.16	-4588.52	-3610.80	21.31	3600.10	-3627.09	-3627.08	**0.00**	2282.15	298800	-3627.09	-3627.08	**0.00**	**18.19**	2798	40	⋯
100-8	-3082.25	-3082.19	0.00	3600.32	-3642.17	-3079.16	15.46	3603.17	−∞	+∞	100.0	3600.10	-3082.19	-3082.19	**0.00**	1607.59	298800	-3082.19	-3082.19	**0.00**	**27.52**	2732	43	⋯
100-9	-3576.59	-3576.48	0.00	3600.18	-4209.41	-3570.29	15.18	3602.89	−∞	+∞	100.0	3600.10	-3576.48	-3576.48	**0.00**	2700.16	298800	-3576.48	-3576.48	**0.00**	**28.84**	2787	52	⋯
1000-1	-33600.67	-33520.60	0.24	3600.26	-39840.77	-33454.28	16.03	3603.12	-42667.89	-33171.48	22.26	3600.10	−∞	+∞	100.00	3600.01	+∞	-33520.97	-33520.94	**0.00**	**396.67**	26936	72	⋯
1000-10	-34140.77	-34058.95	0.24	3600.35	-40483.04	-33990.58	16.04	3602.95	-43352.36	-33693.92	22.28	3600.10	−∞	+∞	100.00	3600.01	+∞	-34060.09	-34060.06	**0.00**	**533.96**	27773	114	⋯
1000-2	-34628.99	-34541.65	0.25	3600.37	-41052.26	-34468.26	16.04	3603.11	-43950.62	-34192.72	22.20	3600.10	−∞	+∞	100.00	3600.01	+∞	-34544.49	-34544.46	**0.00**	**915.15**	27366	88	⋯
1000-3	-34336.18	-34226.33	0.32	3600.47	-40654.99	-34138.72	16.03	3603.23	-43639.41	-33835.75	22.47	3600.10	−∞	+∞	100.00	3600.01	+∞	-34228.59	-34228.55	**0.00**	**400.68**	27225	86	⋯
1000-4	-34865.65	-34756.38	0.31	3600.22	-41296.42	-34666.57	16.05	3602.96	-44379.79	-34352.09	22.60	3600.10	−∞	+∞	100.00	3600.01	+∞	-34759.35	-34759.31	**0.00**	**266.74**	27386	79	⋯
1000-5	-34899.11	-34803.46	0.27	3600.31	-41352.08	-34718.20	16.04	3603.27	-44311.58	-34431.73	22.30	3600.10	−∞	+∞	100.00	3600.01	+∞	-34803.48	-34803.45	**0.00**	**371.97**	27213	74	⋯
1000-6	-36150.74	-36052.71	0.27	3600.47	-42836.50	-35982.71	16.00	3602.91	-45929.38	-35649.21	22.38	3600.10	−∞	+∞	100.00	3600.01	+∞	-36053.76	-36053.73	**0.00**	**383.15**	26875	83	⋯
1000-7	-34162.36	-34078.06	0.25	3600.24	-40503.79	-34007.35	16.04	3602.86	-43407.48	-33736.86	22.28	3600.10	−∞	+∞	100.00	3600.01	+∞	-34078.69	-34078.66	**0.00**	**397.57**	26894	76	⋯
1000-8	-34330.65	-34248.47	0.24	3600.39	-40705.68	-34185.31	16.02	3602.96	-43579.28	-33886.41	22.24	3600.10	−∞	+∞	100.00	3600.01	+∞	-34250.31	-34250.28	**0.00**	**281.02**	26967	67	⋯
1000-9	-35185.78	-35097.26	0.25	3600.48	-41707.97	-35020.62	16.03	3602.96	-44754.78	-34668.90	22.54	3600.10	−∞	+∞	100.00	3600.01	+∞	-35097.28	-35097.25	**0.00**	**422.78**	27082	84	⋯
20-1	-772.43	-772.43	**0.00**	16.15	−∞	+∞	100.0	+∞	-783.45	-772.37	1.41	3600.10	-772.43	-772.43	**0.00**	302.33	59760	-772.43	-772.43	**0.00**	**4.74**	562	29	⋯
20-10	-830.01	-830.01	**0.00**	28.55	−∞	+∞	100.0	+∞	-853.85	-829.88	2.81	3600.10	-830.01	-830.01	**0.00**	246.73	59759	-830.01	-830.01	**0.00**	**6.18**	544	29	⋯
20-2	-473.00	-473.00	**0.00**	215.85	-475.06	-473.00	0.43	3603.28	-480.85	-473.00	1.63	3600.10	-473.00	-473.00	**0.00**	245.97	59760	-473.00	-473.00	**0.00**	**5.68**	554	29	⋯
20-3	-867.87	-867.87	**0.00**	**0.83**	−∞	+∞	100.0	+∞	-881.29	-867.85	1.52	3600.10	-867.87	-867.87	**0.00**	217.48	59760	-867.87	-867.87	**0.00**	2.58	538	25	⋯
20-4	-555.54	-555.54	**0.00**	36.66	−∞	+∞	100.0	+∞	-565.19	-555.51	1.71	3600.10	-555.54	-555.54	**0.00**	247.05	59760	-555.54	-555.54	**0.00**	**6.45**	541	29	⋯
20-5	-875.82	-875.82	**0.00**	341.24	-877.56	-875.82	0.20	3603.21	-891.39	-875.82	1.75	3600.10	-875.82	-875.82	**0.00**	255.32	59760	-875.82	-875.82	**0.00**	**6.73**	544	24	⋯
20-6	-612.83	-612.83	**0.00**	20.07	-617.99	-612.83	0.84	3603.15	-630.83	-612.82	2.85	3600.10	-612.83	-612.83	**0.00**	365.11	59760	-612.83	-612.83	**0.00**	**7.51**	587	29	⋯
20-7	-547.60	-547.60	**0.00**	**2.77**	-549.76	-547.60	0.39	3603.23	-559.94	-547.60	2.20	3600.10	-547.60	-547.60	**0.00**	211.79	59760	-547.60	-547.60	**0.00**	5.35	568	29	⋯
20-8	-624.18	-624.18	**0.00**	30.65	-625.89	-624.18	0.27	3603.22	-634.81	-624.18	1.67	3600.10	-624.18	-624.18	**0.00**	268.83	59759	-624.18	-624.18	**0.00**	**7.59**	591	35	⋯
20-9	-671.95	-671.95	**0.00**	81.83	-679.14	-671.95	1.06	3603.02	-687.91	-671.86	2.33	3600.10	-671.95	-671.95	**0.00**	260.93	59760	-671.95	-671.95	**0.00**	**8.86**	601	31	⋯
200-1	-6666.94	-6660.61	0.09	3600.19	-7891.31	-6648.40	15.75	3603.33	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	-6660.62	-6660.61	**0.00**	**78.58**	5526	59	⋯
200-10	-6900.97	-6893.73	0.10	3600.32	-8170.05	-6880.73	15.78	3602.88	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	-6893.95	-6893.94	**0.00**	**60.14**	5460	50	⋯
200-2	-6842.62	-6831.77	0.16	3600.32	-8093.99	-6816.27	15.79	3603.07	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	-6831.78	-6831.78	**0.00**	**59.23**	5519	66	⋯
200-3	-7505.69	-7503.57	0.03	3600.22	-8895.09	-7495.01	15.74	3602.63	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	-7503.58	-7503.57	**0.00**	**50.85**	5528	48	⋯
200-4	-6933.94	-6922.12	0.17	3600.33	-8197.77	-6914.51	15.65	3603.71	−∞	+∞	100.0	3600.10	-6922.13	-6922.12	**0.00**	3428.71	597600	-6922.13	-6922.12	**0.00**	**45.15**	5446	49	⋯
200-5	-6816.14	-6806.64	0.14	3600.20	-8070.41	-6793.98	15.82	3603.52	−∞	+∞	100.0	3600.10	-6806.73	-6806.73	**0.00**	2722.52	597600	-6806.73	-6806.73	**0.00**	**45.61**	5574	51	⋯
200-6	-6485.14	-6476.61	0.13	3600.34	-7671.46	-6457.79	15.82	3602.71	-8244.01	-6445.70	21.81	3600.10	-6476.61	-6476.61	**0.00**	2410.52	597599	-6476.61	-6476.61	**0.00**	**37.54**	5449	45	⋯
200-7	-6911.84	-6910.93	0.01	3600.23	-8185.01	-6895.45	15.76	3602.88	−∞	+∞	100.0	3600.10	-6910.94	-6910.93	**0.00**	3255.68	597599	-6910.94	-6910.93	**0.00**	**50.10**	5524	51	⋯
200-8	-6921.15	-6908.43	0.18	3600.31	-8188.03	-6902.81	15.70	3603.02	−∞	+∞	100.0	3600.10	-6909.35	-6909.34	**0.00**	2622.02	597599	-6909.35	-6909.34	**0.00**	**44.76**	5450	48	⋯
200-9	-7325.76	-7311.15	0.20	3600.35	-8660.93	-7297.75	15.74	3603.10	−∞	+∞	100.0	3600.10	-7311.16	-7311.15	**0.00**	3201.10	597600	-7311.16	-7311.15	**0.00**	**45.09**	5603	51	⋯

Table 130 **Detailed results for problem NLCKP, cost functions f_5**

Instance	GUROBI DB	PB	GAP	CPU	SCIP DB	PB	GAP	CPU	COUENNE DB	PB	GAP	CPU	NAIVE DB	PB	GAP	CPU	NP	CN24 DB	PB	GAP	CPU	NP	NIT	⋯
10-1	-401.67	-401.67	**0.00**	**2.93**	-401.67	-401.67	**0.00**	3602.57	−∞	+∞	100.0	3600.10	-401.67	-401.67	**0.00**	83.52	14420	-401.67	-401.67	**0.00**	113.38	158	20	⋯
10-10	-244.68	-244.68	**0.00**	**0.42**	-244.68	-244.68	**0.00**	4.68	-244.68	-244.68	**0.00**	7.19	-244.68	-244.68	**0.00**	81.60	14420	-244.68	-244.68	**0.00**	65.07	91	16	⋯
10-2	-286.25	-286.25	**0.00**	**0.36**	-286.25	-286.25	**0.00**	3.13	-286.25	-286.25	**0.00**	20.80	-286.25	-286.25	**0.00**	86.29	14420	-286.25	-286.25	**0.00**	62.16	96	12	⋯
10-3	-220.85	-220.85	**0.00**	**1.28**	-220.85	-220.85	**0.00**	5.86	−∞	+∞	100.0	3600.10	-220.85	-220.85	**0.00**	80.48	14420	-220.85	-220.85	**0.00**	98.21	143	18	⋯
10-4	-386.13	-386.13	**0.00**	**0.31**	-386.13	-386.13	**0.00**	2.52	-386.12	-386.12	**0.00**	0.60	-386.13	-386.13	**0.00**	84.18	14420	-386.13	-386.13	**0.00**	61.41	82	7	⋯
10-5	-373.60	-373.60	**0.00**	**0.45**	-373.60	-373.60	**0.00**	2.74	-373.60	-373.60	**0.00**	12.45	-373.60	-373.60	**0.00**	101.09	14420	-373.60	-373.60	**0.00**	84.72	117	16	⋯
10-6	-204.48	-204.48	**0.00**	**0.43**	-204.48	-204.48	**0.00**	2.74	-204.48	-204.48	**0.00**	212.82	-204.48	-204.48	**0.00**	73.58	14420	-204.48	-204.48	**0.00**	61.66	96	14	⋯
10-7	-191.50	-191.50	**0.00**	**0.60**	-191.50	-191.50	**0.00**	3.41	−∞	+∞	100.0	3600.10	-191.50	-191.50	**0.00**	81.32	14420	-191.50	-191.50	**0.00**	68.15	119	19	⋯
10-8	-235.97	-235.97	**0.00**	**8.91**	-235.97	-235.97	**0.00**	71.29	−∞	+∞	100.0	3600.10	-235.97	-235.97	**0.00**	88.76	14420	-235.97	-235.97	**0.00**	140.05	197	26	⋯
10-9	-204.94	-204.93	**0.00**	**0.51**	-204.94	-204.93	**0.00**	3604.34	−∞	+∞	100.0	3600.10	-204.93	-204.93	**0.00**	80.83	14420	-204.94	-204.93	**0.00**	80.53	130	21	⋯
100-1	-3416.76	-3178.97	6.96	3600.32	-3410.99	-3183.37	6.67	3602.13	-3490.46	-3164.25	9.35	3600.10	-3184.09	-3184.09	**0.00**	1094.17	144200	-3184.09	-3184.09	**0.00**	**786.60**	1150	23	⋯
100-10	-3529.48	-3340.05	5.37	3600.20	-3517.02	-3333.37	5.22	3602.01	-3581.34	-3330.10	7.02	3600.10	-3341.81	-3341.81	**0.00**	937.57	144200	-3341.81	-3341.81	**0.00**	**688.71**	989	22	⋯
100-2	-3324.37	-3097.54	6.82	3600.31	-3308.36	-3092.76	6.52	3602.12	-3395.46	-3083.87	9.18	3600.10	-3099.89	-3099.89	**0.00**	870.70	144200	-3099.89	-3099.89	**0.00**	**765.78**	1114	25	⋯
100-3	-3502.80	-3304.28	5.67	3600.29	-3483.96	-3306.94	5.08	3602.17	-3550.75	-3302.77	6.98	3600.10	-3307.51	-3307.51	**0.00**	824.31	144200	-3307.51	-3307.50	**0.00**	**753.41**	1038	23	⋯
100-4	-3435.38	-3173.33	7.63	3600.31	-3423.91	-3176.34	7.23	3602.11	-3509.49	-3158.30	10.01	3600.10	-3181.57	-3181.56	**0.00**	860.92	144200	-3181.57	-3181.56	**0.00**	**762.76**	1119	32	⋯
100-5	-3220.16	-3018.70	6.26	3600.30	-3186.79	-3020.07	5.23	3602.14	-3276.32	-3001.99	8.37	3600.10	-3022.26	-3022.26	**0.00**	1016.76	144200	-3022.26	-3022.26	**0.00**	**691.59**	1019	24	⋯
100-6	-3282.51	-3027.08	7.78	3600.20	-3268.39	-2997.88	8.28	3602.02	-3327.50	-3010.40	9.53	3600.10	-3033.13	-3033.13	**0.00**	888.47	144200	-3033.13	-3033.13	**0.00**	**778.54**	1179	27	⋯
100-7	-3677.46	-3383.16	8.00	3600.20	-3689.52	-3366.72	8.75	3601.99	-3740.02	-3369.97	9.89	3600.10	-3388.47	-3388.47	**0.00**	817.73	144200	-3388.48	-3388.47	**0.00**	**721.17**	1106	25	⋯
100-8	-3112.62	-2959.86	4.91	3600.31	-3073.66	-2960.91	3.67	3601.99	-3172.29	-2958.13	6.75	3600.10	-2961.17	-2961.17	**0.00**	833.64	144200	-2961.17	-2961.16	**0.00**	**704.39**	1010	24	⋯
100-9	-3656.19	-3363.93	7.99	3600.30	-3667.42	-3357.32	8.46	3602.01	-3733.96	-3353.57	10.19	3600.10	-3374.93	-3374.93	**0.00**	825.11	144200	-3374.93	-3374.93	**0.00**	**812.92**	1226	24	⋯
1000-1	-34923.70	-31676.63	**9.30**	3600.21	-35386.96	-31685.49	10.46	3602.42	−∞	-20178.12	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
1000-10	-35497.29	-32237.73	**9.18**	3600.34	-35951.31	-32280.03	10.21	3602.10	−∞	-18406.40	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
1000-2	-36010.26	-32733.95	**9.10**	3600.46	-36466.12	-32743.70	10.21	3602.51	−∞	-18623.50	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
1000-3	-35752.06	-32133.88	**10.12**	3600.30	-36222.66	-32244.69	10.98	3602.41	−∞	-20692.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
1000-4	-36374.54	-32758.90	**9.94**	3600.24	-36843.14	-32828.63	10.90	3602.20	−∞	-20716.71	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
1000-5	-36281.90	-32666.80	**9.96**	3600.31	-36754.14	-32790.92	10.78	3602.11	−∞	-20916.45	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
1000-6	-37624.52	-33942.18	**9.79**	3600.45	-38122.66	-34007.50	10.79	3602.10	−∞	-30450.96	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
1000-7	-35557.28	-32205.72	**9.43**	3600.29	-36025.19	-32287.64	10.37	3602.10	−∞	-18426.39	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
1000-8	-35688.84	-32366.35	**9.31**	3600.49	-36151.65	-32415.85	10.33	3602.10	−∞	-18501.94	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
1000-9	-36624.42	-33123.07	**9.56**	3600.36	-37112.91	-33067.01	10.90	3602.09	−∞	-19098.29	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	−∞	+∞	100.00	3600.01	+∞	0	⋯
20-1	-737.14	-737.14	**0.00**	**117.01**	-737.14	-737.14	**0.00**	3604.54	-737.14	-737.14	**0.00**	3670.35	-737.14	-737.14	**0.00**	154.10	28840	-737.14	-737.14	**0.00**	187.86	267	25	⋯
20-10	-794.62	-794.62	**0.00**	**27.04**	-794.62	-794.62	**0.00**	54.75	−∞	+∞	100.0	3600.10	-794.62	-794.62	**0.00**	154.04	28840	-794.62	-794.62	**0.00**	162.92	238	19	⋯
20-2	-437.39	-437.39	**0.00**	1479.36	-437.39	-437.39	0.00	3604.23	-437.61	-437.39	0.05	3613.48	-437.39	-437.39	**0.00**	**163.55**	28840	-437.39	-437.39	**0.00**	213.16	312	23	⋯
20-3	-822.90	-822.90	**0.00**	19.17	-822.90	-822.90	**0.00**	**16.02**	−∞	+∞	100.0	3600.10	-822.90	-822.90	**0.00**	179.24	28840	-822.90	-822.90	**0.00**	156.48	219	17	⋯
20-4	-536.56	-536.56	**0.00**	28.70	-536.57	-536.56	**0.00**	3604.59	-536.56	-536.56	**0.00**	**21.54**	-536.56	-536.56	**0.00**	172.91	28840	-536.57	-536.56	**0.00**	164.19	187	15	⋯
20-5	-829.16	-829.16	**0.00**	30.11	-829.16	-829.16	**0.00**	**16.42**	-829.16	-829.16	**0.00**	3652.33	-829.16	-829.16	**0.00**	157.34	28840	-829.16	-829.16	**0.00**	126.40	191	14	⋯
20-6	-583.87	-583.86	**0.00**	**20.43**	-583.87	-583.86	**0.00**	35.03	−∞	+∞	100.0	3600.10	-583.86	-583.86	**0.00**	176.28	28840	-583.86	-583.86	**0.00**	156.03	231	18	⋯
20-7	-503.87	-503.87	**0.00**	**166.30**	-503.87	-503.87	0.00	3604.31	−∞	+∞	100.0	3600.10	-503.87	-503.87	**0.00**	180.74	28840	-503.87	-503.87	**0.00**	208.79	278	25	⋯
20-8	-593.00	-593.00	**0.00**	**51.40**	-593.00	-593.00	**0.00**	53.25	−∞	+∞	100.0	3600.10	-593.00	-593.00	**0.00**	146.29	28840	-593.00	-593.00	**0.00**	156.69	258	19	⋯
20-9	-643.14	-643.14	**0.00**	**8.43**	-643.14	-643.14	**0.00**	15.90	−∞	+∞	100.0	3600.10	-643.14	-643.14	**0.00**	173.27	28840	-643.14	-643.14	**0.00**	156.30	229	18	⋯
200-1	-6871.23	-6310.55	8.16	3600.32	-6924.82	-6291.28	9.15	3602.05	-6910.90	-6264.86	9.35	3600.10	-6312.94	-6312.93	**0.00**	1604.07	288400	-6312.94	-6312.93	**0.00**	**1452.13**	2179	28	⋯
200-10	-7119.36	-6461.46	9.24	3600.33	-7183.83	-6445.30	10.28	3602.05	-7176.89	-6427.30	10.44	3600.10	-6485.06	-6485.06	**0.00**	2495.13	288400	-6485.06	-6485.06	**0.00**	**1450.53**	2194	28	⋯
200-2	-7053.67	-6461.55	8.39	3600.31	-7109.73	-6429.97	9.56	3602.04	-7103.24	-6429.84	9.48	3600.10	-6469.79	-6469.79	**0.00**	2181.77	288400	-6469.79	-6469.79	**0.00**	**1486.15**	2245	37	⋯
200-3	-7719.30	-7064.55	8.48	3600.31	-7785.86	-7049.37	9.46	3602.05	-7773.03	-7003.31	9.90	3600.10	-7079.09	-7079.09	**0.00**	2397.60	288400	-7079.09	-7079.09	**0.00**	**1375.99**	2105	30	⋯
200-4	-7157.27	-6547.26	8.52	3600.33	-7203.35	-6540.63	9.20	3602.04	-7197.86	-6506.02	9.61	3600.10	-6555.49	-6555.48	**0.00**	2529.05	288400	-6555.49	-6555.48	**0.00**	**1590.70**	2303	33	⋯
200-5	-7020.02	-6464.68	7.91	3600.20	-7081.79	-6461.90	8.75	3602.06	-7064.06	-6436.77	8.88	3600.10	-6469.67	-6469.67	**0.00**	2780.42	288400	-6469.67	-6469.67	**0.00**	**1445.16**	2140	25	⋯
200-6	-6693.47	-6109.84	8.72	3600.21	-6762.96	-6084.22	10.04	3602.04	-6732.61	-6067.21	9.88	3600.10	-6123.28	-6123.27	**0.00**	2782.96	288400	-6123.28	-6123.27	**0.00**	**1542.16**	2301	39	⋯
200-7	-7102.84	-6483.94	8.71	3600.30	-7165.07	-6479.74	9.56	3602.03	-7158.41	-6433.01	10.13	3600.10	-6496.98	-6496.98	**0.00**	1934.19	288400	-6496.98	-6496.98	**0.00**	**1485.92**	2221	38	⋯
200-8	-7146.09	-6490.03	9.18	3600.31	-7201.13	-6494.13	9.82	3602.03	-7169.50	-6444.71	10.11	3600.10	-6499.49	-6499.49	**0.00**	1806.69	288400	-6499.50	-6499.49	**0.00**	**1520.96**	2308	34	⋯
200-9	-7578.68	-6870.95	9.34	3600.32	-7650.97	-6868.55	10.23	3602.05	-7631.49	-6845.93	10.29	3600.10	-6889.43	-6889.43	**0.00**	1558.88	288400	-6889.44	-6889.43	**0.00**	**1523.70**	2353	25	⋯

Table 131 Detailed results for problem NLCKP, cost functions f_6

	GUROBI				**SCIP**				**COUENNE**				**NAIVE**					**CN24**						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-1	-445.22	-445.22	**0.00**	0.39	-445.22	-445.22	**0.00**	3.44	-445.22	-445.22	**0.00**	**0.04**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-445.22	-445.22	**0.00**	0.99	127	25	···
10-10	-252.53	-252.53	**0.00**	0.30	-252.53	-252.53	**0.00**	3.23	-252.53	-252.53	**0.00**	**0.04**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-252.53	-252.53	**0.00**	1.00	111	22	···
10-2	-290.62	-290.62	**0.00**	0.36	-290.62	-290.62	**0.00**	3.07	-290.62	-290.62	**0.00**	**0.04**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-290.62	-290.62	**0.00**	1.22	128	29	···
10-3	-238.57	-238.57	**0.00**	0.31	-238.57	-238.57	**0.00**	3.38	-238.57	-238.57	**0.00**	**0.03**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-238.57	-238.57	**0.00**	0.98	135	28	···
10-4	-387.08	-387.08	**0.00**	0.30	-387.08	-387.08	**0.00**	3.07	-387.08	-387.08	**0.00**	**0.03**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-387.08	-387.08	**0.00**	0.83	102	25	···
10-5	-393.31	-393.31	**0.00**	0.30	-393.31	-393.31	**0.00**	3.33	-393.31	-393.31	**0.00**	**0.03**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-393.31	-393.31	**0.00**	0.94	128	29	···
10-6	-211.41	-211.41	**0.00**	0.34	-211.41	-211.41	**0.00**	3.19	-211.41	-211.41	**0.00**	**0.03**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-211.41	-211.41	**0.00**	1.40	151	28	···
10-7	-205.03	-205.02	**0.00**	0.35	-205.02	-205.02	**0.00**	3.04	-205.02	-205.02	**0.00**	**0.06**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-205.02	-205.02	**0.00**	0.89	147	26	···
10-8	-282.57	-282.57	**0.00**	0.38	-282.57	-282.57	**0.00**	3.03	-282.57	-282.57	**0.00**	**0.03**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-282.57	-282.57	**0.00**	1.39	176	29	···
10-9	-222.57	-222.57	**0.00**	0.39	-222.57	-222.57	**0.00**	3.26	-222.57	-222.57	**0.00**	**0.20**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-222.57	-222.57	**0.00**	1.44	182	30	···
100-1	$-\infty$	$+\infty$	100.0	$+\infty$	-3321.01	-3321.01	**0.00**	**3.54**	-3321.01	-3321.00	**0.00**	3610.55	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3321.01	-3321.01	**0.00**	7.01	1430	47	···
100-10	-3430.42	-3430.41	**0.00**	27.86	-3430.41	-3430.41	**0.00**	**3.19**	-3430.41	-3430.41	**0.00**	5.99	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3430.41	-3430.41	**0.00**	9.30	1247	52	···
100-2	$-\infty$	$+\infty$	100.0	$+\infty$	-3242.86	-3242.86	**0.00**	**3.30**	-3242.86	-3242.86	**0.00**	7.20	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3242.86	-3242.86	**0.00**	7.78	1380	47	···
100-3	$-\infty$	$+\infty$	100.0	$+\infty$	-3420.07	-3420.07	**0.00**	**3.25**	-3420.07	-3420.07	**0.00**	28.47	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3420.07	-3420.07	**0.00**	6.88	1276	42	···
100-4	$-\infty$	$+\infty$	100.0	$+\infty$	-3338.00	-3337.99	**0.00**	**3.22**	-3337.99	-3337.99	**0.00**	64.59	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3338.00	-3337.99	**0.00**	8.73	1428	43	···
100-5	-3128.70	-3128.69	**0.00**	1545.30	-3128.69	-3128.69	**0.00**	**3.38**	-3128.69	-3128.69	**0.00**	2490.42	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3128.69	-3128.69	**0.00**	5.98	1294	48	···
100-6	$-\infty$	$+\infty$	100.0	$+\infty$	-3203.89	-3203.89	**0.00**	**3.51**	-3203.89	-3203.89	**0.00**	3608.91	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3203.89	-3203.89	**0.00**	7.11	1366	50	···
100-7	$-\infty$	$+\infty$	100.0	$+\infty$	-3566.44	-3566.43	**0.00**	**3.57**	-3566.43	-3566.43	**0.00**	3607.23	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3566.43	-3566.43	**0.00**	7.45	1523	46	···
100-8	-3045.35	-3045.35	**0.00**	266.45	-3045.35	-3045.35	**0.00**	2.99	-3045.35	-3045.35	**0.00**	**0.85**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3045.35	-3045.35	**0.00**	9.69	1344	50	···
100-9	$-\infty$	$+\infty$	100.0	$+\infty$	-3567.11	-3567.11	**0.00**	**3.08**	-3567.11	-3567.11	**0.00**	2558.37	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3567.11	-3567.11	**0.00**	10.80	1433	40	···
1000-1	-33051.32	-33051.16	**0.00**	3600.37	-33051.18	-33051.16	**0.00**	**4.69**	-33051.20	-33050.68	0.00	3630.35	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33051.18	-33051.15	**0.00**	84.59	13588	111	···
1000-10	-33599.26	-33599.09	**0.00**	3600.39	-33599.11	-33599.09	**0.00**	**4.93**	-33599.15	-33597.86	0.00	3609.37	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33599.11	-33599.08	**0.00**	93.92	13349	91	···
1000-2	-34093.70	-34093.56	**0.00**	3600.37	-34093.57	-34093.56	**0.00**	**5.20**	-34093.61	-34092.88	0.00	3600.34	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-34093.57	-34093.55	**0.00**	76.91	13378	101	···
1000-3	-33789.59	-33789.40	**0.00**	3600.37	-33789.42	-33789.40	**0.00**	**4.78**	-33789.43	-33788.56	0.00	3614.75	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33789.42	-33789.39	**0.00**	89.65	13766	112	···
1000-4	-34423.62	-34423.36	**0.00**	3600.38	-34423.39	-34423.36	**0.00**	**4.53**	-34423.45	-34422.16	0.00	3630.79	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-34423.38	-34423.35	**0.00**	85.97	13797	96	···
1000-5	-34260.52	-34260.35	**0.00**	3600.36	-34260.36	-34260.34	**0.00**	**4.70**	-34260.41	-34259.37	0.00	3626.60	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-34260.37	-34260.33	**0.00**	82.40	13871	90	···
1000-6	-35537.21	-35537.01	**0.00**	3600.36	-35537.03	-35537.01	**0.00**	**4.79**	-35537.06	-35536.33	0.00	3600.49	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-35537.04	-35537.01	**0.00**	82.75	13799	107	···
1000-7	-33642.02	-33641.86	**0.00**	3600.46	-33641.87	-33641.86	**0.00**	**4.87**	-33641.90	-33640.84	0.00	3605.40	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33641.88	-33641.85	**0.00**	76.83	13617	93	···
1000-8	-33743.40	-33743.24	**0.00**	3600.22	-33743.26	-33743.24	**0.00**	**5.00**	-33743.34	-33742.16	0.00	3608.60	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33743.26	-33743.23	**0.00**	75.55	13550	98	···
1000-9	-34639.14	-34638.95	**0.00**	3600.39	-34638.96	-34638.95	**0.00**	**4.45**	-34639.07	-34636.43	0.01	3614.96	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-34638.97	-34638.94	**0.00**	79.70	13586	109	···
20-1	-770.39	-770.39	**0.00**	**0.38**	-770.39	-770.39	**0.00**	3.54	-770.39	-770.39	**0.00**	1.06	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-770.39	-770.39	**0.00**	2.55	275	29	···
20-10	-840.10	-840.10	**0.00**	**0.74**	-840.10	-840.10	**0.00**	3.50	-840.10	-840.10	**0.00**	5.32	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-840.10	-840.10	**0.00**	1.59	271	33	···
20-2	-471.32	-471.32	**0.00**	0.66	-471.32	-471.32	**0.00**	3.32	-471.32	-471.32	**0.00**	**0.64**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-471.32	-471.32	**0.00**	2.18	320	31	···
20-3	-860.41	-860.41	**0.00**	**0.44**	-860.41	-860.41	**0.00**	3.46	-860.41	-860.41	**0.00**	2.69	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-860.41	-860.40	**0.00**	1.41	279	30	···
20-4	-555.47	-555.47	**0.00**	0.39	-555.47	-555.47	**0.00**	3.26	-555.47	-555.47	**0.00**	**0.08**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-555.47	-555.47	**0.00**	1.20	231	28	···
20-5	-876.55	-876.55	**0.00**	**0.58**	-876.55	-876.55	**0.00**	3.30	-876.55	-876.55	**0.00**	0.72	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-876.55	-876.55	**0.00**	1.96	282	34	···
20-6	-617.55	-617.55	**0.00**	**0.48**	-617.55	-617.55	**0.00**	3.27	-617.55	-617.55	**0.00**	3.12	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-617.55	-617.55	**0.00**	2.26	283	31	···
20-7	-548.16	-548.16	**0.00**	0.68	-548.16	-548.16	**0.00**	3.28	-548.16	-548.16	**0.00**	**0.07**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-548.16	-548.16	**0.00**	2.02	328	35	···
20-8	-626.66	-626.66	**0.00**	0.46	-626.66	-626.66	**0.00**	3.33	-626.66	-626.66	**0.00**	**0.07**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-626.66	-626.66	**0.00**	2.26	294	36	···
20-9	-668.08	-668.07	**0.00**	0.40	-668.07	-668.07	**0.00**	3.18	-668.07	-668.07	**0.00**	**0.12**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-668.08	-668.07	**0.00**	1.66	264	31	···
200-1	-6580.68	-6580.66	**0.00**	3600.31	-6580.67	-6580.66	**0.00**	**3.55**	-6580.66	-6580.66	**0.00**	3613.81	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6580.67	-6580.66	**0.00**	13.21	2722	59	···
200-10	-6800.37	-6800.34	**0.00**	3600.33	-6800.35	-6800.34	**0.00**	**3.35**	-6800.35	-6800.33	**0.00**	3618.00	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6800.35	-6800.34	**0.00**	13.00	2764	58	···
200-2	$-\infty$	$+\infty$	100.0	$+\infty$	-6760.79	-6760.79	**0.00**	**3.19**	-6760.79	-6760.79	**0.00**	3607.44	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6760.79	-6760.79	**0.00**	15.07	2777	58	···
200-3	$-\infty$	$+\infty$	100.0	$+\infty$	-7359.39	-7359.39	**0.00**	**3.24**	-7359.39	-7359.39	**0.00**	3605.89	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-7359.39	-7359.38	**0.00**	13.11	2751	61	···
200-4	$-\infty$	$+\infty$	100.0	$+\infty$	-6855.29	-6855.29	**0.00**	**3.16**	-6855.30	-6855.29	**0.00**	3613.65	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6855.29	-6855.29	**0.00**	13.28	2734	56	···
200-5	$-\infty$	$+\infty$	100.0	$+\infty$	-6721.01	-6721.01	**0.00**	**3.15**	-6721.01	-6721.01	**0.00**	3610.83	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6721.01	-6721.01	**0.00**	13.83	2709	46	···
200-6	$-\infty$	$+\infty$	100.0	$+\infty$	-6425.10	-6425.10	**0.00**	**3.36**	-6425.12	-6425.03	0.00	3616.03	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6425.10	-6425.09	**0.00**	12.63	2727	58	···
200-7	-6771.16	-6771.14	**0.00**	3600.30	-6771.14	-6771.14	**0.00**	**3.21**	-6771.16	-6771.09	**0.00**	3614.04	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6771.14	-6771.14	**0.00**	20.17	2901	58	···
200-8	-6814.04	-6814.02	**0.00**	3600.31	-6814.03	-6814.02	**0.00**	**3.58**	-6814.02	-6814.02	**0.00**	159.63	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6814.02	-6814.02	**0.00**	14.85	2796	53	···
200-9	-7262.51	-7262.48	**0.00**	3600.31	-7262.48	-7262.48	**0.00**	**3.52**	-7262.50	-7262.44	**0.00**	3613.00	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-7262.48	-7262.47	**0.00**	15.52	2912	52	···

Table 132 Detailed results for problem NLCKP, cost functions f_7

	GUROBI				SCIP				COUENNE				NAIVE					CN24						⋯
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	⋯
10-1	-375.42	-375.42	0.00	0.36	-375.42	-375.42	0.00	3.70	-375.42	-375.42	0.00	0.16	-375.42	-375.42	0.00	39.26	9360	-375.42	-375.42	0.00	1.32	115	14	⋯
10-10	-240.72	-240.72	0.00	0.29	-240.72	-240.72	0.00	3.44	-240.72	-240.72	0.00	0.04	-240.72	-240.72	0.00	43.35	9360	-240.72	-240.72	0.00	0.85	113	13	⋯
10-2	-284.85	-284.85	0.00	0.27	-284.85	-284.85	0.00	3.58	-284.85	-284.85	0.00	0.05	-284.85	-284.85	0.00	32.84	9360	-284.85	-284.85	0.00	0.79	121	13	⋯
10-3	-209.76	-209.76	0.00	0.29	-209.76	-209.76	0.00	3.53	-209.76	-209.76	0.00	0.12	-209.76	-209.76	0.00	30.98	9360	-209.76	-209.76	0.00	1.68	146	17	⋯
10-4	-385.85	-385.85	0.00	0.29	-385.85	-385.85	0.00	3.10	-385.85	-385.85	0.00	0.05	-385.85	-385.85	0.00	26.54	9360	-385.85	-385.85	0.00	0.64	106	12	⋯
10-5	-361.30	-361.30	0.00	0.29	-361.30	-361.30	0.00	3.48	-361.30	-361.30	0.00	0.03	-361.30	-361.30	0.00	34.09	9360	-361.30	-361.30	0.00	1.13	122	13	⋯
10-6	-198.67	-198.67	0.00	0.29	-198.67	-198.67	0.00	3.28	-198.67	-198.67	0.00	0.04	-198.67	-198.67	0.00	43.93	9360	-198.67	-198.67	0.00	0.72	134	12	⋯
10-7	-181.69	-181.69	0.00	0.29	-181.69	-181.69	0.00	3.45	-181.69	-181.69	0.00	0.07	-181.69	-181.69	0.00	33.49	9360	-181.69	-181.69	0.00	1.25	151	14	⋯
10-8	-219.48	-219.48	0.00	0.31	-219.48	-219.48	0.00	3.26	-219.48	-219.48	0.00	0.05	-219.48	-219.48	0.00	31.14	9360	-219.48	-219.48	0.00	1.22	151	15	⋯
10-9	-199.89	-199.89	0.00	0.29	-199.89	-199.89	0.00	3.21	-199.89	-199.89	0.00	0.23	-199.89	-199.89	0.00	36.34	9360	-199.89	-199.89	0.00	1.12	128	12	⋯
100-1	-3153.54	-3153.54	0.00	0.34	-3153.54	-3153.54	0.00	4.23	-3153.54	-3153.54	0.00	5.52	-3153.54	-3153.54	0.00	804.36	93600	-3153.54	-3153.54	0.00	6.43	1351	20	⋯
100-10	-3324.78	-3324.78	0.00	0.38	-3324.78	-3324.78	0.00	3.80	-3324.78	-3324.78	0.00	8.97	-3324.79	-3324.78	0.00	515.85	93600	-3324.79	-3324.78	0.00	5.03	1238	19	⋯
100-2	-3065.31	-3065.31	0.00	0.35	-3065.31	-3065.31	0.00	3.97	-3065.31	-3065.31	0.00	5.83	-3065.31	-3065.31	0.00	529.11	93600	-3065.31	-3065.31	0.00	5.60	1272	17	⋯
100-3	-3282.69	-3282.69	0.00	0.43	-3282.69	-3282.69	0.00	4.12	-3282.69	-3282.69	0.00	11.80	-3282.70	-3282.69	0.00	531.46	93600	-3282.70	-3282.69	0.00	10.44	1337	25	⋯
100-4	-3152.35	-3152.35	0.00	0.29	-3152.35	-3152.35	0.00	3.90	-3152.35	-3152.35	0.00	7.21	-3152.35	-3152.35	0.00	498.78	93600	-3152.35	-3152.35	0.00	8.72	1322	23	⋯
100-5	-2994.73	-2994.73	0.00	0.37	-2994.73	-2994.73	0.00	3.43	-2994.73	-2994.73	0.00	7.99	-2994.74	-2994.73	0.00	489.55	93600	-2994.74	-2994.73	0.00	5.23	1280	17	⋯
100-6	-2990.85	-2990.85	0.00	0.28	-2990.85	-2990.85	0.00	4.01	-2990.85	-2990.85	0.00	10.07	-2990.85	-2990.85	0.00	900.44	93600	-2990.85	-2990.85	0.00	8.62	1319	20	⋯
100-7	-3352.24	-3352.24	0.00	0.48	-3352.24	-3352.24	0.00	3.93	-3352.24	-3352.24	0.00	7.37	-3352.24	-3352.24	0.00	721.68	93600	-3352.24	-3352.24	0.00	9.82	1400	27	⋯
100-8	-2944.39	-2944.39	0.00	0.27	-2944.39	-2944.39	0.00	3.38	-2944.39	-2944.39	0.00	7.95	-2944.39	-2944.39	0.00	518.82	93600	-2944.39	-2944.39	0.00	6.87	1326	19	⋯
100-9	-3328.77	-3328.77	0.00	0.26	-3328.77	-3328.77	0.00	4.04	-3328.77	-3328.77	0.00	14.44	-3328.77	-3328.77	0.00	730.28	93600	-3328.77	-3328.77	0.00	7.23	1323	17	⋯
1000-1	-31519.40	-31519.37	0.00	6.02	-31519.39	-31519.38	0.00	19.60	-31519.38	-31519.38	0.00	2793.39	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-31519.39	-31519.38	0.00	127.68	13435	45	⋯
1000-10	-32054.26	-32054.26	0.00	13.48	-32054.29	-32054.26	0.00	22.71	-32054.26	-32054.26	0.00	2308.40	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32054.27	-32054.26	0.00	120.80	13337	34	⋯
1000-2	-32569.49	-32569.47	0.00	5.03	-32569.47	-32569.47	0.00	17.00	-32569.47	-32569.47	0.00	1700.61	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32569.48	-32569.47	0.00	84.39	13271	43	⋯
1000-3	-31971.99	-31971.98	0.00	9.69	-31971.98	-31971.98	0.00	19.22	-31971.98	-31971.98	0.00	2364.43	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-31971.99	-31971.98	0.00	136.80	13078	43	⋯
1000-4	-32536.24	-32536.21	0.00	7.57	-32536.24	-32536.21	0.00	68.33	-32536.21	-32536.21	0.00	1175.12	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32536.23	-32536.21	0.00	295.15	12603	47	⋯
1000-5	-32489.59	-32489.57	0.00	9.19	-32489.60	-32489.57	0.00	23.51	$-\infty$	$+\infty$	100.0	3600.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32489.58	-32489.57	0.00	93.49	13186	34	⋯
1000-6	-33740.96	-33740.93	0.00	5.11	-33740.94	-33740.93	0.00	19.31	-33740.93	-33740.93	0.00	2288.11	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33740.94	-33740.93	0.00	129.80	13132	42	⋯
1000-7	-32062.81	-32062.81	0.00	1.31	-32062.83	-32062.81	0.00	16.39	-32062.81	-32062.81	0.00	909.76	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32062.83	-32062.81	0.00	108.07	12819	42	⋯
1000-8	-32161.46	-32161.43	0.00	4.74	-32161.46	-32161.43	0.00	43.87	-32161.43	-32161.43	0.00	3581.70	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32161.45	-32161.43	0.00	181.38	13247	37	⋯
1000-9	-32890.09	-32890.05	0.00	2.30	-32890.08	-32890.05	0.00	17.81	-32890.05	-32890.05	0.00	1067.58	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32890.07	-32890.05	0.00	105.58	13148	29	⋯
20-1	-722.80	-722.80	0.00	0.28	-722.80	-722.80	0.00	3.40	-722.80	-722.80	0.00	0.47	-722.80	-722.80	0.00	46.77	18720	-722.80	-722.80	0.00	1.92	266	14	⋯
20-10	-782.07	-782.07	0.00	0.28	-782.07	-782.07	0.00	3.41	-782.07	-782.07	0.00	0.09	-782.07	-782.07	0.00	49.95	18720	-782.07	-782.07	0.00	1.09	261	13	⋯
20-2	-427.79	-427.79	0.00	0.29	-427.79	-427.79	0.00	3.56	-427.79	-427.79	0.00	0.79	-427.79	-427.79	0.00	55.90	18720	-427.79	-427.79	0.00	1.90	273	15	⋯
20-3	-808.06	-808.06	0.00	0.30	-808.06	-808.06	0.00	3.50	-808.06	-808.06	0.00	0.33	-808.06	-808.06	0.00	55.30	18720	-808.06	-808.06	0.00	1.28	261	13	⋯
20-4	-535.08	-535.08	0.00	0.30	-535.08	-535.08	0.00	3.34	-535.08	-535.08	0.00	0.16	-535.08	-535.08	0.00	64.38	18720	-535.08	-535.08	0.00	1.01	240	12	⋯
20-5	-805.26	-805.26	0.00	0.31	-805.26	-805.26	0.00	3.45	-805.26	-805.26	0.00	0.37	-805.26	-805.26	0.00	56.18	18720	-805.26	-805.26	0.00	2.31	246	15	⋯
20-6	-575.63	-575.63	0.00	0.29	-575.63	-575.63	0.00	3.62	-575.63	-575.63	0.00	0.06	-575.63	-575.63	0.00	55.36	18720	-575.63	-575.63	0.00	1.78	277	13	⋯
20-7	-491.31	-491.31	0.00	0.29	-491.31	-491.31	0.00	3.54	-491.31	-491.31	0.00	0.45	-491.31	-491.31	0.00	57.47	18720	-491.31	-491.31	0.00	2.63	292	18	⋯
20-8	-578.86	-578.86	0.00	0.30	-578.86	-578.86	0.00	3.23	-578.86	-578.86	0.00	0.64	-578.86	-578.86	0.00	55.04	18720	-578.86	-578.86	0.00	1.73	266	13	⋯
20-9	-634.30	-634.30	0.00	0.30	-634.30	-634.30	0.00	3.35	-634.30	-634.30	0.00	0.58	-634.30	-634.30	0.00	51.28	18720	-634.30	-634.30	0.00	1.76	249	13	⋯
200-1	-6273.55	-6273.55	0.00	0.51	-6273.55	-6273.55	0.00	4.13	-6273.55	-6273.55	0.00	13.62	-6273.56	-6273.55	0.00	1585.31	187200	-6273.55	-6273.55	0.00	14.17	2642	20	⋯
200-10	-6430.44	-6430.44	0.00	0.60	-6430.44	-6430.44	0.00	3.53	-6430.44	-6430.44	0.00	61.43	-6430.45	-6430.44	0.00	554.84	187200	-6430.44	-6430.44	0.00	13.74	2660	27	⋯
200-2	-6408.21	-6408.21	0.00	0.63	-6408.21	-6408.21	0.00	4.10	-6408.21	-6408.21	0.00	32.32	-6408.22	-6408.21	0.00	1849.91	187200	-6408.21	-6408.21	0.00	11.34	2689	21	⋯
200-3	-7023.53	-7023.53	0.00	0.30	-7023.53	-7023.53	0.00	4.79	-7023.53	-7023.53	0.00	32.06	-7023.54	-7023.53	0.00	2021.03	187200	-7023.54	-7023.53	0.00	14.19	2571	22	⋯
200-4	-6483.01	-6483.01	0.00	0.50	-6483.01	-6483.01	0.00	4.20	-6483.01	-6483.01	0.00	79.13	-6483.02	-6483.01	0.00	2887.21	187200	-6483.01	-6483.01	0.00	13.47	2687	21	⋯
200-5	-6420.56	-6420.56	0.00	0.35	-6420.56	-6420.56	0.00	4.39	-6420.56	-6420.56	0.00	69.18	-6420.57	-6420.56	0.00	1597.98	187200	-6420.56	-6420.56	0.00	13.01	2603	21	⋯
200-6	-6061.56	-6061.56	0.00	0.65	-6061.56	-6061.56	0.00	4.71	-6061.56	-6061.56	0.00	12.66	-6061.56	-6061.56	0.00	2050.85	187200	-6061.56	-6061.56	0.00	10.51	2691	20	⋯
200-7	-6433.84	-6433.84	0.00	0.48	-6433.84	-6433.84	0.00	5.08	-6433.84	-6433.84	0.00	86.15	-6433.84	-6433.84	0.00	2636.14	187200	-6433.84	-6433.84	0.00	10.91	2703	18	⋯
200-8	-6447.35	-6447.35	0.00	1.20	-6447.35	-6447.35	0.00	4.31	-6447.35	-6447.35	0.00	30.60	-6447.36	-6447.35	0.00	2789.90	187200	-6447.35	-6447.35	0.00	11.64	2688	21	⋯
200-9	-6818.14	-6818.14	0.00	0.53	-6818.14	-6818.14	0.00	3.98	-6818.14	-6818.14	0.00	36.43	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6818.14	-6818.14	0.00	15.04	2676	23	⋯

Table 133 Detailed results for problem NLCKP, cost functions f_8

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-1	-426.93	-426.93	**0.00**	0.29	-426.92	-426.92	**0.00**	3.61	-426.92	-426.92	**0.00**	**0.06**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-426.93	-426.92	**0.00**	1.24	119	30	···
10-10	-249.04	-249.04	**0.00**	0.28	-249.04	-249.04	**0.00**	3.24	-249.04	-249.04	**0.00**	**0.06**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-249.04	-249.04	**0.00**	1.85	111	26	···
10-2	-289.30	-289.30	**0.00**	0.26	-289.30	-289.30	**0.00**	3.31	-289.30	-289.30	**0.00**	**0.07**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-289.30	-289.30	**0.00**	1.66	103	25	···
10-3	-230.77	-230.77	**0.00**	0.29	-230.77	-230.77	**0.00**	3.43	-230.77	-230.77	**0.00**	**0.08**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-230.77	-230.77	**0.00**	1.02	133	30	···
10-4	-387.44	-387.44	**0.00**	0.24	-387.44	-387.44	**0.00**	3.16	-387.44	-387.44	**0.00**	**0.05**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-387.44	-387.44	**0.00**	0.72	89	23	···
10-5	-387.25	-387.25	**0.00**	0.26	-387.25	-387.25	**0.00**	3.28	-387.25	-387.25	**0.00**	**0.07**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-387.25	-387.25	**0.00**	0.85	110	28	···
10-6	-208.77	-208.77	**0.00**	0.29	-208.77	-208.77	**0.00**	3.21	-208.77	-208.77	**0.00**	**0.07**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-208.77	-208.77	**0.00**	1.06	119	27	···
10-7	-199.28	-199.28	**0.00**	0.30	-199.28	-199.28	**0.00**	3.12	-199.28	-199.28	**0.00**	**0.06**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-199.28	-199.28	**0.00**	1.09	127	27	···
10-8	-262.52	-262.52	**0.00**	0.35	-262.52	-262.52	**0.00**	3.18	-262.52	-262.52	**0.00**	**0.06**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-262.52	-262.52	**0.00**	2.36	168	32	···
10-9	-214.41	-214.41	**0.00**	0.31	-214.41	-214.41	**0.00**	3.23	-214.41	-214.41	**0.00**	**0.06**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-214.41	-214.41	**0.00**	1.35	162	30	···
100-1	-3261.87	$+\infty$	100.00	$+\infty$	-3261.85	-3261.85	**0.00**	13.32	-3261.85	-3261.85	**0.00**	104.35	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3261.85	-3261.85	**0.00**	**8.06**	1301	56	···
100-10	-3392.17	$+\infty$	100.00	$+\infty$	-3392.16	-3392.16	**0.00**	**6.38**	-3392.16	-3392.16	**0.00**	26.01	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3392.16	-3392.16	**0.00**	6.88	1130	50	···
100-2	-3181.88	-3181.86	**0.00**	3600.25	-3181.86	-3181.86	**0.00**	9.54	-3181.86	-3181.86	**0.00**	45.49	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3181.86	-3181.86	**0.00**	**7.74**	1242	50	···
100-3	$-\infty$	$+\infty$	100.0	$+\infty$	-3373.52	-3373.52	**0.00**	**4.79**	-3373.52	-3373.52	**0.00**	856.13	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3373.52	-3373.52	**0.00**	5.07	1109	45	···
100-4	$-\infty$	$+\infty$	100.0	$+\infty$	-3271.65	-3271.65	**0.00**	9.31	-3271.65	-3271.65	**0.00**	237.93	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3271.65	-3271.65	**0.00**	**5.68**	1236	49	···
100-5	-3083.19	$+\infty$	100.00	$+\infty$	-3083.18	-3083.18	**0.00**	10.24	-3083.18	-3083.18	**0.00**	828.25	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3083.18	-3083.18	**0.00**	**4.89**	1157	49	···
100-6	$-\infty$	$+\infty$	100.0	$+\infty$	-3130.04	-3130.04	**0.00**	**5.41**	-3130.04	-3130.04	**0.00**	1072.89	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3130.04	-3130.04	**0.00**	6.41	1187	49	···
100-7	$-\infty$	$+\infty$	100.0	$+\infty$	-3487.01	-3487.00	**0.00**	25.58	-3487.00	-3487.00	**0.00**	20.93	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3487.01	-3487.00	**0.00**	**7.26**	1374	53	···
100-8	$-\infty$	$+\infty$	100.0	$+\infty$	-3011.08	-3011.08	**0.00**	**4.07**	-3011.08	-3011.08	**0.00**	10.63	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3011.08	-3011.08	**0.00**	5.15	1182	45	···
100-9	$-\infty$	$+\infty$	100.0	$+\infty$	-3483.93	-3483.93	**0.00**	10.88	-3483.93	-3483.93	**0.00**	669.48	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-3483.93	-3483.93	**0.00**	**8.27**	1270	47	···
1000-1	-32508.27	-32507.83	0.00	3600.27	-32508.01	-32507.82	**0.00**	3603.18	-32508.07	-32507.30	0.00	3629.81	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-32507.84	-32507.81	**0.00**	**79.39**	11976	122	···
1000-10	-33040.64	-33040.16	0.00	3600.41	-33040.33	-33040.16	**0.00**	3603.18	-33040.33	-33039.67	0.00	3625.83	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33040.18	-33040.15	**0.00**	**71.82**	11740	124	···
1000-2	-33537.46	-33537.01	0.00	3600.41	-33537.15	-33537.00	**0.00**	3603.19	-33537.21	-33536.45	0.00	3607.20	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33537.02	-33536.99	**0.00**	**63.31**	11793	105	···
1000-3	-33120.71	-33120.21	0.00	3600.28	-33120.43	-33120.20	**0.00**	3602.90	-33120.40	-33119.41	0.00	3622.33	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33120.22	-33120.19	**0.00**	**73.66**	12392	101	···
1000-4	-33751.44	-33750.95	0.00	3600.28	-33751.17	-33750.94	**0.00**	3603.06	-33751.07	-33750.38	0.00	3627.63	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33750.95	-33750.93	**0.00**	**73.20**	12449	92	···
1000-5	-33625.53	-33625.02	0.00	3600.40	-33625.22	-33625.02	**0.00**	3602.68	-33625.26	-33624.52	0.00	3620.61	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33625.03	-33625.01	**0.00**	**81.30**	12549	112	···
1000-6	-34881.34	-34880.83	0.00	3600.42	-34881.08	-34880.82	**0.00**	3602.93	-34881.02	-34880.07	0.00	3618.85	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-34880.84	-34880.81	**0.00**	**69.17**	12097	111	···
1000-7	-33064.98	-33064.50	0.00	3600.29	-33064.74	-33064.50	**0.00**	3602.94	-33064.63	-33063.80	0.00	3621.71	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33064.51	-33064.49	**0.00**	**70.15**	12259	104	···
1000-8	-33172.64	-33172.16	0.00	3600.26	-33172.37	-33172.16	**0.00**	3602.88	-33172.26	-33172.16	**0.00**	3598.92	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-33172.18	-33172.15	**0.00**	**69.17**	11914	100	···
1000-9	-34006.93	-34006.43	0.00	3600.43	-34006.61	-34006.43	**0.00**	3602.69	-34006.68	-34005.88	0.00	3608.10	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-34006.45	-34006.42	**0.00**	**78.20**	12075	119	···
20-1	-755.05	-755.05	**0.00**	0.54	-755.05	-755.05	**0.00**	3.59	-755.05	-755.05	**0.00**	**0.16**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-755.05	-755.05	**0.00**	2.13	262	34	···
20-10	-820.84	-820.84	**0.00**	0.39	-820.84	-820.84	**0.00**	3.08	-820.84	-820.84	**0.00**	**0.14**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-820.84	-820.84	**0.00**	2.68	227	33	···
20-2	-456.22	-456.22	**0.00**	**0.64**	-456.22	-456.22	**0.00**	3.85	-456.22	-456.22	**0.00**	0.78	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-456.22	-456.22	**0.00**	2.33	302	37	···
20-3	-844.75	-844.75	**0.00**	0.45	-844.75	-844.75	**0.00**	3.55	-844.75	-844.75	**0.00**	**0.15**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-844.75	-844.75	**0.00**	1.79	238	39	···
20-4	-547.71	-547.71	**0.00**	**0.42**	-547.71	-547.71	**0.00**	3.23	-547.71	-547.71	**0.00**	1.01	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-547.71	-547.71	**0.00**	1.44	203	29	···
20-5	-857.24	-857.24	**0.00**	0.55	-857.24	-857.24	**0.00**	3.24	-857.24	-857.24	**0.00**	**0.10**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-857.24	-857.24	**0.00**	2.64	224	31	···
20-6	-603.14	-603.14	**0.00**	1.99	-603.14	-603.14	**0.00**	3.29	-603.14	-603.14	**0.00**	**0.10**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-603.14	-603.14	**0.00**	2.88	273	31	···
20-7	-530.01	-530.01	**0.00**	**0.68**	-530.01	-530.01	**0.00**	3.47	-530.01	-530.01	**0.00**	0.97	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-530.01	-530.01	**0.00**	1.97	285	35	···
20-8	-611.80	-611.80	**0.00**	0.61	-611.80	-611.80	**0.00**	3.34	-611.80	-611.80	**0.00**	**0.10**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-611.80	-611.80	**0.00**	1.75	265	34	···
20-9	-657.55	-657.55	**0.00**	0.52	-657.55	-657.55	**0.00**	3.21	-657.55	-657.55	**0.00**	**0.11**	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-657.55	-657.55	**0.00**	1.72	240	35	···
200-1	-6466.33	-6466.27	**0.00**	3600.26	-6466.28	-6466.27	**0.00**	20.26	-6466.27	-6466.27	**0.00**	3612.68	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6466.27	-6466.27	**0.00**	**10.43**	2373	60	···
200-10	-6663.21	-6663.15	**0.00**	3600.29	-6663.16	-6663.15	**0.00**	47.10	-6663.15	-6663.15	**0.00**	3614.04	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6663.15	-6663.15	**0.00**	**15.10**	2439	65	···
200-2	-6636.40	-6636.33	0.00	3600.27	-6636.33	-6636.33	**0.00**	248.14	-6636.35	-6636.30	**0.00**	3626.67	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6636.33	-6636.33	**0.00**	**14.06**	2432	62	···
200-3	$-\infty$	$+\infty$	100.0	$+\infty$	-7237.32	-7237.31	**0.00**	155.85	-7237.31	-7237.31	**0.00**	183.69	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-7237.31	-7237.31	**0.00**	**14.62**	2377	68	···
200-4	$-\infty$	$+\infty$	100.0	$+\infty$	-6726.34	-6726.33	**0.00**	53.63	-6726.33	-6726.33	**0.00**	975.08	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6726.34	-6726.33	**0.00**	**11.16**	2419	59	···
200-5	-6613.48	$+\infty$	100.00	$+\infty$	-6613.43	-6613.43	**0.00**	3603.70	-6613.44	-6613.41	**0.00**	3629.23	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6613.43	-6613.42	**0.00**	**13.76**	2317	60	···
200-6	$-\infty$	$+\infty$	100.0	$+\infty$	-6297.79	-6297.78	**0.00**	87.65	-6297.78	-6297.78	**0.00**	3340.81	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6297.79	-6297.78	**0.00**	**15.78**	2523	66	···
200-7	-6652.22	-6652.15	0.00	3600.25	-6652.16	-6652.15	**0.00**	296.39	-6652.17	-6652.13	**0.00**	3621.13	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6652.15	-6652.15	**0.00**	**11.98**	2617	69	···
200-8	-6679.92	-6679.85	**0.00**	3600.30	-6679.86	-6679.85	**0.00**	20.00	-6679.89	-6679.81	0.00	3634.48	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-6679.85	-6679.85	**0.00**	**14.60**	2436	58	···
200-9	-7097.76	-7097.68	0.00	3600.27	-7097.68	-7097.67	**0.00**	**3.84**	-7097.68	-7097.67	**0.00**	3618.71	$-\infty$	$+\infty$	100.00	3600.01	$+\infty$	-7097.68	-7097.67	**0.00**	12.83	2576	63	···

Table 134 Detailed results for problem NLCKP, cost functions f_9

	GUROBI				SCIP				COUENNE				NAIVE					CN24						···
Instance	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	DB	PB	GAP	CPU	NP	DB	PB	GAP	CPU	NP	NIT	···
10-1	-421.12	-421.12	0.00	34.24	-421.12	-421.12	0.00	4.67	-421.12	-421.12	0.00	3607.04	−∞	+∞	100.00	3600.01	+∞	-421.12	-421.12	0.00	41.57	211	22	···
10-10	-248.49	-248.49	0.00	0.65	-248.58	-248.58	0.00	3.53	-248.58	-248.58	0.00	1757.99	−∞	+∞	100.00	3600.01	+∞	-248.58	-248.58	0.00	18.51	218	20	···
10-2	-287.89	-287.89	0.00	0.36	-287.89	-287.89	0.00	3.78	-287.89	-287.89	0.00	0.36	−∞	+∞	100.00	3600.01	+∞	-287.89	-287.89	0.00	12.62	177	11	···
10-3	-223.46	-223.46	0.00	23.94	-223.45	-223.45	0.00	4.52	-223.45	-223.45	0.00	9.04	−∞	+∞	100.00	3600.01	+∞	-223.45	-223.45	0.00	20.37	185	19	···
10-4	-385.84	-385.84	0.00	0.32	-385.84	-385.84	0.00	3.80	-385.84	-385.84	0.00	3.16	−∞	+∞	100.00	3600.01	+∞	-385.84	-385.84	0.00	13.01	209	7	···
10-5	-397.77	-397.77	0.00	52.28	-397.76	-397.76	0.00	6.57	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	-397.76	-397.76	0.00	40.74	239	24	···
10-6	-204.38	-204.38	0.00	0.55	-204.38	-204.38	0.00	3.91	-204.38	-204.38	0.00	5.29	−∞	+∞	100.00	3600.01	+∞	-204.38	-204.38	0.00	26.49	182	8	···
10-7	-208.47	-208.47	0.00	21.03	-208.47	-208.47	0.00	3.86	-208.47	-208.47	0.00	94.71	−∞	+∞	100.00	3600.01	+∞	-208.47	-208.47	0.00	15.89	198	24	···
10-8	-228.15	-228.15	0.00	152.85	-228.15	-228.15	0.00	4.88	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	-228.15	-228.15	0.00	27.72	193	20	···
10-9	-230.77	-230.77	0.00	4.01	-230.77	-230.77	0.00	4.53	-230.77	-230.77	0.00	3639.79	−∞	+∞	100.00	3600.01	+∞	-230.77	-230.77	0.00	21.06	216	24	···
100-1	-3613.17	-3292.56	8.87	3600.32	-3604.00	-3196.11	11.32	3603.21	-3611.19	-2820.34	21.90	3600.10	−∞	+∞	100.00	3600.01	+∞	-3322.39	-3322.38	0.00	267.08	2046	27	···
100-10	-3643.02	-3428.28	5.89	3600.19	-3649.48	-3331.95	8.70	3603.44	-3650.51	-3126.58	14.35	3600.10	−∞	+∞	100.00	3600.01	+∞	-3447.91	-3447.91	0.00	277.51	1924	25	···
100-2	-3531.79	-3249.42	8.00	3600.31	-3524.90	-3216.10	8.76	3603.20	-3531.40	-2904.75	17.75	3600.10	−∞	+∞	100.00	3600.01	+∞	-3265.55	-3265.54	0.00	255.29	2124	32	···
100-3	-3674.37	-3419.38	6.94	3600.30	-3665.21	-3331.35	9.11	3602.96	-3675.89	-2933.46	20.20	3600.10	−∞	+∞	100.00	3600.01	+∞	-3424.80	-3424.80	0.00	237.45	2019	29	···
100-4	-3638.49	-3329.44	8.49	3600.32	-3633.46	-3178.14	12.53	3603.01	-3635.23	-2788.08	23.30	3600.10	−∞	+∞	100.00	3600.01	+∞	-3342.73	-3342.73	0.00	236.29	2184	27	···
100-5	-3326.12	-3106.08	6.62	3600.30	-3338.76	-3117.97	6.61	3603.31	-3343.00	-2626.82	21.42	3600.10	−∞	+∞	100.00	3600.01	+∞	-3141.45	-3141.45	0.00	223.92	2036	27	···
100-6	-3517.35	-3157.79	10.22	3600.20	-3513.67	-3087.04	12.14	3603.26	-3516.46	-2731.21	22.33	3600.10	−∞	+∞	100.00	3600.01	+∞	-3196.78	-3196.78	0.00	293.57	2109	32	···
100-7	-3941.87	-3606.18	8.52	3600.32	-3931.69	-3411.23	13.24	3603.29	-3940.59	-2604.44	33.91	3600.10	−∞	+∞	100.00	3600.01	+∞	-3610.13	-3610.13	0.00	239.54	2002	32	···
100-8	-3255.44	-3070.53	5.68	3600.32	-3253.07	-2902.33	10.78	3603.15	-3256.62	-2754.28	15.43	3600.10	−∞	+∞	100.00	3600.01	+∞	-3083.97	-3083.97	0.00	218.01	1959	29	···
100-9	-3923.67	-3527.44	10.10	3600.20	-3904.68	-3399.82	12.93	3603.07	-3921.45	-2901.13	26.02	3600.10	−∞	+∞	100.00	3600.01	+∞	-3544.69	-3544.68	0.00	248.56	2041	31	···
1000-1	-35911.38	-28934.20	19.43	3600.36	-35917.34	-23123.21	35.62	3602.79	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-33286.14	-33286.12	0.00	2727.00	20413	45	···
1000-10	-36468.94	-29607.42	18.81	3600.36	-36489.62	-28913.46	20.76	3602.88	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-33928.82	-33928.79	0.00	2590.77	20616	65	···
1000-2	-36902.14	-29807.06	19.23	3600.38	-36919.68	-29180.04	20.96	3602.88	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-34008.36	-34008.33	0.00	2683.43	20338	51	···
1000-3	-36909.28	-28277.29	23.39	3600.47	-36942.36	-29097.62	21.24	3603.03	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-33962.81	-33962.78	0.00	2853.83	20348	50	···
1000-4	-37563.35	-25125.57	33.11	3600.48	-37581.63	-15924.55	57.63	3603.00	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-34417.13	-34417.10	0.00	2788.65	20009	47	···
1000-5	-37373.38	-29204.28	21.86	3600.47	-37386.23	-30479.98	18.47	3603.06	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-34708.14	-34708.10	0.00	2691.80	20001	58	···
1000-6	-38751.90	-29848.66	22.97	3600.47	-38760.38	-32819.22	15.33	3603.53	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-35738.35	-35738.32	0.00	2636.51	20740	83	···
1000-7	-36521.28	-28377.13	22.30	3600.25	-36529.95	-20243.51	44.58	3603.02	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-33891.32	-33891.29	0.00	2460.69	20365	47	···
1000-8	-36661.02	-27404.87	25.25	3600.47	-36678.98	-21139.49	42.37	3602.81	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-33725.88	-33725.85	0.00	3128.02	20485	76	···
1000-9	-37854.26	-27795.98	26.57	3600.38	-37861.15	-16167.26	57.30	3603.10	−∞	0.00	100.00	3600.10	−∞	+∞	100.00	3600.01	+∞	-34783.08	-34783.04	0.00	2517.40	20192	46	···
20-1	-778.46	-734.99	5.58	3600.27	-745.48	-745.48	0.00	26.56	-784.22	-718.49	8.38	3600.10	−∞	+∞	100.00	3600.01	+∞	-745.48	-745.48	0.00	45.10	382	20	···
20-10	-875.07	-862.55	1.43	3600.22	-862.53	-862.53	0.00	602.36	-907.38	-811.04	10.62	3600.10	−∞	+∞	100.00	3600.01	+∞	-862.53	-862.53	0.00	75.69	491	35	···
20-2	-480.79	-450.19	6.36	3600.29	-451.56	-451.56	0.00	45.33	-499.05	-405.72	18.70	3600.10	−∞	+∞	100.00	3600.01	+∞	-451.56	-451.56	0.00	43.99	418	22	···
20-3	-869.93	-869.63	0.03	3600.29	-869.62	-869.62	0.00	67.28	-890.83	-869.36	2.41	3600.10	−∞	+∞	100.00	3600.01	+∞	-869.62	-869.62	0.00	55.31	435	27	···
20-4	-552.52	-552.52	0.00	390.12	-552.51	-552.51	0.00	13.14	-571.28	-527.13	7.73	3600.10	−∞	+∞	100.00	3600.01	+∞	-552.51	-552.51	0.00	50.87	411	22	···
20-5	-904.79	-861.79	4.75	3600.30	-862.10	-862.10	0.00	31.45	-946.76	-737.26	22.13	3600.10	−∞	+∞	100.00	3600.01	+∞	-862.10	-862.10	0.00	62.72	421	23	···
20-6	-645.15	-607.34	5.86	3600.30	-607.34	-607.34	0.00	39.32	-656.20	-518.92	20.92	3600.10	−∞	+∞	100.00	3600.01	+∞	-607.34	-607.34	0.00	50.33	395	21	···
20-7	-546.46	-523.41	4.22	3600.29	-523.84	-523.84	0.00	20.95	-536.31	-517.04	3.59	3600.10	−∞	+∞	100.00	3600.01	+∞	-523.84	-523.84	0.00	43.27	368	22	···
20-8	-629.65	-627.31	0.37	3600.30	-627.31	-627.31	0.00	55.22	-617.30	-617.30	0.00	3606.54	−∞	+∞	100.00	3600.01	+∞	-627.31	-627.31	0.00	55.31	430	26	···
20-9	-666.26	-666.26	0.00	2507.07	-666.25	-666.25	0.00	32.68	-685.18	-658.15	3.94	3600.10	−∞	+∞	100.00	3600.01	+∞	-666.25	-666.25	0.00	47.13	387	22	···
200-1	-7173.96	-6553.18	8.65	3600.20	-7177.97	-6149.48	14.33	3603.06	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	-6589.05	-6589.05	0.00	432.85	4155	34	···
200-10	-7452.05	-6390.91	14.24	3600.33	-7458.77	-4969.99	33.37	3602.89	-7454.15	-3881.00	47.94	3600.10	−∞	+∞	100.00	3600.01	+∞	-6783.90	-6783.89	0.00	505.08	4138	37	···
200-2	-7383.88	-6390.29	13.46	3600.32	-7387.04	-6431.82	12.93	3603.40	-7385.17	-4232.52	42.69	3600.10	−∞	+∞	100.00	3600.01	+∞	-6802.41	-6802.40	0.00	524.45	4388	37	···
200-3	-8004.83	-6711.18	16.16	3600.33	-8011.45	-6403.63	20.07	3603.08	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	-7432.82	-7432.82	0.00	513.64	4137	37	···
200-4	-7451.20	-6704.03	10.03	3600.33	-7453.72	-6494.35	12.87	3608.72	-7451.47	-5820.82	21.88	3600.10	−∞	+∞	100.00	3600.01	+∞	-6902.98	-6902.97	0.00	538.70	4196	33	···
200-5	-7272.61	-6801.87	6.47	3600.20	-7275.47	-6474.77	11.01	3602.99	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	-6839.37	-6839.37	0.00	533.94	4059	34	···
200-6	-7042.53	-6073.04	13.77	3600.23	-7045.54	-6064.78	13.92	3602.80	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	-6402.27	-6402.27	0.00	518.25	4108	29	···
200-7	-7353.02	-6540.41	11.05	3600.30	-7357.05	-6474.58	11.99	3603.05	−∞	+∞	100.0	3600.10	−∞	+∞	100.00	3600.01	+∞	-6876.98	-6876.97	0.00	493.91	4074	33	···
200-8	-7471.84	-6785.59	9.18	3600.32	-7476.31	-6596.89	11.76	3602.80	-7471.30	-5616.16	24.83	3600.10	−∞	+∞	100.00	3600.01	+∞	-6904.57	-6904.56	0.00	583.63	4154	34	···
200-9	-8052.08	-6947.65	13.72	3600.32	-8058.30	-6638.00	17.63	3603.01	-8056.10	-5782.70	28.22	3600.10	−∞	+∞	100.00	3600.01	+∞	-7249.91	-7249.91	0.00	510.56	4036	34	···

Table 135 Detailed results for problem NLCKP, cost functions f_{11}